

MHT-CET Biology Full Questions

Standard XI prescribed chapters + complete Standard XII | Repaired release v2.1.0

Questions	2,120
Standard XI / XII	436 / 1,684
Difficulty labels	Easy, Medium, Hard, Challenging
Answer-key balance	A = B = C = D = 530
Figures	44 assets linked to 82 questions
Validation	PASS

This review copy presents every deployable record in chapter order. Figure-linked questions include their original release asset. Difficulty labels are visible for editorial review; a production test renderer may hide them from candidates.

Standard 11 | Chapter 01

Biomolecules

BIO-11-CH01-0001 | Easy | Bioelements

The inorganic matter of a living cell is grouped into prime elements, secondary elements and trace elements. Which set of elements forms the prime elements?

- A. Carbon, hydrogen, oxygen and nitrogen
- B. Carbon, hydrogen and oxygen
- C. Nitrogen, phosphorus and potassium
- D. Iron, zinc and copper

BIO-11-CH01-0002 | Medium | Bioelements

Nitrogen and chlorine are both required by living cells, but they are placed in different groups of the inorganic matter of the cell. Which statement places both of them correctly?

- A. Nitrogen is a secondary element and chlorine is a trace element
- B. Nitrogen is a prime element and chlorine is a secondary element
- C. Nitrogen is a prime element and chlorine is a trace element
- D. Nitrogen is a trace element and chlorine is a secondary element

BIO-11-CH01-0003 | Hard | Organic and inorganic matter of the cell

Consider the following three placements in the scheme of biomolecules of a living system. I. Polynucleotides are macromolecules. II. Glycerol is a micromolecule. III. Alkaloids are macromolecules. Which of these placements are correct?

- A. I and III only
- B. II and III only
- C. I, II and III
- D. I and II only

BIO-11-CH01-0004 | Medium | Macromolecules and micromolecules

Which of the following pairs names a macromolecule first and a micromolecule second?

- A. Simple sugars and polysaccharides
- B. Fatty acids and glycerol
- C. Polynucleotides and polysaccharides
- D. Polypeptides and amino acids

BIO-11-CH01-0005 | Medium | Organic and inorganic matter of the cell

The matter of a living cell is first divided into organic and inorganic. Where do essential oils and lectins belong in this scheme?

- A. Organic matter, as micromolecules
- B. Organic matter, as secondary metabolites
- C. Organic matter, as macromolecules
- D. Inorganic matter, along with the trace elements

BIO-11-CH01-0006 | Hard | Secondary metabolites

Consider the following statements about secondary metabolites. I. They are small organic molecules. II. They are essential for the growth, development and reproduction of the organism that produces them. III. They are produced by several bacteria, fungi and plants. Which statements are correct?

- A. I and III only (secondary-metabolite statements)
- B. I and II only (secondary-metabolite statements)
- C. II and III only (secondary-metabolite statements)
- D. I, II and III (secondary-metabolite statements)

BIO-11-CH01-0007 | Easy | General formula of carbohydrates

Carbohydrates have the general formula $(\text{CH}_2\text{O})_n$. What is the ratio of hydrogen atoms to oxygen atoms in such a molecule, and which familiar substance shows the same ratio?

- A. 1:2, the ratio found in water
- B. 2:1, the ratio found in water
- C. 1:1, the ratio found in water
- D. 2:1, the ratio found in carbon dioxide

BIO-11-CH01-0008 | Medium | General formula of carbohydrates

Which of the following molecular formulae is consistent with the general formula $(\text{CH}_2\text{O})_n$ for a carbohydrate?

- A. $\text{C}_5\text{H}_{12}\text{O}_5$
- B. $\text{C}_6\text{H}_{10}\text{O}_6$
- C. $\text{C}_5\text{H}_{10}\text{O}_5$
- D. $\text{C}_4\text{H}_8\text{O}_3$

BIO-11-CH01-0009 | Medium | General formula of carbohydrates

A monosaccharide obeys the general formula $(\text{CH}_2\text{O})_n$ and one of its molecules contains 10 hydrogen atoms. To which series does this sugar belong, and how many oxygen atoms does one molecule contain?

- A. Pentose, 10 oxygen atoms
- B. Hexose, 6 oxygen atoms
- C. Pentose, 5 oxygen atoms
- D. Tetrose, 4 oxygen atoms

BIO-11-CH01-0010 | Hard | Monosaccharides

A monosaccharide named in the classification of carbohydrates contains 7 carbon atoms per molecule and obeys the general formula $(\text{CH}_2\text{O})_n$. Which sugar is it, and how many hydrogen atoms does one of its molecules contain?

- A. Sedoheptulose, 7 hydrogen atoms
- B. Sedoheptulose, 12 hydrogen atoms
- C. Erythrose, 14 hydrogen atoms
- D. Sedoheptulose, 14 hydrogen atoms

BIO-11-CH01-0011 | Easy | Monosaccharides

Glyceraldehyde is the named example of which group of monosaccharides?

- A. Tetrose
- B. Pentose
- C. Heptose
- D. Triose

BIO-11-CH01-0012 | Medium | Monosaccharides

Which of the following pairs a named monosaccharide correctly with the number of carbon atoms in its molecule?

- A. Ribose, 6 carbon atoms
- B. Erythrose, 4 carbon atoms
- C. Galactose, 5 carbon atoms
- D. Sedoheptulose, 6 carbon atoms

BIO-11-CH01-0013 | Challenging | Disaccharides

A carbohydrate Z contains 12 carbon atoms, 22 hydrogen atoms and 11 oxygen atoms per molecule. On hydrolysis Z yields two hexose units, one of which is fructose. Which statement about Z is correct?

- A. Z is maltose, and its hydrogen to oxygen ratio is 2:1
- B. Z is sucrose, and its hydrogen to oxygen ratio is greater than 2:1
- C. Z is lactose, and its hydrogen to oxygen ratio is 1:1
- D. Z is sucrose, and its hydrogen to oxygen ratio is 2:1

BIO-11-CH01-0014 | Easy | Disaccharides

Which of the following pairs consists of a monosaccharide followed by a disaccharide?

- A. Maltose and galactose
- B. Sucrose and lactose
- C. Galactose and maltose
- D. Ribose and fructose

BIO-11-CH01-0015 | Challenging | Disaccharides

Three disaccharides are described. X yields two molecules of the same monosaccharide on hydrolysis. Y yields glucose and galactose. Z is the form in which carbohydrate is transported in plants. Identify X, Y and Z.

- A. X maltose, Y lactose, Z sucrose
- B. X sucrose, Y lactose, Z maltose
- C. X maltose, Y sucrose, Z lactose
- D. X lactose, Y maltose, Z sucrose

BIO-11-CH01-0016 | Easy | Glycosidic bond

Maltose, starch and a nucleoside all contain the same kind of linkage, formed by condensation, holding a sugar unit to the group next to it. Which linkage is this?

- A. Peptide bond
- B. Glycosidic bond
- C. Ester bond
- D. Hydrogen bond

BIO-11-CH01-0017 | Medium | Glycosidic bond

An unbranched polysaccharide chain is built from 250 monosaccharide units joined one after another, each glycosidic bond holding two neighbouring units together. How many glycosidic bonds does this chain contain?

- A. 250
- B. 249
- C. 125
- D. 251

BIO-11-CH01-0018 | Medium | Reducing sugars

Monosaccharides act as reducing sugars in Benedict's test. What change does the copper of Benedict's reagent undergo?

- A. Cu^+ is converted to Cu^{2+}
- B. Cu^{2+} is converted to metallic copper
- C. Cu^+ is converted to metallic copper
- D. Cu^{2+} is converted to Cu^+

BIO-11-CH01-0019 | Hard | Blood glucose

The normal level of glucose in human blood is about 90 mg per 100 ml of blood. Taking this level, what mass of glucose is present in a total blood volume of 5 litres?

- A. 0.9 g
- B. 4.5 g
- C. 45 g
- D. 9 g

BIO-11-CH01-0020 | Easy | Starch

Starch is made up of which two components?

- A. Amylose and glycogen
 - B. Amylopectin and cellulose
 - C. Cellulose and glycogen
 - D. Amylose and amylopectin
-

BIO-11-CH01-0021 | Medium | Starch

Which statement about amylose and amylopectin is correct?

- A. Both are components of starch
 - B. Amylose is a component of starch and amylopectin of glycogen
 - C. Amylose is a component of cellulose and amylopectin of starch
 - D. Both are heteropolysaccharides
-

BIO-11-CH01-0022 | Hard | Cellulose

In cellulose, the sugar units within a single molecule are joined by one kind of bond, while neighbouring cellulose molecules are held to one another by a different kind of bond. Which combination is correct?

- A. Hydrogen bonds within a molecule and glycosidic bonds between molecules
 - B. Glycosidic bonds within a molecule and peptide bonds between molecules
 - C. Glycosidic bonds within a molecule and hydrogen bonds between molecules
 - D. Ester bonds within a molecule and hydrogen bonds between molecules
-

BIO-11-CH01-0023 | Hard | Polysaccharides

Homopolysaccharides are built from a single kind of monosaccharide unit and generally serve the cell as a reserve of carbohydrate. Which of the following is the exception, serving as a structural material instead?

- A. Glycogen
 - B. Starch
 - C. Cellulose
 - D. Amylopectin
-

BIO-11-CH01-0024 | Hard | Polysaccharides

A homopolysaccharide is built from only one kind of monosaccharide unit, while a heteropolysaccharide is built from more than one kind. Which of the following is a heteropolysaccharide?

- A. Glycogen
 - B. Hyaluronic acid
 - C. Starch
 - D. Cellulose
-

BIO-11-CH01-0025 | Easy | Chitin

A student examines the hard outer covering of a beetle and is told that its chief organic component is a polysaccharide. Which polysaccharide is it?

- A. Cellulose
 - B. Glycogen
 - C. Chitin
 - D. Heparin
-

BIO-11-CH01-0026 | Medium | Glycogen

In animals the reserve carbohydrate is stored as a polysaccharide. Which polysaccharide is this, and how is it classified?

- A. Glycogen, a homopolysaccharide
- B. Glycogen, a heteropolysaccharide
- C. Chitin, a homopolysaccharide
- D. Heparin, a heteropolysaccharide

BIO-11-CH01-0027 | Easy | Elemental composition of lipids

Two purified organic compounds are analysed. In compound X the hydrogen to oxygen ratio is exactly 2:1, and in compound Y the same ratio is greater than 2:1. Which identification of the two compounds is correct?

- A. X is a carbohydrate and Y is a lipid
- B. X is a lipid and Y is a carbohydrate
- C. Both X and Y are carbohydrates
- D. Both X and Y are lipids

BIO-11-CH01-0028 | Easy | Fatty acids

Which pair of fatty acids listed in the prescribed account consists only of saturated fatty acids?

- A. Oleic acid and linoleic acid
- B. Palmitic acid and oleic acid
- C. Palmitic acid and stearic acid
- D. Stearic acid and linoleic acid

BIO-11-CH01-0029 | Medium | Fatty acids

Which statement matches the descriptions of fatty acid occurrence given in the prescribed account?

- A. Linoleic acid occurs in nearly all fats and oleic acid occurs in many seed oils
- B. Oleic acid occurs in nearly all fats and linoleic acid occurs in many seed oils
- C. Palmitic acid occurs in nearly all fats and oleic acid occurs in many seed oils
- D. Oleic acid occurs in nearly all fats and stearic acid occurs in many seed oils

BIO-11-CH01-0030 | Easy | Simple lipids

Chemically, the simple lipids are best described as which of the following?

- A. Fatty acids combined with a phosphate group
- B. Fatty acids combined with a sugar residue
- C. Esters of fatty acids with various alcohols
- D. Sterol rings combined with fatty acid chains

BIO-11-CH01-0031 | Medium | Simple lipids

Fats and waxes are both grouped as simple lipids. Which difference between them is correct?

- A. A fat is an ester of fatty acids with glycerol, while a wax is an ester of a long-chain fatty acid with a long-chain alcohol
- B. A fat is an ester of a long-chain fatty acid with a long-chain alcohol, while a wax is an ester of fatty acids with glycerol
- C. A fat contains a phosphate group, while a wax contains glycerol
- D. A fat contains a sugar residue, while a wax contains a sterol nucleus

BIO-11-CH01-0032 | Easy | Simple lipids

The prescribed account gives the fats an alternative name that reflects their alcohol component. That name is:

- A. Phospholipids
- B. Cerebrosides
- C. Phytosterols
- D. Triglycerides

BIO-11-CH01-0033 | Medium | Compound lipids

Compound lipids are of two kinds. Which pairing of a compound lipid with its characteristic additional component is correct?

- A. Phospholipid with a sugar and glycolipid with a phosphate group
- B. Phospholipid with a phosphate group and glycolipid with a sterol nucleus
- C. Phospholipid with a phosphate group and glycolipid with a sugar
- D. Phospholipid with glycerol only and glycolipid with a phosphate group

BIO-11-CH01-0034 | Medium | Derived lipids

In exactly one of the following sets, every substance named is placed among the derived lipids in the prescribed account. Which set is it?

- A. Cholesterol, lecithin and vitamin D
- B. Cholesterol, progesterone and vitamin D
- C. Phytosterol, cerebroside and testosterone
- D. Diosgenin, vanaspati ghee and progesterone

BIO-11-CH01-0035 | Hard | Compound lipids

A compound lipid is named after the extra group it carries in addition to fatty acid and alcohol. Using this rule, a student concludes that cerebroside must contain phosphate because they are compound lipids. Which correction is required?

- A. Cerebroside is a phospholipid and carry phosphate, so no correction is needed
- B. Cerebroside is a derived lipid and carry neither a sugar nor phosphate
- C. Cerebroside is a glycolipid and carry a sugar, not phosphate
- D. Cerebroside is a simple lipid and carry only a fatty acid and an alcohol

BIO-11-CH01-0036 | Medium | Classification of lipids

Three lipids are listed. I is a triglyceride of oleic acid and glycerol, II is lecithin, and III is a phytosterol. Which assignment to the three lipid classes is correct?

- A. I is a compound lipid, II is a simple lipid and III is a derived lipid
- B. I is a simple lipid, II is a compound lipid and III is a derived lipid
- C. I is a derived lipid, II is a compound lipid and III is a simple lipid
- D. I is a simple lipid, II is a derived lipid and III is a compound lipid

BIO-11-CH01-0037 | Medium | Membrane lipids

A purified lipid is dispersed in an aqueous medium and arranges itself into a bilayer of the kind described for the cell membrane. Which class of lipid is being used?

- A. Triglycerides
- B. Waxes
- C. Phytosterols
- D. Phospholipids

BIO-11-CH01-0038 | Hard | Derived lipids

A drug manufacturer needs a plant sterol as the starting material for preparing steroid hormones such as progesterone. Which prescribed plant source provides the appropriate sterol?

- A. Cholesterol from Dioscorea
- B. Diosgenin from Dioscorea
- C. Lecithin from Dioscorea
- D. Diosgenin from flax

BIO-11-CH01-0039 | Hard | Elemental composition of lipids

In carbohydrates the hydrogen to oxygen ratio is exactly 2:1, while in lipids it is greater than 2:1. Two molecules each contain 18 oxygen atoms. Molecule X is a carbohydrate and molecule Y is a lipid. Which statement about their hydrogen atoms is correct?

- A. X has 36 hydrogen atoms and Y has fewer than 36
- B. X has 36 hydrogen atoms and Y has more than 36
- C. X has 9 hydrogen atoms and Y has more than 9
- D. X has 36 hydrogen atoms and Y has exactly 36

BIO-11-CH01-0040 | Hard | Essential fatty acids

A student assumes that every unsaturated fatty acid listed in the prescribed account is an essential fatty acid. Which named unsaturated fatty acid shows this assumption to be wrong?

- A. Oleic acid
- B. Linoleic acid
- C. Linolenic acid
- D. Arachidonic acid

BIO-11-CH01-0041 | Medium | Essential fatty acids

A dietician has to suggest a seed that supplies omega-3 fatty acid. Which seed, listed in the prescribed account for this purpose, should be suggested?

- A. Flax seeds
- B. Mustard seeds
- C. Sesame seeds
- D. Groundnut seeds

BIO-11-CH01-0042 | Challenging | General properties of lipids

Which statement about lipids as a group is correct?

- A. They are insoluble in water, their hydrogen to oxygen ratio is greater than 2:1, and their molecular weight is smaller than that of the macromolecules
- B. They are insoluble in water, their hydrogen to oxygen ratio is exactly 2:1, and their molecular weight is smaller than that of the macromolecules
- C. They are soluble in water, their hydrogen to oxygen ratio is greater than 2:1, and their molecular weight is larger than that of the macromolecules
- D. They are insoluble in water, their hydrogen to oxygen ratio is less than 2:1, and their molecular weight is larger than that of the macromolecules

BIO-11-CH01-0043 | Easy | Introduction to proteins

The term protein was suggested by which scientist, and in which year?

- A. Buchner, in 1830
- B. Berzelius, in 1869
- C. Berzelius, in 1830
- D. Miescher, in 1830

BIO-11-CH01-0044 | Easy | Peptide bond

A peptide bond in a protein is formed between which groups of two amino acid residues?

- A. The carboxyl group of one residue and the amino group of another residue
- B. The amino group of one residue and the amino group of another residue
- C. The carboxyl group of one residue and the carboxyl group of another residue
- D. The side chain of one residue and the carboxyl group of another residue

BIO-11-CH01-0045 | Easy | Introduction to proteins

Which statement gives both the number of amino acid units in a protein and the number of naturally occurring amino acid types correctly?

- A. 100 to 3000 amino acid units, of 20 naturally occurring types
- B. 20 to 300 amino acid units, of 100 naturally occurring types
- C. 100 to 3000 amino acid units, of 30 naturally occurring types
- D. 1000 to 3000 amino acid units, of 20 naturally occurring types

BIO-11-CH01-0046 | Easy | Classification of proteins

Among the simple proteins, the albumins include one protein obtained from pulses. Which is it?

- A. Globulin
- B. Mucin
- C. Legumelin
- D. Peptone

BIO-11-CH01-0047 | Medium | Classification of proteins

Which statement places two named proteins correctly in the prescribed account's classification?

- A. Egg albumin is a conjugated protein and mucin is a simple protein
- B. Serum albumin is a conjugated protein and haemoglobin is a simple protein
- C. Globulin is a conjugated protein and legumelin is a simple protein
- D. Egg albumin is a simple protein and mucin is a conjugated protein

BIO-11-CH01-0048 | Medium | Protein structure

Which comparison between the secondary and the tertiary structure of a protein is correct?

- A. The secondary structure is held by disulphide bonds, while the tertiary structure involves hydrogen bonds
- B. The secondary structure is held by peptide bonds, while the tertiary structure involves hydrogen bonds
- C. The secondary structure is held by hydrogen bonds, while the tertiary structure involves disulphide bonds
- D. Both the secondary and the tertiary structure are held only by peptide bonds

BIO-11-CH01-0049 | Medium | Protein structure

Intermolecular hydrogen bonds link neighbouring peptide chains into pleated sheets. This arrangement is described in which of the following?

- A. Egg albumin
- B. Haemoglobin
- C. Heparin
- D. Silk fibre

BIO-11-CH01-0050 | Medium | Amphoteric nature of proteins

The prescribed account describes proteins as behaving amphoterically, that is as both acid and base, at a stated pH. What is that pH?

- A. pH 2
- B. pH 9.5
- C. pH 7.0
- D. pH 7.4

BIO-11-CH01-0051 | Hard | Amino acids in proteins

Which statement correctly identifies the amino acids listed in the prescribed account for two different roles?

- A. Lysine and cysteine are the basic amino acids, and arginine forms the disulphide bond
- B. Lysine and arginine are the basic amino acids, and cysteine forms the disulphide bond
- C. Arginine and cysteine are the basic amino acids, and lysine forms the disulphide bond
- D. Lysine and arginine are the basic amino acids, and arginine forms the disulphide bond

BIO-11-CH01-0052 | Hard | Protein structure

Hydrogen bonds are stated to hold the alpha helix, the beta helix and the pleated sheet of the secondary structure of a protein. A student therefore expects hydrogen bonds to hold the tertiary structure also. Which bond in fact accounts for the looping, twisting and folding back of the chain at the tertiary level, and which amino acid provides it?

- A. Hydrogen bond, provided by cysteine
- B. Disulphide bond, provided by cysteine
- C. Peptide bond, provided by lysine
- D. Disulphide bond, provided by arginine

BIO-11-CH01-0053 | Medium | Classification of proteins

Haemoglobin consists of the protein globin joined to the non-protein haem group, and is indispensable in respiration. On the basis of this composition, haemoglobin belongs to which class of proteins?

- A. Simple proteins
- B. Derived proteins
- C. Metaproteins
- D. Conjugated proteins

BIO-11-CH01-0054 | Hard | Classification of proteins

Three proteins are listed. I is serum albumin, II is a lipoprotein, and III is a peptone. Which assignment to the three protein classes is correct?

- A. I is a simple protein, II is a conjugated protein and III is a derived protein
- B. I is a conjugated protein, II is a simple protein and III is a derived protein
- C. I is a derived protein, II is a conjugated protein and III is a simple protein
- D. I is a simple protein, II is a derived protein and III is a conjugated protein

BIO-11-CH01-0055 | Hard | Proteins and their functions

Which pairing of a protein with its physiological role is correct?

- A. Fibrinogen with the clotting of blood
- B. Collagen with the clotting of blood
- C. Immunoglobulin IgG with the storage of oxygen in muscle
- D. Myoglobin with the defence of the body against antigens

BIO-11-CH01-0056 | Medium | Peptide bond

A protein may contain up to 3000 amino acid units. If a single unbranched polypeptide chain contains exactly this maximum number of amino acid units, how many peptide bonds does it contain?

- A. 3000
- B. 3001
- C. 2999
- D. 1500

BIO-11-CH01-0057 | Hard | Protein structure

A protein contains 100 to 3000 amino acid units. A protein showing quaternary structure is built from four identical unbranched polypeptide chains, and each chain contains the smallest number of amino acid units in that stated range. How many peptide bonds are present in the complete molecule?

- A. 400
- B. 396
- C. 399
- D. 99

BIO-11-CH01-0058 | Challenging | Classification of proteins

Consider the following statements about proteins. I: Haemoglobin is a conjugated protein consisting of globin and haem. II: Peptones are grouped with the derived proteins. III: The disulphide bonds of the tertiary structure are contributed by cysteine. IV: Egg albumin is a conjugated protein. Which combination of statements is correct?

- A. I, II and III only
- B. I, II and IV only
- C. II, III and IV only
- D. I, III and IV only

BIO-11-CH01-0059 | Hard | History of nucleic acid research

Four landmarks in the study of nucleic acids are listed. I: Chargaff's generalisation about the base ratio of DNA. II: Isolation of nucleic acids from pus cells by Friederich Miescher. III: Identification of the two types of nucleic acid. IV: The work of Fuelgen. Arrange them in chronological order, earliest first.

- A. II, III, IV, I
- B. IV, II, III, I
- C. II, IV, I, III
- D. II, IV, III, I

BIO-11-CH01-0060 | Easy | Nitrogenous bases

Which of the following correctly distinguishes purines from pyrimidines?

- A. Purines are single-ring bases and include adenine and guanine, while pyrimidines are double-ring bases and include cytosine, thymine and uracil
- B. Purines are double-ring bases and include adenine and guanine, while pyrimidines are single-ring bases and include cytosine, thymine and uracil
- C. Purines are double-ring bases and include cytosine and thymine, while pyrimidines are single-ring bases and include adenine, guanine and uracil
- D. Purines are double-ring bases and include adenine and cytosine, while pyrimidines are single-ring bases and include guanine, thymine and uracil

BIO-11-CH01-0061 | Medium | Nucleotide structure

The backbone of a polynucleotide chain is made of alternate sugar and phosphate groups. In each nucleotide the nitrogenous base is attached to one carbon of the pentose sugar, the phosphate group to another, and a third carbon of the sugar is involved in linking the nucleotide into the chain. Which set of positions is correct?

- A. Base at C-5, phosphate at C-1, chain linkage involving C-3
- B. Base at C-1, phosphate at C-3, chain linkage involving C-5
- C. Base at C-3, phosphate at C-5, chain linkage involving C-1
- D. Base at C-1, phosphate at C-5, chain linkage involving C-3

BIO-11-CH01-0062 | Medium | Double helix model

Which statement about the proposal of the double-helical model of DNA is correct?

- A. R. Franklin and M. Wilkins proposed the model using the results obtained by Watson and Crick, who performed the experiments
- B. Watson and Crick proposed the model on the basis of their own experiments, which R. Franklin and M. Wilkins later repeated
- C. Watson and Crick proposed the model using the results obtained by R. Franklin and M. Wilkins, and performed no experiments of their own
- D. Chargaff proposed the model, and Watson, Crick, R. Franklin and M. Wilkins provided the experimental results for it

BIO-11-CH01-0063 | Hard | Structure of the DNA double helix

Consider the following statements about the DNA double helix. I: Its diameter is 20 angstrom. II: The distance between two neighbouring nucleotide pairs is 34 angstrom. III: The two chains are antiparallel, the 3' end of one chain lying beside the 5' end of the other. IV: The helix shows a major groove and a minor groove. Which of the statements are correct?

- A. I, III and IV only
- B. I, II and III only
- C. II, III and IV only
- D. I and II only

BIO-11-CH01-0064 | Easy | Base pairing

In a DNA double helix, how many hydrogen bonds hold an adenine-thymine pair and a guanine-cytosine pair respectively?

- A. Three and two
- B. Two and two
- C. Three and three
- D. Two and three

BIO-11-CH01-0065 | Hard | Chargaff's observations

Chargaff observed that the base ratio $(A+T)/(G+C)$ varies from one species to another but is constant for a given species. DNA is extracted from the liver cells and from the muscle cells of one animal, and also from the liver cells of an animal of a different species. Which result is expected?

- A. The ratio differs between the two tissues of the one animal and is the same in the animal of the other species
- B. The ratio is the same in all three samples, because it has a single fixed value for all living organisms
- C. The ratio is the same in the two tissues of the one animal and may differ in the animal of the other species
- D. The ratio differs in all three samples, because every cell of an organism has its own base ratio

BIO-11-CH01-0066 | Medium | Chargaff arithmetic

A sample of double-stranded DNA contains 22 per cent cytosine. What is the percentage of adenine in this sample?

- A. 22 per cent
- B. 39 per cent
- C. 28 per cent
- D. 56 per cent

BIO-11-CH01-0067 | Hard | Chargaff arithmetic

In a sample of double-stranded DNA the base ratio $(A+T)/(G+C)$ is 1.5. What is the percentage of cytosine in this sample?

- A. 20 per cent
- B. 40 per cent
- C. 30 per cent
- D. 15 per cent

BIO-11-CH01-0068 | Medium | Hydrogen bonding in DNA

A segment of double-stranded DNA contains 40 adenine-thymine pairs and 60 guanine-cytosine pairs. An adenine-thymine pair is held by two hydrogen bonds and a guanine-cytosine pair by three. How many hydrogen bonds hold the two chains of this segment together?

- A. 260
- B. 240
- C. 200
- D. 300

BIO-11-CH01-0069 | Easy | Dimensions of DNA

A double-stranded DNA segment contains 500 base pairs. Use the prescribed approximate axial rise of 3.4 angstrom per base pair. What is the length of this segment?

- A. 3400 angstrom
 - B. 170 angstrom
 - C. 17000 angstrom
 - D. 1700 angstrom
-

BIO-11-CH01-0070 | Medium | Dimensions of DNA

A double-stranded DNA molecule is 1020 angstrom long. One complete turn of the helix measures 34 angstrom. How many complete turns are present in this molecule?

- A. 300
 - B. 51
 - C. 34
 - D. 30
-

BIO-11-CH01-0071 | Challenging | Hydrogen bonding in DNA

A double-stranded DNA segment is 680 angstrom long and contains 60 adenine bases. Use the prescribed approximate axial rise of 3.4 angstrom per base pair. An adenine-thymine pair is held by two hydrogen bonds and a guanine-cytosine pair by three. How many hydrogen bonds hold the two chains of this segment together?

- A. 460
 - B. 600
 - C. 540
 - D. 400
-

BIO-11-CH01-0072 | Hard | Exceptions to base equality

Purines and pyrimidines occur in equal amounts in DNA because the molecule has two chains and every purine of one chain pairs with a pyrimidine of the other. Consider the following claims about bacteriophage phi-x-174. I. Its genetic material is DNA. II. Its DNA has only one chain. III. Its purine and pyrimidine amounts need not be equal. Which of these claims are correct?

- A. I, II and III (phi-x-174 claims)
 - B. I and II only (phi-x-174 claims)
 - C. II and III only (phi-x-174 claims)
 - D. I and III only (phi-x-174 claims)
-

BIO-11-CH01-0073 | Easy | DNA compared with RNA

Which of the following correctly distinguishes DNA from RNA?

- A. DNA is usually single stranded and contains uracil, whereas RNA is usually double stranded and contains thymine in place of uracil
 - B. DNA is usually double stranded and contains uracil, whereas RNA is usually single stranded and contains thymine in place of uracil
 - C. DNA is usually double stranded and contains thymine, whereas RNA is usually single stranded and contains uracil in place of thymine
 - D. DNA is usually single stranded and contains thymine, whereas RNA is usually double stranded and contains uracil in place of thymine
-

BIO-11-CH01-0074 | Hard | Base equality and RNA

The rule that purines and pyrimidines occur in equal amounts applies to DNA. A student extracts the hereditary material of tobacco mosaic virus and predicts that its adenine content will equal its uracil content. The prediction fails. Which explanation is correct?

- A. The hereditary material of this virus is DNA, in which adenine pairs with thymine and not with uracil
- B. The hereditary material of this virus is double-stranded RNA, in which adenine pairs with guanine
- C. The hereditary material of this virus is RNA in which uracil has been replaced by thymine throughout
- D. The hereditary material of this virus is single-stranded RNA, whose bases are not paired with those of a second chain

BIO-11-CH01-0075 | Medium | Forms of RNA in viruses

RNA is generally a single-stranded molecule. In which of the following is RNA present in the double-stranded form?

- A. Tobacco mosaic virus and Reovirus
- B. Reovirus and wound tumour virus
- C. Tobacco mosaic virus and bacteriophage phi-x-174
- D. Bacteriophage phi-x-174 and wound tumour virus

BIO-11-CH01-0076 | Medium | Messenger RNA

Which statement about messenger RNA is correct?

- A. It is a linear polynucleotide forming 80 to 90 per cent of the cellular RNA and carrying initiation and termination codons
- B. It is a clover-leaf shaped molecule forming about 3 per cent of the cellular RNA and carrying initiation and termination codons
- C. It is a linear polynucleotide forming about 3 per cent of the cellular RNA and carrying initiation and termination codons
- D. It is a linear polynucleotide forming about 3 per cent of the cellular RNA and carrying the anticodon loop

BIO-11-CH01-0077 | Hard | Ribosomal RNA

Consider the following statements about ribosomal RNA. I: It forms 50 to 60 per cent of the ribosome. II: It forms about 3 per cent of the cellular RNA. III: It is synthesised in the nucleus. IV: It was studied by Kurland in 1960 and shows secondary coiling. Which of the statements are correct?

- A. I, II and III only (rRNA statements)
- B. II, III and IV only (rRNA statements)
- C. I, III and IV only (rRNA statements)
- D. I and II only (rRNA statements)

BIO-11-CH01-0078 | Challenging | Transfer RNA

Consider the following statements about transfer RNA. I: It contains 70 to 80 nucleotides. II: Its clover-leaf shape was described by Holley in 1965. III: Its four parts are the DHU arm, the amino acid binding site, the anticodon loop and the ribosome recognition site. IV: The amino acid becomes attached at the 5' acceptor stem. Which of the statements are correct?

- A. I, II and IV only
- B. I, II and III only
- C. II, III and IV only
- D. I, III and IV only

BIO-11-CH01-0079 | Easy | Concept of metabolism

Which of the following best describes metabolism?

- A. The sum of all the reactions in which complex molecules are broken down into simpler ones
- B. The sum of all the reactions in which simpler molecules are built up into complex ones
- C. The collection of intermediate molecules available in a cell at a given moment
- D. The sum of all the chemical reactions taking place in an organism

BIO-11-CH01-0080 | Medium | Catabolic and anabolic pathways

Which of the following correctly distinguishes a catabolic pathway from an anabolic pathway?

- A. A catabolic pathway breaks complex molecules down into simpler ones, whereas an anabolic pathway builds complex molecules from simpler ones
- B. A catabolic pathway builds complex molecules from simpler ones, whereas an anabolic pathway breaks complex molecules down into simpler ones
- C. A catabolic pathway builds complex molecules from simpler ones, whereas an anabolic pathway converts one molecule into another of the same size
- D. A catabolic pathway breaks complex molecules down into simpler ones, whereas an anabolic pathway converts one molecule into another of the same size

BIO-11-CH01-0081 | Medium | Metabolic pool

In a cell, molecules formed as intermediates in one reaction can be withdrawn and used in other breaking-down or building-up reactions. The common collection of such molecules available to the cell is called

- A. a catabolic pathway
- B. the metabolic pool
- C. an anabolic pathway
- D. a secondary metabolite

BIO-11-CH01-0082 | Hard | Catabolic and anabolic pathways

Which of the following correctly pairs a cellular event with the type of metabolic pathway to which it belongs?

- A. Formation of a glycosidic bond between two monosaccharides - catabolic
- B. Joining of amino acid residues by peptide bonds - catabolic
- C. Breakdown of a polysaccharide into simple sugars - anabolic
- D. Hydrolysis of sucrose into glucose and fructose - catabolic

BIO-11-CH01-0083 | Easy | Definition of secondary metabolites

Nicotine, cocaine and morphine are examples of secondary metabolites. Which of the following best defines this class of molecules?

- A. Small organic molecules produced by several bacteria, fungi and plants that are essential for their growth, development and reproduction
- B. Small organic molecules produced by several bacteria, fungi and plants that are not essential for their growth, development and reproduction
- C. Small inorganic molecules produced by several bacteria, fungi and plants that are not essential for their growth, development and reproduction
- D. Large organic molecules produced only by green plants that are not essential for their growth, development and reproduction

BIO-11-CH01-0084 | Easy | Examples of secondary metabolites

Which of the following sets contains only secondary metabolites?

- A. Alkaloids, flavonoids and polypeptides
- B. Terpenoids, lectins and nucleotides
- C. Alkaloids, flavonoids and terpenoids
- D. Phenolics, alkaloids and polysaccharides

BIO-11-CH01-0085 | Easy | Sources of secondary metabolites

Morphine, which acts as a pain reliever and a cough suppressant, is obtained from which source?

- A. The yam plant Dioscorea
- B. The cabbage plant
- C. The plant *Papaver somniferum*
- D. The flax plant

BIO-11-CH01-0086 | Medium | Groups of secondary metabolites

Which of the following correctly distinguishes terpenes from phenolics?

- A. Terpenes are synthesised from simple sugars and contain a benzene ring, whereas phenolics are synthesised from mevalonic acid
- B. Terpenes are synthesised from simple sugars, whereas phenolics are synthesised from mevalonic acid and contain a benzene ring
- C. Terpenes are synthesised from mevalonic acid, whereas phenolics are synthesised from simple sugars and contain a benzene ring
- D. Terpenes are synthesised from mevalonic acid and contain sulphur, whereas phenolics are synthesised from simple sugars and contain nitrogen

BIO-11-CH01-0087 | Medium | Classification of secondary metabolites

Consider the following statements about secondary metabolites. I: They are classified on the basis of chemical structure, composition, solubility and pathway of synthesis. II: The three groups recognised are terpenes, phenolics and nitrogen-containing compounds. III: The nitrogen-containing compounds may also contain sulphur. IV: Terpenes are synthesised from simple sugars and contain a benzene ring. Which of the statements are correct?

- A. I, II and IV only
- B. II, III and IV only
- C. I and IV only
- D. I, II and III only

BIO-11-CH01-0088 | Medium | Pathways of synthesis

Secondary metabolites can be grouped according to the pathway by which they are made. Tannins, which give wine and chocolate their astringency, are formed by

- A. the mevalonic acid pathway
- B. the phenylpropanoid pathway
- C. the glucosinolate pathway
- D. the benzene ring pathway

BIO-11-CH01-0089 | Medium | Uses of secondary metabolites

Which of the following pairs a secondary metabolite with the property or use stated for it correctly?

- A. Tannins - used as food preservatives
- B. Glucosinolates - responsible for the astringency of wine and chocolate
- C. Antibiotics - used as food preservatives
- D. Morphine - responsible for the flavour of cabbage and its protection from pests

BIO-11-CH01-0090 | Challenging | Groups and examples together

Which of the following statements about secondary metabolites is correct?

- A. Cannabinol is a phenolic synthesised from simple sugars, and glucosinolates give wine and chocolate their astringency
- B. Cannabinol is a terpene synthesised from simple sugars, and glucosinolates are used as food preservatives
- C. Cannabinol is a nitrogen-containing metabolite synthesised from mevalonic acid, and glucosinolates give chocolate its astringency
- D. Cannabinol is a terpene synthesised from mevalonic acid, and glucosinolates give cabbage its flavour and protect it from pests

BIO-11-CH01-0091 | Easy | Discovery and naming of enzymes

Edward Buchner showed that an extract of yeast containing no living cells could by itself bring about fermentation of sugar. The term enzyme, which came into use from this work, is built from two Greek words whose combined literal meaning is:

- A. In yeast
- B. In sugar
- C. Within the cell
- D. Living ferment

BIO-11-CH01-0092 | Medium | Properties of enzymes

Sucrase can hydrolyse about 100000 times its own weight of sucrose. A purified preparation contains 1 mg of sucrase. The mass of sucrose that this preparation can hydrolyse is about:

- A. 10 g
- B. 100 g
- C. 1 g
- D. 1000 g

BIO-11-CH01-0093 | Medium | Properties of enzymes

An enzyme catalyses the conversion of compound X into compound Y and also catalyses the conversion of Y back into X, and it acts on no substrate other than X and Y. Which two properties of enzymes are illustrated together by this behaviour?

- A. Catalytic property and sensitivity to pH
- B. Reversibility of action and specificity of action
- C. Specificity of action and sensitivity to temperature
- D. Reversibility of action and proteinaceous nature

BIO-11-CH01-0094 | Hard | Properties of enzymes

Rule: the catalytic ability of an enzyme depends on its three-dimensional conformation, which carries the active site. A globular enzyme is treated so that its three-dimensional folding is completely lost, while every peptide bond in the chain remains intact. What is the expected consequence?

- A. The enzyme loses its catalytic ability, because the active site no longer exists
- B. The enzyme retains its catalytic ability, because no peptide bond in the chain has been broken
- C. The enzyme retains its catalytic ability but loses its specificity of action
- D. The enzyme is converted into a cofactor for another enzyme

BIO-11-CH01-0095 | Easy | Composition of enzymes

NAD and FMN occur as non-protein components attached to the protein part of certain conjugated enzymes. Such components are termed:

- A. Coenzymes
- B. Cofactors
- C. Prosthetic groups
- D. Activators

BIO-11-CH01-0096 | Medium | Composition of enzymes

A conjugated enzyme is found to carry two non-protein components: an FMN molecule tightly attached to the protein part, and a zinc ion loosely attached to the protein part. In the terminology used in the prescribed account, these two components are respectively:

- A. A cofactor and a coenzyme
- B. A cofactor and an activator
- C. A coenzyme and an activator
- D. A coenzyme and a cofactor

BIO-11-CH01-0097 | Medium | Composition of enzymes

Which pairing of an enzyme with the inorganic ion listed in the prescribed account as its cofactor is correct?

- A. Catalase and zinc
- B. Peptidases and copper
- C. Peptidases and iron
- D. Catalase and iron

BIO-11-CH01-0098 | Hard | Composition of enzymes

Rule: a conjugated enzyme carries a non-protein component in addition to its protein part, and the different kinds of non-protein component are told apart by the manner of their attachment. In one such enzyme the non-protein component is joined to the protein component by chemical bonds and cannot be separated from it without breaking those bonds. This component is best described as:

- A. A prosthetic group
- B. A coenzyme
- C. A cofactor
- D. An activator

BIO-11-CH01-0099 | Easy | Functional classification of enzymes

Alcohol dehydrogenase is placed in which functional class of enzymes?

- A. Transferases
- B. Hydrolases
- C. Oxidoreductases
- D. Isomerases

BIO-11-CH01-0100 | Medium | Functional classification of enzymes

Which of the following pairings of an enzyme with its functional class is correct?

- A. Xylose isomerase and lyases
- B. Fumarase and lyases
- C. Amylase and transferases
- D. Glucokinase and hydrolases

BIO-11-CH01-0101 | Medium | Functional classification of enzymes

Which functional class of enzymes creates covalent bonds between molecules using the energy obtained from the hydrolysis of a compound?

- A. Lyases
- B. Transferases
- C. Hydrolases
- D. Ligases

BIO-11-CH01-0102 | Challenging | Functional classification of enzymes

Each of the following pairs consists of two enzymes listed in the prescribed account. In exactly one pair the two enzymes belong to different functional classes. Which pair is it?

- A. Alcohol dehydrogenase and xylose isomerase
- B. Aldolase and fumarase
- C. Amylase and nuclease
- D. Glucokinase and glutamate pyruvate transaminase

BIO-11-CH01-0103 | Easy | Mechanism of enzyme action

In the lock and key model put forward by Emil Fischer in 1894, the lock and the key stand respectively for:

- A. The enzyme and the substrate
- B. The substrate and the enzyme
- C. The active site and the product
- D. The product and the substrate

BIO-11-CH01-0104 | Medium | Mechanism of enzyme action

Which statement expresses the essential difference of the induced fit model of Koshland from the earlier lock and key model?

- A. The substrate alters its own shape so as to fit an unchanging active site
- B. The approach of the substrate brings about a change in the conformation of the enzyme
- C. The active site is already exactly complementary to the substrate before the substrate arrives
- D. The enzyme acts on the substrate without forming an enzyme-substrate complex

BIO-11-CH01-0105 | Hard | Mechanism of enzyme action

Rule: in the lock and key model only a key of the correct size and shape fits into the key hole. A student states that this model explains enzyme action by the enzyme changing its shape to accommodate the substrate. The correct evaluation of that statement is that it is:

- A. Correct, because the key hole of the lock must widen a little before the key can enter it
- B. Incorrect, because the lock and key model denies that any enzyme-substrate complex is formed at all
- C. Incorrect, because a change of shape in the enzyme on substrate approach belongs to Koshland's model
- D. Correct, because the two models describe the same conformational change in different words

BIO-11-CH01-0106 | Easy | Michaelis-Menten constant

The Michaelis-Menten constant, K_m , of an enzyme is defined as:

- A. The concentration of substrate, in moles per litre, at which the enzyme becomes completely saturated
- B. The concentration of enzyme, in moles per litre, at which the reaction attains its maximum velocity
- C. The substrate concentration, in moles per litre, that gives half of the maximum velocity
- D. The maximum velocity attained by the reaction when substrate is present in large excess

BIO-11-CH01-0107 | Hard | Michaelis-Menten constant

Enzymes P and Q act on the same substrate. Under identical conditions the K_m of P is 2×10^{-5} moles per litre and the K_m of Q is 4×10^{-3} moles per litre. Which conclusion follows?

- A. Q has the greater affinity, because its K_m value is the numerically larger of the two
- B. P and Q must have equal affinity, because K_m is fixed by the substrate and both act on the same one
- C. Q attains half of its maximum velocity at a lower substrate concentration than P does
- D. P has the greater affinity, because a lower substrate concentration gives half maximal velocity

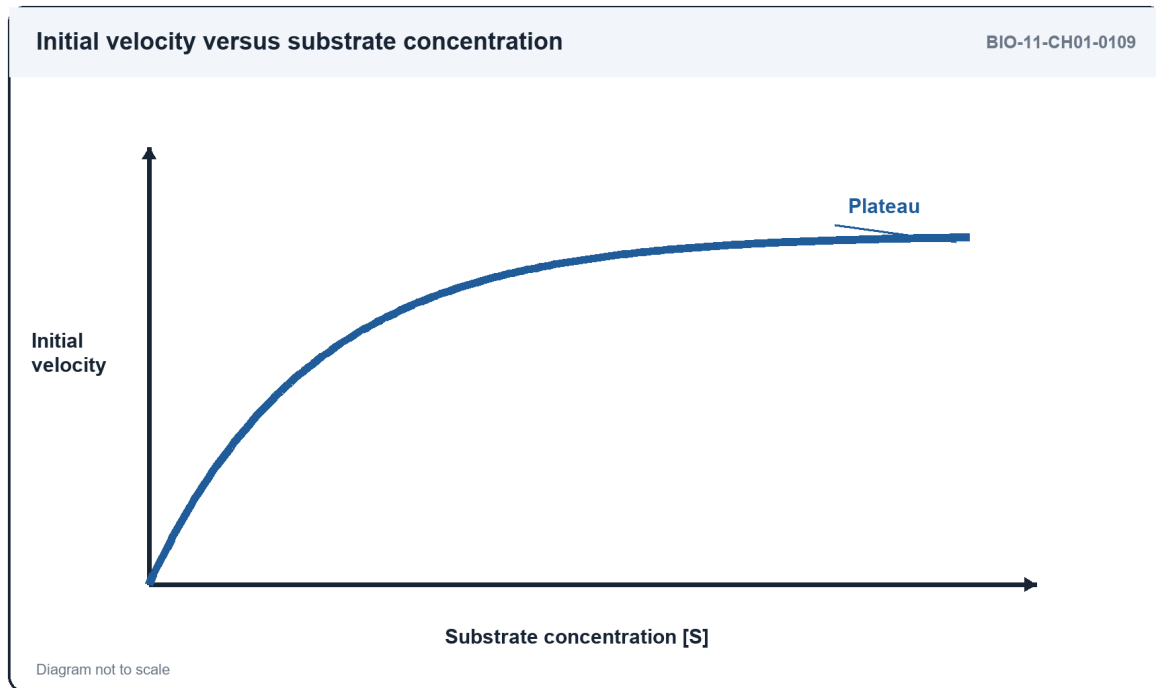
BIO-11-CH01-0108 | Medium | Michaelis-Menten constant

An enzyme has a K_m of 5×10^{-4} moles per litre for its substrate, a value lying within the range of 10^{-5} to 10^{-2} moles reported for most enzymes. In an assay the substrate concentration is held at exactly 5×10^{-4} moles per litre. At this concentration the enzyme is:

- A. 50 per cent saturated with substrate and working at its maximum velocity
- B. 50 per cent saturated with substrate and working at half of its maximum velocity
- C. Completely saturated with substrate and working at half of its maximum velocity
- D. 25 per cent saturated with substrate and working at half of its maximum velocity

BIO-11-CH01-0109 | Hard | Factors affecting enzyme activity

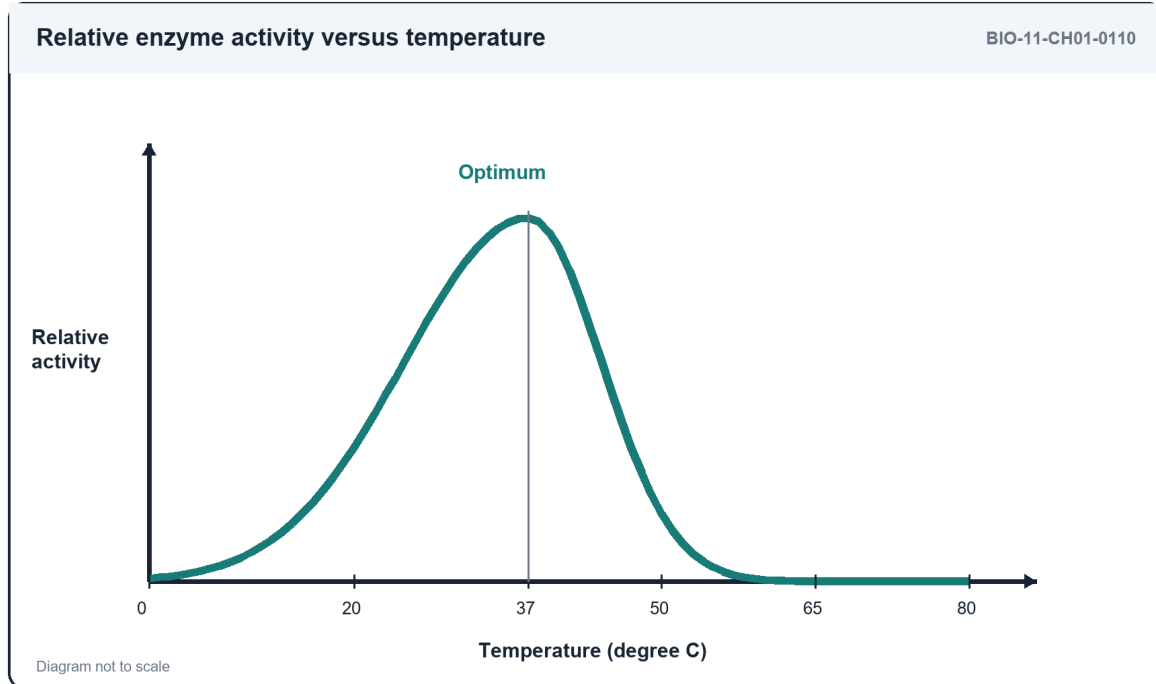
The figure shows initial enzyme velocity plotted against substrate concentration while enzyme quantity, temperature and pH are constant. What accounts for the final, almost horizontal part of the curve?



- A. The enzyme has begun to lose its conformation because the substrate concentration is very high
- B. The substrate has been completely used up, so no further product can be formed in the assay
- C. The reverse reaction has begun to run at a rate equal to that of the forward reaction
- D. All the active sites of the fixed quantity of enzyme present are occupied by substrate

BIO-11-CH01-0110 | Hard | Factors affecting enzyme activity

The figure shows the activity of a mammalian enzyme plotted against temperature while pH and substrate concentration are controlled. Which explanation of the curve is correct?



- A. The optimum is 37 degree C, and above 40 degree C the enzyme denatures rapidly, most enzymes being destroyed at 60 to 70 degree C
- B. The optimum is 65 degree C, and the fall in activity before it is caused by exhaustion of the substrate in the assay mixture
- C. The rise reflects an increasing K_m , and the fall begins once the K_m of the enzyme has been exceeded by the temperature
- D. The steep fall above 37 degree C occurs because heating shifts the pH of the medium away from the optimum for the enzyme

BIO-11-CH01-0111 | Medium | Factors affecting enzyme activity

Two enzymes listed in the prescribed account are assayed over a range of pH values, with the temperature, the enzyme concentration and the substrate concentration identical in the two sets. One enzyme shows its maximum activity at pH 2 and the other at pH 9.5. The two enzymes are respectively:

- A. Pepsin and trypsin
- B. Trypsin and pepsin
- C. Pepsin and amylase
- D. Trypsin and sucrase

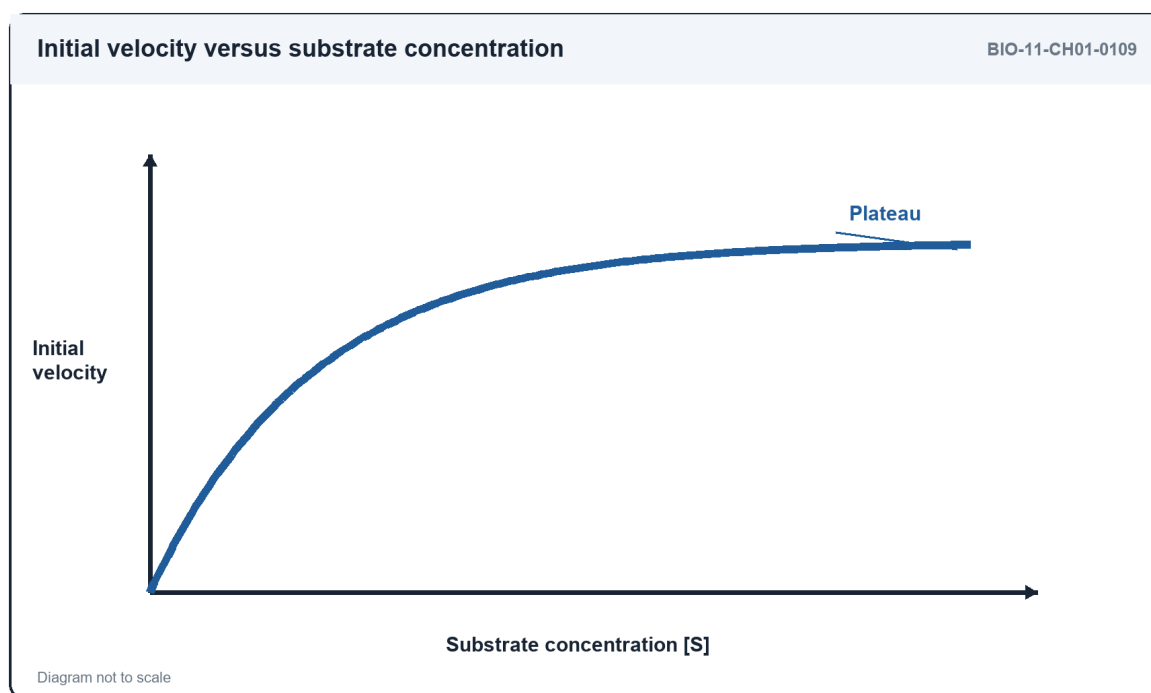
BIO-11-CH01-0112 | Challenging | Factors affecting enzyme activity

An enzyme-catalysed reaction runs steadily at 37 degree C at the optimum pH, with substrate present in large excess. Four further assays are set up, each differing from this control in one respect only. In I a further quantity of the same enzyme is added. In II an inorganic activator is added. In III the temperature is raised to 65 degree C. In IV an inhibitor is added. In how many of the four assays is the activity expected to be LOWER than in the control?

- A. One
- B. Two
- C. Three
- D. Four

BIO-11-CH01-0113 | Easy | Substrate saturation

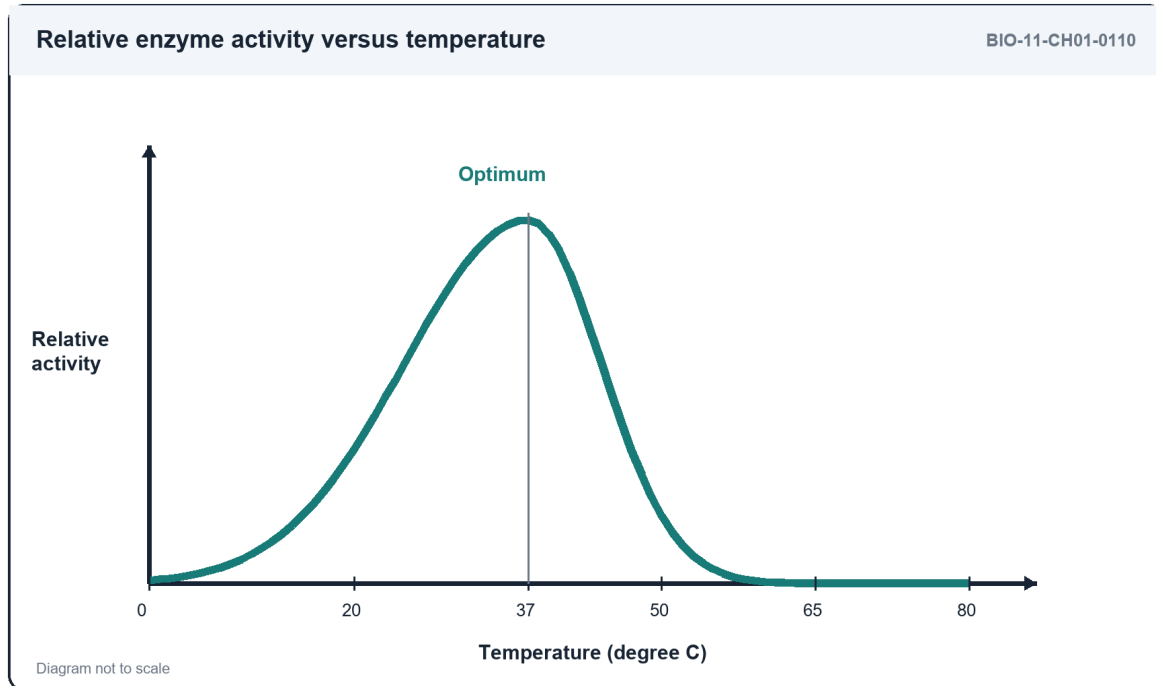
In the enzyme-velocity graph shown, the curve becomes nearly horizontal at high substrate concentration. What does this plateau indicate?



- A. Nearly all active sites of the available enzyme molecules are occupied
- B. The substrate has converted the enzyme into a lipid
- C. The reaction has become independent of enzyme concentration at every substrate level
- D. The enzyme has necessarily been denatured by high temperature

BIO-11-CH01-0114 | Medium | Temperature and enzyme activity

An enzyme follows the temperature-activity curve shown. A sample is heated to 65 degree C and then cooled back to 37 degree C. Why may its original activity fail to return?



- A. Cooling removes every substrate molecule from the solution
- B. High temperature may irreversibly alter the enzyme's three-dimensional active site
- C. The enzyme changes permanently into a carbohydrate at 65 degree C
- D. Cooling forces all enzyme molecules to bind inhibitors

BIO-11-CH01-0115 | Hard | DNA base pairing

A double-stranded DNA segment has 100 base pairs, of which 40 per cent are G-C pairs. How many hydrogen bonds hold the two strands together?

- A. 180
 - B. 200
 - C. 240
 - D. 300
-

BIO-11-CH01-0116 | Challenging | Factors affecting activity

Four assays begin with the same enzyme activity. I receives more enzyme while substrate remains abundant. II receives a specific inhibitor. III is heated to 65 degree C. IV receives more substrate when its enzyme is already saturated. In which assays is activity expected to decrease?

- A. I and IV only
 - B. I, II and III only
 - C. II and IV only
 - D. II and III only
-

Standard 11 | Chapter 02

Respiration and Energy Transfer

BIO-11-CH02-0001 | Easy | Modes of phosphorylation

ATP is formed by phosphorylation, which is stated to occur in three ways. Which set names all three?

- A. Photophosphorylation, substrate-level phosphorylation and oxidative phosphorylation
- B. Photophosphorylation, oxidative phosphorylation and chemiosmosis
- C. Substrate-level phosphorylation, oxidative phosphorylation and terminal oxidation
- D. Photophosphorylation, substrate-level phosphorylation and dehydrogenation

BIO-11-CH02-0002 | Easy | Substrate-level phosphorylation

Which statement correctly defines substrate-level phosphorylation?

- A. Direct phosphorylation of ADP by transfer of a phosphate group from a suitable substrate
- B. Phosphorylation of ADP at the cost of energy released during oxidation of $\text{NADH} + \text{H}^+$ and FADH_2
- C. Phosphorylation of ADP using energy trapped during photosynthesis
- D. Phosphorylation of glucose by transfer of a phosphate group from ATP

BIO-11-CH02-0003 | Medium | Substrate-level versus oxidative phosphorylation

Which statement correctly distinguishes oxidative phosphorylation from substrate-level phosphorylation?

- A. Oxidative phosphorylation uses energy released when $\text{NADH} + \text{H}^+$ and FADH_2 are oxidised, and occurs on the inner mitochondrial membrane
- B. Oxidative phosphorylation transfers a phosphate group straight from a substrate to ADP, and occurs in the cytoplasm
- C. Oxidative phosphorylation occurs in the mitochondrial matrix, while substrate-level phosphorylation occurs on the inner mitochondrial membrane
- D. Oxidative phosphorylation adds a phosphate group to the respiratory substrate rather than to ADP

BIO-11-CH02-0004 | Medium | Substrate-level phosphorylation in the balance sheet

In the aerobic oxidation of one glucose molecule, glycolysis yields 4 ATP by substrate-level phosphorylation, and one GTP (ATP) is formed by substrate-level phosphorylation for every molecule of citric acid oxidised. One glucose gives two turns of the Krebs cycle. How many ATP molecules arise by substrate-level phosphorylation per glucose?

- A. 4
- B. 2
- C. 6
- D. 8

BIO-11-CH02-0005 | Challenging | Oxidative phosphorylation in the balance sheet

The balance sheet for ATP by aerobic oxidation of one glucose molecule accounts each $\text{NADH} + \text{H}^+$ as 3 ATP and each FADH_2 as 2 ATP. Complete aerobic oxidation of one glucose delivers 10 $\text{NADH} + \text{H}^+$ and 2 FADH_2 to the electron transport system, and substrate-level phosphorylation contributes 6 ATP. How many ATP come from oxidative phosphorylation, and what is the gross total?

- A. 30 by oxidative phosphorylation, gross total 38
- B. 34 by oxidative phosphorylation, gross total 40
- C. 34 by oxidative phosphorylation, gross total 38
- D. 24 by oxidative phosphorylation, gross total 30

BIO-11-CH02-0006 | Hard | Identifying the mode of phosphorylation from the reaction

Oxidative phosphorylation is defined as phosphorylation of ADP at the cost of energy released during oxidation of substrates like $\text{NADH} + \text{H}^+$ and FADH_2 , and it is located on the inner mitochondrial membrane. In the Krebs cycle, which runs in the mitochondrial matrix, the conversion of succinyl-CoA to succinate yields one GTP (ATP). By which mode is that ATP formed?

- A. Oxidative phosphorylation
- B. Photophosphorylation
- C. Chemiosmosis through the oxysome
- D. Substrate-level phosphorylation

BIO-11-CH02-0007 | Easy | Site of glycolysis

In which part of the cell does glycolysis take place?

- A. Mitochondrial matrix
- B. Cytoplasm
- C. Inner mitochondrial membrane
- D. Outer chamber of the mitochondrion

BIO-11-CH02-0008 | Easy | Definition and extent of glycolysis

Glycolysis is the breakdown of a glucose molecule into pyruvic acid. In how many steps does it occur, and how many pyruvic acid molecules are formed from one glucose?

- A. Ten steps; two pyruvic acid molecules
- B. Ten steps; one pyruvic acid molecule
- C. Eight steps; two pyruvic acid molecules
- D. Twelve steps; three pyruvic acid molecules

BIO-11-CH02-0009 | Easy | Preparatory phase sequence

Arrange the following glycolytic intermediates in the order in which they are formed: I. Fructose-6-phosphate, II. Glucose-6-phosphate, III. Fructose-1,6-diphosphate, IV. Glyceraldehyde-3-phosphate.

- A. I, II, III, IV
- B. II, I, IV, III
- C. III, II, I, IV
- D. II, I, III, IV

BIO-11-CH02-0010 | Medium | Pay-off phase sequence

Arrange the following glycolytic intermediates in the order in which they are formed: I. 2-Phosphoglycerate, II. 1,3-Bisphosphoglycerate, III. Phosphoenol pyruvate, IV. 3-Phosphoglycerate.

- A. II, I, IV, III
- B. IV, II, I, III
- C. II, IV, I, III
- D. II, IV, III, I

BIO-11-CH02-0011 | Easy | Enzymes of glycolysis

Which enzyme catalyses the conversion of glucose to glucose-6-phosphate?

- A. Hexokinase
- B. Phosphofructokinase
- C. Phosphohexose isomerase
- D. Phosphoglycerate kinase

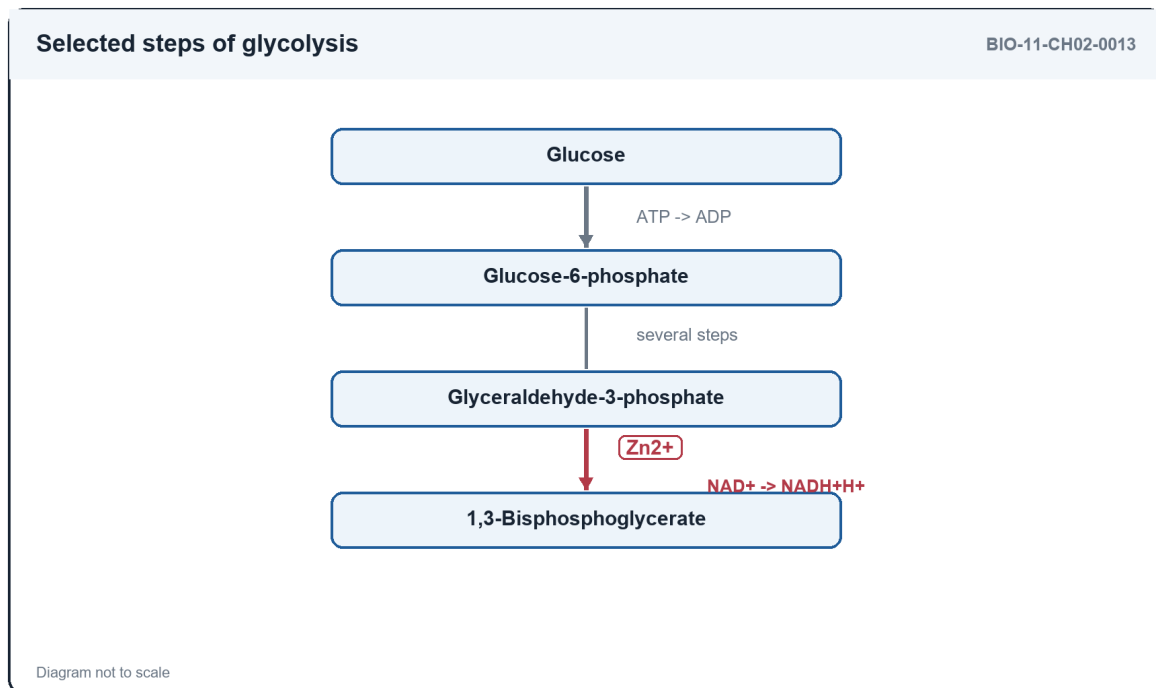
BIO-11-CH02-0012 | Medium | Enzymes of glycolysis

Match each conversion with its enzyme. I. Fructose-1,6-diphosphate to two trioses; II. Dihydroxyacetone phosphate to glyceraldehyde-3-phosphate; III. 2-Phosphoglycerate to phosphoenol pyruvate; IV. Phosphoenol pyruvate to pyruvic acid.

- A. I-Aldolase, II-Triose phosphate isomerase, III-Pyruvate kinase, IV-Enolase
- B. I-Triose phosphate isomerase, II-Aldolase, III-Enolase, IV-Pyruvate kinase
- C. I-Aldolase, II-Triose phosphate isomerase, III-Enolase, IV-Pyruvate kinase
- D. I-Aldolase, II-Phosphohexose isomerase, III-Enolase, IV-Pyruvate kinase

BIO-11-CH02-0013 | Medium | Cofactors of glycolysis

In the glycolysis flow diagram, Zn^{2+} is shown as the cofactor in exactly one conversion. Which conversion is it, and what else is produced in that step?



- A. Glucose to glucose-6-phosphate, with release of ADP
- B. Glyceraldehyde-3-phosphate to 1,3-bisphosphoglycerate, with 2 $\text{NADH}+\text{H}^+$
- C. Fructose-6-phosphate to fructose-1,6-diphosphate, with 2 $\text{NADH}+\text{H}^+$
- D. 2-Phosphoglycerate to phosphoenol pyruvate, with 2 H_2O

BIO-11-CH02-0014 | Hard | Pay-off phase chemistry

In glycolysis, adding a phosphate group to a metabolite usually costs one ATP, as at the hexokinase and phosphofructokinase steps. In the pay-off phase, glyceraldehyde-3-phosphate is converted to 1,3-bisphosphoglycerate and so gains a second phosphate group. What is the source of this phosphate?

- A. ATP, so two more ATP are spent per glucose
- B. GTP, so two GTP are spent per glucose
- C. ADP, which is converted to AMP in the process
- D. Inorganic phosphate, so no further ATP is spent

BIO-11-CH02-0015 | Medium | ATP investment

In the preparatory phase of glycolysis, glucose is phosphorylated twice. How many ATP molecules are consumed per glucose, and which two enzymes catalyse these ATP-consuming steps?

- A. Two ATP; hexokinase and phosphoglycerate kinase
- B. Four ATP; hexokinase and phosphofructokinase
- C. Two ATP; hexokinase and phosphofructokinase
- D. One ATP; hexokinase alone

BIO-11-CH02-0016 | Medium | ATP production

Glycolysis of one glucose consumes 2 ATP and shows a net gain of 2 ATP. How many ATP are produced by substrate-level phosphorylation, and at which conversions?

- A. Two; at 1,3-bisphosphoglycerate to 3-phosphoglycerate and at phosphoenol pyruvate to pyruvic acid
- B. Four; at glucose to glucose-6-phosphate and at fructose-6-phosphate to fructose-1,6-diphosphate
- C. Six; at 1,3-bisphosphoglycerate to 3-phosphoglycerate, at 2-phosphoglycerate to phosphoenol pyruvate and at phosphoenol pyruvate to pyruvic acid
- D. Four; at 1,3-bisphosphoglycerate to 3-phosphoglycerate and at phosphoenol pyruvate to pyruvic acid

BIO-11-CH02-0017 | Hard | Net gain of glycolysis

A tissue converts 5 molecules of glucose completely to pyruvic acid by glycolysis. Glycolysis consumes 2 ATP and produces 4 ATP per glucose. How many pyruvic acid molecules and how many net ATP molecules are formed?

- A. 10 pyruvic acid and 20 net ATP
- B. 10 pyruvic acid and 10 net ATP
- C. 5 pyruvic acid and 10 net ATP
- D. 10 pyruvic acid and 30 net ATP

BIO-11-CH02-0018 | Hard | NADH+H⁺ of glycolysis

Glycolysis produces 2 NADH+H⁺ per glucose, and in the balance sheet for ATP by aerobic oxidation of glucose each NADH+H⁺ is accounted as 3 ATP. If 4 molecules of glucose are oxidised to pyruvic acid, how many NADH+H⁺ are formed and how many ATP would they yield on that accounting?

- A. 8 NADH+H⁺ and 16 ATP
- B. 8 NADH+H⁺ and 24 ATP
- C. 4 NADH+H⁺ and 12 ATP
- D. 8 NADH+H⁺ and 32 ATP

BIO-11-CH02-0019 | Medium | Glycolysis in the ATP balance sheet

In the glycolysis row of the balance sheet for ATP by aerobic oxidation of one glucose molecule, substrate-level phosphorylation contributes 4 ATP and oxidative phosphorylation of the NADH+H⁺ contributes 6 ATP, while 2 ATP are consumed. What are the total and the net benefit entered for glycolysis?

- A. Total 10, net benefit 10
- B. Total 10, net benefit 8
- C. Total 8, net benefit 6
- D. Total 12, net benefit 10

BIO-11-CH02-0020 | Medium | Summary equation of glycolysis

The overall equation of glycolysis is written as: Glucose + 2 ATP + 2 iP + 4 ADP + 2 NAD⁺ → products. Which set appears on the product side?

- A. 2 Pyruvate + 4 ADP + 2 ATP + 2 NADH+H⁺ + 2 H₂O
- B. 2 Pyruvate + 2 ADP + 4 ATP + 2 NAD⁺ + 2 H₂O
- C. 2 Pyruvate + 2 ADP + 4 ATP + 2 NADH+H⁺ + 2 H₂O
- D. 2 Pyruvate + 2 ADP + 4 ATP + 2 NADH+H⁺ + 2 CO₂

BIO-11-CH02-0021 | Easy | Fate of pyruvic acid

Pyruvic acid formed at the end of glycolysis may be converted into which set of compounds?

- A. Lactic acid, ethanol or citric acid
- B. Ethanol, acetyl CoA or oxaloacetate
- C. Lactic acid, ethanol or acetyl CoA
- D. Lactic acid, ethanol or glutamic acid

BIO-11-CH02-0022 | Medium | Connecting link reaction

The conversion of pyruvic acid to acetyl CoA is called the connecting link reaction. Which description of it is correct?

- A. An oxidative decarboxylation by the pyruvate dehydrogenase complex, giving acetyl CoA, CO₂ and NADH+H⁺
- B. A condensation by citrate synthase, giving citrate with release of CoA-SH
- C. A decarboxylation giving acetaldehyde and CO₂, with no coenzyme reduced
- D. A dehydrogenation by succinate dehydrogenase, giving FADH₂

BIO-11-CH02-0023 | Easy | Tissues dependent on glycolysis

In which set of cells and tissues is glycolysis the sole source of energy?

- A. Erythrocytes, renal medulla, brain and sperm
- B. Erythrocytes, liver, brain and sperm
- C. Cardiac muscle, renal cortex, brain and sperm
- D. Erythrocytes, renal medulla, liver and skeletal muscle

BIO-11-CH02-0024 | Hard | Muscle fibre types

Red (dark) and white (pale) muscle fibres differ in their dependence on glycolysis. Which statement about them is correct?

- A. White fibres contain more myoglobin and sustain aerobic oxidation, suiting endurance events such as marathon and swimming
- B. Red fibres contain more myoglobin and sustain aerobic oxidation, suiting endurance events such as marathon and swimming
- C. Red fibres accumulate lactate rapidly and fatigue quickly, suiting sprinting
- D. White fibres contain more myoglobin yet fatigue quickly because they accumulate lactate

BIO-11-CH02-0025 | Medium | Regulation of glycolysis

The rate of glycolysis is stated to be controlled by an interplay of intracellular factors together with certain hormones. Which combination is correct?

- A. ATP consumption, NADH₂ regeneration and regulation of glycolytic enzymes, with glucagon, thyroxine and insulin
- B. Oxygen availability and carbon dioxide removal, with glucagon, epinephrine and insulin
- C. ATP consumption and regeneration of FADH₂, with epinephrine, insulin and cortisol
- D. ATP consumption, NADH₂ regeneration and regulation of glycolytic enzymes, with glucagon, epinephrine and insulin

BIO-11-CH02-0026 | Challenging | Pay-off phase accounting

A cell-free preparation is supplied with glyceraldehyde-3-phosphate instead of glucose, and each molecule is carried through the remaining glycolytic reactions to pyruvic acid. Per glucose these reactions yield 4 ATP and 2 NADH+H⁺ from 2 such trioses. If 6 molecules of glyceraldehyde-3-phosphate are supplied, how many ATP and NADH+H⁺ are formed, and how much ATP is invested?

- A. 12 ATP and 6 NADH+H⁺, with 6 ATP invested
- B. 6 ATP and 6 NADH+H⁺, with no ATP invested
- C. 24 ATP and 12 NADH+H⁺, with no ATP invested
- D. 12 ATP and 6 NADH+H⁺, with no ATP invested

BIO-11-CH02-0027 | Easy | Definition of anaerobic respiration

Which statement correctly defines anaerobic respiration?

- A. Cellular respiration in which molecular oxygen acts as the final electron acceptor
- B. Cellular respiration in which glucose is completely oxidised to carbon dioxide and water
- C. Cellular respiration that is confined to the mitochondrial matrix
- D. Cellular respiration that does not involve oxygen at all, also called fermentation

BIO-11-CH02-0028 | Medium | Lactic acid fermentation

In skeletal muscle the NADH+H⁺ formed during glycolysis is reoxidised to NAD⁺ by donating one proton and two electrons to an acceptor. Which acceptor and which product are involved?

- A. Acetaldehyde is the acceptor and ethanol is the product
- B. Molecular oxygen is the acceptor and water is the product
- C. Pyruvic acid is the acceptor and lactic acid is the product
- D. Oxaloacetate is the acceptor and malate is the product

BIO-11-CH02-0029 | Medium | Alcoholic fermentation

In yeast, pyruvate is first decarboxylated and the product of that decarboxylation is then reduced by NADH+H⁺. Which compound is formed first and which is formed finally?

- A. Acetyl CoA is formed first and ethanol finally
- B. Lactic acid is formed first and ethanol finally
- C. Acetaldehyde is formed first and acetic acid finally
- D. Acetaldehyde is formed first and ethanol finally

BIO-11-CH02-0030 | Medium | Comparison of the two fermentations

Which difference between lactic acid fermentation in muscle and alcoholic fermentation in yeast is correct?

- A. Carbon dioxide is released in alcoholic fermentation but not in lactic acid fermentation
- B. Carbon dioxide is released in lactic acid fermentation but not in alcoholic fermentation
- C. NADH+H⁺ is reoxidised in lactic acid fermentation but not in alcoholic fermentation
- D. Pyruvic acid is the starting compound in alcoholic fermentation but not in lactic acid fermentation

BIO-11-CH02-0031 | Hard | Role of coenzyme reoxidation

In respiration, oxidation of NADH+H⁺ through the electron transport system yields ATP. In fermentation the NADH+H⁺ formed during glycolysis is also oxidised. What is the purpose of this oxidation in fermentation?

- A. Regeneration of NAD⁺ so that glycolysis can continue, with no ATP formed at this step
- B. Production of three ATP for each NADH+H⁺, as in the electron transport system
- C. Production of water by handing the hydrogen to molecular oxygen
- D. Complete oxidation of pyruvic acid to carbon dioxide

BIO-11-CH02-0032 | Medium | ATP yield of fermentation

During fermentation of one glucose molecule, ATP is formed only in the reactions of glycolysis, where 4 ATP are produced and 2 are consumed, and the subsequent conversion of pyruvic acid yields no ATP. What is the net ATP gain per glucose?

- A. 4 ATP (net per glucose)
- B. 2 ATP (net per glucose)
- C. 6 ATP (net per glucose)
- D. 8 ATP (net per glucose)

BIO-11-CH02-0033 | Challenging | Stoichiometry of alcoholic fermentation

Yeast ferments 4 molecules of glucose to ethanol in the absence of oxygen. Glycolysis produces 4 ATP and consumes 2 ATP per glucose, and each of the two pyruvic acid molecules formed per glucose loses one carbon dioxide during alcoholic fermentation. How many net ATP and how many carbon dioxide molecules are formed?

- A. 8 net ATP and 8 CO₂
- B. 16 net ATP and 8 CO₂
- C. 8 net ATP and 4 CO₂
- D. 8 net ATP and 12 CO₂

BIO-11-CH02-0034 | Challenging | Comparison with aerobic respiration

Why is much less energy obtained per glucose molecule in anaerobic respiration than in aerobic respiration?

- A. Glucose is completely oxidised, but the ATP that is formed is at once hydrolysed back to ADP
- B. Glucose is not completely oxidised, and the reduced coenzyme is spent on lactic acid or ethanol
- C. Fewer glucose molecules are able to enter glycolysis when oxygen is absent, so fewer ATP are formed
- D. Anaerobic respiration omits glycolysis altogether and begins at the conversion of pyruvic acid

BIO-11-CH02-0035 | Medium | Try this: fermentation by yeast

In the activity on respiration in yeast, a pinch of dry baker's yeast is added to 10 ml of 10 per cent glucose solution, the mixture is covered with a layer of oil to exclude air and kept at 37 to 38 degree C, and the gas evolved is passed into lime water. Which pair of observations is recorded?

- A. Lime water gradually turns milky and a smell of alcohol develops
- B. Lime water gradually turns milky and the volume of gas steadily declines
- C. Lime water remains clear and a smell of alcohol develops
- D. Lime water gradually turns milky and a smell of lactic acid develops

BIO-11-CH02-0036 | Hard | Try this: germinating seeds in mercury

Seed coats are removed from a few germinating peas, which are placed in a test tube filled with mercury; the tube is inverted in a trough of mercury. As germination proceeds a gas collects at the top of the tube, and when a pellet of potassium hydroxide is introduced it rises to the top, absorbs the gas and the mercury fills the tube again. What does the activity establish?

- A. The gas is oxygen, and the seeds release oxygen as they germinate
- B. The gas is carbon dioxide, and the seeds respire anaerobically in the absence of oxygen
- C. The gas is carbon dioxide, and the seeds respire aerobically using oxygen dissolved in the mercury
- D. The gas is water vapour, which the potassium hydroxide takes up on condensation

BIO-11-CH02-0037 | Easy | Nature of the connecting link reaction

Pyruvic acid formed at the end of glycolysis is converted into acetyl CoA in the step that connects glycolysis with the Krebs cycle. This step removes carbon dioxide and hydrogen from the substrate. What type of reaction is this?

- A. Oxidative decarboxylation
- B. Substrate level phosphorylation
- C. Oxidative phosphorylation
- D. Condensation

BIO-11-CH02-0038 | Easy | Products of the connecting link reaction

One molecule of pyruvic acid is completely converted in the connecting link reaction. Which set names all the products of that single conversion?

- A. Acetyl CoA, carbon dioxide and FADH₂
- B. Acetyl CoA, carbon dioxide and ATP
- C. Acetyl CoA, water and NADH+H⁺
- D. Acetyl CoA, carbon dioxide and NADH+H⁺

BIO-11-CH02-0039 | Medium | Enzyme and coenzymes of the link reaction

Which combination of enzyme and coenzymes carries out the conversion of pyruvic acid to acetyl CoA?

- A. Pyruvate dehydrogenase complex, with FAD⁺ and CoA-SH
- B. Pyruvate dehydrogenase complex, with NAD⁺ and CoA-SH
- C. Citrate synthase, with NAD⁺ and CoA-SH
- D. Isocitrate dehydrogenase, with NAD⁺ and FAD⁺

BIO-11-CH02-0040 | Medium | Site of the connecting link reaction

Where does the conversion of pyruvic acid to acetyl CoA take place in a eukaryotic cell, and where does the same conversion take place in a prokaryotic cell?

- A. In the cytosol in eukaryotes and in the mitochondria in prokaryotes
- B. In the mitochondria in eukaryotes and in the cytosol in prokaryotes
- C. In the mitochondria in eukaryotes and in the mitochondria in prokaryotes
- D. In the cytosol in eukaryotes and in the cytosol in prokaryotes

BIO-11-CH02-0041 | Medium | Vitamin requirement of the link reaction

The pyruvate dehydrogenase complex cannot function in the absence of thiamin (vitamin B₁). A person's diet is severely deficient in thiamin. Which immediate consequence for the connecting link reaction follows?

- A. Pyruvic acid is converted into acetyl CoA without release of carbon dioxide
- B. Acetyl CoA is formed but no NADH+H⁺ accompanies its formation
- C. Pyruvic acid enters the Krebs cycle directly in place of acetyl CoA
- D. Pyruvic acid can no longer be converted into acetyl CoA

BIO-11-CH02-0042 | Medium | ATP accounting for the link reaction

One glucose molecule yields two molecules of pyruvic acid, and each of them undergoes the connecting link reaction, producing one NADH+H⁺ each. Using the prescribed MHT-CET textbook balance sheet each NADH+H⁺ is worth 3 ATP, and this step consumes no ATP. How many ATP does the connecting link reaction contribute per glucose?

- A. 3
- B. 2
- C. 6
- D. 4

BIO-11-CH02-0043 | Easy | Site of the Krebs cycle

In which part of the cell do the reactions of the Krebs cycle take place?

- A. Inner mitochondrial membrane — Krebs-cycle site
- B. Outer chamber of the mitochondrion — Krebs-cycle site
- C. Cytoplasm — Krebs-cycle site
- D. Mitochondrial matrix — Krebs-cycle site

BIO-11-CH02-0044 | Easy | Entry of acetyl CoA

Acetyl CoA entering the Krebs cycle condenses with the acceptor molecule of the cycle, in a reaction catalysed by citrate synthase in which CoA-SH is released. With which compound does acetyl CoA condense?

- A. Alpha-ketoglutarate
- B. Succinate
- C. Oxaloacetate
- D. Malate

BIO-11-CH02-0045 | Easy | Order of intermediates in the first half

Arrange the following species of the Krebs cycle in the order in which they appear in the prescribed pathway sequence: citrate, cis-aconitate, isocitrate, oxalosuccinic acid.

- A. Citrate --> cis-aconitate --> isocitrate --> oxalosuccinic acid
- B. Citrate --> isocitrate --> cis-aconitate --> oxalosuccinic acid
- C. cis-Aconitate --> citrate --> isocitrate --> oxalosuccinic acid
- D. Citrate --> cis-aconitate --> oxalosuccinic acid --> isocitrate

BIO-11-CH02-0046 | Easy | Substrate level phosphorylation in the cycle

Exactly one step of the Krebs cycle produces GTP (ATP) by substrate level phosphorylation. Which conversion is it?

- A. Alpha-ketoglutarate to succinyl-CoA
- B. Succinate to fumarate
- C. Fumarate to malate
- D. Succinyl-CoA to succinate

BIO-11-CH02-0047 | Easy | Regeneration of oxaloacetate

The Krebs cycle ends each turn by restoring the compound that accepts the next acetyl CoA. Which conversion regenerates that acceptor?

- A. Fumarate to malate
- B. Succinate to fumarate
- C. Malate to oxaloacetate
- D. Citrate to cis-aconitate

BIO-11-CH02-0048 | Medium | Order of intermediates in the second half

Starting immediately after succinyl-CoA, arrange the following species in the order in which they appear in the Krebs cycle: fumarate, malate, oxaloacetate, succinate.

- A. Succinate --> malate --> fumarate --> oxaloacetate
- B. Succinate --> fumarate --> malate --> oxaloacetate
- C. Fumarate --> succinate --> malate --> oxaloacetate
- D. Succinate --> fumarate --> oxaloacetate --> malate

BIO-11-CH02-0049 | Medium | Adjacency of intermediates

In the Krebs cycle in the prescribed sequence, which species is formed immediately before alpha-ketoglutarate, and which species is formed immediately after succinyl-CoA?

- A. Isocitrate and succinate
- B. Oxalosuccinic acid and fumarate
- C. Oxalosuccinic acid and succinate
- D. cis-Aconitate and succinate

BIO-11-CH02-0050 | Medium | Enzymes of the Krebs cycle

Which pairing of a Krebs cycle enzyme with the conversion it catalyses is correct?

- A. Citrate synthase with the conversion of citrate to cis-aconitate
- B. Fumarase with the conversion of fumarate to malate
- C. Succinate dehydrogenase with the conversion of malate to oxaloacetate
- D. Aconitase with the conversion of succinyl-CoA to succinate

BIO-11-CH02-0051 | Medium | Decarboxylation steps of the cycle

Carbon dioxide is released at exactly two conversions within one turn of the Krebs cycle. Which pair of conversions are these?

- A. Citrate to cis-aconitate and cis-aconitate to isocitrate
- B. Isocitrate to alpha-ketoglutarate and alpha-ketoglutarate to succinyl-CoA
- C. Succinate to fumarate and fumarate to malate
- D. Alpha-ketoglutarate to succinyl-CoA and succinyl-CoA to succinate

BIO-11-CH02-0052 | Hard | ATP yield per turn

One turn of the Krebs cycle yields 3 NADH+H⁺, 1 FADH₂ and 1 GTP counted as 1 ATP. Using the prescribed MHT-CET balance-sheet convention of 3 ATP per NADH+H⁺ and 2 ATP per FADH₂, how many ATP are attributable to one turn?

- A. 11
- B. 10
- C. 24
- D. 12

BIO-11-CH02-0053 | Medium | Carbon dioxide yield of the cycle

Each turn of the Krebs cycle releases carbon dioxide at its two oxidative decarboxylation steps, and two turns occur for every glucose molecule oxidised. How many molecules of carbon dioxide does the Krebs cycle release per glucose?

- A. 2 CO₂ molecules per glucose
- B. 6 CO₂ molecules per glucose
- C. 4 CO₂ molecules per glucose
- D. 8 CO₂ molecules per glucose

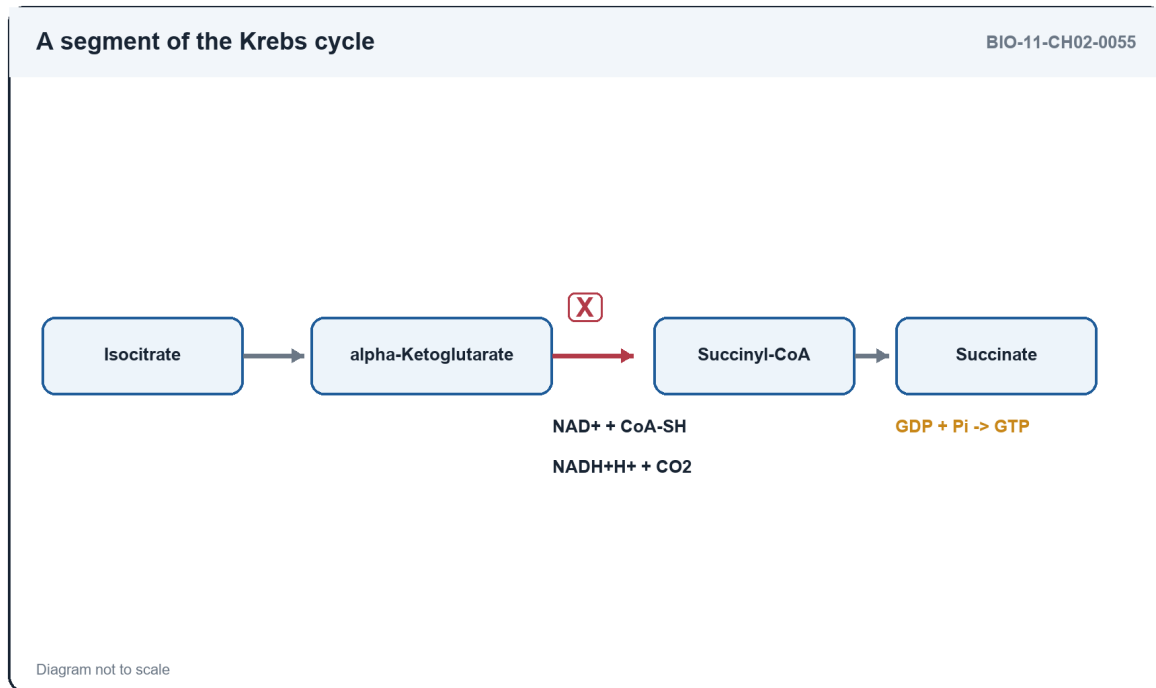
BIO-11-CH02-0054 | Hard | Split of the Krebs contribution by mechanism

The Krebs cycle contributes a total of 24 ATP for every glucose molecule oxidised. Of the mechanisms involved, substrate level phosphorylation supplies 1 GTP counted as 1 ATP in each turn, and two turns occur per glucose. How many of the 24 ATP arise by oxidative phosphorylation?

- A. 23
- B. 20
- C. 22
- D. 18

BIO-11-CH02-0055 | Hard | Reading the labels on a cycle step

In the Krebs-cycle segment shown, conversion X uses NAD⁺ and CoA-SH and releases NADH+H⁺ and carbon dioxide. It produces the compound that next undergoes substrate-level phosphorylation. Which conversion is X?



- A. Alpha-ketoglutarate to succinyl-CoA
- B. Isocitrate to alpha-ketoglutarate
- C. Succinyl-CoA to succinate
- D. Malate to oxaloacetate

BIO-11-CH02-0056 | Medium | Reduced coenzyme yield of the cycle

Each turn of the Krebs cycle produces 3 NADH+H⁺ and 1 FADH₂, and two turns occur for every glucose molecule oxidised. How many molecules of reduced coenzyme in total does the Krebs cycle deliver per glucose?

- A. 4
- B. 6
- C. 10
- D. 8

BIO-11-CH02-0057 | Challenging | Carbon balance of glucose oxidation

One glucose molecule of six carbon atoms is split by glycolysis into two molecules of pyruvic acid of three carbon atoms each. Each pyruvic acid then loses one carbon as carbon dioxide in the connecting link reaction, and each of the two turns of the Krebs cycle releases two more molecules of carbon dioxide. How many of the six carbon atoms of the glucose have left as carbon dioxide once both stages are complete?

- A. 4 carbon atoms
- B. 2 carbon atoms
- C. 6 carbon atoms
- D. 3 carbon atoms

BIO-11-CH02-0058 | Hard | Substrate level phosphorylation across the pathway

In the aerobic oxidation of one glucose molecule, substrate level phosphorylation produces 4 ATP during glycolysis and 1 GTP counted as 1 ATP in each turn of the Krebs cycle, with two turns occurring per glucose. The connecting link reaction contributes none. What is the total ATP produced by substrate level phosphorylation?

- A. 6
- B. 5
- C. 4
- D. 2

BIO-11-CH02-0059 | Challenging | Working back from a measured ATP total

A preparation of mitochondria is supplied with acetyl CoA and yields 60 ATP from Krebs cycle turns alone. Each turn gives 3 NADH+H⁺, 1 FADH₂, 1 GTP counted as 1 ATP, and 2 molecules of carbon dioxide. Each NADH+H⁺ is worth 3 ATP and each FADH₂ is worth 2 ATP. How many turns occurred, and how much carbon dioxide was released?

- A. 5 turns and 5 molecules of carbon dioxide
- B. 5 turns and 10 molecules of carbon dioxide
- C. 6 turns and 12 molecules of carbon dioxide
- D. 10 turns and 20 molecules of carbon dioxide

BIO-11-CH02-0060 | Medium | Hydrogen acceptor at the oxidation steps

Rule: in the Krebs cycle the oxidation steps are catalysed by dehydrogenases, and these hand the removed hydrogen to NAD⁺ to form NADH+H⁺. Exactly one oxidation of the cycle departs from this pattern. Which conversion is it?

- A. Succinate to fumarate
- B. Isocitrate to alpha-ketoglutarate
- C. Malate to oxaloacetate
- D. Alpha-ketoglutarate to succinyl-CoA

BIO-11-CH02-0061 | Hard | An enzyme acting twice in one turn

Rule: in the Krebs cycle most named enzymes catalyse a single conversion of the turn. Which of the following enzymes departs from that pattern by catalysing two consecutive conversions?

- A. Citrate synthase
- B. Fumarase
- C. Malate dehydrogenase
- D. Aconitase

BIO-11-CH02-0062 | Medium | Dehydrogenases of the cycle

Four oxidation steps of the Krebs cycle are catalysed by dehydrogenases. Which of the following named enzymes of the cycle instead catalyses a hydration?

- A. Fumarase
- B. Succinate dehydrogenase
- C. Citrate synthase
- D. Succinyl-CoA synthetase

BIO-11-CH02-0063 | Challenging | Order of intermediates across the whole cycle

Arrange the following species in the order in which they appear in one turn of the Krebs cycle in the prescribed sequence: alpha-ketoglutarate, cis-aconitate, malate, succinyl-CoA.

- A. cis-Aconitate --> succinyl-CoA --> alpha-ketoglutarate --> malate
- B. Alpha-ketoglutarate --> cis-aconitate --> succinyl-CoA --> malate
- C. cis-Aconitate --> alpha-ketoglutarate --> succinyl-CoA --> malate
- D. cis-Aconitate --> alpha-ketoglutarate --> malate --> succinyl-CoA

BIO-11-CH02-0064 | Hard | Net fate of oxaloacetate

Rule: a compound that is used up at the first step of a metabolic sequence has to be supplied continuously from outside that sequence. Oxaloacetate is used up at the first step of the Krebs cycle, yet the cycle can keep running without an outside supply of it. Which feature of the cycle accounts for this?

- A. It is formed from acetyl CoA during the opening condensation step
- B. It is produced from pyruvic acid by the pyruvate dehydrogenase complex
- C. It is re-formed at the last step of the same turn by the oxidation of malate
- D. It is released along with carbon dioxide at the oxidative decarboxylation steps

BIO-11-CH02-0065 | Easy | Meaning of amphibolic

The Krebs cycle is described as an amphibolic pathway. What does this term mean?

- A. A pathway involving only the oxidative breakdown of substrates
- B. A pathway involving catabolism as well as anabolism
- C. A pathway involving only the building up of complex molecules
- D. A pathway that can operate both in the cytoplasm and in the mitochondria

BIO-11-CH02-0066 | Easy | Precursor of aspartic acid

According to the prescribed MHT-CET account, aspartic acid is synthesised using one of the intermediates of the Krebs cycle as its precursor. Which intermediate is it?

- A. Alpha-ketoglutarate
- B. Succinate
- C. Fumarate
- D. Oxaloacetate

BIO-11-CH02-0067 | Medium | Precursor and product pairings

Which pairing of a precursor with the biomolecule synthesised from it is correct as given in the prescribed account?

- A. Alpha-ketoglutarate with aspartic acid
- B. Alpha-ketoglutarate with glutamic acid
- C. Oxaloacetate with fatty acids
- D. Acetyl CoA with aspartic acid

BIO-11-CH02-0068 | Medium | Entry of fats and proteins

Amino acids from protein metabolism and fatty acids from lipid metabolism also take part in the Krebs cycle. Through which compound do they enter it?

- A. Oxaloacetate
- B. Alpha-ketoglutarate
- C. Succinyl-CoA
- D. Acetyl CoA

BIO-11-CH02-0069 | Medium | Consequence of a non-carbohydrate substrate

Aerobic respiration is described as operating through glycolysis, production of acetyl CoA, the Krebs cycle, the electron transfer chain reaction and terminal oxidation. Fatty acids join the pathway as acetyl CoA. Which of these named steps is omitted when fatty acids are used as the respiratory substrate?

- A. Glycolysis
- B. Krebs cycle
- C. Electron transfer chain reaction
- D. Terminal oxidation

BIO-11-CH02-0070 | Hard | Why respiration is not purely catabolic

Rule: a pathway that oxidises its substrates stepwise, releases energy and gives off carbon dioxide is a catabolic pathway. The Krebs cycle does all of these things, yet the prescribed account states calling it catabolic is not entirely correct. Which observation makes it an exception to the rule?

- A. It regenerates oxidised coenzymes that are used again in later turns
- B. It releases its energy in many small steps instead of all at once
- C. It accepts substrates derived from carbohydrates, fats and proteins alike
- D. Some of its intermediates are withdrawn as precursors for the synthesis of other biomolecules

BIO-11-CH02-0071 | Hard | Common oxidative pathway

A cell has no carbohydrate available and is respiring using fatty acids and amino acids as substrates. Which statement about its Krebs cycle is correct?

- A. The cycle stops, because acetyl CoA can be formed only from pyruvic acid
- B. The cycle still operates, because these substrates enter it through acetyl CoA
- C. The cycle operates but releases no carbon dioxide, because carbon dioxide comes only from carbohydrate
- D. The cycle operates only after these substrates have first been converted into glucose

BIO-11-CH02-0072 | Challenging | Cost of withdrawing intermediates

A cell is rapidly synthesising glutamic acid and aspartic acid, and so withdraws large amounts of alpha-ketoglutarate and oxaloacetate from the Krebs cycle. The supply of acetyl CoA to the cell is unchanged. Which statement best explains why the oxidative output of the cycle nevertheless falls?

- A. Alpha-ketoglutarate carries the acetyl group through the cycle, so its removal leaves the acetyl group with no carrier
- B. Both intermediates act as hydrogen acceptors at the oxidation steps, so their removal leaves the dehydrogenases without acceptors
- C. Both intermediates supply the CoA-SH needed at the condensation step, so their removal blocks the formation of citrate
- D. Oxaloacetate must be present at the start of each turn to accept acetyl CoA, so its removal limits the number of turns

BIO-11-CH02-0073 | Easy | Site of the electron transport system

The electron carriers and enzymes that oxidise the NADH+H⁺ and FADH₂ produced during glycolysis, the connecting link reaction and the Krebs cycle are arranged in the form of complexes. On which part of the mitochondrion are they arranged?

- A. Mitochondrial matrix — ETC location
- B. Outer chamber of the mitochondrion — ETC location
- C. Inner mitochondrial membrane — ETC location
- D. Cytoplasm — ETC location

BIO-11-CH02-0074 | Easy | Terminal electron acceptor

In aerobic respiration the electrons handed on from carrier to carrier are finally delivered to one substance. Which substance accepts them?

- A. Molecular oxygen
- B. Ubiquinone
- C. Cytochrome C
- D. NAD⁺

BIO-11-CH02-0075 | Easy | Terminal oxidation and metabolic water

During terminal oxidation a water molecule is formed which is called metabolic water. Which combination gives rise to it?

- A. Reduced molecular oxygen combines with protons
- B. Reduced molecular oxygen combines with electrons
- C. Reduced ubiquinone combines with protons
- D. Reduced cytochrome C combines with protons

BIO-11-CH02-0076 | Easy | Chemiosmosis

Peter Mitchell gave a name to the process in which the transfer of protons across the inner mitochondrial membrane is accompanied by the synthesis of ATP. What is that name?

- A. Substrate-level phosphorylation
- B. Photophosphorylation
- C. Chemiosmosis
- D. Oxidative decarboxylation

BIO-11-CH02-0077 | Easy | Oxysome

One particle of the inner mitochondrial membrane is described as having an F₀ part and an F₁ part and is responsible for the synthesis of ATP. Which particle is it?

- A. NADH dehydrogenase
- B. Oxysome
- C. Cytochrome bc₁ complex
- D. Cytochrome C oxidase

BIO-11-CH02-0078 | Medium | Sequence of electron carriers

Five components of the mitochondrial electron transport system are listed in random order. I: Cytochrome C. II: NADH dehydrogenase. III: Cytochrome C oxidase. IV: Ubiquinone. V: Cytochrome bc₁ complex. Arrange them in the order in which electrons released from NADH+H⁺ pass through them.

- A. II, IV, V, I, III
- B. II, IV, I, V, III
- C. II, V, IV, I, III
- D. IV, II, V, I, III

BIO-11-CH02-0079 | Medium | Sequence of electron carriers

Five components of the mitochondrial electron transport system are listed in random order. I: Ubiquinone. II: Cytochrome C oxidase. III: Succinate dehydrogenase. IV: Cytochrome bc₁ complex. V: Cytochrome C. Arrange them in the order in which electrons released from FADH₂ pass through them.

- A. I, III, IV, V, II
- B. III, I, V, IV, II
- C. III, I, IV, V, II
- D. III, IV, I, V, II

BIO-11-CH02-0080 | Hard | Sequence of electron carriers

The following are listed in random order. I: Metabolic water. II: Cytochrome C oxidase. III: Cytochrome bc1 complex. IV: Molecular oxygen. V: Cytochrome C. Arrange them in the order in which they are involved once ubiquinol has been formed.

- A. III, II, V, IV, I
- B. V, III, II, IV, I
- C. III, V, II, I, IV
- D. III, V, II, IV, I

BIO-11-CH02-0081 | Hard | Sequence of electron carriers

The carriers below are to be arranged against the direction of electron flow, that is beginning with the carrier that hands electrons over to molecular oxygen and ending with the carrier that oxidises $\text{NADH} + \text{H}^+$. I: Ubiquinone. II: Cytochrome C oxidase. III: NADH dehydrogenase. IV: Cytochrome C. V: Cytochrome bc1 complex.

- A. III, I, V, IV, II
- B. II, IV, V, I, III
- C. II, V, IV, I, III
- D. II, IV, I, V, III

BIO-11-CH02-0082 | Medium | Entry of the reduced coenzymes

Which statement correctly matches the two reduced coenzymes of respiration with the complexes that oxidise them?

- A. $\text{NADH} + \text{H}^+$ is oxidised by succinate dehydrogenase and FADH_2 by NADH dehydrogenase
- B. $\text{NADH} + \text{H}^+$ is oxidised by NADH dehydrogenase and FADH_2 by succinate dehydrogenase
- C. $\text{NADH} + \text{H}^+$ is oxidised by NADH dehydrogenase and FADH_2 by the cytochrome bc1 complex
- D. $\text{NADH} + \text{H}^+$ is oxidised by ubiquinone and FADH_2 by succinate dehydrogenase

BIO-11-CH02-0083 | Medium | Ubiquinone as a mobile carrier

Ubiquinone (CoQ) receives electrons from the complexes that oxidise $\text{NADH} + \text{H}^+$ and FADH_2 . What is its reduced form called, and which complex oxidises that reduced form?

- A. Ubiquinol, oxidised by the cytochrome bc1 complex
- B. Ubiquinol, oxidised by cytochrome C oxidase
- C. Ubiquinol, oxidised by succinate dehydrogenase
- D. Ubiquinone, oxidised by the cytochrome bc1 complex

BIO-11-CH02-0084 | Hard | Mobile carriers of the chain

The carriers of the electron transport system are described as arranged on the inner mitochondrial membrane in the form of complexes. One carrier is an exception to this description: it is a small, iron-containing protein that moves between two of those complexes. Which carrier is it?

- A. Ubiquinone
- B. Cytochrome C oxidase
- C. Cytochrome C
- D. Cytochrome bc1 complex

BIO-11-CH02-0085 | Medium | Cytochrome C oxidase

Which carrier of the electron transport system contains cytochrome a and cytochrome a₃ and carries out terminal oxidation by passing electrons to molecular oxygen?

- A. Cytochrome bc1 complex
- B. Cytochrome C oxidase
- C. Cytochrome C
- D. NADH dehydrogenase

BIO-11-CH02-0086 | Medium | ATP from reduced coenzymes

In the balance sheet for ATP by aerobic oxidation of one glucose molecule, oxidative phosphorylation is credited with 3 ATP for each $\text{NADH}+\text{H}^+$ oxidised and 2 ATP for each FADH_2 oxidised. Using the same figures, how many ATP molecules would oxidative phosphorylation yield from 5 molecules of $\text{NADH}+\text{H}^+$ and 2 molecules of FADH_2 ?

- A. 21
- B. 16
- C. 14
- D. 19

BIO-11-CH02-0087 | Hard | ATP accounting for the Krebs cycle

In the balance sheet for ATP by aerobic oxidation of one glucose molecule, the Krebs cycle row records 6 molecules of $\text{NADH}+\text{H}^+$, 2 molecules of FADH_2 and 2 ATP formed by substrate-level phosphorylation. Oxidative phosphorylation is credited with 3 ATP per $\text{NADH}+\text{H}^+$ and 2 ATP per FADH_2 . What total is entered against the Krebs cycle row?

- A. 22
- B. 26
- C. 24
- D. 12

BIO-11-CH02-0088 | Hard | ATP from reduced coenzymes

Oxidative phosphorylation is credited with 3 ATP for each $\text{NADH}+\text{H}^+$ and 2 ATP for each FADH_2 in the prescribed account's ATP balance sheet. A set of reduced coenzymes yields 34 ATP by oxidative phosphorylation, of which 4 ATP come from FADH_2 . How many molecules of each coenzyme were oxidised?

- A. 10 $\text{NADH}+\text{H}^+$ and 2 FADH_2
- B. 15 $\text{NADH}+\text{H}^+$ and 2 FADH_2
- C. 10 $\text{NADH}+\text{H}^+$ and 4 FADH_2
- D. 12 $\text{NADH}+\text{H}^+$ and 2 FADH_2

BIO-11-CH02-0089 | Challenging | ATP yield per reduced coenzyme

Using the prescribed MHT-CET textbook balance sheet for ATP by aerobic oxidation of one glucose molecule, the oxidative phosphorylation of a reduced coenzyme is worked out at 3 ATP for each molecule. One reduced coenzyme in that balance sheet is an exception to this figure. Which coenzyme is it, and what is it credited with?

- A. FADH_2 , credited with 4 ATP for each molecule
- B. FADH_2 , credited with 2 ATP for each molecule
- C. The $\text{NADH}+\text{H}^+$ of glycolysis, credited with 2 ATP for each molecule
- D. The $\text{NADH}+\text{H}^+$ of the connecting link reaction, credited with 2 ATP for each molecule

BIO-11-CH02-0090 | Medium | Proton movement and the oxysome

Protons are released during the oxidation of $\text{NADH}+\text{H}^+$ and FADH_2 by the carriers of the electron transport system. Into which compartment are these protons released, and through which part of the oxysome do they subsequently pass?

- A. Released into the outer chamber, they pass through the F1 part
- B. Released into the inner chamber, they pass through the F0 part
- C. Released into the outer chamber, they pass through the F0 part
- D. Released into the inner chamber, they pass through the F1 part

BIO-11-CH02-0091 | Easy | Regeneration of oxidised coenzymes

One stated significance of the electron transport system is that it regenerates oxidised coenzymes so that they can be recycled. Which coenzymes are regenerated in this way?

- A. $\text{NADH}+\text{H}^+$ and FADH_2
- B. NAD^+ and CoA-SH
- C. NAD^+ and FAD^+
- D. FAD^+ and CoA-SH

BIO-11-CH02-0092 | Medium | Water supplied to the Krebs cycle

Among the stated significances of the electron transport system is the supply of a substance required by another stage of aerobic respiration. Which substance is supplied, and which stage requires it?

- A. Water molecules, required by glycolysis
 - B. Carbon dioxide, required by the Krebs cycle
 - C. Molecular oxygen, required by the Krebs cycle
 - D. Water molecules, required by the Krebs cycle
-

BIO-11-CH02-0093 | Medium | Stepwise release of energy

The electron transport system does not release the energy of the reduced coenzymes all at once. Which statement gives the manner of release and the benefit attributed to it?

- A. Energy is released in a stepwise manner, and this prevents damage of cells
 - B. Energy is released in a single step, and this allows the whole of it to be captured as ATP
 - C. Energy is released in a stepwise manner, and this allows the cell to store it as heat
 - D. Energy is released in a stepwise manner, and this removes the need for molecular oxygen
-

BIO-11-CH02-0094 | Easy | Share of the total ATP

Using the prescribed MHT-CET textbook statement that 34 ATP molecules out of a total of 38 are produced through the electron transport system. On this basis, how many of the 38 ATP molecules arise by means other than the electron transport system?

- A. 6 ATP molecules from non-ETS steps
 - B. 2 ATP molecules from non-ETS steps
 - C. 8 ATP molecules from non-ETS steps
 - D. 4 ATP molecules from non-ETS steps
-

BIO-11-CH02-0095 | Easy | Definition of respiratory quotient

Which statement correctly gives both the meaning of the respiratory quotient and the other name by which it is known?

- A. The ratio of the volume of CO₂ released to the volume of O₂ consumed in respiration, also called the respiratory ratio
 - B. The ratio of the volume of O₂ consumed to the volume of CO₂ released in respiration, also called the respiratory ratio
 - C. The ratio of the volume of CO₂ released to the volume of O₂ consumed in respiration, also called the respiratory index
 - D. The difference between the volume of CO₂ released and the volume of O₂ consumed in respiration, also called the respiratory ratio
-

BIO-11-CH02-0096 | Easy | Formula for respiratory quotient

A respiring tissue releases a volume V₁ of carbon dioxide and consumes a volume V₂ of oxygen during the same interval. Which expression gives its respiratory quotient?

- A. V₂ divided by V₁
 - B. V₁ multiplied by V₂
 - C. V₁ divided by V₂
 - D. V₁ minus V₂
-

BIO-11-CH02-0097 | Medium | RQ of carbohydrate

For which respiratory substrate is the respiratory quotient equal to 1, and what does that value imply about the two gas volumes?

- A. A completely oxidised fat; the volume of CO₂ evolved equals the volume of O₂ consumed
 - B. A completely oxidised carbohydrate; the volume of CO₂ evolved equals the volume of O₂ consumed
 - C. A completely oxidised protein; the volume of CO₂ evolved equals the volume of O₂ consumed
 - D. A completely oxidised carbohydrate; the volume of CO₂ evolved is twice the volume of O₂ consumed
-

BIO-11-CH02-0098 | Medium | RQ from measured gas volumes

A respiring tissue releases 14 mL of carbon dioxide while consuming 20 mL of oxygen during the same interval. What is its respiratory quotient, and which respiratory substrate does that value indicate?

- A. 0.7, indicating protein
- B. 0.7, indicating fat
- C. 0.9, indicating protein
- D. 1, indicating carbohydrate

BIO-11-CH02-0099 | Medium | Gas volume predicted from a known RQ

A tissue respiring on protein consumes 30 mL of oxygen. Taking the respiratory quotient of protein as 0.9, what volume of carbon dioxide does it release in the same interval?

- A. 30 mL
- B. 21 mL
- C. 3 mL
- D. 27 mL

BIO-11-CH02-0100 | Hard | RQ from measured gas volumes

Two respiring samples are observed for the same interval. Sample P releases 25 mL of carbon dioxide and consumes 25 mL of oxygen. Sample Q releases 35 mL of carbon dioxide and consumes 50 mL of oxygen. Which substrates are indicated?

- A. P is respiring on fat and Q on carbohydrate
- B. P is respiring on carbohydrate and Q on protein
- C. P is respiring on protein and Q on fat
- D. P is respiring on carbohydrate and Q on fat

BIO-11-CH02-0101 | Hard | RQ of anaerobic respiration

For a completely oxidised carbohydrate the respiratory quotient is exactly 1, and for fats and proteins it is less than 1. One mode of respiration is an exception to this pattern of finite values. Which mode is it, and what is its respiratory quotient?

- A. Anaerobic respiration, for which the respiratory quotient is infinity
- B. Anaerobic respiration, for which the respiratory quotient is zero
- C. Anaerobic respiration, for which the respiratory quotient is 1
- D. Aerobic respiration on protein, for which the respiratory quotient is infinity

BIO-11-CH02-0102 | Challenging | RQ under aerobic and anaerobic conditions

A tissue is studied twice. With oxygen freely available it releases 36 mL of carbon dioxide while consuming 40 mL of oxygen. Deprived of oxygen, the same tissue releases 12 mL of carbon dioxide and consumes no oxygen at all. What are the two respiratory quotients, in that order?

- A. 0.9 and zero
- B. 0.7 and infinity
- C. 1 and infinity
- D. 0.9 and infinity

BIO-11-CH02-0103 | Easy | Utility of stepwise oxidation

Both anaerobic and aerobic respiration are conducted in many steps. What does a stepwise release of the chemical bond energy make possible?

- A. The utilization of a relatively higher proportion of that energy in ATP synthesis
- B. The utilization of the whole of the released energy in ATP synthesis without any loss
- C. The release of that energy as heat rather than its utilization in ATP synthesis
- D. The synthesis of ATP in the absence of any enzyme once the pathway has begun

BIO-11-CH02-0104 | Medium | Utility of stepwise oxidation

The activities of the enzymes catalysing the different steps of respiration may be enhanced or inhibited by specific compounds. What does this make possible?

- A. Control of the number of steps in the pathway according to the size and age of the cell
- B. Control of the rate of the pathway and of the energy output according to the need of the cell
- C. Operation of the pathway to completion without the formation of any intermediate compound
- D. Maintenance of a constant rate of the pathway and energy output whatever the need of the cell

BIO-11-CH02-0105 | Medium | Utility of stepwise oxidation

A third purpose served by a many-stepped respiratory pathway concerns the compounds formed along the way. What is that purpose?

- A. The same pathway supplies intermediates used in the synthesis of the molecular oxygen it consumes
- B. The same pathway destroys its intermediates so that no other biomolecule can be built from them
- C. The same pathway supplies intermediates whose only further use is the regeneration of the coenzymes
- D. The same pathway supplies intermediates used in the synthesis of other biomolecules such as amino acids

BIO-11-CH02-0106 | Medium | Industrial use of fermentation

Anaerobic respiration is put to use in industry. Which set correctly names both the industries and the commercially produced substances associated with it?

- A. Dairies, bakeries, distilleries, leather and paper industries; alcohol, organic acids, vitamins and pigments
- B. Dairies, bakeries and distilleries only; alcohol, organic acids, vitamins and antibiotics
- C. Dairies, bakeries, distilleries, leather and paper industries; alcohol, organic acids, vitamins and antibiotics
- D. Dairies, bakeries, distilleries, leather and paper industries; alcohol, water, vitamins and antibiotics

BIO-11-CH02-0107 | Hard | Utility of stepwise oxidation

Four statements about a respiratory pathway with many steps are given. I: A stepwise release of chemical bond energy allows a relatively higher proportion of that energy to be used in ATP synthesis. II: The enzymes of the different steps can be enhanced or inhibited by specific compounds, so the rate of the pathway and the energy output can be controlled. III: The same pathway can supply intermediates used in the synthesis of other biomolecules such as amino acids. IV: The many steps allow the pathway to continue without enzymes once it has begun. Which statements are among the purposes served by such stepwise metabolism?

- A. I and II only
- B. II, III and IV
- C. I, II and III
- D. I, III and IV

BIO-11-CH02-0108 | Challenging | Significance of respiration

Which of the following correctly pairs a stated significance of respiration with the feature of respiration that provides it?

- A. Building blocks for the synthesis of complex compounds are provided by the metabolic water of terminal oxidation
- B. The balance between CO₂ and O₂ in the atmosphere is maintained by respiration acting on its own
- C. Building blocks for the synthesis of complex compounds are provided by the intermediates of the Krebs cycle
- D. Insoluble substances are converted into soluble form by the oxidised coenzymes regenerated in the electron transport system

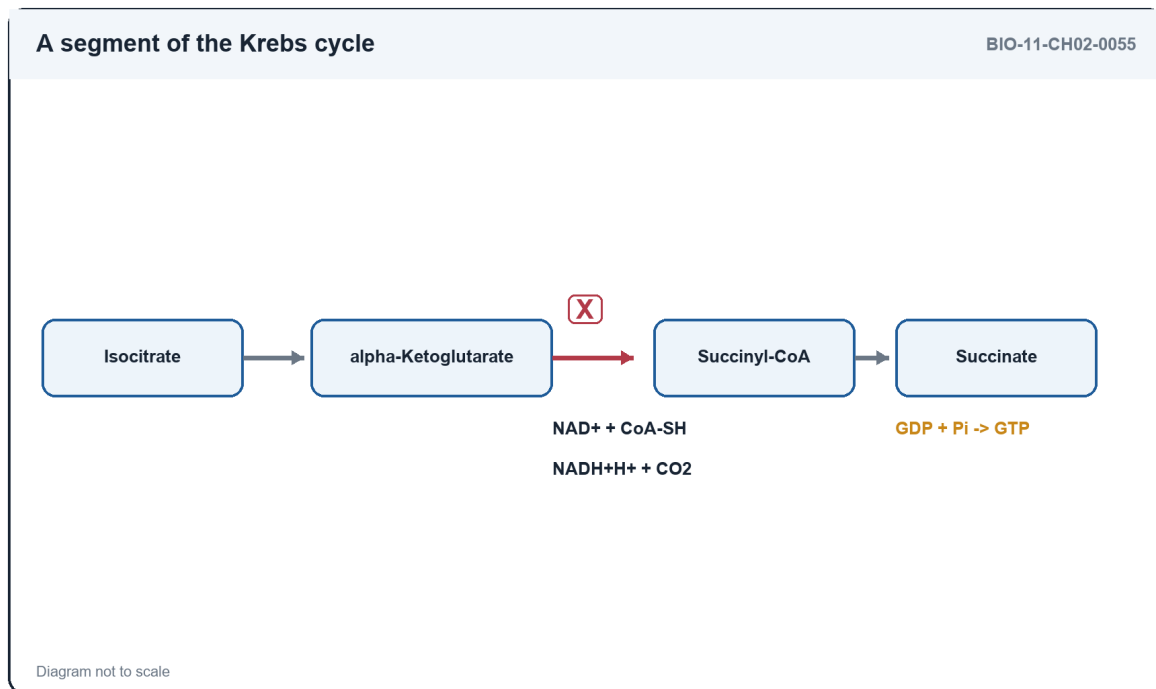
BIO-11-CH02-0109 | Easy | Cellular site

In a eukaryotic cell, where does glycolysis occur?

- A. Cytosol
- B. Mitochondrial matrix
- C. Inner mitochondrial membrane
- D. Outer mitochondrial membrane

BIO-11-CH02-0110 | Medium | Oxidative decarboxylation

In the Krebs-cycle segment shown, reaction X uses NAD^+ and CoA-SH and releases carbon dioxide and $\text{NADH} + \text{H}^+$. Which product is formed?



- A. Citrate
- B. Succinyl-CoA
- C. Fumarate
- D. Oxaloacetate

BIO-11-CH02-0111 | Hard | Experimental interpretation

A respiring tissue consumes 50 ml of oxygen and releases 35 ml of carbon dioxide in the same interval. What is its respiratory quotient and the most likely main substrate?

- A. 1.0; carbohydrate
 - B. 0.5; organic acid
 - C. 0.7; fat
 - D. 1.3; protein
-

BIO-11-CH02-0112 | Challenging | Mixed substrates

Two substrates each consume 100 ml of oxygen. Substrate P has an RQ of 1.0 and substrate Q has an RQ of 0.7. If they are oxidised together, what is the combined RQ?

- A. 0.35
 - B. 0.70
 - C. 1.70
 - D. 0.85
-

Standard 11 | Chapter 03

Human Nutrition

BIO-11-CH03-0001 | Easy | Alimentary canal, general plan

The human alimentary canal is a long tubular structure starting from the mouth and ending with the anus. What is its stated length?

- A. About 8 to 10 metres
- B. About 4 to 6 metres
- C. About 12 to 15 metres
- D. About 1.5 to 2 metres

BIO-11-CH03-0002 | Easy | Buccal cavity

The mouth, also called the oral or buccal cavity, is bounded on different sides by different structures and is internally lined by a mucous membrane. Which structure forms the roof of this cavity?

- A. Tongue
- B. Cheeks
- C. Palate
- D. Fleshy lips

BIO-11-CH03-0003 | Easy | Dentition

Which term is defined as the study of teeth with respect to their number, arrangement, development and similar features?

- A. Thecodont
- B. Diphyodont
- C. Dentition
- D. Heterodont

BIO-11-CH03-0004 | Easy | Tongue

The upper surface of the human tongue bears numerous projections, and some of these projections carry sensory receptors. What are the projections and the receptors called?

- A. Villi, some of which bear taste buds
- B. Papillae, each of which bears a salivary duct
- C. Rugae, some of which bear taste buds
- D. Papillae, some of which bear taste buds

BIO-11-CH03-0005 | Easy | Oesophagus

The oesophagus is a thin muscular tube that passes through the neck and the central aspect of the rib cage, pierces the diaphragm and joins the stomach. What is its stated length?

- A. About 26 cm
- B. About 6 cm
- C. About 45 cm
- D. About 25 cm

BIO-11-CH03-0006 | Easy | Stomach

Which description correctly gives both the shape and the position of the human stomach?

- A. A muscular sac-like 'J' shaped organ in the upper left portion of the abdominal cavity
- B. A muscular sac-like 'U' shaped organ in the upper left portion of the abdominal cavity
- C. A muscular sac-like 'J' shaped organ in the lower right portion of the abdominal cavity
- D. A muscular sac-like 'C' shaped organ in the upper right portion of the abdominal cavity

BIO-11-CH03-0007 | Easy | Large intestine

The elongated, worm like vermiform appendix, which is a vestigial organ in human beings, arises from which part of the large intestine?

- A. Ascending colon
 - B. Transverse colon
 - C. Rectum
 - D. Caecum
-

BIO-11-CH03-0008 | Easy | Large intestine

Which region of the large intestine is described as its posterior region and temporarily stores the undigested waste?

- A. Rectum
 - B. Caecum
 - C. Descending colon
 - D. Anus
-

BIO-11-CH03-0009 | Easy | Buccal cavity

Into which region of the alimentary canal do the salivary glands open?

- A. Pharynx
 - B. Oesophagus
 - C. Stomach
 - D. Buccal cavity
-

BIO-11-CH03-0010 | Medium | Alimentary canal, general plan

Arrange the following regions of the human alimentary canal in the order in which swallowed food passes through them. I. Oesophagus II. Pharynx III. Stomach IV. Buccal cavity

- A. IV, I, II, III
 - B. II, IV, I, III
 - C. IV, II, III, I
 - D. IV, II, I, III
-

BIO-11-CH03-0011 | Medium | Stomach

Arrange the four named parts of the stomach in sequence, beginning with the part at which the oesophagus opens into the stomach and ending with the part that opens into the duodenum. I. Body II. Pylorus III. Cardia IV. Fundus

- A. III, IV, I, II
 - B. III, I, IV, II
 - C. IV, III, I, II
 - D. III, IV, II, I
-

BIO-11-CH03-0012 | Medium | Small intestine

Arrange the following in the order in which chyme leaving the stomach reaches them. I. Ileum II. Duodenum III. Caecum IV. Jejunum

- A. Order: II, I, IV, III
 - B. Order: IV, II, I, III
 - C. Order: II, IV, I, III
 - D. Order: II, IV, III, I
-

BIO-11-CH03-0013 | Medium | Large intestine

Arrange the following in the order in which undigested material passes through them. I. Descending colon II. Caecum III. Ascending colon IV. Transverse colon

- A. II, III, IV, I
- B. II, I, IV, III
- C. III, II, IV, I
- D. II, IV, III, I

BIO-11-CH03-0014 | Medium | Alimentary canal, general plan

An indigestible capsule is swallowed and travels the whole alimentary canal unchanged. Arrange the following in the order in which it passes them. I. Pyloric sphincter II. Cardiac sphincter III. Rectum IV. Jejunum

- A. I, II, IV, III
- B. II, I, IV, III
- C. II, I, III, IV
- D. II, IV, I, III

BIO-11-CH03-0015 | Medium | Dentition

The adult human dental formula is $2,1,2,3 / 2,1,2,3$, the four figures in each half giving the number of incisors, canines, premolars and molars respectively in one half of the upper jaw and one half of the lower jaw. How many premolars are present in the complete adult set?

- A. 8
- B. 4
- C. 12
- D. 16

BIO-11-CH03-0016 | Hard | Dentition

The adult human dental formula is $2,1,2,3 / 2,1,2,3$, the four figures in each half giving the number of incisors, canines, premolars and molars respectively in one half of the upper jaw and one half of the lower jaw, and the complete set contains 32 teeth. What percentage of the complete set is made up of molars?

- A. 25 per cent
- B. 37.5 per cent
- C. 12.5 per cent
- D. 50 per cent

BIO-11-CH03-0017 | Hard | Small intestine

The duodenum is about 26 cm long, the jejunum about 2.5 metres long and the ileum about 3.5 metres long. Taking 1 metre = 100 cm, what is the sum of these three lengths, expressed in centimetres?

- A. 600 cm
- B. 656 cm
- C. 626 cm
- D. 326 cm

BIO-11-CH03-0018 | Hard | Alimentary canal, general plan

The small intestine is about 6 metres long and the large intestine is 1.5 metres long. Taking the whole alimentary canal of a particular adult to be 10 metres long, what fraction of the canal is contributed by the small intestine and the large intestine together?

- A. $\frac{3}{5}$
 - B. $\frac{2}{3}$
 - C. $\frac{3}{4}$
 - D. $\frac{4}{5}$
-

BIO-11-CH03-0019 | Medium | Small intestine

The small intestine is about 6 metres long and 2.5 cm broad, and the large intestine is 1.5 metres in length. What is the ratio of the length of the small intestine to the length of the large intestine?

- A. 3:1
- B. 6:1
- C. 1:4
- D. 4:1

BIO-11-CH03-0020 | Medium | Alimentary canal, general plan

Match each named part in the first list with the organ to which it belongs in the second list. I. Fundus II. Jejunum III. Caecum IV. Pylorus P. Stomach Q. Small intestine R. Large intestine

- A. I-P, II-Q, III-R, IV-P
- B. I-P, II-Q, III-Q, IV-R
- C. I-R, II-Q, III-R, IV-P
- D. I-P, II-R, III-R, IV-Q

BIO-11-CH03-0021 | Medium | Stomach

Which statement correctly distinguishes the cardiac sphincter from the pyloric sphincter?

- A. The cardiac sphincter regulates passage of material from the stomach to the small intestine, while the pyloric sphincter lies at the junction of the oesophagus and the stomach and prevents regurgitation of food
- B. The cardiac sphincter is a band of circular muscles at the junction of the oesophagus and the stomach that prevents regurgitation of food, while the pyloric sphincter regulates passage of material from the stomach to the small intestine
- C. Both sphincters lie at the junction of the oesophagus and the stomach, the cardiac one preventing regurgitation and the pyloric one regulating entry of food
- D. The cardiac sphincter guards the opening of the hepato-pancreatic duct, while the pyloric sphincter prevents regurgitation of food into the oesophagus

BIO-11-CH03-0022 | Medium | Small intestine

Which statement correctly compares the jejunum and the ileum?

- A. The jejunum is about 3.5 metres long and is highly coiled, while the ileum is about 2.5 metres long and is the coiled middle portion
- B. The jejunum is about 2.5 metres long and is the coiled middle portion, while the ileum is about 3.5 metres long and is highly coiled
- C. The jejunum is about 2.5 metres long and is the first part of the small intestine, while the ileum is about 3.5 metres long and is the middle portion
- D. The jejunum is about 26 cm long and is 'U' shaped, while the ileum is about 3.5 metres long and is highly coiled

BIO-11-CH03-0023 | Medium | Dentition

Match each term in the first list with the feature it describes in the second list. I. Thecodont II. Diphyodont III. Heterodont IV. Gomphosis P. Presence of four different types of teeth Q. Each tooth fixed in a separate socket present in the jaw bone R. The type of joint by which a tooth is held in its socket S. Only two sets of teeth in a lifetime

- A. I-Q, II-P, III-S, IV-R
- B. I-Q, II-S, III-P, IV-R
- C. I-S, II-Q, III-P, IV-R
- D. I-R, II-S, III-P, IV-Q

BIO-11-CH03-0024 | Hard | Large intestine

As a rule, every named region of the alimentary canal opens onward into the next region, so that material entering at the mouth can eventually leave at the anus. One named structure of the canal is an exception, being an outgrowth that does not lead onward into any further region. Which is it?

- A. Caecum
- B. Fundus of the stomach
- C. Rectum
- D. Vermiform appendix

BIO-11-CH03-0025 | Medium | Large intestine

The colon of the human large intestine is stated to consist of exactly three parts. Which set names those three parts?

- A. Caecum, ascending colon and transverse colon
- B. Ascending colon, transverse colon and rectum
- C. Ascending colon, transverse colon and descending colon
- D. Caecum, transverse colon and descending colon

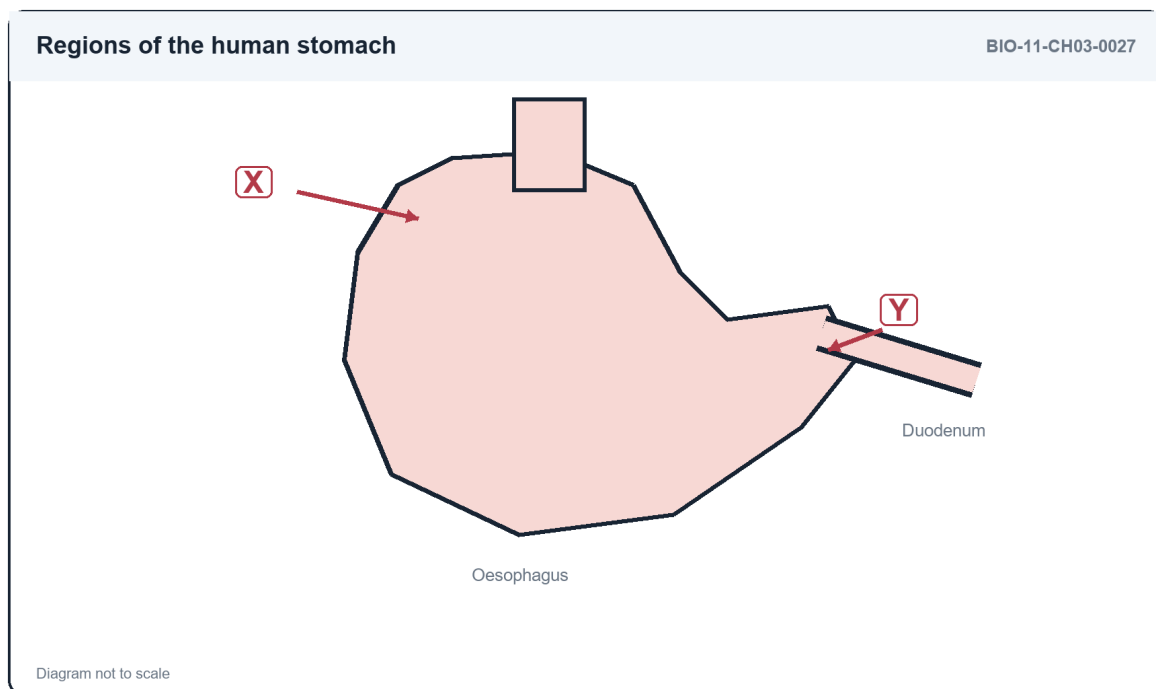
BIO-11-CH03-0026 | Hard | Dentition

The terms thecodont, diphyodont and heterodont describe three different aspects of dentition, namely attachment of the tooth, replacement of the set and variety of tooth form respectively. On this basis, which statement about an adult human being is correct?

- A. The presence of milk teeth and permanent teeth makes the dentition heterodont
- B. The fixing of each tooth in a separate socket makes the dentition diphyodont
- C. The presence of incisors, canines, premolars and molars makes the dentition heterodont
- D. The gomphosis joint between a tooth and the jaw bone makes the dentition heterodont

BIO-11-CH03-0027 | Medium | Stomach

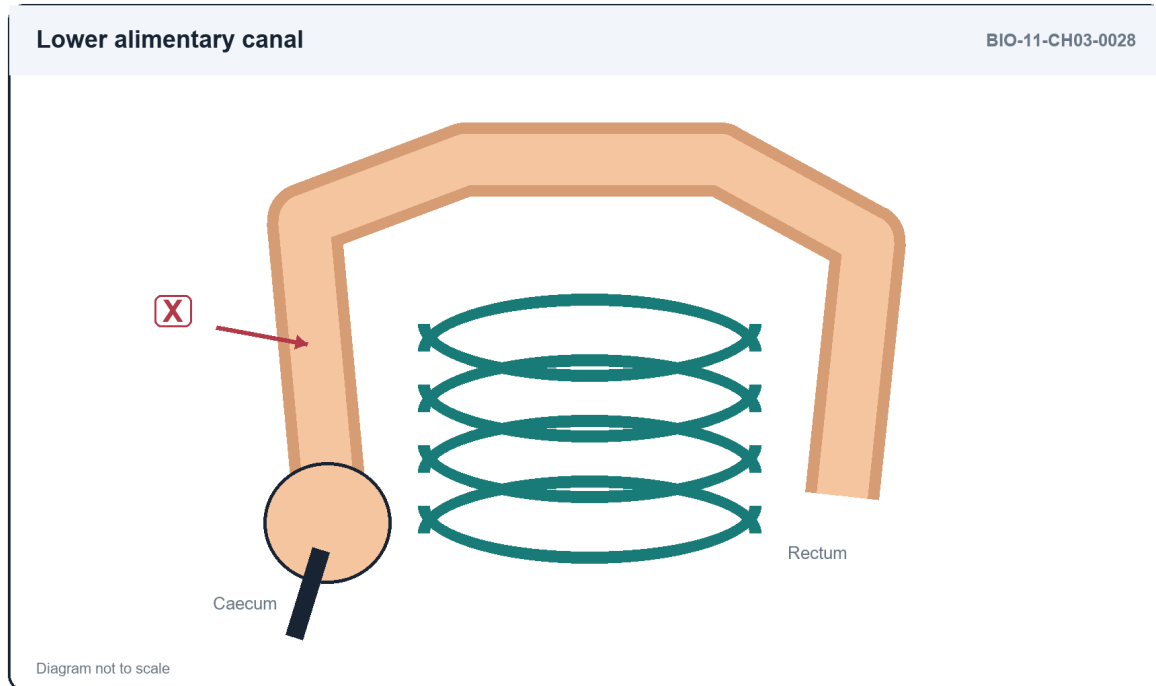
In the stomach diagram, label X marks the dome-shaped region above and to the left of the oesophageal opening. Which region is X?



- A. Cardia
- B. Fundus
- C. Body
- D. Pylorus

BIO-11-CH03-0028 | Easy | Large intestine

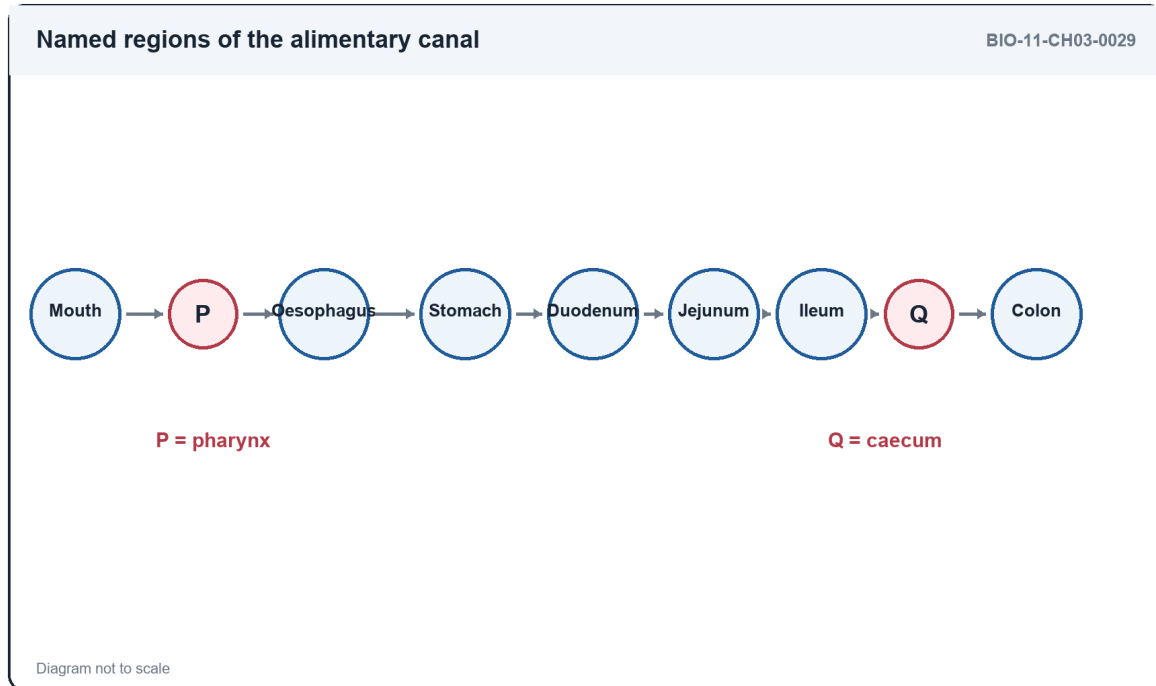
In the lower alimentary-canal diagram, label X marks the part of the large intestine that runs upwards from the caecum. Which part is X?



- A. Transverse colon
- B. Descending colon
- C. Ascending colon
- D. Rectum

BIO-11-CH03-0029 | Hard | Alimentary canal, general plan

In the alimentary-canal sequence shown, P marks the pharynx and Q marks the caecum. How many named regions lie strictly between P and Q?



- A. 4
- B. 5
- C. 6
- D. 7

BIO-11-CH03-0030 | Medium | Pharynx

The buccal cavity leads into a short pharynx. Which statement about the pharynx is correct?

- A. It is a passage for food only, air being carried by a separate tube throughout
- B. It is a common passage for food and air
- C. It is about 25 cm long and pierces the diaphragm before joining the stomach
- D. It is lined by a mucosa that is thrown into irregular folds called rugae

BIO-11-CH03-0031 | Hard | Oesophagus

A person swallows a morsel of food while lying flat on the back, and the morsel still reaches the stomach. Which property of the oesophagus best explains this?

- A. The weight of the morsel carries it down the tube once swallowing is complete
- B. The cardiac sphincter draws the morsel down by remaining permanently open
- C. Rhythmic waves of contraction and relaxation of its muscles push the food along the tube
- D. Churning movements of the oesophageal wall grind the morsel and carry it forward

BIO-11-CH03-0032 | Hard | Small intestine

A surgeon removes the terminal 1.5 metres of a patient's small intestine. Which named part of the small intestine is affected, and what does the removed segment normally lead on to?

- A. The jejunum, which normally leads on to the ileum
- B. The ileum, which normally leads on to the caecum
- C. The duodenum, which normally leads on to the jejunum
- D. The ileum, which normally leads on to the duodenum

BIO-11-CH03-0033 | Hard | Alimentary canal, general plan

Match each stated dimension in the first list with the structure it describes in the second list. I. About 6 cm II. About 26 cm III. About 1.5 metres IV. About 3.5 metres P. Ileum Q. Caecum R. Duodenum S. Large intestine

- A. I-Q, II-R, III-P, IV-S
- B. I-R, II-Q, III-S, IV-P
- C. I-Q, II-R, III-S, IV-P
- D. I-Q, II-S, III-R, IV-P

BIO-11-CH03-0034 | Challenging | Alimentary canal, general plan

For a particular adult the alimentary canal is taken to be 9 metres long. Within it the oesophagus is about 25 cm, the stomach is taken as 30 cm, the small intestine is about 6 metres and the large intestine is 1.5 metres. Taking 1 metre = 100 cm, by how much does the total length of the canal exceed the combined length of these four regions?

- A. 125 cm
- B. 120 cm
- C. 295 cm
- D. 95 cm

BIO-11-CH03-0035 | Challenging | Stomach

Consider the following statements about the human stomach. I. The body forms the large central portion that stores the food. II. The fundus is the dome shaped region above and left of the cardia. III. The pylorus is the narrow posterior region that opens into the jejunum. IV. A band of circular muscles at the junction of the oesophagus and stomach prevents regurgitation of food. Which combination of statements is correct?

- A. I, II and IV only
- B. I, II and III only
- C. II, III and IV only
- D. I and III only

BIO-11-CH03-0036 | Challenging | Dentition

In an adult human the dental formula is 2,1,2,3 / 2,1,2,3, the four figures in each half giving the number of incisors, canines, premolars and molars respectively in one half of the upper jaw and one half of the lower jaw, giving 32 teeth in all. Each tooth is fixed in its own separate socket in the jaw bone by a gomphosis type of joint. If such a person loses all the molars of the lower jaw only, how many teeth and how many tooth-to-socket gomphosis joints remain?

- A. 26 teeth and 32 joints
- B. 20 teeth and 20 joints
- C. 26 teeth and 26 joints
- D. 29 teeth and 29 joints

BIO-11-CH03-0037 | Easy | Layers of the gut wall

The histological structure of the alimentary canal is described layer by layer, beginning with the outermost layer and working inwards towards the lumen. Which sequence gives the four layers in that order?

- A. Mucosa, submucosa, muscularis, serosa
- B. Serosa, submucosa, muscularis, mucosa
- C. Serosa, muscularis, submucosa, mucosa
- D. Muscularis, serosa, submucosa, mucosa

BIO-11-CH03-0038 | Easy | Serosa

The outermost layer of the wall of the alimentary canal is made up of a layer of squamous epithelium together with an inner layer of connective tissue. What name is given to that squamous epithelium?

- A. Mucosa
- B. Brush border
- C. Glisson's capsule
- D. Mesothelium

BIO-11-CH03-0039 | Medium | Submucosa and mucosa

In the wall of the alimentary canal one layer is formed of loose connective tissue and carries blood vessels, lymph vessels and nerves, while another layer lines the lumen and carries mucus-secreting goblet cells. Which pairing of layer with content is correct?

- A. Submucosa carries the vessels and nerves; mucosa carries the goblet cells
- B. Mucosa carries the vessels and nerves; submucosa carries the goblet cells
- C. Serosa carries the vessels and nerves; mucosa carries the goblet cells
- D. Submucosa carries the vessels and nerves; muscularis carries the goblet cells

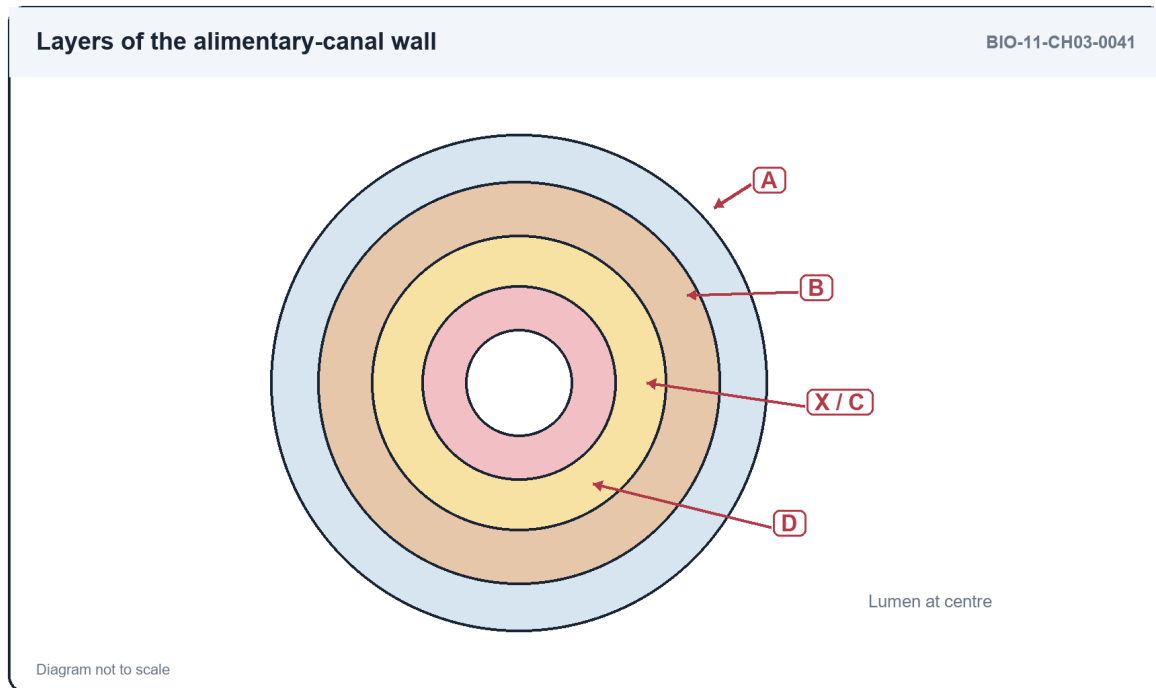
BIO-11-CH03-0040 | Medium | Muscularis

As a rule the muscularis of the alimentary canal is formed of smooth muscle usually arranged in three concentric layers, an outermost longitudinal, a middle circular and an inner oblique layer. Which region departs from this arrangement, and in what way?

- A. The intestine, where the circular layer is absent
- B. The intestine, where the oblique layer is absent
- C. The stomach, where the oblique layer is absent
- D. The stomach, where the longitudinal layer is absent

BIO-11-CH03-0041 | Medium | Layers of the gut wall

In the diagram of the alimentary-canal wall, label X marks the layer immediately outside the mucus-secreting mucosa. Which layer is X?



- A. Muscularis
- B. Submucosa
- C. Serosa
- D. Mesothelium

BIO-11-CH03-0042 | Medium | Mucosa of the small intestine

The mucosa of the small intestine is raised into finger like foldings called villi. Each villus is supplied with a network of capillaries and also with lymph vessels. What name is given to these lymph vessels of the villi?

- A. Sinusoids
- B. Crypts of Lieberkuhn
- C. Lacteals
- D. Microvilli

BIO-11-CH03-0043 | Medium | Mucosa of the small intestine

Between the bases of the villi, the lining of the intestine dips down to form crypts. Which layer forms these crypts, and what are they called?

- A. Submucosa, and they are called crypts of Lieberkuhn
- B. Mucosa, and they are called crypts of Lieberkuhn
- C. Mucosa, and they are called Brunner's glands
- D. Submucosa, and they are called Brunner's glands

BIO-11-CH03-0044 | Hard | Muscularis

A drug abolishes the contractions of the muscularis layer of the small intestine while every digestive gland goes on secreting normally. What is the expected immediate consequence?

- A. The intestinal glands would stop producing their enzymes
- B. The mucosa would stop secreting its mucus into the lumen
- C. Bile juice would no longer be secreted by the liver
- D. Chyme and the digestive juices would remain poorly combined

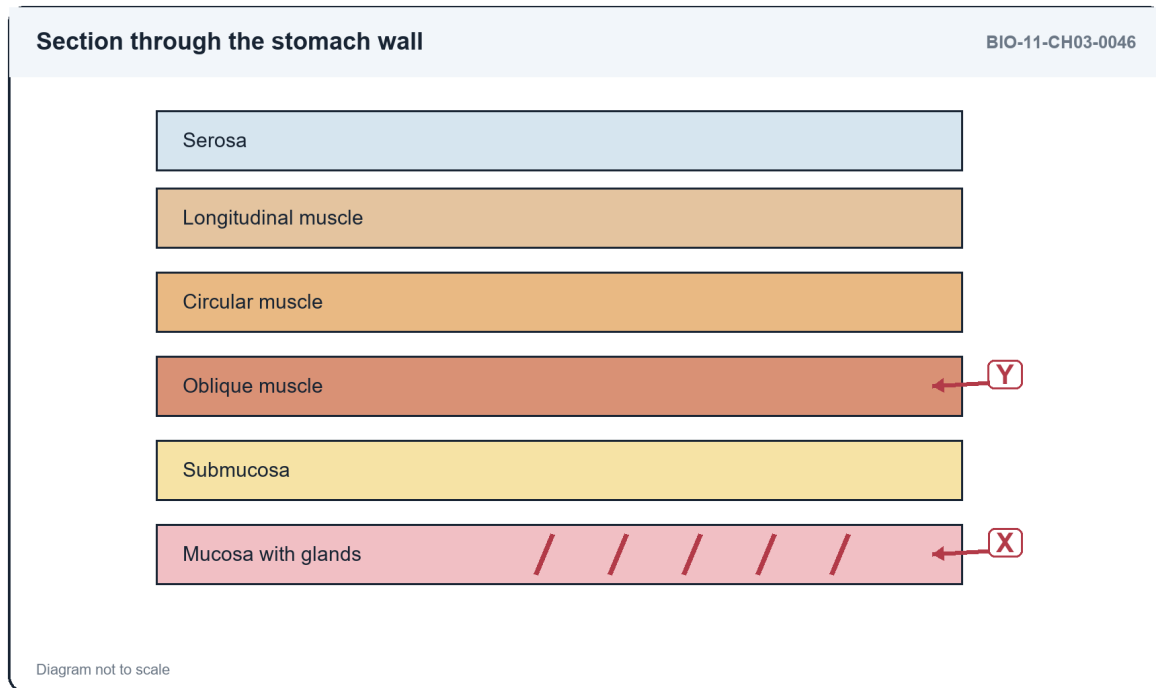
BIO-11-CH03-0045 | Hard | Regional modifications of the mucosa

In two different regions of the alimentary canal the same layer of the wall is modified in two different ways: in one region it is thrown into irregular folds and forms glands that secrete a digestive juice, and in another it is raised into finger like foldings lined by cells bearing microvilli. Which option names the layer and both modifications correctly?

- A. Mucosa, showing rugae in the stomach and villi in the small intestine
- B. Mucosa, showing villi in the stomach and rugae in the small intestine
- C. Submucosa, showing rugae in the stomach and villi in the small intestine
- D. Muscularis, showing rugae in the stomach and villi in the small intestine

BIO-11-CH03-0046 | Hard | Regional structure of the gut wall

The stomach-wall diagram labels the gland-forming innermost layer X and the innermost muscle layer Y. Identify the organ, X and Y.



- A. Stomach; mucosa; oblique muscles
- B. Stomach; submucosa; oblique muscles
- C. Stomach; mucosa; circular muscles
- D. Small intestine; mucosa; oblique muscles

BIO-11-CH03-0047 | Easy | Salivary glands

Three pairs of salivary glands open into the buccal cavity. Which option matches all three pairs with their positions?

- A. Parotid below the lower jaw, submandibular in front of the ear, sublingual below the tongue
- B. Parotid below the tongue, submandibular below the lower jaw, sublingual in front of the ear
- C. Parotid in front of the ear, submandibular below the tongue, sublingual below the lower jaw
- D. Parotid in front of the ear, submandibular below the lower jaw, sublingual below the tongue

BIO-11-CH03-0048 | Easy | Salivary glands

The salivary glands are stated to be three pairs in number, and all of them open into the buccal cavity. Counting each gland of a pair separately, how many salivary glands open into the buccal cavity?

- A. Six
- B. Three
- C. Two
- D. Twelve

BIO-11-CH03-0049 | Medium | Salivary glands

The salivary glands contain two kinds of secretory cells. One kind secretes a fluid containing a digestive enzyme, and the other produces a secretion that lubricates the food and helps swallowing. Which option matches both cell types with what they produce?

- A. Serous cells produce salivary amylase; mucous cells produce mucus
- B. Serous cells produce mucus; mucous cells produce salivary amylase
- C. Serous cells produce lysozyme; mucous cells produce salivary amylase
- D. Serous cells produce salivary amylase; mucous cells produce lysozyme

BIO-11-CH03-0050 | Medium | Composition of saliva

Saliva is stated to contain 98 per cent water and 2 per cent other constituents. In a sample of saliva, what is the ratio of water to all the other constituents taken together?

- A. 98:1
- B. 50:1
- C. 24.5:1
- D. 49:1

BIO-11-CH03-0051 | Easy | Gastric glands

One constituent of gastric juice is described as essential for the absorption of vitamin B12. Which constituent is it?

- A. Pepsinogen
- B. Intrinsic factor
- C. Mucus
- D. Hydrochloric acid

BIO-11-CH03-0052 | Easy | Gastric glands

Each gastric gland has three major types of cells. Which option matches all three of them with their secretions?

- A. Mucus cells secrete mucus; peptic cells secrete hydrochloric acid and intrinsic factor; parietal cells secrete pepsinogen
- B. Mucus cells secrete pepsinogen; peptic cells secrete mucus; parietal cells secrete hydrochloric acid and intrinsic factor
- C. Mucus cells secrete mucus; peptic cells secrete pepsinogen; parietal cells secrete hydrochloric acid and intrinsic factor
- D. Mucus cells secrete intrinsic factor; peptic cells secrete pepsinogen; parietal cells secrete hydrochloric acid and mucus

BIO-11-CH03-0053 | Medium | Gastric glands

Food entering the stomach still carries the enzyme that began acting on starch in the mouth. Which printed action of the hydrochloric acid of gastric juice brings the action of that enzyme to a stop?

- A. It makes the food acidic and thereby stops the action of salivary amylase
- B. It neutralises the food and thereby stops the action of salivary amylase
- C. It converts pepsinogen into pepsin, which then hydrolyses salivary amylase
- D. It lubricates the food and thereby separates salivary amylase from the starch

BIO-11-CH03-0054 | Hard | Gastric glands

In a patient the parietal or oxyntic cells of the gastric glands are selectively destroyed, while the mucus cells and the peptic cells go on working normally. What is the expected consequence for protein digestion in the stomach?

- A. Pepsinogen is no longer secreted, so no pepsin can be formed
- B. Pepsin is secreted directly, so protein digestion is unaffected
- C. Pepsinogen is still secreted but is not converted into pepsin
- D. Pepsinogen is converted into pepsin by the mucus of the gastric juice

BIO-11-CH03-0055 | Medium | Gastric glands

Gastric juice is stated to contain mucus, the inactive enzyme pepsinogen, hydrochloric acid and intrinsic factor. Besides these, the stomach secretes one further enzyme only in a particular group of human beings. Which enzyme is it, and in whom is it secreted?

- A. Rennin, in adults
- B. Pepsin, in infants
- C. Trypsin, in infants
- D. Rennin, in infants

BIO-11-CH03-0056 | Easy | Liver

The liver is a bilobed gland. Each of its lobes is covered by a thin covering. What is this covering called?

- A. Mesothelium
- B. Serosa
- C. Glisson's capsule
- D. Mucosa

BIO-11-CH03-0057 | Medium | Liver

The liver is described as a dark reddish-brown gland lying just below the diaphragm. What mass is given for the liver of an adult human being?

- A. About 1.2 to 1.5 kg
- B. About 0.2 to 0.5 kg
- C. About 2.2 to 2.5 kg
- D. About 12 to 15 kg

BIO-11-CH03-0058 | Medium | Histology of the liver

In the liver-lobule diagram, labels X, Y and Z indicate the central vein, the spaces between hepatic cords and the phagocytic cells within those spaces. Which option names X, Y and Z in order?

Simplified liver lobule

BIO-11-CH03-0058

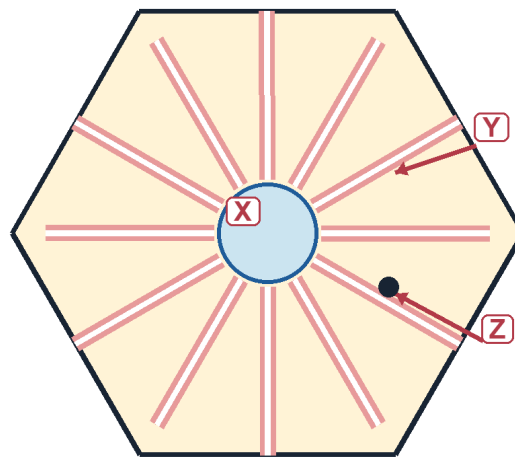


Diagram not to scale

- A. Hepatic portal vein; sinusoids; Kupffer cells
- B. Central vein; sinusoids; Kupffer cells
- C. Central vein; portal areas; Kupffer cells
- D. Central vein; sinusoids; acinar cells

BIO-11-CH03-0059 | Easy | Intestinal glands

In the small intestine, mucus together with the intestinal enzymes constitutes one secretion. By what other name is this intestinal juice known?

- A. Chyme
- B. Chyle
- C. Bolus
- D. Succus entericus

BIO-11-CH03-0060 | Medium | Intestinal glands

The mucus of the intestinal juice is produced by the goblet cells of the mucosa. A further set of glands is stated to help in the action of these goblet cells. What are these glands called, and in which layer of the intestinal wall do they lie?

- A. Brunner's glands, lying in the mucosa
- B. Crypts of Lieberkuhn, lying in the submucosa
- C. Brunner's glands, lying in the submucosa
- D. Crypts of Lieberkuhn, lying in the mucosa

BIO-11-CH03-0061 | Hard | Gall bladder and bile ducts

Trace the path by which bile reaches the duodenum. Which sequence of ducts is correct?

- A. Duct of the gall bladder with pancreatic duct --> common bile duct --> joined by hepatic duct --> hepato-pancreatic duct
- B. Hepatic duct with pancreatic duct --> hepato-pancreatic duct --> joined by duct of the gall bladder --> common bile duct
- C. Duct of the gall bladder with hepatic duct --> common bile duct --> joined by pancreatic duct --> hepato-pancreatic duct
- D. Duct of the gall bladder with hepatic duct --> hepato-pancreatic duct --> joined by pancreatic duct --> common bile duct

BIO-11-CH03-0062 | Hard | Gall bladder

A person has the gall bladder removed, while the liver and pancreas remain intact and the hepato-pancreatic duct remains open. Based on their stated biological roles, what is the expected outcome?

- A. Bile continues to be produced by the liver but is no longer stored before reaching the duodenum
- B. Bile is no longer produced, because the gall bladder is the organ that secretes it
- C. Bile continues to be produced and is now stored in the pancreas before reaching the duodenum
- D. Bile continues to be produced but now reaches the stomach instead of the duodenum

BIO-11-CH03-0063 | Medium | Bile

As a rule the digestive juices owe their digestive action to the enzymes they contain. One of the juices mixed with the food in the small intestine is an exception, since it is stated to contain no digestive enzyme at all. Which juice is this, and how does it act on the chyme?

- A. Bile juice, whose salts hydrolyse fats into fatty acids and monoglycerides
- B. Pancreatic juice, whose salts neutralise the acidity of the chyme and emulsify fats
- C. Intestinal juice, whose mucus neutralises the acidity of the chyme and emulsifies fats
- D. Bile juice, whose salts neutralise the acidity of the chyme and emulsify fats

BIO-11-CH03-0064 | Medium | Bile

Bile juice is a dark green coloured fluid containing pigments, salts, cholesterol and phospholipid. Which option correctly separates its pigments from its salts?

- A. Na-glycocholate and Na-taurocholate are the pigments; bilirubin and biliverdin are the salts
- B. Bilirubin and biliverdin are the pigments; Na-glycocholate and Na-taurocholate are the salts
- C. Bilirubin and biliverdin are the pigments; cholesterol and phospholipid are the salts
- D. Cholesterol and phospholipid are the pigments; Na-glycocholate and Na-taurocholate are the salts

BIO-11-CH03-0065 | Hard | Pancreas

The site at which a secretion is produced, the site into which it is released and the site at which it acts are three different things. Which option states all three correctly for pancreatic juice?

- A. Produced by the islets of Langerhans, released into the duodenum through the pancreatic duct, acting on food in the small intestine
- B. Produced by acinar cells, released into the stomach through the pancreatic duct, acting on food in the stomach
- C. Produced by acinar cells, released into the gall bladder for storage, acting on food in the small intestine
- D. Produced by acinar cells, released into the duodenum through the pancreatic duct, acting on food in the small intestine

BIO-11-CH03-0066 | Medium | Pancreas

The pancreas is a heterocrine gland whose endocrine part consists of islets of Langerhans lying between the acini. These islets contain three types of cells. Which option matches all three of them with their hormones?

- A. alpha-cells secrete insulin, beta-cells secrete glucagon, delta-cells secrete somatostatin
- B. alpha-cells secrete glucagon, beta-cells secrete insulin, delta-cells secrete somatostatin
- C. alpha-cells secrete somatostatin, beta-cells secrete insulin, delta-cells secrete glucagon
- D. alpha-cells secrete glucagon, beta-cells secrete somatostatin, delta-cells secrete insulin

BIO-11-CH03-0067 | Hard | Gall bladder and bile ducts

The common bile duct joins the pancreatic duct to form the hepato-pancreatic duct, whose opening into the duodenum is guarded by the sphincter of Oddi. If this duct were completely blocked just before its opening, which secretions would fail to reach the duodenum?

- A. Bile juice and intestinal juice
- B. Pancreatic juice and intestinal juice
- C. Gastric juice and pancreatic juice
- D. Bile juice and pancreatic juice

BIO-11-CH03-0068 | Challenging | Intestinal glands

Pancreatic juice reaches the duodenum carrying the inactive enzyme trypsinogen, while enterokinase is present in the intestinal juice. Suppose the intestinal glands fail to supply enterokinase while the pancreas goes on secreting normally. What would happen in the small intestine?

- A. Trypsinogen would arrive but would not be converted into trypsin
- B. Trypsinogen would no longer be secreted by the pancreas
- C. Trypsinogen would be converted into trypsin by the bile salts instead
- D. Trypsin would still be formed, since the pancreas secretes it in the active form

BIO-11-CH03-0069 | Easy | Digestion in the buccal cavity

Saliva is made up of 98 per cent water and 2 per cent other constituents. Which constituent of saliva acts as an antibacterial agent and prevents infections?

- A. Salivary amylase
- B. Mucus
- C. Lysozyme
- D. Bicarbonate ions

BIO-11-CH03-0070 | Easy | Digestion in the buccal cavity

About 30 per cent of the starch taken in with food is acted upon by salivary amylase at pH 6.8 in the buccal cavity. Which sugar is formed by this action?

- A. Maltose
- B. Glucose
- C. Fructose
- D. Galactose

BIO-11-CH03-0071 | Medium | Digestion in the buccal cavity

In a certain person the saliva reaching the buccal cavity lacks salivary amylase, while all its other constituents are present in the usual amounts. What is the direct consequence in the buccal cavity?

- A. Food particles cannot be bound together into a bolus
- B. No chemical digestion of food occurs in the buccal cavity
- C. Bacteria present in the food are no longer destroyed
- D. Mastication of food by the teeth is prevented

BIO-11-CH03-0072 | Medium | Digestion in the stomach

Each gastric gland has three major types of cells. Which pairing of cell type with the substance it produces is correct?

- A. Parietal cells produce pepsinogen and peptic cells produce intrinsic factor
- B. Mucus cells produce pepsinogen and parietal cells produce intrinsic factor
- C. Peptic cells produce pepsinogen and mucus cells produce intrinsic factor
- D. Peptic cells produce pepsinogen and parietal cells produce intrinsic factor

BIO-11-CH03-0073 | Easy | Digestion in the stomach

Intrinsic factor is one of the constituents of gastric juice. For which process is it stated to be essential?

- A. Absorption of vitamin B12
- B. Conversion of pepsinogen into pepsin
- C. Emulsification of the fats of the chyme
- D. Protection of the inner lining of the stomach from HCl

BIO-11-CH03-0074 | Medium | Digestion in the stomach

In a person the parietal cells of the gastric glands release no HCl into the stomach, although they continue to release intrinsic factor and the peptic cells continue to release pepsinogen normally. What follows in the stomach?

- A. Vitamin B12 fails to be absorbed for want of intrinsic factor
- B. Pepsinogen remains unconverted to pepsin
- C. Mucus no longer protects the inner lining of the stomach
- D. Proteins are broken down into polypeptides in the stomach

BIO-11-CH03-0075 | Easy | Digestion in the stomach

Pepsin acts on the proteins of the food during gastric digestion. Which products does this action yield?

- A. Polypeptides and dipeptides
- B. Dipeptides and amino acids
- C. Peptones and proteoses
- D. Amino acids and ammonia

BIO-11-CH03-0076 | Medium | Digestion in the stomach

As a rule the gastric juice of the human stomach contains mucus, the inactive enzyme pepsinogen, HCl and intrinsic factor. One further enzyme is secreted by the stomach only under a stated condition. Which enzyme is it, and under what condition?

- A. Pepsin, which the stomach secretes in infants
- B. Rennin, which the stomach secretes in infants
- C. Rennin, which the stomach secretes only in adults
- D. Lysozyme, which the stomach secretes in infants

BIO-11-CH03-0077 | Medium | Digestion in the stomach

Four events of gastric digestion are given. I. Peptones and proteoses are formed from proteins. II. Food is churned by the muscular wall of the stomach. III. Pepsinogen is converted into pepsin by HCl. IV. A semifluid acidic mass called chyme is formed. Arrange them in the order in which they occur.

- A. III, II, I, IV
- B. II, I, III, IV
- C. II, III, IV, I
- D. II, III, I, IV

BIO-11-CH03-0078 | Medium | Digestion in the small intestine

Digestive juices added to the food along the alimentary canal generally supply the enzymes that hydrolyse it. Which of these juices is the exception, containing no digestive enzyme at all?

- A. Pancreatic juice
- B. Intestinal juice
- C. Gastric juice
- D. Bile juice

BIO-11-CH03-0079 | Hard | Digestion in the small intestine

Pancreatic juice is secreted by the acinar cells of the pancreas and carried to the duodenum by the pancreatic duct. Which pair of its enzymes is released in an inactive form?

- A. Pancreatic amylase and pancreatic lipase
- B. Trypsinogen and chymotrypsinogen
- C. Nucleases and pancreatic lipase
- D. Pancreatic amylase and trypsinogen

BIO-11-CH03-0080 | Hard | Digestion in the small intestine

Trypsin and chymotrypsin both act on protein derivatives in the small intestine. Which statement gives the substrate and product of each of them correctly?

- A. Trypsin converts polypeptides into dipeptides, and chymotrypsin converts proteoses and peptones into polypeptides
- B. Trypsin converts proteins into peptones and proteoses, and chymotrypsin converts polypeptides into amino acids
- C. Trypsin converts dipeptides into amino acids, and chymotrypsin converts polypeptides into dipeptides
- D. Trypsin converts proteoses and peptones into polypeptides, and chymotrypsin converts polypeptides into dipeptides

BIO-11-CH03-0081 | Easy | Digestion in the small intestine

Which combination constitutes the intestinal juice, also called succus entericus?

- A. Bile salts together with the intestinal enzymes
- B. Mucus together with hydrochloric acid
- C. Mucus together with the pancreatic enzymes
- D. Mucus together with the intestinal enzymes

BIO-11-CH03-0082 | Hard | Digestion in the small intestine

Enterokinase is one of the enzymes of intestinal juice. Suppose the intestinal juice of a person lacks enterokinase while every other digestive secretion is normal. What is the immediate consequence in the small intestine?

- A. Pepsinogen arriving in the duodenum stays unconverted to pepsin
- B. Pancreatic amylase arriving in the duodenum stays inactive
- C. Bile salts fail to emulsify the fats of the chyme
- D. Trypsinogen arriving in the duodenum stays unconverted to trypsin

BIO-11-CH03-0083 | Hard | Digestion in the small intestine

The common bile duct joins the pancreatic duct to form the hepato-pancreatic duct, whose opening into the duodenum is guarded by the sphincter of Oddi. If the hepato-pancreatic duct is completely blocked, which pair of secretions is prevented from reaching the duodenum?

- A. Pancreatic juice and intestinal juice
- B. Bile juice and pancreatic juice
- C. Bile juice and intestinal juice
- D. Gastric juice and pancreatic juice

BIO-11-CH03-0084 | Medium | Digestion in the small intestine

The intestinal juice of a person contains all its usual enzymes except lactase. Which conversion fails to occur in the small intestine of this person?

- A. Maltose to glucose
- B. Sucrose to glucose and fructose
- C. Lactose to glucose and galactose
- D. Dipeptides to amino acids

BIO-11-CH03-0085 | Medium | Neurohormonal control of digestion

Secretin is one of the chemical messengers that keep the digestive secretions in sequence. Which combination of actions is stated for it?

- A. Inhibition of gastric juice secretion, together with stimulation of bile, pancreatic and intestinal juice secretion
- B. Stimulation of gastric juice secretion, together with inhibition of bile, pancreatic and intestinal juice secretion
- C. Inhibition of gastric juice secretion, together with stimulation of bile juice secretion alone
- D. Stimulation of pancreatic and intestinal juice secretion, together with induction of satiety

BIO-11-CH03-0086 | Challenging | Neurohormonal control of digestion

Four chemical messengers of digestion are listed in the prescribed account: gastrin, secretin, cholecystokinin and gastric inhibiting peptide. Which statement about them is correct?

- A. Gastrin stimulates the secretion of gastric juice, while gastric inhibiting peptide inhibits it
- B. Secretin stimulates the secretion of gastric juice, while gastric inhibiting peptide stimulates the secretion of intestinal juice
- C. Cholecystokinin stimulates the secretion of gastric juice and suppresses the feeling of fullness
- D. Gastrin inhibits the secretion of gastric juice, while the tenth cranial nerve stimulates it

BIO-11-CH03-0087 | Easy | Sites of absorption

Absorption is the passage of the end products of digestion through the mucosal lining of the alimentary canal into blood and lymph. About 90 per cent of it takes place in which region?

- A. Stomach
- B. Large intestine
- C. Small intestine
- D. Buccal cavity

BIO-11-CH03-0088 | Medium | Sites of absorption

The villi of the small intestine are supplied both with a network of blood capillaries and with lymph vessels called lacteals. Which pairing of nutrient class with route is correct?

- A. Glucose, amino acids and water soluble vitamins into the blood capillaries; lipids and fat soluble vitamins into the lacteals
- B. Glucose, amino acids and water soluble vitamins into the lacteals; lipids and fat soluble vitamins into the blood capillaries
- C. Glucose and lipids into the blood capillaries; amino acids and fat soluble vitamins into the lacteals
- D. Amino acids and lipids into the lacteals; glucose and fat soluble vitamins into the blood capillaries

BIO-11-CH03-0089 | Hard | Sites of absorption

The gastric mucosa is impermeable to most substances, so that nutrients reach the intestinal villi unabsorbed. Which set of substances is nevertheless absorbed in the stomach?

- A. Glucose, fructose, galactose, amino acids and water soluble vitamins
- B. Lipids together with the fat soluble vitamins A, D, E and K
- C. Water, electrolytes like sodium and chloride, drugs and some vitamins
- D. A little water, some electrolytes, alcohol and certain drugs like aspirin

BIO-11-CH03-0090 | Hard | Mechanisms of absorption

Match the mechanism of absorption in column I with its description in column II. I. Simple diffusion, II. Facilitated transport, III. Active transport, IV. Osmosis. P. Absorption of sodium against the concentration gradient, requiring energy. Q. Absorption of water along the concentration gradient. R. Absorption of part of the glucose and of chloride ions, depending on the concentration gradient. S. Absorption of fructose with the help of carrier ions like Na⁺.

- A. I-S, II-R, III-P, IV-Q
- B. I-R, II-S, III-P, IV-Q
- C. I-R, II-S, III-Q, IV-P
- D. I-Q, II-S, III-P, IV-R

BIO-11-CH03-0091 | Medium | Absorption of fats

Monoglycerides and fatty acids cannot be absorbed in blood as such. Which statement identifies both micelles and chylomicrons correctly?

- A. Micelles are small protein coated fat globules, and chylomicrons are spherical aggregates formed by bile salts in which monoglycerides and fatty acids dissolve
- B. Micelles are spherical aggregates formed by bile pigments, and chylomicrons are small carbohydrate coated fat globules
- C. Micelles are spherical aggregates formed by bile salts in which monoglycerides and fatty acids dissolve, and chylomicrons are small protein coated fat globules
- D. Micelles are small fat globules coated with phospholipid, and chylomicrons are aggregates formed by bile salts

BIO-11-CH03-0092 | Easy | Assimilation

The absorbed food material finally reaches the tissue and becomes a part of the protoplasm. What is this process called?

- A. Absorption
- B. Assimilation
- C. Egestion
- D. Digestion

BIO-11-CH03-0093 | Medium | Egestion

Undigested waste is converted to faeces in the colon and reaches the rectum. Which combination is stated as the contents of faeces?

- A. Water, inorganic salts, sloughed mucosal cells, bacteria and undigested food
- B. Water, inorganic salts, bile pigments, bacteria and undigested food
- C. Water, inorganic salts, sloughed mucosal cells, digestive enzymes and undigested food
- D. Water, inorganic salts, sloughed mucosal cells, bacteria and absorbed vitamins

BIO-11-CH03-0094 | Challenging | Mechanisms of absorption

Which of these statements about the absorption of a named substance is correct?

- A. Water is absorbed against its concentration gradient and the process requires energy
- B. Sodium is absorbed by simple diffusion, depending on the concentration gradient
- C. Chloride ions are absorbed by facilitated transport with the help of carrier ions
- D. Fructose is absorbed by facilitated transport with the help of carrier ions like Na⁺

BIO-11-CH03-0095 | Easy | Protein energy malnutrition

Kwashiorkor is a protein deficiency disorder arising from protein energy malnutrition. Which set of symptoms is stated for it?

- A. Yellowness of the conjunctiva of the eyes and skin, together with whitish stool
- B. Loose watery stools passed more than three times a day, leading to dehydration
- C. Protruding belly, slender legs, bulging eye and oedema of the lower legs and face
- D. Loss of appetite with acidity, heart burn, regurgitation and upper abdominal pain

BIO-11-CH03-0096 | Medium | Protein energy malnutrition

Kwashiorkor and marasmus both arise from protein energy malnutrition. Which statement distinguishes marasmus from kwashiorkor?

- A. Marasmus is found in children between one and three years of age and shows oedema of the face
- B. Marasmus is found in infants under one year of age and shows no oedema
- C. Marasmus is found in infants under one year of age and shows oedema of the lower legs
- D. Marasmus is found in children between one and three years of age and shows no oedema

BIO-11-CH03-0097 | Medium | Disorders of the digestive system

Which set of conditions is stated as a cause of indigestion?

- A. Unavailability of nutritious food over a prolonged period
- B. Overeating, inadequate enzyme secretion, spicy food and anxiety
- C. Reduction of the frequency of defaecation to less than once per week
- D. Reverse peristaltic movements of the wall of the stomach

BIO-11-CH03-0098 | Hard | Disorders of the digestive system

Match the disorder in column I with its description in column II. I. Constipation, II. Diarrhoea, III. Vomiting, IV. Jaundice. P. Passing loose watery stools more than three times a day. Q. Yellowness of the conjunctiva of the eyes and skin with whitish stool. R. Frequency of defaecation reduced to less than once per week. S. Stomach contents thrown out of the mouth by reverse peristaltic movements.

- A. I-R, II-P, III-S, IV-Q
- B. I-P, II-R, III-S, IV-Q
- C. I-R, II-P, III-Q, IV-S
- D. I-S, II-P, III-R, IV-Q

BIO-11-CH03-0099 | Medium | Calorific value of food

The physiological value of food is the actual energy produced by 1 g of it. These values are 4.0 Kcal/g for carbohydrate, 4.0 Kcal/g for protein and 9.0 Kcal/g for fat. A meal supplies 20 g of carbohydrate, 10 g of protein and 5 g of fat. What is the total energy actually produced by this meal?

- A. 167 Kcal
- B. 181.5 Kcal
- C. 165 Kcal
- D. 185.75 Kcal

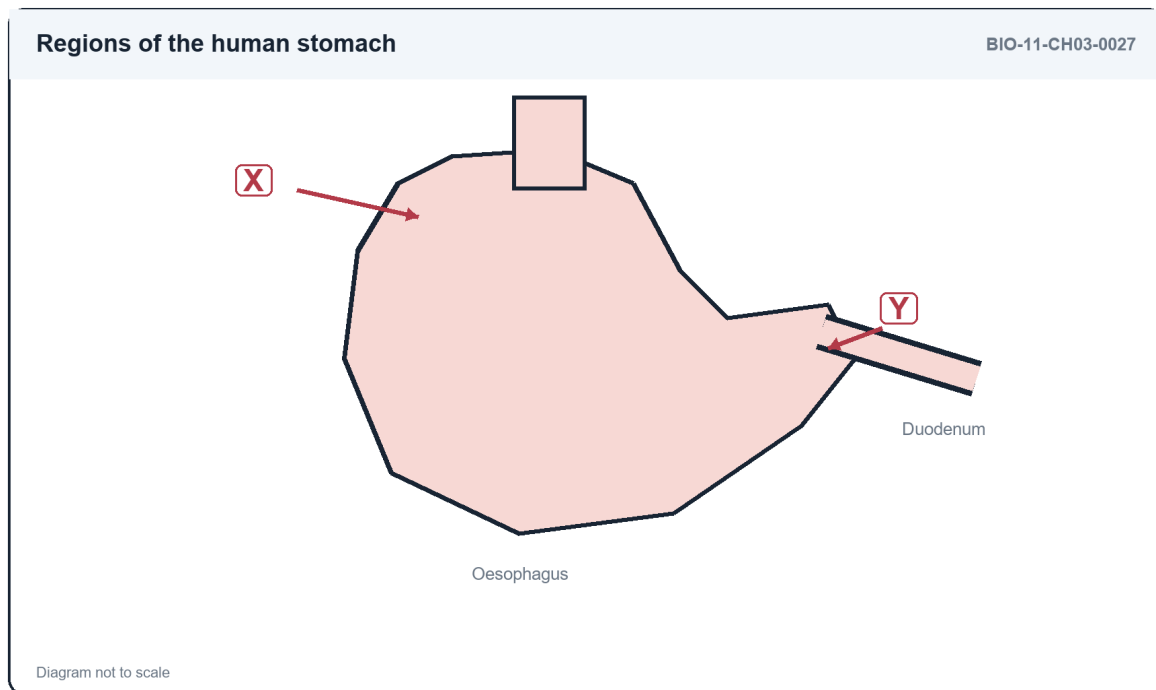
BIO-11-CH03-0100 | Challenging | Calorific value of food

For 1 g of food the gross calorific values are 4.1 Kcal/g for carbohydrate, 5.65 Kcal/g for protein and 9.45 Kcal/g for fat, while the physiological values are 4.0 Kcal/g for carbohydrate, 4.0 Kcal/g for protein and 9.0 Kcal/g for fat. A portion of food contains 50 g of carbohydrate, 20 g of protein and 10 g of fat. By how much does the gross calorific value of this portion exceed its physiological value?

- A. 42.5 Kcal
- B. 38.0 Kcal
- C. 37.5 Kcal
- D. 33.0 Kcal

BIO-11-CH03-0101 | Easy | Stomach regions

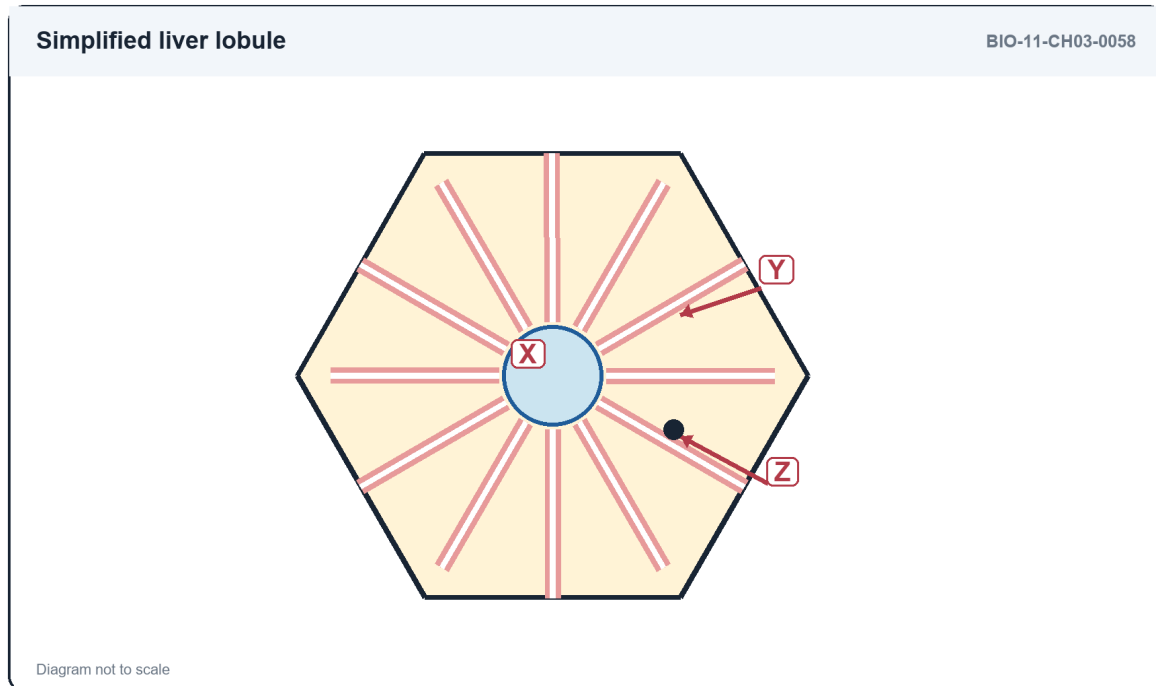
In the stomach diagram, label Y marks the narrow outlet region leading to the duodenum. Which region is Y?



- A. Pylorus (region Y)
- B. Fundus (region Y)
- C. Cardia (region Y)
- D. Body (region Y)

BIO-11-CH03-0102 | Medium | Liver histology

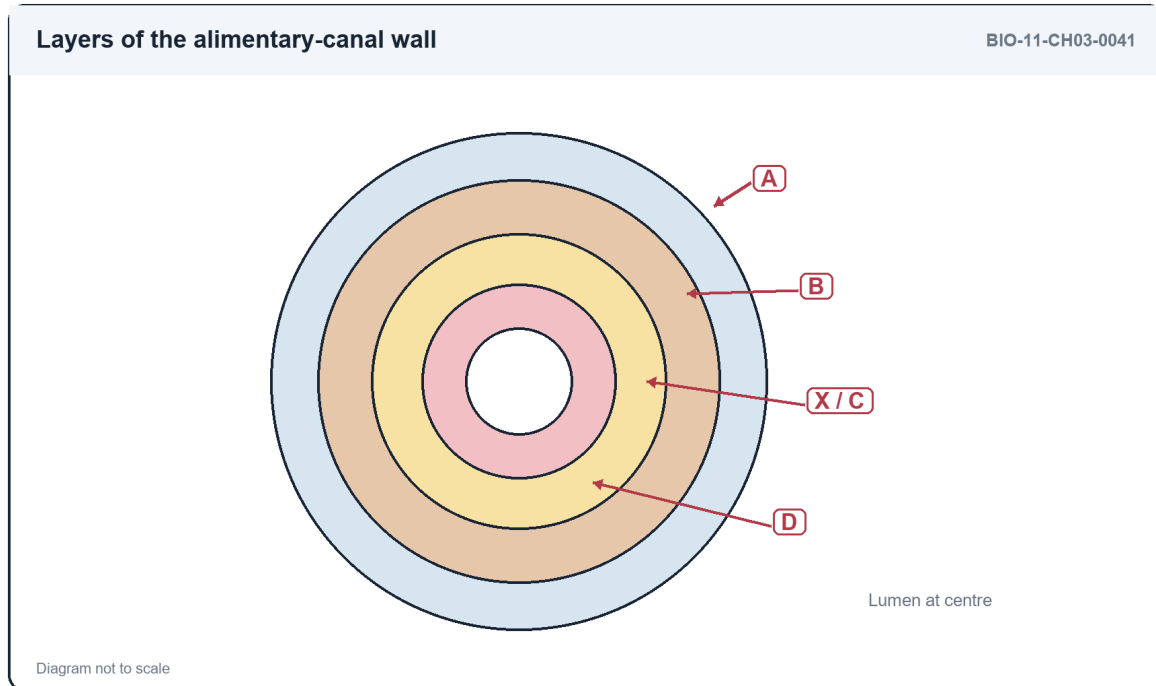
In the liver-lobule diagram, label Y points to a blood-filled space between cords of hepatic cells. What is Y?



- A. Bile canaliculus
- B. Sinusoid
- C. Central vein
- D. Hepatic duct

BIO-11-CH03-0103 | Hard | Wall layers

In the alimentary-canal wall diagram, layer C lies outside the mucosa and contains blood vessels, lymph vessels and nerves. Which layer is C?



- A. Serosa
- B. Muscularis
- C. Submucosa
- D. Mucosal epithelium

BIO-11-CH03-0104 | Challenging | Calorific values

Gross and physiological values are respectively 4.1 and 4.0 Kcal/g for carbohydrate, 5.65 and 4.0 Kcal/g for protein, and 9.45 and 9.0 Kcal/g for fat. By how much does the gross value of a meal containing 60 g carbohydrate, 25 g protein and 15 g fat exceed its physiological value?

- A. 40 Kcal
 - B. 46.5 Kcal
 - C. 50 Kcal
 - D. 54 Kcal
-

Standard 11 | Chapter 04

Excretion and Osmoregulation

BIO-11-CH04-0001 | Easy | Excretion, secretion and egestion

Which one of the following statements defines excretion?

- A. Release of a useful product formed by a gland into a duct or into the blood
- B. Removal of unabsorbed and undigested substances from the alimentary canal
- C. Elimination of metabolic waste products from the body
- D. Breaking down of complex substances into simpler ones inside the cells

BIO-11-CH04-0002 | Easy | Nitrogenous waste products

The body cannot store excess amino acids, so they are broken down by deamination. In this process, which part of the amino acid is separated and which substance is formed?

- A. The carboxyl group is separated and carbon dioxide is formed
- B. The amino group is separated and urea is formed
- C. The amino group is separated and ammonia is formed
- D. The carboxyl group is separated and uric acid is formed

BIO-11-CH04-0003 | Easy | Relative toxicity of nitrogenous wastes

Among the nitrogenous wastes on which animals are classified, which one is the least toxic, so that it can be retained in the body for some time in a concentrated form?

- A. Ammonia
- B. Uric acid
- C. Urea
- D. Guanine

BIO-11-CH04-0004 | Easy | Ureotelism

In ureotelic animals, in which organ and by which cycle is ammonia converted into urea?

- A. In the kidney, by the ornithine cycle
- B. In the liver, by the inosinic acid pathway
- C. In the kidney, by the inosinic acid pathway
- D. In the liver, by the ornithine cycle

BIO-11-CH04-0005 | Easy | Excretory organs of animal groups

Which excretory structures are present in earthworms?

- A. Protonephridia
- B. Malpighian tubules
- C. Metanephridia
- D. Green glands

BIO-11-CH04-0006 | Easy | Guanotelism

Which set of animals eliminates its nitrogenous waste in the form of guanine?

- A. Spiders, land snails and albatrosses
- B. Scorpions, sharks and rays
- C. Penguins, camels and kangaroo rats
- D. Spiders, scorpions and penguins

BIO-11-CH04-0007 | Easy | Osmoregulation

Which one of the following statements describes osmoregulation?

- A. The process of controlling solute concentrations and water balance in the body
- B. The process of eliminating nitrogenous wastes formed inside the body cells
- C. The process of breaking down excess amino acids to release their amino groups
- D. The process of storing excess carbohydrates and fats within the body tissues

BIO-11-CH04-0008 | Easy | Nitrogenous waste products

A nitrogenous excretory product is represented as $\text{H}_2\text{N}-\text{CO}-\text{NH}_2$. Which product is this?

- A. Uric acid
- B. Urea
- C. Ammonia
- D. Creatinine

BIO-11-CH04-0009 | Easy | Pigment wastes

Bilirubin and urochrome are pigments listed among human excretory products. From the breakdown of which substance do they arise, and by which routes are they eliminated?

- A. Haemoglobin; bilirubin through the urine and urochrome through the faeces
- B. Creatinine phosphate; bilirubin through the faeces and urochrome through the urine
- C. Excess amino acids; bilirubin through the urine and urochrome through the faeces
- D. Haemoglobin; bilirubin through the faeces and urochrome through the urine

BIO-11-CH04-0010 | Medium | Relative properties of nitrogenous wastes

Which sequence arranges ammonia, urea and uric acid in decreasing order of their solubility in water?

- A. Uric acid, urea, ammonia
- B. Ammonia, urea, uric acid
- C. Urea, ammonia, uric acid
- D. Uric acid, ammonia, urea

BIO-11-CH04-0011 | Medium | Energy cost of excretory modes

Which statement correctly compares the energy demands of the different modes of nitrogenous excretion?

- A. Excretion of ammonia is an energy saving mechanism, while animals excreting uric acid have to spend more energy
- B. Excretion of uric acid is an energy saving mechanism, while animals excreting ammonia have to spend more energy
- C. Excretion of ammonia and of uric acid both save energy, while the formation of urea alone costs energy
- D. Excretion of ammonia and of uric acid both cost energy, while the formation of urea saves energy

BIO-11-CH04-0012 | Medium | Habitat and excretory product

A bony fish lives in a habitat where water is freely available in large quantity. Which nitrogenous waste will it eliminate, and on what ground?

- A. Urea, because abundant water allows the liver to convert wastes faster
- B. Uric acid, because abundant water helps it to dissolve before elimination
- C. Guanine, because abundant water carries it away from the body surface
- D. Ammonia, because abundant water can dilute it and so reduce its toxicity

BIO-11-CH04-0013 | Medium | Habitat and excretory product

A land snail lives in a dry terrestrial habitat in which water must be conserved. Which nitrogenous waste does it eliminate, and on what ground?

- A. Ammonia, because it is readily soluble and can therefore be flushed out with very little water
- B. Urea, because it is the least toxic of all the wastes and needs no water at all
- C. Uric acid, because it is the least toxic waste and can be retained in a concentrated form with very little water
- D. Guanine, because it dissolves readily and is removed without any loss of water

BIO-11-CH04-0014 | Medium | Energy cost of urea formation

Three ATP molecules are used to produce one molecule of urea. How many ATP molecules are used in producing four molecules of urea?

- A. 3 ATP molecules
- B. 12 ATP molecules
- C. 7 ATP molecules
- D. 9 ATP molecules

BIO-11-CH04-0015 | Medium | Water requirement of nitrogenous wastes

About 50 ml of water is needed to remove 1 g of nitrogen when it is excreted in the form of urea. How much water is needed to remove 6 g of nitrogen in the same form?

- A. 50 ml
- B. 300 ml
- C. 250 ml
- D. 600 ml

BIO-11-CH04-0016 | Medium | Uricotelism

The conversion of ammonia into uric acid by the inosinic acid pathway in the liver is attributed to one particular group of uricotelic animals. Which group is this?

- A. Land snails
- B. Birds
- C. Many reptiles
- D. Some insects

BIO-11-CH04-0017 | Medium | Excretory organs of animal groups

Match each group in the first list with its excretory organ in the second list. First list: P. Insects, Q. Crustaceans, R. Platyhelminthes, S. Unicellular forms. Second list: I. Green glands, II. Contractile vacuoles, III. Malpighian tubules, IV. Protonephridia. Which combination is correct?

- A. P-III, Q-I, R-IV, S-II
- B. P-I, Q-III, R-IV, S-II
- C. P-III, Q-I, R-II, S-IV
- D. P-IV, Q-II, R-III, S-I

BIO-11-CH04-0018 | Medium | Ammonotelism

Which statement explains why ammonia cannot be retained in the body even for a short time?

- A. It is acidic in nature, and a slight fall in pH would disturb all enzyme catalyzed reactions and make the plasma membrane unstable
- B. It is basic in nature, and it would settle in the joints as crystals and damage the excretory organs
- C. It is insoluble in water, and it would accumulate in the tissue fluid and block the excretory organs
- D. It is basic in nature, and a slight rise in pH would disturb all enzyme catalyzed reactions and make the plasma membrane unstable

BIO-11-CH04-0019 | Medium | Creatinine

Plasma creatinine, one of the nitrogenous wastes listed for the human body, is produced from which process?

- A. Deamination of excess amino acids within the liver cells
- B. Breakdown of haemoglobin within the body
- C. Catabolism of creatinine phosphate during skeletal muscle contraction
- D. Conversion of ammonia by the ornithine cycle in the liver

BIO-11-CH04-0020 | Medium | Osmoregulation in marine animals

Marine birds such as the albatross possess salt glands. Where are these glands situated, and by which process do they move salts out of the body?

- A. Near the nostrils, by simple diffusion
- B. At the base of the tail, by active transport
- C. Near the nostrils, by active transport
- D. In the wall of the cloaca, by simple diffusion

BIO-11-CH04-0021 | Medium | Excretion, secretion and egestion

Which one of the following is NOT an act of excretion?

- A. Food residue that has passed the length of the alimentary canal without crossing the gut wall
- B. Bilirubin that leaves the liver and passes out in the faeces
- C. Urochrome that colours the urine and is voided with it
- D. Volatile substances of spices that are breathed out through the lungs

BIO-11-CH04-0022 | Medium | Osmoregulation

Barnacles and clams can handle wide changes in the salinity of their surroundings. How are such organisms described, and how do they differ from organisms of the other type?

- A. They are stenohaline, whereas euryhaline organisms handle only a narrow range of salt concentrations
- B. They are euryhaline, whereas stenohaline organisms handle an even wider range of salt concentrations
- C. They are stenohaline, whereas euryhaline organisms tolerate no change at all in salt concentration
- D. They are euryhaline, whereas stenohaline organisms handle only a narrow range of salt concentrations

BIO-11-CH04-0023 | Medium | Ammonotelism

In ammonotelic animals, through which structures does ammonia leave the body?

- A. General body surface, gills and kidneys
- B. General body surface, lungs and liver
- C. Gills, Malpighian tubules and skin
- D. Kidneys, green glands and lungs

BIO-11-CH04-0024 | Medium | Water requirement of nitrogenous wastes

About 5 to 10 ml of water is needed for the elimination of 1 gm of uric acid. Within what range does the water needed for the elimination of 4 gm of uric acid lie?

- A. 5 ml to 10 ml
- B. 40 ml to 80 ml
- C. 9 ml to 14 ml
- D. 20 ml to 40 ml

BIO-11-CH04-0025 | Hard | Creatinine

A blood report of a male patient shows a creatinine value of 1.92 mg/dl, while the normal range quoted for males is 0.6 to 1.4 mg/dl. By how much does the value exceed the upper limit of the normal male range, and what does such a value indicate?

- A. 0.52 mg/dl above the upper limit, indicating poor kidney function
- B. 0.72 mg/dl above the upper limit, indicating poor kidney function
- C. 1.32 mg/dl above the upper limit, indicating poor kidney function
- D. 0.52 mg/dl above the upper limit, indicating normal kidney function

BIO-11-CH04-0026 | Hard | Water requirement of nitrogenous wastes

Which sequence arranges the three nitrogenous wastes in increasing order of the water needed to eliminate 1 gm of each?

- A. Ammonia, about 5 to 10 ml; urea, about 50 ml; uric acid, about 300 to 500 ml
- B. Uric acid, about 5 to 10 ml; urea, about 50 ml; ammonia, about 300 to 500 ml
- C. Urea, about 5 to 10 ml; uric acid, about 50 ml; ammonia, about 300 to 500 ml
- D. Uric acid, about 5 to 10 ml; ammonia, about 50 ml; urea, about 300 to 500 ml

BIO-11-CH04-0027 | Hard | Relative properties of nitrogenous wastes

A student assumes that because uric acid is the least soluble of the three nitrogenous wastes, its elimination must require the largest volume of water. Which fact shows that this assumption is wrong?

- A. It is the most toxic waste and must be heavily diluted, so about 300 to 500 ml of water per gram is needed
- B. It is converted in the liver before elimination, so about 50 ml of water per gram is needed
- C. It is the least toxic waste and can be retained in concentrated form, so about 5 to 10 ml of water per gram suffices
- D. It dissolves as readily as ammonia once it reaches the excretory organ, so no extra water is needed

BIO-11-CH04-0028 | Hard | Comparative water economy and habitat

Consider the following statements. I: Compared with ammonia, about 100 times less water is needed in human beings when the same nitrogen is removed as urea. II: Camel, kangaroo rat and shark manage with several hundred times less water than would be needed for ammonia. III: The major excretory product of an animal correlates with its phylogenetic relationship rather than with its habitat. Which of these are correct?

- A. I and III only (excretion statements)
- B. II and III only (excretion statements)
- C. I and II only (excretion statements)
- D. I, II and III (excretion statements)

BIO-11-CH04-0029 | Hard | Nitrogenous waste products

The body can store excess carbohydrates and fats but cannot store excess amino acids. What follows from this for the handling of a protein rich meal?

- A. The surplus amino acids are stored in the liver and released again only when the diet is short of protein
- B. The surplus amino acids are excreted unchanged, so that no nitrogenous waste has to be formed at all
- C. The surplus amino acids are deaminated, and the ammonia formed is either excreted as such or converted into urea or uric acid
- D. The surplus amino acids are converted into fat, so that no nitrogenous waste is produced from them

BIO-11-CH04-0030 | Hard | Excretory organs of animal groups

How does the structure of metanephridia differ from that of protonephridia?

- A. Metanephridia are networks of dead end tubes called flame cells, whereas protonephridia are unbranched coiled tubes connected to the body cavity through funnel like nephrostomes
- B. Metanephridia are unbranched coiled tubes connected to the body cavity through funnel like nephrostomes, whereas protonephridia are networks of dead end tubes called flame cells
- C. Metanephridia are blind ended tubules opening into the gut, whereas protonephridia open on the body surface through nephridiopores
- D. Metanephridia are branched tubes ending in green glands, whereas protonephridia are unbranched coiled tubes ending in nephrostomes

BIO-11-CH04-0031 | Hard | Excretion, secretion and egestion

Excretion is the elimination of metabolic waste products formed inside the body cells; egestion is the removal of substances that have never entered the cells; secretion is the release of a product that the body puts to use. Bilirubin and undigested residue both leave the body in the faeces. Using these definitions, which classification is correct?

- A. Bilirubin leaving in the faeces is excretion, while undigested residue leaving in the faeces is egestion
- B. Bilirubin leaving in the faeces is secretion, while undigested residue leaving in the faeces is excretion
- C. Bilirubin leaving in the faeces is egestion, while undigested residue leaving in the faeces is excretion
- D. Bilirubin and undigested residue leaving in the faeces are both egestion, because they share the same route

BIO-11-CH04-0032 | Hard | Osmoregulation in marine animals

Sharks retain a large quantity of urea in their body fluid instead of eliminating all of it. What does this retention achieve?

- A. It makes their blood hypotonic to the surrounding sea water and so speeds up the entry of water into the body
- B. It makes their blood hypertonic to the surrounding sea water and so drives excess salts out through the gills
- C. It makes their blood isotonic to fresh water and so prevents the entry of salts from the sea
- D. It makes their blood isotonic to the surrounding sea water and so prevents the loss of water by exosmosis

BIO-11-CH04-0033 | Challenging | Habitat and excretory product

Four animals are given: a land snail, a shark, a bony fish and a camel. Use the quoted figures of about 300 to 500 ml of water per gram for ammonia, about 50 ml per gram for urea and about 5 to 10 ml per gram for uric acid. Which of these animals eliminates 1 gm of its nitrogenous waste with the smallest quantity of water?

- A. The land snail
- B. The shark
- C. The bony fish
- D. The camel

BIO-11-CH04-0034 | Challenging | Energy cost of excretory modes

Two animals live in the same terrestrial habitat. Animal P eliminates its nitrogenous waste as uric acid and animal Q eliminates it as urea. Which statement about the two is correct?

- A. P spends less energy in producing its waste and needs less water to eliminate it than Q
- B. P spends more energy in producing its waste and needs more water to eliminate it than Q
- C. P spends less energy in producing its waste but needs more water to eliminate it than Q
- D. P spends more energy in producing its waste but needs less water to eliminate it than Q

BIO-11-CH04-0035 | Challenging | Habitat and excretory product

A student states the rule that every animal of a given group excretes the same nitrogenous waste throughout its life. Which pair of observations shows that this rule fails?

- A. Aquatic invertebrates and bony fishes are both ammonotelic, and mammals and most adult amphibians are both ureotelic
- B. Birds and many reptiles are both uricotelic, and sharks and rays are both ureotelic
- C. Spiders and scorpions both excrete guanidine, and land snails and some insects both excrete uric acid
- D. The tadpole of a frog excretes ammonia while adult frogs are mostly ureotelic, and some terrestrial turtles excrete uric acid while others excrete urea or ammonia

BIO-11-CH04-0036 | Challenging | Osmoregulation in marine animals

A marine bird and a shark both live in sea water but meet the osmotic challenge differently. Which statement describes the two strategies correctly?

- A. The bird retains urea in its body fluid to stay isotonic to sea water, while the shark secretes salts by active transport through glands near its nostrils
- B. Both remove excess salts by active transport through glands situated near the nostrils
- C. Both retain urea in the body fluid so that their body fluid remains isotonic to sea water
- D. The bird secretes salts by active transport through glands near its nostrils, while the shark retains urea in its body fluid to stay isotonic to sea water

BIO-11-CH04-0037 | Easy | Organs of the excretory system

The human excretory system is made up of a set of organs that produce urine, convey it, store it and finally discharge it to the exterior. Which description gives the correct number of each of these organs?

- A. One pair of kidneys, one ureter, one pair of urinary bladders and one urethra
- B. One pair of kidneys, one pair of ureters, one urinary bladder and one urethra
- C. One kidney, one pair of ureters, one urinary bladder and one pair of urethrae
- D. One pair of kidneys, one pair of ureters, one pair of urinary bladders and one pair of urethrae

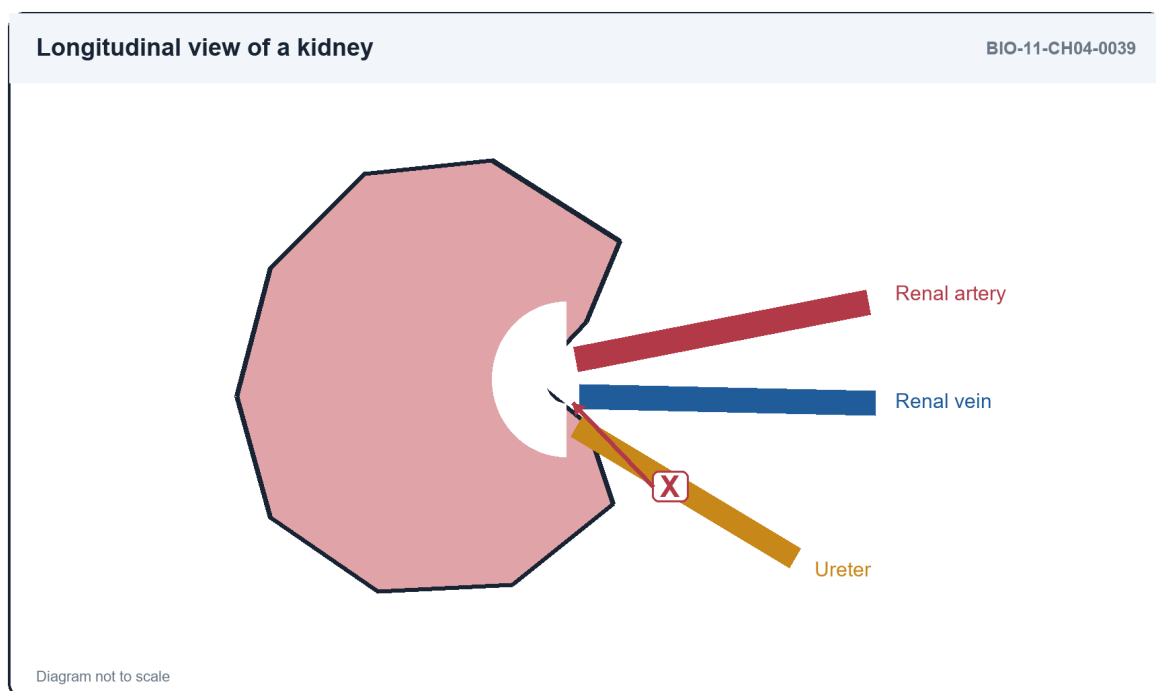
BIO-11-CH04-0038 | Easy | Position of the kidneys

The kidneys are described as retroperitoneal organs lying on either side of the vertebral column. Which statement gives both their relation to the peritoneum and their vertebral extent correctly?

- A. They lie behind the peritoneum, extending from the 12th thoracic to the 3rd lumbar vertebra
- B. They lie in front of the peritoneum, extending from the 12th thoracic to the 3rd lumbar vertebra
- C. They lie behind the peritoneum, extending from the 3rd thoracic to the 12th lumbar vertebra
- D. They lie in front of the peritoneum, extending from the 3rd thoracic to the 12th lumbar vertebra

BIO-11-CH04-0039 | Easy | Gross external form of the kidney

In the kidney diagram, X marks the notch on the concave border through which the renal vessels and ureter pass. What is X, and which vessel enters through it?



- A. The hilum; the renal vein enters through it
- B. The renal pelvis; the renal artery enters through it
- C. The renal pelvis; the renal vein enters through it
- D. The hilum; the renal artery enters through it

BIO-11-CH04-0040 | Easy | Internal regions of the kidney

A longitudinal section of a kidney shows two clearly different internal regions. One of them is the outer, peripheral region and appears red and granular. Which region is this?

- A. Renal medulla
- B. Renal cortex
- C. Renal pelvis
- D. Renal papilla

BIO-11-CH04-0041 | Medium | Tissue layers around the kidney

Three tissue layers surround each kidney, named from the outside inwards as the renal fascia, the adipose capsule and the renal capsule. Which pairing of a layer with its function is correct?

- A. Renal fascia -- shock absorption
- B. Adipose capsule -- shock absorption
- C. Renal capsule -- anchoring the kidney to the abdominal wall
- D. Adipose capsule -- barrier against the spread of infection

BIO-11-CH04-0042 | Medium | Urine collecting spaces of the kidney

Urine leaving the kidney substance at the narrow tip of a renal pyramid passes through a series of collecting spaces before entering the tube that carries it away from the kidney. Which sequence gives that order?

- A. Renal papilla --> major calyx --> minor calyx --> renal pelvis --> ureter
- B. Minor calyx --> renal papilla --> major calyx --> renal pelvis --> ureter
- C. Renal papilla --> minor calyx --> major calyx --> renal pelvis --> ureter
- D. Renal papilla --> minor calyx --> renal pelvis --> major calyx --> ureter

BIO-11-CH04-0043 | Hard | Renal pyramids and renal columns

Take the following as the rule: the renal pyramids are the conical arrangement of the medullary structures, and the renal columns, also called the columns of Bertini, are extensions of the cortex lying between the pyramids. A student examining a longitudinal section concludes that every part of the kidney lying deeper than the outer rim must be medullary. Which observation shows this conclusion to be wrong?

- A. The pyramids themselves are built of cortical tissue and are separated by strips of medulla
- B. Strips of cortical tissue run inwards between adjacent pyramids as the columns of Bertini
- C. The renal capsule sends fibrous strips inwards between adjacent pyramids
- D. The minor calyces send outward extensions of the pelvis between adjacent pyramids

BIO-11-CH04-0044 | Easy | Ureters

The ureters are long muscular tubes that transport urine away from each kidney. From which part of the kidney does a ureter arise, and what is its length?

- A. From the renal pelvis; 4 to 6 cm
- B. From the renal cortex; 25 to 30 cm
- C. From the renal cortex; 4 to 6 cm
- D. From the renal pelvis; 25 to 30 cm

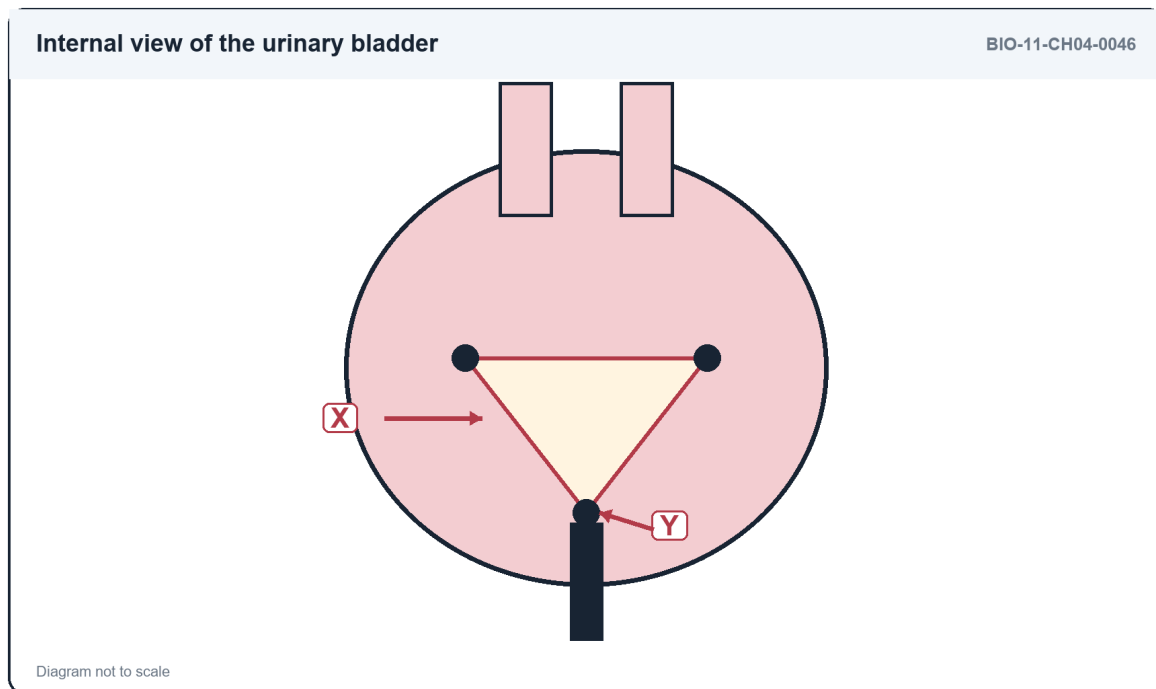
BIO-11-CH04-0045 | Hard | Ureteric openings into the bladder

Take the following as the rule: the two ureters open into the urinary bladder by separate openings, and these openings are NOT guarded by valves. Even so, urine does not flow back into the ureters when a full bladder is compressed. What accounts for this?

- A. The ureters pass obliquely through the wall of the bladder, so compression closes them
- B. The two openings lie so far apart on the bladder wall that urine cannot reach them
- C. A ring of skeletal muscle around each opening contracts as the bladder fills
- D. The transitional epithelium of the bladder seals each opening as the bladder stretches

BIO-11-CH04-0046 | Easy | Trigone of the urinary bladder

In the urinary-bladder diagram, X marks the inverted triangular area at the base and Y marks its apical opening. Name X and Y.



- A. Trigone; the openings of the two ureters lie at the apex
- B. Detrusor; the opening of the urethra lies at the apex
- C. Trigone; the opening of the urethra lies at the apex
- D. Detrusor; the openings of the two ureters lie at the apex

BIO-11-CH04-0047 | Easy | Urinary bladder

The urinary bladder is a hollow muscular organ whose innermost lining helps it to stretch as it fills. Where does the bladder lie, and which epithelium forms this lining?

- A. In the pelvic cavity, posterior to the pubic symphysis; lined by simple squamous epithelium
- B. In the abdominal cavity, anterior to the pubic symphysis; lined by transitional epithelium
- C. In the pelvic cavity, posterior to the pubic symphysis; lined by transitional epithelium
- D. In the abdominal cavity, anterior to the pubic symphysis; lined by simple squamous epithelium

BIO-11-CH04-0048 | Medium | Urethral sphincters

Two urethral sphincters lie between the urinary bladder and the urethra. Which statement compares them correctly with respect to the type of muscle forming them and the control exercised over them?

- A. The internal sphincter is of skeletal muscle and is voluntary, while the external sphincter is of smooth muscle and is involuntary
- B. Both sphincters are of smooth muscle, the internal being involuntary and the external voluntary
- C. The internal sphincter is of smooth muscle and is involuntary, while the external sphincter is of skeletal muscle and is voluntary
- D. Both sphincters are of skeletal muscle, the internal being voluntary and the external involuntary

BIO-11-CH04-0049 | Easy | Nephron as the unit of the kidney

Nephrons are the structural and functional units of the kidney. About how many of them does each human kidney contain?

- A. Almost one million
- B. Almost ten thousand
- C. Almost ten million
- D. Almost one hundred

BIO-11-CH04-0050 | Medium | Overall build of a nephron

A nephron consists of a thin walled tube together with a bunch of capillaries. What is the length of this tube, and what is the capillary bunch called?

- A. 4 to 6 cm long; the capillary bunch is the glomerulus
- B. 4 to 6 cm long; the capillary bunch is the vasa recta
- C. 25 to 30 cm long; the capillary bunch is the glomerulus
- D. 25 to 30 cm long; the capillary bunch is the vasa recta

BIO-11-CH04-0051 | Hard | Path of the filtrate through the nephron

Fluid filtered out of the glomerulus first collects in a space within Bowman's capsule and then travels along the renal tubule. Which sequence gives the order in which it passes through the following parts?

- A. Capsular space --> proximal convoluted tubule --> neck --> descending limb --> ascending limb
- B. Neck --> capsular space --> proximal convoluted tubule --> ascending limb --> descending limb
- C. Capsular space --> neck --> proximal convoluted tubule --> descending limb --> ascending limb
- D. Capsular space --> neck --> descending limb --> proximal convoluted tubule --> ascending limb

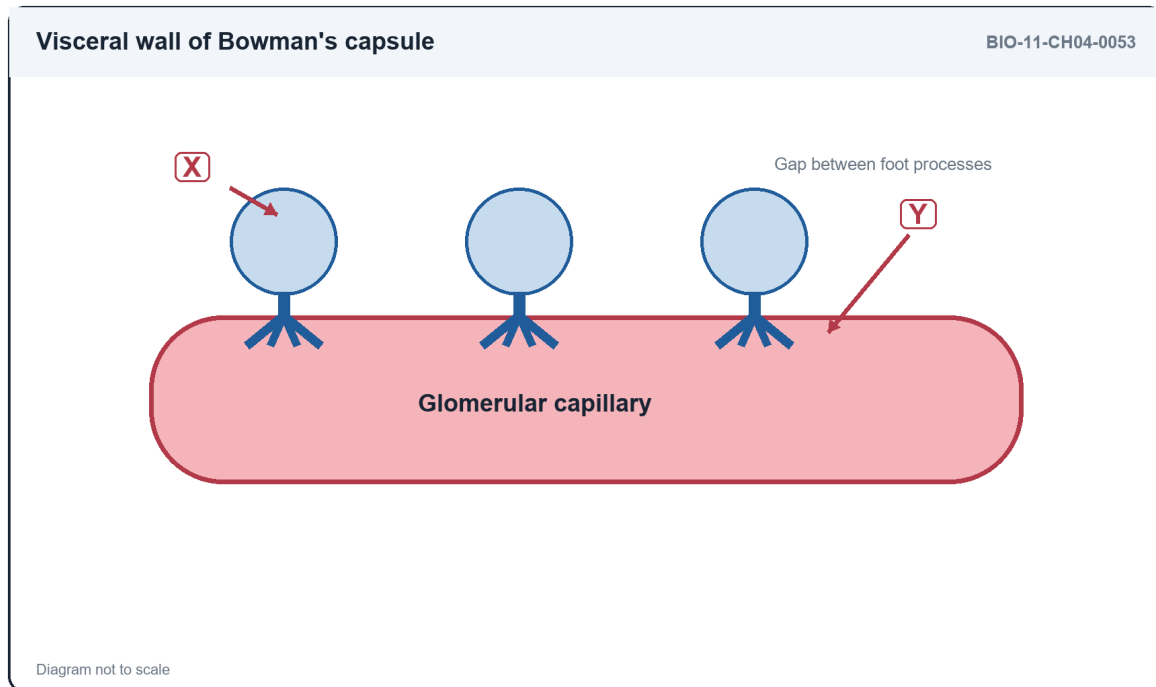
BIO-11-CH04-0052 | Medium | Path of the filtrate through the nephron

After the filtrate has turned upwards in the loop of Henle, it continues through the remaining parts of the nephron and beyond. Which sequence gives that order?

- A. Ascending limb --> collecting tubule --> distal convoluted tubule --> collecting duct
- B. Ascending limb --> distal convoluted tubule --> collecting tubule --> collecting duct
- C. Distal convoluted tubule --> ascending limb --> collecting tubule --> collecting duct
- D. Ascending limb --> distal convoluted tubule --> collecting duct --> collecting tubule

BIO-11-CH04-0053 | Medium | Bowman's capsule and podocytes

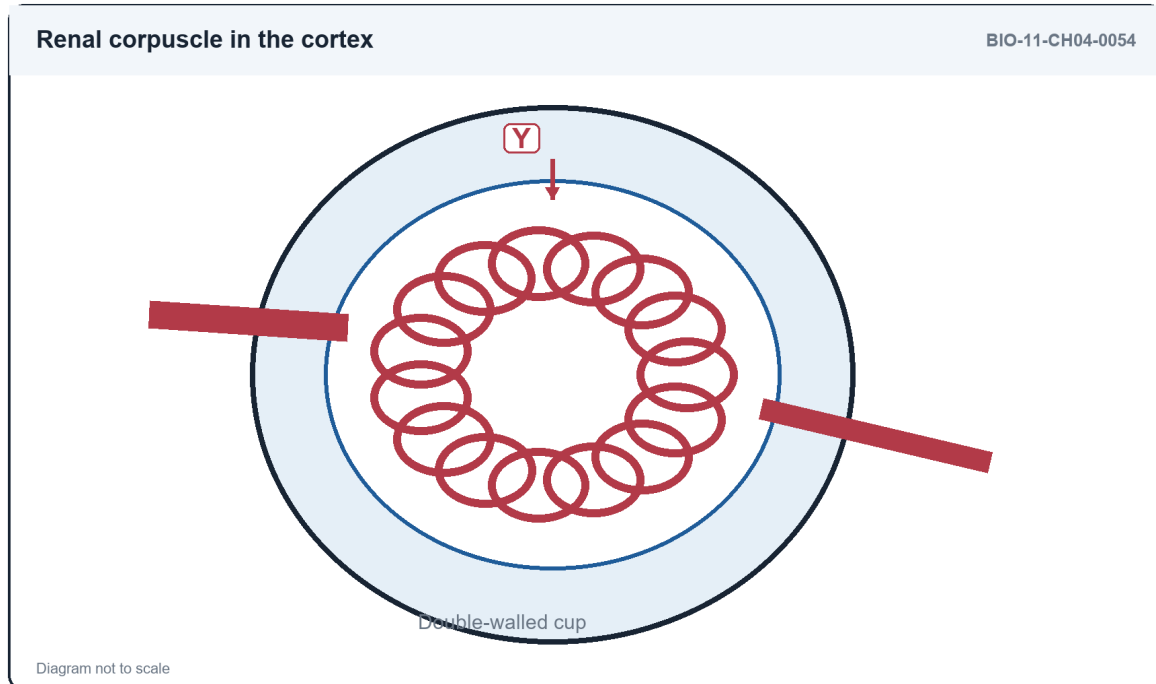
In the Bowman's-capsule diagram, X marks the specialised cells of the visceral wall and Y marks the gaps between their foot processes. What are X and Y?



- A. The parietal wall; the gaps are the filtration slits
- B. The visceral wall; the gaps are the filtration slits
- C. The visceral wall; the gaps are the capsular space
- D. The parietal wall; the gaps are the capsular space

BIO-11-CH04-0054 | Medium | Malpighian body

The figure shows a capillary tuft enclosed by a double-walled cup in the renal cortex. Which name and composition describe the complete structure?



- A. Malpighian body, made of the glomerulus together with Bowman's capsule
- B. Malpighian body, made of the glomerulus together with the afferent and efferent arterioles
- C. Juxtaglomerular apparatus, made of the glomerulus together with Bowman's capsule
- D. Juxtaglomerular apparatus, made of the glomerulus together with the efferent arteriole

BIO-11-CH04-0055 | Medium | Loop of Henle

The loop of Henle is U shaped and its two limbs differ in structure. Which statement compares the descending limb with the ascending limb correctly?

- A. The descending limb is thin walled with simple squamous epithelium and is permeable to water, while the ascending limb is thick walled with simple cuboidal epithelium and is impermeable to water
- B. The descending limb is thick walled with simple cuboidal epithelium and is impermeable to water, while the ascending limb is thin walled with simple squamous epithelium and is permeable to water
- C. The descending limb is thin walled with simple cuboidal epithelium and is impermeable to water, while the ascending limb is thick walled with simple squamous epithelium and is permeable to water
- D. Both limbs are thin walled with simple squamous epithelium, but only the ascending limb is permeable to water

BIO-11-CH04-0056 | Easy | Neck of the nephron

A short part of the nephron lined by ciliated epithelium lies at the urinary pole and links Bowman's capsule with the proximal convoluted tubule. What is this part called?

- A. Macula densa
- B. Neck
- C. Collecting tubule
- D. Renal papilla

BIO-11-CH04-0057 | Challenging | Identification of nephron parts

A part of the nephron is coiled, is lined by simple cuboidal epithelium that bears no brush border, lies in the renal cortex, and carries out water reabsorption, pH regulation and tubular secretion. Which part is it?

- A. Distal convoluted tubule
- B. Proximal convoluted tubule
- C. Collecting tubule
- D. Ascending limb of the loop of Henle

BIO-11-CH04-0058 | Medium | Nephron parts in cortex and medulla

The two internal regions of the kidney each hold a characteristic set of structures. Which grouping assigns the listed structures correctly?

- A. Cortex: Malpighian bodies, convoluted tubules and blood vessels; Medulla: loops of Henle and collecting ducts
- B. Cortex: loops of Henle and collecting ducts; Medulla: Malpighian bodies, convoluted tubules and blood vessels
- C. Cortex: Malpighian bodies, loops of Henle and blood vessels; Medulla: convoluted tubules and collecting ducts
- D. Cortex: convoluted tubules and collecting ducts; Medulla: Malpighian bodies, loops of Henle and blood vessels

BIO-11-CH04-0059 | Hard | Nephron parts in cortex and medulla

Take the following as the rule: the renal cortex contains the Malpighian bodies, the convoluted tubules and blood vessels, while the renal medulla contains the loops of Henle and the collecting ducts. A student argues that once the tubule of a nephron has entered the medulla it keeps going deeper and never comes back. Which part of the nephron shows that this argument is wrong?

- A. Proximal convoluted tubule
- B. Collecting duct
- C. Descending limb of the loop of Henle
- D. Distal convoluted tubule

BIO-11-CH04-0060 | Medium | Types of nephron

Nephrons are of two types. Which statement describes one of these types correctly with respect to both the length of its loop of Henle and its abundance?

- A. Cortical nephrons have a longer loop of Henle that runs deep into the medulla, and they form most of the nephrons
- B. Cortical nephrons have a shorter loop of Henle that extends very little into the medulla, and they form most of the nephrons
- C. Juxtamedullary nephrons have a shorter loop of Henle that extends very little into the medulla, and they form most of the nephrons
- D. Juxtamedullary nephrons have a longer loop of Henle that runs deep into the medulla, and they form most of the nephrons

BIO-11-CH04-0061 | Hard | Vasa recta and nephron type

Take the following two statements as the rule: the vasa recta are the loop shaped capillaries surrounding the loop of Henle of juxtamedullary nephrons, and most of the nephrons of a kidney are cortical nephrons. Which conclusion follows?

- A. Vasa recta are associated with most of the nephrons of the kidney
- B. Vasa recta surround the proximal convoluted tubule of every nephron
- C. Vasa recta take the place of the peritubular capillaries in cortical nephrons
- D. Vasa recta are associated with only a few of the nephrons of the kidney

BIO-11-CH04-0062 | Hard | Path of blood through the kidney

Blood entering the kidney passes through a definite series of vessels before it is carried away. Which sequence gives the order in which it passes through the following?

- A. Renal artery --> efferent arteriole --> glomerulus --> afferent arteriole --> peritubular capillary network --> renal venule
- B. Renal artery --> afferent arteriole --> glomerulus --> peritubular capillary network --> efferent arteriole --> renal venule
- C. Renal artery --> afferent arteriole --> glomerulus --> efferent arteriole --> peritubular capillary network --> renal venule
- D. Renal artery --> afferent arteriole --> efferent arteriole --> glomerulus --> peritubular capillary network --> renal venule

BIO-11-CH04-0063 | Medium | Path of blood through the kidney

A student traces the renal circulation backwards, starting from the vessel that carries blood out of the kidney and working towards the glomerulus. Which sequence gives the vessels in that backward order?

- A. Renal vein --> renal venule --> peritubular capillary network --> afferent arteriole --> glomerulus
- B. Renal vein --> renal venule --> efferent arteriole --> peritubular capillary network --> glomerulus
- C. Renal vein --> peritubular capillary network --> renal venule --> efferent arteriole --> glomerulus
- D. Renal vein --> renal venule --> peritubular capillary network --> efferent arteriole --> glomerulus

BIO-11-CH04-0064 | Hard | From renal blood to glomerular filtrate

A small dissolved molecule enters the kidney in the blood, is filtered out at the glomerulus and then travels along the nephron. Which sequence gives the order in which it meets the following structures?

- A. Renal artery --> afferent arteriole --> capsular space --> glomerulus --> neck --> proximal convoluted tubule
- B. Renal artery --> efferent arteriole --> glomerulus --> capsular space --> neck --> proximal convoluted tubule
- C. Renal artery --> afferent arteriole --> glomerulus --> capsular space --> neck --> proximal convoluted tubule
- D. Renal artery --> afferent arteriole --> glomerulus --> neck --> capsular space --> proximal convoluted tubule

BIO-11-CH04-0065 | Challenging | Blood supply of a juxtamedullary nephron

In a nephron whose long loop of Henle runs deep into the medulla, the loop is surrounded by loop shaped capillaries. Which sequence gives the order in which blood passes through the vessels of this nephron?

- A. Afferent arteriole --> glomerulus --> efferent arteriole --> vasa recta --> renal venule
- B. Afferent arteriole --> glomerulus --> vasa recta --> efferent arteriole --> renal venule
- C. Afferent arteriole --> vasa recta --> glomerulus --> efferent arteriole --> renal venule
- D. Efferent arteriole --> glomerulus --> afferent arteriole --> vasa recta --> renal venule

BIO-11-CH04-0066 | Easy | Juxtaglomerular apparatus

Which two components together form the juxtaglomerular apparatus of a nephron?

- A. The macula densa and the podocytes
- B. The JG cells and the podocytes
- C. The macula densa and the JG cells
- D. The JG cells and the peritubular capillaries

BIO-11-CH04-0067 | Medium | Diameters of the glomerular vessels

Consider the afferent arteriole, the efferent arteriole and the capillaries of the glomerulus. Which statement about their diameters is correct?

- A. The efferent arteriole has the largest diameter, the afferent arteriole is narrower, and the capillaries are narrower than both arterioles
- B. The capillaries have the largest diameter, the afferent arteriole is narrower, and the efferent arteriole is the narrowest
- C. The afferent arteriole has the largest diameter, the efferent arteriole is narrower, and the capillaries are narrower than both arterioles
- D. The afferent and efferent arterioles have equal diameters, and the capillaries are wider than both arterioles

BIO-11-CH04-0068 | Challenging | Integration of nephron type, blood supply and juxtaglomerular apparatus

A nephron is described as follows: its loop of Henle is long and runs deep into the medulla, that loop is surrounded by loop shaped capillaries, and the beginning of its distal convoluted tubule makes contact with the afferent arteriole of the same nephron, the tubule cells there being packed more densely than elsewhere. Which statement about this nephron is correct?

- A. It is a cortical nephron, the capillaries around its loop are the vasa recta, and the densely packed tubule cells form the macula densa
- B. It is a juxtamedullary nephron, the capillaries around its loop are the glomerular capillaries, and the densely packed tubule cells form the JG cells
- C. It is a juxtamedullary nephron, the capillaries around its loop are the vasa recta, and the densely packed tubule cells form the macula densa
- D. It is a cortical nephron, the capillaries around its loop are the glomerular capillaries, and the densely packed tubule cells form the JG cells

BIO-11-CH04-0069 | Easy | The three steps of urine formation

One step of urine formation is defined as the movement of substances from the tubular fluid back into the blood. Which step is this?

- A. Tubular reabsorption
- B. Tubular secretion
- C. Glomerular filtration
- D. Counter current multiplication

BIO-11-CH04-0070 | Medium | Sites of the three steps

Selective reabsorption : Proximal convoluted tubule :: Tubular secretion : _____

- A. Glomerulus and the capsular space of Bowman's capsule
- B. Descending limb and ascending limb of Henle's loop
- C. Afferent arteriole and peritubular capillary network
- D. Distal convoluted tubule and collecting tubule

BIO-11-CH04-0071 | Medium | Effective filtration pressure

In the glomerulus the hydrostatic pressure is about 55 mm Hg, the osmotic pressure of the blood is about 30 mm Hg and the hydrostatic pressure of the filtrate in the capsular space is about 15 mm Hg. Effective filtration pressure equals the hydrostatic pressure in the glomerulus minus the sum of the osmotic pressure of the blood and the filtrate hydrostatic pressure. What is the effective filtration pressure?

- A. 25 mm Hg
- B. 10 mm Hg
- C. 40 mm Hg
- D. 100 mm Hg

BIO-11-CH04-0072 | Hard | Effective filtration pressure

For a nephron the effective filtration pressure is 10 mm Hg and the hydrostatic pressure in the glomerulus is 55 mm Hg. The osmotic pressure of the blood is 30 mm Hg. Effective filtration pressure equals the hydrostatic pressure in the glomerulus minus the sum of the osmotic pressure of the blood and the hydrostatic pressure of the filtrate. What is the hydrostatic pressure of the filtrate in the capsular space?

- A. 45 mm Hg
- B. 25 mm Hg
- C. 20 mm Hg
- D. 15 mm Hg

BIO-11-CH04-0073 | Hard | Glomerular filtration rate

Glomerular filtration proceeds at a rate of 125 ml per minute. Taking a day as 24 hours of 60 minutes each, what volume of glomerular filtrate is formed in one day?

- A. 18 litres
- B. 75 litres
- C. 1800 litres
- D. 180 litres

BIO-11-CH04-0074 | Hard | Selective reabsorption

Glomerular filtrate is formed at 125 ml per minute, and about 99 per cent of the glomerular filtrate is reabsorbed in the proximal and distal convoluted tubules. What volume of filtrate is reabsorbed in one minute?

- A. 124.75 ml
- B. 123.75 ml
- C. 118.75 ml
- D. 126.25 ml

BIO-11-CH04-0075 | Medium | High threshold and low threshold substances

In selective reabsorption, high threshold substances are absorbed actively and low threshold substances are absorbed passively. Which pairing of a substance with the manner of its reabsorption is correct?

- A. Glucose, absorbed passively as a low threshold substance
- B. Vitamin C, absorbed passively as a low threshold substance
- C. Sulphates, absorbed passively as a low threshold substance
- D. Water, absorbed actively as a high threshold substance

BIO-11-CH04-0076 | Medium | Composition of the glomerular filtrate

Glomerular capillary walls are extremely thin and permeable to all major blood components except blood cells and macromolecules such as protein. Which of the following is therefore absent from the glomerular filtrate?

- A. Albumin
- B. Glucose
- C. Urea
- D. Inorganic ions

BIO-11-CH04-0077 | Challenging | Reabsorption contrasted with secretion

In selective reabsorption K^+ is listed among the high threshold substances that are taken back actively from the tubular fluid into the blood. Tubular secretion, by contrast, is the movement of substances from the blood into the tubular fluid. In which region is K^+ moved from the blood into the tubular fluid?

- A. Distal convoluted tubule and collecting tubule
- B. Proximal convoluted tubule and the neck region
- C. Descending limb and ascending limb of Henle's loop
- D. Bowman's capsule and the capsular space

BIO-11-CH04-0078 | Easy | Juxta glomerular apparatus

In each nephron the beginning part of the distal convoluted tubule makes contact with the afferent arteriole of the same nephron. In this region the cells of the wall of the distal convoluted tubule are packed more densely than elsewhere. What is this region called?

- A. JG cells
- B. Podocytes
- C. Peritubular capillaries
- D. Macula densa

BIO-11-CH04-0079 | Medium | Path of the tubular fluid

Arrange the following regions of the renal tubule in the order in which the tubular fluid passes through them: collecting tubule, distal convoluted tubule, Henle's loop, proximal convoluted tubule.

- A. Proximal convoluted tubule --> distal convoluted tubule --> Henle's loop --> collecting tubule
- B. Henle's loop --> proximal convoluted tubule --> distal convoluted tubule --> collecting tubule
- C. Proximal convoluted tubule --> Henle's loop --> collecting tubule --> distal convoluted tubule
- D. Proximal convoluted tubule --> Henle's loop --> distal convoluted tubule --> collecting tubule

BIO-11-CH04-0080 | Medium | Structure and function of the PCT

The cells lining the proximal convoluted tubule are cuboidal and bear a brush border of microvilli, and the convolutions slow the passage of the filtrate. In a nephron in which the brush border of the proximal convoluted tubule is absent while every other structure is normal, which process is directly reduced?

- A. Selective reabsorption of useful molecules from the tubular fluid
- B. Glomerular filtration of plasma into the capsular space
- C. Tubular secretion of K^+ and H^+ into the tubular fluid
- D. Counter current multiplication in Henle's loop

BIO-11-CH04-0081 | Easy | Counter current mechanism

The osmotic gradient established in the renal medulla is maintained by a set of loop shaped capillaries that operate a counter current exchange system. Name these capillaries.

- A. Peritubular capillaries
- B. Glomerular capillaries
- C. Vasa recta
- D. Afferent arteriole

BIO-11-CH04-0082 | Medium | Limbs of Henle's loop

Which pairing of a limb of Henle's loop with its wall thickness, water permeability and lining epithelium is correct?

- A. Descending limb, thick walled, impermeable to water, simple cuboidal epithelium
- B. Ascending limb, thin walled, permeable to water, simple squamous epithelium
- C. Ascending limb, thick walled, permeable to water, simple cuboidal epithelium
- D. Descending limb, thin walled, permeable to water, simple squamous epithelium

BIO-11-CH04-0083 | Medium | Medullary osmotic gradient

In the kidney the osmolarity of the cortex is 300 mOsmol L⁻¹ and that of the deep inner medulla is 1200 mOsmol L⁻¹. The osmolarity of the deep inner medulla is how many times that of the cortex?

- A. 3 times
- B. 4 times
- C. 2 times
- D. 6 times

BIO-11-CH04-0084 | Hard | Urea recycling

During urea recycling, urea diffuses out of the urine into the tissue fluid of the medulla and later re-enters the tubule. One segment of the renal tubule is impermeable to urea and so keeps it trapped in the medullary tissue fluid. Which segment is this?

- A. Proximal convoluted tubule
- B. Thick ascending limb segment
- C. Neck region of the nephron
- D. Descending limb of Henle's loop

BIO-11-CH04-0085 | Medium | Hormonal regulation, sources

Which hormone is correctly paired with the site at which it is produced?

- A. Renin, produced by the macula densa cells of the distal convoluted tubule
- B. Atrial natriuretic peptide, produced by the wall of the collecting duct
- C. Aldosterone, produced by the adrenal cortex
- D. Antidiuretic hormone, produced by the osmoreceptors of the hypothalamus

BIO-11-CH04-0086 | Medium | Renin angiotensin aldosterone system

The JG cells release renin when the blood supply to the afferent arteriole decreases. Renin then acts on angiotensinogen, which is secreted by hepatocytes and circulates in the blood. What is formed directly by this action of renin?

- A. Angiotensin II
- B. Angiotensin I
- C. Aldosterone
- D. Angiotensin converting enzyme

BIO-11-CH04-0087 | Hard | Hormonal regulation, effects

Aldosterone released from the adrenal cortex acts on the distal convoluted tubule and the collecting ducts and increases Na⁺ and water reabsorption. Which hormone acts on the collecting ducts to inhibit Na⁺ and Cl⁻ reabsorption, and also inhibits the release of renin, aldosterone and antidiuretic hormone?

- A. Atrial natriuretic peptide
- B. Antidiuretic hormone
- C. Angiotensin II
- D. Angiotensin converting enzyme

BIO-11-CH04-0088 | Challenging | Antidiuretic hormone

Osmoreceptors in the hypothalamus detect a low water concentration in the blood and the neurohypophysis secretes antidiuretic hormone, which acts on the last part of the distal convoluted tubule and the entire collecting duct. Which set of changes follows?

- A. Permeability of the target cells to water decreases, water is retained in the tubular fluid and dilute urine is formed
- B. Permeability of the target cells to Na⁺ increases, Na⁺ moves into the tissue fluid and blood volume falls
- C. Permeability of the target cells to urea increases, urea moves into the tissue fluid and blood urea falls
- D. Permeability of the target cells to water increases, water moves from the tubular fluid into the tissue fluid and concentrated urine is formed

BIO-11-CH04-0089 | Easy | Colour of normal urine

Normal urine is a pale yellow coloured transparent liquid. This colour is due to which pigment?

- A. Bilirubin
- B. Urochrome
- C. Creatinine
- D. Uric acid

BIO-11-CH04-0090 | Easy | Urine report

According to the prescribed normal urine report, which description is used for the usual reaction of urine?

- A. Alkaline
- B. Neutral
- C. Typically slightly acidic
- D. Amphoteric

BIO-11-CH04-0091 | Hard | Reaction of urine

The glomerular filtrate is alkaline at the time it is formed, yet the urine finally voided is recorded as acidic. Which step of urine formation brings about this change?

- A. Glomerular filtration into the capsular space of Bowman's capsule
- B. Selective reabsorption in the proximal convoluted tubule
- C. Counter current multiplication in the limbs of Henle's loop
- D. Tubular secretion in the distal convoluted tubule and collecting tubule

BIO-11-CH04-0092 | Medium | Specific gravity of urine

A urine report of a healthy person records a specific gravity of 1.02. If the level of antidiuretic hormone in this person rises while everything else remains unchanged, how would the specific gravity of the urine passed behave?

- A. It would decrease
- B. It would increase
- C. It would remain unaffected
- D. It would first fall and then return to 1.02

BIO-11-CH04-0093 | Easy | Skin

Sweat produced by the sweat glands of human skin carries out water, NaCl, lactic acid and urea. For which function is sweat primarily produced?

- A. Thermoregulation of the body
- B. Lubrication and protection of the skin
- C. Osmoregulation of the blood plasma
- D. Elimination of volatile substances from spices

BIO-11-CH04-0094 | Medium | Skin and lungs

Which structure is correctly paired with the substances it gives out?

- A. Sebaceous glands, giving out water, NaCl, lactic acid and urea
- B. Sweat glands, giving out water, NaCl, lactic acid and urea
- C. Sweat glands, giving out sebum for lubrication and protection
- D. Lungs, giving out sebum for lubrication and protection

BIO-11-CH04-0095 | Easy | Haemodialysis

In haemodialysis blood removed from the radial artery is passed through a tube immersed in dialysate fluid, and the wall of this tube acts as the semipermeable membrane. Of what material is this tube made?

- A. Cellophane
- B. Peritoneum
- C. The renal capsule
- D. The basement membrane of the glomerulus

BIO-11-CH04-0096 | Easy | Kidney stones

One type of kidney stone is formed in response to bacterial infection caused by urea splitting bacteria, and such stones grow quickly and become quite large. Which type of stone is this?

- A. Calcium stones
- B. Struvite stones
- C. Uric acid stones
- D. Cystine stones

BIO-11-CH04-0097 | Hard | Uremia

Blood normally contains 0.01 to 0.03 per cent urea, and a value above 0.05 per cent is called uremia. A patient's blood urea is found to be 0.06 per cent. This value is how many times the upper limit of the normal range?

- A. 2 times
- B. 1.2 times
- C. 3 times
- D. 6 times

BIO-11-CH04-0098 | Medium | Acute renal failure

In an adult, oligouria is defined as a urine output of less than 400 ml per day. An adult patient passes 150 ml of urine in 12 hours, and the same rate continues through the rest of the day. What is the output in one day, and does it satisfy the criterion?

- A. 300 ml per day, which does not satisfy the criterion
- B. 150 ml per day, which satisfies the criterion
- C. 600 ml per day, which does not satisfy the criterion
- D. 300 ml per day, which satisfies the criterion

BIO-11-CH04-0099 | Medium | Acute renal failure

In children, oligouria is defined as a urine output of less than 0.5 ml per kilogram of body weight per hour. For a child weighing 20 kg, an hourly urine output below which value satisfies this criterion?

- A. 5 ml per hour
- B. 20 ml per hour
- C. 10 ml per hour
- D. 40 ml per hour

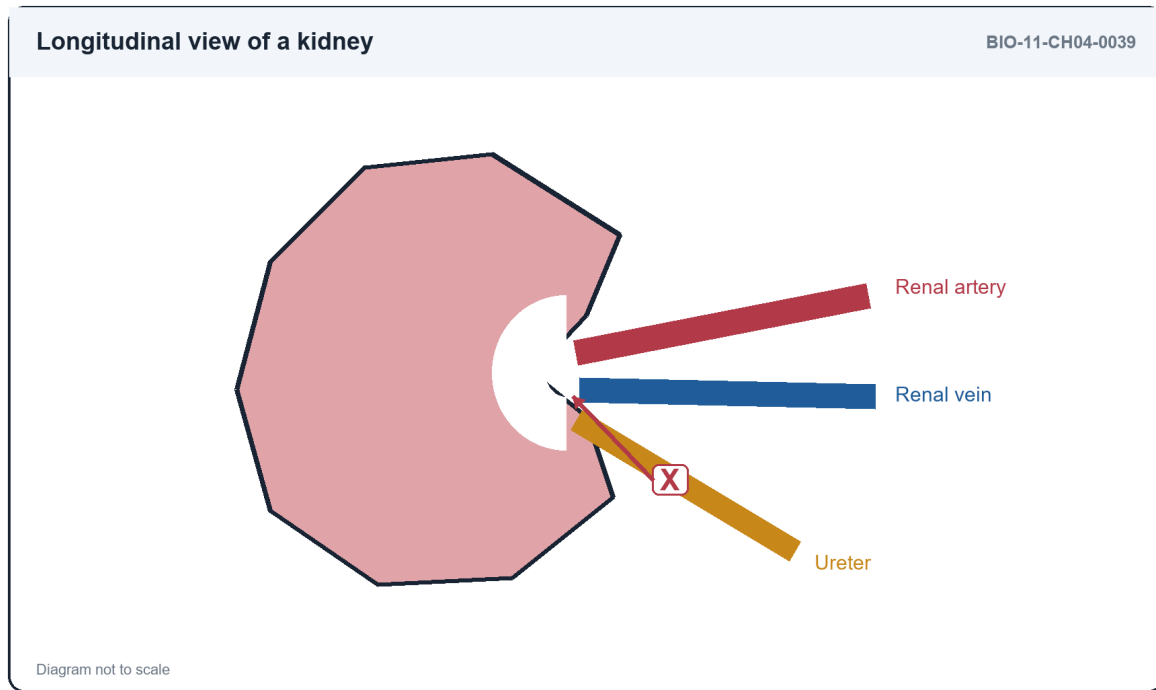
BIO-11-CH04-0100 | Challenging | Dialysis and transplant

Dialysis is regarded as a holding measure until a renal transplant can be performed. Which function of the kidney is a dialysis machine unable to take over, and therefore a reason for preferring a transplant?

- A. Removal of urea and other waste substances from the blood
- B. Removal of excess salts from the blood
- C. Secretion of erythropoietin, renin and calcitriol
- D. Maintenance of the water content of the blood

BIO-11-CH04-0101 | Easy | Kidney hilum

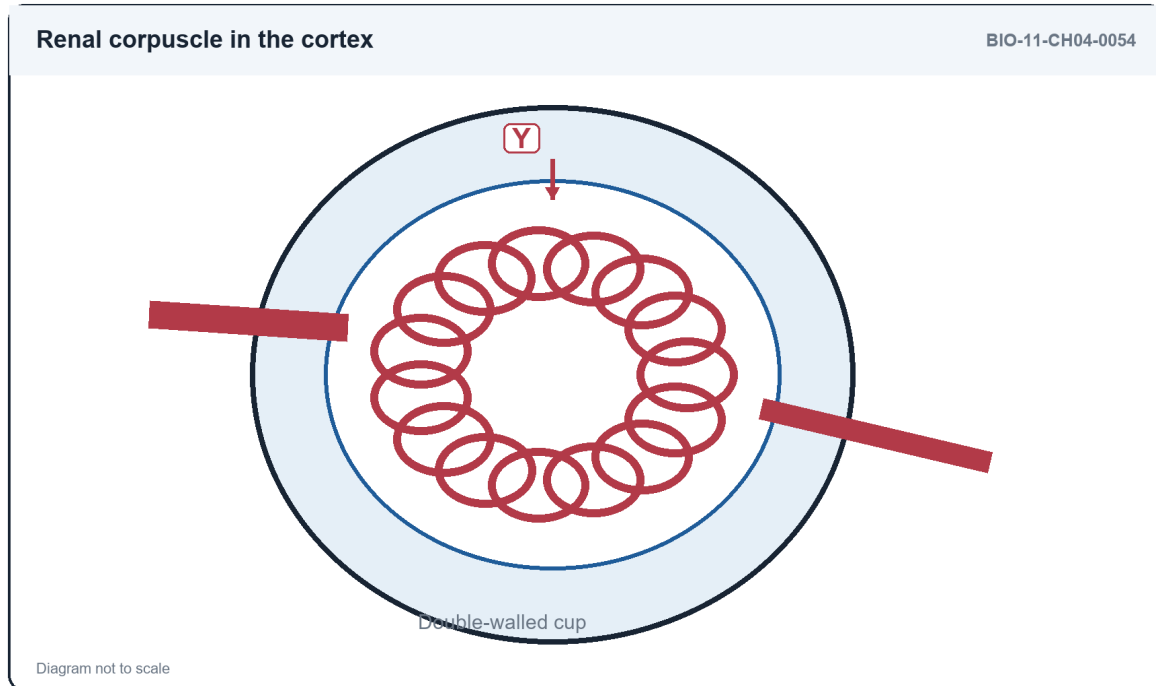
The kidney diagram shows the hilum on the concave border. Which vessel enters the kidney through this notch?



- A. Renal artery
- B. Renal vein
- C. Ureter
- D. Inferior vena cava

BIO-11-CH04-0102 | Medium | Malpighian body

In the renal-capsule diagram, label Y marks the space between the parietal and visceral walls of Bowman's capsule. What is Y?



- A. Glomerular lumen
- B. Capsular space
- C. Collecting duct
- D. Renal pelvis

BIO-11-CH04-0103 | Hard | Ultrafiltration

Glomerular hydrostatic pressure is 60 mm Hg, blood osmotic pressure is 32 mm Hg and capsular filtrate pressure is 18 mm Hg. What is the effective filtration pressure?

- A. 6 mm Hg
 - B. 8 mm Hg
 - C. 10 mm Hg
 - D. 14 mm Hg
-

BIO-11-CH04-0104 | Challenging | Dialysis and transplantation

Dialysis removes wastes and helps regulate water and salts, but it cannot fully replace the kidney's endocrine role. Which set contains only substances whose secretion is not replaced by dialysis?

- A. Urea, uric acid and creatinine
 - B. Water, sodium ions and urea
 - C. Glucose, amino acids and bicarbonate
 - D. Erythropoietin, renin and calcitriol
-

Standard 12 | Chapter 01

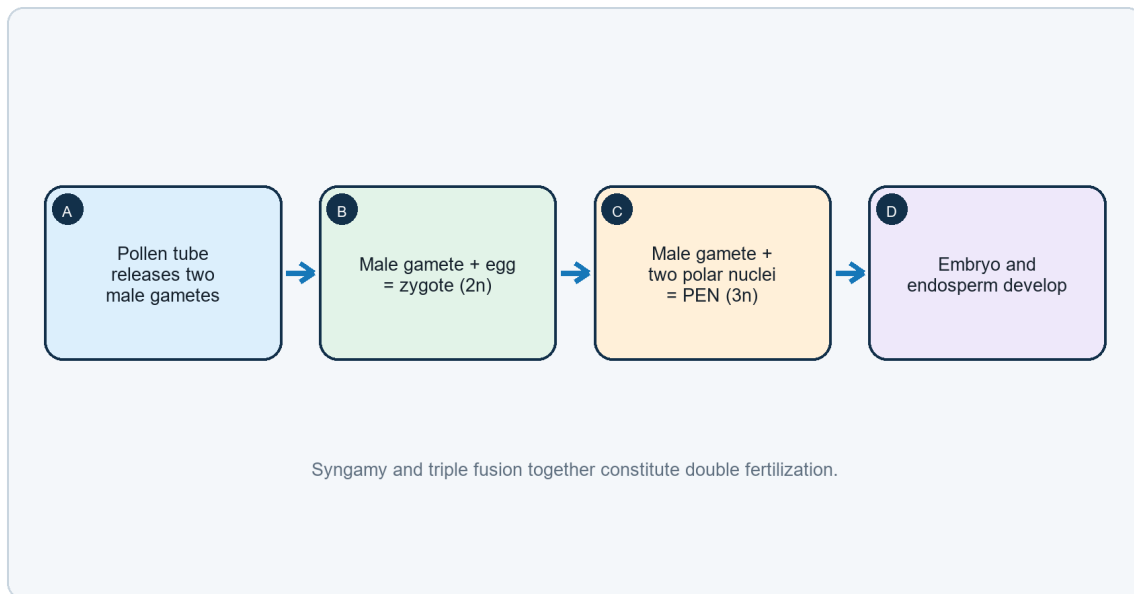
Reproduction in Lower and Higher Plants

BIO-12-CH01-0001 | Medium | Double fertilization in an angiosperm

Which labelled fusion in the figure produces the triploid primary endosperm nucleus?

Double fertilization in an angiosperm

Two male gametes participate in two distinct fusions



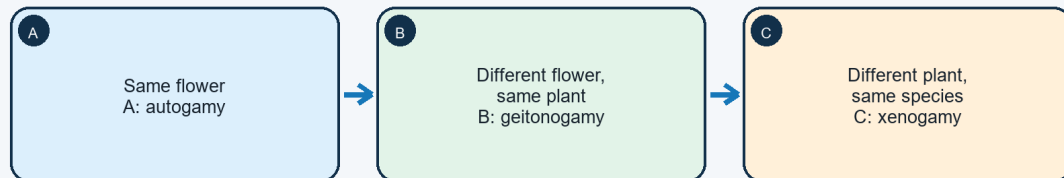
- A. Male gamete with the egg
- B. Egg with a synergid
- C. Two male gametes with each other
- D. Male gamete with the two polar nuclei

BIO-12-CH01-0002 | Hard | Pollination pathways

Which pathway in the figure most directly promotes outcrossing between genetically different plants?

Pollination pathways

Track the source and destination of pollen



Outcrossing is greatest when pollen moves between genetically different plants.

- A. Pathway A: autogamy
- B. Pathway C: xenogamy
- C. Pathway B: geitonogamy
- D. All three pathways equally

BIO-12-CH01-0003 | Hard | Fertilization

A reviewer compares several entries while checking 'Fusion of the second male gamete with the two polar nuclei'. Which entry must be corrected?

- A. Transfer between flowers on different plants of the same species - Xenogamy
- B. Fusion of the second male gamete with the two polar nuclei - Sporopollenin
- C. Pollination carried out by wind - Anemophily
- D. Embryos arising directly from nucellus or integuments - Adventive embryony

BIO-12-CH01-0004 | Hard | Pollination

A reviewer compares several entries while checking 'Transfer of pollen to the stigma of the same flower'. Which entry must be corrected?

- A. Syngamy and triple fusion occurring in one embryo sac - Double fertilization
- B. Gemma formation - Marchantia
- C. Transfer of pollen to the stigma of the same flower - Sporopollenin
- D. Anther wall layer that nourishes developing pollen - Tapetum

BIO-12-CH01-0005 | Medium | Ovule and embryo sac

A source table contains one accurate entry involving 'Junction between the ovule body and its stalk'. Which option preserves a correct biological association?

- A. Fusion of the second male gamete with the two polar nuclei - Porogamy
- B. Embryonic axis below the cotyledons - Hilum
- C. Junction between the ovule body and its stalk - Hilum
- D. Syngamy and triple fusion occurring in one embryo sac - Seed

BIO-12-CH01-0006 | Medium | Asexual reproduction

Complete the statement accurately: Budding under favourable conditions corresponds to _____.

- A. Marchantia
- B. Yeast
- C. Endosperm
- D. Spirogyra

BIO-12-CH01-0007 | Medium | Asexual reproduction

A source table contains one accurate entry involving 'Motile zoospore formation'. Which option preserves a correct biological association?

- A. Motile zoospore formation - Chlamydomonas
- B. Transfer of pollen to the stigma of the same flower - Chlamydomonas
- C. Embryonic axis below the cotyledons - Autogamy
- D. Primary endosperm nucleus generally develops into - Geitonogamy

BIO-12-CH01-0008 | Medium | Post-fertilization development

A source table contains one accurate entry involving 'Ovule after fertilization'. Which option preserves a correct biological association?

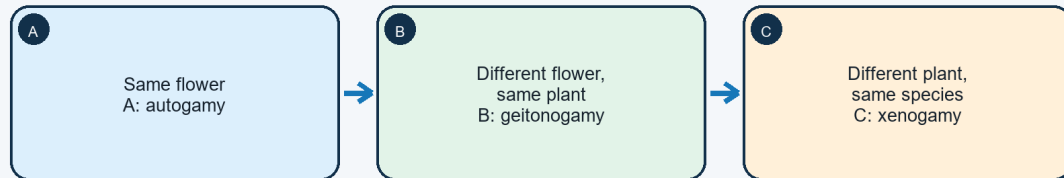
- A. Stalk attaching an ovule to the placenta - Syngamy
- B. Fusion of one male gamete with the egg - A tetrad of haploid microspores
- C. Ovule after fertilization - Seed
- D. Fruit formation without fertilization - Seed

BIO-12-CH01-0009 | Easy | Pollination pathways

In the figure, pathway A transfers pollen within the same flower. What is this process?

Pollination pathways

Track the source and destination of pollen



Outcrossing is greatest when pollen moves between genetically different plants.

- A. Autogamy
- B. Geitonogamy
- C. Xenogamy
- D. Anemophily

BIO-12-CH01-0010 | Easy | Pollination

Which biological term or result is specifically identified by the description 'Pollination carried out by wind'?

- A. Anemophily
- B. Funicle
- C. Porogamy
- D. Seven-celled and eight-nucleate

BIO-12-CH01-0011 | Medium | Microsporogenesis and pollen

A source table contains one accurate entry involving 'Highly resistant material of the pollen exine'. Which option preserves a correct biological association?

- A. Transfer of pollen to the stigma of the same flower - Hypocotyl
- B. Embryonic axis below the cotyledons - Sporopollenin
- C. Highly resistant material of the pollen exine - Sporopollenin
- D. Seed formation without fertilization - Porogamy

BIO-12-CH01-0012 | Medium | Microsporogenesis and pollen

A learner encounters the observation 'Anther wall layer that nourishes developing pollen' while analysing microsporogenesis and pollen. Which conclusion is supported by the prescribed biology?

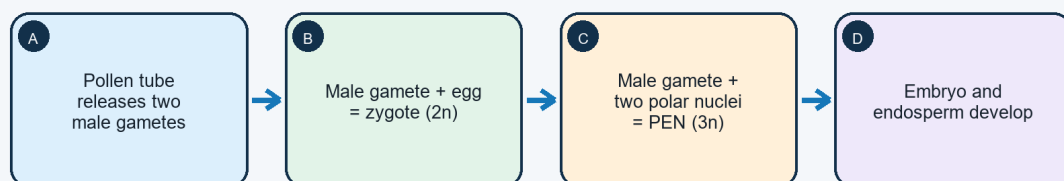
- A. Tapetum
- B. Seven-celled and eight-nucleate
- C. Micropyle
- D. Endosperm

BIO-12-CH01-0013 | Challenging | Double fertilization in an angiosperm

If syngamy occurs but triple fusion is experimentally blocked, which outcome is most likely?

Double fertilization in an angiosperm

Two male gametes participate in two distinct fusions



Syngamy and triple fusion together constitute double fertilization.

- A. Neither zygote nor endosperm can form
- B. A zygote may form, but normal endosperm initiation is absent
- C. Endosperm forms normally without a zygote
- D. Two diploid embryos form in the embryo sac

BIO-12-CH01-0014 | Easy | Ovule and embryo sac

Which biological term or result is specifically identified by the description 'Typical mature angiosperm embryo sac'?

- A. Fruit
- B. Yeast
- C. Micropyle
- D. Seven-celled and eight-nucleate

BIO-12-CH01-0015 | Medium | Special reproductive phenomena

A learner encounters the observation 'Fruit formation without fertilization' while analysing special reproductive phenomena. Which conclusion is supported by the prescribed biology?

- A. Polyembryony
- B. Parthenocarpy
- C. Tapetum
- D. Autogamy

BIO-12-CH01-0016 | Challenging | Ovule and embryo sac

A student submits four conclusions after integrating related chapter concepts.

- I. Junction between the ovule body and its stalk corresponds to Hilum.
- II. Opening left by the integuments at the ovule apex corresponds to Micropyle.
- III. Typical mature angiosperm embryo sac corresponds to Seven-celled and eight-nucleate.
- IV. Product of meiosis in a microspore mother cell corresponds to Hilum.

Which combination should be accepted without correction?

- A. I, II and IV only
- B. I, II and III only
- C. I and III only
- D. II and IV only

BIO-12-CH01-0017 | Medium | Fertilization

Complete the statement accurately: Fusion of one male gamete with the egg corresponds to _____.

- A. Porogamy
- B. Four
- C. Autogamy
- D. Syngamy

BIO-12-CH01-0018 | Medium | Special reproductive phenomena

Complete the statement accurately: Embryos arising directly from nucellus or integuments corresponds to _____.

- A. Apomixis
- B. Polyembryony
- C. Micropyle
- D. Adventive embryony

BIO-12-CH01-0019 | Hard | Post-fertilization development

Two conclusions must both be valid to support the reference concept 'Ovary after fertilization'. Which option contains two accurate associations?

- A. Fusion of one male gamete with the egg - Apomixis; Seed formation without fertilization - Syngamy
- B. Ovary after fertilization - Fruit; Embryos arising directly from nucellus or integuments - Adventive embryony
- C. Typical mature angiosperm embryo sac - Micropyle; Opening left by the integuments at the ovule apex - Seven-celled and eight-nucleate
- D. Pollen-tube entry into an ovule through the micropyle - Syngamy; Fusion of one male gamete with the egg - Porogamy

BIO-12-CH01-0020 | Hard | Pollination

Two conclusions must both be valid to support the reference concept 'Transfer between flowers on different plants of the same species'. Which option contains two accurate associations?

- A. Transfer between flowers on different plants of the same species - Xenogamy; Junction between the ovule body and its stalk - Hilum
- B. Fruit formation without fertilization - Seven-celled and eight-nucleate; Typical mature angiosperm embryo sac - Parthenocarp
- C. Embryos arising directly from nucellus or integuments - Endosperm; Primary endosperm nucleus generally develops into - Adventive embryony
- D. Ovule after fertilization - Adventive embryony; Embryos arising directly from nucellus or integuments - Seed

BIO-12-CH01-0021 | Easy | Fertilization

Which biological term or result is specifically identified by the description 'Pollen-tube entry into an ovule through the micropyle'?

- A. Hilum
- B. Porogamy
- C. Sporopollenin
- D. Seed

BIO-12-CH01-0022 | Medium | Pollination

A learner encounters the observation 'Pollination carried out by wind' while analysing pollination. Which conclusion is supported by the prescribed biology?

- A. Funicle
- B. Anemophily
- C. Seven-celled and eight-nucleate
- D. Double fertilization

BIO-12-CH01-0023 | Hard | Pollination

A reviewer compares several entries while checking 'Transfer between flowers on different plants of the same species'. Which entry must be corrected?

- A. Transfer between flowers on different plants of the same species - Hilum
- B. Embryos arising directly from nucellus or integuments - Adventive embryony
- C. Typical mature angiosperm embryo sac - Seven-celled and eight-nucleate
- D. Pollen-tube entry into an ovule through the micropyle - Porogamy

BIO-12-CH01-0024 | Easy | Ovule and embryo sac

Which biological term or result is specifically identified by the description 'Junction between the ovule body and its stalk'?

- A. Hilum
- B. Double fertilization
- C. Micropyle
- D. Seed

BIO-12-CH01-0025 | Challenging | Microsporogenesis and pollen

During error analysis, four proposed associations are reviewed.

- I. Anther wall layer that nourishes developing pollen corresponds to Sporopollenin.
 - II. Highly resistant material of the pollen exine corresponds to Four.
 - III. Number of pollen sacs in a typical dithecous anther corresponds to Four.
 - IV. Embryos arising directly from nucellus or integuments corresponds to Tapetum.
- Which set contains all and only the valid conclusions?

- A. II only
- B. III only
- C. I and IV only
- D. I and III only

BIO-12-CH01-0026 | Easy | Asexual reproduction

Which description is correctly represented by Spirogyra in this chapter?

- A. Fusion of one male gamete with the egg
- B. Gemma formation
- C. Product of meiosis in a microspore mother cell
- D. Fragmentation in a filamentous alga

BIO-12-CH01-0027 | Medium | Pollination

A learner encounters the observation 'Transfer between flowers on different plants of the same species' while analysing pollination. Which conclusion is supported by the prescribed biology?

- A. Seven-celled and eight-nucleate
- B. Xenogamy
- C. Parthenocarpy
- D. Porogamy

BIO-12-CH01-0028 | Hard | Ovule and embryo sac

A reviewer compares several entries while checking 'Junction between the ovule body and its stalk'. Which entry must be corrected?

- A. Ovule after fertilization - Seed
- B. Budding under favourable conditions - Yeast
- C. Fusion of the second male gamete with the two polar nuclei - Triple fusion
- D. Junction between the ovule body and its stalk - Double fertilization

BIO-12-CH01-0029 | Medium | Fertilization

A source table contains one accurate entry involving 'Syngamy and triple fusion occurring in one embryo sac'. Which option preserves a correct biological association?

- A. Syngamy and triple fusion occurring in one embryo sac - Double fertilization
- B. Stalk attaching an ovule to the placenta - Double fertilization
- C. Fragmentation in a filamentous alga - Geitonogamy
- D. Transfer between different flowers of the same plant - Chlamydomonas

BIO-12-CH01-0030 | Hard | Asexual reproduction

A reviewer compares several entries while checking 'Budding under favourable conditions'. Which entry must be corrected?

- A. Pollen-tube entry into an ovule through the micropyle - Porogamy
- B. Seed formation without fertilization - Apomixis
- C. Budding under favourable conditions - Triple fusion
- D. Junction between the ovule body and its stalk - Hilum

BIO-12-CH01-0031 | Medium | Special reproductive phenomena

Complete the statement accurately: Fruit formation without fertilization corresponds to _____.

- A. Fruit
- B. Parthenocarpy
- C. Autogamy
- D. Adventive embryony

BIO-12-CH01-0032 | Medium | Fertilization

Complete the statement accurately: Pollen-tube entry into an ovule through the micropyle corresponds to _____.

- A. Polyembryony
- B. Hilum
- C. Syngamy
- D. Porogamy

BIO-12-CH01-0033 | Medium | Ovule and embryo sac

Complete the statement accurately: Typical mature angiosperm embryo sac corresponds to _____.

- A. Four
- B. Anemophily
- C. Yeast
- D. Seven-celled and eight-nucleate

BIO-12-CH01-0034 | Easy | Fertilization

Which description is correctly represented by Double fertilization in this chapter?

- A. Transfer between different flowers of the same plant
- B. Occurrence of more than one embryo in a seed
- C. Syngamy and triple fusion occurring in one embryo sac
- D. Motile zoospore formation

BIO-12-CH01-0035 | Hard | Ovule and embryo sac

A reviewer compares several entries while checking 'Opening left by the integuments at the ovule apex'. Which entry must be corrected?

- A. Opening left by the integuments at the ovule apex - Fruit
- B. Anther wall layer that nourishes developing pollen - Tapetum
- C. Embryonic axis below the cotyledons - Hypocotyl
- D. Gemma formation - Marchantia

BIO-12-CH01-0036 | Hard | Asexual reproduction

A reviewer compares several entries while checking 'Gemma formation'. Which entry must be corrected?

- A. Gemma formation - Hypocotyl
- B. Highly resistant material of the pollen exine - Sporopollenin
- C. Occurrence of more than one embryo in a seed - Polyembryony
- D. Fruit formation without fertilization - Parthenocarpy

BIO-12-CH01-0037 | Hard | Special reproductive phenomena

Two conclusions must both be valid to support the reference concept 'Embryos arising directly from nucellus or integuments'. Which option contains two accurate associations?

- A. Transfer between flowers on different plants of the same species - Endosperm; Primary endosperm nucleus generally develops into - Xenogamy
- B. Motile zoospore formation - Xenogamy; Transfer between flowers on different plants of the same species - Chlamydomonas
- C. Fusion of the second male gamete with the two polar nuclei - Hypocotyl; Embryonic axis below the cotyledons - Triple fusion
- D. Embryos arising directly from nucellus or integuments - Adventive embryony; Fusion of one male gamete with the egg - Syngamy

BIO-12-CH01-0038 | Medium | Asexual reproduction

Complete the statement accurately: Fragmentation in a filamentous alga corresponds to _____.

- A. Double fertilization
- B. A tetrad of haploid microspores
- C. Spirogyra
- D. Yeast

BIO-12-CH01-0039 | Medium | Post-fertilization development

A learner encounters the observation 'Ovule after fertilization' while analysing post-fertilization development. Which conclusion is supported by the prescribed biology?

- A. Syngamy
- B. Porogamy
- C. Seed
- D. Anemophily

BIO-12-CH01-0040 | Medium | Special reproductive phenomena

A source table contains one accurate entry involving 'Embryos arising directly from nucellus or integuments'. Which option preserves a correct biological association?

- A. Primary endosperm nucleus generally develops into - Triple fusion
- B. Opening left by the integuments at the ovule apex - Hypocotyl
- C. Fusion of the second male gamete with the two polar nuclei - Adventive embryony
- D. Embryos arising directly from nucellus or integuments - Adventive embryony

BIO-12-CH01-0041 | Easy | Post-fertilization development

Which description is correctly represented by Fruit in this chapter?

- A. Ovary after fertilization
- B. Fusion of one male gamete with the egg
- C. Primary endosperm nucleus generally develops into
- D. Highly resistant material of the pollen exine

BIO-12-CH01-0042 | Hard | Special reproductive phenomena

Two conclusions must both be valid to support the reference concept 'Fruit formation without fertilization'. Which option contains two accurate associations?

- A. Fruit formation without fertilization - Parthenocarpy; Fusion of the second male gamete with the two polar nuclei - Triple fusion
- B. Transfer of pollen to the stigma of the same flower - Yeast; Budding under favourable conditions - Autogamy
- C. Anther wall layer that nourishes developing pollen - Autogamy; Transfer of pollen to the stigma of the same flower - Tapetum
- D. Opening left by the integuments at the ovule apex - A tetrad of haploid microspores; Product of meiosis in a microspore mother cell - Micropyle

BIO-12-CH01-0043 | Easy | Ovule and embryo sac

Which description is correctly represented by Micropyle in this chapter?

- A. Ovary after fertilization
- B. Opening left by the integuments at the ovule apex
- C. Gemma formation
- D. Typical mature angiosperm embryo sac

BIO-12-CH01-0044 | Easy | Fertilization

Which biological term or result is specifically identified by the description 'Fusion of the second male gamete with the two polar nuclei'?

- A. Triple fusion
- B. Xenogamy
- C. Anemophily
- D. Double fertilization

BIO-12-CH01-0045 | Medium | Ovule and embryo sac

Complete the statement accurately: Junction between the ovule body and its stalk corresponds to _____.

- A. Hilum
- B. Triple fusion
- C. Yeast
- D. Geitonogamy

BIO-12-CH01-0046 | Easy | Asexual reproduction

Which biological term or result is specifically identified by the description 'Budding under favourable conditions'?

- A. Apomixis
- B. Seed
- C. Yeast
- D. Endosperm

BIO-12-CH01-0047 | Challenging | Microsporogenesis and pollen

A researcher checks whether four linked observations have been interpreted correctly.

- I. Number of pollen sacs in a typical dithecous anther corresponds to Sporopollenin.
- II. Highly resistant material of the pollen exine corresponds to A tetrad of haploid microspores.
- III. Product of meiosis in a microspore mother cell corresponds to A tetrad of haploid microspores.
- IV. Ovule after fertilization corresponds to Seed.

Which combination survives the biological review?

- A. II and III only
- B. I only
- C. III and IV only
- D. I and III only

BIO-12-CH01-0048 | Challenging | Asexual reproduction

Evaluate the following source-based biological statements.

- I. Motile zoospore formation corresponds to Marchantia.
- II. Gemma formation corresponds to Marchantia.
- III. Budding under favourable conditions corresponds to Xenogamy.
- IV. Transfer between flowers on different plants of the same species corresponds to Chlamydomonas.

Which option identifies exactly the correct statements?

- A. IV only
- B. II only
- C. II and III only
- D. I and IV only

BIO-12-CH01-0049 | Hard | Asexual reproduction

Two conclusions must both be valid to support the reference concept 'Motile zoospore formation'. Which option contains two accurate associations?

- A. Transfer of pollen to the stigma of the same flower - Four; Number of pollen sacs in a typical dithecous anther - Autogamy
- B. Transfer between different flowers of the same plant - Autogamy; Transfer of pollen to the stigma of the same flower - Geitonogamy
- C. Embryonic axis below the cotyledons - Geitonogamy; Transfer between different flowers of the same plant - Hypocotyl
- D. Motile zoospore formation - Chlamydomonas; Seed formation without fertilization - Apomixis

BIO-12-CH01-0050 | Challenging | Asexual reproduction

A student submits four conclusions after integrating related chapter concepts.

- I. Gemma formation corresponds to Chlamydomonas.
 - II. Motile zoospore formation corresponds to Chlamydomonas.
 - III. Fragmentation in a filamentous alga corresponds to Porogamy.
 - IV. Pollen-tube entry into an ovule through the micropyle corresponds to Porogamy.
- Which combination should be accepted without correction?

- A. I and III only
- B. I, III and IV only
- C. II and IV only
- D. I and II only

BIO-12-CH01-0051 | Medium | Fertilization

A learner encounters the observation 'Fusion of one male gamete with the egg' while analysing fertilization. Which conclusion is supported by the prescribed biology?

- A. Adventive embryony
- B. Autogamy
- C. Syngamy
- D. Seed

BIO-12-CH01-0052 | Hard | Special reproductive phenomena

Two conclusions must both be valid to support the reference concept 'Occurrence of more than one embryo in a seed'. Which option contains two accurate associations?

- A. Occurrence of more than one embryo in a seed - Polyembryony; Transfer between different flowers of the same plant - Geitonogamy
- B. Pollination carried out by wind - Four; Number of pollen sacs in a typical dithecous anther - Anemophily
- C. Opening left by the integuments at the ovule apex - Syngamy; Fusion of one male gamete with the egg - Micropyle
- D. Fusion of one male gamete with the egg - Anemophily; Pollination carried out by wind - Syngamy

BIO-12-CH01-0053 | Medium | Asexual reproduction

A learner encounters the observation 'Budding under favourable conditions' while analysing asexual reproduction. Which conclusion is supported by the prescribed biology?

- A. Triple fusion
- B. Yeast
- C. Marchantia
- D. Spirogyra

BIO-12-CH01-0054 | Easy | Ovule and embryo sac

Which description is correctly represented by Funicle in this chapter?

- A. Fusion of one male gamete with the egg
- B. Fruit formation without fertilization
- C. Pollination carried out by wind
- D. Stalk attaching an ovule to the placenta

BIO-12-CH01-0055 | Easy | Post-fertilization development

Which biological term or result is specifically identified by the description 'Embryonic axis below the cotyledons'?

- A. Hypocotyl
- B. Apomixis
- C. Funicle
- D. Syngamy

BIO-12-CH01-0056 | Easy | Special reproductive phenomena

Which description is correctly represented by Polyembryony in this chapter?

- A. Transfer between different flowers of the same plant
- B. Opening left by the integuments at the ovule apex
- C. Occurrence of more than one embryo in a seed
- D. Fusion of one male gamete with the egg

BIO-12-CH01-0057 | Hard | Microsporogenesis and pollen

Two conclusions must both be valid to support the reference concept 'Highly resistant material of the pollen exine'. Which option contains two accurate associations?

- A. Transfer between different flowers of the same plant - Apomixis; Seed formation without fertilization - Geitonogamy
- B. Pollen-tube entry into an ovule through the micropyle - Hypocotyl; Embryonic axis below the cotyledons - Porogamy
- C. Embryonic axis below the cotyledons - Double fertilization; Syngamy and triple fusion occurring in one embryo sac - Hypocotyl
- D. Highly resistant material of the pollen exine - Sporopollenin; Fusion of one male gamete with the egg - Syngamy

BIO-12-CH01-0058 | Challenging | Pollination

During error analysis, four proposed associations are reviewed.

- I. Transfer between different flowers of the same plant corresponds to Geitonogamy.
- II. Transfer of pollen to the stigma of the same flower corresponds to Anemophily.
- III. Pollination carried out by wind corresponds to Double fertilization.
- IV. Syngamy and triple fusion occurring in one embryo sac corresponds to Geitonogamy.

Which set contains all and only the valid conclusions?

- A. I only
- B. I and III only
- C. III and IV only
- D. I, II and IV only

BIO-12-CH01-0059 | Hard | Asexual reproduction

Two conclusions must both be valid to support the reference concept 'Fragmentation in a filamentous alga'. Which option contains two accurate associations?

- A. Fragmentation in a filamentous alga - Spirogyra; Product of meiosis in a microspore mother cell - A tetrad of haploid microspores
- B. Occurrence of more than one embryo in a seed - Sporopollenin; Highly resistant material of the pollen exine - Polyembryony
- C. Syngamy and triple fusion occurring in one embryo sac - Parthenocarpy; Fruit formation without fertilization - Double fertilization
- D. Anther wall layer that nourishes developing pollen - Polyembryony; Occurrence of more than one embryo in a seed - Tapetum

BIO-12-CH01-0060 | Hard | Microsporogenesis and pollen

A reviewer compares several entries while checking 'Number of pollen sacs in a typical dithecous anther'. Which entry must be corrected?

- A. Opening left by the integuments at the ovule apex - Micropyle
- B. Stalk attaching an ovule to the placenta - Funicle
- C. Number of pollen sacs in a typical dithecous anther - Geitonogamy
- D. Seed formation without fertilization - Apomixis

BIO-12-CH01-0061 | Easy | Special reproductive phenomena

Which description is correctly represented by Adventive embryony in this chapter?

- A. Fusion of one male gamete with the egg
- B. Embryos arising directly from nucellus or integuments
- C. Transfer between flowers on different plants of the same species
- D. Primary endosperm nucleus generally develops into

BIO-12-CH01-0062 | Hard | Fertilization

Two conclusions must both be valid to support the reference concept 'Fusion of the second male gamete with the two polar nuclei'. Which option contains two accurate associations?

- A. Transfer between flowers on different plants of the same species - Adventive embryony; Embryos arising directly from nucellus or integuments - Xenogamy
- B. Seed formation without fertilization - Chlamydomonas; Motile zoospore formation - Apomixis
- C. Motile zoospore formation - Anemophily; Pollination carried out by wind - Chlamydomonas
- D. Fusion of the second male gamete with the two polar nuclei - Triple fusion; Highly resistant material of the pollen exine - Sporopollenin

BIO-12-CH01-0063 | Hard | Ovule and embryo sac

Two conclusions must both be valid to support the reference concept 'Opening left by the integuments at the ovule apex'. Which option contains two accurate associations?

- A. Opening left by the integuments at the ovule apex - Micropyle; Ovary after fertilization - Fruit
- B. Anther wall layer that nourishes developing pollen - Hypocotyl; Embryonic axis below the cotyledons - Tapetum
- C. Gemma formation - Four; Number of pollen sacs in a typical dithecous anther - Marchantia
- D. Number of pollen sacs in a typical dithecous anther - Porogamy; Pollen-tube entry into an ovule through the micropyle - Four

BIO-12-CH01-0064 | Hard | Post-fertilization development

A reviewer compares several entries while checking 'Ovary after fertilization'. Which entry must be corrected?

- A. Ovary after fertilization - Adventive embryony
- B. Seed formation without fertilization - Apomixis
- C. Typical mature angiosperm embryo sac - Seven-celled and eight-nucleate
- D. Pollen-tube entry into an ovule through the micropyle - Porogamy

BIO-12-CH01-0065 | Hard | Post-fertilization development

Two conclusions must both be valid to support the reference concept 'Ovule after fertilization'. Which option contains two accurate associations?

- A. Fusion of one male gamete with the egg - Anemophily; Pollination carried out by wind - Syngamy
- B. Pollination carried out by wind - A tetrad of haploid microspores; Product of meiosis in a microspore mother cell - Anemophily
- C. Product of meiosis in a microspore mother cell - Parthenocarpy; Fruit formation without fertilization - A tetrad of haploid microspores
- D. Ovule after fertilization - Seed; Stalk attaching an ovule to the placenta - Funicle

BIO-12-CH01-0066 | Challenging | Post-fertilization development

A researcher checks whether four linked observations have been interpreted correctly.

- I. Ovary after fertilization corresponds to Seed.
- II. Ovule after fertilization corresponds to Seed.
- III. Primary endosperm nucleus generally develops into corresponds to Endosperm.
- IV. Number of pollen sacs in a typical dithecous anther corresponds to Fruit.

Which combination survives the biological review?

- A. II and III only
- B. III only
- C. II only
- D. IV only

BIO-12-CH01-0067 | Medium | Microsporogenesis and pollen

Complete the statement accurately: Anther wall layer that nourishes developing pollen corresponds to _____.

- A. Syngamy
- B. Four
- C. Sporopollenin
- D. Tapetum

BIO-12-CH01-0068 | Medium | Post-fertilization development

Complete the statement accurately: Primary endosperm nucleus generally develops into corresponds to _____.

- A. Spirogyra
- B. Funicle
- C. Hilum
- D. Endosperm

BIO-12-CH01-0069 | Challenging | Pollination

Evaluate the following source-based biological statements.

- I. Transfer of pollen to the stigma of the same flower corresponds to Autogamy.
- II. Transfer between flowers on different plants of the same species corresponds to Xenogamy.
- III. Transfer between different flowers of the same plant corresponds to Geitonogamy.
- IV. Fragmentation in a filamentous alga corresponds to Spirogyra.

Which option identifies exactly the correct statements?

- A. II, III and IV only
- B. II and IV only
- C. I, II, III and IV
- D. I and III only

BIO-12-CH01-0070 | Easy | Pollination

Which description is correctly represented by Xenogamy in this chapter?

- A. Junction between the ovule body and its stalk
- B. Transfer between different flowers of the same plant
- C. Transfer between flowers on different plants of the same species
- D. Typical mature angiosperm embryo sac

BIO-12-CH01-0071 | Medium | Microsporogenesis and pollen

Complete the statement accurately: Number of pollen sacs in a typical dithecous anther corresponds to _____.

- A. Apomixis
- B. Geitonogamy
- C. Tapetum
- D. Four

BIO-12-CH01-0072 | Hard | Microsporogenesis and pollen

Two conclusions must both be valid to support the reference concept 'Number of pollen sacs in a typical dithecous anther'. Which option contains two accurate associations?

- A. Seed formation without fertilization - Micropyle; Opening left by the integuments at the ovule apex - Apomixis
- B. Number of pollen sacs in a typical dithecous anther - Four; Transfer between different flowers of the same plant - Geitonogamy
- C. Typical mature angiosperm embryo sac - Apomixis; Seed formation without fertilization - Seven-celled and eight-nucleate
- D. Stalk attaching an ovule to the placenta - Adventive embryony; Embryos arising directly from nucellus or integuments - Funicle

BIO-12-CH01-0073 | Hard | Post-fertilization development

A reviewer compares several entries while checking 'Embryonic axis below the cotyledons'. Which entry must be corrected?

- A. Budding under favourable conditions - Yeast
- B. Transfer of pollen to the stigma of the same flower - Autogamy
- C. Embryonic axis below the cotyledons - Funicle
- D. Fusion of one male gamete with the egg - Syngamy

BIO-12-CH01-0074 | Medium | Post-fertilization development

A source table contains one accurate entry involving 'Primary endosperm nucleus generally develops into'. Which option preserves a correct biological association?

- A. Transfer between different flowers of the same plant - Four
- B. Fragmentation in a filamentous alga - Yeast
- C. Primary endosperm nucleus generally develops into - Endosperm
- D. Budding under favourable conditions - Funicle

BIO-12-CH01-0075 | Medium | Fertilization

A learner encounters the observation 'Fusion of the second male gamete with the two polar nuclei' while analysing fertilization. Which conclusion is supported by the prescribed biology?

- A. Triple fusion
- B. Anemophily
- C. Chlamydomonas
- D. Syngamy

BIO-12-CH01-0076 | Hard | Pollination

Two conclusions must both be valid to support the reference concept 'Transfer of pollen to the stigma of the same flower'. Which option contains two accurate associations?

- A. Syngamy and triple fusion occurring in one embryo sac - Tapetum; Anther wall layer that nourishes developing pollen - Double fertilization
- B. Gemma formation - Apomixis; Seed formation without fertilization - Marchantia
- C. Transfer of pollen to the stigma of the same flower - Autogamy; Highly resistant material of the pollen exine - Sporopollenin
- D. Fragmentation in a filamentous alga - Double fertilization; Syngamy and triple fusion occurring in one embryo sac - Spirogyra

BIO-12-CH01-0077 | Medium | Special reproductive phenomena

A learner encounters the observation 'Occurrence of more than one embryo in a seed' while analysing special reproductive phenomena. Which conclusion is supported by the prescribed biology?

- A. Polyembryony
- B. Syngamy
- C. Micropyle
- D. Geitonogamy

BIO-12-CH01-0078 | Easy | Asexual reproduction

Which biological term or result is specifically identified by the description 'Gemma formation'?

- A. Sporopollenin
- B. Marchantia
- C. Spirogyra
- D. Autogamy

BIO-12-CH01-0079 | Medium | Ovule and embryo sac

Complete the statement accurately: Opening left by the integuments at the ovule apex corresponds to _____.

- A. Seven-celled and eight-nucleate
- B. Anemophily
- C. Tapetum
- D. Micropyle

BIO-12-CH01-0080 | Medium | Microsporogenesis and pollen

A learner encounters the observation 'Number of pollen sacs in a typical dithecal anther' while analysing microsporogenesis and pollen. Which conclusion is supported by the prescribed biology?

- A. Apomixis
- B. Tapetum
- C. Four
- D. Seven-celled and eight-nucleate

BIO-12-CH01-0081 | Easy | Microsporogenesis and pollen

Which biological term or result is specifically identified by the description 'Number of pollen sacs in a typical dithecal anther'?

- A. Syngamy
- B. Apomixis
- C. Four
- D. Tapetum

BIO-12-CH01-0082 | Medium | Pollination

A learner encounters the observation 'Transfer of pollen to the stigma of the same flower' while analysing pollination. Which conclusion is supported by the prescribed biology?

- A. Xenogamy
- B. Apomixis
- C. Spirogyra
- D. Autogamy

BIO-12-CH01-0083 | Hard | Ovule and embryo sac

Two conclusions must both be valid to support the reference concept 'Stalk attaching an ovule to the placenta'. Which option contains two accurate associations?

- A. Stalk attaching an ovule to the placenta - Funicle; Embryos arising directly from nucellus or integuments - Adventive embryony
- B. Fusion of one male gamete with the egg - Parthenocarpy; Fruit formation without fertilization - Syngamy
- C. Fruit formation without fertilization - Fruit; Ovary after fertilization - Parthenocarpy
- D. Ovule after fertilization - Geitonogamy; Transfer between different flowers of the same plant - Seed

BIO-12-CH01-0084 | Medium | Ovule and embryo sac

A learner encounters the observation 'Typical mature angiosperm embryo sac' while analysing ovule and embryo sac. Which conclusion is supported by the prescribed biology?

- A. Micropyle
- B. Parthenocarpy
- C. Seven-celled and eight-nucleate
- D. Anemophily

BIO-12-CH01-0085 | Medium | Fertilization

A learner encounters the observation 'Syngamy and triple fusion occurring in one embryo sac' while analysing fertilization. Which conclusion is supported by the prescribed biology?

- A. Hilum
- B. Polyembryony
- C. Syngamy
- D. Double fertilization

BIO-12-CH01-0086 | Easy | Fertilization

Which description is correctly represented by Triple fusion in this chapter?

- A. Transfer between flowers on different plants of the same species
- B. Fusion of the second male gamete with the two polar nuclei
- C. Motile zoospore formation
- D. Pollination carried out by wind

BIO-12-CH01-0087 | Medium | Special reproductive phenomena

A learner encounters the observation 'Embryos arising directly from nucellus or integuments' while analysing special reproductive phenomena. Which conclusion is supported by the prescribed biology?

- A. Apomixis
- B. Xenogamy
- C. Adventive embryony
- D. Syngamy

BIO-12-CH01-0088 | Medium | Ovule and embryo sac

A source table contains one accurate entry involving 'Typical mature angiosperm embryo sac'. Which option preserves a correct biological association?

- A. Pollination carried out by wind - Geitonogamy
- B. Typical mature angiosperm embryo sac - Seven-celled and eight-nucleate
- C. Ovary after fertilization - Xenogamy
- D. Transfer between different flowers of the same plant - Seven-celled and eight-nucleate

BIO-12-CH01-0089 | Hard | Ovule and embryo sac

A reviewer compares several entries while checking 'Typical mature angiosperm embryo sac'. Which entry must be corrected?

- A. Typical mature angiosperm embryo sac - Yeast
- B. Ovary after fertilization - Fruit
- C. Transfer between different flowers of the same plant - Geitonogamy
- D. Number of pollen sacs in a typical dithecous anther - Four

BIO-12-CH01-0090 | Medium | Asexual reproduction

A source table contains one accurate entry involving 'Budding under favourable conditions'. Which option preserves a correct biological association?

- A. Budding under favourable conditions - Yeast
- B. Primary endosperm nucleus generally develops into - Porogamy
- C. Ovule after fertilization - Apomixis
- D. Junction between the ovule body and its stalk - Endosperm

BIO-12-CH01-0091 | Medium | Post-fertilization development

A source table contains one accurate entry involving 'Embryonic axis below the cotyledons'. Which option preserves a correct biological association?

- A. Highly resistant material of the pollen exine - Yeast
- B. Budding under favourable conditions - Hypocotyl
- C. Seed formation without fertilization - Tapetum
- D. Embryonic axis below the cotyledons - Hypocotyl

BIO-12-CH01-0092 | Medium | Ovule and embryo sac

Complete the statement accurately: Stalk attaching an ovule to the placenta corresponds to ____.

- A. Parthenocarpy
- B. Funicle
- C. Micropyle
- D. Fruit

BIO-12-CH01-0093 | Medium | Pollination

Complete the statement accurately: Transfer between different flowers of the same plant corresponds to ____.

- A. Spirogyra
- B. Geitonogamy
- C. Adventive embryony
- D. Autogamy

BIO-12-CH01-0094 | Medium | Microsporogenesis and pollen

A source table contains one accurate entry involving 'Anther wall layer that nourishes developing pollen'. Which option preserves a correct biological association?

- A. Primary endosperm nucleus generally develops into - Autogamy
- B. Typical mature angiosperm embryo sac - Seed
- C. Fusion of one male gamete with the egg - Endosperm
- D. Anther wall layer that nourishes developing pollen - Tapetum

BIO-12-CH01-0095 | Medium | Pollination

A learner encounters the observation 'Transfer between different flowers of the same plant' while analysing pollination. Which conclusion is supported by the prescribed biology?

- A. Parthenocarpy
- B. Adventive embryony
- C. Geitonogamy
- D. Anemophily

BIO-12-CH01-0096 | Hard | Special reproductive phenomena

Two conclusions must both be valid to support the reference concept 'Seed formation without fertilization'. Which option contains two accurate associations?

- A. Number of pollen sacs in a typical dithecous anther - Anemophily; Pollination carried out by wind - Four
- B. Typical mature angiosperm embryo sac - Four; Number of pollen sacs in a typical dithecous anther - Seven-celled and eight-nucleate
- C. Motile zoospore formation - Seven-celled and eight-nucleate; Typical mature angiosperm embryo sac - Chlamydomonas
- D. Seed formation without fertilization - Apomixis; Product of meiosis in a microspore mother cell - A tetrad of haploid microspores

BIO-12-CH01-0097 | Medium | Post-fertilization development

Complete the statement accurately: Ovary after fertilization corresponds to _____.

- A. Seed
- B. Fruit
- C. Endosperm
- D. Sporopollenin

BIO-12-CH01-0098 | Medium | Pollination

Complete the statement accurately: Transfer of pollen to the stigma of the same flower corresponds to _____.

- A. Apomixis
- B. Double fertilization
- C. Marchantia
- D. Autogamy

BIO-12-CH01-0099 | Medium | Special reproductive phenomena

A source table contains one accurate entry involving 'Seed formation without fertilization'. Which option preserves a correct biological association?

- A. Number of pollen sacs in a typical dithecous anther - Apomixis
- B. Transfer of pollen to the stigma of the same flower - Chlamydomonas
- C. Product of meiosis in a microspore mother cell - Hypocotyl
- D. Seed formation without fertilization - Apomixis

BIO-12-CH01-0100 | Challenging | Fertilization

A student submits four conclusions after integrating related chapter concepts.

- I. Syngamy and triple fusion occurring in one embryo sac corresponds to Double fertilization.
- II. Fusion of the second male gamete with the two polar nuclei corresponds to Porogamy.
- III. Pollen-tube entry into an ovule through the micropyle corresponds to A tetrad of haploid microspores.
- IV. Product of meiosis in a microspore mother cell corresponds to A tetrad of haploid microspores.

Which combination should be accepted without correction?

- A. I, III and IV only
- B. I and IV only
- C. III only
- D. I, II and IV only

BIO-12-CH01-0101 | Medium | Asexual reproduction

A learner encounters the observation 'Fragmentation in a filamentous alga' while analysing asexual reproduction. Which conclusion is supported by the prescribed biology?

- A. Spirogyra
- B. A tetrad of haploid microspores
- C. Syngamy
- D. Double fertilization

BIO-12-CH01-0102 | Easy | Post-fertilization development

Which description is correctly represented by Seed in this chapter?

- A. Pollination carried out by wind
- B. Primary endosperm nucleus generally develops into
- C. Fusion of one male gamete with the egg
- D. Ovule after fertilization

BIO-12-CH01-0103 | Hard | Fertilization

Two conclusions must both be valid to support the reference concept 'Syngamy and triple fusion occurring in one embryo sac'. Which option contains two accurate associations?

- A. Stalk attaching an ovule to the placenta - Xenogamy; Transfer between flowers on different plants of the same species - Funicle
- B. Syngamy and triple fusion occurring in one embryo sac - Double fertilization; Fragmentation in a filamentous alga - Spirogyra
- C. Transfer between different flowers of the same plant - Hilum; Junction between the ovule body and its stalk - Geitonogamy
- D. Junction between the ovule body and its stalk - Chlamydomonas; Motile zoospore formation - Hilum

BIO-12-CH01-0104 | Hard | Pollination

A reviewer compares several entries while checking 'Pollination carried out by wind'. Which entry must be corrected?

- A. Pollination carried out by wind - Seven-celled and eight-nucleate
- B. Pollen-tube entry into an ovule through the micropyle - Porogamy
- C. Stalk attaching an ovule to the placenta - Funicle
- D. Anther wall layer that nourishes developing pollen - Tapetum

BIO-12-CH01-0105 | Easy | Pollination

Which biological term or result is specifically identified by the description 'Transfer between flowers on different plants of the same species'?

- A. Adventive embryony
- B. Sporopollenin
- C. Xenogamy
- D. Porogamy

BIO-12-CH01-0106 | Medium | Fertilization

Complete the statement accurately: Fusion of the second male gamete with the two polar nuclei corresponds to _____.

- A. Apomixis
- B. Triple fusion
- C. Chlamydomonas
- D. Anemophily

BIO-12-CH01-0107 | Hard | Fertilization

A reviewer compares several entries while checking 'Pollen-tube entry into an ovule through the micropyle'. Which entry must be corrected?

- A. Ovule after fertilization - Seed
- B. Highly resistant material of the pollen exine - Sporopollenin
- C. Ovary after fertilization - Fruit
- D. Pollen-tube entry into an ovule through the micropyle - Hilum

BIO-12-CH01-0108 | Easy | Special reproductive phenomena

Which description is correctly represented by Apomixis in this chapter?

- A. Typical mature angiosperm embryo sac
- B. Occurrence of more than one embryo in a seed
- C. Seed formation without fertilization
- D. Embryos arising directly from nucellus or integuments

BIO-12-CH01-0109 | Medium | Microsporogenesis and pollen

A source table contains one accurate entry involving 'Number of pollen sacs in a typical ditheous anther'. Which option preserves a correct biological association?

- A. Number of pollen sacs in a typical ditheous anther - Four
- B. Transfer between different flowers of the same plant - Apomixis
- C. Typical mature angiosperm embryo sac - Micropyle
- D. Stalk attaching an ovule to the placenta - Four

BIO-12-CH01-0110 | Easy | Microsporogenesis and pollen

Which biological term or result is specifically identified by the description 'Product of meiosis in a microspore mother cell'?

- A. Autogamy
- B. Hilum
- C. A tetrad of haploid microspores
- D. Double fertilization

BIO-12-CH01-0111 | Challenging | Special reproductive phenomena

During error analysis, four proposed associations are reviewed.

- I. Fruit formation without fertilization corresponds to Adventive embryony.
- II. Embryos arising directly from nucellus or integuments corresponds to Adventive embryony.
- III. Seed formation without fertilization corresponds to Apomixis.
- IV. Pollination carried out by wind corresponds to Anemophily.

Which set contains all and only the valid conclusions?

- A. I and IV only
- B. I, II and III only
- C. II, III and IV only
- D. I only

BIO-12-CH01-0112 | Easy | Microsporogenesis and pollen

Which description is correctly represented by Four in this chapter?

- A. Transfer between different flowers of the same plant
- B. Number of pollen sacs in a typical ditheous anther
- C. Highly resistant material of the pollen exine
- D. Opening left by the integuments at the ovule apex

BIO-12-CH01-0113 | Easy | Pollination

Which biological term or result is specifically identified by the description 'Transfer of pollen to the stigma of the same flower'?

- A. Double fertilization
- B. Autogamy
- C. Spirogyra
- D. Geitonogamy

BIO-12-CH01-0114 | Easy | Ovule and embryo sac

Which biological term or result is specifically identified by the description 'Opening left by the integuments at the ovule apex'?

- A. Fruit
- B. Tapetum
- C. Micropyle
- D. Hypocotyl

BIO-12-CH01-0115 | Easy | Microsporogenesis and pollen

Which description is correctly represented by Sporopollenin in this chapter?

- A. Fusion of one male gamete with the egg
- B. Pollen-tube entry into an ovule through the micropyle
- C. Transfer between different flowers of the same plant
- D. Highly resistant material of the pollen exine

BIO-12-CH01-0116 | Easy | Pollination

Which description is correctly represented by Autogamy in this chapter?

- A. Syngamy and triple fusion occurring in one embryo sac
- B. Transfer of pollen to the stigma of the same flower
- C. Highly resistant material of the pollen exine
- D. Gemma formation

BIO-12-CH01-0117 | Easy | Post-fertilization development

Which biological term or result is specifically identified by the description 'Ovary after fertilization'?

- A. Seed
- B. Fruit
- C. Adventive embryony
- D. Sporopollenin

BIO-12-CH01-0118 | Medium | Asexual reproduction

A source table contains one accurate entry involving 'Fragmentation in a filamentous alga'. Which option preserves a correct biological association?

- A. Syngamy and triple fusion occurring in one embryo sac - Tapetum
- B. Fusion of one male gamete with the egg - Parthenocarpy
- C. Product of meiosis in a microspore mother cell - Double fertilization
- D. Fragmentation in a filamentous alga - Spirogyra

BIO-12-CH01-0119 | Challenging | Post-fertilization development

A researcher checks whether four linked observations have been interpreted correctly.

- I. Embryonic axis below the cotyledons corresponds to Hypocotyl.
- II. Ovary after fertilization corresponds to Seed.
- III. Ovule after fertilization corresponds to Seed.
- IV. Pollen-tube entry into an ovule through the micropyle corresponds to Hypocotyl.

Which combination survives the biological review?

- A. I and III only
- B. II only
- C. IV only
- D. I, II and IV only

BIO-12-CH01-0120 | Easy | Special reproductive phenomena

Which description is correctly represented by Parthenocarpy in this chapter?

- A. Fusion of the second male gamete with the two polar nuclei
 - B. Transfer of pollen to the stigma of the same flower
 - C. Fruit formation without fertilization
 - D. Occurrence of more than one embryo in a seed
-

Standard 12 | Chapter 02

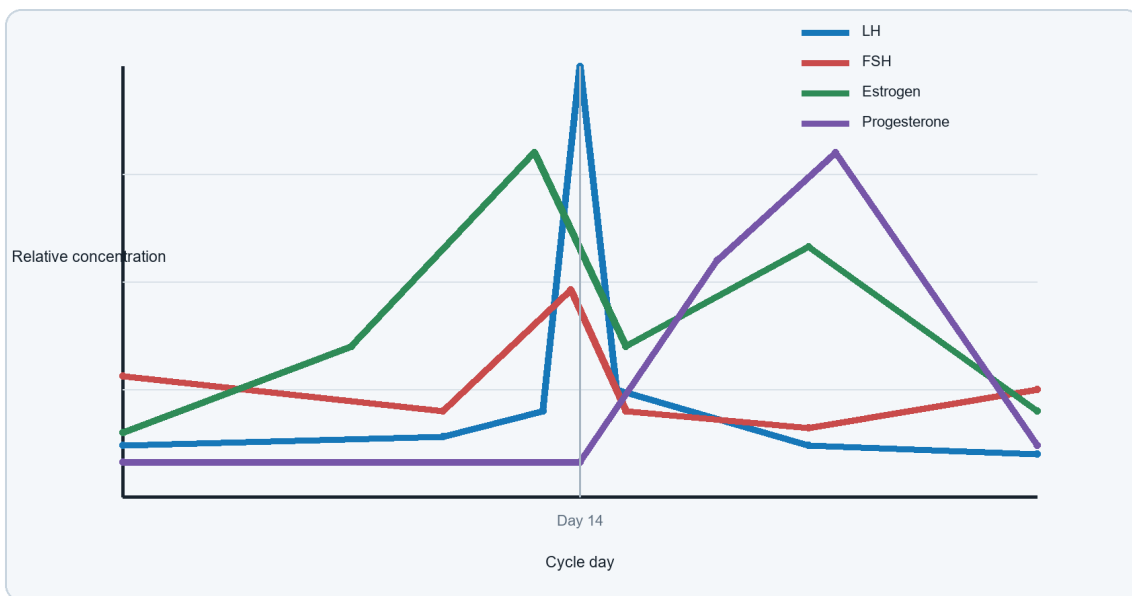
Reproduction in Lower and Higher Animals

BIO-12-CH02-0001 | Easy | Hormonal pattern across a menstrual cycle

The sharp mid-cycle peak in the figure corresponds to which immediate trigger for ovulation?

Hormonal pattern across a menstrual cycle

Relative hormone concentration; day 14 is approximate



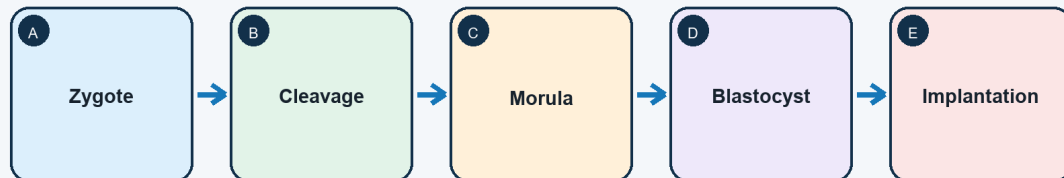
- A. LH surge
- B. Progesterone fall
- C. FSH absence
- D. Prolactin surge

BIO-12-CH02-0002 | Medium | Early human development

Which stage shown in the figure normally begins implantation in the uterine endometrium?

Early human development

Major stages from fertilization to implantation



Cleavage increases cell number without a comparable increase in total embryonic size.

- A. Zygote
- B. Two-cell embryo
- C. Morula before cavitation
- D. Blastocyst

BIO-12-CH02-0003 | Medium | Male reproductive system

Complete the statement accurately: Site of sperm formation corresponds to _____.

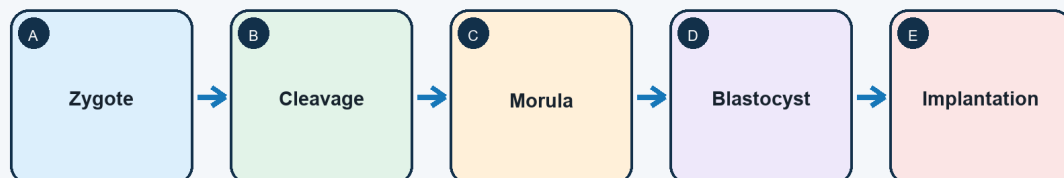
- A. Progesterone
- B. Spermatogonia
- C. Seminiferous tubules
- D. Gastrulation

BIO-12-CH02-0004 | Challenging | Early human development

If cleavage stops immediately after the first zygotic division, which later event in the sequence cannot occur?

Early human development

Major stages from fertilization to implantation



Cleavage increases cell number without a comparable increase in total embryonic size.

- A. Formation of a morula and blastocyst
- B. Fusion of sperm and ovum
- C. Completion of ovulation
- D. Formation of the corpus luteum

BIO-12-CH02-0005 | Medium | Menstrual cycle

A learner encounters the observation 'Hormone chiefly maintaining the secretory endometrium' while analysing menstrual cycle. Which conclusion is supported by the prescribed biology?

- A. Ovary
- B. Freshwater sponge
- C. LH surge
- D. Progesterone

BIO-12-CH02-0006 | Medium | Female reproductive system

Complete the statement accurately: Organ producing ova, estrogen and progesterone corresponds to _____.

- A. Tubectomy
- B. Ovary
- C. Ampulla of the fallopian tube
- D. FSH

BIO-12-CH02-0007 | Medium | Asexual reproduction

A learner encounters the observation 'Budding' while analysing asexual reproduction. Which conclusion is supported by the prescribed biology?

- A. Freshwater sponge
- B. Hydra
- C. Sertoli cells
- D. Vasectomy

BIO-12-CH02-0008 | Medium | Birth control and reproductive health

A source table contains one accurate entry involving 'Surgical interruption of the fallopian tubes'. Which option preserves a correct biological association?

- A. Organ producing ova, estrogen and progesterone - Tubectomy
- B. Embryonic stage with trophoblast and inner cell mass - Primary spermatocyte
- C. Surgical interruption of the fallopian tubes - Tubectomy
- D. Bacterial sexually transmitted infection caused by Neisseria - Ovary

BIO-12-CH02-0009 | Challenging | Male reproductive system

A student submits four conclusions after integrating related chapter concepts.

- I. Cells secreting testicular androgens corresponds to Seminiferous tubules.
 - II. Site of sperm formation corresponds to Seminiferous tubules.
 - III. Duct carrying sperm away from the epididymis corresponds to Vas deferens.
 - IV. Barrier method that also lowers STI transmission corresponds to Leydig cells.
- Which combination should be accepted without correction?

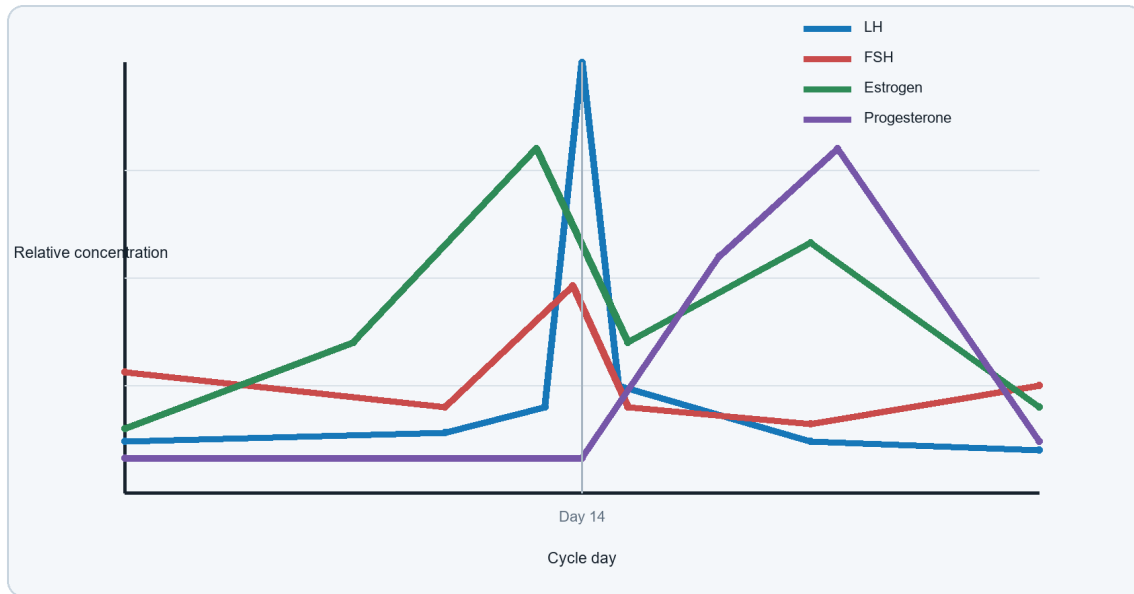
- A. I and III only
- B. III only
- C. II and III only
- D. I, II, III and IV

BIO-12-CH02-0010 | Hard | Hormonal pattern across a menstrual cycle

A normal LH surge is followed by ovulation. Which plotted hormone then chiefly supports the secretory endometrium?

Hormonal pattern across a menstrual cycle

Relative hormone concentration; day 14 is approximate



- A. Progesterone
- B. LH
- C. FSH
- D. Oxytocin

BIO-12-CH02-0011 | Medium | Gametogenesis

Complete the statement accurately: Cap-like structure over the sperm nucleus corresponds to _____.

- A. Amoeba
- B. Blastocyst
- C. Ampulla of the fallopian tube
- D. Acrosome

BIO-12-CH02-0012 | Medium | Birth control and reproductive health

A source table contains one accurate entry involving 'Surgical interruption of the vas deferens'. Which option preserves a correct biological association?

- A. Surgical interruption of the vas deferens - Vasectomy
- B. Process producing the three primary germ layers - Amoeba
- C. Binary fission - Spermatogonia
- D. Usual site of fertilization - Freshwater sponge

BIO-12-CH02-0013 | Medium | Male reproductive system

A source table contains one accurate entry involving 'Duct carrying sperm away from the epididymis'. Which option preserves a correct biological association?

- A. Process producing the three primary germ layers - Infundibulum
- B. Duct carrying sperm away from the epididymis - Vas deferens
- C. Hormone chiefly maintaining the secretory endometrium - Vas deferens
- D. Transfer of an embryo with more than eight blastomeres into the uterus - Ovary

BIO-12-CH02-0014 | Medium | Early development

A learner encounters the observation 'Rapid mitotic divisions of the zygote without overall growth' while analysing early development. Which conclusion is supported by the prescribed biology?

- A. Hydra
- B. FSH
- C. Blastocyst
- D. Cleavage

BIO-12-CH02-0015 | Easy | Female reproductive system

Which biological term or result is specifically identified by the description 'Uterine layer receiving the implanting blastocyst'?

- A. Endometrium
- B. Corpus luteum
- C. Infundibulum
- D. Ovary

BIO-12-CH02-0016 | Easy | Female reproductive system

Which description is correctly represented by Endometrium in this chapter?

- A. Uterine layer receiving the implanting blastocyst
- B. Immediate endocrine trigger for ovulation
- C. Regeneration producing complete individuals
- D. Bacterial sexually transmitted infection caused by Neisseria

BIO-12-CH02-0017 | Medium | Male reproductive system

A source table contains one accurate entry involving 'Cells secreting testicular androgens'. Which option preserves a correct biological association?

- A. Budding - Spermiogenesis
- B. Transformation of spermatids into spermatozoa - Infundibulum
- C. Cells secreting testicular androgens - Leydig cells
- D. Surgical interruption of the fallopian tubes - LH surge

BIO-12-CH02-0018 | Easy | Birth control and reproductive health

Which biological term or result is specifically identified by the description 'Copper-releasing device placed in the uterus'?

- A. Seminiferous tubules
- B. Primary spermatocyte
- C. Copper-T
- D. Condom

BIO-12-CH02-0019 | Easy | Male reproductive system

Which biological term or result is specifically identified by the description 'Cells secreting testicular androgens'?

- A. Leydig cells
- B. IUT
- C. Hydra
- D. Vas deferens

BIO-12-CH02-0020 | Medium | Male reproductive system

A learner encounters the observation 'Cells nourishing developing germ cells' while analysing male reproductive system. Which conclusion is supported by the prescribed biology?

- A. Sertoli cells
- B. ZIFT
- C. Gastrulation
- D. Vas deferens

BIO-12-CH02-0021 | Easy | Gametogenesis

Which biological term or result is specifically identified by the description 'Cell completing meiosis I to form secondary spermatocytes'?

- A. Seminiferous tubules
- B. Primary spermatocyte
- C. Gastrulation
- D. Spermiogenesis

BIO-12-CH02-0022 | Hard | Assisted reproduction and infection

Two conclusions must both be valid to support the reference concept 'Viral syndrome caused by HIV'. Which option contains two accurate associations?

- A. Funnel-like oviduct part bearing fimbriae - Copper-T; Copper-releasing device placed in the uterus - Infundibulum
- B. Hormone chiefly maintaining the secretory endometrium - Infundibulum; Funnel-like oviduct part bearing fimbriae - Progesterone
- C. Site of sperm formation - Planaria; Regeneration producing complete individuals - Seminiferous tubules
- D. Viral syndrome caused by HIV - AIDS; Solid ball of about 8 to 16 blastomeres - Morula

BIO-12-CH02-0023 | Medium | Assisted reproduction and infection

A source table contains one accurate entry involving 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. Which option preserves a correct biological association?

- A. Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT
- B. Solid ball of about 8 to 16 blastomeres - Acrosome
- C. Transformation of spermatids into spermatozoa - Seminiferous tubules
- D. Site of sperm formation - Sertoli cells

BIO-12-CH02-0024 | Easy | Gametogenesis

Which description is correctly represented by Acrosome in this chapter?

- A. Cap-like structure over the sperm nucleus
- B. Process producing the three primary germ layers
- C. Binary fission
- D. Usual site of fertilization

BIO-12-CH02-0025 | Medium | Early development

A learner encounters the observation 'Embryonic stage with trophoblast and inner cell mass' while analysing early development. Which conclusion is supported by the prescribed biology?

- A. Cleavage
- B. IUT
- C. Tubectomy
- D. Blastocyst

BIO-12-CH02-0026 | Hard | Gametogenesis

Two conclusions must both be valid to support the reference concept 'Cap-like structure over the sperm nucleus'. Which option contains two accurate associations?

- A. Surgical interruption of the vas deferens - Vas deferens; Duct carrying sperm away from the epididymis - Vasectomy
- B. Cap-like structure over the sperm nucleus - Acrosome; Binary fission - Amoeba
- C. Immediate endocrine trigger for ovulation - Gastrulation; Process producing the three primary germ layers - LH surge
- D. Process producing the three primary germ layers - Blastocyst; Embryonic stage with trophoblast and inner cell mass - Gastrulation

BIO-12-CH02-0027 | Easy | Female reproductive system

Which description is correctly represented by Infundibulum in this chapter?

- A. Uterine layer receiving the implanting blastocyst
- B. Surgical interruption of the vas deferens
- C. Transfer of an embryo with more than eight blastomeres into the uterus
- D. Funnel-like oviduct part bearing fimbriae

BIO-12-CH02-0028 | Hard | Assisted reproduction and infection

Two conclusions must both be valid to support the reference concept 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. Which option contains two accurate associations?

- A. Solid ball of about 8 to 16 blastomeres - Seminiferous tubules; Site of sperm formation - Morula
- B. Binary fission - Spermiogenesis; Transformation of spermatids into spermatozoa - Amoeba
- C. Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT; Funnel-like oviduct part bearing fimbriae - Infundibulum
- D. Transformation of spermatids into spermatozoa - Morula; Solid ball of about 8 to 16 blastomeres - Spermiogenesis

BIO-12-CH02-0029 | Medium | Menstrual cycle

A learner encounters the observation 'Immediate endocrine trigger for ovulation' while analysing menstrual cycle. Which conclusion is supported by the prescribed biology?

- A. Vasectomy
- B. Endometrium
- C. Progesterone
- D. LH surge

BIO-12-CH02-0030 | Challenging | Assisted reproduction and infection

During error analysis, four proposed associations are reviewed.

- I. Transfer of an early embryo with up to eight blastomeres into the oviduct corresponds to ZIFT.
- II. Bacterial sexually transmitted infection caused by *Neisseria* corresponds to Gonorrhoea.
- III. Viral syndrome caused by HIV corresponds to AIDS.
- IV. Process producing the three primary germ layers corresponds to Gastrulation.

Which set contains all and only the valid conclusions?

- A. I, II, III and IV
- B. II, III and IV only
- C. I and IV only
- D. I, II and IV only

BIO-12-CH02-0031 | Hard | Menstrual cycle

A reviewer compares several entries while checking 'Pituitary hormone promoting follicular growth'. Which entry must be corrected?

- A. Surgical interruption of the fallopian tubes - Tubectomy
- B. Organ producing ova, estrogen and progesterone - Ovary
- C. Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT
- D. Pituitary hormone promoting follicular growth - Primary spermatocyte

BIO-12-CH02-0032 | Easy | Birth control and reproductive health

Which biological term or result is specifically identified by the description 'Surgical interruption of the vas deferens'?

- A. Condom
- B. Vasectomy
- C. Amoeba
- D. Gastrulation

BIO-12-CH02-0033 | Medium | Birth control and reproductive health

Complete the statement accurately: Barrier method that also lowers STI transmission corresponds to _____.

- A. Vasectomy
- B. Cleavage
- C. Condom
- D. Ampulla of the fallopian tube

BIO-12-CH02-0034 | Medium | Birth control and reproductive health

A learner encounters the observation 'Barrier method that also lowers STI transmission' while analysing birth control and reproductive health. Which conclusion is supported by the prescribed biology?

- A. Blastocyst
- B. Condom
- C. IUT
- D. Vasectomy

BIO-12-CH02-0035 | Hard | Male reproductive system

A reviewer compares several entries while checking 'Cells secreting testicular androgens'. Which entry must be corrected?

- A. Cells secreting testicular androgens - IUT
- B. Solid ball of about 8 to 16 blastomeres - Morula
- C. Surgical interruption of the fallopian tubes - Tubectomy
- D. Transformation of spermatids into spermatozoa - Spermiogenesis

BIO-12-CH02-0036 | Medium | Early development

A source table contains one accurate entry involving 'Rapid mitotic divisions of the zygote without overall growth'. Which option preserves a correct biological association?

- A. Cap-like structure over the sperm nucleus - FSH
- B. Rapid mitotic divisions of the zygote without overall growth - Cleavage
- C. Cell completing meiosis I to form secondary spermatocytes - Cleavage
- D. Pituitary hormone promoting follicular growth - Vas deferens

BIO-12-CH02-0037 | Easy | Assisted reproduction and infection

Which biological term or result is specifically identified by the description 'Transfer of an early embryo with up to eight blastomeres into the oviduct'?

- A. Seminiferous tubules
- B. Infundibulum
- C. AIDS
- D. ZIFT

BIO-12-CH02-0038 | Medium | Early development

A learner encounters the observation 'Solid ball of about 8 to 16 blastomeres' while analysing early development. Which conclusion is supported by the prescribed biology?

- A. ZIFT
- B. Morula
- C. Progesterone
- D. Cleavage

BIO-12-CH02-0039 | Hard | Female reproductive system

Two conclusions must both be valid to support the reference concept 'Uterine layer receiving the implanting blastocyst'. Which option contains two accurate associations?

- A. Bacterial sexually transmitted infection caused by Neisseria - Corpus luteum; Temporary ovarian body formed after ovulation - Gonorrhoea
- B. Uterine layer receiving the implanting blastocyst - Endometrium; Regeneration producing complete individuals - Planaria
- C. Immediate endocrine trigger for ovulation - Sertoli cells; Cells nourishing developing germ cells - LH surge
- D. Temporary ovarian body formed after ovulation - Leydig cells; Cells secreting testicular androgens - Corpus luteum

BIO-12-CH02-0040 | Medium | Gametogenesis

A source table contains one accurate entry involving 'Cap-like structure over the sperm nucleus'. Which option preserves a correct biological association?

- A. Embryonic stage with trophoblast and inner cell mass - Vas deferens
- B. Immediate endocrine trigger for ovulation - Blastocyst
- C. Cap-like structure over the sperm nucleus - Acrosome
- D. Usual site of fertilization - Gastrulation

BIO-12-CH02-0041 | Easy | Birth control and reproductive health

Which description is correctly represented by Copper-T in this chapter?

- A. Duct carrying sperm away from the epididymis
- B. Pituitary hormone promoting follicular growth
- C. Copper-releasing device placed in the uterus
- D. Surgical interruption of the fallopian tubes

BIO-12-CH02-0042 | Medium | Birth control and reproductive health

A learner encounters the observation 'Surgical interruption of the fallopian tubes' while analysing birth control and reproductive health. Which conclusion is supported by the prescribed biology?

- A. Condom
- B. Tubectomy
- C. Blastocyst
- D. Gonorrhoea

BIO-12-CH02-0043 | Medium | Asexual reproduction

A source table contains one accurate entry involving 'Budding'. Which option preserves a correct biological association?

- A. Surgical interruption of the fallopian tubes - Hydra
- B. Budding - Hydra
- C. Surgical interruption of the vas deferens - Gastrulation
- D. Uterine layer receiving the implanting blastocyst - Vasectomy

BIO-12-CH02-0044 | Easy | Asexual reproduction

Which description is correctly represented by Hydra in this chapter?

- A. Budding
- B. Gemmule formation
- C. Binary fission
- D. Process producing the three primary germ layers

BIO-12-CH02-0045 | Hard | Male reproductive system

Two conclusions must both be valid to support the reference concept 'Duct carrying sperm away from the epididymis'. Which option contains two accurate associations?

- A. Hormone chiefly maintaining the secretory endometrium - Infundibulum; Funnel-like oviduct part bearing fimbriae - Progesterone
- B. Duct carrying sperm away from the epididymis - Vas deferens; Transfer of an embryo with more than eight blastomeres into the uterus - IUT
- C. Viral syndrome caused by HIV - Gastrulation; Process producing the three primary germ layers - AIDS
- D. Process producing the three primary germ layers - Progesterone; Hormone chiefly maintaining the secretory endometrium - Gastrulation

BIO-12-CH02-0046 | Challenging | Gametogenesis

A researcher checks whether four linked observations have been interpreted correctly.

- I. Transformation of spermatids into spermatozoa corresponds to Spermiogenesis.
- II. Diploid male germ cells multiplying by mitosis corresponds to Primary spermatocyte.
- III. Cell completing meiosis I to form secondary spermatocytes corresponds to Tubectomy.
- IV. Surgical interruption of the fallopian tubes corresponds to Tubectomy.

Which combination survives the biological review?

- A. I, II and III only
- B. I and IV only
- C. I only
- D. IV only

BIO-12-CH02-0047 | Challenging | Birth control and reproductive health

Evaluate the following source-based biological statements.

- I. Barrier method that also lowers STI transmission corresponds to Vasectomy.
- II. Surgical interruption of the vas deferens corresponds to Vasectomy.
- III. Copper-releasing device placed in the uterus corresponds to Copper-T.
- IV. Viral syndrome caused by HIV corresponds to AIDS.

Which option identifies exactly the correct statements?

- A. II only
- B. I and II only
- C. I and III only
- D. II, III and IV only

BIO-12-CH02-0048 | Hard | Female reproductive system

Two conclusions must both be valid to support the reference concept 'Funnel-like oviduct part bearing fimbriae'.

Which option contains two accurate associations?

- A. Duct carrying sperm away from the epididymis - Amoeba; Binary fission - Vas deferens
- B. Binary fission - IUT; Transfer of an embryo with more than eight blastomeres into the uterus - Amoeba
- C. Funnel-like oviduct part bearing fimbriae - Infundibulum; Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea
- D. Cap-like structure over the sperm nucleus - ZIFT; Transfer of an early embryo with up to eight blastomeres into the oviduct - Acrosome

BIO-12-CH02-0049 | Easy | Menstrual cycle

Which biological term or result is specifically identified by the description 'Immediate endocrine trigger for ovulation'?

- A. Ovary
- B. Seminiferous tubules
- C. FSH
- D. LH surge

BIO-12-CH02-0050 | Hard | Asexual reproduction

Two conclusions must both be valid to support the reference concept 'Budding'. Which option contains two accurate associations?

- A. Surgical interruption of the vas deferens - Sertoli cells; Cells nourishing developing germ cells - Vasectomy
- B. Rapid mitotic divisions of the zygote without overall growth - Vasectomy; Surgical interruption of the vas deferens - Cleavage
- C. Cells nourishing developing germ cells - Gastrulation; Process producing the three primary germ layers - Sertoli cells
- D. Budding - Hydra; Uterine layer receiving the implanting blastocyst - Endometrium

BIO-12-CH02-0051 | Medium | Female reproductive system

A learner encounters the observation 'Funnel-like oviduct part bearing fimbriae' while analysing female reproductive system. Which conclusion is supported by the prescribed biology?

- A. Vas deferens
- B. Infundibulum
- C. Vasectomy
- D. IUT

BIO-12-CH02-0052 | Hard | Gametogenesis

A reviewer compares several entries while checking 'Cap-like structure over the sperm nucleus'. Which entry must be corrected?

- A. Cap-like structure over the sperm nucleus - Amoeba
- B. Process producing the three primary germ layers - Gastrulation
- C. Immediate endocrine trigger for ovulation - LH surge
- D. Embryonic stage with trophoblast and inner cell mass - Blastocyst

BIO-12-CH02-0053 | Hard | Female reproductive system

A reviewer compares several entries while checking 'Uterine layer receiving the implanting blastocyst'. Which entry must be corrected?

- A. Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea
- B. Uterine layer receiving the implanting blastocyst - Planaria
- C. Immediate endocrine trigger for ovulation - LH surge
- D. Cells secreting testicular androgens - Leydig cells

BIO-12-CH02-0054 | Challenging | Menstrual cycle

A student submits four conclusions after integrating related chapter concepts.

- I. Hormone chiefly maintaining the secretory endometrium corresponds to Progesterone.
- II. Immediate endocrine trigger for ovulation corresponds to Corpus luteum.
- III. Temporary ovarian body formed after ovulation corresponds to Corpus luteum.
- IV. Budding corresponds to Progesterone.

Which combination should be accepted without correction?

- A. I and III only
- B. I only
- C. II only
- D. II and III only

BIO-12-CH02-0055 | Hard | Asexual reproduction

Two conclusions must both be valid to support the reference concept 'Regeneration producing complete individuals'. Which option contains two accurate associations?

- A. Embryonic stage with trophoblast and inner cell mass - Acrosome; Cap-like structure over the sperm nucleus - Blastocyst
- B. Regeneration producing complete individuals - Planaria; Copper-releasing device placed in the uterus - Copper-T
- C. Transfer of an embryo with more than eight blastomeres into the uterus - Sertoli cells; Cells nourishing developing germ cells - IUT
- D. Cap-like structure over the sperm nucleus - IUT; Transfer of an embryo with more than eight blastomeres into the uterus - Acrosome

BIO-12-CH02-0056 | Medium | Menstrual cycle

Complete the statement accurately: Pituitary hormone promoting follicular growth corresponds to ____.

- A. Tubectomy
- B. Ovary
- C. Primary spermatocyte
- D. FSH

BIO-12-CH02-0057 | Easy | Gametogenesis

Which description is correctly represented by Spermatogonia in this chapter?

- A. Diploid male germ cells multiplying by mitosis
- B. Duct carrying sperm away from the epididymis
- C. Site of sperm formation
- D. Uterine layer receiving the implanting blastocyst

BIO-12-CH02-0058 | Medium | Asexual reproduction

A source table contains one accurate entry involving 'Binary fission'. Which option preserves a correct biological association?

- A. Binary fission - Amoeba
- B. Hormone chiefly maintaining the secretory endometrium - Endometrium
- C. Bacterial sexually transmitted infection caused by Neisseria - Vas deferens
- D. Pituitary hormone promoting follicular growth - AIDS

BIO-12-CH02-0059 | Hard | Gametogenesis

Two conclusions must both be valid to support the reference concept 'Diploid male germ cells multiplying by mitosis'. Which option contains two accurate associations?

- A. Uterine layer receiving the implanting blastocyst - Hydra; Budding - Endometrium
- B. Pituitary hormone promoting follicular growth - Endometrium; Uterine layer receiving the implanting blastocyst - FSH
- C. Diploid male germ cells multiplying by mitosis - Spermatogonia; Duct carrying sperm away from the epididymis - Vas deferens
- D. Site of sperm formation - FSH; Pituitary hormone promoting follicular growth - Seminiferous tubules

BIO-12-CH02-0060 | Easy | Menstrual cycle

Which description is correctly represented by Corpus luteum in this chapter?

- A. Transfer of an embryo with more than eight blastomeres into the uterus
- B. Duct carrying sperm away from the epididymis
- C. Immediate endocrine trigger for ovulation
- D. Temporary ovarian body formed after ovulation

BIO-12-CH02-0061 | Medium | Asexual reproduction

A learner encounters the observation 'Regeneration producing complete individuals' while analysing asexual reproduction. Which conclusion is supported by the prescribed biology?

- A. Leydig cells
- B. Planaria
- C. Copper-T
- D. ZIFT

BIO-12-CH02-0062 | Easy | Asexual reproduction

Which biological term or result is specifically identified by the description 'Gemmule formation'?

- A. Tubectomy
- B. Copper-T
- C. Freshwater sponge
- D. Amoeba

BIO-12-CH02-0063 | Challenging | Early development

During error analysis, four proposed associations are reviewed.

- I. Process producing the three primary germ layers corresponds to Gastrulation.
- II. Rapid mitotic divisions of the zygote without overall growth corresponds to Morula.
- III. Solid ball of about 8 to 16 blastomeres corresponds to Morula.
- IV. Binary fission corresponds to Amoeba.

Which set contains all and only the valid conclusions?

- A. I, II and III only
- B. III only
- C. I and II only
- D. I, III and IV only

BIO-12-CH02-0064 | Hard | Early development

Two conclusions must both be valid to support the reference concept 'Solid ball of about 8 to 16 blastomeres'. Which option contains two accurate associations?

- A. Cells nourishing developing germ cells - Ovary; Organ producing ova, estrogen and progesterone - Sertoli cells
- B. Transfer of an early embryo with up to eight blastomeres into the oviduct - Progesterone; Hormone chiefly maintaining the secretory endometrium - ZIFT
- C. Hormone chiefly maintaining the secretory endometrium - Seminiferous tubules; Site of sperm formation - Progesterone
- D. Solid ball of about 8 to 16 blastomeres - Morula; Diploid male germ cells multiplying by mitosis - Spermatogonia

BIO-12-CH02-0065 | Easy | Assisted reproduction and infection

Which description is correctly represented by ZIFT in this chapter?

- A. Transfer of an early embryo with up to eight blastomeres into the oviduct
- B. Binary fission
- C. Site of sperm formation
- D. Viral syndrome caused by HIV

BIO-12-CH02-0066 | Easy | Female reproductive system

Which description is correctly represented by Ampulla of the fallopian tube in this chapter?

- A. Cells nourishing developing germ cells
- B. Uterine layer receiving the implanting blastocyst
- C. Diploid male germ cells multiplying by mitosis
- D. Usual site of fertilization

BIO-12-CH02-0067 | Hard | Female reproductive system

Two conclusions must both be valid to support the reference concept 'Usual site of fertilization'. Which option contains two accurate associations?

- A. Usual site of fertilization - Ampulla of the fallopian tube; Cells nourishing developing germ cells - Sertoli cells
- B. Diploid male germ cells multiplying by mitosis - Freshwater sponge; Gemmule formation - Spermatogonia
- C. Gemmule formation - Vasectomy; Surgical interruption of the vas deferens - Freshwater sponge
- D. Surgical interruption of the vas deferens - Seminiferous tubules; Site of sperm formation - Vasectomy

BIO-12-CH02-0068 | Hard | Early development

A reviewer compares several entries while checking 'Embryonic stage with trophoblast and inner cell mass'. Which entry must be corrected?

- A. Duct carrying sperm away from the epididymis - Vas deferens
- B. Transfer of an embryo with more than eight blastomeres into the uterus - IUT
- C. Surgical interruption of the fallopian tubes - Tubectomy
- D. Embryonic stage with trophoblast and inner cell mass - Spermiogenesis

BIO-12-CH02-0069 | Hard | Menstrual cycle

Two conclusions must both be valid to support the reference concept 'Temporary ovarian body formed after ovulation'. Which option contains two accurate associations?

- A. Cell completing meiosis I to form secondary spermatocytes - Copper-T; Copper-releasing device placed in the uterus - Primary spermatocyte
- B. Temporary ovarian body formed after ovulation - Corpus luteum; Surgical interruption of the vas deferens - Vasectomy
- C. Duct carrying sperm away from the epididymis - Leydig cells; Cells secreting testicular androgens - Vas deferens
- D. Process producing the three primary germ layers - Vas deferens; Duct carrying sperm away from the epididymis - Gastrulation

BIO-12-CH02-0070 | Medium | Early development

A source table contains one accurate entry involving 'Embryonic stage with trophoblast and inner cell mass'. Which option preserves a correct biological association?

- A. Transfer of an embryo with more than eight blastomeres into the uterus - Tubectomy
- B. Duct carrying sperm away from the epididymis - Blastocyst
- C. Embryonic stage with trophoblast and inner cell mass - Blastocyst
- D. Transformation of spermatids into spermatozoa - IUT

BIO-12-CH02-0071 | Hard | Assisted reproduction and infection

A reviewer compares several entries while checking 'Transfer of an early embryo with up to eight blastomeres into the oviduct'. Which entry must be corrected?

- A. Site of sperm formation - Seminiferous tubules
- B. Binary fission - Amoeba
- C. Transformation of spermatids into spermatozoa - Spermiogenesis
- D. Transfer of an early embryo with up to eight blastomeres into the oviduct - Infundibulum

BIO-12-CH02-0072 | Medium | Assisted reproduction and infection

A source table contains one accurate entry involving 'Viral syndrome caused by HIV'. Which option preserves a correct biological association?

- A. Solid ball of about 8 to 16 blastomeres - Planaria
- B. Copper-releasing device placed in the uterus - AIDS
- C. Viral syndrome caused by HIV - AIDS
- D. Funnel-like oviduct part bearing fimbriae - Sertoli cells

BIO-12-CH02-0073 | Medium | Asexual reproduction

A source table contains one accurate entry involving 'Regeneration producing complete individuals'. Which option preserves a correct biological association?

- A. Copper-releasing device placed in the uterus - ZIFT
- B. Regeneration producing complete individuals - Planaria
- C. Cells secreting testicular androgens - Blastocyst
- D. Cap-like structure over the sperm nucleus - Sertoli cells

BIO-12-CH02-0074 | Easy | Early development

Which description is correctly represented by Gastrulation in this chapter?

- A. Usual site of fertilization
- B. Rapid mitotic divisions of the zygote without overall growth
- C. Solid ball of about 8 to 16 blastomeres
- D. Process producing the three primary germ layers

BIO-12-CH02-0075 | Medium | Assisted reproduction and infection

Complete the statement accurately: Transfer of an early embryo with up to eight blastomeres into the oviduct corresponds to _____.

- A. AIDS
- B. Infundibulum
- C. ZIFT
- D. Spermiogenesis

BIO-12-CH02-0076 | Easy | Male reproductive system

Which description is correctly represented by Vas deferens in this chapter?

- A. Transfer of an embryo with more than eight blastomeres into the uterus
- B. Process producing the three primary germ layers
- C. Duct carrying sperm away from the epididymis
- D. Organ producing ova, estrogen and progesterone

BIO-12-CH02-0077 | Hard | Assisted reproduction and infection

Two conclusions must both be valid to support the reference concept 'Bacterial sexually transmitted infection caused by Neisseria'. Which option contains two accurate associations?

- A. Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea; Regeneration producing complete individuals - Planaria
- B. Uterine layer receiving the implanting blastocyst - Condom; Barrier method that also lowers STI transmission - Endometrium
- C. Funnel-like oviduct part bearing fimbriae - Spermiogenesis; Transformation of spermatids into spermatozoa - Infundibulum
- D. Diploid male germ cells multiplying by mitosis - Copper-T; Copper-releasing device placed in the uterus - Spermatogonia

BIO-12-CH02-0078 | Medium | Gametogenesis

A source table contains one accurate entry involving 'Diploid male germ cells multiplying by mitosis'. Which option preserves a correct biological association?

- A. Duct carrying sperm away from the epididymis - FSH
- B. Uterine layer receiving the implanting blastocyst - AIDS
- C. Diploid male germ cells multiplying by mitosis - Spermatogonia
- D. Pituitary hormone promoting follicular growth - Hydra

BIO-12-CH02-0079 | Easy | Early development

Which biological term or result is specifically identified by the description 'Embryonic stage with trophoblast and inner cell mass'?

- A. Spermiogenesis
- B. Blastocyst
- C. LH surge
- D. Morula

BIO-12-CH02-0080 | Medium | Gametogenesis

A learner encounters the observation 'Cap-like structure over the sperm nucleus' while analysing gametogenesis. Which conclusion is supported by the prescribed biology?

- A. Blastocyst
- B. Spermiogenesis
- C. Ampulla of the fallopian tube
- D. Acrosome

BIO-12-CH02-0081 | Challenging | Early development

A researcher checks whether four linked observations have been interpreted correctly.

- I. Embryonic stage with trophoblast and inner cell mass corresponds to Blastocyst.
- II. Process producing the three primary germ layers corresponds to Gastrulation.
- III. Rapid mitotic divisions of the zygote without overall growth corresponds to AIDS.
- IV. Viral syndrome caused by HIV corresponds to Blastocyst.

Which combination survives the biological review?

- A. II and IV only
- B. I, II and III only
- C. I, II, III and IV
- D. I and II only

BIO-12-CH02-0082 | Medium | Early development

Complete the statement accurately: Solid ball of about 8 to 16 blastomeres corresponds to _____.

- A. Cleavage
- B. Spermatogonia
- C. Morula
- D. Blastocyst

BIO-12-CH02-0083 | Easy | Male reproductive system

Which biological term or result is specifically identified by the description 'Duct carrying sperm away from the epididymis'?

- A. Leydig cells
- B. IUT
- C. Acrosome
- D. Vas deferens

BIO-12-CH02-0084 | Medium | Menstrual cycle

Complete the statement accurately: Hormone chiefly maintaining the secretory endometrium corresponds to _____.

- A. Spermiogenesis
- B. Progesterone
- C. Freshwater sponge
- D. Blastocyst

BIO-12-CH02-0085 | Medium | Male reproductive system

Complete the statement accurately: Cells nourishing developing germ cells corresponds to _____.

- A. Sertoli cells
- B. Spermiogenesis
- C. ZIFT
- D. Spermatogonia

BIO-12-CH02-0086 | Medium | Assisted reproduction and infection

A learner encounters the observation 'Viral syndrome caused by HIV' while analysing assisted reproduction and infection. Which conclusion is supported by the prescribed biology?

- A. Morula
- B. Gonorrhoea
- C. AIDS
- D. IUT

BIO-12-CH02-0087 | Hard | Male reproductive system

A reviewer compares several entries while checking 'Duct carrying sperm away from the epididymis'. Which entry must be corrected?

- A. Cap-like structure over the sperm nucleus - Acrosome
- B. Process producing the three primary germ layers - Gastrulation
- C. Organ producing ova, estrogen and progesterone - Ovary
- D. Duct carrying sperm away from the epididymis - IUT

BIO-12-CH02-0088 | Medium | Gametogenesis

A source table contains one accurate entry involving 'Transformation of spermatids into spermatozoa'. Which option preserves a correct biological association?

- A. Temporary ovarian body formed after ovulation - Spermiogenesis
- B. Transformation of spermatids into spermatozoa - Spermiogenesis
- C. Cells nourishing developing germ cells - Planaria
- D. Viral syndrome caused by HIV - Vasectomy

BIO-12-CH02-0089 | Easy | Assisted reproduction and infection

Which description is correctly represented by Gonorrhoea in this chapter?

- A. Bacterial sexually transmitted infection caused by Neisseria
- B. Uterine layer receiving the implanting blastocyst
- C. Transformation of spermatids into spermatozoa
- D. Regeneration producing complete individuals

BIO-12-CH02-0090 | Medium | Assisted reproduction and infection

A source table contains one accurate entry involving 'Bacterial sexually transmitted infection caused by Neisseria'. Which option preserves a correct biological association?

- A. Barrier method that also lowers STI transmission - Copper-T
- B. Diploid male germ cells multiplying by mitosis - Gonorrhoea
- C. Bacterial sexually transmitted infection caused by Neisseria - Gonorrhoea
- D. Transformation of spermatids into spermatozoa - Condom

BIO-12-CH02-0091 | Medium | Female reproductive system

Complete the statement accurately: Uterine layer receiving the implanting blastocyst corresponds to _____.

- A. Endometrium
- B. Planaria
- C. LH surge
- D. Ovary

BIO-12-CH02-0092 | Medium | Female reproductive system

A learner encounters the observation 'Uterine layer receiving the implanting blastocyst' while analysing female reproductive system. Which conclusion is supported by the prescribed biology?

- A. Infundibulum
- B. LH surge
- C. Corpus luteum
- D. Endometrium

BIO-12-CH02-0093 | Hard | Menstrual cycle

A reviewer compares several entries while checking 'Hormone chiefly maintaining the secretory endometrium'. Which entry must be corrected?

- A. Surgical interruption of the vas deferens - Vasectomy
- B. Hormone chiefly maintaining the secretory endometrium - Spermiogenesis
- C. Embryonic stage with trophoblast and inner cell mass - Blastocyst
- D. Organ producing ova, estrogen and progesterone - Ovary

BIO-12-CH02-0094 | Hard | Birth control and reproductive health

Two conclusions must both be valid to support the reference concept 'Copper-releasing device placed in the uterus'. Which option contains two accurate associations?

- A. Cell completing meiosis I to form secondary spermatocytes - Spermatogonia; Diploid male germ cells multiplying by mitosis - Primary spermatocyte
- B. Diploid male germ cells multiplying by mitosis - Endometrium; Uterine layer receiving the implanting blastocyst - Spermatogonia
- C. Copper-releasing device placed in the uterus - Copper-T; Pituitary hormone promoting follicular growth - FSH
- D. Site of sperm formation - Vas deferens; Duct carrying sperm away from the epididymis - Seminiferous tubules

BIO-12-CH02-0095 | Challenging | Asexual reproduction

Evaluate the following source-based biological statements.

- I. Binary fission corresponds to Amoeba.
- II. Gemmule formation corresponds to Freshwater sponge.
- III. Budding corresponds to Condom.
- IV. Barrier method that also lowers STI transmission corresponds to Condom.

Which option identifies exactly the correct statements?

- A. I, II and IV only
- B. I, II and III only
- C. III and IV only
- D. IV only

BIO-12-CH02-0096 | Hard | Early development

Two conclusions must both be valid to support the reference concept 'Process producing the three primary germ layers'. Which option contains two accurate associations?

- A. Regeneration producing complete individuals - FSH; Pituitary hormone promoting follicular growth - Planaria
- B. Cells secreting testicular androgens - Vas deferens; Duct carrying sperm away from the epididymis - Leydig cells
- C. Pituitary hormone promoting follicular growth - Leydig cells; Cells secreting testicular androgens - FSH
- D. Process producing the three primary germ layers - Gastrulation; Usual site of fertilization - Ampulla of the fallopian tube

BIO-12-CH02-0097 | Hard | Early development

Two conclusions must both be valid to support the reference concept 'Rapid mitotic divisions of the zygote without overall growth'. Which option contains two accurate associations?

- A. Duct carrying sperm away from the epididymis - Primary spermatocyte; Cell completing meiosis I to form secondary spermatocytes - Vas deferens
- B. Cell completing meiosis I to form secondary spermatocytes - Freshwater sponge; Gemmule formation - Primary spermatocyte
- C. Rapid mitotic divisions of the zygote without overall growth - Cleavage; Cap-like structure over the sperm nucleus - Acrosome
- D. Pituitary hormone promoting follicular growth - Hydra; Budding - FSH

BIO-12-CH02-0098 | Easy | Menstrual cycle

Which biological term or result is specifically identified by the description 'Hormone chiefly maintaining the secretory endometrium'?

- A. Freshwater sponge
- B. Progesterone
- C. FSH
- D. Blastocyst

BIO-12-CH02-0099 | Hard | Asexual reproduction

A reviewer compares several entries while checking 'Gemmule formation'. Which entry must be corrected?

- A. Rapid mitotic divisions of the zygote without overall growth - Cleavage
- B. Gemmule formation - Gonorrhoea
- C. Copper-releasing device placed in the uterus - Copper-T
- D. Usual site of fertilization - Ampulla of the fallopian tube

BIO-12-CH02-0100 | Easy | Early development

Which description is correctly represented by Cleavage in this chapter?

- A. Duct carrying sperm away from the epididymis
- B. Solid ball of about 8 to 16 blastomeres
- C. Pituitary hormone promoting follicular growth
- D. Rapid mitotic divisions of the zygote without overall growth

BIO-12-CH02-0101 | Hard | Birth control and reproductive health

A reviewer compares several entries while checking 'Copper-releasing device placed in the uterus'. Which entry must be corrected?

- A. Copper-releasing device placed in the uterus - FSH
- B. Duct carrying sperm away from the epididymis - Vas deferens
- C. Diploid male germ cells multiplying by mitosis - Spermatogonia
- D. Site of sperm formation - Seminiferous tubules

BIO-12-CH02-0102 | Easy | Asexual reproduction

Which description is correctly represented by Freshwater sponge in this chapter?

- A. Gemmule formation
- B. Immediate endocrine trigger for ovulation
- C. Surgical interruption of the fallopian tubes
- D. Copper-releasing device placed in the uterus

BIO-12-CH02-0103 | Hard | Gametogenesis

A reviewer compares several entries while checking 'Cell completing meiosis I to form secondary spermatocytes'. Which entry must be corrected?

- A. Process producing the three primary germ layers - Gastrulation
- B. Budding - Hydra
- C. Cell completing meiosis I to form secondary spermatocytes - Infundibulum
- D. Site of sperm formation - Seminiferous tubules

BIO-12-CH02-0104 | Medium | Birth control and reproductive health

Complete the statement accurately: Surgical interruption of the fallopian tubes corresponds to _____.

- A. Tubectomy
- B. Freshwater sponge
- C. Vasectomy
- D. Acrosome

BIO-12-CH02-0105 | Easy | Assisted reproduction and infection

Which description is correctly represented by AIDS in this chapter?

- A. Viral syndrome caused by HIV
- B. Transfer of an embryo with more than eight blastomeres into the uterus
- C. Regeneration producing complete individuals
- D. Site of sperm formation

BIO-12-CH02-0106 | Medium | Female reproductive system

A learner encounters the observation 'Usual site of fertilization' while analysing female reproductive system. Which conclusion is supported by the prescribed biology?

- A. Endometrium
- B. Ampulla of the fallopian tube
- C. Freshwater sponge
- D. Planaria

BIO-12-CH02-0107 | Medium | Assisted reproduction and infection

A source table contains one accurate entry involving 'Transfer of an embryo with more than eight blastomeres into the uterus'. Which option preserves a correct biological association?

- A. Organ producing ova, estrogen and progesterone - IUT
- B. Transformation of spermatids into spermatozoa - Amoeba
- C. Transfer of an embryo with more than eight blastomeres into the uterus - IUT
- D. Temporary ovarian body formed after ovulation - Spermiogenesis

BIO-12-CH02-0108 | Medium | Menstrual cycle

A learner encounters the observation 'Temporary ovarian body formed after ovulation' while analysing menstrual cycle. Which conclusion is supported by the prescribed biology?

- A. Leydig cells
- B. Progesterone
- C. Corpus luteum
- D. LH surge

BIO-12-CH02-0109 | Challenging | Male reproductive system

A student submits four conclusions after integrating related chapter concepts.

- I. Duct carrying sperm away from the epididymis corresponds to Sertoli cells.
- II. Cells nourishing developing germ cells corresponds to Leydig cells.
- III. Cells secreting testicular androgens corresponds to FSH.
- IV. Pituitary hormone promoting follicular growth corresponds to FSH.

Which combination should be accepted without correction?

- A. II, III and IV only
- B. IV only
- C. II and III only
- D. I and IV only

BIO-12-CH02-0110 | Challenging | Assisted reproduction and infection

During error analysis, four proposed associations are reviewed.

- I. Bacterial sexually transmitted infection caused by Neisseria corresponds to Gonorrhoea.
- II. Transfer of an early embryo with up to eight blastomeres into the oviduct corresponds to ZIFT.
- III. Viral syndrome caused by HIV corresponds to AIDS.
- IV. Pituitary hormone promoting follicular growth corresponds to Gonorrhoea.

Which set contains all and only the valid conclusions?

- A. II only
- B. I, III and IV only
- C. I, II and III only
- D. I and III only

BIO-12-CH02-0111 | Medium | Male reproductive system

A source table contains one accurate entry involving 'Site of sperm formation'. Which option preserves a correct biological association?

- A. Uterine layer receiving the implanting blastocyst - Tubectomy
- B. Organ producing ova, estrogen and progesterone - LH surge
- C. Hormone chiefly maintaining the secretory endometrium - Spermatogonia
- D. Site of sperm formation - Seminiferous tubules

BIO-12-CH02-0112 | Medium | Menstrual cycle

A source table contains one accurate entry involving 'Immediate endocrine trigger for ovulation'. Which option preserves a correct biological association?

- A. Immediate endocrine trigger for ovulation - LH surge
- B. Transfer of an early embryo with up to eight blastomeres into the oviduct - AIDS
- C. Uterine layer receiving the implanting blastocyst - Vasectomy
- D. Organ producing ova, estrogen and progesterone - ZIFT

BIO-12-CH02-0113 | Medium | Gametogenesis

A learner encounters the observation 'Transformation of spermatids into spermatozoa' while analysing gametogenesis. Which conclusion is supported by the prescribed biology?

- A. AIDS
- B. Primary spermatocyte
- C. Vasectomy
- D. Spermiogenesis

BIO-12-CH02-0114 | Easy | Menstrual cycle

Which biological term or result is specifically identified by the description 'Pituitary hormone promoting follicular growth'?

- A. Tubectomy
- B. ZIFT
- C. FSH
- D. Blastocyst

BIO-12-CH02-0115 | Challenging | Female reproductive system

A researcher checks whether four linked observations have been interpreted correctly.

- I. Funnel-like oviduct part bearing fimbriae corresponds to Ampulla of the fallopian tube.
- II. Usual site of fertilization corresponds to Endometrium.
- III. Uterine layer receiving the implanting blastocyst corresponds to Endometrium.
- IV. Temporary ovarian body formed after ovulation corresponds to Infundibulum.

Which combination survives the biological review?

- A. I, II and III only
- B. III only
- C. I and III only
- D. III and IV only

BIO-12-CH02-0116 | Hard | Birth control and reproductive health

A reviewer compares several entries while checking 'Surgical interruption of the vas deferens'. Which entry must be corrected?

- A. Funnel-like oviduct part bearing fimbriae - Infundibulum
- B. Surgical interruption of the vas deferens - Cleavage
- C. Process producing the three primary germ layers - Gastrulation
- D. Usual site of fertilization - Ampulla of the fallopian tube

BIO-12-CH02-0117 | Easy | Early development

Which description is correctly represented by Morula in this chapter?

- A. Organ producing ova, estrogen and progesterone
- B. Hormone chiefly maintaining the secretory endometrium
- C. Solid ball of about 8 to 16 blastomeres
- D. Rapid mitotic divisions of the zygote without overall growth

BIO-12-CH02-0118 | Medium | Female reproductive system

Complete the statement accurately: Funnel-like oviduct part bearing fimbriae corresponds to ____.

- A. Infundibulum
- B. Amoeba
- C. Ampulla of the fallopian tube
- D. Gonorrhoea

BIO-12-CH02-0119 | Hard | Menstrual cycle

A reviewer compares several entries while checking 'Immediate endocrine trigger for ovulation'. Which entry must be corrected?

- A. Organ producing ova, estrogen and progesterone - Ovary
 - B. Transfer of an early embryo with up to eight blastomeres into the oviduct - ZIFT
 - C. Immediate endocrine trigger for ovulation - Seminiferous tubules
 - D. Surgical interruption of the vas deferens - Vasectomy
-

BIO-12-CH02-0120 | Easy | Gametogenesis

Which biological term or result is specifically identified by the description 'Cap-like structure over the sperm nucleus'?

- A. LH surge
 - B. Spermiogenesis
 - C. Acrosome
 - D. Amoeba
-

Standard 12 | Chapter 03

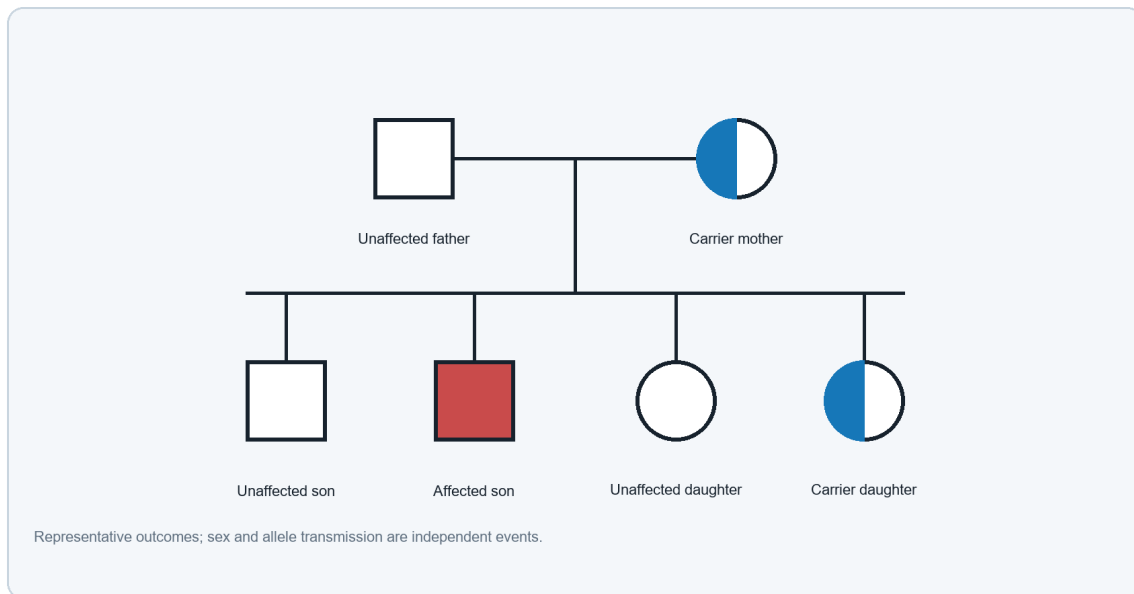
Inheritance and Variation

BIO-12-CH03-0001 | Medium | X-linked recessive inheritance

For the cross shown, what fraction of sons is expected to inherit the recessive affected allele from the carrier mother?

X-linked recessive inheritance

Carrier mother crossed with an unaffected father



- A. One-half of the sons
- B. All sons
- C. One-quarter of the sons
- D. No sons

BIO-12-CH03-0002 | Easy | Monohybrid cross: Tt x Tt

What genotypic ratio is obtained from the Tt x Tt Punnett square?

Monohybrid cross: Tt x Tt

Punnett square under complete dominance

		Parent 2		
		Gametes	T	t
Parent 1	T	TT	Tt	Genotype ratio 1 TT : 2 Tt : 1 tt Phenotype ratio 3 dominant : 1 recessive
	t	Tt	tt	

- A. 3 TT : 1 tt
- B. 1 TT : 2 Tt : 1 tt
- C. 1 Tt : 1 tt
- D. 9 : 3 : 3 : 1

BIO-12-CH03-0003 | Medium | Sex-linked inheritance

Complete the statement accurately: A male transmitting an X-linked allele to corresponds to _____.

- A. All daughters and no sons
- B. Law of segregation
- C. 3:1
- D. Carrier

BIO-12-CH03-0004 | Medium | Sex-linked inheritance

A learner encounters the observation 'A female heterozygous for a recessive X-linked allele' while analysing sex-linked inheritance. Which conclusion is supported by the prescribed biology?

- A. Test cross
- B. Carrier
- C. Colour blindness
- D. Linkage

BIO-12-CH03-0005 | Medium | Crosses

A source table contains one accurate entry involving 'Expected test-cross ratio for a heterozygous monohybrid'. Which option preserves a correct biological association?

- A. Genetic constitution of an organism - Male
- B. Separation of allele pairs during gamete formation - 1:1
- C. Expected test-cross ratio for a heterozygous monohybrid - 1:1
- D. XXY constitution in a male - Codominance

BIO-12-CH03-0006 | Hard | Monohybrid cross: Tt x Tt

Under complete dominance, what is the probability that an offspring in the square shows the recessive phenotype?

Monohybrid cross: Tt x Tt

Punnett square under complete dominance

		Parent 2		
		Gametes		
Parent 1	T	TT	Tt	
	t	Tt	tt	

Genotype ratio
1 TT : 2 Tt : 1 tt

Phenotype ratio
3 dominant : 1 recessive

- A. 1/2
- B. 3/4
- C. 1/4
- D. 1

BIO-12-CH03-0007 | Medium | Sex-linked inheritance

A source table contains one accurate entry involving 'A female heterozygous for a recessive X-linked allele'. Which option preserves a correct biological association?

- A. Genes on the same chromosome tending to be inherited together - Genotype
- B. Cross of an F1 individual with either parent - 1:1
- C. Cross with a homozygous recessive individual - 3:1
- D. A female heterozygous for a recessive X-linked allele - Carrier

BIO-12-CH03-0008 | Hard | Non-Mendelian patterns

Two conclusions must both be valid to support the reference concept 'Heterozygote showing an intermediate phenotype'. Which option contains two accurate associations?

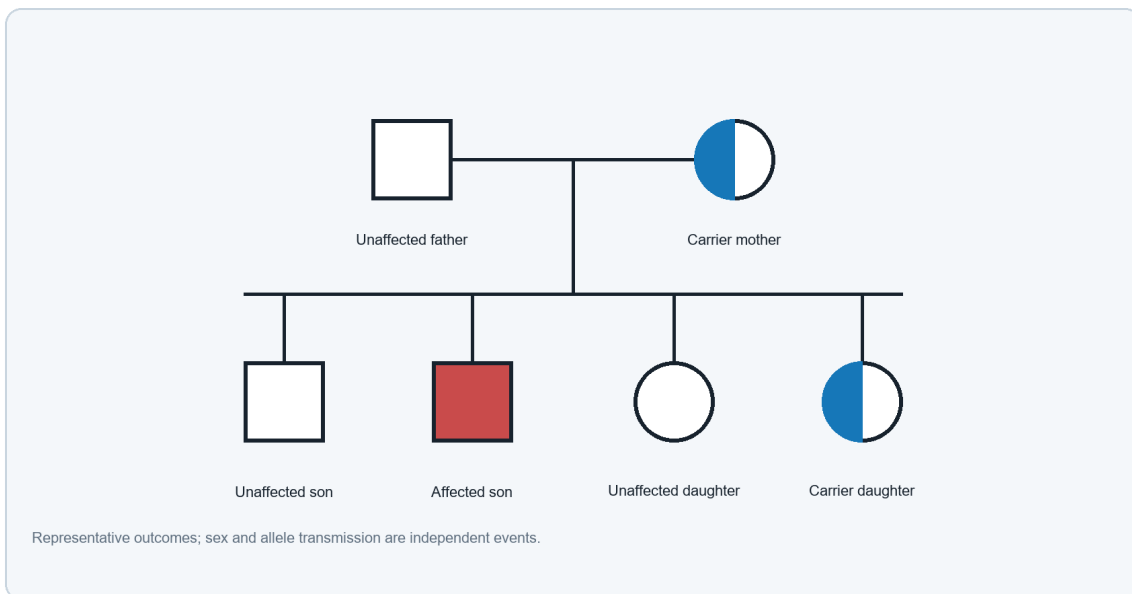
- A. Heterozygote showing an intermediate phenotype - Incomplete dominance; Genes on the same chromosome tending to be inherited together - Linkage
- B. Trisomy of chromosome 21 - 9:3:3:1; Dihybrid F2 phenotypic ratio under independent assortment - Down syndrome
- C. Dihybrid F2 phenotypic ratio under independent assortment - Haploid chromosome number; Number of linkage groups in a species corresponds to its - 9:3:3:1
- D. Observable expression of a character - Down syndrome; Trisomy of chromosome 21 - Phenotype

BIO-12-CH03-0009 | Challenging | X-linked recessive inheritance

Which combined prediction follows from the X-linked recessive cross shown?

X-linked recessive inheritance

Carrier mother crossed with an unaffected father



- A. Affected daughters occur but affected sons do not
- B. All daughters and all sons are affected
- C. Affected sons can occur, but affected daughters are not expected
- D. No child can inherit the recessive allele

BIO-12-CH03-0010 | Easy | Sex determination

Which description is correctly represented by XX in this chapter?

- A. Trisomy of chromosome 21
- B. Human example of multiple alleles cited in the chapter
- C. Heterogametic sex in humans
- D. Sex chromosome system in human females

BIO-12-CH03-0011 | Easy | Sex-linked inheritance

Which description is correctly represented by All daughters and no sons in this chapter?

- A. Monohybrid F2 phenotypic ratio under complete dominance
- B. Trisomy of chromosome 21
- C. A male transmitting an X-linked allele to
- D. A female heterozygous for a recessive X-linked allele

BIO-12-CH03-0012 | Easy | Mendelian inheritance

Which biological term or result is specifically identified by the description 'Monohybrid F2 phenotypic ratio under complete dominance'?

- A. 9:3:3:1
- B. ABO blood grouping
- C. 3:1
- D. Test cross

BIO-12-CH03-0013 | Hard | Chromosomal disorders

Two conclusions must both be valid to support the reference concept 'Monosomy X'. Which option contains two accurate associations?

- A. Monosomy X - Turner syndrome; Exchange between non-sister chromatids of homologous chromosomes - Crossing over
- B. Genetic constitution of an organism - Incomplete dominance; Heterozygote showing an intermediate phenotype - Genotype
- C. Monohybrid F2 phenotypic ratio under complete dominance - Genotype; Genetic constitution of an organism - 3:1
- D. Stage at which crossing over occurs - 3:1; Monohybrid F2 phenotypic ratio under complete dominance - Pachytene of prophase I

BIO-12-CH03-0014 | Easy | Crosses

Which description is correctly represented by 1:1:1:1 in this chapter?

- A. Observable expression of a character
- B. Expected dihybrid test-cross ratio with independent assortment
- C. Cross with a homozygous recessive individual
- D. Classic X-linked human trait involving red-green vision

BIO-12-CH03-0015 | Hard | Crosses

Two conclusions must both be valid to support the reference concept 'Expected dihybrid test-cross ratio with independent assortment'. Which option contains two accurate associations?

- A. Heterozygote showing an intermediate phenotype - Alleles; Alternative forms of the same gene - Incomplete dominance
- B. A female heterozygous for a recessive X-linked allele - 9:3:3:1; Dihybrid F2 phenotypic ratio under independent assortment - Carrier
- C. Dihybrid F2 phenotypic ratio under independent assortment - Phenotype; Observable expression of a character - 9:3:3:1
- D. Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1; Classic X-linked human trait involving red-green vision - Colour blindness

BIO-12-CH03-0016 | Easy | Mendelian inheritance

Which biological term or result is specifically identified by the description 'Dihybrid F2 phenotypic ratio under independent assortment'?

- A. Law of independent assortment
- B. X chromosome
- C. ZW female and ZZ male
- D. 9:3:3:1

BIO-12-CH03-0017 | Hard | Chromosomes linkage and crossing over

Two conclusions must both be valid to support the reference concept 'Exchange between non-sister chromatids of homologous chromosomes'. Which option contains two accurate associations?

- A. Heterozygote showing an intermediate phenotype - XX; Sex chromosome system in human females - Incomplete dominance
- B. Exchange between non-sister chromatids of homologous chromosomes - Crossing over; Classic X-linked human trait involving red-green vision - Colour blindness
- C. Loss or gain of one or a few chromosomes - Incomplete dominance; Heterozygote showing an intermediate phenotype - Aneuploidy
- D. One gene influencing several characters - 9:3:3:1; Dihybrid F2 phenotypic ratio under independent assortment - Pleiotropy

BIO-12-CH03-0018 | Medium | Mendelian inheritance

Complete the statement accurately: Separation of allele pairs during gamete formation corresponds to _____.

- A. 1:1:1:1
- B. X chromosome
- C. 3:1
- D. Law of segregation

BIO-12-CH03-0019 | Medium | Sex determination

A source table contains one accurate entry involving 'Sex determination in honey bees'. Which option preserves a correct biological association?

- A. Sex determination in honey bees - Haplo-diploid type
- B. Monosomy X - Aneuploidy
- C. Expected test-cross ratio for a heterozygous monohybrid - 1:1:1:1
- D. Classic X-linked human trait involving red-green vision - Turner syndrome

BIO-12-CH03-0020 | Medium | Sex determination

A learner encounters the observation 'Sex chromosome system in human females' while analysing sex determination. Which conclusion is supported by the prescribed biology?

- A. X chromosome
- B. XX
- C. ABO blood grouping
- D. Pachytene of prophase I

BIO-12-CH03-0021 | Medium | Genetic terminology

Complete the statement accurately: Observable expression of a character corresponds to _____.

- A. Phenotype
- B. Locus
- C. Crossing over
- D. Pachytene of prophase I

BIO-12-CH03-0022 | Easy | Genetic terminology

Which biological term or result is specifically identified by the description 'Alternative forms of the same gene'?

- A. Alleles
- B. Locus
- C. Genotype
- D. Crossing over

BIO-12-CH03-0023 | Hard | Sex-linked inheritance

Two conclusions must both be valid to support the reference concept 'Chromosome carrying the common haemophilia allele'. Which option contains two accurate associations?

- A. Genes on the same chromosome tending to be inherited together - Pleiotropy; One gene influencing several characters - Linkage
- B. Chromosome carrying the common haemophilia allele - X chromosome; Cross with a homozygous recessive individual - Test cross
- C. Exchange between non-sister chromatids of homologous chromosomes - Down syndrome; Trisomy of chromosome 21 - Crossing over
- D. Monosomy X - Linkage; Genes on the same chromosome tending to be inherited together - Turner syndrome

BIO-12-CH03-0024 | Hard | Non-Mendelian patterns

Two conclusions must both be valid to support the reference concept 'Human example of multiple alleles cited in the chapter'. Which option contains two accurate associations?

- A. Chromosome carrying the common haemophilia allele - Pachytene of prophase I; Stage at which crossing over occurs - X chromosome
- B. Human example of multiple alleles cited in the chapter - ABO blood grouping; Classic X-linked human trait involving red-green vision - Colour blindness
- C. Stage at which crossing over occurs - 1:1; Expected test-cross ratio for a heterozygous monohybrid - Pachytene of prophase I
- D. Monosomy X - Phenotype; Observable expression of a character - Turner syndrome

BIO-12-CH03-0025 | Challenging | Genetic terminology

A student submits four conclusions after integrating related chapter concepts.

- I. Position occupied by a gene on a chromosome corresponds to Locus.
 - II. Genetic constitution of an organism corresponds to Genotype.
 - III. Alternative forms of the same gene corresponds to Law of independent assortment.
 - IV. Independent behaviour of different allele pairs corresponds to Locus.
- Which combination should be accepted without correction?

- A. I and II only
- B. II and IV only
- C. I only
- D. III and IV only

BIO-12-CH03-0026 | Challenging | Sex determination

During error analysis, four proposed associations are reviewed.

- I. Sex chromosome system in human females corresponds to XX.
 - II. Sex determination in honey bees corresponds to Haplo-diploid type.
 - III. Heterogametic sex in humans corresponds to Back cross.
 - IV. Cross of an F1 individual with either parent corresponds to Back cross.
- Which set contains all and only the valid conclusions?

- A. I, II and IV only
- B. II only
- C. I and IV only
- D. IV only

BIO-12-CH03-0027 | Challenging | Chromosomes linkage and crossing over

A researcher checks whether four linked observations have been interpreted correctly.

- I. Number of linkage groups in a species corresponds to its corresponds to Linkage.
- II. Genes on the same chromosome tending to be inherited together corresponds to Crossing over.
- III. Exchange between non-sister chromatids of homologous chromosomes corresponds to Genotype.
- IV. Genetic constitution of an organism corresponds to Genotype.

Which combination survives the biological review?

- A. I, II and III only
- B. I and III only
- C. I, II, III and IV
- D. IV only

BIO-12-CH03-0028 | Hard | Sex determination

Two conclusions must both be valid to support the reference concept 'Sex chromosome system in human females'. Which option contains two accurate associations?

- A. Dihybrid F2 phenotypic ratio under independent assortment - Pachytene of prophase I; Stage at which crossing over occurs - 9:3:3:1
- B. Chromosome carrying the common haemophilia allele - ABO blood grouping; Human example of multiple alleles cited in the chapter - X chromosome
- C. Human example of multiple alleles cited in the chapter - 9:3:3:1; Dihybrid F2 phenotypic ratio under independent assortment - ABO blood grouping
- D. Sex chromosome system in human females - XX; Trisomy of chromosome 21 - Down syndrome

BIO-12-CH03-0029 | Hard | Sex-linked inheritance

A reviewer compares several entries while checking 'Chromosome carrying the common haemophilia allele'. Which entry must be corrected?

- A. Genes on the same chromosome tending to be inherited together - Linkage
- B. Trisomy of chromosome 21 - Down syndrome
- C. Chromosome carrying the common haemophilia allele - Test cross
- D. One gene influencing several characters - Pleiotropy

BIO-12-CH03-0030 | Easy | Chromosomal disorders

Which description is correctly represented by Turner syndrome in this chapter?

- A. XXY constitution in a male
- B. Monohybrid F2 phenotypic ratio under complete dominance
- C. Monosomy X
- D. Genetic constitution of an organism

BIO-12-CH03-0031 | Hard | Sex-linked inheritance

A reviewer compares several entries while checking 'A female heterozygous for a recessive X-linked allele'. Which entry must be corrected?

- A. A female heterozygous for a recessive X-linked allele - Back cross
- B. Monohybrid F2 phenotypic ratio under complete dominance - 3:1
- C. Expected test-cross ratio for a heterozygous monohybrid - 1:1
- D. Both alleles expressed in a heterozygote - Codominance

BIO-12-CH03-0032 | Easy | Non-Mendelian patterns

Which biological term or result is specifically identified by the description 'Human example of multiple alleles cited in the chapter'?

- A. Turner syndrome
- B. ABO blood grouping
- C. Pleiotropy
- D. Colour blindness

BIO-12-CH03-0033 | Hard | Mendelian inheritance

A reviewer compares several entries while checking 'Dihybrid F2 phenotypic ratio under independent assortment'. Which entry must be corrected?

- A. Expected test-cross ratio for a heterozygous monohybrid - 1:1
- B. Dihybrid F2 phenotypic ratio under independent assortment - Back cross
- C. Chromosome carrying the common haemophilia allele - X chromosome
- D. Heterogametic sex in humans - Male

BIO-12-CH03-0034 | Medium | Sex-linked inheritance

A source table contains one accurate entry involving 'A male transmitting an X-linked allele to'. Which option preserves a correct biological association?

- A. Observable expression of a character - Down syndrome
- B. Sex chromosome system in human females - All daughters and no sons
- C. A male transmitting an X-linked allele to - All daughters and no sons
- D. Trisomy of chromosome 21 - 3:1

BIO-12-CH03-0035 | Challenging | Sex-linked inheritance

Evaluate the following source-based biological statements.

- I. A female heterozygous for a recessive X-linked allele corresponds to Carrier.
- II. Chromosome carrying the common haemophilia allele corresponds to X chromosome.
- III. A male transmitting an X-linked allele to corresponds to All daughters and no sons.
- IV. Heterogametic sex in humans corresponds to Carrier.

Which option identifies exactly the correct statements?

- A. I and II only
- B. I, III and IV only
- C. II only
- D. I, II and III only

BIO-12-CH03-0036 | Medium | Chromosomes linkage and crossing over

A learner encounters the observation 'Exchange between non-sister chromatids of homologous chromosomes' while analysing chromosomes linkage and crossing over. Which conclusion is supported by the prescribed biology?

- A. Incomplete dominance
- B. Colour blindness
- C. Linkage
- D. Crossing over

BIO-12-CH03-0037 | Medium | Crosses

Complete the statement accurately: Cross with a homozygous recessive individual corresponds to ____.

- A. Male
- B. Test cross
- C. Back cross
- D. Linkage

BIO-12-CH03-0038 | Medium | Chromosomal disorders

Complete the statement accurately: Trisomy of chromosome 21 corresponds to ____.

- A. Down syndrome
- B. Turner syndrome
- C. Crossing over
- D. Incomplete dominance

BIO-12-CH03-0039 | Easy | Non-Mendelian patterns

Which description is correctly represented by Pleiotropy in this chapter?

- A. Trisomy of chromosome 21
- B. Heterozygote showing an intermediate phenotype
- C. Heterogametic sex in humans
- D. One gene influencing several characters

BIO-12-CH03-0040 | Medium | Sex-linked inheritance

Complete the statement accurately: Classic X-linked human trait involving red-green vision corresponds to _____.

- A. 9:3:3:1
- B. Carrier
- C. Colour blindness
- D. X chromosome

BIO-12-CH03-0041 | Easy | Sex determination

Which biological term or result is specifically identified by the description 'Sex chromosome system in human females'?

- A. 9:3:3:1
- B. XX
- C. Pachytene of prophase I
- D. ABO blood grouping

BIO-12-CH03-0042 | Medium | Chromosomes linkage and crossing over

Complete the statement accurately: Genes on the same chromosome tending to be inherited together corresponds to _____.

- A. 1:1
- B. Haploid chromosome number
- C. Linkage
- D. Locus

BIO-12-CH03-0043 | Medium | Non-Mendelian patterns

Complete the statement accurately: Human example of multiple alleles cited in the chapter corresponds to _____.

- A. Pleiotropy
- B. Pachytene of prophase I
- C. Codominance
- D. ABO blood grouping

BIO-12-CH03-0044 | Easy | Chromosomes linkage and crossing over

Which biological term or result is specifically identified by the description 'Exchange between non-sister chromatids of homologous chromosomes'?

- A. Alleles
- B. Crossing over
- C. Aneuploidy
- D. Colour blindness

BIO-12-CH03-0045 | Medium | Chromosomes linkage and crossing over

Complete the statement accurately: Exchange between non-sister chromatids of homologous chromosomes corresponds to _____.

- A. Colour blindness
- B. XX
- C. Crossing over
- D. Aneuploidy

BIO-12-CH03-0046 | Medium | Mendelian inheritance

Complete the statement accurately: Independent behaviour of different allele pairs corresponds to _____.

- A. 3:1
- B. 1:1:1:1
- C. Law of independent assortment
- D. Law of segregation

BIO-12-CH03-0047 | Hard | Sex determination

A reviewer compares several entries while checking 'Sex determination in honey bees'. Which entry must be corrected?

- A. Sex determination in honey bees - 9:3:3:1
- B. Monosomy X - Turner syndrome
- C. Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1
- D. Expected test-cross ratio for a heterozygous monohybrid - 1:1

BIO-12-CH03-0048 | Medium | Sex determination

Complete the statement accurately: Heterogametic sex in humans corresponds to _____.

- A. Male
- B. Law of segregation
- C. 1:1:1:1
- D. Aneuploidy

BIO-12-CH03-0049 | Hard | Crosses

A reviewer compares several entries while checking 'Cross with a homozygous recessive individual'. Which entry must be corrected?

- A. Monosomy X - Turner syndrome
- B. Cross with a homozygous recessive individual - Male
- C. Monohybrid F₂ phenotypic ratio under complete dominance - 3:1
- D. Genes on the same chromosome tending to be inherited together - Linkage

BIO-12-CH03-0050 | Medium | Crosses

A source table contains one accurate entry involving 'Expected dihybrid test-cross ratio with independent assortment'. Which option preserves a correct biological association?

- A. Exchange between non-sister chromatids of homologous chromosomes - Alleles
- B. Heterozygote showing an intermediate phenotype - 1:1:1:1
- C. A female heterozygous for a recessive X-linked allele - Phenotype
- D. Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1

BIO-12-CH03-0051 | Hard | Mendelian inheritance

A reviewer compares several entries while checking 'Independent behaviour of different allele pairs'. Which entry must be corrected?

- A. Independent behaviour of different allele pairs - Locus
- B. One gene influencing several characters - Pleiotropy
- C. Both alleles expressed in a heterozygote - Codominance
- D. A male transmitting an X-linked allele to - All daughters and no sons

BIO-12-CH03-0052 | Hard | Chromosomal disorders

A reviewer compares several entries while checking 'Trisomy of chromosome 21'. Which entry must be corrected?

- A. Trisomy of chromosome 21 - ABO blood grouping
- B. Separation of allele pairs during gamete formation - Law of segregation
- C. Heterozygote showing an intermediate phenotype - Incomplete dominance
- D. A female heterozygous for a recessive X-linked allele - Carrier

BIO-12-CH03-0053 | Hard | Genetic terminology

A reviewer compares several entries while checking 'Alternative forms of the same gene'. Which entry must be corrected?

- A. Number of linkage groups in a species corresponds to its - Haploid chromosome number
- B. Separation of allele pairs during gamete formation - Law of segregation
- C. Heterogametic sex in humans - Male
- D. Alternative forms of the same gene - Klinefelter syndrome

BIO-12-CH03-0054 | Hard | Non-Mendelian patterns

A reviewer compares several entries while checking 'Human example of multiple alleles cited in the chapter'. Which entry must be corrected?

- A. Chromosome carrying the common haemophilia allele - X chromosome
- B. Human example of multiple alleles cited in the chapter - Colour blindness
- C. Observable expression of a character - Phenotype
- D. Expected test-cross ratio for a heterozygous monohybrid - 1:1

BIO-12-CH03-0055 | Medium | Genetic terminology

A learner encounters the observation 'Position occupied by a gene on a chromosome' while analysing genetic terminology. Which conclusion is supported by the prescribed biology?

- A. 9:3:3:1
- B. Crossing over
- C. Locus
- D. Phenotype

BIO-12-CH03-0056 | Hard | Mendelian inheritance

A reviewer compares several entries while checking 'Monohybrid F₂ phenotypic ratio under complete dominance'. Which entry must be corrected?

- A. Classic X-linked human trait involving red-green vision - Colour blindness
- B. Stage at which crossing over occurs - Pachytene of prophase I
- C. Observable expression of a character - Phenotype
- D. Monohybrid F₂ phenotypic ratio under complete dominance - ZW female and ZZ male

BIO-12-CH03-0057 | Easy | Crosses

Which biological term or result is specifically identified by the description 'Cross with a homozygous recessive individual'?

- A. 1:1
- B. 3:1
- C. Test cross
- D. Male

BIO-12-CH03-0058 | Hard | Crosses

Two conclusions must both be valid to support the reference concept 'Cross with a homozygous recessive individual'. Which option contains two accurate associations?

- A. Exchange between non-sister chromatids of homologous chromosomes - All daughters and no sons; A male transmitting an X-linked allele to - Crossing over
- B. Monosomy X - Linkage; Genes on the same chromosome tending to be inherited together - Turner syndrome
- C. Cross with a homozygous recessive individual - Test cross; Heterogametic sex in humans - Male
- D. A male transmitting an X-linked allele to - Haploid chromosome number; Number of linkage groups in a species corresponds to its - All daughters and no sons

BIO-12-CH03-0059 | Easy | Chromosomal disorders

Which biological term or result is specifically identified by the description 'XXY constitution in a male'?

- A. Aneuploidy
- B. 3:1
- C. Law of segregation
- D. Klinefelter syndrome

BIO-12-CH03-0060 | Medium | Chromosomal disorders

A learner encounters the observation 'Trisomy of chromosome 21' while analysing chromosomal disorders. Which conclusion is supported by the prescribed biology?

- A. Crossing over
- B. Down syndrome
- C. Incomplete dominance
- D. ABO blood grouping

BIO-12-CH03-0061 | Challenging | Chromosomal disorders

A student submits four conclusions after integrating related chapter concepts.

- I. XXY constitution in a male corresponds to Aneuploidy.
 - II. Loss or gain of one or a few chromosomes corresponds to Turner syndrome.
 - III. Monosomy X corresponds to Turner syndrome.
 - IV. Dihybrid F2 phenotypic ratio under independent assortment corresponds to Klinefelter syndrome.
- Which combination should be accepted without correction?

- A. II and III only
- B. I, II and IV only
- C. III only
- D. IV only

BIO-12-CH03-0062 | Easy | Genetic terminology

Which biological term or result is specifically identified by the description 'Genetic constitution of an organism'?

- A. Klinefelter syndrome
- B. 1:1
- C. ABO blood grouping
- D. Genotype

BIO-12-CH03-0063 | Medium | Genetic terminology

A source table contains one accurate entry involving 'Observable expression of a character'. Which option preserves a correct biological association?

- A. Observable expression of a character - Phenotype
- B. Stage at which crossing over occurs - Incomplete dominance
- C. Sex determination in honey bees - ABO blood grouping
- D. Human example of multiple alleles cited in the chapter - Phenotype

BIO-12-CH03-0064 | Medium | Genetic terminology

Complete the statement accurately: Alternative forms of the same gene corresponds to ____.

- A. Law of segregation
- B. Klinefelter syndrome
- C. Locus
- D. Alleles

BIO-12-CH03-0065 | Medium | Sex determination

A source table contains one accurate entry involving 'Sex chromosome system in human females'. Which option preserves a correct biological association?

- A. Sex chromosome system in human females - XX
- B. Monosomy X - XX
- C. Dihybrid F2 phenotypic ratio under independent assortment - Turner syndrome
- D. Stage at which crossing over occurs - All daughters and no sons

BIO-12-CH03-0066 | Medium | Chromosomes linkage and crossing over

A learner encounters the observation 'Genes on the same chromosome tending to be inherited together' while analysing chromosomes linkage and crossing over. Which conclusion is supported by the prescribed biology?

- A. Linkage
- B. All daughters and no sons
- C. Phenotype
- D. Locus

BIO-12-CH03-0067 | Medium | Chromosomal disorders

Complete the statement accurately: XXY constitution in a male corresponds to _____.

- A. ZW female and ZZ male
- B. Law of segregation
- C. All daughters and no sons
- D. Klinefelter syndrome

BIO-12-CH03-0068 | Easy | Non-Mendelian patterns

Which biological term or result is specifically identified by the description 'Both alleles expressed in a heterozygote'?

- A. Codominance
- B. 1:1:1:1
- C. Back cross
- D. 9:3:3:1

BIO-12-CH03-0069 | Medium | Non-Mendelian patterns

A learner encounters the observation 'Human example of multiple alleles cited in the chapter' while analysing non-mendelian patterns. Which conclusion is supported by the prescribed biology?

- A. Pachytene of prophase I
- B. Turner syndrome
- C. ABO blood grouping
- D. Pleiotropy

BIO-12-CH03-0070 | Hard | Sex determination

Two conclusions must both be valid to support the reference concept 'Sex determination in honey bees'. Which option contains two accurate associations?

- A. Sex determination in honey bees - Haplo-diploid type; Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1
- B. Expected dihybrid test-cross ratio with independent assortment - Aneuploidy; Loss or gain of one or a few chromosomes - 1:1:1:1
- C. Monosomy X - 1:1:1:1; Expected dihybrid test-cross ratio with independent assortment - Turner syndrome
- D. Classic X-linked human trait involving red-green vision - 1:1; Expected test-cross ratio for a heterozygous monohybrid - Colour blindness

BIO-12-CH03-0071 | Easy | Chromosomes linkage and crossing over

Which description is correctly represented by Haploid chromosome number in this chapter?

- A. Alternative forms of the same gene
- B. Number of linkage groups in a species corresponds to its
- C. Human example of multiple alleles cited in the chapter
- D. Genetic constitution of an organism

BIO-12-CH03-0072 | Hard | Sex-linked inheritance

Two conclusions must both be valid to support the reference concept 'Classic X-linked human trait involving red-green vision'. Which option contains two accurate associations?

- A. Sex chromosome system in birds - Phenotype; Observable expression of a character - ZW female and ZZ male
- B. Dihybrid F2 phenotypic ratio under independent assortment - ZW female and ZZ male; Sex chromosome system in birds - 9:3:3:1
- C. Observable expression of a character - XX; Sex chromosome system in human females - Phenotype
- D. Classic X-linked human trait involving red-green vision - Colour blindness; Monosomy X - Turner syndrome

BIO-12-CH03-0073 | Easy | Mendelian inheritance

Which description is correctly represented by Law of segregation in this chapter?

- A. Separation of allele pairs during gamete formation
- B. Expected dihybrid test-cross ratio with independent assortment
- C. Position occupied by a gene on a chromosome
- D. Classic X-linked human trait involving red-green vision

BIO-12-CH03-0074 | Medium | Mendelian inheritance

A source table contains one accurate entry involving 'Dihybrid F2 phenotypic ratio under independent assortment'. Which option preserves a correct biological association?

- A. A female heterozygous for a recessive X-linked allele - 9:3:3:1
- B. Cross of an F1 individual with either parent - Male
- C. Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1
- D. Expected test-cross ratio for a heterozygous monohybrid - Linkage

BIO-12-CH03-0075 | Challenging | Genetic terminology

During error analysis, four proposed associations are reviewed.

- I. Alternative forms of the same gene corresponds to Genotype.
- II. Genetic constitution of an organism corresponds to Locus.
- III. Position occupied by a gene on a chromosome corresponds to Locus.
- IV. Expected test-cross ratio for a heterozygous monohybrid corresponds to 1:1.

Which set contains all and only the valid conclusions?

- A. III and IV only
- B. I, II and IV only
- C. I, II and III only
- D. I, II, III and IV

BIO-12-CH03-0076 | Medium | Chromosomal disorders

Complete the statement accurately: Monosomy X corresponds to _____.

- A. Genotype
- B. Turner syndrome
- C. Crossing over
- D. Linkage

BIO-12-CH03-0077 | Easy | Mendelian inheritance

Which description is correctly represented by Law of independent assortment in this chapter?

- A. Independent behaviour of different allele pairs
- B. Expected dihybrid test-cross ratio with independent assortment
- C. Separation of allele pairs during gamete formation
- D. One gene influencing several characters

BIO-12-CH03-0078 | Medium | Sex-linked inheritance

A learner encounters the observation 'Classic X-linked human trait involving red-green vision' while analysing sex-linked inheritance. Which conclusion is supported by the prescribed biology?

- A. Turner syndrome
- B. Carrier
- C. Colour blindness
- D. X chromosome

BIO-12-CH03-0079 | Easy | Non-Mendelian patterns

Which description is correctly represented by Incomplete dominance in this chapter?

- A. Observable expression of a character
- B. One gene influencing several characters
- C. Dihybrid F2 phenotypic ratio under independent assortment
- D. Heterozygote showing an intermediate phenotype

BIO-12-CH03-0080 | Easy | Crosses

Which description is correctly represented by Test cross in this chapter?

- A. Exchange between non-sister chromatids of homologous chromosomes
- B. Heterogametic sex in humans
- C. Cross of an F1 individual with either parent
- D. Cross with a homozygous recessive individual

BIO-12-CH03-0081 | Medium | Chromosomal disorders

A source table contains one accurate entry involving 'Loss or gain of one or a few chromosomes'. Which option preserves a correct biological association?

- A. Exchange between non-sister chromatids of homologous chromosomes - Back cross
- B. A female heterozygous for a recessive X-linked allele - Law of segregation
- C. Loss or gain of one or a few chromosomes - Aneuploidy
- D. Separation of allele pairs during gamete formation - Aneuploidy

BIO-12-CH03-0082 | Hard | Chromosomal disorders

A reviewer compares several entries while checking 'XXY constitution in a male'. Which entry must be corrected?

- A. Sex chromosome system in human females - XX
- B. Monohybrid F2 phenotypic ratio under complete dominance - 3:1
- C. Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1
- D. XXY constitution in a male - Law of segregation

BIO-12-CH03-0083 | Medium | Non-Mendelian patterns

Complete the statement accurately: One gene influencing several characters corresponds to ____.

- A. Pleiotropy
- B. Male
- C. Incomplete dominance
- D. Test cross

BIO-12-CH03-0084 | Medium | Chromosomes linkage and crossing over

A source table contains one accurate entry involving 'Number of linkage groups in a species corresponds to its'. Which option preserves a correct biological association?

- A. Classic X-linked human trait involving red-green vision - Haploid chromosome number
- B. Alternative forms of the same gene - Klinefelter syndrome
- C. Number of linkage groups in a species corresponds to its - Haploid chromosome number
- D. Sex determination in honey bees - Genotype

BIO-12-CH03-0085 | Hard | Chromosomal disorders

Two conclusions must both be valid to support the reference concept 'XXY constitution in a male'. Which option contains two accurate associations?

- A. XXY constitution in a male - Klinefelter syndrome; Separation of allele pairs during gamete formation - Law of segregation
- B. Expected dihybrid test-cross ratio with independent assortment - Test cross; Cross with a homozygous recessive individual - 1:1:1:1
- C. Sex chromosome system in birds - 3:1; Monohybrid F2 phenotypic ratio under complete dominance - ZW female and ZZ male
- D. A male transmitting an X-linked allele to - 1:1:1:1; Expected dihybrid test-cross ratio with independent assortment - All daughters and no sons

BIO-12-CH03-0086 | Easy | Chromosomes linkage and crossing over

Which description is correctly represented by Crossing over in this chapter?

- A. Alternative forms of the same gene
- B. Sex chromosome system in human females
- C. Stage at which crossing over occurs
- D. Exchange between non-sister chromatids of homologous chromosomes

BIO-12-CH03-0087 | Medium | Sex determination

Complete the statement accurately: Sex chromosome system in human females corresponds to _____.

- A. Pachytene of prophase I
- B. Male
- C. XX
- D. ABO blood grouping

BIO-12-CH03-0088 | Medium | Chromosomes linkage and crossing over

Complete the statement accurately: Stage at which crossing over occurs corresponds to _____.

- A. Pachytene of prophase I
- B. Haploid chromosome number
- C. Locus
- D. Crossing over

BIO-12-CH03-0089 | Challenging | Mendelian inheritance

A researcher checks whether four linked observations have been interpreted correctly.

- I. Dihybrid F2 phenotypic ratio under independent assortment corresponds to 3:1.
- II. Monohybrid F2 phenotypic ratio under complete dominance corresponds to 3:1.
- III. Independent behaviour of different allele pairs corresponds to Pachytene of prophase I.
- IV. Stage at which crossing over occurs corresponds to 9:3:3:1.

Which combination survives the biological review?

- A. I only
- B. I, III and IV only
- C. I and III only
- D. II only

BIO-12-CH03-0090 | Hard | Chromosomes linkage and crossing over

A reviewer compares several entries while checking 'Exchange between non-sister chromatids of homologous chromosomes'. Which entry must be corrected?

- A. One gene influencing several characters - Pleiotropy
- B. Heterozygote showing an intermediate phenotype - Incomplete dominance
- C. Exchange between non-sister chromatids of homologous chromosomes - Colour blindness
- D. Alternative forms of the same gene - Alleles

BIO-12-CH03-0091 | Easy | Sex determination

Which biological term or result is specifically identified by the description 'Sex determination in honey bees'?

- A. 1:1:1:1
- B. Haplo-diploid type
- C. XX
- D. Colour blindness

BIO-12-CH03-0092 | Challenging | Mendelian inheritance

Evaluate the following source-based biological statements.

- I. Independent behaviour of different allele pairs corresponds to 9:3:3:1.
- II. Dihybrid F₂ phenotypic ratio under independent assortment corresponds to 9:3:3:1.
- III. Separation of allele pairs during gamete formation corresponds to Back cross.
- IV. Cross of an F₁ individual with either parent corresponds to Back cross.

Which option identifies exactly the correct statements?

- A. I and IV only
- B. II and IV only
- C. II and III only
- D. II, III and IV only

BIO-12-CH03-0093 | Medium | Non-Mendelian patterns

A source table contains one accurate entry involving 'One gene influencing several characters'. Which option preserves a correct biological association?

- A. One gene influencing several characters - Pleiotropy
- B. Heterogametic sex in humans - Aneuploidy
- C. Cross with a homozygous recessive individual - 3:1
- D. Monohybrid F₂ phenotypic ratio under complete dominance - Pleiotropy

BIO-12-CH03-0094 | Easy | Genetic terminology

Which description is correctly represented by Alleles in this chapter?

- A. Number of linkage groups in a species corresponds to its
- B. Position occupied by a gene on a chromosome
- C. Alternative forms of the same gene
- D. XXY constitution in a male

BIO-12-CH03-0095 | Easy | Sex-linked inheritance

Which biological term or result is specifically identified by the description 'Chromosome carrying the common haemophilia allele'?

- A. Colour blindness
- B. Turner syndrome
- C. Test cross
- D. X chromosome

BIO-12-CH03-0096 | Medium | Genetic terminology

A source table contains one accurate entry involving 'Genetic constitution of an organism'. Which option preserves a correct biological association?

- A. Sex chromosome system in birds - Genotype
- B. Genetic constitution of an organism - Genotype
- C. Classic X-linked human trait involving red-green vision - Turner syndrome
- D. XXY constitution in a male - ABO blood grouping

BIO-12-CH03-0097 | Challenging | Non-Mendelian patterns

A student submits four conclusions after integrating related chapter concepts.

- I. One gene influencing several characters corresponds to Pleiotropy.
- II. Human example of multiple alleles cited in the chapter corresponds to Incomplete dominance.
- III. Heterozygote showing an intermediate phenotype corresponds to Aneuploidy.
- IV. Loss or gain of one or a few chromosomes corresponds to Pleiotropy.

Which combination should be accepted without correction?

- A. II and IV only
- B. I only
- C. I, II and III only
- D. II only

BIO-12-CH03-0098 | Medium | Mendelian inheritance

A learner encounters the observation 'Independent behaviour of different allele pairs' while analysing mendelian inheritance. Which conclusion is supported by the prescribed biology?

- A. 3:1
- B. ZW female and ZZ male
- C. All daughters and no sons
- D. Law of independent assortment

BIO-12-CH03-0099 | Medium | Crosses

A source table contains one accurate entry involving 'Cross with a homozygous recessive individual'. Which option preserves a correct biological association?

- A. Monohybrid F2 phenotypic ratio under complete dominance - Linkage
- B. Cross with a homozygous recessive individual - Test cross
- C. Exchange between non-sister chromatids of homologous chromosomes - Haploid chromosome number
- D. A male transmitting an X-linked allele to - Test cross

BIO-12-CH03-0100 | Challenging | Non-Mendelian patterns

During error analysis, four proposed associations are reviewed.

- I. Both alleles expressed in a heterozygote corresponds to Incomplete dominance.
- II. Heterozygote showing an intermediate phenotype corresponds to Incomplete dominance.
- III. One gene influencing several characters corresponds to Pleiotropy.
- IV. Sex determination in honey bees corresponds to Codominance.

Which set contains all and only the valid conclusions?

- A. IV only
- B. II only
- C. I and II only
- D. II and III only

BIO-12-CH03-0101 | Hard | Genetic terminology

Two conclusions must both be valid to support the reference concept 'Alternative forms of the same gene'. Which option contains two accurate associations?

- A. Exchange between non-sister chromatids of homologous chromosomes - Haploid chromosome number; Number of linkage groups in a species corresponds to its - Crossing over
- B. A male transmitting an X-linked allele to - Law of segregation; Separation of allele pairs during gamete formation - All daughters and no sons
- C. Alternative forms of the same gene - Alleles; XXY constitution in a male - Klinefelter syndrome
- D. Heterogametic sex in humans - Test cross; Cross with a homozygous recessive individual - Male

BIO-12-CH03-0102 | Hard | Mendelian inheritance

Two conclusions must both be valid to support the reference concept 'Separation of allele pairs during gamete formation'. Which option contains two accurate associations?

- A. Separation of allele pairs during gamete formation - Law of segregation; Position occupied by a gene on a chromosome - Locus
- B. Chromosome carrying the common haemophilia allele - Colour blindness; Classic X-linked human trait involving red-green vision - X chromosome
- C. Monosomy X - Haploid chromosome number; Number of linkage groups in a species corresponds to its - Turner syndrome
- D. Expected dihybrid test-cross ratio with independent assortment - Turner syndrome; Monosomy X - 1:1:1:1

BIO-12-CH03-0103 | Medium | Genetic terminology

A source table contains one accurate entry involving 'Alternative forms of the same gene'. Which option preserves a correct biological association?

- A. Number of linkage groups in a species corresponds to its - Law of segregation
- B. A male transmitting an X-linked allele to - Male
- C. XXY constitution in a male - Haploid chromosome number
- D. Alternative forms of the same gene - Alleles

BIO-12-CH03-0104 | Challenging | Sex determination

A researcher checks whether four linked observations have been interpreted correctly.

- I. Sex determination in honey bees corresponds to Haplo-diploid type.
 - II. Heterogametic sex in humans corresponds to Male.
 - III. Sex chromosome system in human females corresponds to XX.
 - IV. Genes on the same chromosome tending to be inherited together corresponds to Linkage.
- Which combination survives the biological review?

- A. I only
- B. III and IV only
- C. I, II, III and IV
- D. III only

BIO-12-CH03-0105 | Challenging | Crosses

Evaluate the following source-based biological statements.

- I. Expected dihybrid test-cross ratio with independent assortment corresponds to 1:1:1:1.
 - II. Cross of an F1 individual with either parent corresponds to Test cross.
 - III. Cross with a homozygous recessive individual corresponds to Male.
 - IV. Heterogametic sex in humans corresponds to Male.
- Which option identifies exactly the correct statements?

- A. I and IV only
- B. III only
- C. I, II and IV only
- D. I and III only

BIO-12-CH03-0106 | Hard | Chromosomes linkage and crossing over

Two conclusions must both be valid to support the reference concept 'Number of linkage groups in a species corresponds to its'. Which option contains two accurate associations?

- A. Sex determination in honey bees - Back cross; Cross of an F1 individual with either parent - Haplo-diploid type
- B. Number of linkage groups in a species corresponds to its - Haploid chromosome number; Human example of multiple alleles cited in the chapter - ABO blood grouping
- C. Cross of an F1 individual with either parent - Genotype; Genetic constitution of an organism - Back cross
- D. Alternative forms of the same gene - Colour blindness; Classic X-linked human trait involving red-green vision - Alleles

BIO-12-CH03-0107 | Medium | Sex determination

A learner encounters the observation 'Heterogametic sex in humans' while analysing sex determination. Which conclusion is supported by the prescribed biology?

- A. XX
- B. Male
- C. Incomplete dominance
- D. Law of segregation

BIO-12-CH03-0108 | Easy | Sex-linked inheritance

Which description is correctly represented by X chromosome in this chapter?

- A. Chromosome carrying the common haemophilia allele
- B. Classic X-linked human trait involving red-green vision
- C. Exchange between non-sister chromatids of homologous chromosomes
- D. Cross with a homozygous recessive individual

BIO-12-CH03-0109 | Medium | Sex determination

A learner encounters the observation 'Sex determination in honey bees' while analysing sex determination. Which conclusion is supported by the prescribed biology?

- A. Haplo-diploid type
- B. ZW female and ZZ male
- C. Colour blindness
- D. 9:3:3:1

BIO-12-CH03-0110 | Hard | Genetic terminology

Two conclusions must both be valid to support the reference concept 'Position occupied by a gene on a chromosome'. Which option contains two accurate associations?

- A. Loss or gain of one or a few chromosomes - Haploid chromosome number; Number of linkage groups in a species corresponds to its - Aneuploidy
- B. Position occupied by a gene on a chromosome - Locus; Monosomy X - Turner syndrome
- C. Dihybrid F2 phenotypic ratio under independent assortment - Aneuploidy; Loss or gain of one or a few chromosomes - 9:3:3:1
- D. Expected test-cross ratio for a heterozygous monohybrid - 9:3:3:1; Dihybrid F2 phenotypic ratio under independent assortment - 1:1

BIO-12-CH03-0111 | Medium | Crosses

A learner encounters the observation 'Cross with a homozygous recessive individual' while analysing crosses. Which conclusion is supported by the prescribed biology?

- A. Linkage
- B. Test cross
- C. 1:1
- D. Turner syndrome

BIO-12-CH03-0112 | Medium | Mendelian inheritance

A source table contains one accurate entry involving 'Separation of allele pairs during gamete formation'. Which option preserves a correct biological association?

- A. Chromosome carrying the common haemophilia allele - Incomplete dominance
- B. Classic X-linked human trait involving red-green vision - 1:1:1:1
- C. Monosomy X - Law of segregation
- D. Separation of allele pairs during gamete formation - Law of segregation

BIO-12-CH03-0113 | Hard | Sex determination

A reviewer compares several entries while checking 'Sex chromosome system in human females'. Which entry must be corrected?

- A. Human example of multiple alleles cited in the chapter - ABO blood grouping
- B. Chromosome carrying the common haemophilia allele - X chromosome
- C. Sex chromosome system in human females - Down syndrome
- D. Dihybrid F2 phenotypic ratio under independent assortment - 9:3:3:1

BIO-12-CH03-0114 | Hard | Sex-linked inheritance

Two conclusions must both be valid to support the reference concept 'A male transmitting an X-linked allele to'. Which option contains two accurate associations?

- A. Monohybrid F2 phenotypic ratio under complete dominance - XX; Sex chromosome system in human females - 3:1
- B. Trisomy of chromosome 21 - Back cross; Cross of an F1 individual with either parent - Down syndrome
- C. Sex chromosome system in human females - Linkage; Genes on the same chromosome tending to be inherited together - XX
- D. A male transmitting an X-linked allele to - All daughters and no sons; Observable expression of a character - Phenotype

BIO-12-CH03-0115 | Easy | Chromosomes linkage and crossing over

Which biological term or result is specifically identified by the description 'Number of linkage groups in a species corresponds to its'?

- A. ABO blood grouping
- B. Linkage
- C. Haploid chromosome number
- D. Haplo-diploid type

BIO-12-CH03-0116 | Medium | Mendelian inheritance

A learner encounters the observation 'Monohybrid F2 phenotypic ratio under complete dominance' while analysing mendelian inheritance. Which conclusion is supported by the prescribed biology?

- A. Pachytene of prophase I
- B. ABO blood grouping
- C. 3:1
- D. Colour blindness

BIO-12-CH03-0117 | Medium | Mendelian inheritance

Complete the statement accurately: Dihybrid F2 phenotypic ratio under independent assortment corresponds to _____.

- A. X chromosome
- B. 3:1
- C. Law of independent assortment
- D. 9:3:3:1

BIO-12-CH03-0118 | Easy | Sex-linked inheritance

Which description is correctly represented by Colour blindness in this chapter?

- A. A female heterozygous for a recessive X-linked allele
- B. Classic X-linked human trait involving red-green vision
- C. Dihybrid F2 phenotypic ratio under independent assortment
- D. Observable expression of a character

BIO-12-CH03-0119 | Medium | Crosses

A source table contains one accurate entry involving 'Cross of an F1 individual with either parent'. Which option preserves a correct biological association?

- A. Dihybrid F2 phenotypic ratio under independent assortment - Male
- B. Cross of an F1 individual with either parent - Back cross
- C. Heterogametic sex in humans - X chromosome
- D. One gene influencing several characters - ABO blood grouping

BIO-12-CH03-0120 | Medium | Sex-linked inheritance

Complete the statement accurately: A female heterozygous for a recessive X-linked allele corresponds to _____.

- A. Back cross
- B. Colour blindness
- C. Carrier
- D. Linkage

BIO-12-CH03-0121 | Hard | Genetic terminology

A reviewer compares several entries while checking 'Genetic constitution of an organism'. Which entry must be corrected?

- A. Sex chromosome system in birds - ZW female and ZZ male
- B. Classic X-linked human trait involving red-green vision - Colour blindness
- C. Expected dihybrid test-cross ratio with independent assortment - 1:1:1:1
- D. Genetic constitution of an organism - Klinefelter syndrome

BIO-12-CH03-0122 | Easy | Mendelian inheritance

Which biological term or result is specifically identified by the description 'Independent behaviour of different allele pairs'?

- A. Locus
- B. Law of independent assortment
- C. All daughters and no sons
- D. ZW female and ZZ male

BIO-12-CH03-0123 | Medium | Non-Mendelian patterns

A learner encounters the observation 'Both alleles expressed in a heterozygote' while analysing non-mendelian patterns. Which conclusion is supported by the prescribed biology?

- A. Codominance
- B. 1:1:1:1
- C. Pleiotropy
- D. ABO blood grouping

BIO-12-CH03-0124 | Easy | Sex determination

Which description is correctly represented by Haplo-diploid type in this chapter?

- A. Monosomy X
- B. Sex chromosome system in human females
- C. Sex determination in honey bees
- D. Expected dihybrid test-cross ratio with independent assortment

BIO-12-CH03-0125 | Medium | Non-Mendelian patterns

A learner encounters the observation 'One gene influencing several characters' while analysing non-mendelian patterns. Which conclusion is supported by the prescribed biology?

- A. ABO blood grouping
- B. Pleiotropy
- C. Down syndrome
- D. Male

BIO-12-CH03-0126 | Medium | Chromosomal disorders

A learner encounters the observation 'Monosomy X' while analysing chromosomal disorders. Which conclusion is supported by the prescribed biology?

- A. Linkage
- B. Down syndrome
- C. 3:1
- D. Turner syndrome

BIO-12-CH03-0127 | Hard | Chromosomes linkage and crossing over

A reviewer compares several entries while checking 'Number of linkage groups in a species corresponds to its'. Which entry must be corrected?

- A. Classic X-linked human trait involving red-green vision - Colour blindness
- B. Number of linkage groups in a species corresponds to its - ABO blood grouping
- C. Genetic constitution of an organism - Genotype
- D. Sex determination in honey bees - Haplo-diploid type

BIO-12-CH03-0128 | Medium | Chromosomal disorders

A learner encounters the observation 'XXY constitution in a male' while analysing chromosomal disorders. Which conclusion is supported by the prescribed biology?

- A. Klinefelter syndrome
- B. 3:1
- C. XX
- D. Aneuploidy

BIO-12-CH03-0129 | Hard | Crosses

Two conclusions must both be valid to support the reference concept 'Expected test-cross ratio for a heterozygous monohybrid'. Which option contains two accurate associations?

- A. Loss or gain of one or a few chromosomes - Codominance; Both alleles expressed in a heterozygote - Aneuploidy
- B. Both alleles expressed in a heterozygote - Pachytene of prophase I; Stage at which crossing over occurs - Codominance
- C. Expected test-cross ratio for a heterozygous monohybrid - 1:1; XXY constitution in a male - Klinefelter syndrome
- D. Genetic constitution of an organism - Law of segregation; Separation of allele pairs during gamete formation - Genotype

BIO-12-CH03-0130 | Hard | Non-Mendelian patterns

A reviewer compares several entries while checking 'Both alleles expressed in a heterozygote'. Which entry must be corrected?

- A. Expected test-cross ratio for a heterozygous monohybrid - 1:1
- B. Both alleles expressed in a heterozygote - Haplo-diploid type
- C. Chromosome carrying the common haemophilia allele - X chromosome
- D. Cross of an F1 individual with either parent - Back cross

BIO-12-CH03-0131 | Easy | Genetic terminology

Which description is correctly represented by Locus in this chapter?

- A. Exchange between non-sister chromatids of homologous chromosomes
- B. Position occupied by a gene on a chromosome
- C. Expected test-cross ratio for a heterozygous monohybrid
- D. Genetic constitution of an organism

BIO-12-CH03-0132 | Easy | Non-Mendelian patterns

Which description is correctly represented by ABO blood grouping in this chapter?

- A. Stage at which crossing over occurs
- B. Classic X-linked human trait involving red-green vision
- C. Observable expression of a character
- D. Human example of multiple alleles cited in the chapter

BIO-12-CH03-0133 | Hard | Mendelian inheritance

Two conclusions must both be valid to support the reference concept 'Independent behaviour of different allele pairs'. Which option contains two accurate associations?

- A. Independent behaviour of different allele pairs - Law of independent assortment; Position occupied by a gene on a chromosome - Locus
- B. Sex chromosome system in birds - 1:1:1:1; Expected dihybrid test-cross ratio with independent assortment - ZW female and ZZ male
- C. Both alleles expressed in a heterozygote - Male; Heterogametic sex in humans - Codominance
- D. A male transmitting an X-linked allele to - ZW female and ZZ male; Sex chromosome system in birds - All daughters and no sons

BIO-12-CH03-0134 | Easy | Crosses

Which description is correctly represented by 1:1 in this chapter?

- A. Genetic constitution of an organism
- B. Cross with a homozygous recessive individual
- C. Expected test-cross ratio for a heterozygous monohybrid
- D. Stage at which crossing over occurs

BIO-12-CH03-0135 | Challenging | Sex-linked inheritance

A student submits four conclusions after integrating related chapter concepts.

- I. Chromosome carrying the common haemophilia allele corresponds to Carrier.
- II. A female heterozygous for a recessive X-linked allele corresponds to Carrier.
- III. A male transmitting an X-linked allele to corresponds to All daughters and no sons.
- IV. Sex chromosome system in birds corresponds to ZW female and ZZ male.

Which combination should be accepted without correction?

- A. I and III only
- B. II, III and IV only
- C. II and IV only
- D. II only

BIO-12-CH03-0136 | Medium | Genetic terminology

A learner encounters the observation 'Genetic constitution of an organism' while analysing genetic terminology. Which conclusion is supported by the prescribed biology?

- A. Locus
- B. Klinefelter syndrome
- C. Genotype
- D. 1:1

BIO-12-CH03-0137 | Medium | Non-Mendelian patterns

Complete the statement accurately: Heterozygote showing an intermediate phenotype corresponds to _____.

- A. Incomplete dominance
 - B. 9:3:3:1
 - C. Linkage
 - D. Down syndrome
-

BIO-12-CH03-0138 | Medium | Chromosomes linkage and crossing over

Complete the statement accurately: Number of linkage groups in a species corresponds to its corresponds to _____.

- A. Back cross
 - B. Genotype
 - C. Haploid chromosome number
 - D. Linkage
-

BIO-12-CH03-0139 | Easy | Chromosomal disorders

Which description is correctly represented by Klinefelter syndrome in this chapter?

- A. Sex chromosome system in human females
 - B. XXY constitution in a male
 - C. Separation of allele pairs during gamete formation
 - D. Trisomy of chromosome 21
-

BIO-12-CH03-0140 | Easy | Sex-linked inheritance

Which biological term or result is specifically identified by the description 'A female heterozygous for a recessive X-linked allele'?

- A. 1:1
 - B. Test cross
 - C. Colour blindness
 - D. Carrier
-

Standard 12 | Chapter 04

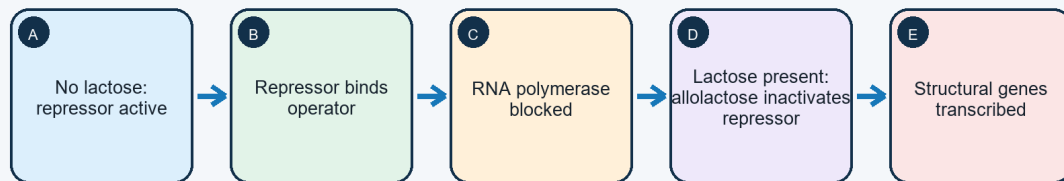
Molecular Basis of Inheritance

BIO-12-CH04-0001 | Medium | Regulation of the lac operon

When lactose is absent, which event in the figure directly prevents transcription of the structural genes?

Regulation of the lac operon

Inducer availability controls access to structural genes



Strongest expression occurs when lactose is available and glucose is scarce.

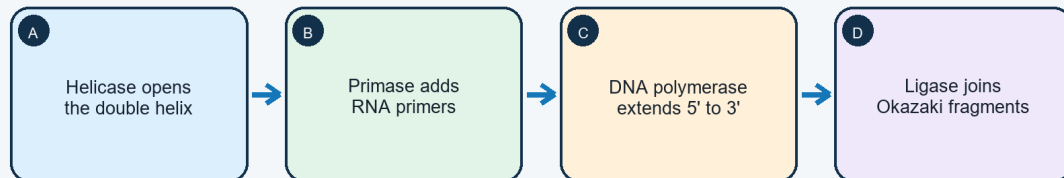
- A. The active repressor binds the operator
- B. Allolactose binds RNA polymerase
- C. The operator is removed from DNA
- D. The structural genes become ribosomal RNA

BIO-12-CH04-0002 | Hard | Events at a DNA replication fork

Why is ligase especially important on the lagging strand shown?

Events at a DNA replication fork

The two new strands are synthesized in different patterns



The lagging strand is synthesized discontinuously because polymerase works only in the 5' to 3' direction.

- A. It joins adjacent Okazaki fragments
- B. It unwinds the parental helix
- C. It synthesizes RNA primers
- D. It reads mRNA codons

BIO-12-CH04-0003 | Medium | DNA structure

Complete the statement accurately: Number of hydrogen bonds in a G-C pair corresponds to _____.

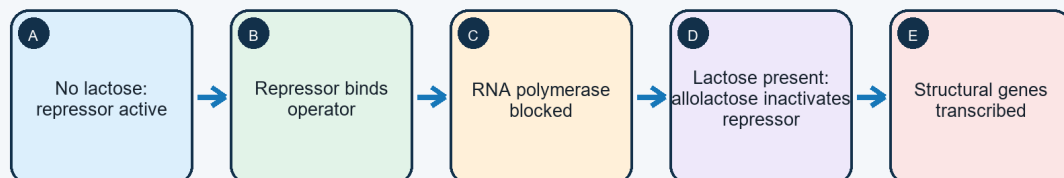
- A. Genome
- B. Adenine
- C. mRNA
- D. Three

BIO-12-CH04-0004 | Challenging | Regulation of the lac operon

Which condition gives the strongest lac-operon expression?

Regulation of the lac operon

Inducer availability controls access to structural genes



Strongest expression occurs when lactose is available and glucose is scarce.

- A. Lactose absent and glucose abundant
- B. Lactose absent and glucose scarce
- C. Lactose present with an active operator-bound repressor
- D. Lactose present and glucose scarce

BIO-12-CH04-0005 | Medium | Gene regulation and genomics

A learner encounters the observation 'Complete DNA content of an organism' while analysing gene regulation and genomics. Which conclusion is supported by the prescribed biology?

- A. Genome
- B. DNA fingerprinting
- C. Deoxyribose
- D. Degeneracy

BIO-12-CH04-0006 | Medium | DNA packaging

A learner encounters the observation 'Core proteins around which DNA is wrapped' while analysing dna packaging. Which conclusion is supported by the prescribed biology?

- A. Friedrich Miescher
- B. Repressor
- C. Histone octamer
- D. Heterochromatin

BIO-12-CH04-0007 | Medium | DNA replication

A learner encounters the observation 'Short DNA segments formed on the lagging strand' while analysing dna replication. Which conclusion is supported by the prescribed biology?

- A. Okazaki fragments
- B. Helicase
- C. Deoxyribose
- D. DNA ligase

BIO-12-CH04-0008 | Hard | Transcription and RNA

A reviewer compares several entries while checking 'RNA carrying codons from DNA to ribosomes'. Which entry must be corrected?

- A. Short DNA segments formed on the lagging strand - Okazaki fragments
- B. RNA carrying codons from DNA to ribosomes - A site
- C. Hershey-Chase label used for phage DNA - Radioactive phosphorus-32
- D. Sugar present in DNA - Deoxyribose

BIO-12-CH04-0009 | Hard | DNA structure

A reviewer compares several entries while checking 'Sugar present in DNA'. Which entry must be corrected?

- A. Complete DNA content of an organism - Genome
- B. Basic repeating unit of chromatin - Nucleosome
- C. Sugar present in DNA - AUG
- D. Mode of DNA replication - Semiconservative

BIO-12-CH04-0010 | Hard | Evidence for genetic material

A reviewer compares several entries while checking 'Hershey-Chase label used for phage DNA'. Which entry must be corrected?

- A. Core proteins around which DNA is wrapped - Histone octamer
- B. Hershey-Chase label used for phage DNA - mRNA
- C. Highly condensed transcriptionally inactive chromatin - Heterochromatin
- D. Purine pairing with thymine - Adenine

BIO-12-CH04-0011 | Medium | DNA packaging

A source table contains one accurate entry involving 'Linker histone'. Which option preserves a correct biological association?

- A. Approximate base pairs per turn of B-DNA - Degeneracy
 - B. Sugar present in DNA - Allolactose
 - C. Linker histone - H1
 - D. Inducer of the lac operon - mRNA
-

BIO-12-CH04-0012 | Medium | DNA packaging

A learner encounters the observation 'Basic repeating unit of chromatin' while analysing dna packaging. Which conclusion is supported by the prescribed biology?

- A. mRNA
 - B. Helicase
 - C. Degeneracy
 - D. Nucleosome
-

BIO-12-CH04-0013 | Medium | Genetic code and translation

A learner encounters the observation 'Property in which several codons specify one amino acid' while analysing genetic code and translation. Which conclusion is supported by the prescribed biology?

- A. Repressor
 - B. Degeneracy
 - C. Genome
 - D. Histone octamer
-

BIO-12-CH04-0014 | Medium | DNA replication

Complete the statement accurately: Mode of DNA replication corresponds to _____.

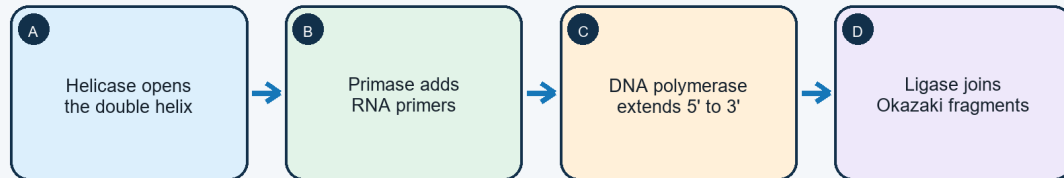
- A. A site
 - B. Radioactive phosphorus-32
 - C. Semiconservative
 - D. mRNA
-

BIO-12-CH04-0015 | Easy | Events at a DNA replication fork

Which enzyme shown first separates the parental DNA strands at the replication fork?

Events at a DNA replication fork

The two new strands are synthesized in different patterns



The lagging strand is synthesized discontinuously because polymerase works only in the 5' to 3' direction.

- A. Helicase
- B. Ligase
- C. Primase
- D. RNA polymerase

BIO-12-CH04-0016 | Hard | DNA replication

A reviewer compares several entries while checking 'Enzyme joining Okazaki fragments'. Which entry must be corrected?

- A. Transforming principle demonstrated in pneumococcus - DNA
- B. Enzyme joining Okazaki fragments - A site
- C. Approximate base pairs per turn of B-DNA - Ten
- D. Highly condensed transcriptionally inactive chromatin - Heterochromatin

BIO-12-CH04-0017 | Challenging | DNA structure

A researcher checks whether four linked observations have been interpreted correctly.

- I. Purine pairing with thymine corresponds to Adenine.
- II. Sugar present in DNA corresponds to Three.
- III. Number of hydrogen bonds in a G-C pair corresponds to Allolactose.
- IV. Inducer of the lac operon corresponds to Adenine.

Which combination survives the biological review?

- A. II only
- B. II, III and IV only
- C. I and III only
- D. I only

BIO-12-CH04-0018 | Challenging | Genetic code and translation

Evaluate the following source-based biological statements.

- I. Ribosomal site accepting an incoming aminoacyl-tRNA corresponds to UAA.
- II. A termination codon corresponds to UAA.
- III. Codon that usually initiates translation corresponds to AUG.
- IV. Hershey-Chase label used for phage DNA corresponds to A site.

Which option identifies exactly the correct statements?

- A. II and III only
- B. III and IV only
- C. I only
- D. II and IV only

BIO-12-CH04-0019 | Hard | Gene regulation and genomics

A reviewer compares several entries while checking 'Regulatory protein that blocks the operator'. Which entry must be corrected?

- A. Regulatory protein that blocks the operator - Nucleosome
- B. RNA carrying codons from DNA to ribosomes - mRNA
- C. Purine pairing with thymine - Adenine
- D. Core proteins around which DNA is wrapped - Histone octamer

BIO-12-CH04-0020 | Medium | DNA structure

A source table contains one accurate entry involving 'Approximate base pairs per turn of B-DNA'. Which option preserves a correct biological association?

- A. Scientist who first isolated nuclein - Repressor
- B. RNA forming a major structural and catalytic part of ribosomes - Allolactose
- C. RNA carrying codons from DNA to ribosomes - rRNA
- D. Approximate base pairs per turn of B-DNA - Ten

BIO-12-CH04-0021 | Easy | DNA replication

Which description is correctly represented by Helicase in this chapter?

- A. Mode of DNA replication
- B. Enzyme that abolished transformation in the Avery experiment
- C. Enzyme unwinding the DNA double helix
- D. Scientist who first isolated nuclein

BIO-12-CH04-0022 | Hard | Genetic code and translation

Two conclusions must both be valid to support the reference concept 'Ribosomal site accepting an incoming aminoacyl-tRNA'. Which option contains two accurate associations?

- A. Inducer of the lac operon - rRNA; RNA forming a major structural and catalytic part of ribosomes - Allolactose
- B. Ribosomal site accepting an incoming aminoacyl-tRNA - A site; Approximate base pairs per turn of B-DNA - Ten
- C. Enzyme that abolished transformation in the Avery experiment - Coding strand; DNA strand with a sequence corresponding to RNA except T for U - DNase
- D. RNA bearing an anticodon and amino acid - DNA ligase; Enzyme joining Okazaki fragments - tRNA

BIO-12-CH04-0023 | Hard | Transcription and RNA

A reviewer compares several entries while checking 'RNA bearing an anticodon and amino acid'. Which entry must be corrected?

- A. Highly condensed transcriptionally inactive chromatin - Heterochromatin
- B. RNA bearing an anticodon and amino acid - DNA ligase
- C. Hershey-Chase label used for phage DNA - Radioactive phosphorus-32
- D. Core proteins around which DNA is wrapped - Histone octamer

BIO-12-CH04-0024 | Hard | DNA packaging

A reviewer compares several entries while checking 'Highly condensed transcriptionally inactive chromatin'. Which entry must be corrected?

- A. Technique identifying individuals from variable DNA regions - DNA fingerprinting
- B. Scientist who first isolated nuclein - Friedrich Miescher
- C. RNA bearing an anticodon and amino acid - tRNA
- D. Highly condensed transcriptionally inactive chromatin - Degeneracy

BIO-12-CH04-0025 | Easy | Transcription and RNA

Which biological term or result is specifically identified by the description 'RNA forming a major structural and catalytic part of ribosomes'?

- A. rRNA
- B. Heterochromatin
- C. H1
- D. tRNA

BIO-12-CH04-0026 | Hard | DNA structure

Two conclusions must both be valid to support the reference concept 'Approximate base pairs per turn of B-DNA'. Which option contains two accurate associations?

- A. Ribosomal site accepting an incoming aminoacyl-tRNA - Allolactose; Inducer of the lac operon - A site
- B. RNA forming a major structural and catalytic part of ribosomes - A site; Ribosomal site accepting an incoming aminoacyl-tRNA - rRNA
- C. Approximate base pairs per turn of B-DNA - Ten; Scientist who first isolated nuclein - Friedrich Miescher
- D. Regulatory protein that blocks the operator - rRNA; RNA forming a major structural and catalytic part of ribosomes - Repressor

BIO-12-CH04-0027 | Medium | Transcription and RNA

Complete the statement accurately: RNA carrying codons from DNA to ribosomes corresponds to _____.

- A. DNA
- B. mRNA
- C. rRNA
- D. tRNA

BIO-12-CH04-0028 | Medium | DNA replication

A source table contains one accurate entry involving 'Enzyme joining Okazaki fragments'. Which option preserves a correct biological association?

- A. Transforming principle demonstrated in pneumococcus - Ten
- B. Approximate base pairs per turn of B-DNA - DNA ligase
- C. Enzyme joining Okazaki fragments - DNA ligase
- D. Highly condensed transcriptionally inactive chromatin - DNA

BIO-12-CH04-0029 | Easy | Gene regulation and genomics

Which biological term or result is specifically identified by the description 'Inducer of the lac operon'?

- A. Adenine
- B. Allolactose
- C. Genome
- D. H1

BIO-12-CH04-0030 | Hard | Genetic code and translation

A reviewer compares several entries while checking 'A termination codon'. Which entry must be corrected?

- A. Sugar present in DNA - Deoxyribose
- B. Highly condensed transcriptionally inactive chromatin - Heterochromatin
- C. A termination codon - Allolactose
- D. Technique identifying individuals from variable DNA regions - DNA fingerprinting

BIO-12-CH04-0031 | Hard | Genetic code and translation

Two conclusions must both be valid to support the reference concept 'Codon that usually initiates translation'. Which option contains two accurate associations?

- A. Codon that usually initiates translation - AUG; DNA strand with a sequence corresponding to RNA except T for U - Coding strand
- B. Scientist who first isolated nuclein - Helicase; Enzyme unwinding the DNA double helix - Friedrich Miescher
- C. RNA carrying codons from DNA to ribosomes - Three; Number of hydrogen bonds in a G-C pair - mRNA
- D. Enzyme unwinding the DNA double helix - mRNA; RNA carrying codons from DNA to ribosomes - Helicase

BIO-12-CH04-0032 | Challenging | DNA replication

A student submits four conclusions after integrating related chapter concepts.

- I. Enzyme joining Okazaki fragments corresponds to DNA ligase.
 - II. Enzyme unwinding the DNA double helix corresponds to Helicase.
 - III. Short DNA segments formed on the lagging strand corresponds to Okazaki fragments.
 - IV. Property in which several codons specify one amino acid corresponds to Degeneracy.
- Which combination should be accepted without correction?

- A. I and II only
- B. I and III only
- C. I only
- D. I, II, III and IV

BIO-12-CH04-0033 | Easy | Transcription and RNA

Which biological term or result is specifically identified by the description 'RNA carrying codons from DNA to ribosomes'?

- A. rRNA
- B. Okazaki fragments
- C. UAA
- D. mRNA

BIO-12-CH04-0034 | Easy | Evidence for genetic material

Which biological term or result is specifically identified by the description 'Hershey-Chase label used for phage DNA'?

- A. Radioactive phosphorus-32
- B. Histone octamer
- C. Friedrich Miescher
- D. Ten

BIO-12-CH04-0035 | Easy | DNA replication

Which description is correctly represented by Okazaki fragments in this chapter?

- A. Sugar present in DNA
- B. A termination codon
- C. Short DNA segments formed on the lagging strand
- D. Basic repeating unit of chromatin

BIO-12-CH04-0036 | Medium | DNA replication

Complete the statement accurately: Enzyme unwinding the DNA double helix corresponds to ____.

- A. Three
- B. Helicase
- C. DNase
- D. Okazaki fragments

BIO-12-CH04-0037 | Medium | Genetic code and translation

Complete the statement accurately: Ribosomal site accepting an incoming aminoacyl-tRNA corresponds to ____.

- A. Ten
- B. rRNA
- C. A site
- D. DNA ligase

BIO-12-CH04-0038 | Medium | Gene regulation and genomics

A source table contains one accurate entry involving 'Complete DNA content of an organism'. Which option preserves a correct biological association?

- A. Property in which several codons specify one amino acid - tRNA
- B. A termination codon - DNA ligase
- C. Complete DNA content of an organism - Genome
- D. Sugar present in DNA - UAA

BIO-12-CH04-0039 | Medium | Genetic code and translation

Complete the statement accurately: A termination codon corresponds to ____.

- A. Deoxyribose
- B. UAA
- C. DNA fingerprinting
- D. Allolactose

BIO-12-CH04-0040 | Medium | Genetic code and translation

A source table contains one accurate entry involving 'A termination codon'. Which option preserves a correct biological association?

- A. Inducer of the lac operon - Deoxyribose
- B. A termination codon - UAA
- C. Highly condensed transcriptionally inactive chromatin - Friedrich Miescher
- D. Scientist who first isolated nuclein - UAA

BIO-12-CH04-0041 | Challenging | DNA replication

During error analysis, four proposed associations are reviewed.

- I. Short DNA segments formed on the lagging strand corresponds to Okazaki fragments.
- II. Enzyme unwinding the DNA double helix corresponds to DNA ligase.
- III. Enzyme joining Okazaki fragments corresponds to Degeneracy.
- IV. Property in which several codons specify one amino acid corresponds to Degeneracy.

Which set contains all and only the valid conclusions?

- A. I, II and IV only
- B. II only
- C. I and IV only
- D. I, III and IV only

BIO-12-CH04-0042 | Easy | DNA packaging

Which description is correctly represented by Nucleosome in this chapter?

- A. Basic repeating unit of chromatin
- B. RNA carrying codons from DNA to ribosomes
- C. Linker histone
- D. Enzyme unwinding the DNA double helix

BIO-12-CH04-0043 | Hard | DNA replication

A reviewer compares several entries while checking 'Mode of DNA replication'. Which entry must be corrected?

- A. RNA carrying codons from DNA to ribosomes - mRNA
- B. Enzyme that abolished transformation in the Avery experiment - DNase
- C. Property in which several codons specify one amino acid - Degeneracy
- D. Mode of DNA replication - Radioactive phosphorus-32

BIO-12-CH04-0044 | Easy | DNA replication

Which description is correctly represented by DNA ligase in this chapter?

- A. Enzyme unwinding the DNA double helix
- B. A termination codon
- C. Mode of DNA replication
- D. Enzyme joining Okazaki fragments

BIO-12-CH04-0045 | Hard | DNA packaging

Two conclusions must both be valid to support the reference concept 'Basic repeating unit of chromatin'. Which option contains two accurate associations?

- A. Transforming principle demonstrated in pneumococcus - mRNA; RNA carrying codons from DNA to ribosomes - DNA
- B. Enzyme unwinding the DNA double helix - DNA; Transforming principle demonstrated in pneumococcus - Helicase
- C. Basic repeating unit of chromatin - Nucleosome; Property in which several codons specify one amino acid - Degeneracy
- D. Inducer of the lac operon - Three; Number of hydrogen bonds in a G-C pair - Allolactose

BIO-12-CH04-0046 | Medium | Evidence for genetic material

A learner encounters the observation 'Scientist who first isolated nuclein' while analysing evidence for genetic material. Which conclusion is supported by the prescribed biology?

- A. DNA
- B. Friedrich Miescher
- C. Adenine
- D. tRNA

BIO-12-CH04-0047 | Medium | Evidence for genetic material

Complete the statement accurately: Transforming principle demonstrated in pneumococcus corresponds to _____.

- A. DNase
- B. Okazaki fragments
- C. Semiconservative
- D. DNA

BIO-12-CH04-0048 | Medium | DNA structure

A source table contains one accurate entry involving 'Number of hydrogen bonds in a G-C pair'. Which option preserves a correct biological association?

- A. Number of hydrogen bonds in a G-C pair - Three
- B. RNA carrying codons from DNA to ribosomes - DNA
- C. Transforming principle demonstrated in pneumococcus - DNase
- D. Complete DNA content of an organism - mRNA

BIO-12-CH04-0049 | Medium | Gene regulation and genomics

Complete the statement accurately: Complete DNA content of an organism corresponds to _____.

- A. Repressor
- B. DNA fingerprinting
- C. UAA
- D. Genome

BIO-12-CH04-0050 | Hard | Genetic code and translation

A reviewer compares several entries while checking 'Ribosomal site accepting an incoming aminoacyl-tRNA'. Which entry must be corrected?

- A. Enzyme joining Okazaki fragments - DNA ligase
- B. RNA forming a major structural and catalytic part of ribosomes - rRNA
- C. RNA bearing an anticodon and amino acid - tRNA
- D. Ribosomal site accepting an incoming aminoacyl-tRNA - Ten

BIO-12-CH04-0051 | Hard | Gene regulation and genomics

A reviewer compares several entries while checking 'Technique identifying individuals from variable DNA regions'. Which entry must be corrected?

- A. Enzyme joining Okazaki fragments - DNA ligase
- B. Approximate base pairs per turn of B-DNA - Ten
- C. Technique identifying individuals from variable DNA regions - Friedrich Miescher
- D. Transforming principle demonstrated in pneumococcus - DNA

BIO-12-CH04-0052 | Medium | DNA structure

Complete the statement accurately: Sugar present in DNA corresponds to _____.

- A. Genome
- B. mRNA
- C. rRNA
- D. Deoxyribose

BIO-12-CH04-0053 | Hard | DNA structure

Two conclusions must both be valid to support the reference concept 'Sugar present in DNA'. Which option contains two accurate associations?

- A. Sugar present in DNA - Deoxyribose; Codon that usually initiates translation - AUG
- B. Mode of DNA replication - DNase; Enzyme that abolished transformation in the Avery experiment - Semiconservative
- C. RNA forming a major structural and catalytic part of ribosomes - Genome; Complete DNA content of an organism - rRNA
- D. RNA carrying codons from DNA to ribosomes - rRNA; RNA forming a major structural and catalytic part of ribosomes - mRNA

BIO-12-CH04-0054 | Medium | DNA packaging

Complete the statement accurately: Highly condensed transcriptionally inactive chromatin corresponds to _____.

- A. Degeneracy
- B. Heterochromatin
- C. Friedrich Miescher
- D. DNA fingerprinting

BIO-12-CH04-0055 | Hard | Genetic code and translation

Two conclusions must both be valid to support the reference concept 'Property in which several codons specify one amino acid'. Which option contains two accurate associations?

- A. Property in which several codons specify one amino acid - Degeneracy; Regulatory protein that blocks the operator - Repressor
- B. Core proteins around which DNA is wrapped - Adenine; Purine pairing with thymine - Histone octamer
- C. Technique identifying individuals from variable DNA regions - Helicase; Enzyme unwinding the DNA double helix - DNA fingerprinting
- D. RNA bearing an anticodon and amino acid - Genome; Complete DNA content of an organism - tRNA

BIO-12-CH04-0056 | Easy | Evidence for genetic material

Which description is correctly represented by DNase in this chapter?

- A. RNA bearing an anticodon and amino acid
- B. Enzyme that abolished transformation in the Avery experiment
- C. Scientist who first isolated nuclein
- D. Codon that usually initiates translation

BIO-12-CH04-0057 | Medium | DNA structure

A learner encounters the observation 'Sugar present in DNA' while analysing dna structure. Which conclusion is supported by the prescribed biology?

- A. Genome
- B. Deoxyribose
- C. Adenine
- D. mRNA

BIO-12-CH04-0058 | Medium | DNA packaging

A learner encounters the observation 'Linker histone' while analysing dna packaging. Which conclusion is supported by the prescribed biology?

- A. Ten
- B. Allolactose
- C. Histone octamer
- D. H1

BIO-12-CH04-0059 | Easy | Genetic code and translation

Which description is correctly represented by AUG in this chapter?

- A. Enzyme unwinding the DNA double helix
- B. RNA carrying codons from DNA to ribosomes
- C. Codon that usually initiates translation
- D. Number of hydrogen bonds in a G-C pair

BIO-12-CH04-0060 | Easy | Gene regulation and genomics

Which description is correctly represented by Genome in this chapter?

- A. A termination codon
- B. Complete DNA content of an organism
- C. Technique identifying individuals from variable DNA regions
- D. RNA bearing an anticodon and amino acid

BIO-12-CH04-0061 | Medium | DNA structure

A source table contains one accurate entry involving 'Purine pairing with thymine'. Which option preserves a correct biological association?

- A. Ribosomal site accepting an incoming aminoacyl-tRNA - Radioactive phosphorus-32
- B. Hershey-Chase label used for phage DNA - Adenine
- C. Enzyme unwinding the DNA double helix - A site
- D. Purine pairing with thymine - Adenine

BIO-12-CH04-0062 | Challenging | Evidence for genetic material

A researcher checks whether four linked observations have been interpreted correctly.

- I. Hershey-Chase label used for phage DNA corresponds to DNA.
- II. Transforming principle demonstrated in pneumococcus corresponds to DNA.
- III. Enzyme that abolished transformation in the Avery experiment corresponds to DNase.
- IV. Ribosomal site accepting an incoming aminoacyl-tRNA corresponds to A site.

Which combination survives the biological review?

- A. II, III and IV only
- B. I, II and III only
- C. III only
- D. I only

BIO-12-CH04-0063 | Easy | Gene regulation and genomics

Which biological term or result is specifically identified by the description 'Technique identifying individuals from variable DNA regions'?

- A. Friedrich Miescher
- B. Helicase
- C. DNA
- D. DNA fingerprinting

BIO-12-CH04-0064 | Hard | Evidence for genetic material

Two conclusions must both be valid to support the reference concept 'Hershey-Chase label used for phage DNA'.

Which option contains two accurate associations?

- A. Hershey-Chase label used for phage DNA - Radioactive phosphorus-32; RNA carrying codons from DNA to ribosomes - mRNA
- B. Purine pairing with thymine - Ten; Approximate base pairs per turn of B-DNA - Adenine
- C. Approximate base pairs per turn of B-DNA - Histone octamer; Core proteins around which DNA is wrapped - Ten
- D. Core proteins around which DNA is wrapped - A site; Ribosomal site accepting an incoming aminoacyl-tRNA - Histone octamer

BIO-12-CH04-0065 | Medium | Transcription and RNA

Complete the statement accurately: RNA bearing an anticodon and amino acid corresponds to _____.

- A. tRNA
- B. mRNA
- C. Radioactive phosphorus-32
- D. Adenine

BIO-12-CH04-0066 | Challenging | Evidence for genetic material

Evaluate the following source-based biological statements.

- I. Scientist who first isolated nuclein corresponds to Friedrich Miescher.
- II. Transforming principle demonstrated in pneumococcus corresponds to DNase.
- III. Enzyme that abolished transformation in the Avery experiment corresponds to DNase.
- IV. RNA carrying codons from DNA to ribosomes corresponds to Friedrich Miescher.

Which option identifies exactly the correct statements?

- A. I, II and III only
- B. I and III only
- C. II and IV only
- D. II only

BIO-12-CH04-0067 | Hard | Gene regulation and genomics

Two conclusions must both be valid to support the reference concept 'Complete DNA content of an organism'. Which option contains two accurate associations?

- A. Enzyme joining Okazaki fragments - Friedrich Miescher; Scientist who first isolated nuclein - DNA ligase
- B. Complete DNA content of an organism - Genome; Codon that usually initiates translation - AUG
- C. Sugar present in DNA - Degeneracy; Property in which several codons specify one amino acid - Deoxyribose
- D. Property in which several codons specify one amino acid - UAA; A termination codon - Degeneracy

BIO-12-CH04-0068 | Hard | Evidence for genetic material

Two conclusions must both be valid to support the reference concept 'Enzyme that abolished transformation in the Avery experiment'. Which option contains two accurate associations?

- A. Basic repeating unit of chromatin - Ten; Approximate base pairs per turn of B-DNA - Nucleosome
- B. Approximate base pairs per turn of B-DNA - AUG; Codon that usually initiates translation - Ten
- C. Ribosomal site accepting an incoming aminoacyl-tRNA - Deoxyribose; Sugar present in DNA - A site
- D. Enzyme that abolished transformation in the Avery experiment - DNase; RNA bearing an anticodon and amino acid - tRNA

BIO-12-CH04-0069 | Medium | DNA replication

Complete the statement accurately: Enzyme joining Okazaki fragments corresponds to _____.

- A. A site
- B. Semiconservative
- C. Heterochromatin
- D. DNA ligase

BIO-12-CH04-0070 | Medium | Evidence for genetic material

A learner encounters the observation 'Hershey-Chase label used for phage DNA' while analysing evidence for genetic material. Which conclusion is supported by the prescribed biology?

- A. DNase
- B. Radioactive phosphorus-32
- C. A site
- D. Ten

BIO-12-CH04-0071 | Medium | Gene regulation and genomics

Complete the statement accurately: Technique identifying individuals from variable DNA regions corresponds to _____.

- A. Helicase
- B. DNA fingerprinting
- C. Ten
- D. Allolactose

BIO-12-CH04-0072 | Easy | Genetic code and translation

Which biological term or result is specifically identified by the description 'Ribosomal site accepting an incoming aminoacyl-tRNA'?

- A. Allolactose
- B. DNA ligase
- C. A site
- D. AUG

BIO-12-CH04-0073 | Easy | DNA structure

Which description is correctly represented by Ten in this chapter?

- A. Scientist who first isolated nuclein
- B. RNA forming a major structural and catalytic part of ribosomes
- C. Regulatory protein that blocks the operator
- D. Approximate base pairs per turn of B-DNA

BIO-12-CH04-0074 | Medium | DNA packaging

A source table contains one accurate entry involving 'Basic repeating unit of chromatin'. Which option preserves a correct biological association?

- A. Enzyme that abolished transformation in the Avery experiment - DNA
- B. Inducer of the lac operon - Nucleosome
- C. Transforming principle demonstrated in pneumococcus - Allolactose
- D. Basic repeating unit of chromatin - Nucleosome

BIO-12-CH04-0075 | Easy | Transcription and RNA

Which biological term or result is specifically identified by the description 'RNA bearing an anticodon and amino acid'?

- A. Heterochromatin
- B. tRNA
- C. DNA ligase
- D. mRNA

BIO-12-CH04-0076 | Easy | Genetic code and translation

Which biological term or result is specifically identified by the description 'A termination codon'?

- A. Genome
- B. Deoxyribose
- C. UAA
- D. DNA fingerprinting

BIO-12-CH04-0077 | Hard | DNA replication

Two conclusions must both be valid to support the reference concept 'Short DNA segments formed on the lagging strand'. Which option contains two accurate associations?

- A. RNA forming a major structural and catalytic part of ribosomes - DNA fingerprinting; Technique identifying individuals from variable DNA regions - rRNA
- B. Sugar present in DNA - UAA; A termination codon - Deoxyribose
- C. Short DNA segments formed on the lagging strand - Okazaki fragments; Highly condensed transcriptionally inactive chromatin - Heterochromatin
- D. Technique identifying individuals from variable DNA regions - Allolactose; Inducer of the lac operon - DNA fingerprinting

BIO-12-CH04-0078 | Easy | DNA structure

Which biological term or result is specifically identified by the description 'Sugar present in DNA'?

- A. Adenine
- B. AUG
- C. Ten
- D. Deoxyribose

BIO-12-CH04-0079 | Easy | Evidence for genetic material

Which biological term or result is specifically identified by the description 'Transforming principle demonstrated in pneumococcus'?

- A. DNA fingerprinting
- B. DNA
- C. Okazaki fragments
- D. Semiconservative

BIO-12-CH04-0080 | Medium | DNA replication

A learner encounters the observation 'Enzyme unwinding the DNA double helix' while analysing dna replication. Which conclusion is supported by the prescribed biology?

- A. Semiconservative
- B. Three
- C. Helicase
- D. Heterochromatin

BIO-12-CH04-0081 | Medium | Evidence for genetic material

Complete the statement accurately: Hershey-Chase label used for phage DNA corresponds to ____.

- A. Radioactive phosphorus-32
- B. Ten
- C. DNase
- D. mRNA

BIO-12-CH04-0082 | Hard | DNA replication

Two conclusions must both be valid to support the reference concept 'Enzyme joining Okazaki fragments'. Which option contains two accurate associations?

- A. Approximate base pairs per turn of B-DNA - DNA fingerprinting; Technique identifying individuals from variable DNA regions - Ten
- B. Transforming principle demonstrated in pneumococcus - DNase; Enzyme that abolished transformation in the Avery experiment - DNA
- C. Highly condensed transcriptionally inactive chromatin - UAA; A termination codon - Heterochromatin
- D. Enzyme joining Okazaki fragments - DNA ligase; Ribosomal site accepting an incoming aminoacyl-tRNA - A site

BIO-12-CH04-0083 | Medium | Evidence for genetic material

A learner encounters the observation 'Enzyme that abolished transformation in the Avery experiment' while analysing evidence for genetic material. Which conclusion is supported by the prescribed biology?

- A. AUG
- B. Nucleosome
- C. Radioactive phosphorus-32
- D. DNase

BIO-12-CH04-0084 | Easy | DNA replication

Which biological term or result is specifically identified by the description 'Enzyme joining Okazaki fragments'?

- A. DNA ligase
- B. Helicase
- C. Semiconservative
- D. DNase

BIO-12-CH04-0085 | Hard | Gene regulation and genomics

A reviewer compares several entries while checking 'Inducer of the lac operon'. Which entry must be corrected?

- A. Highly condensed transcriptionally inactive chromatin - Heterochromatin
- B. Scientist who first isolated nuclein - Friedrich Miescher
- C. Inducer of the lac operon - A site
- D. Purine pairing with thymine - Adenine

BIO-12-CH04-0086 | Challenging | DNA structure

A student submits four conclusions after integrating related chapter concepts.

- I. Number of hydrogen bonds in a G-C pair corresponds to Three.
- II. Sugar present in DNA corresponds to Ten.
- III. Approximate base pairs per turn of B-DNA corresponds to Ten.
- IV. Ribosomal site accepting an incoming aminoacyl-tRNA corresponds to A site.

Which combination should be accepted without correction?

- A. II and III only
- B. II, III and IV only
- C. I and II only
- D. I, III and IV only

BIO-12-CH04-0087 | Medium | Gene regulation and genomics

A learner encounters the observation 'Regulatory protein that blocks the operator' while analysing gene regulation and genomics. Which conclusion is supported by the prescribed biology?

- A. Adenine
- B. Repressor
- C. Nucleosome
- D. DNA fingerprinting

BIO-12-CH04-0088 | Hard | Transcription and RNA

A reviewer compares several entries while checking 'RNA forming a major structural and catalytic part of ribosomes'. Which entry must be corrected?

- A. RNA forming a major structural and catalytic part of ribosomes - Ten
- B. Linker histone - H1
- C. Purine pairing with thymine - Adenine
- D. Basic repeating unit of chromatin - Nucleosome

BIO-12-CH04-0089 | Easy | DNA replication

Which biological term or result is specifically identified by the description 'Mode of DNA replication'?

- A. Helicase
- B. mRNA
- C. Semiconservative
- D. Adenine

BIO-12-CH04-0090 | Challenging | Transcription and RNA

During error analysis, four proposed associations are reviewed.

- I. RNA forming a major structural and catalytic part of ribosomes corresponds to rRNA.
- II. DNA strand with a sequence corresponding to RNA except T for U corresponds to Coding strand.
- III. RNA bearing an anticodon and amino acid corresponds to DNA.
- IV. Transforming principle demonstrated in pneumococcus corresponds to rRNA.

Which set contains all and only the valid conclusions?

- A. I and IV only
- B. I and II only
- C. I, II and IV only
- D. I, II and III only

BIO-12-CH04-0091 | Medium | Transcription and RNA

Complete the statement accurately: RNA forming a major structural and catalytic part of ribosomes corresponds to _____.

- A. rRNA
- B. Ten
- C. tRNA
- D. mRNA

BIO-12-CH04-0092 | Easy | DNA packaging

Which biological term or result is specifically identified by the description 'Highly condensed transcriptionally inactive chromatin'?

- A. Friedrich Miescher
- B. Heterochromatin
- C. Nucleosome
- D. Three

BIO-12-CH04-0093 | Medium | Evidence for genetic material

A source table contains one accurate entry involving 'Enzyme that abolished transformation in the Avery experiment'. Which option preserves a correct biological association?

- A. Enzyme that abolished transformation in the Avery experiment - DNase
- B. RNA bearing an anticodon and amino acid - Nucleosome
- C. Ribosomal site accepting an incoming aminoacyl-tRNA - DNase
- D. Regulatory protein that blocks the operator - Ten

BIO-12-CH04-0094 | Easy | Transcription and RNA

Which description is correctly represented by tRNA in this chapter?

- A. DNA strand with a sequence corresponding to RNA except T for U
- B. Highly condensed transcriptionally inactive chromatin
- C. RNA bearing an anticodon and amino acid
- D. Hershey-Chase label used for phage DNA

BIO-12-CH04-0095 | Medium | DNA replication

A learner encounters the observation 'Mode of DNA replication' while analysing dna replication. Which conclusion is supported by the prescribed biology?

- A. Okazaki fragments
 - B. Radioactive phosphorus-32
 - C. Semiconservative
 - D. DNase
-

BIO-12-CH04-0096 | Medium | Genetic code and translation

Complete the statement accurately: Property in which several codons specify one amino acid corresponds to _____.

- A. Degeneracy
 - B. Adenine
 - C. tRNA
 - D. Repressor
-

BIO-12-CH04-0097 | Medium | Genetic code and translation

A source table contains one accurate entry involving 'Codon that usually initiates translation'. Which option preserves a correct biological association?

- A. DNA strand with a sequence corresponding to RNA except T for U - Helicase
 - B. RNA carrying codons from DNA to ribosomes - tRNA
 - C. Codon that usually initiates translation - AUG
 - D. Number of hydrogen bonds in a G-C pair - Allolactose
-

BIO-12-CH04-0098 | Easy | Evidence for genetic material

Which description is correctly represented by DNA in this chapter?

- A. Highly condensed transcriptionally inactive chromatin
 - B. Short DNA segments formed on the lagging strand
 - C. Inducer of the lac operon
 - D. Transforming principle demonstrated in pneumococcus
-

BIO-12-CH04-0099 | Easy | Gene regulation and genomics

Which biological term or result is specifically identified by the description 'Regulatory protein that blocks the operator'?

- A. DNA fingerprinting
 - B. Repressor
 - C. Radioactive phosphorus-32
 - D. Adenine
-

BIO-12-CH04-0100 | Medium | Genetic code and translation

A learner encounters the observation 'Codon that usually initiates translation' while analysing genetic code and translation. Which conclusion is supported by the prescribed biology?

- A. Helicase
 - B. Friedrich Miescher
 - C. AUG
 - D. UAA
-

BIO-12-CH04-0101 | Hard | DNA replication

Two conclusions must both be valid to support the reference concept 'Enzyme unwinding the DNA double helix'. Which option contains two accurate associations?

- A. Inducer of the lac operon - Adenine; Purine pairing with thymine - Allolactose
- B. Scientist who first isolated nuclein - Allolactose; Inducer of the lac operon - Friedrich Miescher
- C. Number of hydrogen bonds in a G-C pair - Friedrich Miescher; Scientist who first isolated nuclein - Three
- D. Enzyme unwinding the DNA double helix - Helicase; Enzyme that abolished transformation in the Avery experiment - DNase

BIO-12-CH04-0102 | Medium | DNA packaging

A source table contains one accurate entry involving 'Highly condensed transcriptionally inactive chromatin'. Which option preserves a correct biological association?

- A. Technique identifying individuals from variable DNA regions - tRNA
- B. Highly condensed transcriptionally inactive chromatin - Heterochromatin
- C. DNA strand with a sequence corresponding to RNA except T for U - Heterochromatin
- D. Scientist who first isolated nuclein - Three

BIO-12-CH04-0103 | Medium | Transcription and RNA

A source table contains one accurate entry involving 'RNA forming a major structural and catalytic part of ribosomes'. Which option preserves a correct biological association?

- A. RNA forming a major structural and catalytic part of ribosomes - rRNA
- B. Highly condensed transcriptionally inactive chromatin - Nucleosome
- C. Linker histone - Adenine
- D. Short DNA segments formed on the lagging strand - rRNA

BIO-12-CH04-0104 | Easy | Genetic code and translation

Which description is correctly represented by Degeneracy in this chapter?

- A. RNA bearing an anticodon and amino acid
- B. A termination codon
- C. Property in which several codons specify one amino acid
- D. Core proteins around which DNA is wrapped

BIO-12-CH04-0105 | Hard | DNA structure

Two conclusions must both be valid to support the reference concept 'Number of hydrogen bonds in a G-C pair'. Which option contains two accurate associations?

- A. Enzyme joining Okazaki fragments - mRNA; RNA carrying codons from DNA to ribosomes - DNA ligase
- B. Number of hydrogen bonds in a G-C pair - Three; Complete DNA content of an organism - Genome
- C. RNA carrying codons from DNA to ribosomes - DNA fingerprinting; Technique identifying individuals from variable DNA regions - mRNA
- D. Transforming principle demonstrated in pneumococcus - Allolactose; Inducer of the lac operon - DNA

BIO-12-CH04-0106 | Medium | DNA replication

A source table contains one accurate entry involving 'Enzyme unwinding the DNA double helix'. Which option preserves a correct biological association?

- A. Enzyme unwinding the DNA double helix - Helicase
- B. Inducer of the lac operon - Degeneracy
- C. Number of hydrogen bonds in a G-C pair - Allolactose
- D. Purine pairing with thymine - Helicase

BIO-12-CH04-0107 | Hard | DNA structure

A reviewer compares several entries while checking 'Approximate base pairs per turn of B-DNA'. Which entry must be corrected?

- A. RNA forming a major structural and catalytic part of ribosomes - rRNA
- B. Approximate base pairs per turn of B-DNA - Friedrich Miescher
- C. RNA carrying codons from DNA to ribosomes - mRNA
- D. Regulatory protein that blocks the operator - Repressor

BIO-12-CH04-0108 | Easy | DNA structure

Which description is correctly represented by Three in this chapter?

- A. Purine pairing with thymine
- B. Number of hydrogen bonds in a G-C pair
- C. RNA carrying codons from DNA to ribosomes
- D. Technique identifying individuals from variable DNA regions

BIO-12-CH04-0109 | Hard | Evidence for genetic material

A reviewer compares several entries while checking 'Transforming principle demonstrated in pneumococcus'. Which entry must be corrected?

- A. Mode of DNA replication - Semiconservative
- B. Short DNA segments formed on the lagging strand - Okazaki fragments
- C. Transforming principle demonstrated in pneumococcus - Heterochromatin
- D. Inducer of the lac operon - Allolactose

BIO-12-CH04-0110 | Medium | Gene regulation and genomics

A learner encounters the observation 'Inducer of the lac operon' while analysing gene regulation and genomics. Which conclusion is supported by the prescribed biology?

- A. H1
- B. Genome
- C. Repressor
- D. Allolactose

BIO-12-CH04-0111 | Easy | DNA structure

Which description is correctly represented by Deoxyribose in this chapter?

- A. Sugar present in DNA
- B. Codon that usually initiates translation
- C. Complete DNA content of an organism
- D. Purine pairing with thymine

BIO-12-CH04-0112 | Medium | Transcription and RNA

A learner encounters the observation 'RNA bearing an anticodon and amino acid' while analysing transcription and rna. Which conclusion is supported by the prescribed biology?

- A. Radioactive phosphorus-32
- B. mRNA
- C. tRNA
- D. Coding strand

BIO-12-CH04-0113 | Challenging | Transcription and RNA

A researcher checks whether four linked observations have been interpreted correctly.

- I. DNA strand with a sequence corresponding to RNA except T for U corresponds to Coding strand.
- II. RNA carrying codons from DNA to ribosomes corresponds to mRNA.
- III. RNA bearing an anticodon and amino acid corresponds to UAA.
- IV. A termination codon corresponds to UAA.

Which combination survives the biological review?

- A. IV only
- B. I and II only
- C. I only
- D. I, II and IV only

BIO-12-CH04-0114 | Challenging | Gene regulation and genomics

Evaluate the following source-based biological statements.

- I. Regulatory protein that blocks the operator corresponds to Genome.
- II. Complete DNA content of an organism corresponds to Allolactose.
- III. Inducer of the lac operon corresponds to UAA.
- IV. A termination codon corresponds to UAA.

Which option identifies exactly the correct statements?

- A. III and IV only
- B. I and III only
- C. I only
- D. IV only

BIO-12-CH04-0115 | Hard | DNA packaging

A reviewer compares several entries while checking 'Core proteins around which DNA is wrapped'. Which entry must be corrected?

- A. Property in which several codons specify one amino acid - Degeneracy
- B. Core proteins around which DNA is wrapped - Friedrich Miescher
- C. Enzyme unwinding the DNA double helix - Helicase
- D. RNA bearing an anticodon and amino acid - tRNA

BIO-12-CH04-0116 | Medium | Evidence for genetic material

Complete the statement accurately: Scientist who first isolated nuclein corresponds to _____.

- A. Friedrich Miescher
- B. DNase
- C. DNA
- D. Allolactose

BIO-12-CH04-0117 | Medium | Gene regulation and genomics

A learner encounters the observation 'Technique identifying individuals from variable DNA regions' while analysing gene regulation and genomics. Which conclusion is supported by the prescribed biology?

- A. DNA fingerprinting
- B. Friedrich Miescher
- C. mRNA
- D. Helicase

BIO-12-CH04-0118 | Medium | Evidence for genetic material

A source table contains one accurate entry involving 'Transforming principle demonstrated in pneumococcus'. Which option preserves a correct biological association?

- A. DNA strand with a sequence corresponding to RNA except T for U - DNA
- B. Short DNA segments formed on the lagging strand - DNA fingerprinting
- C. Transforming principle demonstrated in pneumococcus - DNA
- D. Inducer of the lac operon - Semiconservative

BIO-12-CH04-0119 | Challenging | Genetic code and translation

A student submits four conclusions after integrating related chapter concepts.

- I. Property in which several codons specify one amino acid corresponds to Degeneracy.
 - II. Codon that usually initiates translation corresponds to AUG.
 - III. A termination codon corresponds to UAA.
 - IV. Technique identifying individuals from variable DNA regions corresponds to Degeneracy.
- Which combination should be accepted without correction?

- A. I, II and III only
- B. I, III and IV only
- C. I only
- D. II, III and IV only

BIO-12-CH04-0120 | Medium | Gene regulation and genomics

A source table contains one accurate entry involving 'Regulatory protein that blocks the operator'. Which option preserves a correct biological association?

- A. A termination codon - mRNA
- B. Regulatory protein that blocks the operator - Repressor
- C. Core proteins around which DNA is wrapped - Repressor
- D. Basic repeating unit of chromatin - UAA

BIO-12-CH04-0121 | Easy | DNA packaging

Which biological term or result is specifically identified by the description 'Core proteins around which DNA is wrapped'?

- A. Histone octamer
- B. tRNA
- C. Helicase
- D. Repressor

BIO-12-CH04-0122 | Medium | DNA packaging

A source table contains one accurate entry involving 'Core proteins around which DNA is wrapped'. Which option preserves a correct biological association?

- A. Core proteins around which DNA is wrapped - Histone octamer
- B. RNA bearing an anticodon and amino acid - Helicase
- C. Regulatory protein that blocks the operator - Radioactive phosphorus-32
- D. Hershey-Chase label used for phage DNA - Degeneracy

BIO-12-CH04-0123 | Hard | Transcription and RNA

Two conclusions must both be valid to support the reference concept 'RNA bearing an anticodon and amino acid'. Which option contains two accurate associations?

- A. Hershey-Chase label used for phage DNA - Adenine; Purine pairing with thymine - Radioactive phosphorus-32
- B. Core proteins around which DNA is wrapped - A site; Ribosomal site accepting an incoming aminoacyl-tRNA - Histone octamer
- C. RNA bearing an anticodon and amino acid - tRNA; Enzyme joining Okazaki fragments - DNA ligase
- D. Purine pairing with thymine - DNA fingerprinting; Technique identifying individuals from variable DNA regions - Adenine

BIO-12-CH04-0124 | Medium | DNA structure

A learner encounters the observation 'Purine pairing with thymine' while analysing dna structure. Which conclusion is supported by the prescribed biology?

- A. A site
- B. Nucleosome
- C. Adenine
- D. Allolactose

BIO-12-CH04-0125 | Easy | DNA packaging

Which biological term or result is specifically identified by the description 'Basic repeating unit of chromatin'?

- A. DNase
- B. H1
- C. Histone octamer
- D. Nucleosome

BIO-12-CH04-0126 | Easy | Genetic code and translation

Which description is correctly represented by A site in this chapter?

- A. Property in which several codons specify one amino acid
- B. Enzyme joining Okazaki fragments
- C. Ribosomal site accepting an incoming aminoacyl-tRNA
- D. Codon that usually initiates translation

BIO-12-CH04-0127 | Hard | DNA packaging

A reviewer compares several entries while checking 'Basic repeating unit of chromatin'. Which entry must be corrected?

- A. RNA carrying codons from DNA to ribosomes - mRNA
- B. Enzyme that abolished transformation in the Avery experiment - DNase
- C. Transforming principle demonstrated in pneumococcus - DNA
- D. Basic repeating unit of chromatin - Degeneracy

BIO-12-CH04-0128 | Medium | Transcription and RNA

A source table contains one accurate entry involving 'DNA strand with a sequence corresponding to RNA except T for U'. Which option preserves a correct biological association?

- A. DNA strand with a sequence corresponding to RNA except T for U - Coding strand
- B. Codon that usually initiates translation - UAA
- C. A termination codon - Radioactive phosphorus-32
- D. Hershey-Chase label used for phage DNA - Allolactose

BIO-12-CH04-0129 | Easy | Evidence for genetic material

Which description is correctly represented by Radioactive phosphorus-32 in this chapter?

- A. Core proteins around which DNA is wrapped
- B. Purine pairing with thymine
- C. Hershey-Chase label used for phage DNA
- D. Ribosomal site accepting an incoming aminoacyl-tRNA

BIO-12-CH04-0130 | Hard | Gene regulation and genomics

A reviewer compares several entries while checking 'Complete DNA content of an organism'. Which entry must be corrected?

- A. Enzyme joining Okazaki fragments - DNA ligase
- B. Complete DNA content of an organism - AUG
- C. Property in which several codons specify one amino acid - Degeneracy
- D. Sugar present in DNA - Deoxyribose

BIO-12-CH04-0131 | Easy | DNA structure

Which biological term or result is specifically identified by the description 'Approximate base pairs per turn of B-DNA'?

- A. Three
 - B. Repressor
 - C. Friedrich Miescher
 - D. Ten
-

BIO-12-CH04-0132 | Challenging | DNA packaging

During error analysis, four proposed associations are reviewed.

- I. Highly condensed transcriptionally inactive chromatin corresponds to Histone octamer.
- II. Core proteins around which DNA is wrapped corresponds to Nucleosome.
- III. Basic repeating unit of chromatin corresponds to Nucleosome.
- IV. Hershey-Chase label used for phage DNA corresponds to Heterochromatin.

Which set contains all and only the valid conclusions?

- A. III only
 - B. III and IV only
 - C. II and IV only
 - D. II and III only
-

Standard 12 | Chapter 05

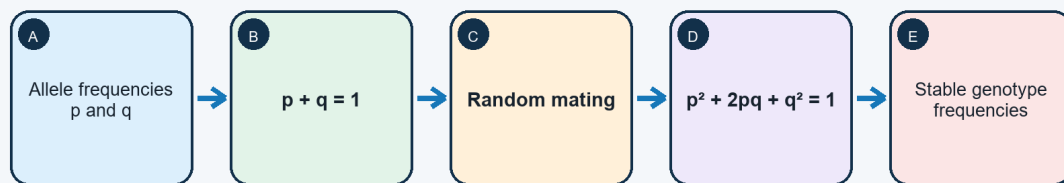
Origin and Evolution of Life

BIO-12-CH05-0001 | Challenging | Hardy-Weinberg equilibrium

A severe bottleneck changes allele frequencies randomly. Which Hardy-Weinberg condition is most directly violated?

Hardy-Weinberg equilibrium

Expected genotype frequencies in a two-allele population



Mutation, migration, selection, drift or non-random mating can disturb the equilibrium.

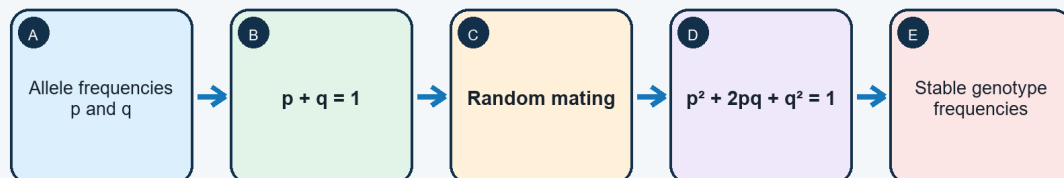
- A. Allele frequencies must sum to one
- B. Absence of genetic drift in a large population
- C. Diploid organisms possess two alleles
- D. Dominant alleles are always common

BIO-12-CH05-0002 | Medium | Hardy-Weinberg equilibrium

If $p = 0.7$ and $q = 0.3$, what heterozygote frequency is expected at equilibrium?

Hardy-Weinberg equilibrium

Expected genotype frequencies in a two-allele population



Mutation, migration, selection, drift or non-random mating can disturb the equilibrium.

- A. $p^2 = 0.49$
- B. $q^2 = 0.09$
- C. $p + q = 0.21$
- D. $2pq = 0.42$

BIO-12-CH05-0003 | Medium | Evidence of evolution

Complete the statement accurately: Developmental similarities among embryos corresponds to _____.

- A. Gene flow
- B. Embryological evidence
- C. Genetic drift
- D. Holocene

BIO-12-CH05-0004 | Challenging | Geological history

Evaluate the following source-based biological statements.

- I. Era known for dominance of reptiles corresponds to Mesozoic.
- II. Current geological epoch corresponds to Holocene.
- III. Era in which mammals diversified extensively corresponds to Inheritance of acquired characters.
- IV. Lamarckian mechanism corresponds to Mesozoic.

Which option identifies exactly the correct statements?

- A. I, II, III and IV
- B. II and IV only
- C. II only
- D. I and II only

BIO-12-CH05-0005 | Medium | Origin of life

A source table contains one accurate entry involving 'Miller-Urey experiment produced'. Which option preserves a correct biological association?

- A. Modern synthetic theory combines selection with - Homo sapiens
- B. Same basic structure with different functions - Reproductive isolation
- C. Hardy-Weinberg sum of allele frequencies - Population genetics
- D. Miller-Urey experiment produced - Organic molecules including amino acids

BIO-12-CH05-0006 | Medium | Human evolution

A learner encounters the observation 'Genus of the famous fossil Lucy' while analysing human evolution. Which conclusion is supported by the prescribed biology?

- A. Australopithecus
- B. Mutation
- C. Holocene
- D. Adaptive radiation

BIO-12-CH05-0007 | Challenging | Evidence of evolution

A student submits four conclusions after integrating related chapter concepts.

- I. Reduced structures with little current function corresponds to Vestigial organs.
- II. Same basic structure with different functions corresponds to Homologous organs.
- III. Different origin but similar function corresponds to Biogenesis.
- IV. Theory proposing life came from pre-existing life corresponds to Biogenesis.

Which combination should be accepted without correction?

- A. II only
- B. I, II and IV only
- C. IV only
- D. I only

BIO-12-CH05-0008 | Medium | Geological history

A learner encounters the observation 'Period associated with the first birds' while analysing geological history. Which conclusion is supported by the prescribed biology?

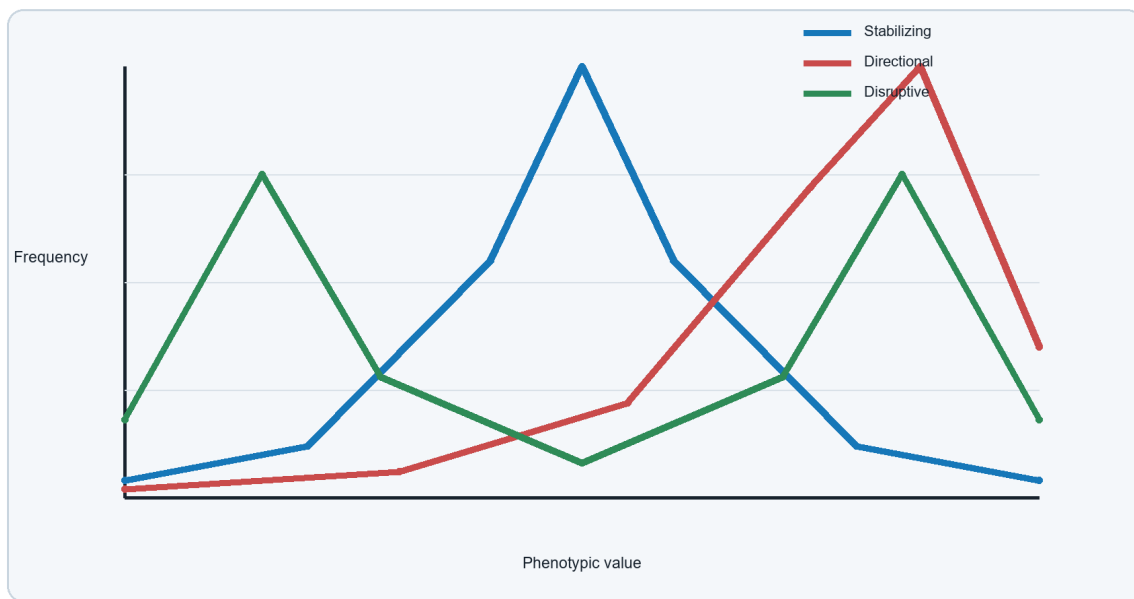
- A. Jurassic
- B. Cenozoic
- C. Chemical evolution
- D. Mutation

BIO-12-CH05-0009 | Easy | Three modes of natural selection

Which curve represents selection favouring intermediate phenotypes and reducing both extremes?

Three modes of natural selection

Trait-frequency distributions before and after selection



- A. Directional selection
- B. Stabilizing selection
- C. Disruptive selection
- D. Artificial hybridization

BIO-12-CH05-0010 | Medium | Speciation and adaptation

A source table contains one accurate entry involving 'Barrier preventing gene flow between populations'. Which option preserves a correct biological association?

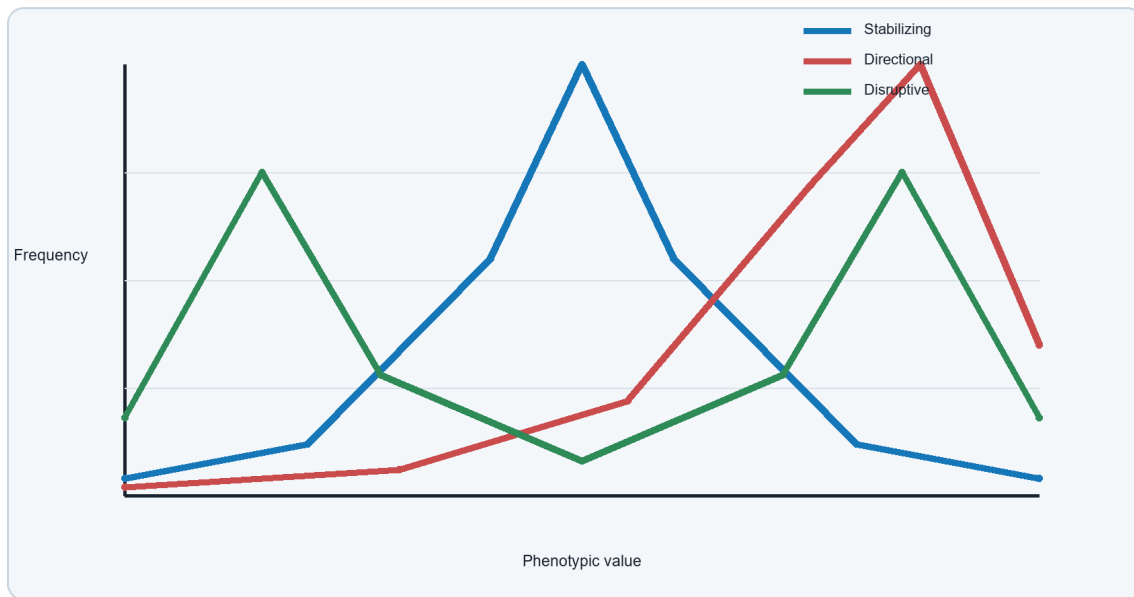
- A. Early human associated with systematic tool use - Reproductive isolation
- B. Barrier preventing gene flow between populations - Reproductive isolation
- C. Random change in allele frequency, strongest in small populations - Inheritance of acquired characters
- D. Current geological epoch - Genetic drift

BIO-12-CH05-0011 | Hard | Three modes of natural selection

Which selection mode in the figure can split a population by favouring both phenotypic extremes?

Three modes of natural selection

Trait-frequency distributions before and after selection



- A. Stabilizing selection
- B. Directional selection
- C. No selection
- D. Disruptive selection

BIO-12-CH05-0012 | Easy | Population genetics

Which description is correctly represented by Gene flow in this chapter?

- A. Random change in allele frequency, strongest in small populations
- B. Current geological epoch
- C. Same basic structure with different functions
- D. Movement of alleles between populations

BIO-12-CH05-0013 | Medium | Geological history

A learner encounters the observation 'Current geological epoch' while analysing geological history. Which conclusion is supported by the prescribed biology?

- A. Holocene
- B. Embryological evidence
- C. Population genetics
- D. Cenozoic

BIO-12-CH05-0014 | Challenging | Population genetics

During error analysis, four proposed associations are reviewed.

- I. Hardy-Weinberg genotype equation corresponds to $p + q = 1$.
- II. Hardy-Weinberg sum of allele frequencies corresponds to Gene flow.
- III. Movement of alleles between populations corresponds to Stabilizing selection.
- IV. Selection favouring the average phenotype corresponds to Stabilizing selection.

Which set contains all and only the valid conclusions?

- A. I, II and III only
- B. I, II and IV only
- C. III only
- D. IV only

BIO-12-CH05-0015 | Easy | Origin of life

Which description is correctly represented by Coacervates in this chapter?

- A. Genus of the famous fossil Lucy
- B. Early aggregates of organic molecules in water
- C. Hardy-Weinberg genotype equation
- D. Era in which mammals diversified extensively

BIO-12-CH05-0016 | Medium | Population genetics

A learner encounters the observation 'Random change in allele frequency, strongest in small populations' while analysing population genetics. Which conclusion is supported by the prescribed biology?

- A. Embryological evidence
- B. $p^2 + 2pq + q^2 = 1$
- C. Genetic drift
- D. Stabilizing selection

BIO-12-CH05-0017 | Hard | Geological history

A reviewer compares several entries while checking 'Current geological epoch'. Which entry must be corrected?

- A. Current geological epoch - Stabilizing selection
- B. Random change in allele frequency, strongest in small populations - Genetic drift
- C. Modern synthetic theory combines selection with - Population genetics
- D. Reduced structures with little current function - Vestigial organs

BIO-12-CH05-0018 | Medium | Evolutionary theories

Complete the statement accurately: Darwinian mechanism corresponds to _____.

- A. Population genetics
- B. Coacervates
- C. Natural selection
- D. Inheritance of acquired characters

BIO-12-CH05-0019 | Medium | Human evolution

A source table contains one accurate entry involving 'Early human associated with systematic tool use'. Which option preserves a correct biological association?

- A. Early human associated with systematic tool use - Homo habilis
- B. Selection favouring the average phenotype - Cenozoic
- C. Era in which mammals diversified extensively - Biogenesis
- D. De Vries emphasized - $p^2 + 2pq + q^2 = 1$

BIO-12-CH05-0020 | Easy | Human evolution

Which description is correctly represented by Homo sapiens in this chapter?

- A. Genus of the famous fossil Lucy
- B. Scientific name of modern humans
- C. Same basic structure with different functions
- D. De Vries emphasized

BIO-12-CH05-0021 | Hard | Evidence of evolution

A reviewer compares several entries while checking 'Reduced structures with little current function'. Which entry must be corrected?

- A. Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$
- B. Reduced structures with little current function - Coacervates
- C. Era in which mammals diversified extensively - Cenozoic
- D. De Vries emphasized - Mutation

BIO-12-CH05-0022 | Medium | Population genetics

Complete the statement accurately: Hardy-Weinberg sum of allele frequencies corresponds to _____.

- A. Holocene
- B. $p^2 + 2pq + q^2 = 1$
- C. $p + q = 1$
- D. Embryological evidence

BIO-12-CH05-0023 | Hard | Speciation and adaptation

Two conclusions must both be valid to support the reference concept 'Selection favouring both extreme phenotypes'. Which option contains two accurate associations?

- A. Different origin but similar function - Biogenesis; Theory proposing life came from pre-existing life - Analogous organs
- B. Selection favouring both extreme phenotypes - Disruptive selection; De Vries emphasized - Mutation
- C. Theory proposing life came from pre-existing life - Population genetics; Modern synthetic theory combines selection with - Biogenesis
- D. Random change in allele frequency, strongest in small populations - Analogous organs; Different origin but similar function - Genetic drift

BIO-12-CH05-0024 | Hard | Speciation and adaptation

Two conclusions must both be valid to support the reference concept 'Formation of different species from a common ancestor in one region'. Which option contains two accurate associations?

- A. Era in which mammals diversified extensively - Embryological evidence; Developmental similarities among embryos - Cenozoic
- B. Formation of different species from a common ancestor in one region - Adaptive radiation; Genus of the famous fossil Lucy - Australopithecus
- C. Reduced structures with little current function - Biogenesis; Theory proposing life came from pre-existing life - Vestigial organs
- D. Developmental similarities among embryos - Chemical evolution; Oparin-Haldane proposal for origin of life - Embryological evidence

BIO-12-CH05-0025 | Hard | Human evolution

A reviewer compares several entries while checking 'Early human associated with systematic tool use'. Which entry must be corrected?

- A. Miller-Urey experiment produced - Organic molecules including amino acids
- B. De Vries emphasized - Mutation
- C. Early human associated with systematic tool use - Population genetics
- D. Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$

BIO-12-CH05-0026 | Easy | Population genetics

Which description is correctly represented by $p + q = 1$ in this chapter?

- A. Hardy-Weinberg genotype equation
- B. Hardy-Weinberg sum of allele frequencies
- C. Random change in allele frequency, strongest in small populations
- D. Different origin but similar function

BIO-12-CH05-0027 | Medium | Human evolution

A learner encounters the observation 'Scientific name of modern humans' while analysing human evolution. Which conclusion is supported by the prescribed biology?

- A. Homo sapiens
- B. Population genetics
- C. Australopithecus
- D. Chemical evolution

BIO-12-CH05-0028 | Medium | Evidence of evolution

A source table contains one accurate entry involving 'Reduced structures with little current function'. Which option preserves a correct biological association?

- A. Reduced structures with little current function - Vestigial organs
- B. Human species associated with Java man - Cenozoic
- C. Oparin-Haldane proposal for origin of life - Homo erectus
- D. Hardy-Weinberg genotype equation - Gene flow

BIO-12-CH05-0029 | Easy | Evolutionary theories

Which biological term or result is specifically identified by the description 'Modern synthetic theory combines selection with'?

- A. Chemical evolution
- B. Population genetics
- C. Homo habilis
- D. Natural selection

BIO-12-CH05-0030 | Medium | Geological history

Complete the statement accurately: Era in which mammals diversified extensively corresponds to _____.

- A. Coacervates
- B. Jurassic
- C. Cenozoic
- D. Homologous organs

BIO-12-CH05-0031 | Medium | Evolutionary theories

A learner encounters the observation 'Darwinian mechanism' while analysing evolutionary theories. Which conclusion is supported by the prescribed biology?

- A. Inheritance of acquired characters
- B. Coacervates
- C. Natural selection
- D. Homo erectus

BIO-12-CH05-0032 | Hard | Origin of life

A reviewer compares several entries while checking 'Early aggregates of organic molecules in water'. Which entry must be corrected?

- A. Different origin but similar function - Analogous organs
- B. Early aggregates of organic molecules in water - Cenozoic
- C. Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$
- D. Genus of the famous fossil Lucy - Australopithecus

BIO-12-CH05-0033 | Hard | Human evolution

Two conclusions must both be valid to support the reference concept 'Human species associated with Java man'. Which option contains two accurate associations?

- A. Theory proposing life came from pre-existing life - Homologous organs; Same basic structure with different functions - Biogenesis
- B. Reduced structures with little current function - $p + q = 1$; Hardy-Weinberg sum of allele frequencies - Vestigial organs
- C. Human species associated with Java man - Homo erectus; Darwinian mechanism - Natural selection
- D. Same basic structure with different functions - Jurassic; Period associated with the first birds - Homologous organs

BIO-12-CH05-0034 | Easy | Evidence of evolution

Which description is correctly represented by Homologous organs in this chapter?

- A. Developmental similarities among embryos
- B. Random change in allele frequency, strongest in small populations
- C. Same basic structure with different functions
- D. Selection favouring both extreme phenotypes

BIO-12-CH05-0035 | Medium | Evidence of evolution

A learner encounters the observation 'Developmental similarities among embryos' while analysing evidence of evolution. Which conclusion is supported by the prescribed biology?

- A. Embryological evidence
- B. Vestigial organs
- C. Holocene
- D. Genetic drift

BIO-12-CH05-0036 | Easy | Population genetics

Which biological term or result is specifically identified by the description 'Movement of alleles between populations'?

- A. Coacervates
- B. Gene flow
- C. Holocene
- D. Genetic drift

BIO-12-CH05-0037 | Easy | Speciation and adaptation

Which biological term or result is specifically identified by the description 'Selection favouring both extreme phenotypes'?

- A. Genetic drift
- B. Mutation
- C. Disruptive selection
- D. Biogenesis

BIO-12-CH05-0038 | Hard | Population genetics

Two conclusions must both be valid to support the reference concept 'Movement of alleles between populations'. Which option contains two accurate associations?

- A. Genus of the famous fossil Lucy - Jurassic; Period associated with the first birds - Australopithecus
- B. Same basic structure with different functions - Holocene; Current geological epoch - Homologous organs
- C. Movement of alleles between populations - Gene flow; Oparin-Haldane proposal for origin of life - Chemical evolution
- D. Human species associated with Java man - Australopithecus; Genus of the famous fossil Lucy - Homo erectus

BIO-12-CH05-0039 | Hard | Evolutionary theories

A reviewer compares several entries while checking 'Lamarckian mechanism'. Which entry must be corrected?

- A. Reduced structures with little current function - Vestigial organs
- B. Same basic structure with different functions - Homologous organs
- C. Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$
- D. Lamarckian mechanism - Australopithecus

BIO-12-CH05-0040 | Medium | Geological history

A source table contains one accurate entry involving 'Current geological epoch'. Which option preserves a correct biological association?

- A. Modern synthetic theory combines selection with - Vestigial organs
- B. Current geological epoch - Holocene
- C. Reduced structures with little current function - Coacervates
- D. Genus of the famous fossil Lucy - Holocene

BIO-12-CH05-0041 | Medium | Origin of life

A learner encounters the observation 'Theory proposing life came from pre-existing life' while analysing origin of life. Which conclusion is supported by the prescribed biology?

- A. Inheritance of acquired characters
- B. Mutation
- C. Genetic drift
- D. Biogenesis

BIO-12-CH05-0042 | Easy | Speciation and adaptation

Which description is correctly represented by Disruptive selection in this chapter?

- A. Selection favouring both extreme phenotypes
- B. Random change in allele frequency, strongest in small populations
- C. De Vries emphasized
- D. Theory proposing life came from pre-existing life

BIO-12-CH05-0043 | Challenging | Human evolution

A researcher checks whether four linked observations have been interpreted correctly.

- I. Early human associated with systematic tool use corresponds to Homo habilis.
- II. Human species associated with Java man corresponds to Homo erectus.
- III. Scientific name of modern humans corresponds to Homo sapiens.
- IV. Different origin but similar function corresponds to Homo habilis.

Which combination survives the biological review?

- A. I, II and III only
- B. I, II, III and IV
- C. III only
- D. I, II and IV only

BIO-12-CH05-0044 | Easy | Geological history

Which biological term or result is specifically identified by the description 'Era known for dominance of reptiles'?

- A. Genetic drift
- B. Holocene
- C. Homo habilis
- D. Mesozoic

BIO-12-CH05-0045 | Easy | Speciation and adaptation

Which biological term or result is specifically identified by the description 'Barrier preventing gene flow between populations'?

- A. Homologous organs
- B. Holocene
- C. Reproductive isolation
- D. Inheritance of acquired characters

BIO-12-CH05-0046 | Hard | Speciation and adaptation

A reviewer compares several entries while checking 'Selection favouring both extreme phenotypes'. Which entry must be corrected?

- A. Selection favouring both extreme phenotypes - Mutation
- B. Modern synthetic theory combines selection with - Population genetics
- C. Theory proposing life came from pre-existing life - Biogenesis
- D. Era known for dominance of reptiles - Mesozoic

BIO-12-CH05-0047 | Challenging | Origin of life

Evaluate the following source-based biological statements.

- I. Early aggregates of organic molecules in water corresponds to Chemical evolution.
- II. Oparin-Haldane proposal for origin of life corresponds to Biogenesis.
- III. Theory proposing life came from pre-existing life corresponds to Biogenesis.
- IV. Darwinian mechanism corresponds to Coacervates.

Which option identifies exactly the correct statements?

- A. I and III only
- B. I, II and III only
- C. II only
- D. III only

BIO-12-CH05-0048 | Hard | Origin of life

A reviewer compares several entries while checking 'Oparin-Haldane proposal for origin of life'. Which entry must be corrected?

- A. Movement of alleles between populations - Gene flow
- B. Random change in allele frequency, strongest in small populations - Genetic drift
- C. Oparin-Haldane proposal for origin of life - Stabilizing selection
- D. Selection favouring both extreme phenotypes - Disruptive selection

BIO-12-CH05-0049 | Medium | Population genetics

A source table contains one accurate entry involving 'Random change in allele frequency, strongest in small populations'. Which option preserves a correct biological association?

- A. Developmental similarities among embryos - Cenozoic
- B. Random change in allele frequency, strongest in small populations - Genetic drift
- C. Era in which mammals diversified extensively - Genetic drift
- D. Selection favouring the average phenotype - Organic molecules including amino acids

BIO-12-CH05-0050 | Hard | Human evolution

Two conclusions must both be valid to support the reference concept 'Scientific name of modern humans'. Which option contains two accurate associations?

- A. Reduced structures with little current function - Biogenesis; Theory proposing life came from pre-existing life - Vestigial organs
- B. Scientific name of modern humans - Homo sapiens; De Vries emphasized - Mutation
- C. Modern synthetic theory combines selection with - Homologous organs; Same basic structure with different functions - Population genetics
- D. Same basic structure with different functions - Vestigial organs; Reduced structures with little current function - Homologous organs

BIO-12-CH05-0051 | Medium | Origin of life

Complete the statement accurately: Miller-Urey experiment produced corresponds to _____.

- A. Coacervates
- B. Population genetics
- C. Organic molecules including amino acids
- D. Chemical evolution

BIO-12-CH05-0052 | Challenging | Geological history

A student submits four conclusions after integrating related chapter concepts.

- I. Era in which mammals diversified extensively corresponds to Mesozoic.
 - II. Era known for dominance of reptiles corresponds to Holocene.
 - III. Current geological epoch corresponds to Holocene.
 - IV. Miller-Urey experiment produced corresponds to Organic molecules including amino acids.
- Which combination should be accepted without correction?

- A. III and IV only
- B. I, II and III only
- C. I, II, III and IV
- D. I, III and IV only

BIO-12-CH05-0053 | Medium | Speciation and adaptation

Complete the statement accurately: Selection favouring the average phenotype corresponds to _____.

- A. Stabilizing selection
- B. $p^2 + 2pq + q^2 = 1$
- C. Reproductive isolation
- D. Population genetics

BIO-12-CH05-0054 | Hard | Human evolution

A reviewer compares several entries while checking 'Scientific name of modern humans'. Which entry must be corrected?

- A. Same basic structure with different functions - Homologous organs
- B. Oparin-Haldane proposal for origin of life - Chemical evolution
- C. Miller-Urey experiment produced - Organic molecules including amino acids
- D. Scientific name of modern humans - Mutation

BIO-12-CH05-0055 | Easy | Evidence of evolution

Which biological term or result is specifically identified by the description 'Different origin but similar function'?

- A. Homologous organs
- B. Analogous organs
- C. Australopithecus
- D. Gene flow

BIO-12-CH05-0056 | Medium | Geological history

Complete the statement accurately: Current geological epoch corresponds to _____.

- A. Inheritance of acquired characters
- B. Mesozoic
- C. Holocene
- D. Homo erectus

BIO-12-CH05-0057 | Medium | Population genetics

A source table contains one accurate entry involving 'Movement of alleles between populations'. Which option preserves a correct biological association?

- A. Same basic structure with different functions - Coacervates
- B. Early aggregates of organic molecules in water - Australopithecus
- C. Human species associated with Java man - Jurassic
- D. Movement of alleles between populations - Gene flow

BIO-12-CH05-0058 | Easy | Population genetics

Which biological term or result is specifically identified by the description 'Random change in allele frequency, strongest in small populations'?

- A. Genetic drift
- B. Embryological evidence
- C. Australopithecus
- D. $p + q = 1$

BIO-12-CH05-0059 | Medium | Speciation and adaptation

A learner encounters the observation 'Selection favouring the average phenotype' while analysing speciation and adaptation. Which conclusion is supported by the prescribed biology?

- A. Jurassic
- B. Vestigial organs
- C. Mutation
- D. Stabilizing selection

BIO-12-CH05-0060 | Medium | Origin of life

A learner encounters the observation 'Miller-Urey experiment produced' while analysing origin of life. Which conclusion is supported by the prescribed biology?

- A. Coacervates
- B. $p + q = 1$
- C. Organic molecules including amino acids
- D. Reproductive isolation

BIO-12-CH05-0061 | Hard | Evidence of evolution

A reviewer compares several entries while checking 'Different origin but similar function'. Which entry must be corrected?

- A. Movement of alleles between populations - Gene flow
- B. Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$
- C. Period associated with the first birds - Jurassic
- D. Different origin but similar function - Adaptive radiation

BIO-12-CH05-0062 | Medium | Evolutionary theories

A source table contains one accurate entry involving 'Darwinian mechanism'. Which option preserves a correct biological association?

- A. Selection favouring the average phenotype - Jurassic
- B. Early human associated with systematic tool use - Natural selection
- C. Different origin but similar function - Homo habilis
- D. Darwinian mechanism - Natural selection

BIO-12-CH05-0063 | Medium | Human evolution

Complete the statement accurately: Early human associated with systematic tool use corresponds to _____.

- A. Homo habilis
- B. Mutation
- C. Australopithecus
- D. Organic molecules including amino acids

BIO-12-CH05-0064 | Easy | Geological history

Which biological term or result is specifically identified by the description 'Current geological epoch'?

- A. Holocene
- B. Embryological evidence
- C. Genetic drift
- D. Stabilizing selection

BIO-12-CH05-0065 | Medium | Evolutionary theories

A source table contains one accurate entry involving 'Lamarckian mechanism'. Which option preserves a correct biological association?

- A. Genus of the famous fossil Lucy - Holocene
- B. Reduced structures with little current function - $p^2 + 2pq + q^2 = 1$
- C. Same basic structure with different functions - Inheritance of acquired characters
- D. Lamarckian mechanism - Inheritance of acquired characters

BIO-12-CH05-0066 | Hard | Population genetics

A reviewer compares several entries while checking 'Random change in allele frequency, strongest in small populations'. Which entry must be corrected?

- A. Developmental similarities among embryos - Embryological evidence
 - B. Era in which mammals diversified extensively - Cenozoic
 - C. Random change in allele frequency, strongest in small populations - Australopithecus
 - D. Early aggregates of organic molecules in water - Coacervates
-

BIO-12-CH05-0067 | Hard | Geological history

A reviewer compares several entries while checking 'Era known for dominance of reptiles'. Which entry must be corrected?

- A. Modern synthetic theory combines selection with - Population genetics
 - B. Movement of alleles between populations - Gene flow
 - C. Random change in allele frequency, strongest in small populations - Genetic drift
 - D. Era known for dominance of reptiles - Australopithecus
-

BIO-12-CH05-0068 | Medium | Evolutionary theories

A learner encounters the observation 'Lamarckian mechanism' while analysing evolutionary theories. Which conclusion is supported by the prescribed biology?

- A. Australopithecus
 - B. Inheritance of acquired characters
 - C. Natural selection
 - D. Adaptive radiation
-

BIO-12-CH05-0069 | Medium | Evidence of evolution

Complete the statement accurately: Different origin but similar function corresponds to _____.

- A. Jurassic
 - B. Analogous organs
 - C. Adaptive radiation
 - D. Vestigial organs
-

BIO-12-CH05-0070 | Easy | Human evolution

Which description is correctly represented by Homo erectus in this chapter?

- A. Same basic structure with different functions
 - B. Reduced structures with little current function
 - C. Human species associated with Java man
 - D. Genus of the famous fossil Lucy
-

BIO-12-CH05-0071 | Easy | Evolutionary theories

Which description is correctly represented by Natural selection in this chapter?

- A. Darwinian mechanism
 - B. Selection favouring the average phenotype
 - C. Different origin but similar function
 - D. Modern synthetic theory combines selection with
-

BIO-12-CH05-0072 | Easy | Evolutionary theories

Which biological term or result is specifically identified by the description 'Lamarckian mechanism'?

- A. $p^2 + 2pq + q^2 = 1$
 - B. Inheritance of acquired characters
 - C. Adaptive radiation
 - D. Holocene
-

BIO-12-CH05-0073 | Medium | Speciation and adaptation

A source table contains one accurate entry involving 'Selection favouring the average phenotype'. Which option preserves a correct biological association?

- A. Period associated with the first birds - Mutation
- B. Reduced structures with little current function - Genetic drift
- C. De Vries emphasized - Cenozoic
- D. Selection favouring the average phenotype - Stabilizing selection

BIO-12-CH05-0074 | Medium | Speciation and adaptation

Complete the statement accurately: Formation of different species from a common ancestor in one region corresponds to _____.

- A. Biogenesis
- B. Stabilizing selection
- C. Vestigial organs
- D. Adaptive radiation

BIO-12-CH05-0075 | Medium | Evidence of evolution

A learner encounters the observation 'Reduced structures with little current function' while analysing evidence of evolution. Which conclusion is supported by the prescribed biology?

- A. Vestigial organs
- B. $p^2 + 2pq + q^2 = 1$
- C. Homologous organs
- D. Homo erectus

BIO-12-CH05-0076 | Medium | Population genetics

A source table contains one accurate entry involving 'Hardy-Weinberg sum of allele frequencies'. Which option preserves a correct biological association?

- A. Current geological epoch - Analogous organs
- B. Era in which mammals diversified extensively - Embryological evidence
- C. Hardy-Weinberg sum of allele frequencies - $p + q = 1$
- D. Human species associated with Java man - Mutation

BIO-12-CH05-0077 | Challenging | Origin of life

During error analysis, four proposed associations are reviewed.

- I. Miller-Urey experiment produced corresponds to Chemical evolution.
- II. Oparin-Haldane proposal for origin of life corresponds to Chemical evolution.
- III. Theory proposing life came from pre-existing life corresponds to Stabilizing selection.
- IV. Selection favouring the average phenotype corresponds to Organic molecules including amino acids.

Which set contains all and only the valid conclusions?

- A. III and IV only
- B. III only
- C. II only
- D. IV only

BIO-12-CH05-0078 | Medium | Evolutionary theories

Complete the statement accurately: Modern synthetic theory combines selection with corresponds to _____.

- A. Population genetics
- B. Genetic drift
- C. Chemical evolution
- D. Natural selection

BIO-12-CH05-0079 | Medium | Human evolution

Complete the statement accurately: Genus of the famous fossil Lucy corresponds to _____.

- A. Homo sapiens
- B. Embryological evidence
- C. Australopithecus
- D. Mutation

BIO-12-CH05-0080 | Easy | Human evolution

Which biological term or result is specifically identified by the description 'Early human associated with systematic tool use'?

- A. Stabilizing selection
- B. Homo habilis
- C. Cenozoic
- D. Organic molecules including amino acids

BIO-12-CH05-0081 | Hard | Evolutionary theories

Two conclusions must both be valid to support the reference concept 'Darwinian mechanism'. Which option contains two accurate associations?

- A. Darwinian mechanism - Natural selection; Hardy-Weinberg sum of allele frequencies - $p + q = 1$
- B. Different origin but similar function - Stabilizing selection; Selection favouring the average phenotype - Analogous organs
- C. Early human associated with systematic tool use - Jurassic; Period associated with the first birds - Homo habilis
- D. Early aggregates of organic molecules in water - Analogous organs; Different origin but similar function - Coacervates

BIO-12-CH05-0082 | Easy | Geological history

Which description is correctly represented by Mesozoic in this chapter?

- A. Early human associated with systematic tool use
- B. Modern synthetic theory combines selection with
- C. Genus of the famous fossil Lucy
- D. Era known for dominance of reptiles

BIO-12-CH05-0083 | Medium | Geological history

Complete the statement accurately: Period associated with the first birds corresponds to _____.

- A. Vestigial organs
- B. Jurassic
- C. Mesozoic
- D. Population genetics

BIO-12-CH05-0084 | Hard | Speciation and adaptation

A reviewer compares several entries while checking 'Barrier preventing gene flow between populations'. Which entry must be corrected?

- A. Same basic structure with different functions - Homologous organs
- B. Miller-Urey experiment produced - Organic molecules including amino acids
- C. Random change in allele frequency, strongest in small populations - Genetic drift
- D. Barrier preventing gene flow between populations - Holocene

BIO-12-CH05-0085 | Easy | Origin of life

Which biological term or result is specifically identified by the description 'Oparin-Haldane proposal for origin of life'?

- A. Organic molecules including amino acids
- B. Stabilizing selection
- C. Biogenesis
- D. Chemical evolution

BIO-12-CH05-0086 | Hard | Geological history

Two conclusions must both be valid to support the reference concept 'Era known for dominance of reptiles'. Which option contains two accurate associations?

- A. Era known for dominance of reptiles - Mesozoic; Genus of the famous fossil Lucy - Australopithecus
- B. Early human associated with systematic tool use - Gene flow; Movement of alleles between populations - Homo habilis
- C. Modern synthetic theory combines selection with - Genetic drift; Random change in allele frequency, strongest in small populations - Population genetics
- D. Random change in allele frequency, strongest in small populations - Homo habilis; Early human associated with systematic tool use - Genetic drift

BIO-12-CH05-0087 | Hard | Evidence of evolution

Two conclusions must both be valid to support the reference concept 'Same basic structure with different functions'. Which option contains two accurate associations?

- A. Random change in allele frequency, strongest in small populations - Gene flow; Movement of alleles between populations - Genetic drift
- B. De Vries emphasized - Mesozoic; Era known for dominance of reptiles - Mutation
- C. Same basic structure with different functions - Homologous organs; Hardy-Weinberg genotype equation - $p^2 + 2pq + q^2 = 1$
- D. Darwinian mechanism - Mutation; De Vries emphasized - Natural selection

BIO-12-CH05-0088 | Hard | Origin of life

A reviewer compares several entries while checking 'Miller-Urey experiment produced'. Which entry must be corrected?

- A. Random change in allele frequency, strongest in small populations - Genetic drift
- B. Barrier preventing gene flow between populations - Reproductive isolation
- C. Modern synthetic theory combines selection with - Population genetics
- D. Miller-Urey experiment produced - $p + q = 1$

BIO-12-CH05-0089 | Easy | Evidence of evolution

Which biological term or result is specifically identified by the description 'Reduced structures with little current function'?

- A. Homo erectus
- B. Chemical evolution
- C. $p^2 + 2pq + q^2 = 1$
- D. Vestigial organs

BIO-12-CH05-0090 | Easy | Origin of life

Which biological term or result is specifically identified by the description 'Early aggregates of organic molecules in water'?

- A. Cenozoic
- B. Analogous organs
- C. Coacervates
- D. Biogenesis

BIO-12-CH05-0091 | Hard | Population genetics

A reviewer compares several entries while checking 'Movement of alleles between populations'. Which entry must be corrected?

- A. Same basic structure with different functions - Homologous organs
- B. Movement of alleles between populations - Chemical evolution
- C. Current geological epoch - Holocene
- D. Early aggregates of organic molecules in water - Coacervates

BIO-12-CH05-0092 | Medium | Evidence of evolution

A learner encounters the observation 'Different origin but similar function' while analysing evidence of evolution. Which conclusion is supported by the prescribed biology?

- A. Jurassic
- B. Australopithecus
- C. Homologous organs
- D. Analogous organs

BIO-12-CH05-0093 | Medium | Speciation and adaptation

A learner encounters the observation 'Formation of different species from a common ancestor in one region' while analysing speciation and adaptation. Which conclusion is supported by the prescribed biology?

- A. Biogenesis
- B. Disruptive selection
- C. Adaptive radiation
- D. Homo erectus

BIO-12-CH05-0094 | Hard | Origin of life

Two conclusions must both be valid to support the reference concept 'Early aggregates of organic molecules in water'. Which option contains two accurate associations?

- A. Early aggregates of organic molecules in water - Coacervates; Era in which mammals diversified extensively - Cenozoic
- B. Same basic structure with different functions - $p + q = 1$; Hardy-Weinberg sum of allele frequencies - Homologous organs
- C. Genus of the famous fossil Lucy - Homologous organs; Same basic structure with different functions - Australopithecus
- D. Hardy-Weinberg genotype equation - Analogous organs; Different origin but similar function - $p^2 + 2pq + q^2 = 1$

BIO-12-CH05-0095 | Challenging | Evolutionary theories

A researcher checks whether four linked observations have been interpreted correctly.

- I. Modern synthetic theory combines selection with corresponds to Natural selection.
- II. Darwinian mechanism corresponds to Natural selection.
- III. De Vries emphasized corresponds to Australopithecus.
- IV. Genus of the famous fossil Lucy corresponds to Australopithecus.

Which combination survives the biological review?

- A. II and IV only
- B. IV only
- C. I and II only
- D. I, II and III only

BIO-12-CH05-0096 | Easy | Origin of life

Which biological term or result is specifically identified by the description 'Miller-Urey experiment produced'?

- A. Population genetics
- B. Homologous organs
- C. Organic molecules including amino acids
- D. Homo sapiens

BIO-12-CH05-0097 | Hard | Population genetics

Two conclusions must both be valid to support the reference concept 'Hardy-Weinberg sum of allele frequencies'. Which option contains two accurate associations?

- A. Developmental similarities among embryos - Analogous organs; Different origin but similar function - Embryological evidence
- B. Different origin but similar function - Homo erectus; Human species associated with Java man - Analogous organs
- C. Hardy-Weinberg sum of allele frequencies - $p + q = 1$; Era in which mammals diversified extensively - Cenozoic
- D. Current geological epoch - Embryological evidence; Developmental similarities among embryos - Holocene

BIO-12-CH05-0098 | Medium | Origin of life

Complete the statement accurately: Early aggregates of organic molecules in water corresponds to _____.

- A. Cenozoic
- B. Coacervates
- C. Biogenesis
- D. $p^2 + 2pq + q^2 = 1$

BIO-12-CH05-0099 | Challenging | Speciation and adaptation

Evaluate the following source-based biological statements.

- I. Barrier preventing gene flow between populations corresponds to Reproductive isolation.
- II. Selection favouring the average phenotype corresponds to Adaptive radiation.
- III. Formation of different species from a common ancestor in one region corresponds to Biogenesis.
- IV. Theory proposing life came from pre-existing life corresponds to Reproductive isolation.

Which option identifies exactly the correct statements?

- A. I only
- B. II and III only
- C. I, II and IV only
- D. III and IV only

BIO-12-CH05-0100 | Easy | Speciation and adaptation

Which description is correctly represented by Adaptive radiation in this chapter?

- A. Formation of different species from a common ancestor in one region
 - B. Reduced structures with little current function
 - C. Genus of the famous fossil Lucy
 - D. Theory proposing life came from pre-existing life
-

Standard 12 | Chapter 06

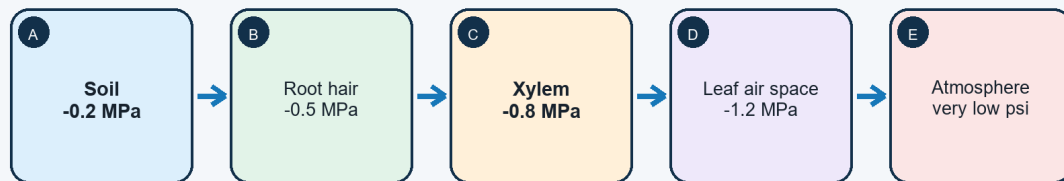
Plant Water Relation

BIO-12-CH06-0001 | Hard | Water-potential gradient through a plant

What immediate whole-plant effect follows widespread stomatal closure during water stress?

Water-potential gradient through a plant

Water moves along a decreasing water-potential pathway



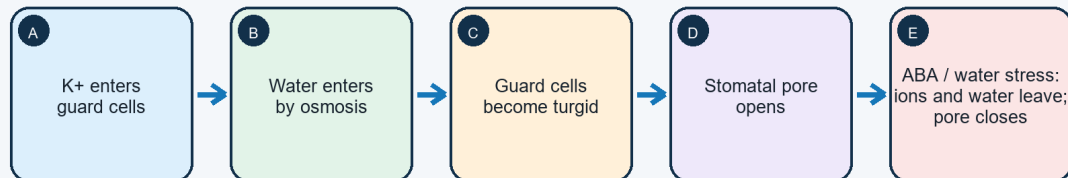
- A. Increased transpiration and increased carbon-dioxide entry
- B. Reduced transpiration pull and reduced carbon-dioxide entry
- C. Reversal of xylem flow toward soil
- D. Permanent conversion of xylem into phloem

BIO-12-CH06-0002 | Medium | Guard-cell control of a stoma

Which guard-cell state in the sequence is directly associated with an open stomatal pore?

Guard-cell control of a stoma

Ion movement changes guard-cell water content and aperture



Closure limits water loss but also restricts carbon-dioxide entry.

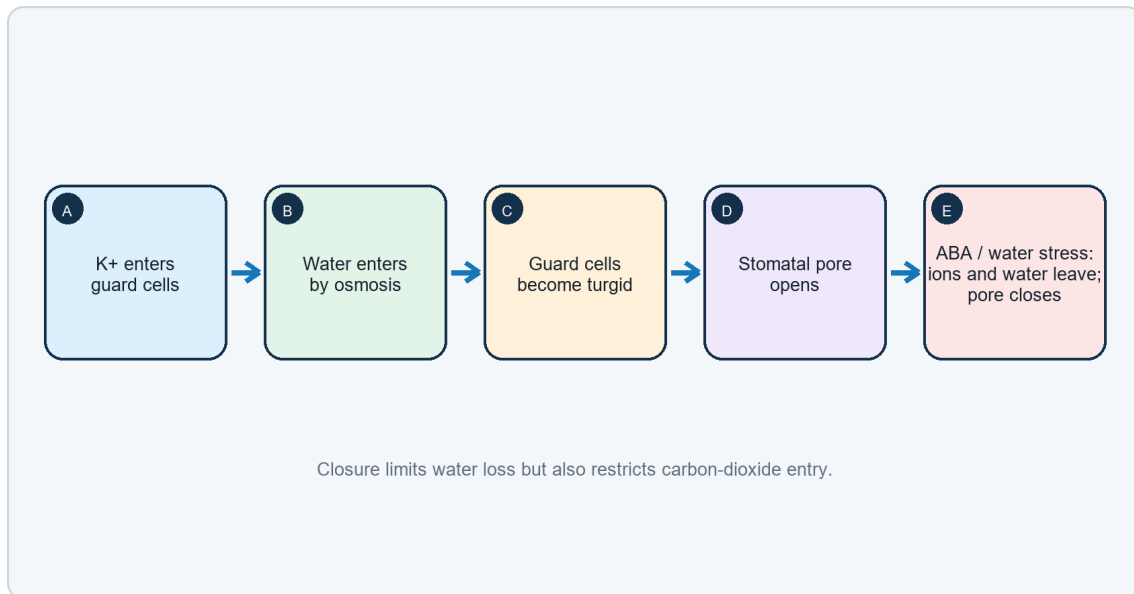
- A. Turgid guard cells
- B. Flaccid guard cells
- C. Dead guard cells
- D. Guard cells lacking water and potassium

BIO-12-CH06-0003 | Challenging | Guard-cell control of a stoma

ABA promotes ion loss from guard cells. Which causal chain best matches the figure?

Guard-cell control of a stoma

Ion movement changes guard-cell water content and aperture



- A. Water enters guard cells, turgor rises and the pore closes
- B. Water exits guard cells, turgor falls and the pore closes
- C. Chlorophyll leaves guard cells and the pore opens
- D. Xylem pressure rises and permanently locks the pore open

BIO-12-CH06-0004 | Medium | Ascent of sap

A source table contains one accurate entry involving 'Theory chiefly explaining ascent of sap'. Which option preserves a correct biological association?

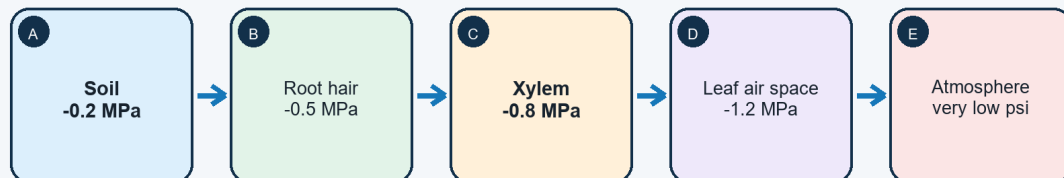
- A. Movement through connected cytoplasm via plasmodesmata - Cuticular transpiration
- B. Principal water-absorbing region of a young root - Lenticular transpiration
- C. Water loss directly through the cuticle - Adhesion
- D. Theory chiefly explaining ascent of sap - Cohesion-tension theory

BIO-12-CH06-0005 | Easy | Water-potential gradient through a plant

According to the gradient, water moves spontaneously from soil toward the root because it moves from

Water-potential gradient through a plant

Water moves along a decreasing water-potential pathway



Transpiration maintains tension in the continuous water column of xylem.

- A. higher to lower water potential
- B. lower to higher water potential
- C. higher to lower solute concentration only
- D. xylem to soil against the gradient

BIO-12-CH06-0006 | Hard | Water potential and osmosis

Two conclusions must both be valid to support the reference concept 'Shrinkage of a protoplast in a hypertonic solution'. Which option contains two accurate associations?

- A. Typical sink tissue - Stomata; Major route of transpiration - Developing fruit or storage organ
- B. Main tissue conducting water upward - Root-hair zone; Principal water-absorbing region of a young root - Xylem
- C. Shrinkage of a protoplast in a hypertonic solution - Plasmolysis; Energy-dependent uptake of mineral ions - Active absorption
- D. Movement through connected cytoplasm via plasmodesmata - Transpiration pull; Force generated by evaporation from leaves - Symplast pathway

BIO-12-CH06-0007 | Easy | Root absorption

Which description is correctly represented by Symplast pathway in this chapter?

- A. Movement through connected cytoplasm via plasmodesmata
- B. Pressure-driven movement through sieve tubes
- C. Barrier forcing water through living cells at the endodermis
- D. Net movement of water through a selectively permeable membrane

BIO-12-CH06-0008 | Hard | Mineral and food transport

A reviewer compares several entries while checking 'Energy-dependent uptake of mineral ions'. Which entry must be corrected?

- A. Attraction between water molecules - Cohesion
- B. Guard-cell shape typical of grasses - Dumb-bell shaped
- C. Ion whose accumulation promotes guard-cell opening - Potassium
- D. Energy-dependent uptake of mineral ions - Capillary water

BIO-12-CH06-0009 | Hard | Mineral and food transport

Two conclusions must both be valid to support the reference concept 'Typical sink tissue'. Which option contains two accurate associations?

- A. Main tissue conducting water upward - Capillary water; Soil water most readily available to plants - Xylem
- B. Passive movement through cell walls and intercellular spaces - Solute potential; Component of water potential produced by dissolved solute - Apoplast pathway
- C. Water potential of pure water at standard conditions - Apoplast pathway; Passive movement through cell walls and intercellular spaces - Zero
- D. Typical sink tissue - Developing fruit or storage organ; Cells controlling the stomatal pore - Guard cells

BIO-12-CH06-0010 | Medium | Mineral and food transport

Complete the statement accurately: Pressure-driven movement through sieve tubes corresponds to _____.

- A. Mass flow
- B. Developing fruit or storage organ
- C. Mature photosynthetic leaf
- D. Cohesion-tension theory

BIO-12-CH06-0011 | Easy | Water potential and osmosis

Which description is correctly represented by Plasmolysis in this chapter?

- A. Energy-dependent uptake of mineral ions
- B. Force generated by evaporation from leaves
- C. Shrinkage of a protoplast in a hypertonic solution
- D. Water potential of pure water at standard conditions

BIO-12-CH06-0012 | Easy | Water potential and osmosis

Which biological term or result is specifically identified by the description 'Water potential of pure water at standard conditions'?

- A. Osmosis
- B. Zero
- C. Plasmolysis
- D. Stomata

BIO-12-CH06-0013 | Hard | Water potential and osmosis

A reviewer compares several entries while checking 'Water potential of pure water at standard conditions'. Which entry must be corrected?

- A. Theory chiefly explaining ascent of sap - Cohesion-tension theory
- B. Positive pressure developed in roots - Root pressure
- C. Loss of water as vapour from aerial plant parts - Transpiration
- D. Water potential of pure water at standard conditions - Stomata

BIO-12-CH06-0014 | Hard | Root absorption

Two conclusions must both be valid to support the reference concept 'Passive movement through cell walls and intercellular spaces'. Which option contains two accurate associations?

- A. Cells controlling the stomatal pore - Cohesion; Attraction between water molecules - Guard cells
- B. Positive pressure developed in roots - Transpiration; Loss of water as vapour from aerial plant parts - Root pressure
- C. Passive movement through cell walls and intercellular spaces - Apoplast pathway; Guard-cell shape typical of dicots - Kidney-shaped
- D. Component of water potential produced by dissolved solute - Guard cells; Cells controlling the stomatal pore - Solute potential

BIO-12-CH06-0015 | Hard | Stomatal apparatus

A reviewer compares several entries while checking 'Guard-cell shape typical of grasses'. Which entry must be corrected?

- A. Passive movement through cell walls and intercellular spaces - Apoplast pathway
- B. Source tissue in phloem transport - Mature photosynthetic leaf
- C. Soil water held tightly to particle surfaces - Hygroscopic water
- D. Guard-cell shape typical of grasses - Capillary water

BIO-12-CH06-0016 | Easy | Root absorption

Which biological term or result is specifically identified by the description 'Principal water-absorbing region of a young root'?

- A. Plasmolysis
- B. Root-hair zone
- C. Active absorption
- D. Symplast pathway

BIO-12-CH06-0017 | Challenging | Transpiration

During error analysis, four proposed associations are reviewed.

- I. Water loss directly through the cuticle corresponds to Lenticular transpiration.
 - II. Water loss through lenticels corresponds to Transpiration.
 - III. Loss of water as vapour from aerial plant parts corresponds to Transpiration.
 - IV. Component of water potential produced by dissolved solute corresponds to Cuticular transpiration.
- Which set contains all and only the valid conclusions?

- A. II and III only
- B. III only
- C. III and IV only
- D. I and IV only

BIO-12-CH06-0018 | Hard | Root absorption

A reviewer compares several entries while checking 'Principal water-absorbing region of a young root'. Which entry must be corrected?

- A. Cells controlling the stomatal pore - Guard cells
- B. Water loss directly through the cuticle - Cuticular transpiration
- C. Principal water-absorbing region of a young root - Plasmolysis
- D. Energy-dependent uptake of mineral ions - Active absorption

BIO-12-CH06-0019 | Medium | Water potential and osmosis

Complete the statement accurately: Component of water potential produced by dissolved solute corresponds to _____.

- A. Solute potential
- B. Hygroscopic water
- C. Plasmolysis
- D. Zero

BIO-12-CH06-0020 | Medium | Root absorption

A source table contains one accurate entry involving 'Passive movement through cell walls and intercellular spaces'. Which option preserves a correct biological association?

- A. Component of water potential produced by dissolved solute - Cohesion
- B. Loss of water as vapour from aerial plant parts - Guard cells
- C. Attraction between water molecules - Apoplast pathway
- D. Passive movement through cell walls and intercellular spaces - Apoplast pathway

BIO-12-CH06-0021 | Hard | Root absorption

Two conclusions must both be valid to support the reference concept 'Movement through connected cytoplasm via plasmodesmata'. Which option contains two accurate associations?

- A. Guard-cell shape typical of grasses - Cuticular transpiration; Water loss directly through the cuticle - Dumb-bell shaped
- B. Pressure-driven movement through sieve tubes - Dumb-bell shaped; Guard-cell shape typical of grasses - Mass flow
- C. Shrinkage of a protoplast in a hypertonic solution - Adhesion; Attraction of water to a different surface - Plasmolysis
- D. Movement through connected cytoplasm via plasmodesmata - Symplast pathway; Net movement of water through a selectively permeable membrane - Osmosis

BIO-12-CH06-0022 | Medium | Root absorption

A source table contains one accurate entry involving 'Movement through connected cytoplasm via plasmodesmata'. Which option preserves a correct biological association?

- A. Source tissue in phloem transport - Adhesion
- B. Guard-cell shape typical of grasses - Symplast pathway
- C. Movement through connected cytoplasm via plasmodesmata - Symplast pathway
- D. Attraction of water to a different surface - Dumb-bell shaped

BIO-12-CH06-0023 | Medium | Transpiration

Complete the statement accurately: Loss of water as vapour from aerial plant parts corresponds to _____.

- A. Stomata
- B. Transpiration
- C. Plasmolysis
- D. Osmosis

BIO-12-CH06-0024 | Medium | Stomatal apparatus

Complete the statement accurately: Cells controlling the stomatal pore corresponds to _____.

- A. Developing fruit or storage organ
- B. Mature photosynthetic leaf
- C. Kidney-shaped
- D. Guard cells

BIO-12-CH06-0025 | Medium | Root absorption

Complete the statement accurately: Principal water-absorbing region of a young root corresponds to _____.

- A. Apoplast pathway
- B. Root-hair zone
- C. Mature photosynthetic leaf
- D. Symplast pathway

BIO-12-CH06-0026 | Medium | Properties and forms of water

A source table contains one accurate entry involving 'Attraction between water molecules'. Which option preserves a correct biological association?

- A. Attraction between water molecules - Cohesion
- B. Barrier forcing water through living cells at the endodermis - Lenticular transpiration
- C. Passive movement through cell walls and intercellular spaces - Cohesion
- D. Component of water potential produced by dissolved solute - Apoplast pathway

BIO-12-CH06-0027 | Medium | Root absorption

A source table contains one accurate entry involving 'Principal water-absorbing region of a young root'. Which option preserves a correct biological association?

- A. Energy-dependent uptake of mineral ions - Guard cells
- B. Cells controlling the stomatal pore - Root-hair zone
- C. Principal water-absorbing region of a young root - Root-hair zone
- D. Typical sink tissue - Mature photosynthetic leaf

BIO-12-CH06-0028 | Challenging | Root absorption

A researcher checks whether four linked observations have been interpreted correctly.

- I. Passive movement through cell walls and intercellular spaces corresponds to Casparian strip.
- II. Barrier forcing water through living cells at the endodermis corresponds to Root-hair zone.
- III. Principal water-absorbing region of a young root corresponds to Root-hair zone.
- IV. Component of water potential produced by dissolved solute corresponds to Solute potential.

Which combination survives the biological review?

- A. III and IV only
- B. I, III and IV only
- C. I only
- D. I, II and IV only

BIO-12-CH06-0029 | Medium | Stomatal apparatus

A source table contains one accurate entry involving 'Ion whose accumulation promotes guard-cell opening'. Which option preserves a correct biological association?

- A. Barrier forcing water through living cells at the endodermis - Apoplast pathway
- B. Water potential of pure water at standard conditions - Potassium
- C. Ion whose accumulation promotes guard-cell opening - Potassium
- D. Passive movement through cell walls and intercellular spaces - Zero

BIO-12-CH06-0030 | Easy | Properties and forms of water

Which description is correctly represented by Cohesion in this chapter?

- A. Barrier forcing water through living cells at the endodermis
- B. Attraction between water molecules
- C. Soil water most readily available to plants
- D. Water loss through lenticels

BIO-12-CH06-0031 | Hard | Ascent of sap

Two conclusions must both be valid to support the reference concept 'Positive pressure developed in roots'. Which option contains two accurate associations?

- A. Source tissue in phloem transport - Mass flow; Pressure-driven movement through sieve tubes - Mature photosynthetic leaf
- B. Water potential of pure water at standard conditions - Adhesion; Attraction of water to a different surface - Zero
- C. Positive pressure developed in roots - Root pressure; Attraction between water molecules - Cohesion
- D. Attraction of water to a different surface - Mature photosynthetic leaf; Source tissue in phloem transport - Adhesion

BIO-12-CH06-0032 | Easy | Water potential and osmosis

Which biological term or result is specifically identified by the description 'Net movement of water through a selectively permeable membrane'?

- A. Solute potential
- B. Dumb-bell shaped
- C. Osmosis
- D. Symplast pathway

BIO-12-CH06-0033 | Medium | Mineral and food transport

Complete the statement accurately: Energy-dependent uptake of mineral ions corresponds to _____.

- A. Kidney-shaped
- B. Active absorption
- C. Potassium
- D. Capillary water

BIO-12-CH06-0034 | Hard | Stomatal apparatus

Two conclusions must both be valid to support the reference concept 'Guard-cell shape typical of grasses'. Which option contains two accurate associations?

- A. Source tissue in phloem transport - Apoplast pathway; Passive movement through cell walls and intercellular spaces - Mature photosynthetic leaf
- B. Passive movement through cell walls and intercellular spaces - Root pressure; Positive pressure developed in roots - Apoplast pathway
- C. Soil water held tightly to particle surfaces - Transpiration pull; Force generated by evaporation from leaves - Hygroscopic water
- D. Guard-cell shape typical of grasses - Dumb-bell shaped; Soil water most readily available to plants - Capillary water

BIO-12-CH06-0035 | Easy | Transpiration

Which biological term or result is specifically identified by the description 'Loss of water as vapour from aerial plant parts'?

- A. Transpiration
- B. Hygroscopic water
- C. Plasmolysis
- D. Osmosis

BIO-12-CH06-0036 | Medium | Water potential and osmosis

A learner encounters the observation 'Net movement of water through a selectively permeable membrane' while analysing water potential and osmosis. Which conclusion is supported by the prescribed biology?

- A. Osmosis
- B. Plasmolysis
- C. Solute potential
- D. Active absorption

BIO-12-CH06-0037 | Medium | Transpiration

A source table contains one accurate entry involving 'Water loss through lenticels'. Which option preserves a correct biological association?

- A. Theory chiefly explaining ascent of sap - Guard cells
- B. Water loss through lenticels - Lenticular transpiration
- C. Cells controlling the stomatal pore - Capillary water
- D. Principal water-absorbing region of a young root - Apoplast pathway

BIO-12-CH06-0038 | Hard | Mineral and food transport

A reviewer compares several entries while checking 'Pressure-driven movement through sieve tubes'. Which entry must be corrected?

- A. Barrier forcing water through living cells at the endodermis - Casparian strip
- B. Theory chiefly explaining ascent of sap - Cohesion-tension theory
- C. Pressure-driven movement through sieve tubes - Xylem
- D. Component of water potential produced by dissolved solute - Solute potential

BIO-12-CH06-0039 | Easy | Stomatal apparatus

Which biological term or result is specifically identified by the description 'Ion whose accumulation promotes guard-cell opening'?

- A. Root pressure
- B. Potassium
- C. Guard cells
- D. Dumb-bell shaped

BIO-12-CH06-0040 | Medium | Mineral and food transport

A learner encounters the observation 'Typical sink tissue' while analysing mineral and food transport. Which conclusion is supported by the prescribed biology?

- A. Zero
- B. Guard cells
- C. Developing fruit or storage organ
- D. Apoplast pathway

BIO-12-CH06-0041 | Easy | Stomatal apparatus

Which description is correctly represented by Dumb-bell shaped in this chapter?

- A. Guard-cell shape typical of dicots
- B. Typical sink tissue
- C. Guard-cell shape typical of grasses
- D. Ion whose accumulation promotes guard-cell opening

BIO-12-CH06-0042 | Easy | Transpiration

Which description is correctly represented by Transpiration in this chapter?

- A. Soil water held tightly to particle surfaces
- B. Water loss through lenticels
- C. Major route of transpiration
- D. Loss of water as vapour from aerial plant parts

BIO-12-CH06-0043 | Challenging | Properties and forms of water

Evaluate the following source-based biological statements.

- I. Soil water held tightly to particle surfaces corresponds to Cohesion.
- II. Attraction between water molecules corresponds to Cohesion.
- III. Attraction of water to a different surface corresponds to Casparian strip.
- IV. Barrier forcing water through living cells at the endodermis corresponds to Hygroscopic water.

Which option identifies exactly the correct statements?

- A. II only
- B. I and III only
- C. I only
- D. II and III only

BIO-12-CH06-0044 | Hard | Transpiration

A reviewer compares several entries while checking 'Loss of water as vapour from aerial plant parts'. Which entry must be corrected?

- A. Movement through connected cytoplasm via plasmodesmata - Symplast pathway
- B. Loss of water as vapour from aerial plant parts - Casparian strip
- C. Soil water held tightly to particle surfaces - Hygroscopic water
- D. Net movement of water through a selectively permeable membrane - Osmosis

BIO-12-CH06-0045 | Medium | Water potential and osmosis

A source table contains one accurate entry involving 'Shrinkage of a protoplast in a hypertonic solution'. Which option preserves a correct biological association?

- A. Movement through connected cytoplasm via plasmodesmata - Developing fruit or storage organ
- B. Force generated by evaporation from leaves - Stomata
- C. Shrinkage of a protoplast in a hypertonic solution - Plasmolysis
- D. Main tissue conducting water upward - Plasmolysis

BIO-12-CH06-0046 | Medium | Transpiration

A source table contains one accurate entry involving 'Water loss directly through the cuticle'. Which option preserves a correct biological association?

- A. Soil water most readily available to plants - Kidney-shaped
- B. Soil water held tightly to particle surfaces - Cuticular transpiration
- C. Water loss directly through the cuticle - Cuticular transpiration
- D. Movement through connected cytoplasm via plasmodesmata - Xylem

BIO-12-CH06-0047 | Hard | Stomatal apparatus

A reviewer compares several entries while checking 'Ion whose accumulation promotes guard-cell opening'. Which entry must be corrected?

- A. Passive movement through cell walls and intercellular spaces - Apoplast pathway
- B. Ion whose accumulation promotes guard-cell opening - Symplast pathway
- C. Barrier forcing water through living cells at the endodermis - Casparian strip
- D. Attraction of water to a different surface - Adhesion

BIO-12-CH06-0048 | Medium | Properties and forms of water

A learner encounters the observation 'Attraction between water molecules' while analysing properties and forms of water. Which conclusion is supported by the prescribed biology?

- A. Capillary water
- B. Lenticular transpiration
- C. Adhesion
- D. Cohesion

BIO-12-CH06-0049 | Easy | Stomatal apparatus

Which biological term or result is specifically identified by the description 'Guard-cell shape typical of grasses'?

- A. Dumb-bell shaped
- B. Kidney-shaped
- C. Mature photosynthetic leaf
- D. Potassium

BIO-12-CH06-0050 | Easy | Mineral and food transport

Which description is correctly represented by Developing fruit or storage organ in this chapter?

- A. Typical sink tissue
- B. Water potential of pure water at standard conditions
- C. Source tissue in phloem transport
- D. Passive movement through cell walls and intercellular spaces

BIO-12-CH06-0051 | Challenging | Ascent of sap

A student submits four conclusions after integrating related chapter concepts.

- I. Positive pressure developed in roots corresponds to Cohesion-tension theory.
- II. Theory chiefly explaining ascent of sap corresponds to Cohesion-tension theory.
- III. Main tissue conducting water upward corresponds to Root-hair zone.
- IV. Principal water-absorbing region of a young root corresponds to Root-hair zone.

Which combination should be accepted without correction?

- A. I, III and IV only
- B. II and IV only
- C. I, II and III only
- D. I, II, III and IV

BIO-12-CH06-0052 | Medium | Transpiration

A learner encounters the observation 'Water loss directly through the cuticle' while analysing transpiration. Which conclusion is supported by the prescribed biology?

- A. Osmosis
- B. Cuticular transpiration
- C. Symplast pathway
- D. Transpiration pull

BIO-12-CH06-0053 | Easy | Properties and forms of water

Which biological term or result is specifically identified by the description 'Attraction between water molecules'?

- A. Lenticular transpiration
- B. Transpiration
- C. Adhesion
- D. Cohesion

BIO-12-CH06-0054 | Medium | Properties and forms of water

Complete the statement accurately: Attraction between water molecules corresponds to _____.

- A. Cohesion
- B. Transpiration
- C. Casparian strip
- D. Capillary water

BIO-12-CH06-0055 | Easy | Mineral and food transport

Which biological term or result is specifically identified by the description 'Pressure-driven movement through sieve tubes'?

- A. Stomata
- B. Mature photosynthetic leaf
- C. Mass flow
- D. Solute potential

BIO-12-CH06-0056 | Medium | Stomatal apparatus

Complete the statement accurately: Ion whose accumulation promotes guard-cell opening corresponds to _____.

- A. Potassium
- B. Symplast pathway
- C. Apoplast pathway
- D. Casparian strip

BIO-12-CH06-0057 | Medium | Ascent of sap

A learner encounters the observation 'Main tissue conducting water upward' while analysing ascent of sap. Which conclusion is supported by the prescribed biology?

- A. Hygroscopic water
- B. Apoplast pathway
- C. Dumb-bell shaped
- D. Xylem

BIO-12-CH06-0058 | Hard | Properties and forms of water

Two conclusions must both be valid to support the reference concept 'Attraction between water molecules'. Which option contains two accurate associations?

- A. Net movement of water through a selectively permeable membrane - Apoplast pathway; Passive movement through cell walls and intercellular spaces - Osmosis
- B. Component of water potential produced by dissolved solute - Osmosis; Net movement of water through a selectively permeable membrane - Solute potential
- C. Attraction between water molecules - Cohesion; Barrier forcing water through living cells at the endodermis - Casparian strip
- D. Water loss through lenticels - Solute potential; Component of water potential produced by dissolved solute - Lenticular transpiration

BIO-12-CH06-0059 | Medium | Ascent of sap

A source table contains one accurate entry involving 'Force generated by evaporation from leaves'. Which option preserves a correct biological association?

- A. Attraction of water to a different surface - Developing fruit or storage organ
- B. Typical sink tissue - Guard cells
- C. Source tissue in phloem transport - Adhesion
- D. Force generated by evaporation from leaves - Transpiration pull

BIO-12-CH06-0060 | Medium | Water potential and osmosis

A learner encounters the observation 'Component of water potential produced by dissolved solute' while analysing water potential and osmosis. Which conclusion is supported by the prescribed biology?

- A. Plasmolysis
- B. Solute potential
- C. Zero
- D. Apoplast pathway

BIO-12-CH06-0061 | Challenging | Transpiration

During error analysis, four proposed associations are reviewed.

- I. Loss of water as vapour from aerial plant parts corresponds to Transpiration.
- II. Water loss through lenticels corresponds to Stomata.
- III. Major route of transpiration corresponds to Capillary water.
- IV. Soil water most readily available to plants corresponds to Transpiration.

Which set contains all and only the valid conclusions?

- A. I only
- B. I, II and III only
- C. I, II, III and IV
- D. II and III only

BIO-12-CH06-0062 | Easy | Root absorption

Which description is correctly represented by Apoplast pathway in this chapter?

- A. Positive pressure developed in roots
- B. Passive movement through cell walls and intercellular spaces
- C. Component of water potential produced by dissolved solute
- D. Movement through connected cytoplasm via plasmodesmata

BIO-12-CH06-0063 | Challenging | Water potential and osmosis

A researcher checks whether four linked observations have been interpreted correctly.

- I. Component of water potential produced by dissolved solute corresponds to Zero.
- II. Water potential of pure water at standard conditions corresponds to Zero.
- III. Shrinkage of a protoplast in a hypertonic solution corresponds to Plasmolysis.
- IV. Pressure-driven movement through sieve tubes corresponds to Solute potential.

Which combination survives the biological review?

- A. II only
- B. I, II, III and IV
- C. I, III and IV only
- D. II and III only

BIO-12-CH06-0064 | Easy | Ascent of sap

Which description is correctly represented by Transpiration pull in this chapter?

- A. Force generated by evaporation from leaves
- B. Source tissue in phloem transport
- C. Guard-cell shape typical of dicots
- D. Theory chiefly explaining ascent of sap

BIO-12-CH06-0065 | Easy | Mineral and food transport

Which biological term or result is specifically identified by the description 'Energy-dependent uptake of mineral ions'?

- A. Active absorption
- B. Kidney-shaped
- C. Developing fruit or storage organ
- D. Mature photosynthetic leaf

BIO-12-CH06-0066 | Hard | Ascent of sap

Two conclusions must both be valid to support the reference concept 'Force generated by evaporation from leaves'. Which option contains two accurate associations?

- A. Soil water most readily available to plants - Guard cells; Cells controlling the stomatal pore - Capillary water
- B. Attraction of water to a different surface - Kidney-shaped; Guard-cell shape typical of dicots - Adhesion
- C. Typical sink tissue - Capillary water; Soil water most readily available to plants - Developing fruit or storage organ
- D. Force generated by evaporation from leaves - Transpiration pull; Source tissue in phloem transport - Mature photosynthetic leaf

BIO-12-CH06-0067 | Medium | Stomatal apparatus

Complete the statement accurately: Guard-cell shape typical of grasses corresponds to _____.

- A. Dumb-bell shaped
- B. Capillary water
- C. Root pressure
- D. Kidney-shaped

BIO-12-CH06-0068 | Medium | Ascent of sap

A source table contains one accurate entry involving 'Positive pressure developed in roots'. Which option preserves a correct biological association?

- A. Pressure-driven movement through sieve tubes - Lenticular transpiration
- B. Net movement of water through a selectively permeable membrane - Root pressure
- C. Positive pressure developed in roots - Root pressure
- D. Attraction of water to a different surface - Mass flow

BIO-12-CH06-0069 | Challenging | Root absorption

Evaluate the following source-based biological statements.

- I. Movement through connected cytoplasm via plasmodesmata corresponds to Symplast pathway.
- II. Principal water-absorbing region of a young root corresponds to Root-hair zone.
- III. Barrier forcing water through living cells at the endodermis corresponds to Casparian strip.
- IV. Guard-cell shape typical of grasses corresponds to Dumb-bell shaped.

Which option identifies exactly the correct statements?

- A. I, II and IV only
- B. I, II and III only
- C. III and IV only
- D. I, II, III and IV

BIO-12-CH06-0070 | Hard | Mineral and food transport

A reviewer compares several entries while checking 'Source tissue in phloem transport'. Which entry must be corrected?

- A. Source tissue in phloem transport - Apoplast pathway
- B. Attraction between water molecules - Cohesion
- C. Movement through connected cytoplasm via plasmodesmata - Symplast pathway
- D. Guard-cell shape typical of grasses - Dumb-bell shaped

BIO-12-CH06-0071 | Medium | Properties and forms of water

Complete the statement accurately: Soil water held tightly to particle surfaces corresponds to _____.

- A. Hygroscopic water
- B. Cohesion-tension theory
- C. Transpiration
- D. Cohesion

BIO-12-CH06-0072 | Easy | Transpiration

Which description is correctly represented by Lenticular transpiration in this chapter?

- A. Loss of water as vapour from aerial plant parts
- B. Soil water most readily available to plants
- C. Water loss through lenticels
- D. Theory chiefly explaining ascent of sap

BIO-12-CH06-0073 | Medium | Mineral and food transport

Complete the statement accurately: Typical sink tissue corresponds to _____.

- A. Developing fruit or storage organ
- B. Zero
- C. Mass flow
- D. Mature photosynthetic leaf

BIO-12-CH06-0074 | Hard | Water potential and osmosis

A reviewer compares several entries while checking 'Net movement of water through a selectively permeable membrane'. Which entry must be corrected?

- A. Attraction between water molecules - Cohesion
- B. Source tissue in phloem transport - Mature photosynthetic leaf
- C. Net movement of water through a selectively permeable membrane - Active absorption
- D. Movement through connected cytoplasm via plasmodesmata - Symplast pathway

BIO-12-CH06-0075 | Hard | Ascent of sap

Two conclusions must both be valid to support the reference concept 'Theory chiefly explaining ascent of sap'. Which option contains two accurate associations?

- A. Movement through connected cytoplasm via plasmodesmata - Lenticular transpiration; Water loss through lenticels - Symplast pathway
- B. Theory chiefly explaining ascent of sap - Cohesion-tension theory; Principal water-absorbing region of a young root - Root-hair zone
- C. Guard-cell shape typical of grasses - Adhesion; Attraction of water to a different surface - Dumb-bell shaped
- D. Attraction of water to a different surface - Capillary water; Soil water most readily available to plants - Adhesion

BIO-12-CH06-0076 | Easy | Mineral and food transport

Which biological term or result is specifically identified by the description 'Source tissue in phloem transport'?

- A. Cohesion
- B. Mature photosynthetic leaf
- C. Symplast pathway
- D. Zero

BIO-12-CH06-0077 | Medium | Transpiration

A source table contains one accurate entry involving 'Loss of water as vapour from aerial plant parts'. Which option preserves a correct biological association?

- A. Main tissue conducting water upward - Transpiration
- B. Shrinkage of a protoplast in a hypertonic solution - Symplast pathway
- C. Barrier forcing water through living cells at the endodermis - Plasmolysis
- D. Loss of water as vapour from aerial plant parts - Transpiration

BIO-12-CH06-0078 | Hard | Properties and forms of water

A reviewer compares several entries while checking 'Attraction between water molecules'. Which entry must be corrected?

- A. Attraction between water molecules - Casparian strip
- B. Net movement of water through a selectively permeable membrane - Osmosis
- C. Component of water potential produced by dissolved solute - Solute potential
- D. Passive movement through cell walls and intercellular spaces - Apoplast pathway

BIO-12-CH06-0079 | Easy | Ascent of sap

Which description is correctly represented by Cohesion-tension theory in this chapter?

- A. Guard-cell shape typical of grasses
- B. Force generated by evaporation from leaves
- C. Water loss through lenticels
- D. Theory chiefly explaining ascent of sap

BIO-12-CH06-0080 | Medium | Water potential and osmosis

A source table contains one accurate entry involving 'Net movement of water through a selectively permeable membrane'. Which option preserves a correct biological association?

- A. Energy-dependent uptake of mineral ions - Dumb-bell shaped
- B. Net movement of water through a selectively permeable membrane - Osmosis
- C. Main tissue conducting water upward - Mature photosynthetic leaf
- D. Movement through connected cytoplasm via plasmodesmata - Xylem

BIO-12-CH06-0081 | Medium | Root absorption

A learner encounters the observation 'Passive movement through cell walls and intercellular spaces' while analysing root absorption. Which conclusion is supported by the prescribed biology?

- A. Symplast pathway
- B. Apoplast pathway
- C. Root pressure
- D. Transpiration

BIO-12-CH06-0082 | Medium | Properties and forms of water

A source table contains one accurate entry involving 'Attraction of water to a different surface'. Which option preserves a correct biological association?

- A. Shrinkage of a protoplast in a hypertonic solution - Adhesion
- B. Source tissue in phloem transport - Zero
- C. Water potential of pure water at standard conditions - Plasmolysis
- D. Attraction of water to a different surface - Adhesion

BIO-12-CH06-0083 | Hard | Transpiration

Two conclusions must both be valid to support the reference concept 'Loss of water as vapour from aerial plant parts'. Which option contains two accurate associations?

- A. Soil water held tightly to particle surfaces - Plasmolysis; Shrinkage of a protoplast in a hypertonic solution - Hygroscopic water
- B. Loss of water as vapour from aerial plant parts - Transpiration; Barrier forcing water through living cells at the endodermis - Casparian strip
- C. Main tissue conducting water upward - Active absorption; Energy-dependent uptake of mineral ions - Xylem
- D. Net movement of water through a selectively permeable membrane - Symplast pathway; Movement through connected cytoplasm via plasmodesmata - Osmosis

BIO-12-CH06-0084 | Medium | Stomatal apparatus

A learner encounters the observation 'Cells controlling the stomatal pore' while analysing stomatal apparatus. Which conclusion is supported by the prescribed biology?

- A. Mature photosynthetic leaf
- B. Dumb-bell shaped
- C. Guard cells
- D. Mass flow

BIO-12-CH06-0085 | Challenging | Stomatal apparatus

A student submits four conclusions after integrating related chapter concepts.

- I. Guard-cell shape typical of grasses corresponds to Dumb-bell shaped.
- II. Guard-cell shape typical of dicots corresponds to Potassium.
- III. Ion whose accumulation promotes guard-cell opening corresponds to Root-hair zone.
- IV. Principal water-absorbing region of a young root corresponds to Root-hair zone.

Which combination should be accepted without correction?

- A. I, II, III and IV
- B. II only
- C. III only
- D. I and IV only

BIO-12-CH06-0086 | Medium | Mineral and food transport

A learner encounters the observation 'Pressure-driven movement through sieve tubes' while analysing mineral and food transport. Which conclusion is supported by the prescribed biology?

- A. Mature photosynthetic leaf
- B. Xylem
- C. Mass flow
- D. Solute potential

BIO-12-CH06-0087 | Easy | Properties and forms of water

Which description is correctly represented by Capillary water in this chapter?

- A. Source tissue in phloem transport
- B. Energy-dependent uptake of mineral ions
- C. Soil water most readily available to plants
- D. Theory chiefly explaining ascent of sap

BIO-12-CH06-0088 | Hard | Transpiration

Two conclusions must both be valid to support the reference concept 'Water loss through lenticels'. Which option contains two accurate associations?

- A. Soil water most readily available to plants - Hygroscopic water; Soil water held tightly to particle surfaces - Capillary water
- B. Principal water-absorbing region of a young root - Guard cells; Cells controlling the stomatal pore - Root-hair zone
- C. Soil water held tightly to particle surfaces - Cohesion; Attraction between water molecules - Hygroscopic water
- D. Water loss through lenticels - Lenticular transpiration; Theory chiefly explaining ascent of sap - Cohesion-tension theory

Standard 12 | Chapter 07

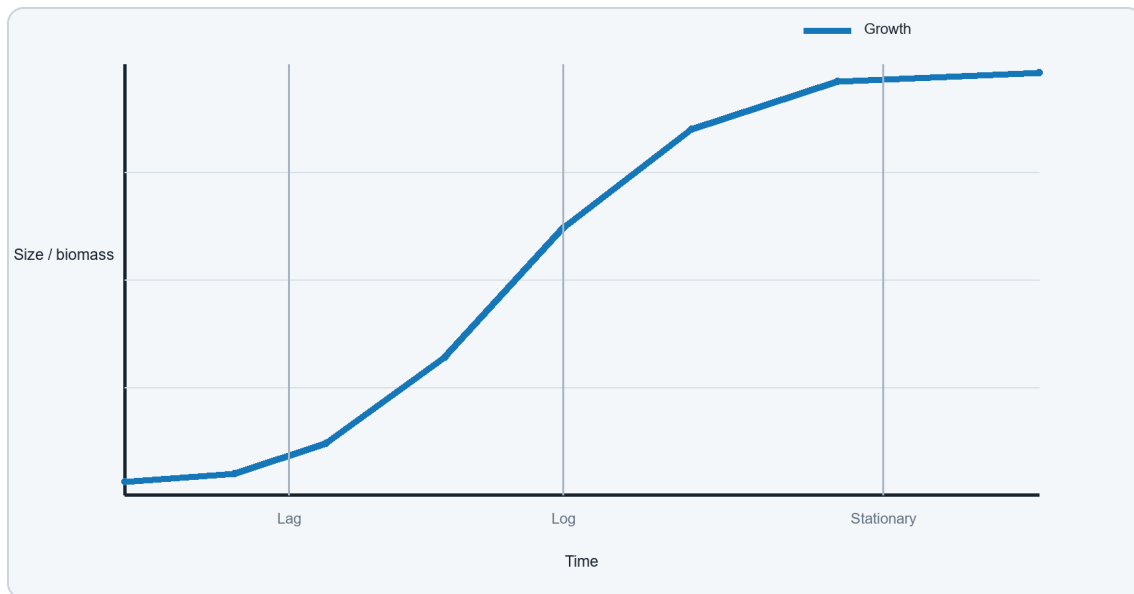
Plant Growth and Mineral Nutrition

BIO-12-CH07-0001 | Hard | Sigmoid growth curve

Why does the curve approach a plateau rather than continue exponentially?

Sigmoid growth curve

Typical phases of growth under limiting conditions



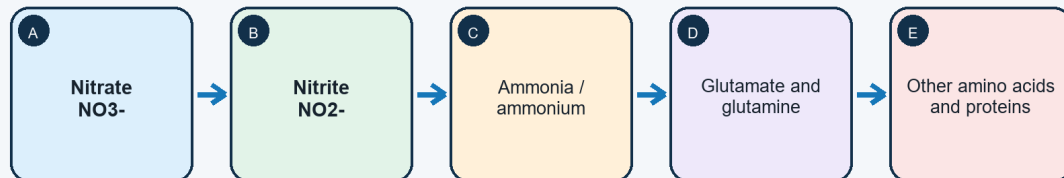
- A. All cells immediately stop metabolism
- B. Carrying capacity increases without limit
- C. Resources become limiting and growth rate falls
- D. The organism loses all genetic material

BIO-12-CH07-0002 | Challenging | Nitrogen assimilation in a plant

Molybdenum deficiency reduces nitrate-reductase activity. Which downstream effect is expected first?

Nitrogen assimilation in a plant

Reduction of nitrate precedes amino-acid synthesis



Nitrate reductase requires molybdenum-containing cofactors.

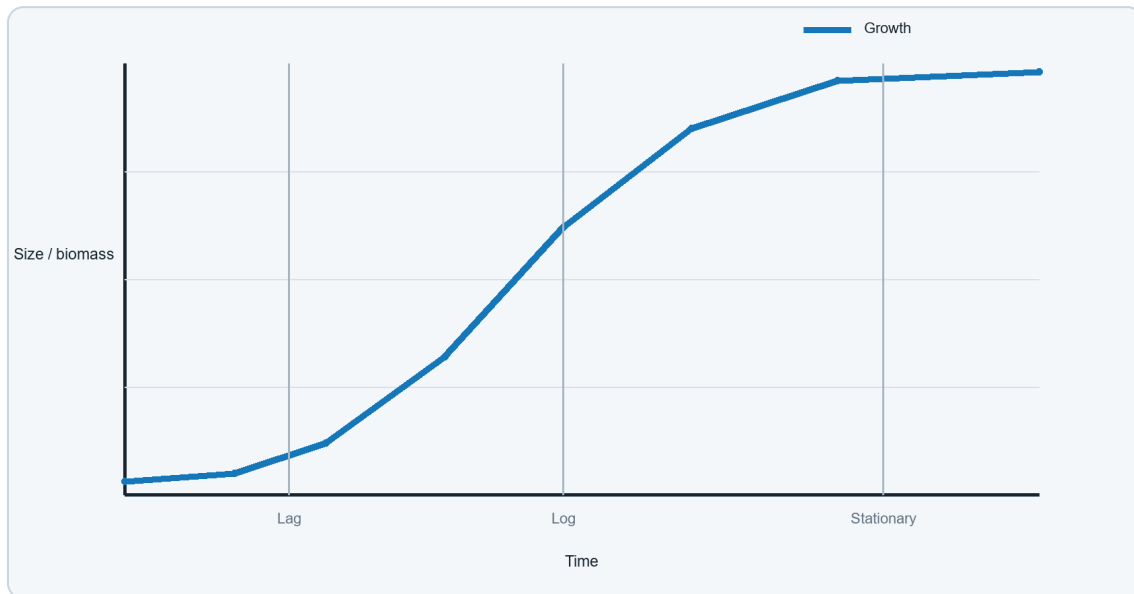
- A. Impaired nitrate assimilation and reduced amino-acid synthesis
- B. Faster conversion of nitrate to protein
- C. Immediate conversion of nitrate into atmospheric nitrogen
- D. Selective increase in chlorophyll without nitrogen uptake

BIO-12-CH07-0003 | Easy | Sigmoid growth curve

During which part of the sigmoid curve is absolute growth rate greatest?

Sigmoid growth curve

Typical phases of growth under limiting conditions



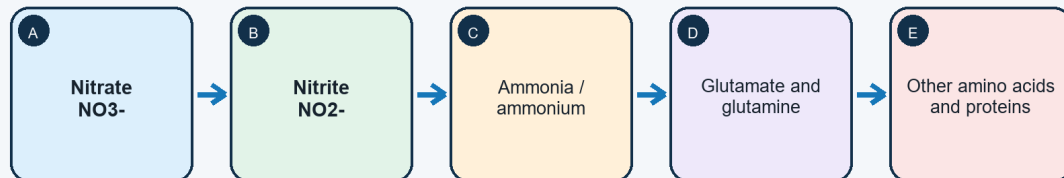
- A. Initial lag only
- B. Log or exponential phase
- C. Stationary phase
- D. After all cells die

BIO-12-CH07-0004 | Medium | Nitrogen assimilation in a plant

Which enzyme catalyses the first reduction step from nitrate to nitrite?

Nitrogen assimilation in a plant

Reduction of nitrate precedes amino-acid synthesis



Nitrate reductase requires molybdenum-containing cofactors.

- A. Nitrogenase
- B. Rubisco
- C. Nitrate reductase
- D. Pepsin

BIO-12-CH07-0005 | Medium | Photoperiodism and vernalization

A learner encounters the observation 'Response of flowering to relative day and night length' while analysing photoperiodism and vernalization. Which conclusion is supported by the prescribed biology?

- A. Geometric growth
- B. Vernalization
- C. Photoperiodism
- D. Nitrogen fixation

BIO-12-CH07-0006 | Easy | Essential mineral elements

Which biological term or result is specifically identified by the description 'Element important in ATP and nucleic acids'?

- A. Phosphorus
- B. Phytochrome
- C. Nitrogen
- D. Absolute growth rate

BIO-12-CH07-0007 | Hard | Plant hormones II

A reviewer compares several entries while checking 'Hormone associated with stomatal closure and stress'. Which entry must be corrected?

- A. Central atom of chlorophyll - Magnesium
- B. Growth occurring throughout life in meristematic regions - Indeterminate growth
- C. Hormone associated with stomatal closure and stress - Gibberellin
- D. Increase by a constant amount per unit time - Arithmetic growth

BIO-12-CH07-0008 | Hard | Photoperiodism and vernalization

A reviewer compares several entries while checking 'Cold treatment promoting flowering'. Which entry must be corrected?

- A. Element required for functioning of stomata and enzyme activation - Potassium
- B. Phase in which cells acquire specialized form and function - Maturation phase
- C. Element required for synthesis of amino acids and nucleic acids - Nitrogen
- D. Cold treatment promoting flowering - Sigmoid curve

BIO-12-CH07-0009 | Medium | Growth fundamentals

A source table contains one accurate entry involving 'Growth measurable as increase per unit time'. Which option preserves a correct biological association?

- A. Growth per unit initial size per unit time - Arithmetic growth
- B. Increase proportional to the existing size - Growth rate
- C. Hormone delaying leaf senescence - Relative growth rate
- D. Growth measurable as increase per unit time - Growth rate

BIO-12-CH07-0010 | Hard | Growth rates

Two conclusions must both be valid to support the reference concept 'Increase proportional to the existing size'. Which option contains two accurate associations?

- A. Hormone associated with stomatal closure and stress - Ethylene; Gaseous hormone promoting fruit ripening - Absciscic acid
- B. Increase proportional to the existing size - Geometric growth; Permanent irreversible increase in size - Growth
- C. Conversion of atmospheric nitrogen to ammonia - Absciscic acid; Hormone associated with stomatal closure and stress - Nitrogen fixation
- D. Gaseous hormone promoting fruit ripening - Development; Overall sequence from germination to senescence - Ethylene

BIO-12-CH07-0011 | Medium | Plant hormones II

Complete the statement accurately: Hormone associated with stomatal closure and stress corresponds to _____.

- A. Magnesium
- B. Absciscic acid
- C. Auxin
- D. Gibberellin

BIO-12-CH07-0012 | Medium | Essential mineral elements

A source table contains one accurate entry involving 'Central atom of chlorophyll'. Which option preserves a correct biological association?

- A. Overall sequence from germination to senescence - Magnesium
- B. Permanent irreversible increase in size - Maturation phase
- C. Central atom of chlorophyll - Magnesium
- D. Typical growth curve of a whole organ - Denitrification

BIO-12-CH07-0013 | Easy | Nitrogen metabolism

Which description is correctly represented by Nitrification in this chapter?

- A. Element required for synthesis of amino acids and nucleic acids
- B. Symbiotic nitrogen-fixing bacterium in legume nodules
- C. Conversion of ammonia to nitrite and nitrate
- D. Pigment involved in photoperiod perception

BIO-12-CH07-0014 | Challenging | Plant hormones I

A researcher checks whether four linked observations have been interpreted correctly.

- I. Hormone promoting cell division corresponds to Gibberellin.
- II. Hormone promoting stem elongation and bolting corresponds to Gibberellin.
- III. Hormone promoting cell elongation and apical dominance corresponds to Arithmetic growth.
- IV. Increase by a constant amount per unit time corresponds to Arithmetic growth.

Which combination survives the biological review?

- A. I and IV only
- B. I and II only
- C. I, III and IV only
- D. II and IV only

BIO-12-CH07-0015 | Hard | Nitrogen metabolism

A reviewer compares several entries while checking 'Reduction of nitrate to atmospheric nitrogen'. Which entry must be corrected?

- A. Permanent irreversible increase in size - Growth
- B. Reduction of nitrate to atmospheric nitrogen - Vernalization
- C. Element required for functioning of stomata and enzyme activation - Potassium
- D. Plant flowering when day length is shorter than a critical value - Short-day plant

BIO-12-CH07-0016 | Medium | Phases and curves

A learner encounters the observation 'Typical growth curve of a whole organ' while analysing phases and curves. Which conclusion is supported by the prescribed biology?

- A. Ethylene
- B. Phytochrome
- C. Vernalization
- D. Sigmoid curve

BIO-12-CH07-0017 | Hard | Growth fundamentals

A reviewer compares several entries while checking 'Growth occurring throughout life in meristematic regions'. Which entry must be corrected?

- A. Growth occurring throughout life in meristematic regions - 2,4-D
- B. Conversion of atmospheric nitrogen to ammonia - Nitrogen fixation
- C. Element required for synthesis of amino acids and nucleic acids - Nitrogen
- D. Hormone promoting cell elongation and apical dominance - Auxin

BIO-12-CH07-0018 | Hard | Plant hormones I

Two conclusions must both be valid to support the reference concept 'Gaseous hormone promoting fruit ripening'. Which option contains two accurate associations?

- A. Overall sequence from germination to senescence - Denitrification; Reduction of nitrate to atmospheric nitrogen - Development
- B. Gaseous hormone promoting fruit ripening - Ethylene; Hormone associated with stomatal closure and stress - Abscissic acid
- C. Element important in ATP and nucleic acids - Geometric growth; Increase proportional to the existing size - Phosphorus
- D. Reduction of nitrate to atmospheric nitrogen - Cytokinin; Hormone delaying leaf senescence - Denitrification

BIO-12-CH07-0019 | Easy | Phases and curves

Which description is correctly represented by Elongation phase in this chapter?

- A. Overall sequence from germination to senescence
- B. Phase dominated by cell enlargement
- C. Element required for synthesis of amino acids and nucleic acids
- D. Growth phase dominated by active cell division

BIO-12-CH07-0020 | Medium | Growth rates

Complete the statement accurately: Increase by a constant amount per unit time corresponds to ____.

- A. Arithmetic growth
- B. Denitrification
- C. Elongation phase
- D. Magnesium

BIO-12-CH07-0021 | Easy | Photoperiodism and vernalization

Which biological term or result is specifically identified by the description 'Pigment involved in photoperiod perception'?

- A. Short-day plant
- B. Photoperiodism
- C. Phytochrome
- D. Growth

BIO-12-CH07-0022 | Hard | Essential mineral elements

Two conclusions must both be valid to support the reference concept 'Element important in ATP and nucleic acids'. Which option contains two accurate associations?

- A. Conversion of atmospheric nitrogen to ammonia - Photoperiodism; Response of flowering to relative day and night length - Nitrogen fixation
- B. Element important in ATP and nucleic acids - Phosphorus; Gaseous hormone promoting fruit ripening - Ethylene
- C. Total growth per unit time - Relative growth rate; Growth per unit initial size per unit time - Absolute growth rate
- D. Response of flowering to relative day and night length - Absolute growth rate; Total growth per unit time - Photoperiodism

BIO-12-CH07-0023 | Easy | Plant hormones II

Which biological term or result is specifically identified by the description 'Hormone delaying leaf senescence'?

- A. 2,4-D
- B. Cytokinin
- C. Geometric growth
- D. Absciscic acid

BIO-12-CH07-0024 | Hard | Growth rates

Two conclusions must both be valid to support the reference concept 'Growth per unit initial size per unit time'. Which option contains two accurate associations?

- A. Growth per unit initial size per unit time - Relative growth rate; Hormone promoting cell division - Cytokinin
- B. Hormone delaying leaf senescence - Auxin; Hormone used to promote rooting in stem cuttings - Cytokinin
- C. Growth phase dominated by active cell division - Maturation phase; Phase in which cells acquire specialized form and function - Meristematic phase
- D. Hormone used to promote rooting in stem cuttings - Growth rate; Growth measurable as increase per unit time - Auxin

BIO-12-CH07-0025 | Medium | Essential mineral elements

A learner encounters the observation 'Element required for functioning of stomata and enzyme activation' while analysing essential mineral elements. Which conclusion is supported by the prescribed biology?

- A. Potassium
- B. Geometric growth
- C. Magnesium
- D. Cytokinin

BIO-12-CH07-0026 | Easy | Growth rates

Which description is correctly represented by Geometric growth in this chapter?

- A. Overall sequence from germination to senescence
- B. Gaseous hormone promoting fruit ripening
- C. Hormone associated with stomatal closure and stress
- D. Increase proportional to the existing size

BIO-12-CH07-0027 | Medium | Nitrogen metabolism

A source table contains one accurate entry involving 'Conversion of atmospheric nitrogen to ammonia'. Which option preserves a correct biological association?

- A. Response of flowering to relative day and night length - Absciscic acid
- B. Conversion of atmospheric nitrogen to ammonia - Nitrogen fixation
- C. Growth phase dominated by active cell division - 2,4-D
- D. Hormone used to promote rooting in stem cuttings - Nitrogen fixation

BIO-12-CH07-0028 | Challenging | Nitrogen metabolism

Evaluate the following source-based biological statements.

- I. Conversion of atmospheric nitrogen to ammonia corresponds to Nitrogen fixation.
- II. Symbiotic nitrogen-fixing bacterium in legume nodules corresponds to Denitrification.
- III. Reduction of nitrate to atmospheric nitrogen corresponds to Phosphorus.
- IV. Element important in ATP and nucleic acids corresponds to Nitrogen fixation.

Which option identifies exactly the correct statements?

- A. I only
- B. I and II only
- C. III only
- D. II, III and IV only

BIO-12-CH07-0029 | Hard | Plant hormones II

A reviewer compares several entries while checking 'Hormone delaying leaf senescence'. Which entry must be corrected?

- A. Conversion of atmospheric nitrogen to ammonia - Nitrogen fixation
- B. Central atom of chlorophyll - Magnesium
- C. Growth occurring throughout life in meristematic regions - Indeterminate growth
- D. Hormone delaying leaf senescence - Denitrification

BIO-12-CH07-0030 | Medium | Plant hormones II

A source table contains one accurate entry involving 'Hormone associated with stomatal closure and stress'. Which option preserves a correct biological association?

- A. Hormone promoting stem elongation and bolting - Magnesium
- B. Hormone associated with stomatal closure and stress - Absciscic acid
- C. Increase by a constant amount per unit time - Absciscic acid
- D. Symbiotic nitrogen-fixing bacterium in legume nodules - Absolute growth rate

BIO-12-CH07-0031 | Easy | Growth fundamentals

Which description is correctly represented by Development in this chapter?

- A. Growth per unit initial size per unit time
- B. Permanent irreversible increase in size
- C. Overall sequence from germination to senescence
- D. Reduction of nitrate to atmospheric nitrogen

BIO-12-CH07-0032 | Hard | Plant hormones II

Two conclusions must both be valid to support the reference concept 'Synthetic auxin widely used as a selective weedicide'. Which option contains two accurate associations?

- A. Increase proportional to the existing size - Relative growth rate; Growth per unit initial size per unit time - Geometric growth
- B. Total growth per unit time - Nitrogen; Element required for synthesis of amino acids and nucleic acids - Absolute growth rate
- C. Synthetic auxin widely used as a selective weedicide - 2,4-D; Typical growth curve of a whole organ - Sigmoid curve
- D. Element required for synthesis of amino acids and nucleic acids - Geometric growth; Increase proportional to the existing size - Nitrogen

BIO-12-CH07-0033 | Medium | Plant hormones II

A source table contains one accurate entry involving 'Synthetic auxin widely used as a selective weedicide'. Which option preserves a correct biological association?

- A. Total growth per unit time - Geometric growth
- B. Hormone promoting cell elongation and apical dominance - Nitrogen
- C. Element required for synthesis of amino acids and nucleic acids - Relative growth rate
- D. Synthetic auxin widely used as a selective weedicide - 2,4-D

BIO-12-CH07-0034 | Easy | Growth fundamentals

Which description is correctly represented by Indeterminate growth in this chapter?

- A. Overall sequence from germination to senescence
- B. Permanent irreversible increase in size
- C. Conversion of atmospheric nitrogen to ammonia
- D. Growth occurring throughout life in meristematic regions

BIO-12-CH07-0035 | Medium | Phases and curves

Complete the statement accurately: Typical growth curve of a whole organ corresponds to _____.

- A. Maturation phase
- B. Vernalization
- C. Sigmoid curve
- D. Phytochrome

BIO-12-CH07-0036 | Easy | Plant hormones I

Which description is correctly represented by Ethylene in this chapter?

- A. Hormone associated with stomatal closure and stress
- B. Gaseous hormone promoting fruit ripening
- C. Hormone promoting cell elongation and apical dominance
- D. Element important in ATP and nucleic acids

BIO-12-CH07-0037 | Medium | Growth rates

A learner encounters the observation 'Increase by a constant amount per unit time' while analysing growth rates. Which conclusion is supported by the prescribed biology?

- A. Denitrification
- B. Elongation phase
- C. Arithmetic growth
- D. 2,4-D

BIO-12-CH07-0038 | Medium | Photoperiodism and vernalization

Complete the statement accurately: Response of flowering to relative day and night length corresponds to _____.

- A. Cytokinin
- B. Nitrogen fixation
- C. Photoperiodism
- D. Vernalization

BIO-12-CH07-0039 | Easy | Phases and curves

Which description is correctly represented by Meristematic phase in this chapter?

- A. Phase dominated by cell enlargement
- B. Growth phase dominated by active cell division
- C. Typical growth curve of a whole organ
- D. Cold treatment promoting flowering

BIO-12-CH07-0040 | Medium | Phases and curves

A learner encounters the observation 'Phase dominated by cell enlargement' while analysing phases and curves. Which conclusion is supported by the prescribed biology?

- A. Elongation phase
- B. Development
- C. Sigmoid curve
- D. Auxin

BIO-12-CH07-0041 | Easy | Growth fundamentals

Which biological term or result is specifically identified by the description 'Growth occurring throughout life in meristematic regions'?

- A. Growth
- B. Development
- C. Indeterminate growth
- D. Nitrogen fixation

BIO-12-CH07-0042 | Hard | Phases and curves

A reviewer compares several entries while checking 'Growth phase dominated by active cell division'. Which entry must be corrected?

- A. Hormone promoting cell division - Cytokinin
- B. Growth phase dominated by active cell division - Vernalization
- C. Hormone promoting stem elongation and bolting - Gibberellin
- D. Response of flowering to relative day and night length - Photoperiodism

BIO-12-CH07-0043 | Medium | Plant hormones I

A learner encounters the observation 'Hormone promoting cell elongation and apical dominance' while analysing plant hormones i. Which conclusion is supported by the prescribed biology?

- A. Auxin
- B. Growth rate
- C. Development
- D. Nitrogen

BIO-12-CH07-0044 | Easy | Plant hormones II

Which biological term or result is specifically identified by the description 'Hormone used to promote rooting in stem cuttings'?

- A. 2,4-D
- B. Auxin
- C. Indeterminate growth
- D. Rhizobium

BIO-12-CH07-0045 | Easy | Plant hormones II

Which biological term or result is specifically identified by the description 'Hormone associated with stomatal closure and stress'?

- A. Denitrification
- B. Absciscic acid
- C. Indeterminate growth
- D. Gibberellin

BIO-12-CH07-0046 | Medium | Growth rates

Complete the statement accurately: Increase proportional to the existing size corresponds to _____.

- A. Growth
- B. Development
- C. Geometric growth
- D. Absciscic acid

BIO-12-CH07-0047 | Hard | Phases and curves

Two conclusions must both be valid to support the reference concept 'Growth phase dominated by active cell division'. Which option contains two accurate associations?

- A. Growth phase dominated by active cell division - Meristematic phase; Cold treatment promoting flowering - Vernalization
- B. Reduction of nitrate to atmospheric nitrogen - Gibberellin; Hormone promoting stem elongation and bolting - Denitrification
- C. Hormone promoting stem elongation and bolting - Cytokinin; Hormone promoting cell division - Gibberellin
- D. Response of flowering to relative day and night length - Ethylene; Gaseous hormone promoting fruit ripening - Photoperiodism

BIO-12-CH07-0048 | Easy | Nitrogen metabolism

Which biological term or result is specifically identified by the description 'Reduction of nitrate to atmospheric nitrogen'?

- A. Vernalization
- B. Short-day plant
- C. Ethylene
- D. Denitrification

BIO-12-CH07-0049 | Easy | Nitrogen metabolism

Which biological term or result is specifically identified by the description 'Conversion of atmospheric nitrogen to ammonia'?

- A. Nitrogen fixation
- B. Denitrification
- C. Photoperiodism
- D. Nitrification

BIO-12-CH07-0050 | Easy | Essential mineral elements

Which biological term or result is specifically identified by the description 'Element required for functioning of stomata and enzyme activation'?

- A. Cytokinin
- B. Nitrogen
- C. Absciscic acid
- D. Potassium

BIO-12-CH07-0051 | Hard | Essential mineral elements

A reviewer compares several entries while checking 'Element required for functioning of stomata and enzyme activation'. Which entry must be corrected?

- A. Response of flowering to relative day and night length - Photoperiodism
- B. Hormone promoting cell division - Cytokinin
- C. Element required for functioning of stomata and enzyme activation - Geometric growth
- D. Synthetic auxin widely used as a selective weedicide - 2,4-D

BIO-12-CH07-0052 | Medium | Nitrogen metabolism

Complete the statement accurately: Symbiotic nitrogen-fixing bacterium in legume nodules corresponds to _____.

- A. Indeterminate growth
- B. 2,4-D
- C. Gibberellin
- D. Rhizobium

BIO-12-CH07-0053 | Hard | Nitrogen metabolism

A reviewer compares several entries while checking 'Conversion of atmospheric nitrogen to ammonia'. Which entry must be corrected?

- A. Hormone delaying leaf senescence - Cytokinin
- B. Overall sequence from germination to senescence - Development
- C. Hormone associated with stomatal closure and stress - Absciscic acid
- D. Conversion of atmospheric nitrogen to ammonia - Photoperiodism

BIO-12-CH07-0054 | Medium | Phases and curves

A source table contains one accurate entry involving 'Phase dominated by cell enlargement'. Which option preserves a correct biological association?

- A. Phase dominated by cell enlargement - Elongation phase
- B. Hormone delaying leaf senescence - Phosphorus
- C. Symbiotic nitrogen-fixing bacterium in legume nodules - Elongation phase
- D. Element required for synthesis of amino acids and nucleic acids - Rhizobium

BIO-12-CH07-0055 | Medium | Growth fundamentals

A learner encounters the observation 'Growth measurable as increase per unit time' while analysing growth fundamentals. Which conclusion is supported by the prescribed biology?

- A. Maturation phase
- B. Cytokinin
- C. Auxin
- D. Growth rate

BIO-12-CH07-0056 | Challenging | Plant hormones II

A student submits four conclusions after integrating related chapter concepts.

- I. Synthetic auxin widely used as a selective weedicide corresponds to Cytokinin.
- II. Hormone delaying leaf senescence corresponds to Cytokinin.
- III. Hormone associated with stomatal closure and stress corresponds to Absciscic acid.
- IV. Typical growth curve of a whole organ corresponds to 2,4-D.

Which combination should be accepted without correction?

- A. II and III only
- B. I, II, III and IV
- C. I and III only
- D. I, II and III only

BIO-12-CH07-0057 | Medium | Plant hormones I

A source table contains one accurate entry involving 'Hormone promoting cell elongation and apical dominance'. Which option preserves a correct biological association?

- A. Hormone promoting cell elongation and apical dominance - Auxin
- B. Conversion of atmospheric nitrogen to ammonia - Auxin
- C. Element required for synthesis of amino acids and nucleic acids - Vernalization
- D. Growth measurable as increase per unit time - Nitrogen fixation

BIO-12-CH07-0058 | Easy | Essential mineral elements

Which description is correctly represented by Phosphorus in this chapter?

- A. Conversion of atmospheric nitrogen to ammonia
- B. Pigment involved in photoperiod perception
- C. Gaseous hormone promoting fruit ripening
- D. Element important in ATP and nucleic acids

BIO-12-CH07-0059 | Hard | Growth fundamentals

Two conclusions must both be valid to support the reference concept 'Growth occurring throughout life in meristematic regions'. Which option contains two accurate associations?

- A. Growth occurring throughout life in meristematic regions - Indeterminate growth; Synthetic auxin widely used as a selective weedicide - 2,4-D
- B. Conversion of atmospheric nitrogen to ammonia - Auxin; Hormone promoting cell elongation and apical dominance - Nitrogen fixation
- C. Element required for synthesis of amino acids and nucleic acids - Geometric growth; Increase proportional to the existing size - Nitrogen
- D. Increase proportional to the existing size - Sigmoid curve; Typical growth curve of a whole organ - Geometric growth

BIO-12-CH07-0060 | Medium | Photoperiodism and vernalization

A source table contains one accurate entry involving 'Pigment involved in photoperiod perception'. Which option preserves a correct biological association?

- A. Permanent irreversible increase in size - Meristematic phase
- B. Hormone promoting stem elongation and bolting - Magnesium
- C. Pigment involved in photoperiod perception - Phytochrome
- D. Growth phase dominated by active cell division - Phytochrome

BIO-12-CH07-0061 | Challenging | Growth rates

During error analysis, four proposed associations are reviewed.

- I. Growth per unit initial size per unit time corresponds to Relative growth rate.
- II. Increase proportional to the existing size corresponds to Geometric growth.
- III. Total growth per unit time corresponds to Absolute growth rate.
- IV. Conversion of ammonia to nitrite and nitrate corresponds to Nitrification.

Which set contains all and only the valid conclusions?

- A. I only
- B. I, II, III and IV
- C. I and III only
- D. I and II only

BIO-12-CH07-0062 | Challenging | Phases and curves

A researcher checks whether four linked observations have been interpreted correctly.

- I. Typical growth curve of a whole organ corresponds to Sigmoid curve.
- II. Phase dominated by cell enlargement corresponds to Meristematic phase.
- III. Growth phase dominated by active cell division corresponds to Cytokinin.
- IV. Hormone promoting cell division corresponds to Cytokinin.

Which combination survives the biological review?

- A. II, III and IV only
- B. III and IV only
- C. IV only
- D. I and IV only

BIO-12-CH07-0063 | Medium | Growth fundamentals

A learner encounters the observation 'Growth occurring throughout life in meristematic regions' while analysing growth fundamentals. Which conclusion is supported by the prescribed biology?

- A. Nitrogen
- B. Indeterminate growth
- C. Development
- D. 2,4-D

BIO-12-CH07-0064 | Medium | Essential mineral elements

Complete the statement accurately: Element important in ATP and nucleic acids corresponds to ____.

- A. Phosphorus
- B. Magnesium
- C. Nitrogen
- D. Absolute growth rate

BIO-12-CH07-0065 | Medium | Photoperiodism and vernalization

A learner encounters the observation 'Pigment involved in photoperiod perception' while analysing photoperiodism and vernalization. Which conclusion is supported by the prescribed biology?

- A. Phytochrome
- B. Short-day plant
- C. Photoperiodism
- D. Ethylene

BIO-12-CH07-0066 | Medium | Plant hormones II

Complete the statement accurately: Hormone delaying leaf senescence corresponds to _____.

- A. Denitrification
- B. 2,4-D
- C. Geometric growth
- D. Cytokinin

BIO-12-CH07-0067 | Hard | Phases and curves

Two conclusions must both be valid to support the reference concept 'Phase dominated by cell enlargement'. Which option contains two accurate associations?

- A. Cold treatment promoting flowering - Nitrogen; Element required for synthesis of amino acids and nucleic acids - Vernalization
- B. Symbiotic nitrogen-fixing bacterium in legume nodules - Phosphorus; Element important in ATP and nucleic acids - Rhizobium
- C. Overall sequence from germination to senescence - Vernalization; Cold treatment promoting flowering - Development
- D. Phase dominated by cell enlargement - Elongation phase; Hormone used to promote rooting in stem cuttings - Auxin

BIO-12-CH07-0068 | Hard | Essential mineral elements

A reviewer compares several entries while checking 'Element important in ATP and nucleic acids'. Which entry must be corrected?

- A. Growth per unit initial size per unit time - Relative growth rate
- B. Element important in ATP and nucleic acids - Ethylene
- C. Response of flowering to relative day and night length - Photoperiodism
- D. Total growth per unit time - Absolute growth rate

BIO-12-CH07-0069 | Medium | Growth fundamentals

Complete the statement accurately: Permanent irreversible increase in size corresponds to _____.

- A. Growth
- B. Auxin
- C. Growth rate
- D. Potassium

BIO-12-CH07-0070 | Medium | Nitrogen metabolism

A source table contains one accurate entry involving 'Symbiotic nitrogen-fixing bacterium in legume nodules'. Which option preserves a correct biological association?

- A. Pigment involved in photoperiod perception - Rhizobium
- B. Total growth per unit time - Sigmoid curve
- C. Typical growth curve of a whole organ - Phytochrome
- D. Symbiotic nitrogen-fixing bacterium in legume nodules - Rhizobium

BIO-12-CH07-0071 | Medium | Phases and curves

Complete the statement accurately: Growth phase dominated by active cell division corresponds to ____.

- A. Gibberellin
- B. Sigmoid curve
- C. Meristematic phase
- D. Elongation phase

BIO-12-CH07-0072 | Medium | Essential mineral elements

A learner encounters the observation 'Central atom of chlorophyll' while analysing essential mineral elements. Which conclusion is supported by the prescribed biology?

- A. Absciscic acid
- B. Nitrogen fixation
- C. Nitrogen
- D. Magnesium

BIO-12-CH07-0073 | Medium | Nitrogen metabolism

Complete the statement accurately: Reduction of nitrate to atmospheric nitrogen corresponds to ____.

- A. Potassium
- B. Vernalization
- C. Denitrification
- D. Ethylene

BIO-12-CH07-0074 | Hard | Photoperiodism and vernalization

Two conclusions must both be valid to support the reference concept 'Cold treatment promoting flowering'. Which option contains two accurate associations?

- A. Symbiotic nitrogen-fixing bacterium in legume nodules - Nitrogen; Element required for synthesis of amino acids and nucleic acids - Rhizobium
- B. Phase in which cells acquire specialized form and function - Arithmetic growth; Increase by a constant amount per unit time - Maturation phase
- C. Cold treatment promoting flowering - Vernalization; Typical growth curve of a whole organ - Sigmoid curve
- D. Element required for functioning of stomata and enzyme activation - Rhizobium; Symbiotic nitrogen-fixing bacterium in legume nodules - Potassium

BIO-12-CH07-0075 | Medium | Plant hormones I

Complete the statement accurately: Hormone promoting cell elongation and apical dominance corresponds to ____.

- A. Gibberellin
- B. Auxin
- C. Magnesium
- D. Development

BIO-12-CH07-0076 | Easy | Plant hormones II

Which description is correctly represented by 2,4-D in this chapter?

- A. Hormone delaying leaf senescence
- B. Synthetic auxin widely used as a selective weedicide
- C. Element required for synthesis of amino acids and nucleic acids
- D. Increase proportional to the existing size

BIO-12-CH07-0077 | Medium | Growth rates

A learner encounters the observation 'Total growth per unit time' while analysing growth rates. Which conclusion is supported by the prescribed biology?

- A. Development
- B. Auxin
- C. Elongation phase
- D. Absolute growth rate

BIO-12-CH07-0078 | Medium | Plant hormones II

A source table contains one accurate entry involving 'Hormone used to promote rooting in stem cuttings'. Which option preserves a correct biological association?

- A. Cold treatment promoting flowering - Indeterminate growth
- B. Element important in ATP and nucleic acids - Auxin
- C. Symbiotic nitrogen-fixing bacterium in legume nodules - Phytochrome
- D. Hormone used to promote rooting in stem cuttings - Auxin

BIO-12-CH07-0079 | Medium | Growth fundamentals

Complete the statement accurately: Overall sequence from germination to senescence corresponds to _____.

- A. Magnesium
- B. Cytokinin
- C. Development
- D. Denitrification

BIO-12-CH07-0080 | Easy | Photoperiodism and vernalization

Which biological term or result is specifically identified by the description 'Cold treatment promoting flowering'?

- A. Potassium
- B. Maturation phase
- C. Sigmoid curve
- D. Vernalization

BIO-12-CH07-0081 | Medium | Photoperiodism and vernalization

Complete the statement accurately: Cold treatment promoting flowering corresponds to _____.

- A. Potassium
- B. Vernalization
- C. Short-day plant
- D. Phytochrome

BIO-12-CH07-0082 | Hard | Plant hormones I

Two conclusions must both be valid to support the reference concept 'Hormone promoting cell elongation and apical dominance'. Which option contains two accurate associations?

- A. Element required for synthesis of amino acids and nucleic acids - Growth rate; Growth measurable as increase per unit time - Nitrogen
- B. Hormone promoting cell elongation and apical dominance - Auxin; Central atom of chlorophyll - Magnesium
- C. Overall sequence from germination to senescence - Nitrogen; Element required for synthesis of amino acids and nucleic acids - Development
- D. Cold treatment promoting flowering - Nitrogen fixation; Conversion of atmospheric nitrogen to ammonia - Vernalization

BIO-12-CH07-0083 | Hard | Plant hormones I

A reviewer compares several entries while checking 'Gaseous hormone promoting fruit ripening'. Which entry must be corrected?

- A. Overall sequence from germination to senescence - Development
- B. Increase proportional to the existing size - Geometric growth
- C. Gaseous hormone promoting fruit ripening - Absciscic acid
- D. Element required for synthesis of amino acids and nucleic acids - Nitrogen

BIO-12-CH07-0084 | Challenging | Essential mineral elements

Evaluate the following source-based biological statements.

- I. Element important in ATP and nucleic acids corresponds to Nitrogen.
- II. Element required for synthesis of amino acids and nucleic acids corresponds to Nitrogen.
- III. Central atom of chlorophyll corresponds to Magnesium.
- IV. Total growth per unit time corresponds to Absolute growth rate.

Which option identifies exactly the correct statements?

- A. I and IV only
- B. I, II and III only
- C. II, III and IV only
- D. I and II only

BIO-12-CH07-0085 | Challenging | Photoperiodism and vernalization

A student submits four conclusions after integrating related chapter concepts.

- I. Pigment involved in photoperiod perception corresponds to Phytochrome.
- II. Plant flowering when day length is shorter than a critical value corresponds to Vernalization.
- III. Cold treatment promoting flowering corresponds to Vernalization.
- IV. Gaseous hormone promoting fruit ripening corresponds to Phytochrome.

Which combination should be accepted without correction?

- A. I and III only
- B. IV only
- C. I, II, III and IV
- D. III only

BIO-12-CH07-0086 | Medium | Essential mineral elements

A source table contains one accurate entry involving 'Element required for synthesis of amino acids and nucleic acids'. Which option preserves a correct biological association?

- A. Element required for synthesis of amino acids and nucleic acids - Nitrogen
- B. Hormone used to promote rooting in stem cuttings - Cytokinin
- C. Growth occurring throughout life in meristematic regions - Growth rate
- D. Growth phase dominated by active cell division - Nitrogen

BIO-12-CH07-0087 | Hard | Growth rates

A reviewer compares several entries while checking 'Total growth per unit time'. Which entry must be corrected?

- A. Hormone promoting cell elongation and apical dominance - Auxin
- B. Hormone associated with stomatal closure and stress - Absciscic acid
- C. Element required for synthesis of amino acids and nucleic acids - Nitrogen
- D. Total growth per unit time - Nitrification

BIO-12-CH07-0088 | Hard | Growth fundamentals

Two conclusions must both be valid to support the reference concept 'Overall sequence from germination to senescence'. Which option contains two accurate associations?

- A. Overall sequence from germination to senescence - Development; Conversion of ammonia to nitrite and nitrate - Nitrification
- B. Reduction of nitrate to atmospheric nitrogen - Relative growth rate; Growth per unit initial size per unit time - Denitrification
- C. Response of flowering to relative day and night length - Vernalization; Cold treatment promoting flowering - Photoperiodism
- D. Growth per unit initial size per unit time - Photoperiodism; Response of flowering to relative day and night length - Relative growth rate

BIO-12-CH07-0089 | Challenging | Plant hormones I

During error analysis, four proposed associations are reviewed.

- I. Gaseous hormone promoting fruit ripening corresponds to Ethylene.
 - II. Hormone promoting cell elongation and apical dominance corresponds to Gibberellin.
 - III. Hormone promoting stem elongation and bolting corresponds to Gibberellin.
 - IV. Reduction of nitrate to atmospheric nitrogen corresponds to Denitrification.
- Which set contains all and only the valid conclusions?

- A. IV only
- B. II and IV only
- C. I, III and IV only
- D. I, II, III and IV

BIO-12-CH07-0090 | Easy | Plant hormones I

Which biological term or result is specifically identified by the description 'Gaseous hormone promoting fruit ripening'?

- A. Ethylene
- B. Nitrogen
- C. Development
- D. Absciscic acid

BIO-12-CH07-0091 | Easy | Growth rates

Which biological term or result is specifically identified by the description 'Total growth per unit time'?

- A. Absciscic acid
- B. Nitrification
- C. Absolute growth rate
- D. Auxin

BIO-12-CH07-0092 | Medium | Nitrogen metabolism

A source table contains one accurate entry involving 'Conversion of ammonia to nitrite and nitrate'. Which option preserves a correct biological association?

- A. Element required for synthesis of amino acids and nucleic acids - Auxin
- B. Conversion of ammonia to nitrite and nitrate - Nitrification
- C. Increase proportional to the existing size - Absolute growth rate
- D. Permanent irreversible increase in size - Growth rate

BIO-12-CH07-0093 | Medium | Growth rates

A source table contains one accurate entry involving 'Increase by a constant amount per unit time'. Which option preserves a correct biological association?

- A. Reduction of nitrate to atmospheric nitrogen - Ethylene
- B. Increase by a constant amount per unit time - Arithmetic growth
- C. Central atom of chlorophyll - Elongation phase
- D. Phase dominated by cell enlargement - Photoperiodism

BIO-12-CH07-0094 | Medium | Plant hormones I

A source table contains one accurate entry involving 'Hormone promoting stem elongation and bolting'. Which option preserves a correct biological association?

- A. Hormone promoting stem elongation and bolting - Gibberellin
- B. Phase dominated by cell enlargement - 2,4-D
- C. Total growth per unit time - Nitrogen
- D. Synthetic auxin widely used as a selective weedicide - Short-day plant

BIO-12-CH07-0095 | Hard | Photoperiodism and vernalization

A reviewer compares several entries while checking 'Pigment involved in photoperiod perception'. Which entry must be corrected?

- A. Gaseous hormone promoting fruit ripening - Ethylene
- B. Hormone promoting stem elongation and bolting - Gibberellin
- C. Pigment involved in photoperiod perception - Cytokinin
- D. Growth phase dominated by active cell division - Meristematic phase

BIO-12-CH07-0096 | Challenging | Growth fundamentals

A researcher checks whether four linked observations have been interpreted correctly.

- I. Permanent irreversible increase in size corresponds to Growth.
- II. Overall sequence from germination to senescence corresponds to Development.
- III. Growth occurring throughout life in meristematic regions corresponds to Magnesium.
- IV. Central atom of chlorophyll corresponds to Growth.

Which combination survives the biological review?

- A. II only
- B. II and III only
- C. I and IV only
- D. I and II only

BIO-12-CH07-0097 | Easy | Growth rates

Which description is correctly represented by Relative growth rate in this chapter?

- A. Growth per unit initial size per unit time
- B. Growth measurable as increase per unit time
- C. Increase proportional to the existing size
- D. Hormone promoting cell division

BIO-12-CH07-0098 | Hard | Plant hormones II

A reviewer compares several entries while checking 'Hormone used to promote rooting in stem cuttings'. Which entry must be corrected?

- A. Element important in ATP and nucleic acids - Phosphorus
- B. Symbiotic nitrogen-fixing bacterium in legume nodules - Rhizobium
- C. Growth occurring throughout life in meristematic regions - Indeterminate growth
- D. Hormone used to promote rooting in stem cuttings - Auxin

BIO-12-CH07-0099 | Easy | Phases and curves

Which biological term or result is specifically identified by the description 'Growth phase dominated by active cell division'?

- A. Meristematic phase
- B. Denitrification
- C. Photoperiodism
- D. Elongation phase

BIO-12-CH07-0100 | Easy | Plant hormones I

Which description is correctly represented by Auxin in this chapter?

- A. Central atom of chlorophyll
 - B. Hormone promoting cell elongation and apical dominance
 - C. Cold treatment promoting flowering
 - D. Growth measurable as increase per unit time
-

Standard 12 | Chapter 08

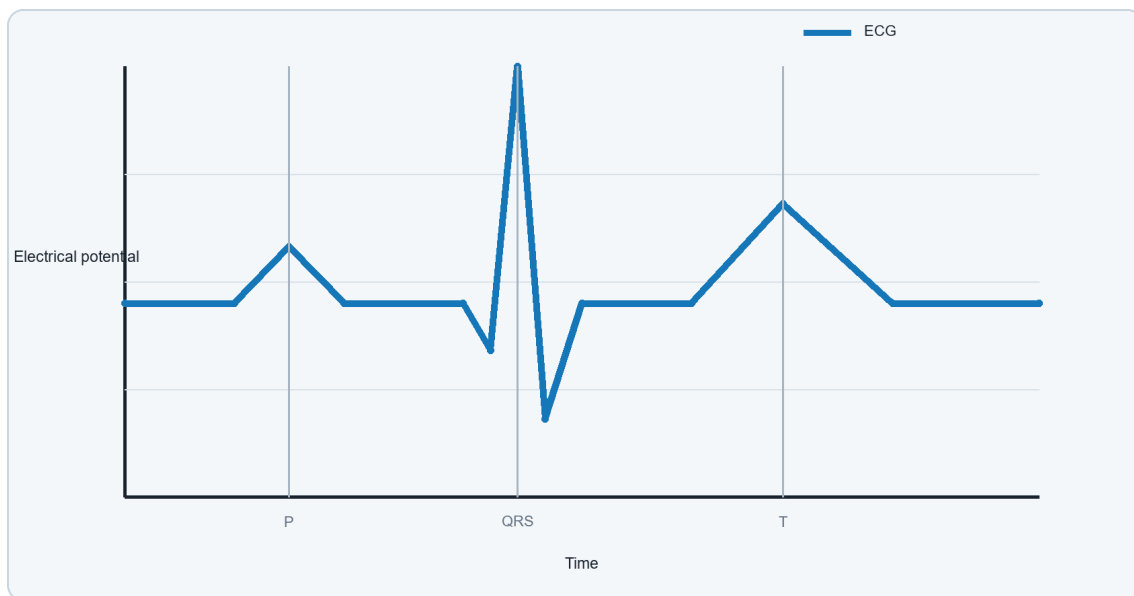
Respiration and Circulation

BIO-12-CH08-0001 | Medium | Electrical events in one cardiac cycle

Which ECG component in the figure represents ventricular depolarization?

Electrical events in one cardiac cycle

Simplified ECG sequence



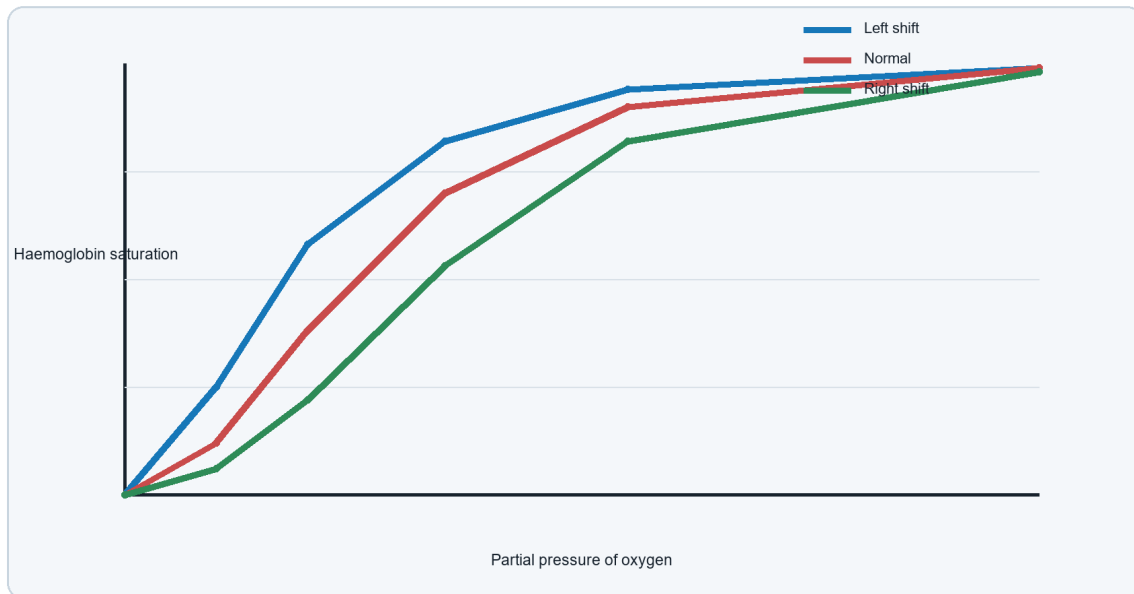
- A. QRS complex
- B. P wave
- C. T wave
- D. Interval after T only

BIO-12-CH08-0002 | Hard | Oxyhaemoglobin dissociation curves

At the same tissue oxygen pressure, what is the principal effect of a right shift?

Oxyhaemoglobin dissociation curves

A right shift promotes oxygen unloading in active tissues



- A. Less oxygen is unloaded to tissues
- B. More oxygen is unloaded to tissues
- C. Haemoglobin affinity becomes permanently infinite
- D. Oxygen cannot bind in the lungs

BIO-12-CH08-0003 | Medium | Respiratory organs

A source table contains one accurate entry involving 'Respiratory surface in earthworm'. Which option preserves a correct biological association?

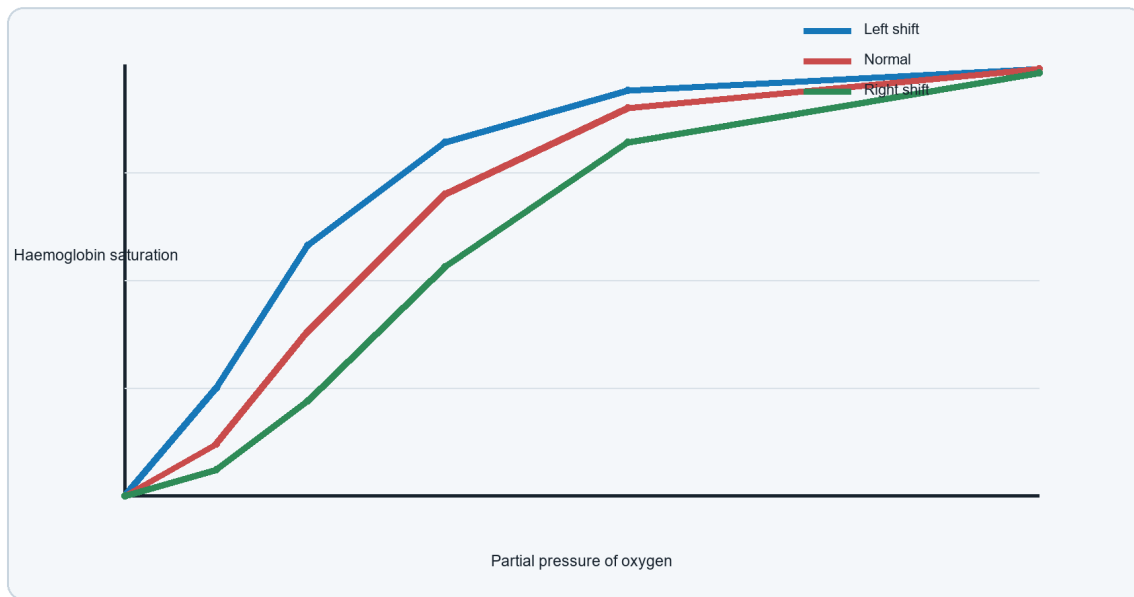
- A. ECG wave representing atrial depolarization - 0.8 second
- B. Respiratory surface in earthworm - Moist skin
- C. Anucleate biconcave cells transporting respiratory gases - Leucocytes
- D. Vessels containing valves to prevent backflow - P wave

BIO-12-CH08-0004 | Easy | Oxyhaemoglobin dissociation curves

Why is the normal oxyhaemoglobin dissociation curve sigmoid?

Oxyhaemoglobin dissociation curves

A right shift promotes oxygen unloading in active tissues



- A. Oxygen is insoluble in plasma
- B. Each haemoglobin binds only one oxygen molecule
- C. Carbon dioxide irreversibly destroys haemoglobin
- D. Cooperative binding of oxygen to haemoglobin

BIO-12-CH08-0005 | Easy | Human respiratory tract

Which description is correctly represented by Alveoli in this chapter?

- A. Microscopic sites of gaseous exchange
- B. Voice box
- C. Enzyme accelerating carbonic acid formation in red cells
- D. Change in thoracic volume during inspiration

BIO-12-CH08-0006 | Hard | Heart anatomy

A reviewer compares several entries while checking 'Vessel carrying blood from right ventricle to lungs'. Which entry must be corrected?

- A. ECG complex representing ventricular depolarization - QRS complex
- B. Fluid returning tissue proteins and absorbed fats to blood - Lymph
- C. Vessel carrying blood from right ventricle to lungs - Lungs
- D. ECG wave representing atrial depolarization - P wave

BIO-12-CH08-0007 | Medium | Blood cells

Complete the statement accurately: Cells chiefly responsible for defence corresponds to _____.

- A. Sinoatrial node
- B. Leucocytes
- C. Platelets
- D. Veins

BIO-12-CH08-0008 | Hard | Gas transport

A reviewer compares several entries while checking 'Major form of carbon dioxide transport'. Which entry must be corrected?

- A. Cell fragments important in clotting - Platelets
- B. Major form of carbon dioxide transport - QRS complex
- C. Change in thoracic volume during inspiration - Increase
- D. ECG wave representing atrial depolarization - P wave

BIO-12-CH08-0009 | Easy | Respiratory organs

Which biological term or result is specifically identified by the description 'Respiratory organs in insects'?

- A. 0.8 second
- B. Platelets
- C. Tracheal system
- D. Bicuspid or mitral valve

BIO-12-CH08-0010 | Hard | Mechanics of breathing

Two conclusions must both be valid to support the reference concept 'Change in intrapulmonary pressure during inspiration'. Which option contains two accurate associations?

- A. Valve between left atrium and left ventricle - Pharynx; Common passage for food and air - Bicuspid or mitral valve
- B. Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure; Shift promoting oxygen unloading at high carbon dioxide - Bohr effect
- C. ECG wave representing atrial depolarization - QRS complex; ECG complex representing ventricular depolarization - P wave
- D. Vessels carrying blood away from the heart - Larynx; Voice box - Arteries

BIO-12-CH08-0011 | Medium | Mechanics of breathing

A source table contains one accurate entry involving 'Main muscle of quiet inspiration'. Which option preserves a correct biological association?

- A. ECG complex representing ventricular depolarization - Diaphragm
- B. Vessel carrying blood from right ventricle to lungs - Trachea
- C. Windpipe supported by cartilaginous rings - Bicarbonate ions
- D. Main muscle of quiet inspiration - Diaphragm

BIO-12-CH08-0012 | Medium | Heart anatomy

A learner encounters the observation 'Vessel carrying blood from right ventricle to lungs' while analysing heart anatomy. Which conclusion is supported by the prescribed biology?

- A. Pulmonary artery
- B. Lungs
- C. Sinoatrial node
- D. Bicuspid or mitral valve

BIO-12-CH08-0013 | Medium | Mechanics of breathing

Complete the statement accurately: Main muscle of quiet inspiration corresponds to _____.

- A. Decrease below atmospheric pressure
- B. Increase
- C. Diaphragm
- D. Systolic pressure

BIO-12-CH08-0014 | Easy | Heart anatomy

Which description is correctly represented by Sinoatrial node in this chapter?

- A. Approximate duration of one cardiac cycle at 72 beats per minute
- B. Valve between left atrium and left ventricle
- C. Respiratory surface in earthworm
- D. Natural pacemaker of the heart

BIO-12-CH08-0015 | Medium | Human respiratory tract

Complete the statement accurately: Windpipe supported by cartilaginous rings corresponds to _____.

- A. Decrease below atmospheric pressure
- B. Trachea
- C. Leucocytes
- D. Mostly passive

BIO-12-CH08-0016 | Medium | Gas transport

A source table contains one accurate entry involving 'Major form of oxygen transport in blood'. Which option preserves a correct biological association?

- A. Major form of oxygen transport in blood - Oxyhaemoglobin
- B. Pressure recorded during ventricular systole - Tracheal system
- C. Main muscle of quiet inspiration - Decrease below atmospheric pressure
- D. Voice box - P wave

BIO-12-CH08-0017 | Medium | Respiratory organs

A learner encounters the observation 'Respiratory organs in most fishes' while analysing respiratory organs. Which conclusion is supported by the prescribed biology?

- A. Gills
- B. Lungs
- C. Mostly passive
- D. Carbonic anhydrase

BIO-12-CH08-0018 | Hard | Heart anatomy

Two conclusions must both be valid to support the reference concept 'Natural pacemaker of the heart'. Which option contains two accurate associations?

- A. Cells chiefly responsible for defence - Moist skin; Respiratory surface in earthworm - Leucocytes
- B. Natural pacemaker of the heart - Sinoatrial node; Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second
- C. ECG complex representing ventricular depolarization - Lungs; Primary respiratory organs in adult humans - QRS complex
- D. Windpipe supported by cartilaginous rings - Leucocytes; Cells chiefly responsible for defence - Trachea

BIO-12-CH08-0019 | Easy | Cardiac cycle and ECG

Which description is correctly represented by 0.8 second in this chapter?

- A. Approximate duration of one cardiac cycle at 72 beats per minute
- B. Vessels containing valves to prevent backflow
- C. Respiratory surface in earthworm
- D. Valve between right atrium and right ventricle

BIO-12-CH08-0020 | Hard | Respiratory organs

Two conclusions must both be valid to support the reference concept 'Respiratory organs in insects'. Which option contains two accurate associations?

- A. Respiratory organs in insects - Tracheal system; Vessel carrying blood from right ventricle to lungs - Pulmonary artery
- B. Approximate duration of one cardiac cycle at 72 beats per minute - Mostly passive; Quiet expiration at rest - 0.8 second
- C. Cell fragments important in clotting - 0.8 second; Approximate duration of one cardiac cycle at 72 beats per minute - Platelets
- D. Valve between left atrium and left ventricle - Platelets; Cell fragments important in clotting - Bicuspid or mitral valve

BIO-12-CH08-0021 | Hard | Human respiratory tract

Two conclusions must both be valid to support the reference concept 'Microscopic sites of gaseous exchange'. Which option contains two accurate associations?

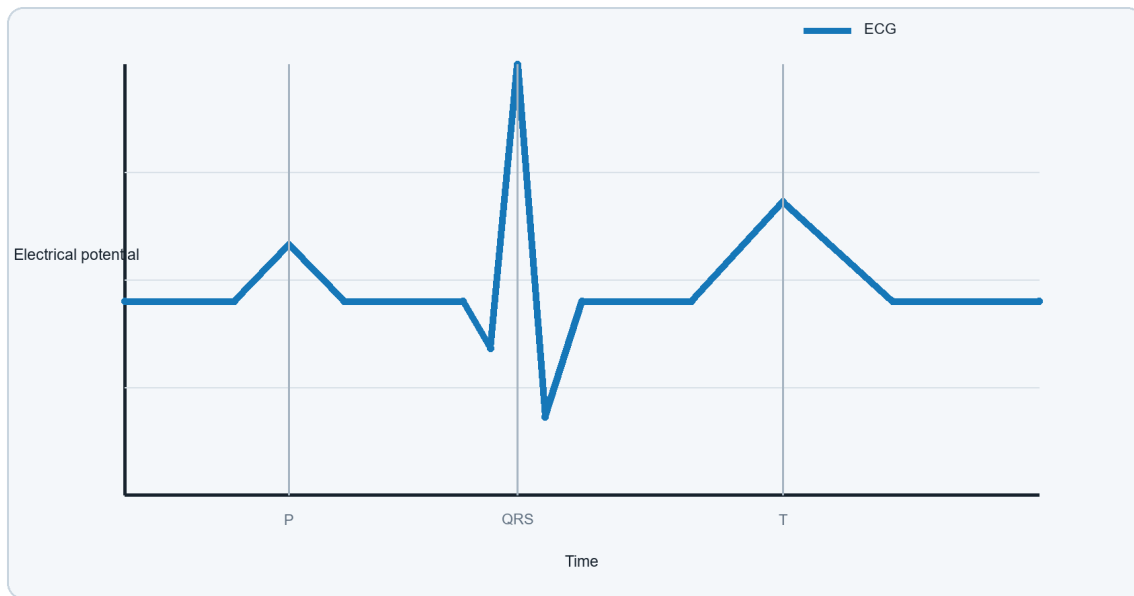
- A. Change in intrapulmonary pressure during inspiration - Diaphragm; Main muscle of quiet inspiration - Decrease below atmospheric pressure
- B. Major form of oxygen transport in blood - Decrease below atmospheric pressure; Change in intrapulmonary pressure during inspiration - Oxyhaemoglobin
- C. Valve between left atrium and left ventricle - Increase; Change in thoracic volume during inspiration - Bicuspid or mitral valve
- D. Microscopic sites of gaseous exchange - Alveoli; Vessels carrying blood away from the heart - Arteries

BIO-12-CH08-0022 | Challenging | Electrical events in one cardiac cycle

What is the functional value of the atrioventricular nodal delay before the QRS complex?

Electrical events in one cardiac cycle

Simplified ECG sequence



- A. It prevents atrial depolarization
- B. It causes semilunar valves to remain permanently open
- C. It initiates oxygen binding by haemoglobin
- D. It allows atrial emptying and ventricular filling before ventricular contraction

BIO-12-CH08-0023 | Easy | Human respiratory tract

Which biological term or result is specifically identified by the description 'Windpipe supported by cartilaginous rings'?

- A. Alveoli
- B. P wave
- C. Trachea
- D. Pharynx

BIO-12-CH08-0024 | Hard | Human respiratory tract

A reviewer compares several entries while checking 'Common passage for food and air'. Which entry must be corrected?

- A. Anucleate biconcave cells transporting respiratory gases - Erythrocytes
- B. Common passage for food and air - Moist skin
- C. Major form of oxygen transport in blood - Oxyhaemoglobin
- D. Major form of carbon dioxide transport - Bicarbonate ions

BIO-12-CH08-0025 | Easy | Cardiac cycle and ECG

Which description is correctly represented by P wave in this chapter?

- A. Approximate duration of one cardiac cycle at 72 beats per minute
- B. Major form of oxygen transport in blood
- C. Fluid returning tissue proteins and absorbed fats to blood
- D. ECG wave representing atrial depolarization

BIO-12-CH08-0026 | Hard | Respiratory organs

A reviewer compares several entries while checking 'Respiratory surface in earthworm'. Which entry must be corrected?

- A. Respiratory surface in earthworm - Decrease below atmospheric pressure
- B. Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second
- C. Most abundant granulocyte and active phagocyte - Neutrophil
- D. ECG wave representing atrial depolarization - P wave

BIO-12-CH08-0027 | Medium | Blood cells

Complete the statement accurately: Cell fragments important in clotting corresponds to ____.

- A. Larynx
- B. Trachea
- C. Platelets
- D. Oxyhaemoglobin

BIO-12-CH08-0028 | Medium | Cardiac cycle and ECG

A learner encounters the observation 'ECG complex representing ventricular depolarization' while analysing cardiac cycle and ecg. Which conclusion is supported by the prescribed biology?

- A. Gills
- B. Increase
- C. QRS complex
- D. T wave

BIO-12-CH08-0029 | Medium | Heart anatomy

A source table contains one accurate entry involving 'Vessel carrying blood from right ventricle to lungs'. Which option preserves a correct biological association?

- A. Vessel carrying blood from right ventricle to lungs - Pulmonary artery
- B. Fluid returning tissue proteins and absorbed fats to blood - P wave
- C. Primary respiratory organs in adult humans - QRS complex
- D. Microscopic sites of gaseous exchange - Lymph

BIO-12-CH08-0030 | Hard | Mechanics of breathing

A reviewer compares several entries while checking 'Main muscle of quiet inspiration'. Which entry must be corrected?

- A. Main muscle of quiet inspiration - Moist skin
- B. Pressure recorded during ventricular systole - Systolic pressure
- C. Respiratory organs in insects - Tracheal system
- D. Windpipe supported by cartilaginous rings - Trachea

BIO-12-CH08-0031 | Medium | Gas transport

A learner encounters the observation 'Shift promoting oxygen unloading at high carbon dioxide' while analysing gas transport. Which conclusion is supported by the prescribed biology?

- A. Carbonic anhydrase
- B. Larynx
- C. Bohr effect
- D. Bicarbonate ions

BIO-12-CH08-0032 | Medium | Vessels pressure and lymph

A learner encounters the observation 'Fluid returning tissue proteins and absorbed fats to blood' while analysing vessels pressure and lymph. Which conclusion is supported by the prescribed biology?

- A. Pulmonary artery
- B. Arteries
- C. Veins
- D. Lymph

BIO-12-CH08-0033 | Medium | Human respiratory tract

A source table contains one accurate entry involving 'Common passage for food and air'. Which option preserves a correct biological association?

- A. Common passage for food and air - Pharynx
- B. Major form of carbon dioxide transport - Oxyhaemoglobin
- C. Major form of oxygen transport in blood - Pharynx
- D. Respiratory surface in earthworm - Erythrocytes

BIO-12-CH08-0034 | Hard | Cardiac cycle and ECG

Two conclusions must both be valid to support the reference concept 'Approximate duration of one cardiac cycle at 72 beats per minute'. Which option contains two accurate associations?

- A. Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second; Vessels containing valves to prevent backflow - Veins
- B. Valve between left atrium and left ventricle - Leucocytes; Cells chiefly responsible for defence - Bicuspid or mitral valve
- C. Primary respiratory organs in adult humans - Bicuspid or mitral valve; Valve between left atrium and left ventricle - Lungs
- D. Respiratory surface in earthworm - Tricuspid valve; Valve between right atrium and right ventricle - Moist skin

BIO-12-CH08-0035 | Challenging | Vessels pressure and lymph

A student submits four conclusions after integrating related chapter concepts.

- I. Vessels carrying blood away from the heart corresponds to Arteries.
 - II. Fluid returning tissue proteins and absorbed fats to blood corresponds to Veins.
 - III. Vessels containing valves to prevent backflow corresponds to Pulmonary artery.
 - IV. Vessel carrying blood from right ventricle to lungs corresponds to Pulmonary artery.
- Which combination should be accepted without correction?

- A. II and IV only
- B. I only
- C. III only
- D. I and IV only

BIO-12-CH08-0036 | Medium | Gas transport

Complete the statement accurately: Enzyme accelerating carbonic acid formation in red cells corresponds to _____.

- A. Oxyhaemoglobin
- B. P wave
- C. Carbonic anhydrase
- D. Tricuspid valve

BIO-12-CH08-0037 | Medium | Vessels pressure and lymph

A learner encounters the observation 'Vessels containing valves to prevent backflow' while analysing vessels pressure and lymph. Which conclusion is supported by the prescribed biology?

- A. Erythrocytes
- B. Larynx
- C. Veins
- D. Neutrophil

BIO-12-CH08-0038 | Challenging | Human respiratory tract

During error analysis, four proposed associations are reviewed.

I. Windpipe supported by cartilaginous rings corresponds to Larynx.

II. Voice box corresponds to Larynx.

III. Common passage for food and air corresponds to Pharynx.

IV. ECG complex representing ventricular depolarization corresponds to QRS complex.

Which set contains all and only the valid conclusions?

- A. I, II, III and IV
- B. II and III only
- C. III and IV only
- D. II, III and IV only

BIO-12-CH08-0039 | Medium | Respiratory organs

Complete the statement accurately: Respiratory organs in insects corresponds to _____.

- A. Mostly passive
- B. Bicuspid or mitral valve
- C. 0.8 second
- D. Tracheal system

BIO-12-CH08-0040 | Medium | Human respiratory tract

A source table contains one accurate entry involving 'Voice box'. Which option preserves a correct biological association?

- A. Quiet expiration at rest - Larynx
- B. Voice box - Larynx
- C. Vessels containing valves to prevent backflow - Moist skin
- D. Respiratory surface in earthworm - Systolic pressure

BIO-12-CH08-0041 | Hard | Heart anatomy

Two conclusions must both be valid to support the reference concept 'Valve between left atrium and left ventricle'.

Which option contains two accurate associations?

- A. Fluid returning tissue proteins and absorbed fats to blood - QRS complex; ECG complex representing ventricular depolarization - Lymph
- B. Enzyme accelerating carbonic acid formation in red cells - Lymph; Fluid returning tissue proteins and absorbed fats to blood - Carbonic anhydrase
- C. ECG complex representing ventricular depolarization - Neutrophil; Most abundant granulocyte and active phagocyte - QRS complex
- D. Valve between left atrium and left ventricle - Bicuspid or mitral valve; Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure

BIO-12-CH08-0042 | Challenging | Mechanics of breathing

A researcher checks whether four linked observations have been interpreted correctly.

- I. Change in intrapulmonary pressure during inspiration corresponds to Decrease below atmospheric pressure.
- II. Change in thoracic volume during inspiration corresponds to Mostly passive.
- III. Quiet expiration at rest corresponds to Mostly passive.
- IV. ECG complex representing ventricular depolarization corresponds to Decrease below atmospheric pressure.

Which combination survives the biological review?

- A. I only
- B. III only
- C. I and III only
- D. I and II only

BIO-12-CH08-0043 | Medium | Cardiac cycle and ECG

Complete the statement accurately: Approximate duration of one cardiac cycle at 72 beats per minute corresponds to _____.

- A. Tricuspid valve
- B. Moist skin
- C. 0.8 second
- D. Veins

BIO-12-CH08-0044 | Easy | Human respiratory tract

Which biological term or result is specifically identified by the description 'Common passage for food and air'?

- A. Decrease below atmospheric pressure
- B. Bicarbonate ions
- C. Erythrocytes
- D. Pharynx

BIO-12-CH08-0045 | Easy | Mechanics of breathing

Which description is correctly represented by Decrease below atmospheric pressure in this chapter?

- A. Valve between left atrium and left ventricle
- B. Shift promoting oxygen unloading at high carbon dioxide
- C. Change in intrapulmonary pressure during inspiration
- D. Vessels carrying blood away from the heart

BIO-12-CH08-0046 | Hard | Gas transport

Two conclusions must both be valid to support the reference concept 'Enzyme accelerating carbonic acid formation in red cells'. Which option contains two accurate associations?

- A. ECG wave representing ventricular repolarization - Leucocytes; Cells chiefly responsible for defence - T wave
- B. Enzyme accelerating carbonic acid formation in red cells - Carbonic anhydrase; Voice box - Larynx
- C. Most abundant granulocyte and active phagocyte - Platelets; Cell fragments important in clotting - Neutrophil
- D. ECG wave representing atrial depolarization - Tricuspid valve; Valve between right atrium and right ventricle - P wave

BIO-12-CH08-0047 | Easy | Heart anatomy

Which biological term or result is specifically identified by the description 'Vessel carrying blood from right ventricle to lungs'?

- A. Lymph
- B. Sinoatrial node
- C. Pulmonary artery
- D. QRS complex

BIO-12-CH08-0048 | Medium | Human respiratory tract

A learner encounters the observation 'Voice box' while analysing human respiratory tract. Which conclusion is supported by the prescribed biology?

- A. Trachea
- B. Moist skin
- C. Larynx
- D. QRS complex

BIO-12-CH08-0049 | Challenging | Human respiratory tract

Evaluate the following source-based biological statements.

- I. Voice box corresponds to Larynx.
- II. Windpipe supported by cartilaginous rings corresponds to Alveoli.
- III. Microscopic sites of gaseous exchange corresponds to Alveoli.
- IV. Pressure recorded during ventricular systole corresponds to Systolic pressure.

Which option identifies exactly the correct statements?

- A. I, II and IV only
- B. I, III and IV only
- C. I and IV only
- D. I, II, III and IV

BIO-12-CH08-0050 | Easy | Heart anatomy

Which biological term or result is specifically identified by the description 'Valve between right atrium and right ventricle'?

- A. Lymph
- B. Bohr effect
- C. Tricuspid valve
- D. Platelets

BIO-12-CH08-0051 | Hard | Respiratory organs

A reviewer compares several entries while checking 'Respiratory organs in most fishes'. Which entry must be corrected?

- A. Shift promoting oxygen unloading at high carbon dioxide - Bohr effect
- B. Enzyme accelerating carbonic acid formation in red cells - Carbonic anhydrase
- C. Respiratory organs in most fishes - Neutrophil
- D. Natural pacemaker of the heart - Sinoatrial node

BIO-12-CH08-0052 | Medium | Gas transport

A learner encounters the observation 'Major form of oxygen transport in blood' while analysing gas transport. Which conclusion is supported by the prescribed biology?

- A. Diaphragm
- B. Oxyhaemoglobin
- C. Larynx
- D. Systolic pressure

BIO-12-CH08-0053 | Medium | Respiratory organs

A source table contains one accurate entry involving 'Respiratory organs in insects'. Which option preserves a correct biological association?

- A. Cell fragments important in clotting - Mostly passive
- B. Natural pacemaker of the heart - Tracheal system
- C. Respiratory organs in insects - Tracheal system
- D. Quiet expiration at rest - QRS complex

BIO-12-CH08-0054 | Hard | Blood cells

Two conclusions must both be valid to support the reference concept 'Anucleate biconcave cells transporting respiratory gases'. Which option contains two accurate associations?

- A. Anucleate biconcave cells transporting respiratory gases - Erythrocytes; Respiratory organs in most fishes - Gills
- B. Valve between left atrium and left ventricle - Lymph; Fluid returning tissue proteins and absorbed fats to blood - Bicuspid or mitral valve
- C. Vessels containing valves to prevent backflow - Tricuspid valve; Valve between right atrium and right ventricle - Veins
- D. Change in intrapulmonary pressure during inspiration - Veins; Vessels containing valves to prevent backflow - Decrease below atmospheric pressure

BIO-12-CH08-0055 | Easy | Human respiratory tract

Which description is correctly represented by Pharynx in this chapter?

- A. Change in intrapulmonary pressure during inspiration
- B. Common passage for food and air
- C. Microscopic sites of gaseous exchange
- D. Major form of carbon dioxide transport

BIO-12-CH08-0056 | Hard | Human respiratory tract

Two conclusions must both be valid to support the reference concept 'Common passage for food and air'. Which option contains two accurate associations?

- A. Major form of carbon dioxide transport - Decrease below atmospheric pressure; Change in intrapulmonary pressure during inspiration - Bicarbonate ions
- B. Anucleate biconcave cells transporting respiratory gases - Bicarbonate ions; Major form of carbon dioxide transport - Erythrocytes
- C. Fluid returning tissue proteins and absorbed fats to blood - Erythrocytes; Anucleate biconcave cells transporting respiratory gases - Lymph
- D. Common passage for food and air - Pharynx; Respiratory surface in earthworm - Moist skin

BIO-12-CH08-0057 | Challenging | Vessels pressure and lymph

A student submits four conclusions after integrating related chapter concepts.

- I. Vessels containing valves to prevent backflow corresponds to Veins.
- II. Pressure recorded during ventricular systole corresponds to Systolic pressure.
- III. Vessels carrying blood away from the heart corresponds to 0.8 second.
- IV. Approximate duration of one cardiac cycle at 72 beats per minute corresponds to Veins.

Which combination should be accepted without correction?

- A. II and IV only
- B. I and II only
- C. IV only
- D. II, III and IV only

BIO-12-CH08-0058 | Easy | Blood cells

Which description is correctly represented by Erythrocytes in this chapter?

- A. Anucleate biconcave cells transporting respiratory gases
- B. Vessels containing valves to prevent backflow
- C. Valve between right atrium and right ventricle
- D. Change in intrapulmonary pressure during inspiration

BIO-12-CH08-0059 | Medium | Mechanics of breathing

A learner encounters the observation 'Change in thoracic volume during inspiration' while analysing mechanics of breathing. Which conclusion is supported by the prescribed biology?

- A. Increase
- B. Lymph
- C. Trachea
- D. Carbonic anhydrase

BIO-12-CH08-0060 | Medium | Human respiratory tract

A learner encounters the observation 'Common passage for food and air' while analysing human respiratory tract. Which conclusion is supported by the prescribed biology?

- A. Lymph
- B. Larynx
- C. Moist skin
- D. Pharynx

BIO-12-CH08-0061 | Easy | Cardiac cycle and ECG

Which biological term or result is specifically identified by the description 'ECG wave representing ventricular repolarization'?

- A. Systolic pressure
- B. T wave
- C. Erythrocytes
- D. Neutrophil

BIO-12-CH08-0062 | Easy | Respiratory organs

Which description is correctly represented by Tracheal system in this chapter?

- A. Respiratory organs in most fishes
- B. Approximate duration of one cardiac cycle at 72 beats per minute
- C. Quiet expiration at rest
- D. Respiratory organs in insects

BIO-12-CH08-0063 | Hard | Mechanics of breathing

Two conclusions must both be valid to support the reference concept 'Quiet expiration at rest'. Which option contains two accurate associations?

- A. Respiratory organs in most fishes - 0.8 second; Approximate duration of one cardiac cycle at 72 beats per minute - Gills
- B. Microscopic sites of gaseous exchange - Pharynx; Common passage for food and air - Alveoli
- C. Most abundant granulocyte and active phagocyte - Gills; Respiratory organs in most fishes - Neutrophil
- D. Quiet expiration at rest - Mostly passive; Valve between right atrium and right ventricle - Tricuspid valve

BIO-12-CH08-0064 | Hard | Gas transport

A reviewer compares several entries while checking 'Shift promoting oxygen unloading at high carbon dioxide'. Which entry must be corrected?

- A. Common passage for food and air - Pharynx
- B. Quiet expiration at rest - Mostly passive
- C. Shift promoting oxygen unloading at high carbon dioxide - Veins
- D. Voice box - Larynx

BIO-12-CH08-0065 | Easy | Blood cells

Which biological term or result is specifically identified by the description 'Cells chiefly responsible for defence'?

- A. Leucocytes
- B. Veins
- C. Sinoatrial node
- D. Decrease below atmospheric pressure

BIO-12-CH08-0066 | Hard | Vessels pressure and lymph

Two conclusions must both be valid to support the reference concept 'Fluid returning tissue proteins and absorbed fats to blood'. Which option contains two accurate associations?

- A. Fluid returning tissue proteins and absorbed fats to blood - Lymph; Vessel carrying blood from right ventricle to lungs - Pulmonary artery
- B. Microscopic sites of gaseous exchange - Leucocytes; Cells chiefly responsible for defence - Alveoli
- C. Cell fragments important in clotting - 0.8 second; Approximate duration of one cardiac cycle at 72 beats per minute - Platelets
- D. Approximate duration of one cardiac cycle at 72 beats per minute - Bicarbonate ions; Major form of carbon dioxide transport - 0.8 second

BIO-12-CH08-0067 | Medium | Vessels pressure and lymph

A source table contains one accurate entry involving 'Pressure recorded during ventricular systole'. Which option preserves a correct biological association?

- A. Change in thoracic volume during inspiration - T wave
- B. Vessel carrying blood from right ventricle to lungs - Pharynx
- C. Primary respiratory organs in adult humans - Systolic pressure
- D. Pressure recorded during ventricular systole - Systolic pressure

BIO-12-CH08-0068 | Medium | Cardiac cycle and ECG

A source table contains one accurate entry involving 'ECG wave representing ventricular repolarization'. Which option preserves a correct biological association?

- A. Vessels containing valves to prevent backflow - Arteries
- B. ECG wave representing ventricular repolarization - T wave
- C. Quiet expiration at rest - Tricuspid valve
- D. Vessels carrying blood away from the heart - T wave

BIO-12-CH08-0069 | Easy | Blood cells

Which biological term or result is specifically identified by the description 'Most abundant granulocyte and active phagocyte'?

- A. Neutrophil
- B. Carbonic anhydrase
- C. Larynx
- D. Leucocytes

BIO-12-CH08-0070 | Medium | Mechanics of breathing

A learner encounters the observation 'Main muscle of quiet inspiration' while analysing mechanics of breathing. Which conclusion is supported by the prescribed biology?

- A. Diaphragm
- B. Pulmonary artery
- C. Tracheal system
- D. Increase

BIO-12-CH08-0071 | Medium | Mechanics of breathing

Complete the statement accurately: Change in intrapulmonary pressure during inspiration corresponds to _____.

- A. Increase
- B. Decrease below atmospheric pressure
- C. Larynx
- D. Arteries

BIO-12-CH08-0072 | Hard | Vessels pressure and lymph

A reviewer compares several entries while checking 'Vessels carrying blood away from the heart'. Which entry must be corrected?

- A. Valve between right atrium and right ventricle - Tricuspid valve
- B. Vessels carrying blood away from the heart - P wave
- C. Quiet expiration at rest - Mostly passive
- D. Main muscle of quiet inspiration - Diaphragm

BIO-12-CH08-0073 | Challenging | Blood cells

During error analysis, four proposed associations are reviewed.

- I. Most abundant granulocyte and active phagocyte corresponds to Neutrophil.
- II. Anucleate biconcave cells transporting respiratory gases corresponds to Erythrocytes.
- III. Cells chiefly responsible for defence corresponds to Lymph.
- IV. Fluid returning tissue proteins and absorbed fats to blood corresponds to Lymph.

Which set contains all and only the valid conclusions?

- A. I, III and IV only
- B. III and IV only
- C. II only
- D. I, II and IV only

BIO-12-CH08-0074 | Medium | Heart anatomy

A source table contains one accurate entry involving 'Valve between right atrium and right ventricle'. Which option preserves a correct biological association?

- A. Fluid returning tissue proteins and absorbed fats to blood - Platelets
- B. Major form of oxygen transport in blood - Trachea
- C. Shift promoting oxygen unloading at high carbon dioxide - Oxyhaemoglobin
- D. Valve between right atrium and right ventricle - Tricuspid valve

BIO-12-CH08-0075 | Medium | Blood cells

Complete the statement accurately: Anucleate biconcave cells transporting respiratory gases corresponds to _____.

- A. Decrease below atmospheric pressure
- B. Erythrocytes
- C. Leucocytes
- D. QRS complex

BIO-12-CH08-0076 | Easy | Gas transport

Which description is correctly represented by Carbonic anhydrase in this chapter?

- A. Major form of oxygen transport in blood
- B. ECG wave representing atrial depolarization
- C. Enzyme accelerating carbonic acid formation in red cells
- D. Most abundant granulocyte and active phagocyte

BIO-12-CH08-0077 | Easy | Mechanics of breathing

Which biological term or result is specifically identified by the description 'Main muscle of quiet inspiration'?

- A. Decrease below atmospheric pressure
- B. Moist skin
- C. Diaphragm
- D. Increase

BIO-12-CH08-0078 | Easy | Gas transport

Which biological term or result is specifically identified by the description 'Enzyme accelerating carbonic acid formation in red cells'?

- A. Neutrophil
- B. Carbonic anhydrase
- C. Oxyhaemoglobin
- D. Larynx

BIO-12-CH08-0079 | Medium | Gas transport

Complete the statement accurately: Major form of oxygen transport in blood corresponds to _____.

- A. Oxyhaemoglobin
- B. Systolic pressure
- C. P wave
- D. Carbonic anhydrase

BIO-12-CH08-0080 | Hard | Human respiratory tract

A reviewer compares several entries while checking 'Windpipe supported by cartilaginous rings'. Which entry must be corrected?

- A. ECG wave representing atrial depolarization - P wave
- B. Quiet expiration at rest - Mostly passive
- C. Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure
- D. Windpipe supported by cartilaginous rings - Pulmonary artery

BIO-12-CH08-0081 | Easy | Respiratory organs

Which biological term or result is specifically identified by the description 'Respiratory organs in most fishes'?

- A. Bohr effect
- B. Lungs
- C. Gills
- D. Carbonic anhydrase

BIO-12-CH08-0082 | Hard | Cardiac cycle and ECG

Two conclusions must both be valid to support the reference concept 'ECG wave representing atrial depolarization'. Which option contains two accurate associations?

- A. ECG wave representing atrial depolarization - P wave; Windpipe supported by cartilaginous rings - Trachea
- B. Fluid returning tissue proteins and absorbed fats to blood - Alveoli; Microscopic sites of gaseous exchange - Lymph
- C. Microscopic sites of gaseous exchange - Sinoatrial node; Natural pacemaker of the heart - Alveoli
- D. Major form of oxygen transport in blood - Tracheal system; Respiratory organs in insects - Oxyhaemoglobin

BIO-12-CH08-0083 | Easy | Gas transport

Which biological term or result is specifically identified by the description 'Major form of oxygen transport in blood'?

- A. Bicarbonate ions
- B. Tracheal system
- C. Oxyhaemoglobin
- D. Larynx

BIO-12-CH08-0084 | Hard | Cardiac cycle and ECG

A reviewer compares several entries while checking 'ECG wave representing ventricular repolarization'. Which entry must be corrected?

- A. Pressure recorded during ventricular systole - Systolic pressure
- B. Anucleate biconcave cells transporting respiratory gases - Erythrocytes
- C. Vessels containing valves to prevent backflow - Veins
- D. ECG wave representing ventricular repolarization - Neutrophil

BIO-12-CH08-0085 | Easy | Vessels pressure and lymph

Which biological term or result is specifically identified by the description 'Vessels carrying blood away from the heart'?

- A. Arteries
- B. Mostly passive
- C. Diaphragm
- D. P wave

BIO-12-CH08-0086 | Medium | Vessels pressure and lymph

A learner encounters the observation 'Vessels carrying blood away from the heart' while analysing vessels pressure and lymph. Which conclusion is supported by the prescribed biology?

- A. P wave
- B. Arteries
- C. Mostly passive
- D. Veins

BIO-12-CH08-0087 | Hard | Vessels pressure and lymph

A reviewer compares several entries while checking 'Vessels containing valves to prevent backflow'. Which entry must be corrected?

- A. Voice box - Larynx
- B. Vessels containing valves to prevent backflow - Erythrocytes
- C. Windpipe supported by cartilaginous rings - Trachea
- D. Common passage for food and air - Pharynx

BIO-12-CH08-0088 | Easy | Vessels pressure and lymph

Which description is correctly represented by Lymph in this chapter?

- A. Cell fragments important in clotting
- B. Vessel carrying blood from right ventricle to lungs
- C. Vessels containing valves to prevent backflow
- D. Fluid returning tissue proteins and absorbed fats to blood

BIO-12-CH08-0089 | Easy | Heart anatomy

Which description is correctly represented by Bicuspid or mitral valve in this chapter?

- A. Change in intrapulmonary pressure during inspiration
- B. Valve between left atrium and left ventricle
- C. Cell fragments important in clotting
- D. Shift promoting oxygen unloading at high carbon dioxide

BIO-12-CH08-0090 | Hard | Vessels pressure and lymph

Two conclusions must both be valid to support the reference concept 'Pressure recorded during ventricular systole'. Which option contains two accurate associations?

- A. Pressure recorded during ventricular systole - Systolic pressure; Major form of oxygen transport in blood - Oxyhaemoglobin
- B. ECG wave representing ventricular repolarization - Pharynx; Common passage for food and air - T wave
- C. Primary respiratory organs in adult humans - Erythrocytes; Anucleate biconcave cells transporting respiratory gases - Lungs
- D. Vessel carrying blood from right ventricle to lungs - T wave; ECG wave representing ventricular repolarization - Pulmonary artery

BIO-12-CH08-0091 | Easy | Vessels pressure and lymph

Which description is correctly represented by Systolic pressure in this chapter?

- A. Fluid returning tissue proteins and absorbed fats to blood
- B. Common passage for food and air
- C. ECG wave representing ventricular repolarization
- D. Pressure recorded during ventricular systole

BIO-12-CH08-0092 | Medium | Gas transport

Complete the statement accurately: Shift promoting oxygen unloading at high carbon dioxide corresponds to _____.

- A. Diaphragm
- B. Bohr effect
- C. Pharynx
- D. Veins

BIO-12-CH08-0093 | Hard | Heart anatomy

A reviewer compares several entries while checking 'Valve between right atrium and right ventricle'. Which entry must be corrected?

- A. Vessels carrying blood away from the heart - Arteries
- B. Windpipe supported by cartilaginous rings - Trachea
- C. Valve between right atrium and right ventricle - Lymph
- D. Major form of oxygen transport in blood - Oxyhaemoglobin

BIO-12-CH08-0094 | Hard | Gas transport

A reviewer compares several entries while checking 'Enzyme accelerating carbonic acid formation in red cells'. Which entry must be corrected?

- A. Enzyme accelerating carbonic acid formation in red cells - Larynx
- B. Cell fragments important in clotting - Platelets
- C. Valve between right atrium and right ventricle - Tricuspid valve
- D. ECG wave representing ventricular repolarization - T wave

BIO-12-CH08-0095 | Medium | Cardiac cycle and ECG

A source table contains one accurate entry involving 'ECG complex representing ventricular depolarization'. Which option preserves a correct biological association?

- A. Change in thoracic volume during inspiration - Erythrocytes
- B. Anucleate biconcave cells transporting respiratory gases - Pulmonary artery
- C. Cell fragments important in clotting - Gills
- D. ECG complex representing ventricular depolarization - QRS complex

BIO-12-CH08-0096 | Challenging | Cardiac cycle and ECG

A researcher checks whether four linked observations have been interpreted correctly.

- I. ECG complex representing ventricular depolarization corresponds to 0.8 second.
- II. Approximate duration of one cardiac cycle at 72 beats per minute corresponds to P wave.
- III. ECG wave representing atrial depolarization corresponds to Tracheal system.
- IV. Respiratory organs in insects corresponds to Tracheal system.

Which combination survives the biological review?

- A. I and II only
- B. III only
- C. IV only
- D. I, II and III only

BIO-12-CH08-0097 | Hard | Cardiac cycle and ECG

A reviewer compares several entries while checking 'Approximate duration of one cardiac cycle at 72 beats per minute'. Which entry must be corrected?

- A. Valve between right atrium and right ventricle - Tricuspid valve
- B. Major form of oxygen transport in blood - Oxyhaemoglobin
- C. Approximate duration of one cardiac cycle at 72 beats per minute - Veins
- D. Respiratory surface in earthworm - Moist skin

BIO-12-CH08-0098 | Hard | Blood cells

A reviewer compares several entries while checking 'Cells chiefly responsible for defence'. Which entry must be corrected?

- A. Cells chiefly responsible for defence - Veins
- B. Natural pacemaker of the heart - Sinoatrial node
- C. Change in intrapulmonary pressure during inspiration - Decrease below atmospheric pressure
- D. Pressure recorded during ventricular systole - Systolic pressure

BIO-12-CH08-0099 | Easy | Vessels pressure and lymph

Which biological term or result is specifically identified by the description 'Vessels containing valves to prevent backflow'?

- A. Neutrophil
- B. Trachea
- C. Erythrocytes
- D. Veins

BIO-12-CH08-0100 | Challenging | Mechanics of breathing

Evaluate the following source-based biological statements.

- I. Main muscle of quiet inspiration corresponds to Diaphragm.
- II. Change in thoracic volume during inspiration corresponds to Increase.
- III. Change in intrapulmonary pressure during inspiration corresponds to Decrease below atmospheric pressure.
- IV. Most abundant granulocyte and active phagocyte corresponds to Diaphragm.

Which option identifies exactly the correct statements?

- A. IV only
- B. I, II and III only
- C. II and III only
- D. II only

BIO-12-CH08-0101 | Hard | Respiratory organs

A reviewer compares several entries while checking 'Respiratory organs in insects'. Which entry must be corrected?

- A. Respiratory organs in insects - Pulmonary artery
- B. Approximate duration of one cardiac cycle at 72 beats per minute - 0.8 second
- C. Valve between left atrium and left ventricle - Bicuspid or mitral valve
- D. Natural pacemaker of the heart - Sinoatrial node

BIO-12-CH08-0102 | Easy | Gas transport

Which biological term or result is specifically identified by the description 'Major form of carbon dioxide transport'?

- A. Bicarbonate ions
- B. Carbonic anhydrase
- C. QRS complex
- D. Bicuspid or mitral valve

BIO-12-CH08-0103 | Medium | Human respiratory tract

A learner encounters the observation 'Windpipe supported by cartilaginous rings' while analysing human respiratory tract. Which conclusion is supported by the prescribed biology?

- A. Leucocytes
- B. Mostly passive
- C. Pharynx
- D. Trachea

BIO-12-CH08-0104 | Easy | Blood cells

Which description is correctly represented by Neutrophil in this chapter?

- A. Vessel carrying blood from right ventricle to lungs
- B. Most abundant granulocyte and active phagocyte
- C. ECG complex representing ventricular depolarization
- D. Voice box

BIO-12-CH08-0105 | Challenging | Heart anatomy

A student submits four conclusions after integrating related chapter concepts.

- I. Vessel carrying blood from right ventricle to lungs corresponds to Bicuspid or mitral valve.
- II. Valve between left atrium and left ventricle corresponds to Tricuspid valve.
- III. Valve between right atrium and right ventricle corresponds to Tricuspid valve.
- IV. Cell fragments important in clotting corresponds to Pulmonary artery.

Which combination should be accepted without correction?

- A. I and IV only
- B. III only
- C. I, II and IV only
- D. II, III and IV only

BIO-12-CH08-0106 | Easy | Mechanics of breathing

Which biological term or result is specifically identified by the description 'Change in thoracic volume during inspiration'?

- A. Decrease below atmospheric pressure
- B. Trachea
- C. Lymph
- D. Increase

BIO-12-CH08-0107 | Challenging | Heart anatomy

During error analysis, four proposed associations are reviewed.

- I. Natural pacemaker of the heart corresponds to Pulmonary artery.
- II. Vessel carrying blood from right ventricle to lungs corresponds to Tricuspid valve.
- III. Valve between right atrium and right ventricle corresponds to Tricuspid valve.
- IV. Anucleate biconcave cells transporting respiratory gases corresponds to Erythrocytes.

Which set contains all and only the valid conclusions?

- A. II, III and IV only
- B. III and IV only
- C. I, II, III and IV
- D. III only

BIO-12-CH08-0108 | Medium | Respiratory organs

Complete the statement accurately: Primary respiratory organs in adult humans corresponds to _____.

- A. QRS complex
- B. Decrease below atmospheric pressure
- C. Gills
- D. Lungs

BIO-12-CH08-0109 | Medium | Blood cells

A source table contains one accurate entry involving 'Cell fragments important in clotting'. Which option preserves a correct biological association?

- A. Major form of oxygen transport in blood - Mostly passive
- B. Vessels carrying blood away from the heart - Bohr effect
- C. Cell fragments important in clotting - Platelets
- D. Common passage for food and air - Platelets

BIO-12-CH08-0110 | Medium | Cardiac cycle and ECG

Complete the statement accurately: ECG wave representing atrial depolarization corresponds to _____.

- A. P wave
- B. Trachea
- C. Sinoatrial node
- D. QRS complex

BIO-12-CH08-0111 | Medium | Vessels pressure and lymph

A source table contains one accurate entry involving 'Vessels carrying blood away from the heart'. Which option preserves a correct biological association?

- A. Voice box - QRS complex
- B. Vessels carrying blood away from the heart - Arteries
- C. Quiet expiration at rest - Diaphragm
- D. ECG wave representing atrial depolarization - Larynx

BIO-12-CH08-0112 | Medium | Cardiac cycle and ECG

Complete the statement accurately: ECG wave representing ventricular repolarization corresponds to _____.

- A. Veins
- B. QRS complex
- C. T wave
- D. Mostly passive

BIO-12-CH08-0113 | Hard | Blood cells

Two conclusions must both be valid to support the reference concept 'Most abundant granulocyte and active phagocyte'. Which option contains two accurate associations?

- A. Most abundant granulocyte and active phagocyte - Neutrophil; Voice box - Larynx
- B. Enzyme accelerating carbonic acid formation in red cells - P wave; ECG wave representing atrial depolarization - Carbonic anhydrase
- C. Vessel carrying blood from right ventricle to lungs - Systolic pressure; Pressure recorded during ventricular systole - Pulmonary artery
- D. ECG wave representing atrial depolarization - Tracheal system; Respiratory organs in insects - P wave

BIO-12-CH08-0114 | Medium | Heart anatomy

Complete the statement accurately: Vessel carrying blood from right ventricle to lungs corresponds to _____.

- A. Pulmonary artery
- B. Bicuspid or mitral valve
- C. Sinoatrial node
- D. Lymph

BIO-12-CH08-0115 | Medium | Heart anatomy

Complete the statement accurately: Natural pacemaker of the heart corresponds to _____.

- A. Bicuspid or mitral valve
- B. Sinoatrial node
- C. Trachea
- D. Moist skin

BIO-12-CH08-0116 | Medium | Respiratory organs

Complete the statement accurately: Respiratory organs in most fishes corresponds to _____.

- A. Mostly passive
- B. Lungs
- C. Tracheal system
- D. Gills

BIO-12-CH08-0117 | Medium | Vessels pressure and lymph

Complete the statement accurately: Pressure recorded during ventricular systole corresponds to _____.

- A. Systolic pressure
- B. Pulmonary artery
- C. T wave
- D. Pharynx

BIO-12-CH08-0118 | Medium | Blood cells

A source table contains one accurate entry involving 'Anucleate biconcave cells transporting respiratory gases'. Which option preserves a correct biological association?

- A. Anucleate biconcave cells transporting respiratory gases - Erythrocytes
- B. Vessels containing valves to prevent backflow - QRS complex
- C. Change in intrapulmonary pressure during inspiration - Tricuspid valve
- D. Valve between left atrium and left ventricle - Erythrocytes

BIO-12-CH08-0119 | Hard | Gas transport

A reviewer compares several entries while checking 'Major form of oxygen transport in blood'. Which entry must be corrected?

- A. Cells chiefly responsible for defence - Leucocytes
- B. ECG wave representing atrial depolarization - P wave
- C. Major form of oxygen transport in blood - Systolic pressure
- D. Respiratory organs in insects - Tracheal system

BIO-12-CH08-0120 | Challenging | Gas transport

A researcher checks whether four linked observations have been interpreted correctly.

- I. Shift promoting oxygen unloading at high carbon dioxide corresponds to Carbonic anhydrase.
- II. Enzyme accelerating carbonic acid formation in red cells corresponds to Carbonic anhydrase.
- III. Major form of carbon dioxide transport corresponds to Alveoli.
- IV. Microscopic sites of gaseous exchange corresponds to Bohr effect.

Which combination survives the biological review?

- A. III only
- B. II only
- C. I only
- D. II and IV only

BIO-12-CH08-0121 | Medium | Vessels pressure and lymph

Complete the statement accurately: Vessels containing valves to prevent backflow corresponds to ____.

- A. Trachea
- B. Veins
- C. Arteries
- D. Neutrophil

BIO-12-CH08-0122 | Medium | Blood cells

Complete the statement accurately: Most abundant granulocyte and active phagocyte corresponds to ____.

- A. Leucocytes
- B. Larynx
- C. Neutrophil
- D. Systolic pressure

BIO-12-CH08-0123 | Medium | Gas transport

A source table contains one accurate entry involving 'Shift promoting oxygen unloading at high carbon dioxide'. Which option preserves a correct biological association?

- A. Quiet expiration at rest - Larynx
- B. Primary respiratory organs in adult humans - Bohr effect
- C. Vessels containing valves to prevent backflow - Diaphragm
- D. Shift promoting oxygen unloading at high carbon dioxide - Bohr effect

BIO-12-CH08-0124 | Easy | Respiratory organs

Which biological term or result is specifically identified by the description 'Respiratory surface in earthworm'?

- A. Decrease below atmospheric pressure
- B. Veins
- C. Tracheal system
- D. Moist skin

BIO-12-CH08-0125 | Medium | Heart anatomy

Complete the statement accurately: Valve between left atrium and left ventricle corresponds to _____.

- A. Carbonic anhydrase
 - B. Sinoatrial node
 - C. Bicuspid or mitral valve
 - D. Lymph
-

BIO-12-CH08-0126 | Medium | Mechanics of breathing

A learner encounters the observation 'Quiet expiration at rest' while analysing mechanics of breathing. Which conclusion is supported by the prescribed biology?

- A. Pharynx
 - B. Alveoli
 - C. Diaphragm
 - D. Mostly passive
-

BIO-12-CH08-0127 | Easy | Mechanics of breathing

Which description is correctly represented by Mostly passive in this chapter?

- A. Change in intrapulmonary pressure during inspiration
 - B. Valve between right atrium and right ventricle
 - C. Quiet expiration at rest
 - D. Vessel carrying blood from right ventricle to lungs
-

BIO-12-CH08-0128 | Medium | Heart anatomy

A learner encounters the observation 'Valve between right atrium and right ventricle' while analysing heart anatomy. Which conclusion is supported by the prescribed biology?

- A. Tricuspid valve
 - B. Pulmonary artery
 - C. Bicuspid or mitral valve
 - D. Lymph
-

BIO-12-CH08-0129 | Hard | Mechanics of breathing

A reviewer compares several entries while checking 'Change in thoracic volume during inspiration'. Which entry must be corrected?

- A. Natural pacemaker of the heart - Sinoatrial node
 - B. Fluid returning tissue proteins and absorbed fats to blood - Lymph
 - C. Change in thoracic volume during inspiration - Leucocytes
 - D. Major form of oxygen transport in blood - Oxyhaemoglobin
-

BIO-12-CH08-0130 | Easy | Gas transport

Which biological term or result is specifically identified by the description 'Shift promoting oxygen unloading at high carbon dioxide'?

- A. Diaphragm
 - B. Pharynx
 - C. Mostly passive
 - D. Bohr effect
-

BIO-12-CH08-0131 | Medium | Blood cells

A learner encounters the observation 'Anucleate biconcave cells transporting respiratory gases' while analysing blood cells. Which conclusion is supported by the prescribed biology?

- A. Platelets
- B. Erythrocytes
- C. QRS complex
- D. Gills

BIO-12-CH08-0132 | Challenging | Gas transport

Evaluate the following source-based biological statements.

- I. Enzyme accelerating carbonic acid formation in red cells corresponds to Oxyhaemoglobin.
- II. Major form of oxygen transport in blood corresponds to Oxyhaemoglobin.
- III. Shift promoting oxygen unloading at high carbon dioxide corresponds to Systolic pressure.
- IV. Pressure recorded during ventricular systole corresponds to Systolic pressure.

Which option identifies exactly the correct statements?

- A. III and IV only
 - B. I and II only
 - C. II and IV only
 - D. I, II and IV only
-

Standard 12 | Chapter 09

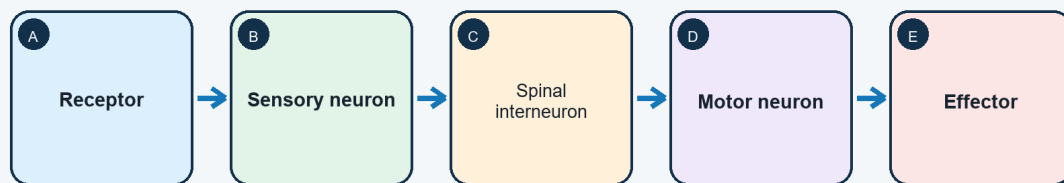
Control and Co-ordination

BIO-12-CH09-0001 | Hard | Reflex arc

Damage to the sensory root of the reflex pathway would most directly cause

Reflex arc

Neural pathway producing a rapid involuntary response



The sensory neuron carries an afferent impulse; the motor neuron carries an efferent impulse.

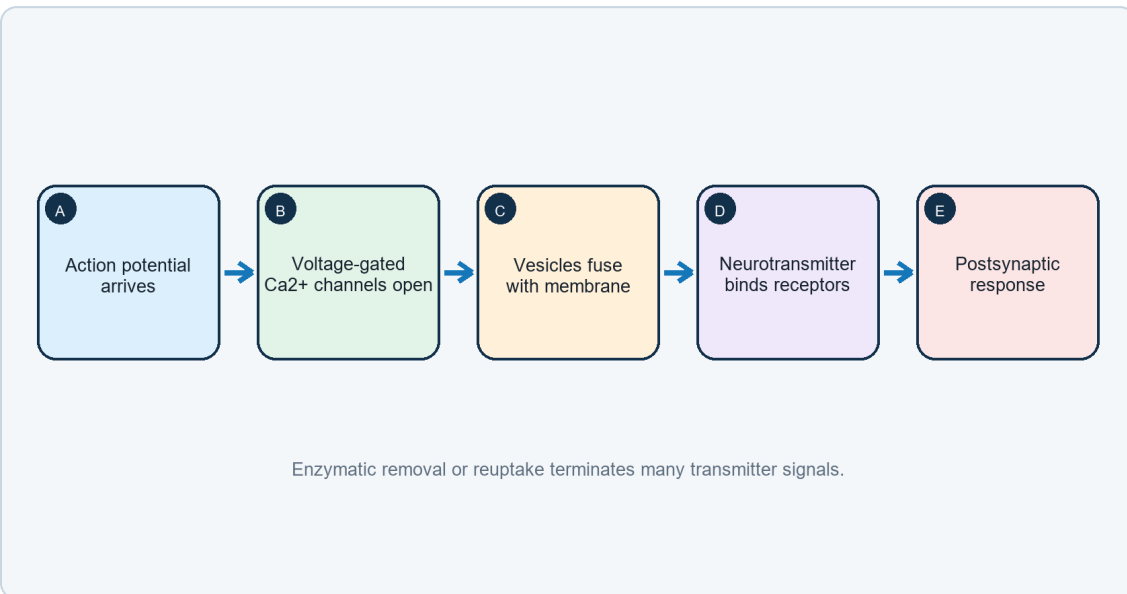
- A. loss of afferent input from the receptor
- B. continuous motor output with normal sensation
- C. faster neurotransmitter removal
- D. increased hormone secretion by the pituitary

BIO-12-CH09-0002 | Challenging | Chemical synaptic transmission

If acetylcholinesterase is inhibited at a cholinergic synapse, what is the most likely immediate effect?

Chemical synaptic transmission

Sequence at a presynaptic terminal



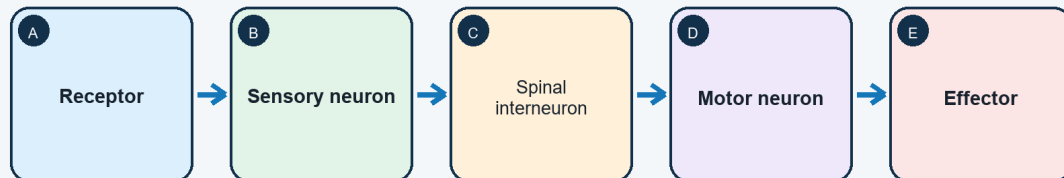
- A. Prolonged activation of postsynaptic receptors
- B. Failure to release any neurotransmitter
- C. Instant destruction of all receptors
- D. Conversion of acetylcholine into DNA

BIO-12-CH09-0003 | Easy | Reflex arc

Which neuron carries the impulse from the receptor toward the spinal cord in the reflex arc?

Reflex arc

Neural pathway producing a rapid involuntary response



The sensory neuron carries an afferent impulse; the motor neuron carries an efferent impulse.

- A. Motor neuron
- B. Postganglionic autonomic neuron only
- C. Neurosecretory cell
- D. Sensory neuron

BIO-12-CH09-0004 | Easy | Central nervous system

Which biological term or result is specifically identified by the description 'Protective membranes around brain and spinal cord'?

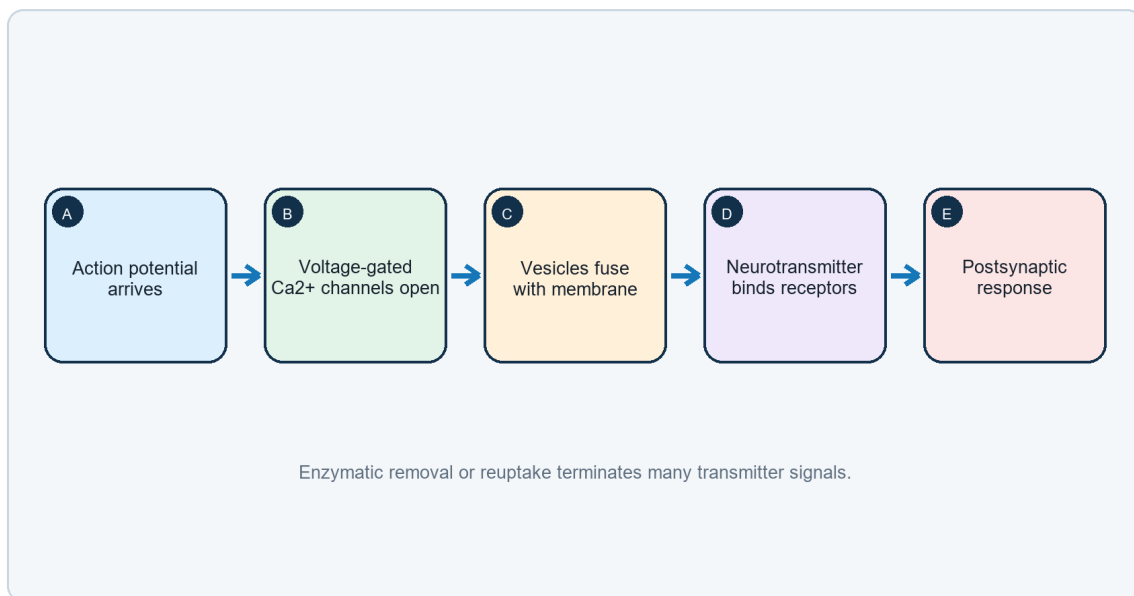
- A. Reflex action
- B. Axon
- C. Meninges
- D. Diabetes mellitus

BIO-12-CH09-0005 | Medium | Chemical synaptic transmission

What directly triggers synaptic-vesicle fusion at the presynaptic terminal?

Chemical synaptic transmission

Sequence at a presynaptic terminal



- A. Sodium removal from the synaptic cleft
- B. Calcium-ion entry
- C. DNA replication
- D. Myelin breakdown

BIO-12-CH09-0006 | Medium | Neuron structure

A source table contains one accurate entry involving 'Neuron processes usually receiving impulses'. Which option preserves a correct biological association?

- A. Junction between two neurons - Neurotransmitter
 - B. Neuron processes usually receiving impulses - Dendrites
 - C. Thyroid hormone containing iodine - Synapse
 - D. Receptors detecting light - Adrenaline
-

BIO-12-CH09-0007 | Easy | Neuron structure

Which biological term or result is specifically identified by the description 'Neuron processes usually receiving impulses'?

- A. Dendrites
 - B. Axon
 - C. Thyroxine
 - D. Medulla oblongata
-

BIO-12-CH09-0008 | Easy | Impulse and synapse

Which description is correctly represented by Resting potential in this chapter?

- A. Membrane state with inside negative relative to outside
 - B. Protective membranes around brain and spinal cord
 - C. Neural pathway mediating a reflex
 - D. Iodine-deficiency enlargement of thyroid
-

BIO-12-CH09-0009 | Medium | Nervous organization

A learner encounters the observation 'Diffuse nerve network without a central brain' while analysing nervous organization. Which conclusion is supported by the prescribed biology?

- A. Simple goitre
 - B. Diabetes mellitus
 - C. Planaria
 - D. Hydra
-

BIO-12-CH09-0010 | Easy | Endocrine disorders

Which biological term or result is specifically identified by the description 'Excess growth hormone in adults'?

- A. Dwarfism
 - B. Neuron
 - C. Depolarization
 - D. Acromegaly
-

BIO-12-CH09-0011 | Easy | Pancreas adrenal and gonads

Which biological term or result is specifically identified by the description 'Pancreatic hormone lowering blood glucose'?

- A. Insulin
 - B. Reflex action
 - C. Simple goitre
 - D. Diabetes mellitus
-

BIO-12-CH09-0012 | Medium | Receptors and reflexes

A learner encounters the observation 'Rapid automatic response through a neural pathway' while analysing receptors and reflexes. Which conclusion is supported by the prescribed biology?

- A. Calcitonin
- B. Medulla oblongata
- C. Reflex action
- D. Dendrites

BIO-12-CH09-0013 | Medium | Pituitary and thyroid

Complete the statement accurately: Hormone lowering blood calcium corresponds to _____.

- A. Calcitonin
- B. Testosterone
- C. Growth hormone
- D. Dendrites

BIO-12-CH09-0014 | Easy | Pancreas adrenal and gonads

Which biological term or result is specifically identified by the description 'Adrenal medullary fight-or-flight hormone'?

- A. Depolarization
- B. Reflex action
- C. Adrenaline
- D. Planaria

BIO-12-CH09-0015 | Medium | Central nervous system

A learner encounters the observation 'Protective membranes around brain and spinal cord' while analysing central nervous system. Which conclusion is supported by the prescribed biology?

- A. Meninges
- B. Reflex action
- C. TSH
- D. Diabetes mellitus

BIO-12-CH09-0016 | Challenging | Receptors and reflexes

During error analysis, four proposed associations are reviewed.

- I. Rapid automatic response through a neural pathway corresponds to Reflex action.
- II. Receptors detecting sound and pressure corresponds to Mechanoreceptors.
- III. Neural pathway mediating a reflex corresponds to Reflex arc.
- IV. Iodine-deficiency enlargement of thyroid corresponds to Reflex action.

Which set contains all and only the valid conclusions?

- A. II, III and IV only
- B. I only
- C. I, II and III only
- D. I, II and IV only

BIO-12-CH09-0017 | Medium | Endocrine disorders

A source table contains one accurate entry involving 'Iodine-deficiency enlargement of thyroid'. Which option preserves a correct biological association?

- A. Iodine-deficiency enlargement of thyroid - Simple goitre
- B. Region controlling breathing and cardiovascular reflexes - Synapse
- C. Insulating sheath around many axons - Medulla oblongata
- D. Membrane state with inside negative relative to outside - Cerebrum

BIO-12-CH09-0018 | Hard | Neuron structure

A reviewer compares several entries while checking 'Neuron processes usually receiving impulses'. Which entry must be corrected?

- A. Receptors detecting light - Photoreceptors
- B. Adrenal medullary fight-or-flight hormone - Adrenaline
- C. Junction between two neurons - Synapse
- D. Neuron processes usually receiving impulses - Medulla oblongata

BIO-12-CH09-0019 | Medium | Neuron structure

Complete the statement accurately: Neuron processes usually receiving impulses corresponds to _____.

- A. Dendrites
- B. Axon
- C. Adrenaline
- D. Synapse

BIO-12-CH09-0020 | Medium | Neuron structure

A learner encounters the observation 'Gaps between adjacent myelin segments' while analysing neuron structure. Which conclusion is supported by the prescribed biology?

- A. Glucagon
- B. Axon
- C. Nodes of Ranvier
- D. Cerebellum

BIO-12-CH09-0021 | Hard | Impulse and synapse

Two conclusions must both be valid to support the reference concept 'Membrane state with inside negative relative to outside'. Which option contains two accurate associations?

- A. Hormone lowering blood calcium - Reflex arc; Neural pathway mediating a reflex - Calcitonin
- B. Membrane state with inside negative relative to outside - Resting potential; Brain region coordinating posture and balance - Cerebellum
- C. Protective membranes around brain and spinal cord - Photoreceptors; Receptors detecting light - Meninges
- D. Iodine-deficiency enlargement of thyroid - Meninges; Protective membranes around brain and spinal cord - Simple goitre

BIO-12-CH09-0022 | Hard | Pancreas adrenal and gonads

Two conclusions must both be valid to support the reference concept 'Principal androgen secreted by testes'. Which option contains two accurate associations?

- A. Receptors detecting light - Mechanoreceptors; Receptors detecting sound and pressure - Photoreceptors
- B. Neural pathway mediating a reflex - Planaria; Ladder-like nervous system - Reflex arc
- C. Principal androgen secreted by testes - Testosterone; Pituitary hormone promoting body growth - Growth hormone
- D. Thyroid hormone containing iodine - Reflex action; Rapid automatic response through a neural pathway - Thyroxine

BIO-12-CH09-0023 | Hard | Nervous organization

A reviewer compares several entries while checking 'Structural and functional unit of nervous tissue'. Which entry must be corrected?

- A. Structural and functional unit of nervous tissue - Synapse
- B. Thyroid hormone containing iodine - Thyroxine
- C. Protective membranes around brain and spinal cord - Meninges
- D. Pancreatic hormone raising blood glucose - Glucagon

BIO-12-CH09-0024 | Medium | Neuron structure

A source table contains one accurate entry involving 'Gaps between adjacent myelin segments'. Which option preserves a correct biological association?

- A. Pancreatic hormone raising blood glucose - Resting potential
- B. Insulin deficiency or ineffective action - Hydra
- C. Gaps between adjacent myelin segments - Nodes of Ranvier
- D. Brain region coordinating posture and balance - Mechanoreceptors

BIO-12-CH09-0025 | Medium | Pancreas adrenal and gonads

Complete the statement accurately: Pancreatic hormone raising blood glucose corresponds to _____.

- A. Glucagon
- B. Hydra
- C. Reflex arc
- D. Photoreceptors

BIO-12-CH09-0026 | Hard | Neuron structure

A reviewer compares several entries while checking 'Gaps between adjacent myelin segments'. Which entry must be corrected?

- A. Insulin deficiency or ineffective action - Diabetes mellitus
- B. Diffuse nerve network without a central brain - Hydra
- C. Membrane state with inside negative relative to outside - Resting potential
- D. Gaps between adjacent myelin segments - Glucagon

BIO-12-CH09-0027 | Medium | Receptors and reflexes

A source table contains one accurate entry involving 'Receptors detecting sound and pressure'. Which option preserves a correct biological association?

- A. Adrenal medullary fight-or-flight hormone - Mechanoreceptors
- B. Receptors detecting sound and pressure - Mechanoreceptors
- C. Protective membranes around brain and spinal cord - Depolarization
- D. Childhood deficiency of growth hormone - Medulla oblongata

BIO-12-CH09-0028 | Hard | Central nervous system

A reviewer compares several entries while checking 'Protective membranes around brain and spinal cord'. Which entry must be corrected?

- A. Pituitary hormone stimulating the thyroid - TSH
- B. Rapid automatic response through a neural pathway - Reflex action
- C. Long process conducting impulses away from the cell body - Axon
- D. Protective membranes around brain and spinal cord - Simple goitre

BIO-12-CH09-0029 | Hard | Pancreas adrenal and gonads

Two conclusions must both be valid to support the reference concept 'Pancreatic hormone lowering blood glucose'. Which option contains two accurate associations?

- A. Iodine-deficiency enlargement of thyroid - Neuroglia; Supporting cells of nervous tissue - Simple goitre
- B. Ladder-like nervous system - Resting potential; Membrane state with inside negative relative to outside - Planaria
- C. Pancreatic hormone lowering blood glucose - Insulin; Rapid automatic response through a neural pathway - Reflex action
- D. Insulin deficiency or ineffective action - Planaria; Ladder-like nervous system - Diabetes mellitus

BIO-12-CH09-0030 | Medium | Receptors and reflexes

A source table contains one accurate entry involving 'Neural pathway mediating a reflex'. Which option preserves a correct biological association?

- A. Insulating sheath around many axons - Glucagon
- B. Pancreatic hormone raising blood glucose - Nodes of Ranvier
- C. Gaps between adjacent myelin segments - Diabetes mellitus
- D. Neural pathway mediating a reflex - Reflex arc

BIO-12-CH09-0031 | Medium | Pancreas adrenal and gonads

Complete the statement accurately: Pancreatic hormone lowering blood glucose corresponds to _____.

- A. Diabetes mellitus
- B. Neuroglia
- C. Insulin
- D. Cerebrum

BIO-12-CH09-0032 | Challenging | Central nervous system

A researcher checks whether four linked observations have been interpreted correctly.

- I. Region controlling breathing and cardiovascular reflexes corresponds to Cerebrum.
- II. Largest part of the human brain corresponds to Meninges.
- III. Protective membranes around brain and spinal cord corresponds to Meninges.
- IV. Hormone lowering blood calcium corresponds to Medulla oblongata.

Which combination survives the biological review?

- A. III only
- B. III and IV only
- C. I and II only
- D. IV only

BIO-12-CH09-0033 | Medium | Nervous organization

A learner encounters the observation 'Supporting cells of nervous tissue' while analysing nervous organization. Which conclusion is supported by the prescribed biology?

- A. Cerebrum
- B. Neuroglia
- C. Resting potential
- D. Synapse

BIO-12-CH09-0034 | Hard | Pituitary and thyroid

A reviewer compares several entries while checking 'Pituitary hormone stimulating the thyroid'. Which entry must be corrected?

- A. Insulating sheath around many axons - Myelin sheath
- B. Pituitary hormone stimulating the thyroid - Reflex arc
- C. Receptors detecting sound and pressure - Mechanoreceptors
- D. Region controlling breathing and cardiovascular reflexes - Medulla oblongata

BIO-12-CH09-0035 | Medium | Nervous organization

Complete the statement accurately: Supporting cells of nervous tissue corresponds to _____.

- A. Neuroglia
- B. Adrenaline
- C. Planaria
- D. Resting potential

BIO-12-CH09-0036 | Hard | Pancreas adrenal and gonads

A reviewer compares several entries while checking 'Adrenal medullary fight-or-flight hormone'. Which entry must be corrected?

- A. Insulating sheath around many axons - Myelin sheath
- B. Adrenal medullary fight-or-flight hormone - Depolarization
- C. Ladder-like nervous system - Planaria
- D. Structural and functional unit of nervous tissue - Neuron

BIO-12-CH09-0037 | Easy | Endocrine disorders

Which biological term or result is specifically identified by the description 'Insulin deficiency or ineffective action'?

- A. Diabetes mellitus
- B. Growth hormone
- C. Simple goitre
- D. Meninges

BIO-12-CH09-0038 | Medium | Impulse and synapse

Complete the statement accurately: Rapid reversal of membrane polarity corresponds to _____.

- A. Reflex action
- B. Depolarization
- C. Synapse
- D. Resting potential

BIO-12-CH09-0039 | Medium | Neuron structure

A learner encounters the observation 'Neuron processes usually receiving impulses' while analysing neuron structure. Which conclusion is supported by the prescribed biology?

- A. Dendrites
- B. Nodes of Ranvier
- C. Photoreceptors
- D. Medulla oblongata

BIO-12-CH09-0040 | Hard | Receptors and reflexes

Two conclusions must both be valid to support the reference concept 'Receptors detecting light'. Which option contains two accurate associations?

- A. Receptors detecting light - Photoreceptors; Pituitary hormone promoting body growth - Growth hormone
- B. Largest part of the human brain - Insulin; Pancreatic hormone lowering blood glucose - Cerebrum
- C. Pituitary hormone stimulating the thyroid - Cerebrum; Largest part of the human brain - TSH
- D. Thyroid hormone containing iodine - TSH; Pituitary hormone stimulating the thyroid - Thyroxine

BIO-12-CH09-0041 | Hard | Endocrine disorders

A reviewer compares several entries while checking 'Insulin deficiency or ineffective action'. Which entry must be corrected?

- A. Chemical released from a presynaptic terminal - Neurotransmitter
- B. Protective membranes around brain and spinal cord - Meninges
- C. Insulin deficiency or ineffective action - Planaria
- D. Largest part of the human brain - Cerebrum

BIO-12-CH09-0042 | Hard | Impulse and synapse

Two conclusions must both be valid to support the reference concept 'Junction between two neurons'. Which option contains two accurate associations?

- A. Adrenal medullary fight-or-flight hormone - Hydra; Diffuse nerve network without a central brain - Adrenaline
- B. Junction between two neurons - Synapse; Rapid automatic response through a neural pathway - Reflex action
- C. Thyroid hormone containing iodine - Meninges; Protective membranes around brain and spinal cord - Thyroxine
- D. Insulating sheath around many axons - Thyroxine; Thyroid hormone containing iodine - Myelin sheath

BIO-12-CH09-0043 | Easy | Impulse and synapse

Which description is correctly represented by Neurotransmitter in this chapter?

- A. Neuron processes usually receiving impulses
- B. Chemical released from a presynaptic terminal
- C. Junction between two neurons
- D. Excess growth hormone in adults

BIO-12-CH09-0044 | Hard | Nervous organization

Two conclusions must both be valid to support the reference concept 'Supporting cells of nervous tissue'. Which option contains two accurate associations?

- A. Adrenal medullary fight-or-flight hormone - Cerebrum; Largest part of the human brain - Adrenaline
- B. Childhood deficiency of growth hormone - Photoreceptors; Receptors detecting light - Dwarfism
- C. Supporting cells of nervous tissue - Neuroglia; Pancreatic hormone lowering blood glucose - Insulin
- D. Membrane state with inside negative relative to outside - Dwarfism; Childhood deficiency of growth hormone - Resting potential

BIO-12-CH09-0045 | Hard | Pituitary and thyroid

A reviewer compares several entries while checking 'Pituitary hormone promoting body growth'. Which entry must be corrected?

- A. Pituitary hormone promoting body growth - Myelin sheath
- B. Rapid automatic response through a neural pathway - Reflex action
- C. Long process conducting impulses away from the cell body - Axon
- D. Neuron processes usually receiving impulses - Dendrites

BIO-12-CH09-0046 | Medium | Pancreas adrenal and gonads

A learner encounters the observation 'Pancreatic hormone raising blood glucose' while analysing pancreas adrenal and gonads. Which conclusion is supported by the prescribed biology?

- A. Reflex arc
- B. Glucagon
- C. Hydra
- D. Testosterone

BIO-12-CH09-0047 | Medium | Impulse and synapse

Complete the statement accurately: Chemical released from a presynaptic terminal corresponds to _____.

- A. Meninges
- B. Photoreceptors
- C. Neurotransmitter
- D. Acromegaly

BIO-12-CH09-0048 | Hard | Impulse and synapse

A reviewer compares several entries while checking 'Junction between two neurons'. Which entry must be corrected?

- A. Thyroid hormone containing iodine - Thyroxine
- B. Adrenal medullary fight-or-flight hormone - Adrenaline
- C. Junction between two neurons - Reflex action
- D. Insulin deficiency or ineffective action - Diabetes mellitus

BIO-12-CH09-0049 | Easy | Endocrine disorders

Which biological term or result is specifically identified by the description 'Childhood deficiency of growth hormone'?

- A. Simple goitre
- B. Acromegaly
- C. TSH
- D. Dwarfism

BIO-12-CH09-0050 | Easy | Pancreas adrenal and gonads

Which description is correctly represented by Testosterone in this chapter?

- A. Rapid automatic response through a neural pathway
- B. Thyroid hormone containing iodine
- C. Principal androgen secreted by testes
- D. Ladder-like nervous system

BIO-12-CH09-0051 | Medium | Endocrine disorders

Complete the statement accurately: Insulin deficiency or ineffective action corresponds to ____.

- A. Diabetes mellitus
- B. Acromegaly
- C. Growth hormone
- D. Planaria

BIO-12-CH09-0052 | Medium | Nervous organization

A source table contains one accurate entry involving 'Ladder-like nervous system'. Which option preserves a correct biological association?

- A. Thyroid hormone containing iodine - TSH
- B. Excess growth hormone in adults - Mechanoreceptors
- C. Hormone lowering blood calcium - Thyroxine
- D. Ladder-like nervous system - Planaria

BIO-12-CH09-0053 | Medium | Pancreas adrenal and gonads

A learner encounters the observation 'Adrenal medullary fight-or-flight hormone' while analysing pancreas adrenal and gonads. Which conclusion is supported by the prescribed biology?

- A. Glucagon
- B. Adrenaline
- C. Neuron
- D. Reflex action

BIO-12-CH09-0054 | Medium | Pituitary and thyroid

A source table contains one accurate entry involving 'Pituitary hormone promoting body growth'. Which option preserves a correct biological association?

- A. Adrenal medullary fight-or-flight hormone - Reflex action
- B. Insulating sheath around many axons - Dendrites
- C. Diffuse nerve network without a central brain - Growth hormone
- D. Pituitary hormone promoting body growth - Growth hormone

BIO-12-CH09-0055 | Medium | Pancreas adrenal and gonads

A source table contains one accurate entry involving 'Principal androgen secreted by testes'. Which option preserves a correct biological association?

- A. Principal androgen secreted by testes - Testosterone
- B. Thyroid hormone containing iodine - Photoreceptors
- C. Neural pathway mediating a reflex - Thyroxine
- D. Ladder-like nervous system - Reflex action

BIO-12-CH09-0056 | Challenging | Pancreas adrenal and gonads

Evaluate the following source-based biological statements.

- I. Principal androgen secreted by testes corresponds to Glucagon.
- II. Pancreatic hormone raising blood glucose corresponds to Insulin.
- III. Pancreatic hormone lowering blood glucose corresponds to Insulin.
- IV. Thyroid hormone containing iodine corresponds to Thyroxine.

Which option identifies exactly the correct statements?

- A. I, II and III only
- B. III and IV only
- C. I, II, III and IV
- D. II and III only

BIO-12-CH09-0057 | Medium | Impulse and synapse

Complete the statement accurately: Membrane state with inside negative relative to outside corresponds to ____.

- A. Resting potential
- B. Neurotransmitter
- C. Simple goitre
- D. Depolarization

BIO-12-CH09-0058 | Easy | Nervous organization

Which biological term or result is specifically identified by the description 'Structural and functional unit of nervous tissue'?

- A. Testosterone
- B. Neuroglia
- C. Glucagon
- D. Neuron

BIO-12-CH09-0059 | Challenging | Receptors and reflexes

A student submits four conclusions after integrating related chapter concepts.

- I. Neural pathway mediating a reflex corresponds to Mechanoreceptors.
- II. Receptors detecting sound and pressure corresponds to Mechanoreceptors.
- III. Rapid automatic response through a neural pathway corresponds to Cerebrum.
- IV. Largest part of the human brain corresponds to Reflex arc.

Which combination should be accepted without correction?

- A. I and IV only
- B. IV only
- C. II only
- D. III and IV only

BIO-12-CH09-0060 | Medium | Receptors and reflexes

A learner encounters the observation 'Receptors detecting sound and pressure' while analysing receptors and reflexes. Which conclusion is supported by the prescribed biology?

- A. Neurotransmitter
- B. Depolarization
- C. Mechanoreceptors
- D. Reflex action

BIO-12-CH09-0061 | Medium | Neuron structure

A source table contains one accurate entry involving 'Long process conducting impulses away from the cell body'. Which option preserves a correct biological association?

- A. Insulin deficiency or ineffective action - Hydra
- B. Diffuse nerve network without a central brain - Axon
- C. Hormone lowering blood calcium - Adrenaline
- D. Long process conducting impulses away from the cell body - Axon

BIO-12-CH09-0062 | Hard | Nervous organization

Two conclusions must both be valid to support the reference concept 'Structural and functional unit of nervous tissue'. Which option contains two accurate associations?

- A. Protective membranes around brain and spinal cord - Testosterone; Principal androgen secreted by testes - Meninges
- B. Insulin deficiency or ineffective action - Meninges; Protective membranes around brain and spinal cord - Diabetes mellitus
- C. Structural and functional unit of nervous tissue - Neuron; Junction between two neurons - Synapse
- D. Principal androgen secreted by testes - Glucagon; Pancreatic hormone raising blood glucose - Testosterone

BIO-12-CH09-0063 | Medium | Central nervous system

A learner encounters the observation 'Brain region coordinating posture and balance' while analysing central nervous system. Which conclusion is supported by the prescribed biology?

- A. Testosterone
- B. Cerebellum
- C. Neurotransmitter
- D. Dendrites

BIO-12-CH09-0064 | Hard | Pancreas adrenal and gonads

A reviewer compares several entries while checking 'Pancreatic hormone lowering blood glucose'. Which entry must be corrected?

- A. Supporting cells of nervous tissue - Neuroglia
- B. Ladder-like nervous system - Planaria
- C. Largest part of the human brain - Cerebrum
- D. Pancreatic hormone lowering blood glucose - Reflex action

BIO-12-CH09-0065 | Medium | Pituitary and thyroid

Complete the statement accurately: Pituitary hormone promoting body growth corresponds to _____.

- A. Adrenaline
- B. Growth hormone
- C. Myelin sheath
- D. Calcitonin

BIO-12-CH09-0066 | Easy | Pituitary and thyroid

Which description is correctly represented by TSH in this chapter?

- A. Region controlling breathing and cardiovascular reflexes
- B. Pituitary hormone promoting body growth
- C. Pituitary hormone stimulating the thyroid
- D. Neural pathway mediating a reflex

BIO-12-CH09-0067 | Easy | Impulse and synapse

Which biological term or result is specifically identified by the description 'Membrane state with inside negative relative to outside'?

- A. Neurotransmitter
- B. Calcitonin
- C. Cerebellum
- D. Resting potential

BIO-12-CH09-0068 | Challenging | Impulse and synapse

During error analysis, four proposed associations are reviewed.

- I. Rapid reversal of membrane polarity corresponds to Neurotransmitter.
- II. Chemical released from a presynaptic terminal corresponds to Neurotransmitter.
- III. Junction between two neurons corresponds to Cerebrum.
- IV. Largest part of the human brain corresponds to Cerebrum.

Which set contains all and only the valid conclusions?

- A. II and III only
- B. II and IV only
- C. II only
- D. III and IV only

BIO-12-CH09-0069 | Medium | Impulse and synapse

A source table contains one accurate entry involving 'Membrane state with inside negative relative to outside'. Which option preserves a correct biological association?

- A. Brain region coordinating posture and balance - Reflex arc
- B. Iodine-deficiency enlargement of thyroid - Photoreceptors
- C. Membrane state with inside negative relative to outside - Resting potential
- D. Receptors detecting light - Resting potential

BIO-12-CH09-0070 | Easy | Pancreas adrenal and gonads

Which description is correctly represented by Insulin in this chapter?

- A. Largest part of the human brain
- B. Principal androgen secreted by testes
- C. Insulin deficiency or ineffective action
- D. Pancreatic hormone lowering blood glucose

BIO-12-CH09-0071 | Medium | Pituitary and thyroid

A source table contains one accurate entry involving 'Pituitary hormone stimulating the thyroid'. Which option preserves a correct biological association?

- A. Neural pathway mediating a reflex - Testosterone
- B. Pituitary hormone stimulating the thyroid - TSH
- C. Insulin deficiency or ineffective action - Mechanoreceptors
- D. Receptors detecting sound and pressure - Myelin sheath

BIO-12-CH09-0072 | Medium | Impulse and synapse

A source table contains one accurate entry involving 'Chemical released from a presynaptic terminal'. Which option preserves a correct biological association?

- A. Receptors detecting light - Neuron
- B. Chemical released from a presynaptic terminal - Neurotransmitter
- C. Excess growth hormone in adults - Photoreceptors
- D. Pituitary hormone promoting body growth - Meninges

BIO-12-CH09-0073 | Medium | Endocrine disorders

A source table contains one accurate entry involving 'Insulin deficiency or ineffective action'. Which option preserves a correct biological association?

- A. Pituitary hormone promoting body growth - Adrenaline
- B. Adrenal medullary fight-or-flight hormone - Reflex action
- C. Insulin deficiency or ineffective action - Diabetes mellitus
- D. Ladder-like nervous system - Growth hormone

BIO-12-CH09-0074 | Medium | Receptors and reflexes

A learner encounters the observation 'Neural pathway mediating a reflex' while analysing receptors and reflexes. Which conclusion is supported by the prescribed biology?

- A. Glucagon
- B. Mechanoreceptors
- C. Reflex arc
- D. Insulin

BIO-12-CH09-0075 | Medium | Neuron structure

A source table contains one accurate entry involving 'Insulating sheath around many axons'. Which option preserves a correct biological association?

- A. Receptors detecting light - Thyroxine
- B. Pituitary hormone promoting body growth - Depolarization
- C. Rapid reversal of membrane polarity - Photoreceptors
- D. Insulating sheath around many axons - Myelin sheath

BIO-12-CH09-0076 | Hard | Receptors and reflexes

A reviewer compares several entries while checking 'Rapid automatic response through a neural pathway'. Which entry must be corrected?

- A. Hormone lowering blood calcium - Calcitonin
- B. Structural and functional unit of nervous tissue - Neuron
- C. Rapid automatic response through a neural pathway - Medulla oblongata
- D. Neuron processes usually receiving impulses - Dendrites

BIO-12-CH09-0077 | Hard | Neuron structure

A reviewer compares several entries while checking 'Long process conducting impulses away from the cell body'. Which entry must be corrected?

- A. Insulin deficiency or ineffective action - Diabetes mellitus
- B. Long process conducting impulses away from the cell body - Calcitonin
- C. Diffuse nerve network without a central brain - Hydra
- D. Structural and functional unit of nervous tissue - Neuron

BIO-12-CH09-0078 | Easy | Pituitary and thyroid

Which biological term or result is specifically identified by the description 'Pituitary hormone promoting body growth'?

- A. Reflex action
- B. Myelin sheath
- C. Growth hormone
- D. Dendrites

BIO-12-CH09-0079 | Medium | Central nervous system

Complete the statement accurately: Region controlling breathing and cardiovascular reflexes corresponds to _____.

- A. Diabetes mellitus
- B. Medulla oblongata
- C. Meninges
- D. Resting potential

BIO-12-CH09-0080 | Easy | Endocrine disorders

Which description is correctly represented by Diabetes mellitus in this chapter?

- A. Adrenal medullary fight-or-flight hormone
- B. Protective membranes around brain and spinal cord
- C. Largest part of the human brain
- D. Insulin deficiency or ineffective action

BIO-12-CH09-0081 | Easy | Nervous organization

Which biological term or result is specifically identified by the description 'Diffuse nerve network without a central brain'?

- A. Nodes of Ranvier
- B. Hydra
- C. Planaria
- D. Neuroglia

BIO-12-CH09-0082 | Hard | Pituitary and thyroid

Two conclusions must both be valid to support the reference concept 'Pituitary hormone promoting body growth'. Which option contains two accurate associations?

- A. Rapid automatic response through a neural pathway - Axon; Long process conducting impulses away from the cell body - Reflex action
- B. Long process conducting impulses away from the cell body - Hydra; Diffuse nerve network without a central brain - Axon
- C. Pituitary hormone promoting body growth - Growth hormone; Insulating sheath around many axons - Myelin sheath
- D. Neuron processes usually receiving impulses - Reflex action; Rapid automatic response through a neural pathway - Dendrites

BIO-12-CH09-0083 | Easy | Receptors and reflexes

Which description is correctly represented by Reflex arc in this chapter?

- A. Pancreatic hormone lowering blood glucose
- B. Gaps between adjacent myelin segments
- C. Insulating sheath around many axons
- D. Neural pathway mediating a reflex

BIO-12-CH09-0084 | Easy | Central nervous system

Which biological term or result is specifically identified by the description 'Region controlling breathing and cardiovascular reflexes'?

- A. Dendrites
- B. Medulla oblongata
- C. Diabetes mellitus
- D. Cerebrum

BIO-12-CH09-0085 | Medium | Pituitary and thyroid

Complete the statement accurately: Thyroid hormone containing iodine corresponds to _____.

- A. Growth hormone
- B. Nodes of Ranvier
- C. Axon
- D. Thyroxine

BIO-12-CH09-0086 | Medium | Central nervous system

Complete the statement accurately: Brain region coordinating posture and balance corresponds to _____.

- A. Reflex action
- B. Cerebellum
- C. Cerebrum
- D. Meninges

BIO-12-CH09-0087 | Easy | Neuron structure

Which description is correctly represented by Nodes of Ranvier in this chapter?

- A. Insulating sheath around many axons
- B. Membrane state with inside negative relative to outside
- C. Long process conducting impulses away from the cell body
- D. Gaps between adjacent myelin segments

BIO-12-CH09-0088 | Easy | Central nervous system

Which description is correctly represented by Medulla oblongata in this chapter?

- A. Region controlling breathing and cardiovascular reflexes
- B. Largest part of the human brain
- C. Long process conducting impulses away from the cell body
- D. Diffuse nerve network without a central brain

BIO-12-CH09-0089 | Easy | Nervous organization

Which description is correctly represented by Neuron in this chapter?

- A. Principal androgen secreted by testes
- B. Structural and functional unit of nervous tissue
- C. Pancreatic hormone raising blood glucose
- D. Supporting cells of nervous tissue

BIO-12-CH09-0090 | Medium | Receptors and reflexes

A source table contains one accurate entry involving 'Rapid automatic response through a neural pathway'. Which option preserves a correct biological association?

- A. Rapid automatic response through a neural pathway - Reflex action
- B. Pancreatic hormone raising blood glucose - Reflex action
- C. Hormone lowering blood calcium - Glucagon
- D. Gaps between adjacent myelin segments - Calcitonin

BIO-12-CH09-0091 | Hard | Impulse and synapse

A reviewer compares several entries while checking 'Membrane state with inside negative relative to outside'. Which entry must be corrected?

- A. Iodine-deficiency enlargement of thyroid - Simple goitre
- B. Hormone lowering blood calcium - Calcitonin
- C. Membrane state with inside negative relative to outside - Cerebellum
- D. Neural pathway mediating a reflex - Reflex arc

BIO-12-CH09-0092 | Challenging | Neuron structure

A researcher checks whether four linked observations have been interpreted correctly.

- I. Insulating sheath around many axons corresponds to Myelin sheath.
- II. Gaps between adjacent myelin segments corresponds to Dendrites.
- III. Neuron processes usually receiving impulses corresponds to Dwarfism.
- IV. Childhood deficiency of growth hormone corresponds to Myelin sheath.

Which combination survives the biological review?

- A. I, II, III and IV
- B. III only
- C. II, III and IV only
- D. I only

BIO-12-CH09-0093 | Medium | Nervous organization

A source table contains one accurate entry involving 'Structural and functional unit of nervous tissue'. Which option preserves a correct biological association?

- A. Insulin deficiency or ineffective action - Testosterone
- B. Thyroid hormone containing iodine - Neuron
- C. Structural and functional unit of nervous tissue - Neuron
- D. Principal androgen secreted by testes - Thyroxine

BIO-12-CH09-0094 | Easy | Neuron structure

Which biological term or result is specifically identified by the description 'Gaps between adjacent myelin segments'?

- A. Diabetes mellitus
- B. Axon
- C. Glucagon
- D. Nodes of Ranvier

BIO-12-CH09-0095 | Hard | Central nervous system

Two conclusions must both be valid to support the reference concept 'Protective membranes around brain and spinal cord'. Which option contains two accurate associations?

- A. Protective membranes around brain and spinal cord - Meninges; Iodine-deficiency enlargement of thyroid - Simple goitre
- B. Long process conducting impulses away from the cell body - TSH; Pituitary hormone stimulating the thyroid - Axon
- C. Pituitary hormone stimulating the thyroid - Neuron; Structural and functional unit of nervous tissue - TSH
- D. Insulin deficiency or ineffective action - Reflex action; Rapid automatic response through a neural pathway - Diabetes mellitus

BIO-12-CH09-0096 | Challenging | Pituitary and thyroid

Evaluate the following source-based biological statements.

- I. Hormone lowering blood calcium corresponds to TSH.
- II. Pituitary hormone stimulating the thyroid corresponds to TSH.
- III. Pituitary hormone promoting body growth corresponds to Growth hormone.
- IV. Structural and functional unit of nervous tissue corresponds to Calcitonin.

Which option identifies exactly the correct statements?

- A. III and IV only
- B. II only
- C. I and II only
- D. II and III only

BIO-12-CH09-0097 | Challenging | Endocrine disorders

A student submits four conclusions after integrating related chapter concepts.

- I. Iodine-deficiency enlargement of thyroid corresponds to Simple goitre.
- II. Excess growth hormone in adults corresponds to Acromegaly.
- III. Insulin deficiency or ineffective action corresponds to Diabetes mellitus.
- IV. Rapid automatic response through a neural pathway corresponds to Reflex action.

Which combination should be accepted without correction?

- A. I, II, III and IV
- B. III and IV only
- C. I, III and IV only
- D. II, III and IV only

BIO-12-CH09-0098 | Medium | Pancreas adrenal and gonads

Complete the statement accurately: Principal androgen secreted by testes corresponds to ____.

- A. Testosterone
- B. Glucagon
- C. Reflex arc
- D. Growth hormone

BIO-12-CH09-0099 | Medium | Impulse and synapse

Complete the statement accurately: Junction between two neurons corresponds to ____.

- A. Neurotransmitter
- B. Reflex action
- C. Synapse
- D. Depolarization

BIO-12-CH09-0100 | Easy | Impulse and synapse

Which description is correctly represented by Synapse in this chapter?

- A. Rapid reversal of membrane polarity
- B. Insulating sheath around many axons
- C. Insulin deficiency or ineffective action
- D. Junction between two neurons

BIO-12-CH09-0101 | Medium | Pituitary and thyroid

A learner encounters the observation 'Hormone lowering blood calcium' while analysing pituitary and thyroid. Which conclusion is supported by the prescribed biology?

- A. TSH
- B. Neuron
- C. Growth hormone
- D. Calcitonin

BIO-12-CH09-0102 | Medium | Pancreas adrenal and gonads

Complete the statement accurately: Adrenal medullary fight-or-flight hormone corresponds to _____.

- A. Depolarization
- B. Adrenaline
- C. Reflex arc
- D. Neuron

BIO-12-CH09-0103 | Medium | Central nervous system

A source table contains one accurate entry involving 'Largest part of the human brain'. Which option preserves a correct biological association?

- A. Junction between two neurons - Simple goitre
- B. Largest part of the human brain - Cerebrum
- C. Neuron processes usually receiving impulses - Synapse
- D. Principal androgen secreted by testes - Cerebrum

BIO-12-CH09-0104 | Challenging | Nervous organization

During error analysis, four proposed associations are reviewed.

- I. Structural and functional unit of nervous tissue corresponds to Neuron.
- II. Diffuse nerve network without a central brain corresponds to Planaria.
- III. Ladder-like nervous system corresponds to Adrenaline.
- IV. Adrenal medullary fight-or-flight hormone corresponds to Adrenaline.

Which set contains all and only the valid conclusions?

- A. II, III and IV only
- B. I and III only
- C. II and III only
- D. I and IV only

BIO-12-CH09-0105 | Medium | Receptors and reflexes

Complete the statement accurately: Rapid automatic response through a neural pathway corresponds to _____.

- A. Reflex action
- B. Calcitonin
- C. Medulla oblongata
- D. Neuron

BIO-12-CH09-0106 | Easy | Pituitary and thyroid

Which description is correctly represented by Growth hormone in this chapter?

- A. Adrenal medullary fight-or-flight hormone
- B. Neuron processes usually receiving impulses
- C. Pituitary hormone promoting body growth
- D. Long process conducting impulses away from the cell body

BIO-12-CH09-0107 | Medium | Central nervous system

A source table contains one accurate entry involving 'Protective membranes around brain and spinal cord'. Which option preserves a correct biological association?

- A. Pituitary hormone stimulating the thyroid - Testosterone
- B. Structural and functional unit of nervous tissue - Meninges
- C. Protective membranes around brain and spinal cord - Meninges
- D. Insulin deficiency or ineffective action - Axon

BIO-12-CH09-0108 | Hard | Central nervous system

A reviewer compares several entries while checking 'Region controlling breathing and cardiovascular reflexes'. Which entry must be corrected?

- A. Pituitary hormone stimulating the thyroid - TSH
- B. Region controlling breathing and cardiovascular reflexes - Resting potential
- C. Insulin deficiency or ineffective action - Diabetes mellitus
- D. Long process conducting impulses away from the cell body - Axon

BIO-12-CH09-0109 | Medium | Endocrine disorders

A learner encounters the observation 'Insulin deficiency or ineffective action' while analysing endocrine disorders. Which conclusion is supported by the prescribed biology?

- A. Cerebrum
- B. Acromegaly
- C. Diabetes mellitus
- D. Planaria

BIO-12-CH09-0110 | Challenging | Pituitary and thyroid

A researcher checks whether four linked observations have been interpreted correctly.

- I. Pituitary hormone promoting body growth corresponds to TSH.
- II. Pituitary hormone stimulating the thyroid corresponds to TSH.
- III. Hormone lowering blood calcium corresponds to Calcitonin.
- IV. Long process conducting impulses away from the cell body corresponds to Axon.

Which combination survives the biological review?

- A. II, III and IV only
- B. I, II and IV only
- C. I, II and III only
- D. IV only

BIO-12-CH09-0111 | Hard | Impulse and synapse

Two conclusions must both be valid to support the reference concept 'Chemical released from a presynaptic terminal'. Which option contains two accurate associations?

- A. Excess growth hormone in adults - Meninges; Protective membranes around brain and spinal cord - Acromegaly
- B. Chemical released from a presynaptic terminal - Neurotransmitter; Neuron processes usually receiving impulses - Dendrites
- C. Receptors detecting light - Axon; Long process conducting impulses away from the cell body - Photoreceptors
- D. Protective membranes around brain and spinal cord - Photoreceptors; Receptors detecting light - Meninges

BIO-12-CH09-0112 | Easy | Receptors and reflexes

Which description is correctly represented by Photoreceptors in this chapter?

- A. Pituitary hormone stimulating the thyroid
- B. Neural pathway mediating a reflex
- C. Pituitary hormone promoting body growth
- D. Receptors detecting light

BIO-12-CH09-0113 | Medium | Nervous organization

A learner encounters the observation 'Ladder-like nervous system' while analysing nervous organization. Which conclusion is supported by the prescribed biology?

- A. Thyroxine
- B. Neuroglia
- C. Planaria
- D. Calcitonin

BIO-12-CH09-0114 | Easy | Central nervous system

Which description is correctly represented by Meninges in this chapter?

- A. Largest part of the human brain
- B. Rapid automatic response through a neural pathway
- C. Protective membranes around brain and spinal cord
- D. Long process conducting impulses away from the cell body

BIO-12-CH09-0115 | Easy | Receptors and reflexes

Which biological term or result is specifically identified by the description 'Rapid automatic response through a neural pathway'?

- A. Reflex action
- B. Neuron
- C. Dendrites
- D. Mechanoreceptors

BIO-12-CH09-0116 | Hard | Neuron structure

Two conclusions must both be valid to support the reference concept 'Gaps between adjacent myelin segments'. Which option contains two accurate associations?

- A. Membrane state with inside negative relative to outside - Hydra; Diffuse nerve network without a central brain - Resting potential
- B. Diffuse nerve network without a central brain - Cerebellum; Brain region coordinating posture and balance - Hydra
- C. Brain region coordinating posture and balance - Neuron; Structural and functional unit of nervous tissue - Cerebellum
- D. Gaps between adjacent myelin segments - Nodes of Ranvier; Pancreatic hormone raising blood glucose - Glucagon

BIO-12-CH09-0117 | Easy | Pituitary and thyroid

Which biological term or result is specifically identified by the description 'Pituitary hormone stimulating the thyroid'?

- A. TSH
- B. Reflex arc
- C. Diabetes mellitus
- D. Testosterone

BIO-12-CH09-0118 | Challenging | Nervous organization

Evaluate the following source-based biological statements.

- I. Diffuse nerve network without a central brain corresponds to Hydra.
- II. Supporting cells of nervous tissue corresponds to Neuron.
- III. Structural and functional unit of nervous tissue corresponds to Neuron.
- IV. Insulin deficiency or ineffective action corresponds to Hydra.

Which option identifies exactly the correct statements?

- A. I, II, III and IV
- B. IV only
- C. II and III only
- D. I and III only

BIO-12-CH09-0119 | Hard | Receptors and reflexes

Two conclusions must both be valid to support the reference concept 'Neural pathway mediating a reflex'. Which option contains two accurate associations?

- A. Pancreatic hormone raising blood glucose - Medulla oblongata; Region controlling breathing and cardiovascular reflexes - Glucagon
- B. Neural pathway mediating a reflex - Reflex arc; Insulating sheath around many axons - Myelin sheath
- C. Junction between two neurons - Diabetes mellitus; Insulin deficiency or ineffective action - Synapse
- D. Region controlling breathing and cardiovascular reflexes - Nodes of Ranvier; Gaps between adjacent myelin segments - Medulla oblongata

BIO-12-CH09-0120 | Easy | Impulse and synapse

Which biological term or result is specifically identified by the description 'Junction between two neurons'?

- A. Synapse
- B. Hydra
- C. Depolarization
- D. Neurotransmitter

BIO-12-CH09-0121 | Challenging | Endocrine disorders

A student submits four conclusions after integrating related chapter concepts.

- I. Childhood deficiency of growth hormone corresponds to Dwarfism.
 - II. Iodine-deficiency enlargement of thyroid corresponds to Diabetes mellitus.
 - III. Insulin deficiency or ineffective action corresponds to Diabetes mellitus.
 - IV. Insulating sheath around many axons corresponds to Myelin sheath.
- Which combination should be accepted without correction?

- A. I and IV only
- B. I, III and IV only
- C. II and IV only
- D. IV only

BIO-12-CH09-0122 | Hard | Nervous organization

A reviewer compares several entries while checking 'Diffuse nerve network without a central brain'. Which entry must be corrected?

- A. Excess growth hormone in adults - Acromegaly
- B. Insulin deficiency or ineffective action - Diabetes mellitus
- C. Brain region coordinating posture and balance - Cerebellum
- D. Diffuse nerve network without a central brain - Dendrites

BIO-12-CH09-0123 | Easy | Nervous organization

Which description is correctly represented by Neuroglia in this chapter?

- A. Adrenal medullary fight-or-flight hormone
- B. Supporting cells of nervous tissue
- C. Pancreatic hormone lowering blood glucose
- D. Ladder-like nervous system

BIO-12-CH09-0124 | Hard | Central nervous system

Two conclusions must both be valid to support the reference concept 'Region controlling breathing and cardiovascular reflexes'. Which option contains two accurate associations?

- A. Diffuse nerve network without a central brain - Dendrites; Neuron processes usually receiving impulses - Hydra
- B. Insulin deficiency or ineffective action - TSH; Pituitary hormone stimulating the thyroid - Diabetes mellitus
- C. Long process conducting impulses away from the cell body - Diabetes mellitus; Insulin deficiency or ineffective action - Axon
- D. Region controlling breathing and cardiovascular reflexes - Medulla oblongata; Membrane state with inside negative relative to outside - Resting potential

BIO-12-CH09-0125 | Easy | Neuron structure

Which biological term or result is specifically identified by the description 'Long process conducting impulses away from the cell body'?

- A. Adrenaline
- B. Neuroglia
- C. Diabetes mellitus
- D. Axon

BIO-12-CH09-0126 | Hard | Receptors and reflexes

A reviewer compares several entries while checking 'Receptors detecting light'. Which entry must be corrected?

- A. Receptors detecting light - Growth hormone
- B. Excess growth hormone in adults - Acromegaly
- C. Iodine-deficiency enlargement of thyroid - Simple goitre
- D. Pituitary hormone stimulating the thyroid - TSH

BIO-12-CH09-0127 | Hard | Endocrine disorders

Two conclusions must both be valid to support the reference concept 'Insulin deficiency or ineffective action'. Which option contains two accurate associations?

- A. Protective membranes around brain and spinal cord - Growth hormone; Pituitary hormone promoting body growth - Meninges
- B. Chemical released from a presynaptic terminal - Reflex action; Rapid automatic response through a neural pathway - Neurotransmitter
- C. Largest part of the human brain - Adrenaline; Adrenal medullary fight-or-flight hormone - Cerebrum
- D. Insulin deficiency or ineffective action - Diabetes mellitus; Ladder-like nervous system - Planaria

BIO-12-CH09-0128 | Medium | Impulse and synapse

A learner encounters the observation 'Chemical released from a presynaptic terminal' while analysing impulse and synapse. Which conclusion is supported by the prescribed biology?

- A. Depolarization
- B. Neurotransmitter
- C. Acromegaly
- D. Growth hormone

BIO-12-CH09-0129 | Medium | Endocrine disorders

A source table contains one accurate entry involving 'Excess growth hormone in adults'. Which option preserves a correct biological association?

- A. Excess growth hormone in adults - Acromegaly
- B. Ladder-like nervous system - Testosterone
- C. Principal androgen secreted by testes - Cerebellum
- D. Structural and functional unit of nervous tissue - Mechanoreceptors

BIO-12-CH09-0130 | Hard | Pituitary and thyroid

Two conclusions must both be valid to support the reference concept 'Pituitary hormone stimulating the thyroid'. Which option contains two accurate associations?

- A. Principal androgen secreted by testes - Mechanoreceptors; Receptors detecting sound and pressure - Testosterone
- B. Insulin deficiency or ineffective action - Testosterone; Principal androgen secreted by testes - Diabetes mellitus
- C. Pituitary hormone stimulating the thyroid - TSH; Neural pathway mediating a reflex - Reflex arc
- D. Insulating sheath around many axons - Simple goitre; Iodine-deficiency enlargement of thyroid - Myelin sheath

BIO-12-CH09-0131 | Hard | Endocrine disorders

A reviewer compares several entries while checking 'Childhood deficiency of growth hormone'. Which entry must be corrected?

- A. Pancreatic hormone lowering blood glucose - Insulin
- B. Childhood deficiency of growth hormone - Growth hormone
- C. Largest part of the human brain - Cerebrum
- D. Pituitary hormone stimulating the thyroid - TSH

BIO-12-CH09-0132 | Medium | Nervous organization

Complete the statement accurately: Structural and functional unit of nervous tissue corresponds to _____.

- A. Neuron
 - B. Planaria
 - C. Neuroglia
 - D. Synapse
-

Standard 12 | Chapter 10

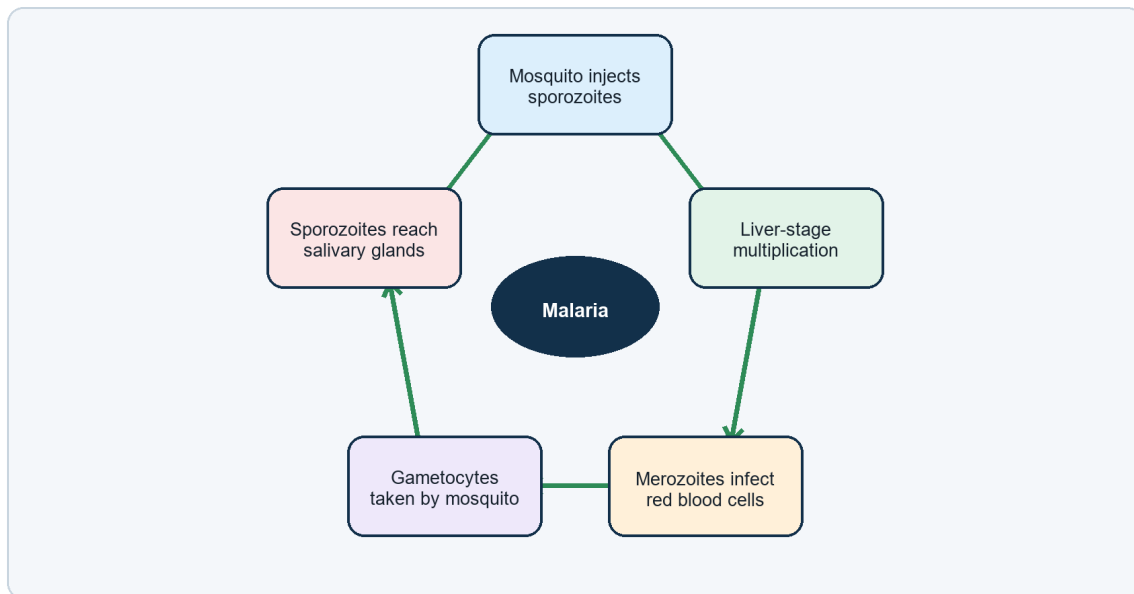
Human Health and Diseases

BIO-12-CH10-0001 | Challenging | Key transfers in the malaria life cycle

A drug prevents sporozoites from reaching mosquito salivary glands. What is the most direct epidemiological effect?

Key transfers in the malaria life cycle

Alternation between human and female *Anopheles* mosquito



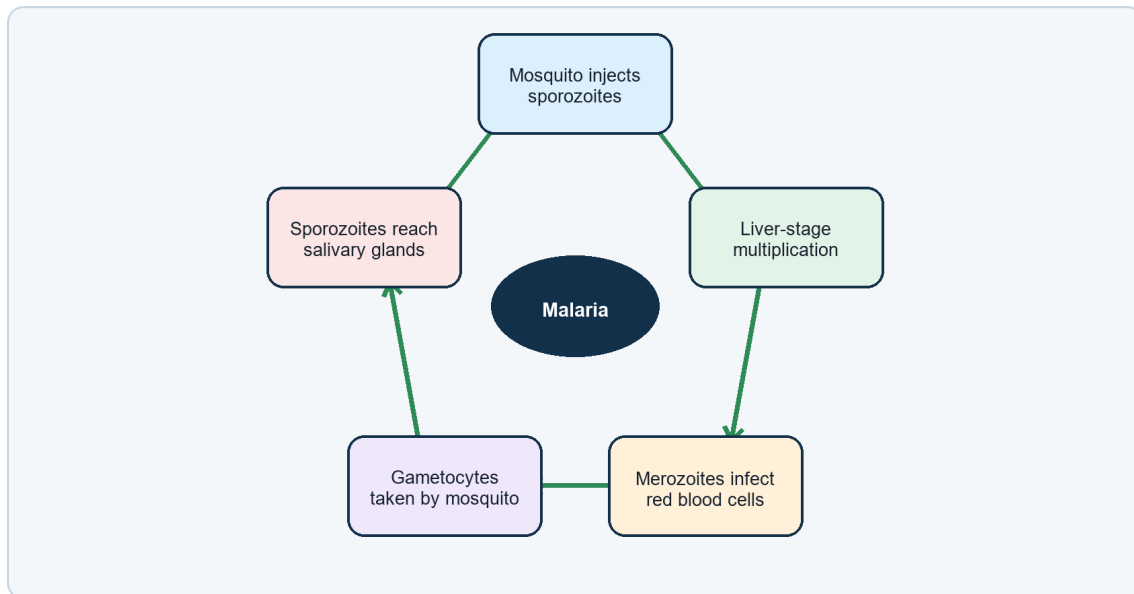
- A. Transmission from mosquito to human is greatly reduced
- B. Human red cells release more merozoites
- C. All human gametocytes become sporozoites
- D. Mosquitoes can transmit malaria without biting

BIO-12-CH10-0002 | Medium | Key transfers in the malaria life cycle

Which infective stage is injected into humans by an infected female Anopheles mosquito?

Key transfers in the malaria life cycle

Alternation between human and female Anopheles mosquito



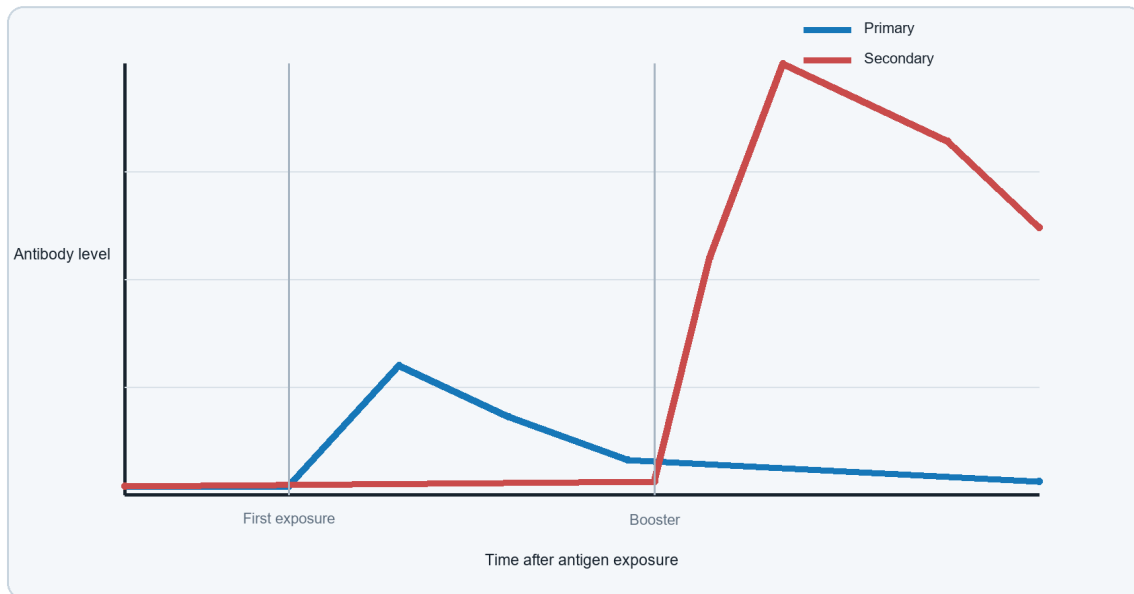
- A. Merozoites
- B. Sporozoites
- C. Gametocytes
- D. Trophozoites inside red cells

BIO-12-CH10-0003 | Hard | Primary and secondary antibody responses

What is the immunological basis of the booster effect shown?

Primary and secondary antibody responses

Response after first exposure and booster exposure



- A. Immediate loss of all plasma cells
- B. Rapid clonal expansion of antigen-specific memory cells
- C. Nonspecific phagocytosis alone
- D. A permanent fall in antibody affinity

BIO-12-CH10-0004 | Medium | Immunity

A learner encounters the observation 'Specific defence developing after exposure' while analysing immunity. Which conclusion is supported by the prescribed biology?

- A. Acquired immunity
- B. Vaccination
- C. Innate immunity
- D. Carcinogens

BIO-12-CH10-0005 | Medium | Cancer and immunity disorders

A source table contains one accurate entry involving 'Spread of malignant cells to distant sites'. Which option preserves a correct biological association?

- A. Administration of antigenic material to induce immune memory - Active immunity
- B. Stimulant derived from coca plant - Vaccination
- C. Control approach that interrupts a pathogen's vector - Salmonella typhi
- D. Spread of malignant cells to distant sites - Metastasis

BIO-12-CH10-0006 | Challenging | Antibody structure

A researcher checks whether four linked observations have been interpreted correctly.

- I. Basic shape of an antibody molecule corresponds to Y-shaped.
- II. Stem region interacting with immune components corresponds to Constant Fc region.
- III. Chains present in each antibody molecule corresponds to Two heavy and two light chains.
- IV. Control approach that interrupts a pathogen's vector corresponds to Vector control.

Which combination survives the biological review?

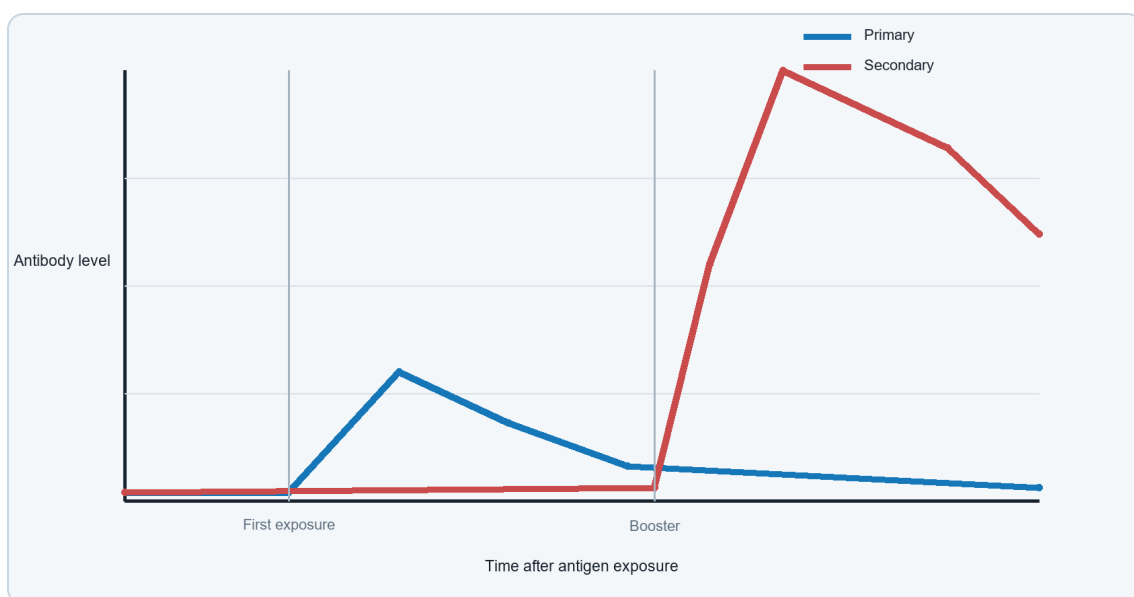
- A. I, II and IV only
- B. I, II, III and IV
- C. III and IV only
- D. II and IV only

BIO-12-CH10-0007 | Easy | Primary and secondary antibody responses

Why is the secondary antibody response higher and faster than the primary response?

Primary and secondary antibody responses

Response after first exposure and booster exposure



- A. All antibodies become hormones
- B. The antigen no longer has epitopes

- C. Innate immunity permanently replaces adaptive immunity
 - D. Memory lymphocytes respond rapidly
-

BIO-12-CH10-0008 | Medium | Viral and helminth diseases

Complete the statement accurately: Filarial worm transmitted by mosquitoes corresponds to _____.

- A. HIV
- B. Vector control
- C. Cannabis sativa
- D. Wuchereria

BIO-12-CH10-0009 | Medium | Bacterial and protozoan diseases

Complete the statement accurately: Protozoan causing malaria corresponds to _____.

- A. Female Anopheles
- B. ELISA
- C. Plasmodium
- D. Streptococcus pneumoniae

BIO-12-CH10-0010 | Challenging | Bacterial and protozoan diseases

Evaluate the following source-based biological statements.

- I. Causative bacterium of pneumonia commonly cited in the chapter corresponds to Streptococcus pneumoniae.
- II. Protozoan causing malaria corresponds to Female Anopheles.
- III. Mosquito transmitting malaria corresponds to Wuchereria.
- IV. Filarial worm transmitted by mosquitoes corresponds to Wuchereria.

Which option identifies exactly the correct statements?

- A. I, II and IV only
- B. II and III only
- C. I and IV only
- D. I, II and III only

BIO-12-CH10-0011 | Challenging | Immunity

A student submits four conclusions after integrating related chapter concepts.

- I. Non-specific defence present from birth corresponds to Acquired immunity.
- II. Specific defence developing after exposure corresponds to Acquired immunity.
- III. Immediate immunity produced by transferred antibodies corresponds to Passive immunity.
- IV. Virus that targets helper T lymphocytes corresponds to HIV.

Which combination should be accepted without correction?

- A. I only
- B. III and IV only
- C. I and IV only
- D. II, III and IV only

BIO-12-CH10-0012 | Easy | Drugs and addiction

Which description is correctly represented by Morphine in this chapter?

- A. Responsible adolescent practice that prevents substance misuse
- B. Repeated compulsive substance use despite harm
- C. Non-specific defence present from birth
- D. Opioid obtained from poppy latex

BIO-12-CH10-0013 | Medium | Drugs and addiction

Complete the statement accurately: Repeated compulsive substance use despite harm corresponds to _____.

- A. Passive immunity
- B. Morphine
- C. Addiction
- D. Cannabis sativa

BIO-12-CH10-0014 | Medium | Antibody structure

Complete the statement accurately: Antigen-binding regions corresponds to _____.

- A. Variable regions
- B. Y-shaped
- C. Morphine
- D. Two heavy and two light chains

BIO-12-CH10-0015 | Easy | Immunity

Which biological term or result is specifically identified by the description 'Non-specific defence present from birth'?

- A. Innate immunity
- B. Variable regions
- C. Metastasis
- D. Acquired immunity

BIO-12-CH10-0016 | Challenging | Cancer and immunity disorders

During error analysis, four proposed associations are reviewed.

- I. Spread of malignant cells to distant sites corresponds to Metastasis.
- II. Uncontrolled abnormal cell division corresponds to Carcinogens.
- III. Agents capable of causing cancer corresponds to Carcinogens.
- IV. Control approach that interrupts a pathogen's vector corresponds to Metastasis.

Which set contains all and only the valid conclusions?

- A. II and IV only
- B. III only
- C. I and III only
- D. IV only

BIO-12-CH10-0017 | Hard | Viral and helminth diseases

Two conclusions must both be valid to support the reference concept 'Roundworm causing ascariasis'. Which option contains two accurate associations?

- A. Stem region interacting with immune components - Passive immunity; Immediate immunity produced by transferred antibodies - Constant Fc region
- B. Roundworm causing ascariasis - Ascaris; Opioid obtained from poppy latex - Morphine
- C. Causative bacterium of typhoid - Female Anopheles; Mosquito transmitting malaria - Salmonella typhi
- D. Cannabinoid source plant - Constant Fc region; Stem region interacting with immune components - Cannabis sativa

BIO-12-CH10-0018 | Hard | Viral and helminth diseases

A reviewer compares several entries while checking 'Virus that targets helper T lymphocytes'. Which entry must be corrected?

- A. Administration of antigenic material to induce immune memory - Vaccination
- B. Control approach that interrupts a pathogen's vector - Vector control
- C. Uncontrolled abnormal cell division - Cancer
- D. Virus that targets helper T lymphocytes - Metastasis

BIO-12-CH10-0019 | Medium | Antibody structure

A learner encounters the observation 'Stem region interacting with immune components' while analysing antibody structure. Which conclusion is supported by the prescribed biology?

- A. Constant Fc region
- B. Carcinogens
- C. Two heavy and two light chains
- D. Addiction

BIO-12-CH10-0020 | Easy | Bacterial and protozoan diseases

Which description is correctly represented by Plasmodium in this chapter?

- A. Roundworm causing ascariasis
- B. Protozoan causing malaria
- C. Agents capable of causing cancer
- D. Diagnostic test commonly used to detect HIV antibodies

BIO-12-CH10-0021 | Medium | Cancer and immunity disorders

Complete the statement accurately: Agents capable of causing cancer corresponds to _____.

- A. Carcinogens
- B. Metastasis
- C. Variable regions
- D. Y-shaped

BIO-12-CH10-0022 | Easy | Immunity

Which description is correctly represented by Acquired immunity in this chapter?

- A. Causative bacterium of typhoid
- B. Cannabinoid source plant
- C. Specific defence developing after exposure
- D. Non-specific defence present from birth

BIO-12-CH10-0023 | Challenging | Cancer and immunity disorders

A researcher checks whether four linked observations have been interpreted correctly.

- I. Agents capable of causing cancer corresponds to Carcinogens.
 - II. Spread of malignant cells to distant sites corresponds to Cancer.
 - III. Uncontrolled abnormal cell division corresponds to Cancer.
 - IV. Specific defence developing after exposure corresponds to Acquired immunity.
- Which combination survives the biological review?

- A. IV only
- B. I, II, III and IV
- C. I, III and IV only
- D. II, III and IV only

BIO-12-CH10-0024 | Easy | Viral and helminth diseases

Which biological term or result is specifically identified by the description 'Virus that targets helper T lymphocytes'?

- A. Vaccination
- B. Innate immunity
- C. Metastasis
- D. HIV

BIO-12-CH10-0025 | Medium | Cancer and immunity disorders

A learner encounters the observation 'Agents capable of causing cancer' while analysing cancer and immunity disorders. Which conclusion is supported by the prescribed biology?

- A. Metastasis
- B. ELISA
- C. Passive immunity
- D. Carcinogens

BIO-12-CH10-0026 | Easy | Viral and helminth diseases

Which description is correctly represented by Ascaris in this chapter?

- A. Cannabinoid source plant
- B. Immediate immunity produced by transferred antibodies
- C. Roundworm causing ascariasis
- D. Opioid obtained from poppy latex

BIO-12-CH10-0027 | Medium | Drugs and addiction

Complete the statement accurately: Stimulant derived from coca plant corresponds to _____.

- A. Acquired immunity
- B. Vector control
- C. Y-shaped
- D. Cocaine

BIO-12-CH10-0028 | Hard | Viral and helminth diseases

A reviewer compares several entries while checking 'Filarial worm transmitted by mosquitoes'. Which entry must be corrected?

- A. Cannabinoid source plant - Cannabis sativa
- B. Filarial worm transmitted by mosquitoes - Addiction
- C. Opioid obtained from poppy latex - Morphine
- D. Agents capable of causing cancer - Carcinogens

BIO-12-CH10-0029 | Medium | Immunity

Complete the statement accurately: Specific defence developing after exposure corresponds to _____.

- A. Active immunity
- B. Acquired immunity
- C. Salmonella typhi
- D. Vaccination

BIO-12-CH10-0030 | Hard | Bacterial and protozoan diseases

Two conclusions must both be valid to support the reference concept 'Protozoan causing malaria'. Which option contains two accurate associations?

- A. Spread of malignant cells to distant sites - Acquired immunity; Specific defence developing after exposure - Metastasis
- B. Cannabinoid source plant - ELISA; Diagnostic test commonly used to detect HIV antibodies - Cannabis sativa
- C. Protozoan causing malaria - Plasmodium; Agents capable of causing cancer - Carcinogens
- D. Non-specific defence present from birth - Cannabis sativa; Cannabinoid source plant - Innate immunity

BIO-12-CH10-0031 | Hard | Cancer and immunity disorders

A reviewer compares several entries while checking 'Specialists who study diagnosis and treatment of cancer'. Which entry must be corrected?

- A. Specialists who study diagnosis and treatment of cancer - Wuchereria
- B. Immunity produced by a person's own immune response - Active immunity
- C. Mosquito vector of dengue - Aedes
- D. Administration of antigenic material to induce immune memory - Vaccination

BIO-12-CH10-0032 | Medium | Prevention and public health

Complete the statement accurately: Administration of antigenic material to induce immune memory corresponds to _____.

- A. Vaccination
- B. Innate immunity
- C. Y-shaped
- D. ELISA

BIO-12-CH10-0033 | Hard | Immunity

A reviewer compares several entries while checking 'Non-specific defence present from birth'. Which entry must be corrected?

- A. Spread of malignant cells to distant sites - Metastasis
- B. Non-specific defence present from birth - Y-shaped
- C. Stem region interacting with immune components - Constant Fc region
- D. Uncontrolled abnormal cell division - Cancer

BIO-12-CH10-0034 | Hard | Drugs and addiction

Two conclusions must both be valid to support the reference concept 'Cannabinoid source plant'. Which option contains two accurate associations?

- A. Causative bacterium of pneumonia commonly cited in the chapter - Active immunity; Immunity produced by a person's own immune response - Streptococcus pneumoniae
- B. Cannabinoid source plant - Cannabis sativa; Basic shape of an antibody molecule - Y-shaped
- C. Uncontrolled abnormal cell division - Female Anopheles; Mosquito transmitting malaria - Cancer
- D. Protozoan causing malaria - Streptococcus pneumoniae; Causative bacterium of pneumonia commonly cited in the chapter - Plasmodium

BIO-12-CH10-0035 | Hard | Prevention and public health

Two conclusions must both be valid to support the reference concept 'Control approach that interrupts a pathogen's vector'. Which option contains two accurate associations?

- A. Cannabinoid source plant - Streptococcus pneumoniae; Causative bacterium of pneumonia commonly cited in the chapter - Cannabis sativa
- B. Causative bacterium of pneumonia commonly cited in the chapter - Cocaine; Stimulant derived from coca plant - Streptococcus pneumoniae
- C. Protozoan causing malaria - Female Anopheles; Mosquito transmitting malaria - Plasmodium
- D. Control approach that interrupts a pathogen's vector - Vector control; Uncontrolled abnormal cell division - Cancer

BIO-12-CH10-0036 | Medium | Immunity

A source table contains one accurate entry involving 'Immediate immunity produced by transferred antibodies'. Which option preserves a correct biological association?

- A. Administration of antigenic material to induce immune memory - Salmonella typhi
- B. Immediate immunity produced by transferred antibodies - Passive immunity
- C. Causative bacterium of typhoid - Cocaine
- D. Stem region interacting with immune components - Cannabis sativa

BIO-12-CH10-0037 | Easy | Bacterial and protozoan diseases

Which biological term or result is specifically identified by the description 'Protozoan causing malaria'?

- A. Cannabis sativa
- B. Carcinogens
- C. Innate immunity
- D. Plasmodium

BIO-12-CH10-0038 | Medium | Antibody structure

A source table contains one accurate entry involving 'Antigen-binding regions'. Which option preserves a correct biological association?

- A. Opioid obtained from poppy latex - Vector control
- B. Mosquito vector of dengue - Active immunity
- C. Filarial worm transmitted by mosquitoes - Variable regions
- D. Antigen-binding regions - Variable regions

BIO-12-CH10-0039 | Medium | Bacterial and protozoan diseases

Complete the statement accurately: Mosquito transmitting malaria corresponds to _____.

- A. Wuchereria
- B. Ascaris
- C. Female Anopheles
- D. Passive immunity

BIO-12-CH10-0040 | Medium | Viral and helminth diseases

A learner encounters the observation 'Filarial worm transmitted by mosquitoes' while analysing viral and helminth diseases. Which conclusion is supported by the prescribed biology?

- A. Wuchereria
- B. ELISA
- C. Vector control
- D. Carcinogens

BIO-12-CH10-0041 | Easy | Immunity

Which biological term or result is specifically identified by the description 'Immediate immunity produced by transferred antibodies'?

- A. Vaccination
- B. Acquired immunity
- C. Salmonella typhi
- D. Passive immunity

BIO-12-CH10-0042 | Easy | Prevention and public health

Which description is correctly represented by Vector control in this chapter?

- A. Control approach that interrupts a pathogen's vector
- B. Uncontrolled abnormal cell division
- C. Specific defence developing after exposure
- D. Stimulant derived from coca plant

BIO-12-CH10-0043 | Hard | Drugs and addiction

A reviewer compares several entries while checking 'Stimulant derived from coca plant'. Which entry must be corrected?

- A. Uncontrolled abnormal cell division - Cancer
- B. Immediate immunity produced by transferred antibodies - Passive immunity
- C. Stimulant derived from coca plant - Aedes
- D. Specific defence developing after exposure - Acquired immunity

BIO-12-CH10-0044 | Challenging | Antibody structure

Evaluate the following source-based biological statements.

- I. Antigen-binding regions corresponds to Variable regions.
- II. Stem region interacting with immune components corresponds to Constant Fc region.
- III. Chains present in each antibody molecule corresponds to ELISA.
- IV. Diagnostic test commonly used to detect HIV antibodies corresponds to Variable regions.

Which option identifies exactly the correct statements?

- A. II, III and IV only
- B. II only
- C. I and II only
- D. II and III only

BIO-12-CH10-0045 | Easy | Prevention and public health

Which biological term or result is specifically identified by the description 'Responsible adolescent practice that prevents substance misuse'?

- A. Acquired immunity
- B. Avoiding peer pressure and seeking guidance
- C. ELISA
- D. Carcinogens

BIO-12-CH10-0046 | Easy | Immunity

Which description is correctly represented by Innate immunity in this chapter?

- A. Non-specific defence present from birth
- B. Control approach that interrupts a pathogen's vector
- C. Basic shape of an antibody molecule
- D. Antigen-binding regions

BIO-12-CH10-0047 | Hard | Cancer and immunity disorders

Two conclusions must both be valid to support the reference concept 'Specialists who study diagnosis and treatment of cancer'. Which option contains two accurate associations?

- A. Immediate immunity produced by transferred antibodies - HIV; Virus that targets helper T lymphocytes - Passive immunity
- B. Mosquito vector of dengue - Passive immunity; Immediate immunity produced by transferred antibodies - Aedes
- C. Administration of antigenic material to induce immune memory - Active immunity; Immunity produced by a person's own immune response - Vaccination
- D. Specialists who study diagnosis and treatment of cancer - Oncologists; Filarial worm transmitted by mosquitoes - Wuchereria

BIO-12-CH10-0048 | Challenging | Bacterial and protozoan diseases

A student submits four conclusions after integrating related chapter concepts.

- I. Protozoan causing malaria corresponds to Plasmodium.
- II. Causative bacterium of typhoid corresponds to Salmonella typhi.
- III. Causative bacterium of pneumonia commonly cited in the chapter corresponds to Constant Fc region.
- IV. Stem region interacting with immune components corresponds to Constant Fc region.

Which combination should be accepted without correction?

- A. I, II and IV only
- B. II and III only
- C. I, II, III and IV
- D. I only

BIO-12-CH10-0049 | Medium | Bacterial and protozoan diseases

A source table contains one accurate entry involving 'Causative bacterium of pneumonia commonly cited in the chapter'. Which option preserves a correct biological association?

- A. Antigen-binding regions - Streptococcus pneumoniae
- B. Basic shape of an antibody molecule - Variable regions
- C. Cannabinoid source plant - Vector control
- D. Causative bacterium of pneumonia commonly cited in the chapter - Streptococcus pneumoniae

BIO-12-CH10-0050 | Hard | Bacterial and protozoan diseases

A reviewer compares several entries while checking 'Causative bacterium of pneumonia commonly cited in the chapter'. Which entry must be corrected?

- A. Control approach that interrupts a pathogen's vector - Vector control
- B. Causative bacterium of pneumonia commonly cited in the chapter - Oncologists
- C. Roundworm causing ascariasis - Ascaris
- D. Cannabinoid source plant - Cannabis sativa

BIO-12-CH10-0051 | Medium | Drugs and addiction

A source table contains one accurate entry involving 'Cannabinoid source plant'. Which option preserves a correct biological association?

- A. Causative bacterium of pneumonia commonly cited in the chapter - Cannabis sativa
- B. Cannabinoid source plant - Cannabis sativa
- C. Mosquito transmitting malaria - Plasmodium
- D. Protozoan causing malaria - Active immunity

BIO-12-CH10-0052 | Easy | Antibody structure

Which description is correctly represented by Constant Fc region in this chapter?

- A. Chains present in each antibody molecule
- B. Repeated compulsive substance use despite harm
- C. Stem region interacting with immune components
- D. Antigen-binding regions

BIO-12-CH10-0053 | Easy | Bacterial and protozoan diseases

Which biological term or result is specifically identified by the description 'Causative bacterium of pneumonia commonly cited in the chapter'?

- A. Streptococcus pneumoniae
- B. Ascaris
- C. Oncologists
- D. Plasmodium

BIO-12-CH10-0054 | Easy | Viral and helminth diseases

Which biological term or result is specifically identified by the description 'Filarial worm transmitted by mosquitoes'?

- A. ELISA
- B. Cannabis sativa
- C. Addiction
- D. Wuchereria

BIO-12-CH10-0055 | Easy | Bacterial and protozoan diseases

Which description is correctly represented by Salmonella typhi in this chapter?

- A. Causative bacterium of typhoid
- B. Antigen-binding regions
- C. Control approach that interrupts a pathogen's vector
- D. Causative bacterium of pneumonia commonly cited in the chapter

BIO-12-CH10-0056 | Easy | Prevention and public health

Which description is correctly represented by Vaccination in this chapter?

- A. Non-specific defence present from birth
- B. Repeated compulsive substance use despite harm
- C. Administration of antigenic material to induce immune memory
- D. Diagnostic test commonly used to detect HIV antibodies

BIO-12-CH10-0057 | Medium | Drugs and addiction

A learner encounters the observation 'Cannabinoid source plant' while analysing drugs and addiction. Which conclusion is supported by the prescribed biology?

- A. Morphine
- B. Y-shaped
- C. Variable regions
- D. Cannabis sativa

BIO-12-CH10-0058 | Medium | Bacterial and protozoan diseases

A source table contains one accurate entry involving 'Mosquito transmitting malaria'. Which option preserves a correct biological association?

- A. Basic shape of an antibody molecule - Ascaris
- B. Roundworm causing ascariasis - Wuchereria
- C. Mosquito transmitting malaria - Female Anopheles
- D. Administration of antigenic material to induce immune memory - Cannabis sativa

BIO-12-CH10-0059 | Easy | Viral and helminth diseases

Which description is correctly represented by Wuchereria in this chapter?

- A. Repeated compulsive substance use despite harm
- B. Virus that targets helper T lymphocytes
- C. Diagnostic test commonly used to detect HIV antibodies
- D. Filarial worm transmitted by mosquitoes

BIO-12-CH10-0060 | Hard | Antibody structure

A reviewer compares several entries while checking 'Basic shape of an antibody molecule'. Which entry must be corrected?

- A. Causative bacterium of pneumonia commonly cited in the chapter - Streptococcus pneumoniae
- B. Basic shape of an antibody molecule - Female Anopheles
- C. Immediate immunity produced by transferred antibodies - Passive immunity
- D. Repeated compulsive substance use despite harm - Addiction

BIO-12-CH10-0061 | Medium | Immunity

A learner encounters the observation 'Non-specific defence present from birth' while analysing immunity. Which conclusion is supported by the prescribed biology?

- A. Vector control
- B. Y-shaped
- C. Metastasis
- D. Innate immunity

BIO-12-CH10-0062 | Easy | Drugs and addiction

Which description is correctly represented by Addiction in this chapter?

- A. Immediate immunity produced by transferred antibodies
- B. Mosquito transmitting malaria
- C. Causative bacterium of typhoid
- D. Repeated compulsive substance use despite harm

BIO-12-CH10-0063 | Hard | Antibody structure

A reviewer compares several entries while checking 'Antigen-binding regions'. Which entry must be corrected?

- A. Antigen-binding regions - Aedes
- B. Non-specific defence present from birth - Innate immunity
- C. Control approach that interrupts a pathogen's vector - Vector control
- D. Opioid obtained from poppy latex - Morphine

BIO-12-CH10-0064 | Challenging | Prevention and public health

During error analysis, four proposed associations are reviewed.

- I. Control approach that interrupts a pathogen's vector corresponds to ELISA.
 - II. Diagnostic test commonly used to detect HIV antibodies corresponds to Vaccination.
 - III. Administration of antigenic material to induce immune memory corresponds to *Streptococcus pneumoniae*.
 - IV. Causative bacterium of pneumonia commonly cited in the chapter corresponds to *Streptococcus pneumoniae*.
- Which set contains all and only the valid conclusions?

- A. IV only
- B. I, II, III and IV
- C. II and IV only
- D. I and II only

BIO-12-CH10-0065 | Hard | Cancer and immunity disorders

A reviewer compares several entries while checking 'Spread of malignant cells to distant sites'. Which entry must be corrected?

- A. Causative bacterium of typhoid - *Salmonella typhi*
- B. Spread of malignant cells to distant sites - Cocaine
- C. Immediate immunity produced by transferred antibodies - Passive immunity
- D. Immunity produced by a person's own immune response - Active immunity

BIO-12-CH10-0066 | Medium | Cancer and immunity disorders

Complete the statement accurately: Uncontrolled abnormal cell division corresponds to _____.

- A. Cancer
- B. Female Anopheles
- C. Ascaris
- D. Constant Fc region

BIO-12-CH10-0067 | Hard | Drugs and addiction

Two conclusions must both be valid to support the reference concept 'Opioid obtained from poppy latex'. Which option contains two accurate associations?

- A. Specialists who study diagnosis and treatment of cancer - Acquired immunity; Specific defence developing after exposure - Oncologists
- B. Mosquito vector of dengue - Avoiding peer pressure and seeking guidance; Responsible adolescent practice that prevents substance misuse - *Aedes*
- C. Opioid obtained from poppy latex - Morphine; Non-specific defence present from birth - Innate immunity
- D. Immunity produced by a person's own immune response - Carcinogens; Agents capable of causing cancer - Active immunity

BIO-12-CH10-0068 | Easy | Cancer and immunity disorders

Which biological term or result is specifically identified by the description 'Spread of malignant cells to distant sites'?

- A. Vector control
- B. Metastasis
- C. Passive immunity
- D. Carcinogens

BIO-12-CH10-0069 | Easy | Antibody structure

Which biological term or result is specifically identified by the description 'Antigen-binding regions'?

- A. Variable regions
- B. *Aedes*
- C. Morphine
- D. Innate immunity

BIO-12-CH10-0070 | Easy | Cancer and immunity disorders

Which description is correctly represented by Carcinogens in this chapter?

- A. Diagnostic test commonly used to detect HIV antibodies
- B. Protozoan causing malaria
- C. Basic shape of an antibody molecule
- D. Agents capable of causing cancer

BIO-12-CH10-0071 | Medium | Cancer and immunity disorders

Complete the statement accurately: Specialists who study diagnosis and treatment of cancer corresponds to _____.

- A. Carcinogens
- B. Oncologists
- C. Active immunity
- D. Wuchereria

BIO-12-CH10-0072 | Medium | Drugs and addiction

A learner encounters the observation 'Repeated compulsive substance use despite harm' while analysing drugs and addiction. Which conclusion is supported by the prescribed biology?

- A. Addiction
- B. Salmonella typhi
- C. Cannabis sativa
- D. Morphine

BIO-12-CH10-0073 | Easy | Antibody structure

Which description is correctly represented by Two heavy and two light chains in this chapter?

- A. Repeated compulsive substance use despite harm
- B. Chains present in each antibody molecule
- C. Responsible adolescent practice that prevents substance misuse
- D. Specific defence developing after exposure

BIO-12-CH10-0074 | Medium | Drugs and addiction

Complete the statement accurately: Cannabinoid source plant corresponds to _____.

- A. Variable regions
- B. Y-shaped
- C. Female Anopheles
- D. Cannabis sativa

BIO-12-CH10-0075 | Hard | Immunity

Two conclusions must both be valid to support the reference concept 'Non-specific defence present from birth'. Which option contains two accurate associations?

- A. Spread of malignant cells to distant sites - Cancer; Uncontrolled abnormal cell division - Metastasis
- B. Uncontrolled abnormal cell division - Morphine; Opioid obtained from poppy latex - Cancer
- C. Control approach that interrupts a pathogen's vector - Variable regions; Antigen-binding regions - Vector control
- D. Non-specific defence present from birth - Innate immunity; Basic shape of an antibody molecule - Y-shaped

BIO-12-CH10-0076 | Hard | Viral and helminth diseases

Two conclusions must both be valid to support the reference concept 'Filarial worm transmitted by mosquitoes'. Which option contains two accurate associations?

- A. Filarial worm transmitted by mosquitoes - Wuchereria; Repeated compulsive substance use despite harm - Addiction
- B. Opioid obtained from poppy latex - Cancer; Uncontrolled abnormal cell division - Morphine
- C. Agents capable of causing cancer - Vector control; Control approach that interrupts a pathogen's vector - Carcinogens
- D. Control approach that interrupts a pathogen's vector - Morphine; Opioid obtained from poppy latex - Vector control

BIO-12-CH10-0077 | Medium | Viral and helminth diseases

A source table contains one accurate entry involving 'Virus that targets helper T lymphocytes'. Which option preserves a correct biological association?

- A. Administration of antigenic material to induce immune memory - Cannabis sativa
- B. Virus that targets helper T lymphocytes - HIV
- C. Causative bacterium of typhoid - HIV
- D. Spread of malignant cells to distant sites - Vector control

BIO-12-CH10-0078 | Medium | Immunity

A source table contains one accurate entry involving 'Specific defence developing after exposure'. Which option preserves a correct biological association?

- A. Agents capable of causing cancer - Addiction
- B. Mosquito transmitting malaria - Acquired immunity
- C. Administration of antigenic material to induce immune memory - Salmonella typhi
- D. Specific defence developing after exposure - Acquired immunity

BIO-12-CH10-0079 | Medium | Bacterial and protozoan diseases

Complete the statement accurately: Causative bacterium of pneumonia commonly cited in the chapter corresponds to _____.

- A. Vector control
- B. Streptococcus pneumoniae
- C. Oncologists
- D. Plasmodium

BIO-12-CH10-0080 | Medium | Cancer and immunity disorders

A learner encounters the observation 'Specialists who study diagnosis and treatment of cancer' while analysing cancer and immunity disorders. Which conclusion is supported by the prescribed biology?

- A. Vaccination
- B. Cocaine
- C. Oncologists
- D. Cancer

BIO-12-CH10-0081 | Hard | Antibody structure

Two conclusions must both be valid to support the reference concept 'Chains present in each antibody molecule'. Which option contains two accurate associations?

- A. Responsible adolescent practice that prevents substance misuse - Acquired immunity; Specific defence developing after exposure - Avoiding peer pressure and seeking guidance
- B. Chains present in each antibody molecule - Two heavy and two light chains; Stimulant derived from coca plant - Cocaine
- C. Spread of malignant cells to distant sites - ELISA; Diagnostic test commonly used to detect HIV antibodies - Metastasis
- D. Specific defence developing after exposure - Addiction; Repeated compulsive substance use despite harm - Acquired immunity

BIO-12-CH10-0082 | Medium | Drugs and addiction

A source table contains one accurate entry involving 'Stimulant derived from coca plant'. Which option preserves a correct biological association?

- A. Uncontrolled abnormal cell division - Cocaine
- B. Control approach that interrupts a pathogen's vector - Passive immunity
- C. Immediate immunity produced by transferred antibodies - Cancer
- D. Stimulant derived from coca plant - Cocaine

BIO-12-CH10-0083 | Hard | Cancer and immunity disorders

Two conclusions must both be valid to support the reference concept 'Uncontrolled abnormal cell division'. Which option contains two accurate associations?

- A. Mosquito transmitting malaria - Morphine; Opioid obtained from poppy latex - Female Anopheles
- B. Uncontrolled abnormal cell division - Cancer; Virus that targets helper T lymphocytes - HIV
- C. Stem region interacting with immune components - Female Anopheles; Mosquito transmitting malaria - Constant Fc region
- D. Administration of antigenic material to induce immune memory - Constant Fc region; Stem region interacting with immune components - Vaccination

BIO-12-CH10-0084 | Easy | Cancer and immunity disorders

Which biological term or result is specifically identified by the description 'Specialists who study diagnosis and treatment of cancer'?

- A. Carcinogens
- B. Cocaine
- C. Oncologists
- D. Wuchereria

BIO-12-CH10-0085 | Challenging | Drugs and addiction

A researcher checks whether four linked observations have been interpreted correctly.

- I. Stimulant derived from coca plant corresponds to Cocaine.
- II. Opioid obtained from poppy latex corresponds to Morphine.
- III. Repeated compulsive substance use despite harm corresponds to Addiction.
- IV. Specific defence developing after exposure corresponds to Cocaine.

Which combination survives the biological review?

- A. I, II, III and IV
- B. I, II and IV only
- C. I, II and III only
- D. III only

BIO-12-CH10-0086 | Medium | Immunity

Complete the statement accurately: Non-specific defence present from birth corresponds to _____.

- A. Constant Fc region
- B. Innate immunity
- C. Acquired immunity
- D. Y-shaped

BIO-12-CH10-0087 | Medium | Prevention and public health

A learner encounters the observation 'Administration of antigenic material to induce immune memory' while analysing prevention and public health. Which conclusion is supported by the prescribed biology?

- A. Avoiding peer pressure and seeking guidance
- B. Oncologists
- C. Vaccination
- D. Addiction

BIO-12-CH10-0088 | Challenging | Drugs and addiction

Evaluate the following source-based biological statements.

- I. Repeated compulsive substance use despite harm corresponds to Morphine.
 - II. Opioid obtained from poppy latex corresponds to Cannabis sativa.
 - III. Cannabinoid source plant corresponds to Cannabis sativa.
 - IV. Immediate immunity produced by transferred antibodies corresponds to Addiction.
- Which option identifies exactly the correct statements?

- A. I and II only
- B. II and IV only
- C. III only
- D. III and IV only

BIO-12-CH10-0089 | Medium | Viral and helminth diseases

A source table contains one accurate entry involving 'Roundworm causing ascariasis'. Which option preserves a correct biological association?

- A. Cannabinoid source plant - Passive immunity
- B. Stem region interacting with immune components - Plasmodium
- C. Protozoan causing malaria - Female Anopheles
- D. Roundworm causing ascariasis - Ascaris

BIO-12-CH10-0090 | Challenging | Viral and helminth diseases

A student submits four conclusions after integrating related chapter concepts.

- I. Roundworm causing ascariasis corresponds to Aedes.
 - II. Mosquito vector of dengue corresponds to HIV.
 - III. Virus that targets helper T lymphocytes corresponds to HIV.
 - IV. Cannabinoid source plant corresponds to Cannabis sativa.
- Which combination should be accepted without correction?

- A. III and IV only
- B. I, II and III only
- C. II, III and IV only
- D. II and IV only

BIO-12-CH10-0091 | Easy | Viral and helminth diseases

Which description is correctly represented by Aedes in this chapter?

- A. Mosquito vector of dengue
- B. Immunity produced by a person's own immune response
- C. Responsible adolescent practice that prevents substance misuse
- D. Filarial worm transmitted by mosquitoes

BIO-12-CH10-0092 | Hard | Bacterial and protozoan diseases

A reviewer compares several entries while checking 'Protozoan causing malaria'. Which entry must be corrected?

- A. Non-specific defence present from birth - Innate immunity
- B. Roundworm causing ascariasis - Ascaris
- C. Cannabinoid source plant - Cannabis sativa
- D. Protozoan causing malaria - Carcinogens

BIO-12-CH10-0093 | Medium | Prevention and public health

A learner encounters the observation 'Responsible adolescent practice that prevents substance misuse' while analysing prevention and public health. Which conclusion is supported by the prescribed biology?

- A. Avoiding peer pressure and seeking guidance
- B. Acquired immunity
- C. Metastasis
- D. ELISA

BIO-12-CH10-0094 | Medium | Prevention and public health

A source table contains one accurate entry involving 'Responsible adolescent practice that prevents substance misuse'. Which option preserves a correct biological association?

- A. Chains present in each antibody molecule - Avoiding peer pressure and seeking guidance
- B. Agents capable of causing cancer - Metastasis
- C. Responsible adolescent practice that prevents substance misuse - Avoiding peer pressure and seeking guidance
- D. Non-specific defence present from birth - Two heavy and two light chains

BIO-12-CH10-0095 | Easy | Cancer and immunity disorders

Which description is correctly represented by Cancer in this chapter?

- A. Roundworm causing ascariasis
- B. Spread of malignant cells to distant sites
- C. Uncontrolled abnormal cell division
- D. Administration of antigenic material to induce immune memory

BIO-12-CH10-0096 | Medium | Viral and helminth diseases

A source table contains one accurate entry involving 'Mosquito vector of dengue'. Which option preserves a correct biological association?

- A. Mosquito vector of dengue - Aedes
- B. Immediate immunity produced by transferred antibodies - Vaccination
- C. Causative bacterium of pneumonia commonly cited in the chapter - Aedes
- D. Stem region interacting with immune components - Active immunity

BIO-12-CH10-0097 | Hard | Prevention and public health

Two conclusions must both be valid to support the reference concept 'Administration of antigenic material to induce immune memory'. Which option contains two accurate associations?

- A. Basic shape of an antibody molecule - Oncologists; Specialists who study diagnosis and treatment of cancer - Y-shaped
- B. Non-specific defence present from birth - Cannabis sativa; Cannabinoid source plant - Innate immunity
- C. Administration of antigenic material to induce immune memory - Vaccination; Repeated compulsive substance use despite harm - Addiction
- D. Causative bacterium of typhoid - Carcinogens; Agents capable of causing cancer - Salmonella typhi

BIO-12-CH10-0098 | Medium | Prevention and public health

A learner encounters the observation 'Control approach that interrupts a pathogen's vector' while analysing prevention and public health. Which conclusion is supported by the prescribed biology?

- A. Streptococcus pneumoniae
- B. Vector control
- C. ELISA
- D. Acquired immunity

BIO-12-CH10-0099 | Easy | Cancer and immunity disorders

Which description is correctly represented by Oncologists in this chapter?

- A. Specialists who study diagnosis and treatment of cancer
- B. Immunity produced by a person's own immune response
- C. Filarial worm transmitted by mosquitoes
- D. Agents capable of causing cancer

BIO-12-CH10-0100 | Hard | Antibody structure

Two conclusions must both be valid to support the reference concept 'Stem region interacting with immune components'. Which option contains two accurate associations?

- A. Protozoan causing malaria - Active immunity; Immunity produced by a person's own immune response - Plasmodium
- B. Immunity produced by a person's own immune response - Female Anopheles; Mosquito transmitting malaria - Active immunity
- C. Repeated compulsive substance use despite harm - Oncologists; Specialists who study diagnosis and treatment of cancer - Addiction
- D. Stem region interacting with immune components - Constant Fc region; Agents capable of causing cancer - Carcinogens

BIO-12-CH10-0101 | Hard | Bacterial and protozoan diseases

Two conclusions must both be valid to support the reference concept 'Causative bacterium of typhoid'. Which option contains two accurate associations?

- A. Causative bacterium of typhoid - Salmonella typhi; Diagnostic test commonly used to detect HIV antibodies - ELISA
- B. Responsible adolescent practice that prevents substance misuse - Variable regions; Antigen-binding regions - Avoiding peer pressure and seeking guidance
- C. Control approach that interrupts a pathogen's vector - Passive immunity; Immediate immunity produced by transferred antibodies - Vector control
- D. Antigen-binding regions - HIV; Virus that targets helper T lymphocytes - Variable regions

BIO-12-CH10-0102 | Hard | Prevention and public health

A reviewer compares several entries while checking 'Responsible adolescent practice that prevents substance misuse'. Which entry must be corrected?

- A. Virus that targets helper T lymphocytes - HIV
- B. Responsible adolescent practice that prevents substance misuse - Acquired immunity
- C. Chains present in each antibody molecule - Two heavy and two light chains
- D. Non-specific defence present from birth - Innate immunity

BIO-12-CH10-0103 | Hard | Immunity

A reviewer compares several entries while checking 'Immediate immunity produced by transferred antibodies'. Which entry must be corrected?

- A. Administration of antigenic material to induce immune memory - Vaccination
- B. Immediate immunity produced by transferred antibodies - Carcinogens
- C. Specialists who study diagnosis and treatment of cancer - Oncologists
- D. Causative bacterium of typhoid - Salmonella typhi

BIO-12-CH10-0104 | Easy | Drugs and addiction

Which biological term or result is specifically identified by the description 'Stimulant derived from coca plant'?

- A. Cocaine
- B. Morphine
- C. Acquired immunity
- D. Cannabis sativa

BIO-12-CH10-0105 | Hard | Prevention and public health

A reviewer compares several entries while checking 'Administration of antigenic material to induce immune memory'. Which entry must be corrected?

- A. Specialists who study diagnosis and treatment of cancer - Oncologists
- B. Cannabinoid source plant - Cannabis sativa
- C. Administration of antigenic material to induce immune memory - Addiction
- D. Causative bacterium of typhoid - Salmonella typhi

BIO-12-CH10-0106 | Medium | Cancer and immunity disorders

A learner encounters the observation 'Uncontrolled abnormal cell division' while analysing cancer and immunity disorders. Which conclusion is supported by the prescribed biology?

- A. Oncologists
- B. Metastasis
- C. Vaccination
- D. Cancer

BIO-12-CH10-0107 | Medium | Antibody structure

A source table contains one accurate entry involving 'Stem region interacting with immune components'. Which option preserves a correct biological association?

- A. Mosquito transmitting malaria - Constant Fc region
- B. Stem region interacting with immune components - Constant Fc region
- C. Protozoan causing malaria - Female Anopheles
- D. Agents capable of causing cancer - Oncologists

BIO-12-CH10-0108 | Hard | Immunity

Two conclusions must both be valid to support the reference concept 'Specific defence developing after exposure'. Which option contains two accurate associations?

- A. Causative bacterium of typhoid - Cannabis sativa; Cannabinoid source plant - Salmonella typhi
- B. Repeated compulsive substance use despite harm - Salmonella typhi; Causative bacterium of typhoid - Addiction
- C. Specific defence developing after exposure - Acquired immunity; Agents capable of causing cancer - Carcinogens
- D. Administration of antigenic material to induce immune memory - Addiction; Repeated compulsive substance use despite harm - Vaccination

BIO-12-CH10-0109 | Easy | Prevention and public health

Which biological term or result is specifically identified by the description 'Administration of antigenic material to induce immune memory'?

- A. Y-shaped
- B. Vaccination
- C. Innate immunity
- D. Avoiding peer pressure and seeking guidance

BIO-12-CH10-0110 | Medium | Prevention and public health

A learner encounters the observation 'Diagnostic test commonly used to detect HIV antibodies' while analysing prevention and public health. Which conclusion is supported by the prescribed biology?

- A. Avoiding peer pressure and seeking guidance
- B. Plasmodium
- C. ELISA
- D. Carcinogens

BIO-12-CH10-0111 | Medium | Antibody structure

A source table contains one accurate entry involving 'Chains present in each antibody molecule'. Which option preserves a correct biological association?

- A. Repeated compulsive substance use despite harm - ELISA
- B. Mosquito transmitting malaria - Acquired immunity
- C. Chains present in each antibody molecule - Two heavy and two light chains
- D. Responsible adolescent practice that prevents substance misuse - Addiction

BIO-12-CH10-0112 | Medium | Antibody structure

A learner encounters the observation 'Antigen-binding regions' while analysing antibody structure. Which conclusion is supported by the prescribed biology?

- A. Aedes
- B. Variable regions
- C. Innate immunity
- D. Active immunity

BIO-12-CH10-0113 | Easy | Antibody structure

Which biological term or result is specifically identified by the description 'Basic shape of an antibody molecule'?

- A. Streptococcus pneumoniae
- B. Addiction
- C. Y-shaped
- D. Constant Fc region

BIO-12-CH10-0114 | Hard | Drugs and addiction

Two conclusions must both be valid to support the reference concept 'Repeated compulsive substance use despite harm'. Which option contains two accurate associations?

- A. Repeated compulsive substance use despite harm - Addiction; Causative bacterium of typhoid - Salmonella typhi
- B. Specialists who study diagnosis and treatment of cancer - Passive immunity; Immediate immunity produced by transferred antibodies - Oncologists
- C. Immediate immunity produced by transferred antibodies - Female Anopheles; Mosquito transmitting malaria - Passive immunity
- D. Agents capable of causing cancer - ELISA; Diagnostic test commonly used to detect HIV antibodies - Carcinogens

BIO-12-CH10-0115 | Medium | Viral and helminth diseases

Complete the statement accurately: Virus that targets helper T lymphocytes corresponds to _____.

- A. Vaccination
- B. Aedes
- C. Innate immunity
- D. HIV

BIO-12-CH10-0116 | Medium | Antibody structure

Complete the statement accurately: Chains present in each antibody molecule corresponds to _____.

- A. Two heavy and two light chains
- B. Avoiding peer pressure and seeking guidance
- C. Constant Fc region
- D. Acquired immunity

BIO-12-CH10-0117 | Medium | Bacterial and protozoan diseases

A learner encounters the observation 'Causative bacterium of pneumonia commonly cited in the chapter' while analysing bacterial and protozoan diseases. Which conclusion is supported by the prescribed biology?

- A. Y-shaped
- B. Vector control
- C. Streptococcus pneumoniae
- D. Female Anopheles

BIO-12-CH10-0118 | Medium | Viral and helminth diseases

A learner encounters the observation 'Mosquito vector of dengue' while analysing viral and helminth diseases. Which conclusion is supported by the prescribed biology?

- A. Aedes
 - B. Ascaris
 - C. Constant Fc region
 - D. Passive immunity
-

BIO-12-CH10-0119 | Medium | Immunity

A source table contains one accurate entry involving 'Non-specific defence present from birth'. Which option preserves a correct biological association?

- A. Basic shape of an antibody molecule - Variable regions
 - B. Antigen-binding regions - Metastasis
 - C. Non-specific defence present from birth - Innate immunity
 - D. Uncontrolled abnormal cell division - Innate immunity
-

BIO-12-CH10-0120 | Hard | Cancer and immunity disorders

Two conclusions must both be valid to support the reference concept 'Agents capable of causing cancer'. Which option contains two accurate associations?

- A. Virus that targets helper T lymphocytes - Morphine; Opioid obtained from poppy latex - HIV
 - B. Agents capable of causing cancer - Carcinogens; Immediate immunity produced by transferred antibodies - Passive immunity
 - C. Antigen-binding regions - Plasmodium; Protozoan causing malaria - Variable regions
 - D. Diagnostic test commonly used to detect HIV antibodies - HIV; Virus that targets helper T lymphocytes - ELISA
-

Standard 12 | Chapter 11

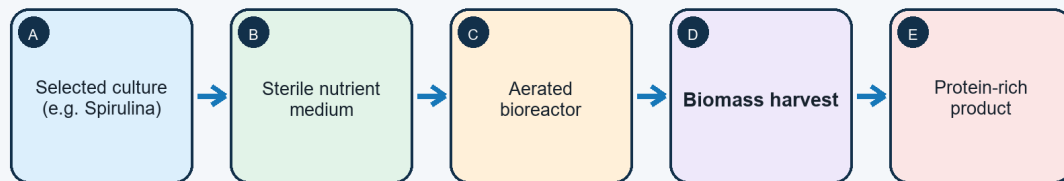
Enhancement of Food Production

BIO-12-CH11-0001 | Medium | Single-cell protein production

Which organism shown is widely used as a single-cell protein source?

Single-cell protein production

Controlled biomass production in a bioreactor



Contamination or poor aeration reduces culture productivity and product consistency.

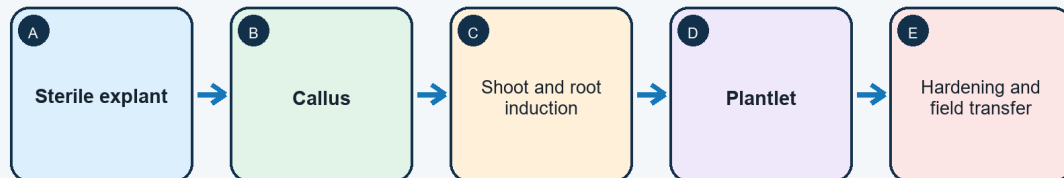
- A. Spirulina
- B. Plasmodium
- C. Taenia
- D. Rhizopus causing bread spoilage only

BIO-12-CH11-0002 | Hard | Plant micropropagation

Which culture variable most directly shifts organogenesis between shoot and root formation?

Plant micropropagation

Clonal plants produced through tissue culture



The auxin-to-cytokinin balance influences root and shoot organogenesis.

- A. Blood glucose concentration
- B. Atmospheric noble-gas ratio
- C. Antibody titre
- D. Auxin-to-cytokinin ratio

BIO-12-CH11-0003 | Hard | Plant breeding objectives

Two conclusions must both be valid to support the reference concept 'Improvement of protein, oil, vitamin or mineral content'. Which option contains two accurate associations?

- A. Improvement of protein, oil, vitamin or mineral content - Quality improvement; Fungal antagonist used against plant pathogens - Trichoderma
- B. Microbe producing citric acid - Bacillus thuringiensis; Bacterial biocontrol agent producing insecticidal crystal proteins - Aspergillus niger
- C. Bacterial biocontrol agent producing insecticidal crystal proteins - Floccs; Microbial aggregates in aeration tanks - Bacillus thuringiensis
- D. Microbial aggregates in aeration tanks - Penicillium; Fungus associated with penicillin production - Floccs

BIO-12-CH11-0004 | Hard | Tissue culture

Two conclusions must both be valid to support the reference concept 'Small tissue piece used to start a culture'. Which option contains two accurate associations?

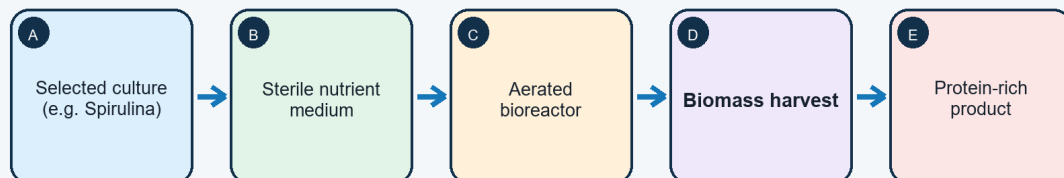
- A. Methanogen commonly found in cattle rumen - Multi-location evaluation; Testing a new line at several places before release - Methanobacterium
- B. Testing parents for desirable characters - Quality improvement; Improvement of protein, oil, vitamin or mineral content - Evaluation and selection
- C. Development of crops tolerating drought or salinity - Evaluation and selection; Testing parents for desirable characters - Abiotic stress resistance
- D. Small tissue piece used to start a culture - Explant; Development of crop varieties with higher output - Increased yield

BIO-12-CH11-0005 | Challenging | Single-cell protein production

If sterility and aeration fail simultaneously in an SCP bioreactor, which outcome is most likely?

Single-cell protein production

Controlled biomass production in a bioreactor



Contamination or poor aeration reduces culture productivity and product consistency.

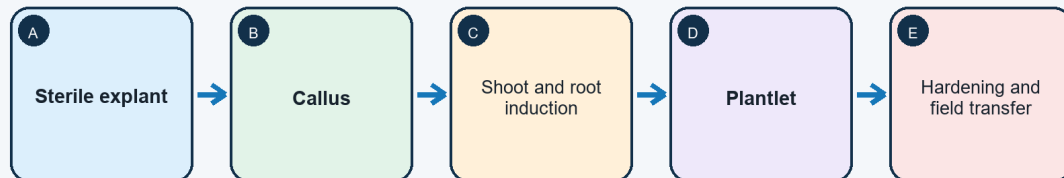
- A. Pure-culture yield necessarily doubles
- B. Contamination rises and useful biomass yield falls
- C. The culture becomes genetically identical to a plantlet
- D. Protein production becomes independent of nutrients

BIO-12-CH11-0006 | Easy | Plant micropropagation

Which starting material is placed on sterile culture medium in micropropagation?

Plant micropropagation

Clonal plants produced through tissue culture



The auxin-to-cytokinin balance influences root and shoot organogenesis.

- A. Mature pollen only
- B. Explant
- C. Whole field soil
- D. Animal plasma

BIO-12-CH11-0007 | Easy | Industrial microbes

Which description is correctly represented by *Saccharomyces cerevisiae* in this chapter?

- A. Fungal antagonist used against plant pathogens
- B. Microbial biomass used as protein-rich food or feed
- C. Scientific management of farm animals
- D. Yeast used in ethanol production

BIO-12-CH11-0008 | Hard | Biofortification and SCP

Two conclusions must both be valid to support the reference concept 'Breeding crops for improved nutritional quality'. Which option contains two accurate associations?

- A. Microbial aggregates in aeration tanks - Dairy farming; Management of milk-producing animals - Flocs
- B. Breeding crops for improved nutritional quality - Biofortification; Breeding for resistance to pathogens - Disease resistance
- C. Management of milk-producing animals - *Lactobacillus*; Bacterium producing lactic acid - Dairy farming
- D. Symbiotic nitrogen fixer in legume roots - Flocs; Microbial aggregates in aeration tanks - *Rhizobium*

BIO-12-CH11-0009 | Medium | Biofortification and SCP

Complete the statement accurately: Alga widely used as single-cell protein corresponds to _____.

- A. Biofortification
- B. Fortified maize
- C. *Spirulina*
- D. Apiculture

BIO-12-CH11-0010 | Easy | Plant breeding process

Which biological term or result is specifically identified by the description 'Crossing genetically distinct selected parents'?

- A. Callus
- B. Increased yield
- C. Fortified maize
- D. Hybridization

BIO-12-CH11-0011 | Medium | Plant breeding process

A learner encounters the observation 'Testing parents for desirable characters' while analysing plant breeding process. Which conclusion is supported by the prescribed biology?

- A. Evaluation and selection
- B. *Methanobacterium*
- C. Biofortification
- D. Explant

BIO-12-CH11-0012 | Easy | Plant breeding objectives

Which biological term or result is specifically identified by the description 'Development of crops tolerating drought or salinity'?

- A. Quality improvement
- B. Totipotency
- C. Abiotic stress resistance
- D. Flocs

BIO-12-CH11-0013 | Challenging | Tissue culture

A researcher checks whether four linked observations have been interpreted correctly.

- I. Unorganized mass of cultured cells corresponds to Callus.
- II. Small tissue piece used to start a culture corresponds to Explant.
- III. Rapid clonal production through tissue culture corresponds to Biogas.
- IV. Anaerobic digesters produce methane-rich corresponds to Biogas.

Which combination survives the biological review?

- A. I, II and IV only
- B. II and III only
- C. IV only
- D. I, III and IV only

BIO-12-CH11-0014 | Easy | Plant breeding objectives

Which description is correctly represented by Quality improvement in this chapter?

- A. Unorganized mass of cultured cells
- B. Improvement of protein, oil, vitamin or mineral content
- C. Breeding crops for improved nutritional quality
- D. Fungal antagonist used against plant pathogens

BIO-12-CH11-0015 | Easy | Biocontrol and biofertilizers

Which biological term or result is specifically identified by the description 'Bacterial biocontrol agent producing insecticidal crystal proteins'?

- A. Micropropagation
- B. Trichoderma
- C. *Bacillus thuringiensis*
- D. Mycorrhiza

BIO-12-CH11-0016 | Medium | Biocontrol and biofertilizers

Complete the statement accurately: Association of fungi with plant roots improving mineral uptake corresponds to _____.

- A. Germplasm collection
- B. Mycorrhiza
- C. *Rhizobium*
- D. Fortified maize

BIO-12-CH11-0017 | Hard | Animal husbandry

A reviewer compares several entries while checking 'Rearing of honey bees'. Which entry must be corrected?

- A. Development of crops tolerating drought or salinity - Abiotic stress resistance
- B. Rearing of honey bees - Hybridization
- C. Improvement of protein, oil, vitamin or mineral content - Quality improvement
- D. Indicator of organic pollution reduced during secondary treatment - Biochemical oxygen demand

BIO-12-CH11-0018 | Easy | Tissue culture

Which biological term or result is specifically identified by the description 'Unorganized mass of cultured cells'?

- A. Callus
- B. *Rhizobium*
- C. Totipotency
- D. Micropropagation

BIO-12-CH11-0019 | Medium | Biofortification and SCP

A source table contains one accurate entry involving 'Alga widely used as single-cell protein'. Which option preserves a correct biological association?

- A. Management of milk-producing animals - Rhizobium
- B. Symbiotic nitrogen fixer in legume roots - Apiculture
- C. Indicator of organic pollution reduced during secondary treatment - Poultry farming
- D. Alga widely used as single-cell protein - Spirulina

BIO-12-CH11-0020 | Challenging | Plant breeding process

Evaluate the following source-based biological statements.

- I. Testing parents for desirable characters corresponds to Multi-location evaluation.
- II. Testing a new line at several places before release corresponds to Hybridization.
- III. Crossing genetically distinct selected parents corresponds to Abiotic stress resistance.
- IV. Development of crops tolerating drought or salinity corresponds to Abiotic stress resistance.

Which option identifies exactly the correct statements?

- A. I, III and IV only
- B. II and III only
- C. IV only
- D. I only

BIO-12-CH11-0021 | Medium | Biocontrol and biofertilizers

Complete the statement accurately: Symbiotic nitrogen fixer in legume roots corresponds to _____.

- A. Multi-location evaluation
- B. Explant
- C. Mycorrhiza
- D. Rhizobium

BIO-12-CH11-0022 | Easy | Sewage and biogas

Which description is correctly represented by Flocs in this chapter?

- A. Crossing genetically distinct selected parents
- B. Indicator of organic pollution reduced during secondary treatment
- C. Breeding for resistance to pathogens
- D. Microbial aggregates in aeration tanks

BIO-12-CH11-0023 | Medium | Industrial microbes

A source table contains one accurate entry involving 'Fungus associated with penicillin production'. Which option preserves a correct biological association?

- A. Management of milk-producing animals - Penicillium
- B. Microbial aggregates in aeration tanks - Germplasm collection
- C. Fungus associated with penicillin production - Penicillium
- D. Rearing of honey bees - Explant

BIO-12-CH11-0024 | Hard | Tissue culture

A reviewer compares several entries while checking 'Rapid clonal production through tissue culture'. Which entry must be corrected?

- A. Methanogen commonly found in cattle rumen - Methanobacterium
- B. Rapid clonal production through tissue culture - Disease resistance
- C. Microbial biomass used as protein-rich food or feed - Single-cell protein
- D. Bacterial biocontrol agent producing insecticidal crystal proteins - Bacillus thuringiensis

BIO-12-CH11-0025 | Medium | Sewage and biogas

Complete the statement accurately: Microbial aggregates in aeration tanks corresponds to _____.

- A. Rhizobium
- B. Flocs
- C. Quality improvement
- D. Methanobacterium

BIO-12-CH11-0026 | Medium | Plant breeding process

A learner encounters the observation 'Testing a new line at several places before release' while analysing plant breeding process. Which conclusion is supported by the prescribed biology?

- A. Germplasm collection
- B. Biofortification
- C. Hybridization
- D. Multi-location evaluation

BIO-12-CH11-0027 | Medium | Plant breeding objectives

A learner encounters the observation 'Development of crop varieties with higher output' while analysing plant breeding objectives. Which conclusion is supported by the prescribed biology?

- A. Penicillium
- B. Increased yield
- C. *Aspergillus niger*
- D. Single-cell protein

BIO-12-CH11-0028 | Hard | Biocontrol and biofertilizers

A reviewer compares several entries while checking 'Bacterial biocontrol agent producing insecticidal crystal proteins'. Which entry must be corrected?

- A. Microbial biomass used as protein-rich food or feed - Single-cell protein
- B. Rapid clonal production through tissue culture - Micropropagation
- C. Small tissue piece used to start a culture - Explant
- D. Bacterial biocontrol agent producing insecticidal crystal proteins - Poultry farming

BIO-12-CH11-0029 | Medium | Industrial microbes

A learner encounters the observation 'Microbe producing citric acid' while analysing industrial microbes. Which conclusion is supported by the prescribed biology?

- A. Dairy farming
- B. *Lactobacillus*
- C. Animal husbandry
- D. *Aspergillus niger*

BIO-12-CH11-0030 | Medium | Plant breeding process

A learner encounters the observation 'Collection of diverse cultivated and wild forms' while analysing plant breeding process. Which conclusion is supported by the prescribed biology?

- A. Hybridization
- B. Biochemical oxygen demand
- C. Germplasm collection
- D. Methanobacterium

BIO-12-CH11-0031 | Medium | Sewage and biogas

A source table contains one accurate entry involving 'Indicator of organic pollution reduced during secondary treatment'. Which option preserves a correct biological association?

- A. Indicator of organic pollution reduced during secondary treatment - Biochemical oxygen demand
- B. Fungus associated with penicillin production - Biochemical oxygen demand
- C. Testing parents for desirable characters - Animal husbandry
- D. Scientific management of farm animals - Hybridization

BIO-12-CH11-0032 | Medium | Sewage and biogas

A learner encounters the observation 'Anaerobic digesters produce methane-rich' while analysing sewage and biogas. Which conclusion is supported by the prescribed biology?

- A. Biofortification
- B. Evaluation and selection
- C. Biogas
- D. Hybridization

BIO-12-CH11-0033 | Medium | Animal husbandry

Complete the statement accurately: Rearing of birds for eggs and meat corresponds to _____.

- A. Poultry farming
- B. Flocs
- C. Animal husbandry
- D. Mycorrhiza

BIO-12-CH11-0034 | Medium | Plant breeding process

A learner encounters the observation 'Crossing genetically distinct selected parents' while analysing plant breeding process. Which conclusion is supported by the prescribed biology?

- A. Biogas
- B. Fortified maize
- C. Germplasm collection
- D. Hybridization

BIO-12-CH11-0035 | Medium | Animal husbandry

Complete the statement accurately: Scientific management of farm animals corresponds to _____.

- A. Mycorrhiza
- B. Biochemical oxygen demand
- C. Animal husbandry
- D. Apiculture

BIO-12-CH11-0036 | Easy | Biofortification and SCP

Which biological term or result is specifically identified by the description 'Microbial biomass used as protein-rich food or feed'?

- A. Hybridization
- B. Single-cell protein
- C. Fortified maize
- D. *Saccharomyces cerevisiae*

BIO-12-CH11-0037 | Easy | Tissue culture

Which description is correctly represented by Explant in this chapter?

- A. Methanogen commonly found in cattle rumen
- B. Unorganized mass of cultured cells
- C. Improvement of protein, oil, vitamin or mineral content
- D. Small tissue piece used to start a culture

BIO-12-CH11-0038 | Medium | Tissue culture

A learner encounters the observation 'Unorganized mass of cultured cells' while analysing tissue culture. Which conclusion is supported by the prescribed biology?

- A. Micropropagation
- B. Callus
- C. *Saccharomyces cerevisiae*
- D. Totipotency

BIO-12-CH11-0039 | Hard | Plant breeding objectives

A reviewer compares several entries while checking 'Development of crops tolerating drought or salinity'. Which entry must be corrected?

- A. Association of fungi with plant roots improving mineral uptake - Mycorrhiza
- B. Alga widely used as single-cell protein - Spirulina
- C. Development of crops tolerating drought or salinity - Totipotency
- D. Symbiotic nitrogen fixer in legume roots - Rhizobium

BIO-12-CH11-0040 | Hard | Sewage and biogas

Two conclusions must both be valid to support the reference concept 'Microbial aggregates in aeration tanks'. Which option contains two accurate associations?

- A. Microbial aggregates in aeration tanks - Flocs; Bacterium producing lactic acid - *Lactobacillus*
- B. Breeding for resistance to pathogens - Quality improvement; Improvement of protein, oil, vitamin or mineral content - Disease resistance
- C. Symbiotic nitrogen fixer in legume roots - Disease resistance; Breeding for resistance to pathogens - Rhizobium
- D. Crossing genetically distinct selected parents - Rhizobium; Symbiotic nitrogen fixer in legume roots - Hybridization

BIO-12-CH11-0041 | Easy | Animal husbandry

Which biological term or result is specifically identified by the description 'Management of milk-producing animals'?

- A. Germplasm collection
- B. Apiculture
- C. Dairy farming
- D. Quality improvement

BIO-12-CH11-0042 | Easy | Industrial microbes

Which description is correctly represented by *Lactobacillus* in this chapter?

- A. Bacterium producing lactic acid
- B. Yeast used in ethanol production
- C. Management of milk-producing animals
- D. Fungal antagonist used against plant pathogens

BIO-12-CH11-0043 | Challenging | Sewage and biogas

A student submits four conclusions after integrating related chapter concepts.

- I. Microbial aggregates in aeration tanks corresponds to Flocs.
- II. Anaerobic digesters produce methane-rich corresponds to Biogas.
- III. Methanogen commonly found in cattle rumen corresponds to *Methanobacterium*.
- IV. Scientific management of farm animals corresponds to Flocs.

Which combination should be accepted without correction?

- A. II and III only
- B. I and IV only
- C. I only
- D. I, II and III only

BIO-12-CH11-0044 | Medium | Plant breeding objectives

A source table contains one accurate entry involving 'Development of crop varieties with higher output'. Which option preserves a correct biological association?

- A. Development of crop varieties with higher output - Increased yield
- B. Microbe producing citric acid - Multi-location evaluation
- C. Fungus associated with penicillin production - Explant
- D. Symbiotic nitrogen fixer in legume roots - Mycorrhiza

BIO-12-CH11-0045 | Medium | Tissue culture

A learner encounters the observation 'Rapid clonal production through tissue culture' while analysing tissue culture. Which conclusion is supported by the prescribed biology?

- A. Micropropagation
- B. Disease resistance
- C. Methanobacterium
- D. Floccs

BIO-12-CH11-0046 | Hard | Industrial microbes

Two conclusions must both be valid to support the reference concept 'Fungus associated with penicillin production'. Which option contains two accurate associations?

- A. Small tissue piece used to start a culture - Germplasm collection; Collection of diverse cultivated and wild forms - Explant
- B. Rearing of honey bees - Floccs; Microbial aggregates in aeration tanks - Apiculture
- C. Fungus associated with penicillin production - Penicillium; Testing parents for desirable characters - Evaluation and selection
- D. Management of milk-producing animals - Rhizobium; Symbiotic nitrogen fixer in legume roots - Dairy farming

BIO-12-CH11-0047 | Easy | Biocontrol and biofertilizers

Which description is correctly represented by Rhizobium in this chapter?

- A. Testing a new line at several places before release
- B. Symbiotic nitrogen fixer in legume roots
- C. Fungal antagonist used against plant pathogens
- D. Association of fungi with plant roots improving mineral uptake

BIO-12-CH11-0048 | Hard | Biofortification and SCP

Two conclusions must both be valid to support the reference concept 'Alga widely used as single-cell protein'. Which option contains two accurate associations?

- A. Rearing of birds for eggs and meat - Multi-location evaluation; Testing a new line at several places before release - Poultry farming
- B. Rearing of honey bees - Poultry farming; Rearing of birds for eggs and meat - Apiculture
- C. Symbiotic nitrogen fixer in legume roots - Biochemical oxygen demand; Indicator of organic pollution reduced during secondary treatment - Rhizobium
- D. Alga widely used as single-cell protein - Spirulina; Management of milk-producing animals - Dairy farming

BIO-12-CH11-0049 | Easy | Plant breeding objectives

Which description is correctly represented by Increased yield in this chapter?

- A. Development of crop varieties with higher output
- B. Microbe producing citric acid
- C. Symbiotic nitrogen fixer in legume roots
- D. Fungus associated with penicillin production

BIO-12-CH11-0050 | Hard | Animal husbandry

A reviewer compares several entries while checking 'Management of milk-producing animals'. Which entry must be corrected?

- A. Collection of diverse cultivated and wild forms - Germplasm collection
- B. Management of milk-producing animals - Quality improvement
- C. Rapid clonal production through tissue culture - Micropropagation
- D. Association of fungi with plant roots improving mineral uptake - Mycorrhiza

BIO-12-CH11-0051 | Easy | Plant breeding process

Which biological term or result is specifically identified by the description 'Testing parents for desirable characters'?

- A. Rhizobium
- B. Evaluation and selection
- C. *Aspergillus niger*
- D. Explant

BIO-12-CH11-0052 | Medium | Plant breeding process

A source table contains one accurate entry involving 'Collection of diverse cultivated and wild forms'. Which option preserves a correct biological association?

- A. Collection of diverse cultivated and wild forms - Germplasm collection
- B. Management of milk-producing animals - Totipotency
- C. Microbial aggregates in aeration tanks - Germplasm collection
- D. Scientific management of farm animals - Biochemical oxygen demand

BIO-12-CH11-0053 | Easy | Animal husbandry

Which biological term or result is specifically identified by the description 'Rearing of honey bees'?

- A. Dairy farming
- B. *Bacillus thuringiensis*
- C. Apiculture
- D. Biochemical oxygen demand

BIO-12-CH11-0054 | Medium | Biocontrol and biofertilizers

A learner encounters the observation 'Bacterial biocontrol agent producing insecticidal crystal proteins' while analysing biocontrol and biofertilizers. Which conclusion is supported by the prescribed biology?

- A. Mycorrhiza
- B. *Bacillus thuringiensis*
- C. Explant
- D. Micropropagation

BIO-12-CH11-0055 | Challenging | Biocontrol and biofertilizers

During error analysis, four proposed associations are reviewed.

- I. Fungal antagonist used against plant pathogens corresponds to *Bacillus thuringiensis*.
- II. Bacterial biocontrol agent producing insecticidal crystal proteins corresponds to *Rhizobium*.
- III. Symbiotic nitrogen fixer in legume roots corresponds to *Rhizobium*.
- IV. Breeding for resistance to pathogens corresponds to *Trichoderma*.

Which set contains all and only the valid conclusions?

- A. III and IV only
- B. I and IV only
- C. III only
- D. I and II only

BIO-12-CH11-0056 | Hard | Plant breeding objectives

Two conclusions must both be valid to support the reference concept 'Development of crop varieties with higher output'. Which option contains two accurate associations?

- A. Microbial biomass used as protein-rich food or feed - Mycorrhiza; Association of fungi with plant roots improving mineral uptake - Single-cell protein
- B. Fungus associated with penicillin production - *Aspergillus niger*; Microbe producing citric acid - *Penicillium*
- C. Microbe producing citric acid - Explant; Small tissue piece used to start a culture - *Aspergillus niger*
- D. Development of crop varieties with higher output - Increased yield; Symbiotic nitrogen fixer in legume roots - *Rhizobium*

BIO-12-CH11-0057 | Easy | Tissue culture

Which biological term or result is specifically identified by the description 'Rapid clonal production through tissue culture'?

- A. Single-cell protein
- B. Hybridization
- C. Micropropagation
- D. Callus

BIO-12-CH11-0058 | Hard | Sewage and biogas

Two conclusions must both be valid to support the reference concept 'Methanogen commonly found in cattle rumen'. Which option contains two accurate associations?

- A. Rearing of honey bees - Abiotic stress resistance; Development of crops tolerating drought or salinity - Apiculture
- B. Methanogen commonly found in cattle rumen - *Methanobacterium*; Microbe producing citric acid - *Aspergillus niger*
- C. Development of crops tolerating drought or salinity - Single-cell protein; Microbial biomass used as protein-rich food or feed - Abiotic stress resistance
- D. Microbial biomass used as protein-rich food or feed - Hybridization; Crossing genetically distinct selected parents - Single-cell protein

BIO-12-CH11-0059 | Easy | Biocontrol and biofertilizers

Which biological term or result is specifically identified by the description 'Association of fungi with plant roots improving mineral uptake'?

- A. *Rhizobium*
- B. Fortified maize
- C. Mycorrhiza
- D. *Bacillus thuringiensis*

BIO-12-CH11-0060 | Medium | Industrial microbes

A source table contains one accurate entry involving 'Microbe producing citric acid'. Which option preserves a correct biological association?

- A. Microbial aggregates in aeration tanks - Micropropagation
- B. Biofortified crop with twice the amount of lysine and tryptophan - Evaluation and selection
- C. Indicator of organic pollution reduced during secondary treatment - Dairy farming
- D. Microbe producing citric acid - *Aspergillus niger*

BIO-12-CH11-0061 | Medium | Biocontrol and biofertilizers

A learner encounters the observation 'Fungal antagonist used against plant pathogens' while analysing biocontrol and biofertilizers. Which conclusion is supported by the prescribed biology?

- A. *Spirulina*
- B. *Bacillus thuringiensis*
- C. *Trichoderma*
- D. *Lactobacillus*

BIO-12-CH11-0062 | Medium | Tissue culture

A learner encounters the observation 'Capacity of a plant cell to regenerate a whole plant' while analysing tissue culture. Which conclusion is supported by the prescribed biology?

- A. Totipotency
- B. Disease resistance
- C. Animal husbandry
- D. Flocs

BIO-12-CH11-0063 | Medium | Sewage and biogas

Complete the statement accurately: Methanogen commonly found in cattle rumen corresponds to _____.

- A. Biofortification
- B. Apiculture
- C. Methanobacterium
- D. Evaluation and selection

BIO-12-CH11-0064 | Medium | Tissue culture

A learner encounters the observation 'Small tissue piece used to start a culture' while analysing tissue culture. Which conclusion is supported by the prescribed biology?

- A. Explant
- B. Quality improvement
- C. Methanobacterium
- D. Abiotic stress resistance

BIO-12-CH11-0065 | Hard | Plant breeding process

A reviewer compares several entries while checking 'Testing parents for desirable characters'. Which entry must be corrected?

- A. Testing parents for desirable characters - Biofortification
- B. Symbiotic nitrogen fixer in legume roots - Rhizobium
- C. Methanogen commonly found in cattle rumen - Methanobacterium
- D. Microbe producing citric acid - Aspergillus niger

BIO-12-CH11-0066 | Easy | Plant breeding process

Which description is correctly represented by Evaluation and selection in this chapter?

- A. Methanogen commonly found in cattle rumen
- B. Crossing genetically distinct selected parents
- C. Microbe producing citric acid
- D. Testing parents for desirable characters

BIO-12-CH11-0067 | Hard | Plant breeding process

Two conclusions must both be valid to support the reference concept 'Testing parents for desirable characters'. Which option contains two accurate associations?

- A. Testing parents for desirable characters - Evaluation and selection; Breeding crops for improved nutritional quality - Biofortification
- B. Fungal antagonist used against plant pathogens - Quality improvement; Improvement of protein, oil, vitamin or mineral content - Trichoderma
- C. Microbe producing citric acid - Rhizobium; Symbiotic nitrogen fixer in legume roots - Aspergillus niger
- D. Methanogen commonly found in cattle rumen - Explant; Small tissue piece used to start a culture - Methanobacterium

BIO-12-CH11-0068 | Hard | Animal husbandry

Two conclusions must both be valid to support the reference concept 'Management of milk-producing animals'. Which option contains two accurate associations?

- A. Fungal antagonist used against plant pathogens - Rhizobium; Symbiotic nitrogen fixer in legume roots - Trichoderma
- B. Association of fungi with plant roots improving mineral uptake - Micropropagation; Rapid clonal production through tissue culture - Mycorrhiza
- C. Rapid clonal production through tissue culture - Lactobacillus; Bacterium producing lactic acid - Micropropagation
- D. Management of milk-producing animals - Dairy farming; Improvement of protein, oil, vitamin or mineral content - Quality improvement

BIO-12-CH11-0069 | Easy | Biofortification and SCP

Which description is correctly represented by Spirulina in this chapter?

- A. Alga widely used as single-cell protein
- B. Indicator of organic pollution reduced during secondary treatment
- C. Rearing of honey bees
- D. Management of milk-producing animals

BIO-12-CH11-0070 | Hard | Tissue culture

A reviewer compares several entries while checking 'Unorganized mass of cultured cells'. Which entry must be corrected?

- A. Symbiotic nitrogen fixer in legume roots - Rhizobium
- B. Collection of diverse cultivated and wild forms - Germplasm collection
- C. Unorganized mass of cultured cells - Single-cell protein
- D. Testing a new line at several places before release - Multi-location evaluation

BIO-12-CH11-0071 | Hard | Sewage and biogas

A reviewer compares several entries while checking 'Indicator of organic pollution reduced during secondary treatment'. Which entry must be corrected?

- A. Fungus associated with penicillin production - Penicillium
- B. Indicator of organic pollution reduced during secondary treatment - Evaluation and selection
- C. Scientific management of farm animals - Animal husbandry
- D. Crossing genetically distinct selected parents - Hybridization

BIO-12-CH11-0072 | Medium | Biofortification and SCP

A learner encounters the observation 'Microbial biomass used as protein-rich food or feed' while analysing biofortification and scp. Which conclusion is supported by the prescribed biology?

- A. Disease resistance
- B. *Saccharomyces cerevisiae*
- C. *Bacillus thuringiensis*
- D. Single-cell protein

BIO-12-CH11-0073 | Easy | Biofortification and SCP

Which description is correctly represented by Biofortification in this chapter?

- A. Breeding crops for improved nutritional quality
- B. Breeding for resistance to pathogens
- C. Symbiotic nitrogen fixer in legume roots
- D. Alga widely used as single-cell protein

BIO-12-CH11-0074 | Challenging | Plant breeding objectives

A researcher checks whether four linked observations have been interpreted correctly.

- I. Development of crop varieties with higher output corresponds to Quality improvement.
- II. Improvement of protein, oil, vitamin or mineral content corresponds to Disease resistance.
- III. Breeding for resistance to pathogens corresponds to Disease resistance.
- IV. Rapid clonal production through tissue culture corresponds to Micropropagation.

Which combination survives the biological review?

- A. I, II and IV only
- B. III and IV only
- C. I, II and III only
- D. I only

BIO-12-CH11-0075 | Hard | Biofortification and SCP

A reviewer compares several entries while checking 'Microbial biomass used as protein-rich food or feed'. Which entry must be corrected?

- A. Methanogen commonly found in cattle rumen - Methanobacterium
- B. Microbial biomass used as protein-rich food or feed - Saccharomyces cerevisiae
- C. Crossing genetically distinct selected parents - Hybridization
- D. Rapid clonal production through tissue culture - Micropropagation

BIO-12-CH11-0076 | Medium | Biocontrol and biofertilizers

Complete the statement accurately: Fungal antagonist used against plant pathogens corresponds to _____.

- A. Explant
- B. Spirulina
- C. Mycorrhiza
- D. Trichoderma

BIO-12-CH11-0077 | Medium | Tissue culture

Complete the statement accurately: Unorganized mass of cultured cells corresponds to _____.

- A. Callus
- B. Totipotency
- C. Single-cell protein
- D. Micropropagation

BIO-12-CH11-0078 | Easy | Industrial microbes

Which description is correctly represented by Penicillium in this chapter?

- A. Microbial aggregates in aeration tanks
- B. Yeast used in ethanol production
- C. Fungus associated with penicillin production
- D. Microbe producing citric acid

BIO-12-CH11-0079 | Hard | Tissue culture

A reviewer compares several entries while checking 'Capacity of a plant cell to regenerate a whole plant'. Which entry must be corrected?

- A. Breeding for resistance to pathogens - Disease resistance
- B. Capacity of a plant cell to regenerate a whole plant - Penicillium
- C. Collection of diverse cultivated and wild forms - Germplasm collection
- D. Fungal antagonist used against plant pathogens - Trichoderma

BIO-12-CH11-0080 | Medium | Animal husbandry

A learner encounters the observation 'Rearing of birds for eggs and meat' while analysing animal husbandry. Which conclusion is supported by the prescribed biology?

- A. Dairy farming
- B. Animal husbandry
- C. Poultry farming
- D. Evaluation and selection

BIO-12-CH11-0081 | Medium | Industrial microbes

A source table contains one accurate entry involving 'Yeast used in ethanol production'. Which option preserves a correct biological association?

- A. Development of crops tolerating drought or salinity - Biogas
- B. Biofortified crop with twice the amount of lysine and tryptophan - Abiotic stress resistance
- C. Microbial biomass used as protein-rich food or feed - Animal husbandry
- D. Yeast used in ethanol production - *Saccharomyces cerevisiae*

BIO-12-CH11-0082 | Challenging | Industrial microbes

Evaluate the following source-based biological statements.

- I. Fungus associated with penicillin production corresponds to *Aspergillus niger*.
- II. Microbe producing citric acid corresponds to *Aspergillus niger*.
- III. Yeast used in ethanol production corresponds to *Methanobacterium*.
- IV. Methanogen commonly found in cattle rumen corresponds to *Penicillium*.

Which option identifies exactly the correct statements?

- A. II only
- B. III only
- C. II, III and IV only
- D. III and IV only

BIO-12-CH11-0083 | Medium | Plant breeding objectives

Complete the statement accurately: Breeding for resistance to pathogens corresponds to ____.

- A. *Methanobacterium*
- B. *Bacillus thuringiensis*
- C. Disease resistance
- D. Increased yield

BIO-12-CH11-0084 | Hard | Biocontrol and biofertilizers

A reviewer compares several entries while checking 'Association of fungi with plant roots improving mineral uptake'. Which entry must be corrected?

- A. Biofortified crop with twice the amount of lysine and tryptophan - Fortified maize
- B. Association of fungi with plant roots improving mineral uptake - Apiculture
- C. Collection of diverse cultivated and wild forms - Germplasm collection
- D. Rearing of birds for eggs and meat - Poultry farming

BIO-12-CH11-0085 | Challenging | Sewage and biogas

A student submits four conclusions after integrating related chapter concepts.

- I. Indicator of organic pollution reduced during secondary treatment corresponds to Flocs.
- II. Microbial aggregates in aeration tanks corresponds to Flocs.
- III. Anaerobic digesters produce methane-rich corresponds to Totipotency.
- IV. Capacity of a plant cell to regenerate a whole plant corresponds to Totipotency.

Which combination should be accepted without correction?

- A. I and III only
- B. III and IV only
- C. II and IV only
- D. I and IV only

BIO-12-CH11-0086 | Medium | Sewage and biogas

A source table contains one accurate entry involving 'Microbial aggregates in aeration tanks'. Which option preserves a correct biological association?

- A. Crossing genetically distinct selected parents - Disease resistance
- B. Microbial aggregates in aeration tanks - Flocs
- C. Fungus associated with penicillin production - Flocs
- D. Bacterium producing lactic acid - Rhizobium

BIO-12-CH11-0087 | Hard | Plant breeding process

A reviewer compares several entries while checking 'Crossing genetically distinct selected parents'. Which entry must be corrected?

- A. Microbial biomass used as protein-rich food or feed - Single-cell protein
- B. Development of crop varieties with higher output - Increased yield
- C. Crossing genetically distinct selected parents - Fortified maize
- D. Rearing of birds for eggs and meat - Poultry farming

BIO-12-CH11-0088 | Medium | Industrial microbes

Complete the statement accurately: Fungus associated with penicillin production corresponds to _____.

- A. Penicillium
- B. Apiculture
- C. Flocs
- D. Germplasm collection

BIO-12-CH11-0089 | Medium | Plant breeding objectives

A source table contains one accurate entry involving 'Improvement of protein, oil, vitamin or mineral content'. Which option preserves a correct biological association?

- A. Fungal antagonist used against plant pathogens - Callus
- B. Improvement of protein, oil, vitamin or mineral content - Quality improvement
- C. Breeding crops for improved nutritional quality - *Aspergillus niger*
- D. Unorganized mass of cultured cells - *Bacillus thuringiensis*

BIO-12-CH11-0090 | Hard | Industrial microbes

Two conclusions must both be valid to support the reference concept 'Bacterium producing lactic acid'. Which option contains two accurate associations?

- A. Bacterium producing lactic acid - *Lactobacillus*; Rearing of honey bees - Apiculture
- B. Development of crop varieties with higher output - Mycorrhiza; Association of fungi with plant roots improving mineral uptake - Increased yield
- C. Alga widely used as single-cell protein - Increased yield; Development of crop varieties with higher output - *Spirulina*
- D. Fungal antagonist used against plant pathogens - Dairy farming; Management of milk-producing animals - *Trichoderma*

BIO-12-CH11-0091 | Easy | Tissue culture

Which biological term or result is specifically identified by the description 'Capacity of a plant cell to regenerate a whole plant'?

- A. Explant
- B. Totipotency
- C. Animal husbandry
- D. Trichoderma

BIO-12-CH11-0092 | Medium | Biofortification and SCP

A learner encounters the observation 'Alga widely used as single-cell protein' while analysing biofortification and scp. Which conclusion is supported by the prescribed biology?

- A. Apiculture
- B. Dairy farming
- C. Fortified maize
- D. Spirulina

BIO-12-CH11-0093 | Hard | Biocontrol and biofertilizers

Two conclusions must both be valid to support the reference concept 'Symbiotic nitrogen fixer in legume roots'. Which option contains two accurate associations?

- A. Small tissue piece used to start a culture - Micropropagation; Rapid clonal production through tissue culture - Explant
- B. Development of crop varieties with higher output - Hybridization; Crossing genetically distinct selected parents - Increased yield
- C. Symbiotic nitrogen fixer in legume roots - Rhizobium; Bacterium producing lactic acid - Lactobacillus
- D. Testing a new line at several places before release - Aspergillus niger; Microbe producing citric acid - Multi-location evaluation

BIO-12-CH11-0094 | Medium | Plant breeding objectives

Complete the statement accurately: Development of crop varieties with higher output corresponds to ____.

- A. Increased yield
- B. Rhizobium
- C. Disease resistance
- D. Mycorrhiza

BIO-12-CH11-0095 | Challenging | Biofortification and SCP

During error analysis, four proposed associations are reviewed.

- I. Breeding crops for improved nutritional quality corresponds to Biofortification.
- II. Biofortified crop with twice the amount of lysine and tryptophan corresponds to Spirulina.
- III. Alga widely used as single-cell protein corresponds to Totipotency.
- IV. Capacity of a plant cell to regenerate a whole plant corresponds to Biofortification.

Which set contains all and only the valid conclusions?

- A. IV only
- B. II only
- C. II and IV only
- D. I only

BIO-12-CH11-0096 | Challenging | Animal husbandry

A researcher checks whether four linked observations have been interpreted correctly.

- I. Rearing of birds for eggs and meat corresponds to Dairy farming.
- II. Management of milk-producing animals corresponds to Dairy farming.
- III. Rearing of honey bees corresponds to Apiculture.
- IV. Development of crop varieties with higher output corresponds to Poultry farming.

Which combination survives the biological review?

- A. I, II and IV only
 - B. I and III only
 - C. I, III and IV only
 - D. II and III only
-

BIO-12-CH11-0097 | Medium | Animal husbandry

Complete the statement accurately: Management of milk-producing animals corresponds to _____.

- A. Dairy farming
 - B. Lactobacillus
 - C. Micropropagation
 - D. Apiculture
-

BIO-12-CH11-0098 | Easy | Animal husbandry

Which description is correctly represented by Dairy farming in this chapter?

- A. Rearing of honey bees
 - B. Management of milk-producing animals
 - C. Improvement of protein, oil, vitamin or mineral content
 - D. Bacterium producing lactic acid
-

BIO-12-CH11-0099 | Easy | Sewage and biogas

Which description is correctly represented by Methanobacterium in this chapter?

- A. Anaerobic digesters produce methane-rich
 - B. Development of crops tolerating drought or salinity
 - C. Rearing of honey bees
 - D. Methanogen commonly found in cattle rumen
-

BIO-12-CH11-0100 | Medium | Animal husbandry

A learner encounters the observation 'Rearing of honey bees' while analysing animal husbandry. Which conclusion is supported by the prescribed biology?

- A. Animal husbandry
 - B. Quality improvement
 - C. Apiculture
 - D. Single-cell protein
-

Standard 12 | Chapter 12

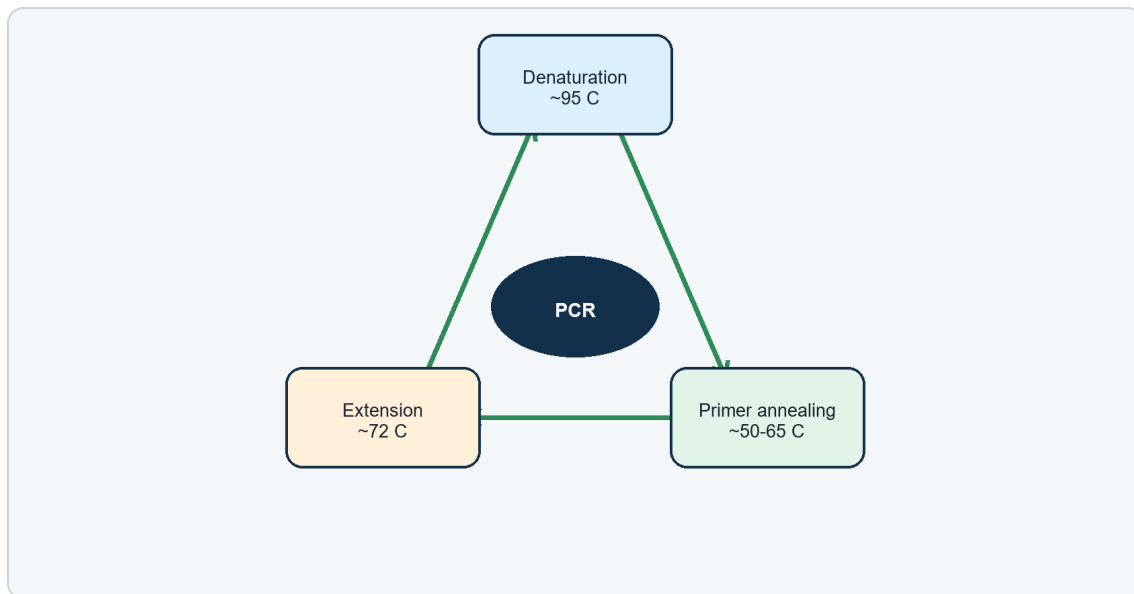
Biotechnology

BIO-12-CH12-0001 | Easy | Polymerase chain reaction

Which PCR step separates the two DNA strands?

Polymerase chain reaction

One thermal cycle of DNA amplification



- A. Annealing
- B. Extension
- C. Denaturation
- D. Ligation into a chromosome

BIO-12-CH12-0002 | Easy | rDNA methodology

Which description is correctly represented by Cloning in this chapter?

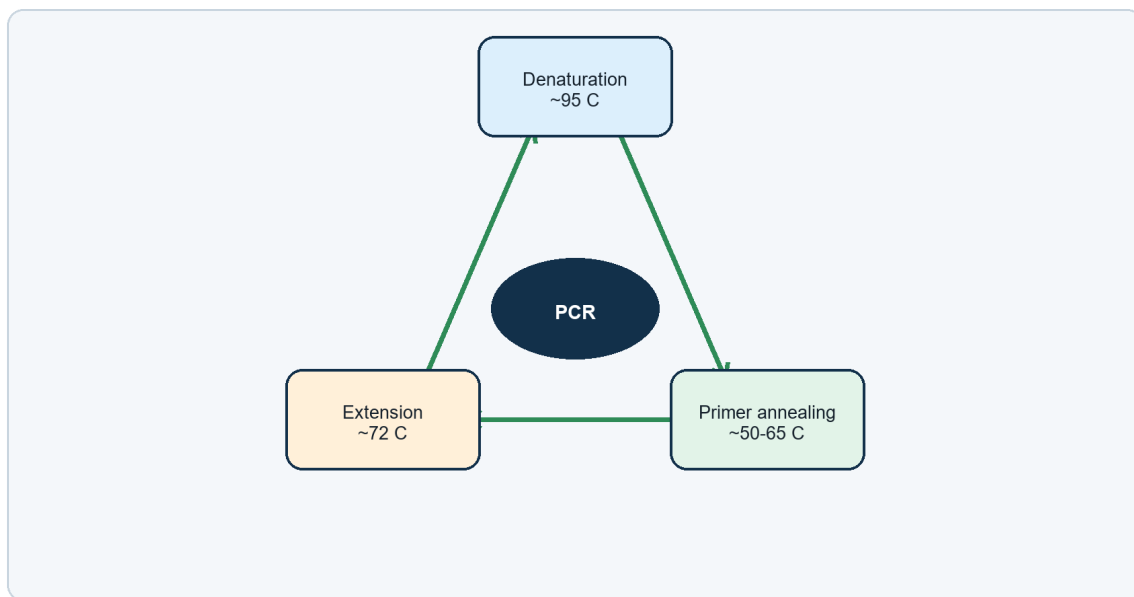
- A. Enzymes cutting DNA at specific recognition sequences
- B. Bt crop protection targets
- C. Production of many identical copies from one transformed cell
- D. Separation of DNA fragments according to size

BIO-12-CH12-0003 | Hard | Polymerase chain reaction

Starting from one double-stranded target molecule, what is the ideal product count after n complete PCR cycles?

Polymerase chain reaction

One thermal cycle of DNA amplification



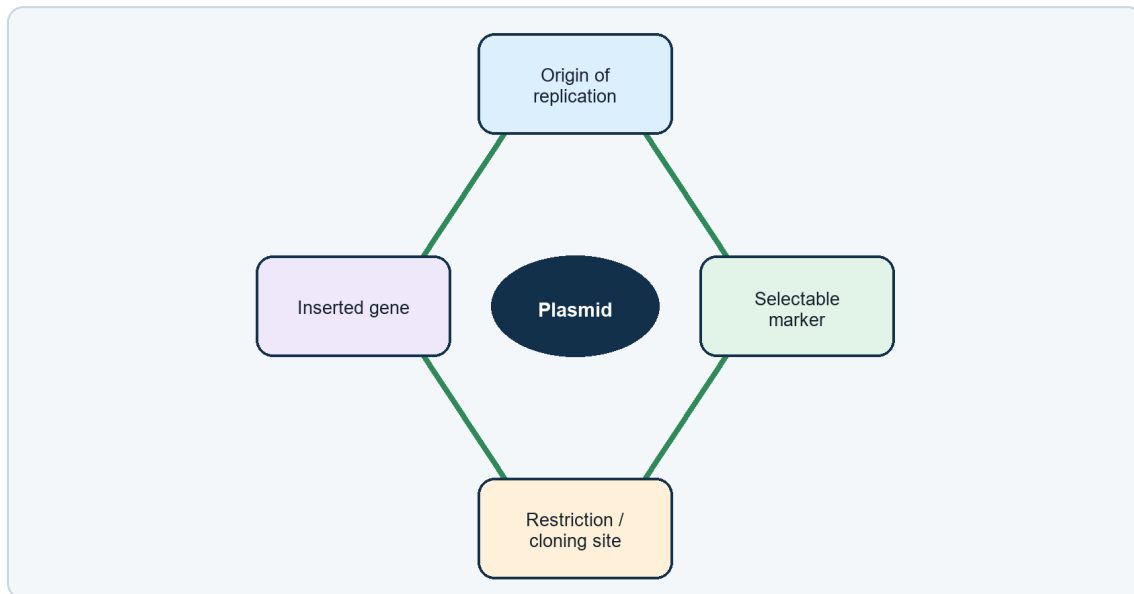
- A. Exactly n
- B. Approximately 2 to the power n
- C. Approximately n squared
- D. Always two molecules

BIO-12-CH12-0004 | Medium | **Simplified plasmid cloning vector**

Which vector region allows a recombinant plasmid to replicate inside a host cell?

Simplified plasmid cloning vector

Functional regions used in recombinant-DNA selection



- A. Inserted gene alone
- B. Restriction enzyme protein
- C. Origin of replication
- D. Antibody-binding site

BIO-12-CH12-0005 | Easy | Bioethics and ownership

Which description is correctly represented by Biopiracy in this chapter?

- A. Ethical concern about altered organisms entering nature
- B. PCR step in which polymerase synthesizes DNA
- C. Small circular bacterial DNA commonly used as a vector
- D. Unfair commercial use of biological resources or traditional knowledge

BIO-12-CH12-0006 | Medium | Medical applications

A learner encounters the observation 'First therapeutic product made using recombinant DNA technology' while analysing medical applications. Which conclusion is supported by the prescribed biology?

- A. Human insulin
- B. PCR
- C. PCR-based diagnosis
- D. Gene therapy

BIO-12-CH12-0007 | Medium | Bioethics and ownership

Complete the statement accurately: Ethical concern about altered organisms entering nature corresponds to ____.

- A. Environmental release of GMOs
- B. DNA ligase
- C. Biopatent
- D. Extension

BIO-12-CH12-0008 | Hard | Restriction and joining enzymes

Two conclusions must both be valid to support the reference concept 'Enzymes cutting DNA at specific recognition sequences'. Which option contains two accurate associations?

- A. Gene allowing identification of transformed cells - Monoclonal antibody; Antibody produced from a single clone of cells - Selectable marker
- B. Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases; Amplification of a chosen DNA segment in vitro - PCR
- C. Site permitting vector replication in a host - Specific insect pests; Bt crop protection targets - Origin of replication
- D. Antibody produced from a single clone of cells - Origin of replication; Site permitting vector replication in a host - Monoclonal antibody

BIO-12-CH12-0009 | Easy | Core principles

Which description is correctly represented by Bioprocess engineering in this chapter?

- A. Maintenance of sterile conditions for industrial microbial growth
- B. DNA formed by joining sequences from different sources
- C. Organism carrying and expressing an introduced gene
- D. Site permitting vector replication in a host

BIO-12-CH12-0010 | Medium | Restriction and joining enzymes

A learner encounters the observation 'Enzymes cutting DNA at specific recognition sequences' while analysing restriction and joining enzymes. Which conclusion is supported by the prescribed biology?

- A. Selectable marker
- B. PCR
- C. Transformation
- D. Restriction endonucleases

BIO-12-CH12-0011 | Medium | Core principles

A source table contains one accurate entry involving 'DNA formed by joining sequences from different sources'. Which option preserves a correct biological association?

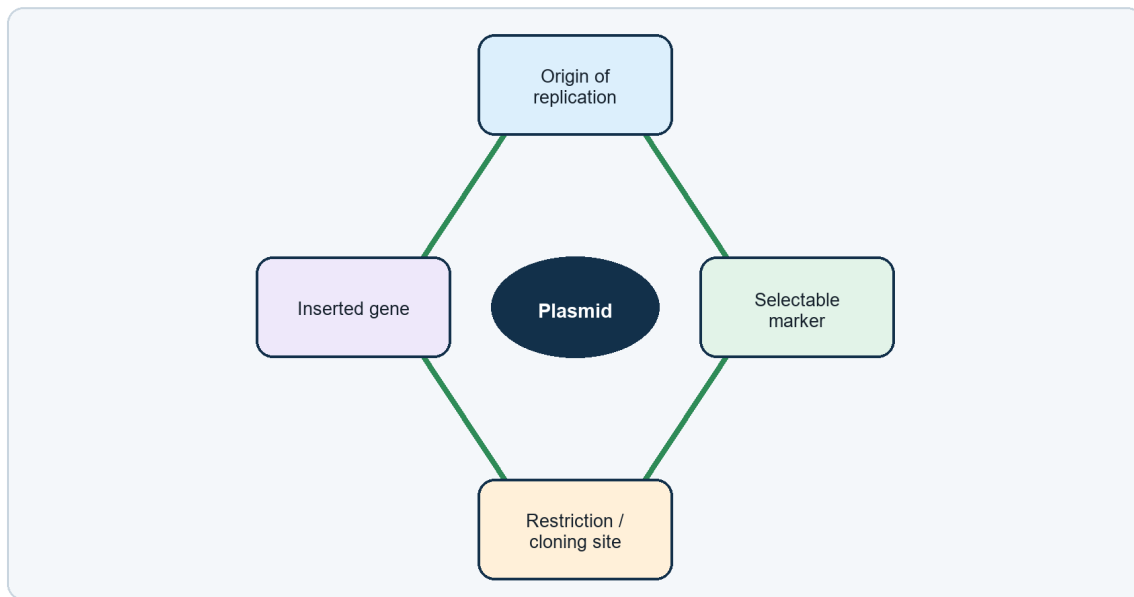
- A. Treatment by introducing a functional gene - Selectable marker
- B. DNA formed by joining sequences from different sources - Recombinant DNA
- C. Early molecular diagnosis using amplification - Recombinant DNA
- D. Vector region containing several unique restriction sites - Biopatent

BIO-12-CH12-0012 | Challenging | Simplified plasmid cloning vector

An insert disrupts a marker gene at the cloning site. How can this support recombinant screening?

Simplified plasmid cloning vector

Functional regions used in recombinant-DNA selection



- A. The insert makes every colony identical without testing
- B. The origin of replication is no longer needed
- C. Nonrecombinant plasmids lose all DNA
- D. Insertional inactivation changes the marker phenotype of recombinant colonies

BIO-12-CH12-0013 | Medium | Cloning vectors

A source table contains one accurate entry involving 'Vector region containing several unique restriction sites'. Which option preserves a correct biological association?

- A. Separation of DNA fragments according to size - *Bacillus thuringiensis*
- B. Legal protection for a novel biotechnological invention - Gel electrophoresis
- C. Restriction sequence reading the same in opposite directions - Multiple cloning site
- D. Vector region containing several unique restriction sites - Multiple cloning site

BIO-12-CH12-0014 | Easy | PCR and analysis

Which description is correctly represented by Denaturation in this chapter?

- A. PCR step in which polymerase synthesizes DNA
- B. PCR step separating DNA strands
- C. Nutritionally enhanced transgenic crop example
- D. Enzymes cutting DNA at specific recognition sequences

BIO-12-CH12-0015 | Hard | Bioethics and ownership

Two conclusions must both be valid to support the reference concept 'Unfair commercial use of biological resources or traditional knowledge'. Which option contains two accurate associations?

- A. Enzymes cutting DNA at specific recognition sequences - Recombinant DNA; DNA formed by joining sequences from different sources - Restriction endonucleases
- B. PCR step in which polymerase synthesizes DNA - Bt cotton; Transgenic plant known for insect resistance - Extension
- C. Unfair commercial use of biological resources or traditional knowledge - Biopiracy; Small circular bacterial DNA commonly used as a vector - Plasmid
- D. DNA formed by joining sequences from different sources - DNA ligase; Enzyme sealing phosphodiester bonds between DNA fragments - Recombinant DNA

BIO-12-CH12-0016 | Easy | PCR and analysis

Which description is correctly represented by Extension in this chapter?

- A. Single-stranded ends produced by staggered restriction cuts
- B. PCR step in which polymerase synthesizes DNA
- C. Separation of DNA fragments according to size
- D. Vector region containing several unique restriction sites

BIO-12-CH12-0017 | Challenging | PCR and analysis

A student submits four conclusions after integrating related chapter concepts.

- I. PCR step allowing primers to bind corresponds to Extension.
- II. PCR step in which polymerase synthesizes DNA corresponds to Denaturation.
- III. PCR step separating DNA strands corresponds to Denaturation.
- IV. Early molecular diagnosis using amplification corresponds to PCR-based diagnosis.

Which combination should be accepted without correction?

- A. I only
- B. IV only
- C. III and IV only
- D. II and III only

BIO-12-CH12-0018 | Medium | Medical applications

A source table contains one accurate entry involving 'Early molecular diagnosis using amplification'. Which option preserves a correct biological association?

- A. Bt toxin gene source - Bt cotton
- B. DNA formed by joining sequences from different sources - PCR-based diagnosis
- C. Small circular bacterial DNA commonly used as a vector - Recombinant DNA
- D. Early molecular diagnosis using amplification - PCR-based diagnosis

BIO-12-CH12-0019 | Medium | Cloning vectors

A learner encounters the observation 'Small circular bacterial DNA commonly used as a vector' while analysing cloning vectors. Which conclusion is supported by the prescribed biology?

- A. Origin of replication
- B. Transgenic organism
- C. Plasmid
- D. Bt cotton

BIO-12-CH12-0020 | Hard | PCR and analysis

Two conclusions must both be valid to support the reference concept 'PCR step allowing primers to bind'. Which option contains two accurate associations?

- A. Bt toxin gene source - Transgenic organism; Organism carrying and expressing an introduced gene - *Bacillus thuringiensis*
- B. PCR step allowing primers to bind - Annealing; Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase
- C. Indian committee that decides validity and safety of GMO research - Origin of replication; Site permitting vector replication in a host - GEAC
- D. Site permitting vector replication in a host - Monoclonal antibody; Antibody produced from a single clone of cells - Origin of replication

BIO-12-CH12-0021 | Medium | Agricultural applications

A learner encounters the observation 'Bt toxin gene source' while analysing agricultural applications. Which conclusion is supported by the prescribed biology?

- A. Gene therapy
- B. Sticky ends
- C. Golden rice
- D. *Bacillus thuringiensis*

BIO-12-CH12-0022 | Easy | Restriction and joining enzymes

Which biological term or result is specifically identified by the description 'Restriction sequence reading the same in opposite directions'?

- A. Transgenic organism
- B. Golden rice
- C. Monoclonal antibody
- D. Palindromic sequence

BIO-12-CH12-0023 | Challenging | Agricultural applications

During error analysis, four proposed associations are reviewed.

- I. Bt toxin gene source corresponds to Golden rice.
 - II. Nutritionally enhanced transgenic crop example corresponds to Golden rice.
 - III. Transgenic plant known for insect resistance corresponds to Bioprocess engineering.
 - IV. Maintenance of sterile conditions for industrial microbial growth corresponds to *Bacillus thuringiensis*.
- Which set contains all and only the valid conclusions?

- A. I and III only
- B. II and IV only
- C. II only
- D. I, II and III only

BIO-12-CH12-0024 | Challenging | Medical applications

A researcher checks whether four linked observations have been interpreted correctly.

- I. Early molecular diagnosis using amplification corresponds to Human insulin.
- II. First therapeutic product made using recombinant DNA technology corresponds to Human insulin.
- III. Treatment by introducing a functional gene corresponds to Golden rice.
- IV. Nutritionally enhanced transgenic crop example corresponds to Golden rice.

Which combination survives the biological review?

- A. II and IV only
- B. III and IV only
- C. I, III and IV only
- D. IV only

BIO-12-CH12-0025 | Medium | Bioethics and ownership

A learner encounters the observation 'Legal protection for a novel biotechnological invention' while analysing bioethics and ownership. Which conclusion is supported by the prescribed biology?

- A. Biopatent
- B. Environmental release of GMOs
- C. Biopiracy
- D. Bt cotton

BIO-12-CH12-0026 | Easy | Medical applications

Which description is correctly represented by Monoclonal antibody in this chapter?

- A. Early molecular diagnosis using amplification
- B. Nutritionally enhanced transgenic crop example
- C. Treatment by introducing a functional gene
- D. Antibody produced from a single clone of cells

BIO-12-CH12-0027 | Hard | Restriction and joining enzymes

A reviewer compares several entries while checking 'Restriction sequence reading the same in opposite directions'. Which entry must be corrected?

- A. Organism carrying and expressing an introduced gene - Transgenic organism
- B. Restriction sequence reading the same in opposite directions - Monoclonal antibody
- C. Bt crop protection targets - Specific insect pests
- D. PCR step in which polymerase synthesizes DNA - Extension

BIO-12-CH12-0028 | Hard | PCR and analysis

Two conclusions must both be valid to support the reference concept 'PCR step in which polymerase synthesizes DNA'. Which option contains two accurate associations?

- A. Bt toxin gene source - Biopatent; Legal protection for a novel biotechnological invention - *Bacillus thuringiensis*
- B. Separation of DNA fragments according to size - Sticky ends; Single-stranded ends produced by staggered restriction cuts - Gel electrophoresis
- C. PCR step in which polymerase synthesizes DNA - Extension; Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase
- D. Amplification of a chosen DNA segment in vitro - Multiple cloning site; Vector region containing several unique restriction sites - PCR

BIO-12-CH12-0029 | Challenging | Medical applications

Evaluate the following source-based biological statements.

- I. First therapeutic product made using recombinant DNA technology corresponds to Human insulin.
- II. Treatment by introducing a functional gene corresponds to Monoclonal antibody.
- III. Antibody produced from a single clone of cells corresponds to Plasmid.
- IV. Small circular bacterial DNA commonly used as a vector corresponds to Human insulin.

Which option identifies exactly the correct statements?

- A. II and III only
- B. I only
- C. I, II and III only
- D. III only

BIO-12-CH12-0030 | Medium | rDNA methodology

A learner encounters the observation 'Production of many identical copies from one transformed cell' while analysing rDNA methodology. Which conclusion is supported by the prescribed biology?

- A. Restriction endonucleases
- B. Cloning
- C. Transformation
- D. Gel electrophoresis

BIO-12-CH12-0031 | Hard | Core principles

Two conclusions must both be valid to support the reference concept 'Maintenance of sterile conditions for industrial microbial growth'. Which option contains two accurate associations?

- A. Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering; Site permitting vector replication in a host - Origin of replication
- B. Enzymes cutting DNA at specific recognition sequences - Multiple cloning site; Vector region containing several unique restriction sites - Restriction endonucleases
- C. Early molecular diagnosis using amplification - Selectable marker; Gene allowing identification of transformed cells - PCR-based diagnosis
- D. Ethical concern about altered organisms entering nature - DNA ligase; Enzyme sealing phosphodiester bonds between DNA fragments - Environmental release of GMOs

BIO-12-CH12-0032 | Medium | rDNA methodology

Complete the statement accurately: Separation of DNA fragments according to size corresponds to ____.

- A. Human insulin
- B. PCR-based diagnosis
- C. PCR
- D. Gel electrophoresis

BIO-12-CH12-0033 | Medium | Core principles

Complete the statement accurately: DNA formed by joining sequences from different sources corresponds to ____.

- A. Multiple cloning site
- B. Genetic engineering
- C. Recombinant DNA
- D. Human insulin

BIO-12-CH12-0034 | Hard | Agricultural applications

A reviewer compares several entries while checking 'Bt toxin gene source'. Which entry must be corrected?

- A. Bt toxin gene source - Human insulin
- B. Production of many identical copies from one transformed cell - Cloning
- C. Small circular bacterial DNA commonly used as a vector - Plasmid
- D. Single-stranded ends produced by staggered restriction cuts - Sticky ends

BIO-12-CH12-0035 | Medium | Restriction and joining enzymes

Complete the statement accurately: Single-stranded ends produced by staggered restriction cuts corresponds to _____.

- A. Sticky ends
- B. DNA ligase
- C. PCR
- D. Palindromic sequence

BIO-12-CH12-0036 | Easy | Medical applications

Which biological term or result is specifically identified by the description 'Antibody produced from a single clone of cells'?

- A. PCR-based diagnosis
- B. Gene therapy
- C. Monoclonal antibody
- D. Taq polymerase

BIO-12-CH12-0037 | Hard | rDNA methodology

A reviewer compares several entries while checking 'Production of many identical copies from one transformed cell'. Which entry must be corrected?

- A. PCR step allowing primers to bind - Annealing
- B. Legal protection for a novel biotechnological invention - Biopatent
- C. Single-stranded ends produced by staggered restriction cuts - Sticky ends
- D. Production of many identical copies from one transformed cell - Specific insect pests

BIO-12-CH12-0038 | Easy | Bioethics and ownership

Which biological term or result is specifically identified by the description 'Indian committee that decides validity and safety of GMO research'?

- A. Environmental release of GMOs
- B. Gene therapy
- C. GEAC
- D. Specific insect pests

BIO-12-CH12-0039 | Hard | Medical applications

A reviewer compares several entries while checking 'Antibody produced from a single clone of cells'. Which entry must be corrected?

- A. Antibody produced from a single clone of cells - Cloning
- B. DNA formed by joining sequences from different sources - Recombinant DNA
- C. Thermostable polymerase commonly used in PCR - Taq polymerase
- D. Nutritionally enhanced transgenic crop example - Golden rice

BIO-12-CH12-0040 | Hard | Core principles

A reviewer compares several entries while checking 'Deliberate alteration of genetic material'. Which entry must be corrected?

- A. Ethical concern about altered organisms entering nature - Environmental release of GMOs
- B. Thermostable polymerase commonly used in PCR - Taq polymerase
- C. Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase
- D. Deliberate alteration of genetic material - Bacillus thuringiensis

BIO-12-CH12-0041 | Challenging | Agricultural applications

A student submits four conclusions after integrating related chapter concepts.

- I. Nutritionally enhanced transgenic crop example corresponds to *Bacillus thuringiensis*.
 - II. Bt toxin gene source corresponds to *Bacillus thuringiensis*.
 - III. Transgenic plant known for insect resistance corresponds to Bt cotton.
 - IV. Organism carrying and expressing an introduced gene corresponds to Golden rice.
- Which combination should be accepted without correction?

- A. III only
- B. II, III and IV only
- C. II and III only
- D. I, II and III only

BIO-12-CH12-0042 | Medium | Bioethics and ownership

Complete the statement accurately: Legal protection for a novel biotechnological invention corresponds to _____.

- A. Sticky ends
- B. Environmental release of GMOs
- C. Biopatent
- D. Transgenic organism

BIO-12-CH12-0043 | Easy | rDNA methodology

Which description is correctly represented by Gel electrophoresis in this chapter?

- A. Treatment by introducing a functional gene
- B. Separation of DNA fragments according to size
- C. Amplification of a chosen DNA segment in vitro
- D. PCR step separating DNA strands

BIO-12-CH12-0044 | Medium | Core principles

A source table contains one accurate entry involving 'Organism carrying and expressing an introduced gene'. Which option preserves a correct biological association?

- A. Organism carrying and expressing an introduced gene - Transgenic organism
- B. Unfair commercial use of biological resources or traditional knowledge - GEAC
- C. Introduction of recombinant DNA into a host cell - Biopatent
- D. Indian committee that decides validity and safety of GMO research - Selectable marker

BIO-12-CH12-0045 | Easy | rDNA methodology

Which biological term or result is specifically identified by the description 'Production of many identical copies from one transformed cell'?

- A. Restriction endonucleases
- B. Biopatent
- C. Sticky ends
- D. Cloning

BIO-12-CH12-0046 | Easy | Medical applications

Which description is correctly represented by PCR-based diagnosis in this chapter?

- A. Early molecular diagnosis using amplification
- B. Transgenic plant known for insect resistance
- C. Thermostable polymerase commonly used in PCR
- D. Bt toxin gene source

BIO-12-CH12-0047 | Medium | Restriction and joining enzymes

A source table contains one accurate entry involving 'Single-stranded ends produced by staggered restriction cuts'. Which option preserves a correct biological association?

- A. First therapeutic product made using recombinant DNA technology - Bioprocess engineering
- B. Amplification of a chosen DNA segment in vitro - Recombinant DNA
- C. Single-stranded ends produced by staggered restriction cuts - Sticky ends
- D. Organism carrying and expressing an introduced gene - Human insulin

BIO-12-CH12-0048 | Hard | Cloning vectors

A reviewer compares several entries while checking 'Vector region containing several unique restriction sites'. Which entry must be corrected?

- A. Legal protection for a novel biotechnological invention - Biopatent
- B. Antibody produced from a single clone of cells - Monoclonal antibody
- C. Vector region containing several unique restriction sites - Biopiracy
- D. Separation of DNA fragments according to size - Gel electrophoresis

BIO-12-CH12-0049 | Hard | Agricultural applications

Two conclusions must both be valid to support the reference concept 'Bt crop protection targets'. Which option contains two accurate associations?

- A. Ethical concern about altered organisms entering nature - Recombinant DNA; DNA formed by joining sequences from different sources - Environmental release of GMOs
- B. Bt crop protection targets - Specific insect pests; Indian committee that decides validity and safety of GMO research - GEAC
- C. Organism carrying and expressing an introduced gene - Environmental release of GMOs; Ethical concern about altered organisms entering nature - Transgenic organism
- D. DNA formed by joining sequences from different sources - Gel electrophoresis; Separation of DNA fragments according to size - Recombinant DNA

BIO-12-CH12-0050 | Easy | Agricultural applications

Which description is correctly represented by Specific insect pests in this chapter?

- A. Organism carrying and expressing an introduced gene
- B. Nutritionally enhanced transgenic crop example
- C. Ethical concern about altered organisms entering nature
- D. Bt crop protection targets

BIO-12-CH12-0051 | Hard | PCR and analysis

Two conclusions must both be valid to support the reference concept 'Thermostable polymerase commonly used in PCR'. Which option contains two accurate associations?

- A. Organism carrying and expressing an introduced gene - Environmental release of GMOs; Ethical concern about altered organisms entering nature - Transgenic organism
- B. Site permitting vector replication in a host - Plasmid; Small circular bacterial DNA commonly used as a vector - Origin of replication
- C. Thermostable polymerase commonly used in PCR - Taq polymerase; Unfair commercial use of biological resources or traditional knowledge - Biopiracy
- D. Single-stranded ends produced by staggered restriction cuts - Transgenic organism; Organism carrying and expressing an introduced gene - Sticky ends

BIO-12-CH12-0052 | Medium | PCR and analysis

A learner encounters the observation 'Thermostable polymerase commonly used in PCR' while analysing pcr and analysis. Which conclusion is supported by the prescribed biology?

- A. Extension
- B. Taq polymerase
- C. Sticky ends
- D. Environmental release of GMOs

BIO-12-CH12-0053 | Medium | PCR and analysis

Complete the statement accurately: PCR step separating DNA strands corresponds to _____.

- A. Restriction endonucleases
- B. Golden rice
- C. Denaturation
- D. Human insulin

BIO-12-CH12-0054 | Medium | Restriction and joining enzymes

A source table contains one accurate entry involving 'Enzymes cutting DNA at specific recognition sequences'. Which option preserves a correct biological association?

- A. Bt crop protection targets - Restriction endonucleases
- B. Amplification of a chosen DNA segment in vitro - Selectable marker
- C. Introduction of recombinant DNA into a host cell - Monoclonal antibody
- D. Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases

BIO-12-CH12-0055 | Medium | Restriction and joining enzymes

A source table contains one accurate entry involving 'Restriction sequence reading the same in opposite directions'. Which option preserves a correct biological association?

- A. PCR step in which polymerase synthesizes DNA - Palindromic sequence
- B. Antibody produced from a single clone of cells - Specific insect pests
- C. Restriction sequence reading the same in opposite directions - Palindromic sequence
- D. Deliberate alteration of genetic material - Bt cotton

BIO-12-CH12-0056 | Medium | Bioethics and ownership

A learner encounters the observation 'Indian committee that decides validity and safety of GMO research' while analysing bioethics and ownership. Which conclusion is supported by the prescribed biology?

- A. Specific insect pests
- B. Bacillus thuringiensis
- C. Selectable marker
- D. GEAC

BIO-12-CH12-0057 | Hard | PCR and analysis

Two conclusions must both be valid to support the reference concept 'PCR step separating DNA strands'. Which option contains two accurate associations?

- A. Site permitting vector replication in a host - Human insulin; First therapeutic product made using recombinant DNA technology - Origin of replication
- B. Production of many identical copies from one transformed cell - Biopatent; Legal protection for a novel biotechnological invention - Cloning
- C. Nutritionally enhanced transgenic crop example - Cloning; Production of many identical copies from one transformed cell - Golden rice
- D. PCR step separating DNA strands - Denaturation; Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering

BIO-12-CH12-0058 | Medium | Medical applications

A source table contains one accurate entry involving 'First therapeutic product made using recombinant DNA technology'. Which option preserves a correct biological association?

- A. Amplification of a chosen DNA segment in vitro - DNA ligase
- B. First therapeutic product made using recombinant DNA technology - Human insulin
- C. Enzyme sealing phosphodiester bonds between DNA fragments - Transformation
- D. Thermostable polymerase commonly used in PCR - Extension

BIO-12-CH12-0059 | Medium | PCR and analysis

Complete the statement accurately: PCR step allowing primers to bind corresponds to _____.

- A. Golden rice
- B. Extension
- C. Annealing
- D. Denaturation

BIO-12-CH12-0060 | Hard | Medical applications

Two conclusions must both be valid to support the reference concept 'Antibody produced from a single clone of cells'. Which option contains two accurate associations?

- A. PCR step allowing primers to bind - GEAC; Indian committee that decides validity and safety of GMO research - Annealing
- B. Antibody produced from a single clone of cells - Monoclonal antibody; Production of many identical copies from one transformed cell - Cloning
- C. DNA formed by joining sequences from different sources - Selectable marker; Gene allowing identification of transformed cells - Recombinant DNA
- D. Thermostable polymerase commonly used in PCR - Golden rice; Nutritionally enhanced transgenic crop example - Taq polymerase

BIO-12-CH12-0061 | Easy | PCR and analysis

Which description is correctly represented by Taq polymerase in this chapter?

- A. PCR step in which polymerase synthesizes DNA
- B. Organism carrying and expressing an introduced gene
- C. Thermostable polymerase commonly used in PCR
- D. Indian committee that decides validity and safety of GMO research

BIO-12-CH12-0062 | Easy | Agricultural applications

Which biological term or result is specifically identified by the description 'Nutritionally enhanced transgenic crop example'?

- A. Bt cotton
- B. Sticky ends
- C. Golden rice
- D. Restriction endonucleases

BIO-12-CH12-0063 | Hard | rDNA methodology

Two conclusions must both be valid to support the reference concept 'Separation of DNA fragments according to size'. Which option contains two accurate associations?

- A. Separation of DNA fragments according to size - Gel electrophoresis; PCR step separating DNA strands - Denaturation
- B. First therapeutic product made using recombinant DNA technology - Plasmid; Small circular bacterial DNA commonly used as a vector - Human insulin
- C. Treatment by introducing a functional gene - Human insulin; First therapeutic product made using recombinant DNA technology - Gene therapy
- D. Early molecular diagnosis using amplification - Origin of replication; Site permitting vector replication in a host - PCR-based diagnosis

BIO-12-CH12-0064 | Easy | Core principles

Which biological term or result is specifically identified by the description 'Deliberate alteration of genetic material'?

- A. Taq polymerase
- B. *Bacillus thuringiensis*
- C. Genetic engineering
- D. DNA ligase

BIO-12-CH12-0065 | Medium | PCR and analysis

Complete the statement accurately: Thermostable polymerase commonly used in PCR corresponds to _____.

- A. Sticky ends
- B. Taq polymerase
- C. Biopiracy
- D. GEAC

BIO-12-CH12-0066 | Medium | Cloning vectors

Complete the statement accurately: Vector region containing several unique restriction sites corresponds to _____.

- A. Biopiracy
- B. Biopatent
- C. Origin of replication
- D. Multiple cloning site

BIO-12-CH12-0067 | Hard | Agricultural applications

A reviewer compares several entries while checking 'Nutritionally enhanced transgenic crop example'. Which entry must be corrected?

- A. Gene allowing identification of transformed cells - Selectable marker
- B. Nutritionally enhanced transgenic crop example - Recombinant DNA
- C. Single-stranded ends produced by staggered restriction cuts - Sticky ends
- D. Enzymes cutting DNA at specific recognition sequences - Restriction endonucleases

BIO-12-CH12-0068 | Easy | Bioethics and ownership

Which biological term or result is specifically identified by the description 'Ethical concern about altered organisms entering nature'?

- A. Environmental release of GMOs
- B. Golden rice
- C. Extension
- D. PCR

BIO-12-CH12-0069 | Easy | Cloning vectors

Which description is correctly represented by Origin of replication in this chapter?

- A. Transgenic plant known for insect resistance
- B. Site permitting vector replication in a host
- C. Gene allowing identification of transformed cells
- D. Enzymes cutting DNA at specific recognition sequences

BIO-12-CH12-0070 | Medium | Agricultural applications

A source table contains one accurate entry involving 'Bt toxin gene source'. Which option preserves a correct biological association?

- A. Treatment by introducing a functional gene - Sticky ends
- B. Small circular bacterial DNA commonly used as a vector - Monoclonal antibody
- C. Bt toxin gene source - *Bacillus thuringiensis*
- D. First therapeutic product made using recombinant DNA technology - Gene therapy

BIO-12-CH12-0071 | Easy | Agricultural applications

Which description is correctly represented by Bt cotton in this chapter?

- A. Transgenic plant known for insect resistance
- B. Nutritionally enhanced transgenic crop example
- C. Antibody produced from a single clone of cells
- D. Bt toxin gene source

BIO-12-CH12-0072 | Hard | Cloning vectors

Two conclusions must both be valid to support the reference concept 'Small circular bacterial DNA commonly used as a vector'. Which option contains two accurate associations?

- A. Transgenic plant known for insect resistance - Palindromic sequence; Restriction sequence reading the same in opposite directions - Bt cotton
- B. Nutritionally enhanced transgenic crop example - Bt cotton; Transgenic plant known for insect resistance - Golden rice
- C. Enzymes cutting DNA at specific recognition sequences - Transgenic organism; Organism carrying and expressing an introduced gene - Restriction endonucleases
- D. Small circular bacterial DNA commonly used as a vector - Plasmid; Amplification of a chosen DNA segment in vitro - PCR

BIO-12-CH12-0073 | Hard | Agricultural applications

Two conclusions must both be valid to support the reference concept 'Transgenic plant known for insect resistance'. Which option contains two accurate associations?

- A. Amplification of a chosen DNA segment in vitro - Denaturation; PCR step separating DNA strands - PCR
- B. Single-stranded ends produced by staggered restriction cuts - PCR; Amplification of a chosen DNA segment in vitro - Sticky ends
- C. Transgenic plant known for insect resistance - Bt cotton; Antibody produced from a single clone of cells - Monoclonal antibody
- D. Organism carrying and expressing an introduced gene - DNA ligase; Enzyme sealing phosphodiester bonds between DNA fragments - Transgenic organism

BIO-12-CH12-0074 | Medium | Cloning vectors

A learner encounters the observation 'Site permitting vector replication in a host' while analysing cloning vectors. Which conclusion is supported by the prescribed biology?

- A. Restriction endonucleases
- B. Gel electrophoresis
- C. Selectable marker
- D. Origin of replication

BIO-12-CH12-0075 | Easy | Cloning vectors

Which biological term or result is specifically identified by the description 'Gene allowing identification of transformed cells'?

- A. DNA ligase
- B. Selectable marker
- C. Multiple cloning site
- D. GEAC

BIO-12-CH12-0076 | Easy | Core principles

Which biological term or result is specifically identified by the description 'Maintenance of sterile conditions for industrial microbial growth'?

- A. Bioprocess engineering
- B. Recombinant DNA
- C. PCR-based diagnosis
- D. Selectable marker

BIO-12-CH12-0077 | Medium | rDNA methodology

A source table contains one accurate entry involving 'Introduction of recombinant DNA into a host cell'. Which option preserves a correct biological association?

- A. Introduction of recombinant DNA into a host cell - Transformation
- B. PCR step separating DNA strands - Bioprocess engineering
- C. First therapeutic product made using recombinant DNA technology - Transformation
- D. Maintenance of sterile conditions for industrial microbial growth - Genetic engineering

BIO-12-CH12-0078 | Easy | PCR and analysis

Which description is correctly represented by Annealing in this chapter?

- A. Site permitting vector replication in a host
- B. Antibody produced from a single clone of cells
- C. Enzyme sealing phosphodiester bonds between DNA fragments
- D. PCR step allowing primers to bind

BIO-12-CH12-0079 | Challenging | rDNA methodology

During error analysis, four proposed associations are reviewed.

- I. Production of many identical copies from one transformed cell corresponds to Cloning.
- II. Separation of DNA fragments according to size corresponds to Gel electrophoresis.
- III. Amplification of a chosen DNA segment in vitro corresponds to PCR.
- IV. Legal protection for a novel biotechnological invention corresponds to Biopatent.

Which set contains all and only the valid conclusions?

- A. II and III only
- B. III and IV only
- C. I, II, III and IV
- D. I, II and IV only

BIO-12-CH12-0080 | Hard | Cloning vectors

Two conclusions must both be valid to support the reference concept 'Site permitting vector replication in a host'. Which option contains two accurate associations?

- A. Enzymes cutting DNA at specific recognition sequences - Bt cotton; Transgenic plant known for insect resistance - Restriction endonucleases
- B. Single-stranded ends produced by staggered restriction cuts - Restriction endonucleases; Enzymes cutting DNA at specific recognition sequences - Sticky ends
- C. Bt crop protection targets - Golden rice; Nutritionally enhanced transgenic crop example - Specific insect pests
- D. Site permitting vector replication in a host - Origin of replication; Restriction sequence reading the same in opposite directions - Palindromic sequence

BIO-12-CH12-0081 | Medium | rDNA methodology

Complete the statement accurately: Amplification of a chosen DNA segment in vitro corresponds to _____.

- A. GEAC
- B. PCR
- C. Restriction endonucleases
- D. Transformation

BIO-12-CH12-0082 | Medium | Medical applications

A learner encounters the observation 'Treatment by introducing a functional gene' while analysing medical applications. Which conclusion is supported by the prescribed biology?

- A. Origin of replication
- B. Gene therapy
- C. Recombinant DNA
- D. Human insulin

BIO-12-CH12-0083 | Challenging | Cloning vectors

A researcher checks whether four linked observations have been interpreted correctly.

- I. Gene allowing identification of transformed cells corresponds to Selectable marker.
- II. Vector region containing several unique restriction sites corresponds to Origin of replication.
- III. Site permitting vector replication in a host corresponds to Transgenic organism.
- IV. Organism carrying and expressing an introduced gene corresponds to Transgenic organism.

Which combination survives the biological review?

- A. I and IV only
- B. I, II and III only
- C. II, III and IV only
- D. I, III and IV only

BIO-12-CH12-0084 | Hard | Bioethics and ownership

A reviewer compares several entries while checking 'Ethical concern about altered organisms entering nature'. Which entry must be corrected?

- A. Restriction sequence reading the same in opposite directions - Palindromic sequence
- B. Ethical concern about altered organisms entering nature - Extension
- C. Nutritionally enhanced transgenic crop example - Golden rice
- D. Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase

BIO-12-CH12-0085 | Challenging | Restriction and joining enzymes

Evaluate the following source-based biological statements.

- I. Single-stranded ends produced by staggered restriction cuts corresponds to Restriction endonucleases.
- II. Enzymes cutting DNA at specific recognition sequences corresponds to Restriction endonucleases.
- III. Enzyme sealing phosphodiester bonds between DNA fragments corresponds to DNA ligase.
- IV. Site permitting vector replication in a host corresponds to Origin of replication.

Which option identifies exactly the correct statements?

- A. I, II and IV only
- B. III only
- C. II, III and IV only
- D. I and II only

BIO-12-CH12-0086 | Medium | Agricultural applications

A learner encounters the observation 'Transgenic plant known for insect resistance' while analysing agricultural applications. Which conclusion is supported by the prescribed biology?

- A. Bt cotton
- B. PCR-based diagnosis
- C. Specific insect pests
- D. Monoclonal antibody

BIO-12-CH12-0087 | Medium | rDNA methodology

A learner encounters the observation 'Separation of DNA fragments according to size' while analysing rdna methodology. Which conclusion is supported by the prescribed biology?

- A. Plasmid
- B. Gel electrophoresis
- C. PCR-based diagnosis
- D. Gene therapy

BIO-12-CH12-0088 | Medium | PCR and analysis

A learner encounters the observation 'PCR step in which polymerase synthesizes DNA' while analysing pcr and analysis. Which conclusion is supported by the prescribed biology?

- A. Extension
- B. Gel electrophoresis
- C. Sticky ends
- D. PCR

BIO-12-CH12-0089 | Challenging | Bioethics and ownership

A student submits four conclusions after integrating related chapter concepts.

- I. Ethical concern about altered organisms entering nature corresponds to Environmental release of GMOs.
- II. Legal protection for a novel biotechnological invention corresponds to GEAC.
- III. Indian committee that decides validity and safety of GMO research corresponds to GEAC.
- IV. Nutritionally enhanced transgenic crop example corresponds to Environmental release of GMOs.

Which combination should be accepted without correction?

- A. I and III only
- B. I and II only
- C. III only
- D. III and IV only

BIO-12-CH12-0090 | Hard | rDNA methodology

Two conclusions must both be valid to support the reference concept 'Introduction of recombinant DNA into a host cell'. Which option contains two accurate associations?

- A. Maintenance of sterile conditions for industrial microbial growth - Bt cotton; Transgenic plant known for insect resistance - Bioprocess engineering
- B. Transgenic plant known for insect resistance - Genetic engineering; Deliberate alteration of genetic material - Bt cotton
- C. Antibody produced from a single clone of cells - Bioprocess engineering; Maintenance of sterile conditions for industrial microbial growth - Monoclonal antibody
- D. Introduction of recombinant DNA into a host cell - Transformation; PCR step separating DNA strands - Denaturation

BIO-12-CH12-0091 | Medium | Core principles

A source table contains one accurate entry involving 'Maintenance of sterile conditions for industrial microbial growth'. Which option preserves a correct biological association?

- A. Vector region containing several unique restriction sites - Selectable marker
- B. Enzymes cutting DNA at specific recognition sequences - PCR-based diagnosis
- C. Early molecular diagnosis using amplification - Environmental release of GMOs
- D. Maintenance of sterile conditions for industrial microbial growth - Bioprocess engineering

BIO-12-CH12-0092 | Medium | Restriction and joining enzymes

Complete the statement accurately: Enzymes cutting DNA at specific recognition sequences corresponds to _____.

- A. Restriction endonucleases
- B. DNA ligase
- C. Origin of replication
- D. Monoclonal antibody

BIO-12-CH12-0093 | Medium | rDNA methodology

Complete the statement accurately: Introduction of recombinant DNA into a host cell corresponds to _____.

- A. PCR
- B. Genetic engineering
- C. Denaturation
- D. Transformation

BIO-12-CH12-0094 | Medium | Core principles

Complete the statement accurately: Maintenance of sterile conditions for industrial microbial growth corresponds to _____.

- A. Bioprocess engineering
- B. Multiple cloning site
- C. Restriction endonucleases
- D. Origin of replication

BIO-12-CH12-0095 | Medium | Cloning vectors

A source table contains one accurate entry involving 'Site permitting vector replication in a host'. Which option preserves a correct biological association?

- A. Bt crop protection targets - Origin of replication
- B. Single-stranded ends produced by staggered restriction cuts - Bt cotton
- C. Transgenic plant known for insect resistance - Specific insect pests
- D. Site permitting vector replication in a host - Origin of replication

BIO-12-CH12-0096 | Hard | Core principles

A reviewer compares several entries while checking 'Maintenance of sterile conditions for industrial microbial growth'. Which entry must be corrected?

- A. Early molecular diagnosis using amplification - PCR-based diagnosis
- B. Maintenance of sterile conditions for industrial microbial growth - Origin of replication
- C. Ethical concern about altered organisms entering nature - Environmental release of GMOs
- D. Vector region containing several unique restriction sites - Multiple cloning site

BIO-12-CH12-0097 | Medium | Agricultural applications

A source table contains one accurate entry involving 'Nutritionally enhanced transgenic crop example'. Which option preserves a correct biological association?

- A. Nutritionally enhanced transgenic crop example - Golden rice
- B. Single-stranded ends produced by staggered restriction cuts - GEAC
- C. Gene allowing identification of transformed cells - Golden rice
- D. Enzymes cutting DNA at specific recognition sequences - Cloning

BIO-12-CH12-0098 | Easy | Cloning vectors

Which biological term or result is specifically identified by the description 'Vector region containing several unique restriction sites'?

- A. Selectable marker
- B. Multiple cloning site
- C. Gel electrophoresis
- D. Biopiracy

BIO-12-CH12-0099 | Easy | Agricultural applications

Which biological term or result is specifically identified by the description 'Bt toxin gene source'?

- A. Plasmid
- B. Cloning
- C. Bt cotton
- D. *Bacillus thuringiensis*

BIO-12-CH12-0100 | Easy | rDNA methodology

Which description is correctly represented by Transformation in this chapter?

- A. PCR step separating DNA strands
- B. Deliberate alteration of genetic material
- C. Maintenance of sterile conditions for industrial microbial growth
- D. Introduction of recombinant DNA into a host cell

BIO-12-CH12-0101 | Challenging | Core principles

During error analysis, four proposed associations are reviewed.

- I. Deliberate alteration of genetic material corresponds to Genetic engineering.
- II. Organism carrying and expressing an introduced gene corresponds to Bioprocess engineering.
- III. Maintenance of sterile conditions for industrial microbial growth corresponds to Bioprocess engineering.
- IV. Transgenic plant known for insect resistance corresponds to Bt cotton.

Which set contains all and only the valid conclusions?

- A. II and IV only
- B. I, III and IV only
- C. III only
- D. I only

BIO-12-CH12-0102 | Hard | Bioethics and ownership

A reviewer compares several entries while checking 'Indian committee that decides validity and safety of GMO research'. Which entry must be corrected?

- A. Restriction sequence reading the same in opposite directions - Palindromic sequence
- B. Indian committee that decides validity and safety of GMO research - Gene therapy
- C. Bt toxin gene source - *Bacillus thuringiensis*
- D. Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase

BIO-12-CH12-0103 | Medium | Core principles

A learner encounters the observation 'Organism carrying and expressing an introduced gene' while analysing core principles. Which conclusion is supported by the prescribed biology?

- A. Transgenic organism
- B. Selectable marker
- C. Genetic engineering
- D. Recombinant DNA

BIO-12-CH12-0104 | Hard | Cloning vectors

A reviewer compares several entries while checking 'Gene allowing identification of transformed cells'. Which entry must be corrected?

- A. Gene allowing identification of transformed cells - Specific insect pests
- B. Enzyme sealing phosphodiester bonds between DNA fragments - DNA ligase
- C. Bt toxin gene source - *Bacillus thuringiensis*
- D. Amplification of a chosen DNA segment in vitro - PCR

BIO-12-CH12-0105 | Medium | Agricultural applications

Complete the statement accurately: Transgenic plant known for insect resistance corresponds to ____.

- A. *Bacillus thuringiensis*
- B. Bt cotton
- C. Sticky ends
- D. Monoclonal antibody

BIO-12-CH12-0106 | Hard | rDNA methodology

Two conclusions must both be valid to support the reference concept 'Production of many identical copies from one transformed cell'. Which option contains two accurate associations?

- A. Production of many identical copies from one transformed cell - Cloning; Bt crop protection targets - Specific insect pests
- B. PCR step allowing primers to bind - Denaturation; PCR step separating DNA strands - Annealing
- C. Enzymes cutting DNA at specific recognition sequences - Biopatent; Legal protection for a novel biotechnological invention - Restriction endonucleases
- D. Single-stranded ends produced by staggered restriction cuts - Annealing; PCR step allowing primers to bind - Sticky ends

BIO-12-CH12-0107 | Hard | Medical applications

Two conclusions must both be valid to support the reference concept 'Early molecular diagnosis using amplification'. Which option contains two accurate associations?

- A. DNA formed by joining sequences from different sources - Palindromic sequence; Restriction sequence reading the same in opposite directions - Recombinant DNA
- B. Transgenic plant known for insect resistance - Plasmid; Small circular bacterial DNA commonly used as a vector - Bt cotton
- C. Early molecular diagnosis using amplification - PCR-based diagnosis; Bt toxin gene source - Bacillus thuringiensis
- D. Small circular bacterial DNA commonly used as a vector - Taq polymerase; Thermostable polymerase commonly used in PCR - Plasmid

BIO-12-CH12-0108 | Easy | Restriction and joining enzymes

Which description is correctly represented by Restriction endonucleases in this chapter?

- A. Antibody produced from a single clone of cells
- B. Single-stranded ends produced by staggered restriction cuts
- C. Enzymes cutting DNA at specific recognition sequences
- D. Site permitting vector replication in a host

BIO-12-CH12-0109 | Easy | Restriction and joining enzymes

Which biological term or result is specifically identified by the description 'Enzymes cutting DNA at specific recognition sequences'?

- A. PCR
- B. Restriction endonucleases
- C. Palindromic sequence
- D. Bt cotton

BIO-12-CH12-0110 | Medium | Bioethics and ownership

Complete the statement accurately: Unfair commercial use of biological resources or traditional knowledge corresponds to _____.

- A. Biopiracy
- B. Recombinant DNA
- C. Environmental release of GMOs
- D. Restriction endonucleases

BIO-12-CH12-0111 | Medium | Agricultural applications

A learner encounters the observation 'Bt crop protection targets' while analysing agricultural applications. Which conclusion is supported by the prescribed biology?

- A. Denaturation
- B. Environmental release of GMOs
- C. Biopiracy
- D. Specific insect pests

BIO-12-CH12-0112 | Medium | Medical applications

Complete the statement accurately: Treatment by introducing a functional gene corresponds to _____.

- A. Gene therapy
 - B. Origin of replication
 - C. Plasmid
 - D. PCR-based diagnosis
-

Standard 12 | Chapter 13

Organisms and Populations

BIO-12-CH13-0001 | Medium | Age structure of populations

Which age pyramid indicates a rapidly expanding population?

Age structure of populations

Relative widths of three age groups



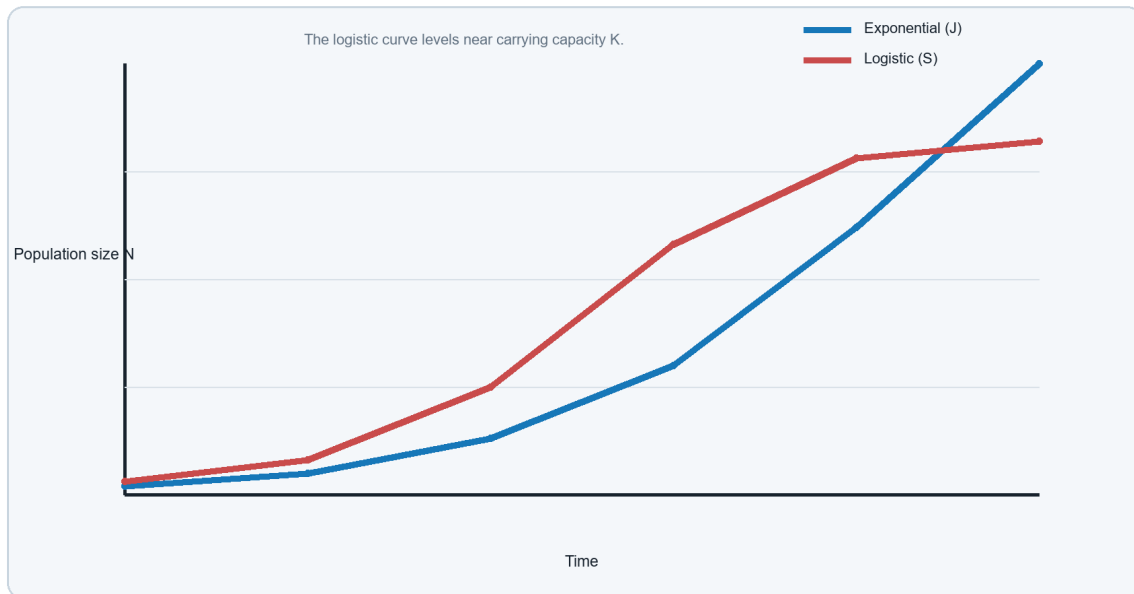
- A. The top-heavy declining pyramid
- B. A pyramid with no pre-reproductive group
- C. All shapes necessarily show zero growth
- D. The pyramid with a broad pre-reproductive base

BIO-12-CH13-0002 | Hard | Population growth models

Why does logistic population growth slow as N approaches K ?

Population growth models

Exponential growth compared with logistic growth



- A. Environmental resistance increases as resources become limiting
- B. Birth rate must become infinite
- C. Every individual changes species
- D. Competition disappears completely

BIO-12-CH13-0003 | Medium | Age pyramids

A source table contains one accurate entry involving 'Broad-base age pyramid'. Which option preserves a correct biological association?

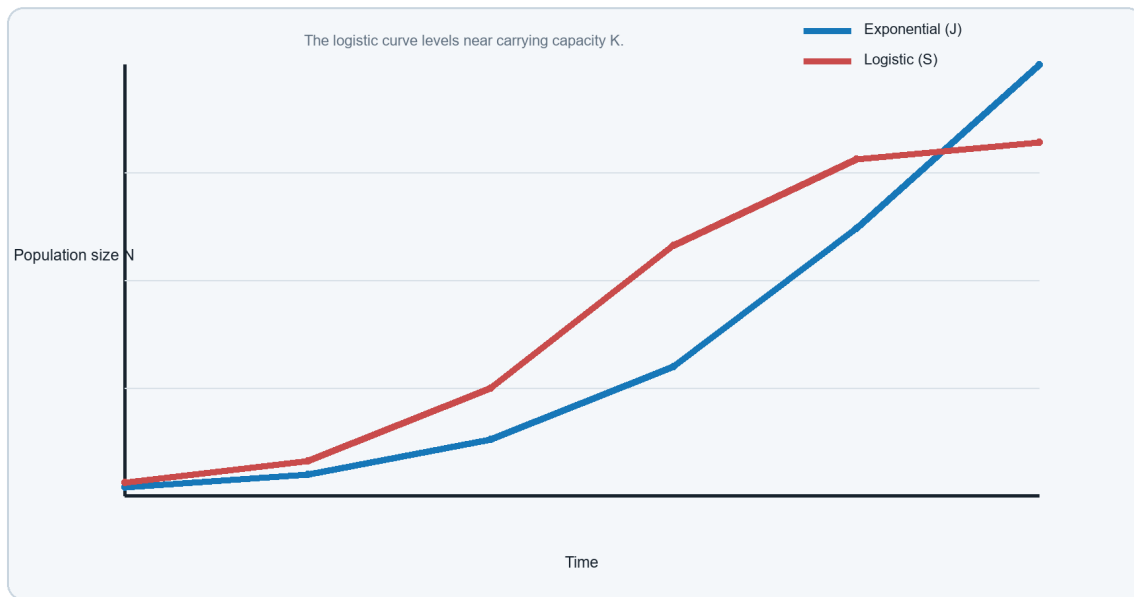
- A. One benefits and the other is unaffected - Exponential growth
- B. Broad-base age pyramid - Expanding population
- C. Intrinsic rate of natural increase - Expanding population
- D. Proportion of individuals in different age groups - Dormancy

BIO-12-CH13-0004 | Easy | Population growth models

In the logistic model, what does K represent?

Population growth models

Exponential growth compared with logistic growth



- A. Intrinsic mutation rate
- B. Carrying capacity
- C. Number of chromosomes
- D. Minimum viable age

BIO-12-CH13-0005 | Medium | Negative interactions

A source table contains one accurate entry involving 'One benefits while harming a host'. Which option preserves a correct biological association?

- A. One benefits while harming a host - Parasitism
- B. Individuals before reproductive maturity - Stable population
- C. Organisms maintaining a nearly constant internal temperature - Exponential growth
- D. Number of deaths during a given period - Parasitism

BIO-12-CH13-0006 | Hard | Positive interactions

A reviewer compares several entries while checking 'Lichen partnership'. Which entry must be corrected?

- A. Narrow-base age pyramid - Declining population
- B. Number of individuals per unit area or volume - Population density
- C. Lichen partnership - Logistic growth
- D. One organism kills and consumes another - Predation

BIO-12-CH13-0007 | Hard | Positive interactions

Two conclusions must both be valid to support the reference concept 'Both species benefit'. Which option contains two accurate associations?

- A. Organisms maintaining a nearly constant internal temperature - Biochemical cold adaptation; Production of antifreeze compounds in cold-water organisms - Regulators
- B. Number of births during a given period - Mortality; Number of deaths during a given period - Natality
- C. Both species benefit - Mutualism; Proportion of individuals in different age groups - Age structure
- D. Number of deaths during a given period - Regulators; Organisms maintaining a nearly constant internal temperature - Mortality

BIO-12-CH13-0008 | Medium | Age pyramids

A source table contains one accurate entry involving 'Nearly bell-shaped age pyramid'. Which option preserves a correct biological association?

- A. J-shaped population growth under unlimited resources - Amensalism
- B. Nearly bell-shaped age pyramid - Stable population
- C. One is inhibited while the other is unaffected - Commensalism
- D. One benefits while harming a host - Biotic potential

BIO-12-CH13-0009 | Easy | Negative interactions

Which biological term or result is specifically identified by the description 'Both species are adversely affected while sharing a limiting resource'?

- A. Biochemical cold adaptation
- B. Exponential growth
- C. Natality
- D. Competition

BIO-12-CH13-0010 | Medium | Population attributes

A learner encounters the observation 'Number of births during a given period' while analysing population attributes. Which conclusion is supported by the prescribed biology?

- A. Opuntia
- B. Biochemical cold adaptation
- C. Natality
- D. Population density

BIO-12-CH13-0011 | Hard | Age pyramids

A reviewer compares several entries while checking 'Nearly bell-shaped age pyramid'. Which entry must be corrected?

- A. Nearly bell-shaped age pyramid - Exponential growth
- B. One benefits while harming a host - Parasitism
- C. One organism kills and consumes another - Predation
- D. One benefits and the other is unaffected - Commensalism

BIO-12-CH13-0012 | Medium | Abiotic factors

A learner encounters the observation 'Organisms whose body temperature changes with surroundings' while analysing abiotic factors. Which conclusion is supported by the prescribed biology?

- A. Fungus and alga or cyanobacterium
- B. Dormancy
- C. Conformers
- D. Opuntia

BIO-12-CH13-0013 | Medium | Population growth

Complete the statement accurately: Maximum population an environment can sustain corresponds to _____.

- A. Logistic growth
- B. Pre-reproductive group
- C. Mutualism
- D. Carrying capacity

BIO-12-CH13-0014 | Medium | Positive interactions

Complete the statement accurately: Both species benefit corresponds to _____.

- A. Mutualism
- B. Biotic potential
- C. Commensalism
- D. Fungus and alga or cyanobacterium

BIO-12-CH13-0015 | Medium | Abiotic factors

Complete the statement accurately: Organisms maintaining a nearly constant internal temperature corresponds to _____.

- A. Parasitism
- B. Regulators
- C. Conformers
- D. Exponential growth

BIO-12-CH13-0016 | Easy | Population growth

Which biological term or result is specifically identified by the description 'Intrinsic rate of natural increase'?

- A. Regulators
- B. Biotic potential
- C. Exponential growth
- D. Symbiosis

BIO-12-CH13-0017 | Hard | Age pyramids

A reviewer compares several entries while checking 'Individuals before reproductive maturity'. Which entry must be corrected?

- A. Individuals before reproductive maturity - Commensalism
- B. One benefits while harming a host - Parasitism
- C. Number of births during a given period - Natality
- D. State of reduced activity during an unfavourable period - Dormancy

BIO-12-CH13-0018 | Medium | Population growth

A learner encounters the observation 'Intrinsic rate of natural increase' while analysing population growth. Which conclusion is supported by the prescribed biology?

- A. Regulators
- B. Parasitism
- C. Biotic potential
- D. Biochemical cold adaptation

BIO-12-CH13-0019 | Easy | Adaptations

Which biological term or result is specifically identified by the description 'Desert plant whose leaves are reduced to spines'?

- A. Opuntia
- B. Commensalism
- C. Regulators
- D. CAM

BIO-12-CH13-0020 | Medium | Adaptations

A source table contains one accurate entry involving 'Photosynthetic pathway that lets desert plants keep stomata closed by day'. Which option preserves a correct biological association?

- A. Lichen partnership - Expanding population
- B. Broad-base age pyramid - CAM
- C. One organism kills and consumes another - Fungus and alga or cyanobacterium
- D. Photosynthetic pathway that lets desert plants keep stomata closed by day - CAM

BIO-12-CH13-0021 | Challenging | Age structure of populations

A population has fewer pre-reproductive than reproductive individuals. What future trend is most likely if rates remain similar?

Age structure of populations

Relative widths of three age groups



- A. Population size will tend to decline
- B. Population size must grow exponentially
- C. Population instantly reaches infinite carrying capacity
- D. Age structure has no demographic implication

BIO-12-CH13-0022 | Medium | Positive interactions

A source table contains one accurate entry involving 'Lichen partnership'. Which option preserves a correct biological association?

- A. Lichen partnership - Fungus and alga or cyanobacterium
- B. One is inhibited while the other is unaffected - Population density
- C. Number of individuals per unit area or volume - Fungus and alga or cyanobacterium
- D. One organism kills and consumes another - Declining population

BIO-12-CH13-0023 | Easy | Age pyramids

Which biological term or result is specifically identified by the description 'Nearly bell-shaped age pyramid'?

- A. Commensalism
- B. Predation
- C. Expanding population
- D. Stable population

BIO-12-CH13-0024 | Challenging | Population attributes

Evaluate the following source-based biological statements.

- I. Proportion of individuals in different age groups corresponds to Age structure.
- II. Number of individuals per unit area or volume corresponds to Population density.
- III. Number of births during a given period corresponds to Natality.
- IV. Narrow-base age pyramid corresponds to Declining population.

Which option identifies exactly the correct statements?

- A. I, II, III and IV
- B. III and IV only
- C. I, II and IV only
- D. II, III and IV only

BIO-12-CH13-0025 | Challenging | Positive interactions

A student submits four conclusions after integrating related chapter concepts.

- I. Both species benefit corresponds to Mutualism.
- II. Close association often benefiting both partners corresponds to Commensalism.
- III. One benefits and the other is unaffected corresponds to Age structure.
- IV. Proportion of individuals in different age groups corresponds to Age structure.

Which combination should be accepted without correction?

- A. I and IV only
- B. II, III and IV only
- C. II and III only
- D. III only

BIO-12-CH13-0026 | Medium | Positive interactions

A source table contains one accurate entry involving 'One benefits and the other is unaffected'. Which option preserves a correct biological association?

- A. State of reduced activity during an unfavourable period - Commensalism
- B. Both species are adversely affected while sharing a limiting resource - Population density
- C. One benefits and the other is unaffected - Commensalism
- D. Organisms whose body temperature changes with surroundings - Dormancy

BIO-12-CH13-0027 | Hard | Age pyramids

A reviewer compares several entries while checking 'Narrow-base age pyramid'. Which entry must be corrected?

- A. State of reduced activity during an unfavourable period - Dormancy
- B. Both species benefit - Mutualism
- C. Close association often benefiting both partners - Symbiosis
- D. Narrow-base age pyramid - Conformers

BIO-12-CH13-0028 | Medium | Population growth

A source table contains one accurate entry involving 'J-shaped population growth under unlimited resources'. Which option preserves a correct biological association?

- A. Photosynthetic pathway that lets desert plants keep stomata closed by day - Dormancy
- B. Number of individuals per unit area or volume - Pre-reproductive group
- C. One organism kills and consumes another - CAM
- D. J-shaped population growth under unlimited resources - Exponential growth

BIO-12-CH13-0029 | Medium | Age pyramids

A learner encounters the observation 'Broad-base age pyramid' while analysing age pyramids. Which conclusion is supported by the prescribed biology?

- A. Expanding population
- B. Opuntia
- C. Stable population
- D. Age structure

BIO-12-CH13-0030 | Medium | Negative interactions

A source table contains one accurate entry involving 'One is inhibited while the other is unaffected'. Which option preserves a correct biological association?

- A. State of reduced activity during an unfavourable period - Biotic potential
- B. Intrinsic rate of natural increase - Water-conservation adaptations
- C. One is inhibited while the other is unaffected - Amensalism
- D. Concentrated urine and reduced sweating in desert mammals - Expanding population

BIO-12-CH13-0031 | Hard | Negative interactions

Two conclusions must both be valid to support the reference concept 'One organism kills and consumes another'. Which option contains two accurate associations?

- A. Close association often benefiting both partners - Biochemical cold adaptation; Production of antifreeze compounds in cold-water organisms - Symbiosis
- B. State of reduced activity during an unfavourable period - Symbiosis; Close association often benefiting both partners - Dormancy
- C. Narrow-base age pyramid - Dormancy; State of reduced activity during an unfavourable period - Declining population
- D. One organism kills and consumes another - Predation; Individuals before reproductive maturity - Pre-reproductive group

BIO-12-CH13-0032 | Medium | Positive interactions

Complete the statement accurately: One benefits and the other is unaffected corresponds to _____.

- A. Conformers
- B. Parasitism
- C. Commensalism
- D. Population density

BIO-12-CH13-0033 | Challenging | Population growth

During error analysis, four proposed associations are reviewed.

- I. S-shaped growth under resource limitation corresponds to Carrying capacity.
- II. Maximum population an environment can sustain corresponds to Carrying capacity.
- III. J-shaped population growth under unlimited resources corresponds to Exponential growth.
- IV. One benefits while harming a host corresponds to Parasitism.

Which set contains all and only the valid conclusions?

- A. II only
- B. I only
- C. I, II and IV only
- D. II, III and IV only

BIO-12-CH13-0034 | Challenging | Population growth

A researcher checks whether four linked observations have been interpreted correctly.

- I. Intrinsic rate of natural increase corresponds to Biotic potential.
- II. J-shaped population growth under unlimited resources corresponds to Carrying capacity.
- III. Maximum population an environment can sustain corresponds to Carrying capacity.
- IV. Number of individuals per unit area or volume corresponds to Biotic potential.

Which combination survives the biological review?

- A. IV only
- B. I only
- C. II and IV only
- D. I and III only

BIO-12-CH13-0035 | Hard | Positive interactions

A reviewer compares several entries while checking 'Both species benefit'. Which entry must be corrected?

- A. Both species benefit - Age structure
- B. Intrinsic rate of natural increase - Biotic potential
- C. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation
- D. Number of births during a given period - Natality

BIO-12-CH13-0036 | Easy | Population growth

Which biological term or result is specifically identified by the description 'J-shaped population growth under unlimited resources'?

- A. Biotic potential
- B. Carrying capacity
- C. Water-conservation adaptations
- D. Exponential growth

BIO-12-CH13-0037 | Medium | Negative interactions

A learner encounters the observation 'One organism kills and consumes another' while analysing negative interactions. Which conclusion is supported by the prescribed biology?

- A. Dormancy
- B. Declining population
- C. Biochemical cold adaptation
- D. Predation

BIO-12-CH13-0038 | Medium | Population attributes

Complete the statement accurately: Number of deaths during a given period corresponds to _____.

- A. Biochemical cold adaptation
- B. Commensalism
- C. Mortality
- D. Conformers

BIO-12-CH13-0039 | Medium | Population growth

A source table contains one accurate entry involving 'S-shaped growth under resource limitation'. Which option preserves a correct biological association?

- A. S-shaped growth under resource limitation - Logistic growth
- B. Number of deaths during a given period - Natality
- C. Close association often benefiting both partners - Logistic growth
- D. Number of individuals per unit area or volume - Mortality

BIO-12-CH13-0040 | Easy | Population growth

Which biological term or result is specifically identified by the description 'Maximum population an environment can sustain'?

- A. Mutualism
- B. Exponential growth
- C. Expanding population
- D. Carrying capacity

BIO-12-CH13-0041 | Easy | Abiotic factors

Which description is correctly represented by Dormancy in this chapter?

- A. State of reduced activity during an unfavourable period
- B. Seasonal movement to a favourable habitat
- C. One benefits and the other is unaffected
- D. Organisms maintaining a nearly constant internal temperature

BIO-12-CH13-0042 | Hard | Population growth

A reviewer compares several entries while checking 'J-shaped population growth under unlimited resources'. Which entry must be corrected?

- A. J-shaped population growth under unlimited resources - Population density
- B. One organism kills and consumes another - Predation
- C. Individuals before reproductive maturity - Pre-reproductive group
- D. State of reduced activity during an unfavourable period - Dormancy

BIO-12-CH13-0043 | Medium | Adaptations

A source table contains one accurate entry involving 'Concentrated urine and reduced sweating in desert mammals'. Which option preserves a correct biological association?

- A. Organisms whose body temperature changes with surroundings - Expanding population
- B. Concentrated urine and reduced sweating in desert mammals - Water-conservation adaptations
- C. One benefits and the other is unaffected - Predation
- D. Organisms maintaining a nearly constant internal temperature - Commensalism

BIO-12-CH13-0044 | Challenging | Abiotic factors

Evaluate the following source-based biological statements.

- I. Organisms whose body temperature changes with surroundings corresponds to Conformers.
- II. State of reduced activity during an unfavourable period corresponds to Regulators.
- III. Organisms maintaining a nearly constant internal temperature corresponds to Regulators.
- IV. One benefits and the other is unaffected corresponds to Commensalism.

Which option identifies exactly the correct statements?

- A. II and III only
- B. I only
- C. I, III and IV only
- D. III only

BIO-12-CH13-0045 | Hard | Population growth

A reviewer compares several entries while checking 'Intrinsic rate of natural increase'. Which entry must be corrected?

- A. Intrinsic rate of natural increase - Symbiosis
- B. Organisms maintaining a nearly constant internal temperature - Regulators
- C. One benefits while harming a host - Parasitism
- D. Photosynthetic pathway that lets desert plants keep stomata closed by day - CAM

BIO-12-CH13-0046 | Easy | Positive interactions

Which biological term or result is specifically identified by the description 'Close association often benefiting both partners'?

- A. Fungus and alga or cyanobacterium
- B. Symbiosis
- C. Mutualism
- D. Migration

BIO-12-CH13-0047 | Easy | Negative interactions

Which description is correctly represented by Amensalism in this chapter?

- A. One benefits while harming a host
- B. State of reduced activity during an unfavourable period
- C. Intrinsic rate of natural increase
- D. One is inhibited while the other is unaffected

BIO-12-CH13-0048 | Medium | Age pyramids

A learner encounters the observation 'Individuals before reproductive maturity' while analysing age pyramids. Which conclusion is supported by the prescribed biology?

- A. Commensalism
- B. Parasitism
- C. Symbiosis
- D. Pre-reproductive group

BIO-12-CH13-0049 | Medium | Positive interactions

A source table contains one accurate entry involving 'Both species benefit'. Which option preserves a correct biological association?

- A. Number of deaths during a given period - Biochemical cold adaptation
- B. Both species benefit - Mutualism
- C. Production of antifreeze compounds in cold-water organisms - Mutualism
- D. Number of births during a given period - Regulators

BIO-12-CH13-0050 | Hard | Population attributes

A reviewer compares several entries while checking 'Proportion of individuals in different age groups'. Which entry must be corrected?

- A. Photosynthetic pathway that lets desert plants keep stomata closed by day - CAM
- B. Close association often benefiting both partners - Symbiosis
- C. Proportion of individuals in different age groups - Carrying capacity
- D. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation

BIO-12-CH13-0051 | Medium | Adaptations

A learner encounters the observation 'Photosynthetic pathway that lets desert plants keep stomata closed by day' while analysing adaptations. Which conclusion is supported by the prescribed biology?

- A. Migration
- B. Fungus and alga or cyanobacterium
- C. Water-conservation adaptations
- D. CAM

BIO-12-CH13-0052 | Easy | Positive interactions

Which biological term or result is specifically identified by the description 'Both species benefit'?

- A. Mortality
- B. Mutualism
- C. Biotic potential
- D. Natality

BIO-12-CH13-0053 | Medium | Negative interactions

Complete the statement accurately: One benefits while harming a host corresponds to _____.

- A. Pre-reproductive group
- B. Parasitism
- C. Amensalism
- D. Stable population

BIO-12-CH13-0054 | Easy | Adaptations

Which biological term or result is specifically identified by the description 'Photosynthetic pathway that lets desert plants keep stomata closed by day'?

- A. Mortality
- B. Predation
- C. Migration
- D. CAM

BIO-12-CH13-0055 | Medium | Population attributes

Complete the statement accurately: Number of individuals per unit area or volume corresponds to _____.

- A. Biochemical cold adaptation
- B. Age structure
- C. Population density
- D. Water-conservation adaptations

BIO-12-CH13-0056 | Easy | Negative interactions

Which description is correctly represented by Predation in this chapter?

- A. Production of antifreeze compounds in cold-water organisms
- B. One organism kills and consumes another
- C. State of reduced activity during an unfavourable period
- D. Close association often benefiting both partners

BIO-12-CH13-0057 | Medium | Abiotic factors

Complete the statement accurately: Seasonal movement to a favourable habitat corresponds to _____.

- A. Regulators
- B. Competition
- C. Conformers
- D. Migration

BIO-12-CH13-0058 | Hard | Abiotic factors

A reviewer compares several entries while checking 'Seasonal movement to a favourable habitat'. Which entry must be corrected?

- A. Seasonal movement to a favourable habitat - Pre-reproductive group
- B. Proportion of individuals in different age groups - Age structure
- C. Both species are adversely affected while sharing a limiting resource - Competition
- D. One organism kills and consumes another - Predation

BIO-12-CH13-0059 | Medium | Population attributes

A learner encounters the observation 'Proportion of individuals in different age groups' while analysing population attributes. Which conclusion is supported by the prescribed biology?

- A. Population density
- B. Age structure
- C. Logistic growth
- D. Carrying capacity

BIO-12-CH13-0060 | Medium | Positive interactions

A learner encounters the observation 'Lichen partnership' while analysing positive interactions. Which conclusion is supported by the prescribed biology?

- A. Symbiosis
- B. Fungus and alga or cyanobacterium
- C. Mortality
- D. Predation

BIO-12-CH13-0061 | Hard | Negative interactions

A reviewer compares several entries while checking 'Both species are adversely affected while sharing a limiting resource'. Which entry must be corrected?

- A. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation
- B. Broad-base age pyramid - Expanding population
- C. Both species are adversely affected while sharing a limiting resource - Exponential growth
- D. Number of births during a given period - Natality

BIO-12-CH13-0062 | Hard | Adaptations

A reviewer compares several entries while checking 'Desert plant whose leaves are reduced to spines'. Which entry must be corrected?

- A. Individuals before reproductive maturity - Pre-reproductive group
- B. One benefits and the other is unaffected - Commensalism
- C. Desert plant whose leaves are reduced to spines - Regulators
- D. Intrinsic rate of natural increase - Biotic potential

BIO-12-CH13-0063 | Hard | Population attributes

A reviewer compares several entries while checking 'Number of births during a given period'. Which entry must be corrected?

- A. Number of births during a given period - Dormancy
- B. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation
- C. Organisms maintaining a nearly constant internal temperature - Regulators
- D. Desert plant whose leaves are reduced to spines - Opuntia

BIO-12-CH13-0064 | Hard | Abiotic factors

Two conclusions must both be valid to support the reference concept 'Organisms whose body temperature changes with surroundings'. Which option contains two accurate associations?

- A. Organisms whose body temperature changes with surroundings - Conformers; S-shaped growth under resource limitation - Logistic growth
- B. Proportion of individuals in different age groups - Competition; Both species are adversely affected while sharing a limiting resource - Age structure
- C. Narrow-base age pyramid - Amensalism; One is inhibited while the other is unaffected - Declining population
- D. Both species are adversely affected while sharing a limiting resource - Declining population; Narrow-base age pyramid - Competition

BIO-12-CH13-0065 | Easy | Negative interactions

Which biological term or result is specifically identified by the description 'One organism kills and consumes another'?

- A. Amensalism
- B. Declining population
- C. Symbiosis
- D. Predation

BIO-12-CH13-0066 | Hard | Age pyramids

Two conclusions must both be valid to support the reference concept 'Broad-base age pyramid'. Which option contains two accurate associations?

- A. J-shaped population growth under unlimited resources - Opuntia; Desert plant whose leaves are reduced to spines - Exponential growth
- B. Desert plant whose leaves are reduced to spines - Biotic potential; Intrinsic rate of natural increase - Opuntia
- C. Intrinsic rate of natural increase - Water-conservation adaptations; Concentrated urine and reduced sweating in desert mammals - Biotic potential
- D. Broad-base age pyramid - Expanding population; Proportion of individuals in different age groups - Age structure

BIO-12-CH13-0067 | Challenging | Abiotic factors

A student submits four conclusions after integrating related chapter concepts.

- I. State of reduced activity during an unfavourable period corresponds to Dormancy.
- II. Organisms whose body temperature changes with surroundings corresponds to Conformers.
- III. Organisms maintaining a nearly constant internal temperature corresponds to Carrying capacity.
- IV. Maximum population an environment can sustain corresponds to Dormancy.

Which combination should be accepted without correction?

- A. II and IV only
- B. I and II only
- C. I and IV only
- D. I, III and IV only

BIO-12-CH13-0068 | Medium | Abiotic factors

A source table contains one accurate entry involving 'Organisms whose body temperature changes with surroundings'. Which option preserves a correct biological association?

- A. Proportion of individuals in different age groups - Declining population
- B. Desert plant whose leaves are reduced to spines - Competition
- C. Organisms whose body temperature changes with surroundings - Conformers
- D. S-shaped growth under resource limitation - Opuntia

BIO-12-CH13-0069 | Hard | Abiotic factors

Two conclusions must both be valid to support the reference concept 'State of reduced activity during an unfavourable period'. Which option contains two accurate associations?

- A. Production of antifreeze compounds in cold-water organisms - Mutualism; Both species benefit - Biochemical cold adaptation
- B. Individuals before reproductive maturity - Biochemical cold adaptation; Production of antifreeze compounds in cold-water organisms - Pre-reproductive group
- C. State of reduced activity during an unfavourable period - Dormancy; S-shaped growth under resource limitation - Logistic growth
- D. One benefits and the other is unaffected - Pre-reproductive group; Individuals before reproductive maturity - Commensalism

BIO-12-CH13-0070 | Medium | Abiotic factors

Complete the statement accurately: State of reduced activity during an unfavourable period corresponds to _____.

- A. Commensalism
- B. Dormancy
- C. Pre-reproductive group
- D. Mutualism

BIO-12-CH13-0071 | Easy | Population attributes

Which biological term or result is specifically identified by the description 'Proportion of individuals in different age groups'?

- A. Age structure
- B. Logistic growth
- C. Population density
- D. Mortality

BIO-12-CH13-0072 | Medium | Population attributes

A source table contains one accurate entry involving 'Proportion of individuals in different age groups'. Which option preserves a correct biological association?

- A. Maximum population an environment can sustain - Symbiosis
- B. Proportion of individuals in different age groups - Age structure
- C. Photosynthetic pathway that lets desert plants keep stomata closed by day - Biochemical cold adaptation
- D. Nearly bell-shaped age pyramid - Age structure

BIO-12-CH13-0073 | Hard | Population growth

A reviewer compares several entries while checking 'Maximum population an environment can sustain'. Which entry must be corrected?

- A. Individuals before reproductive maturity - Pre-reproductive group
- B. Maximum population an environment can sustain - Mutualism
- C. Proportion of individuals in different age groups - Age structure
- D. Broad-base age pyramid - Expanding population

BIO-12-CH13-0074 | Hard | Adaptations

A reviewer compares several entries while checking 'Photosynthetic pathway that lets desert plants keep stomata closed by day'. Which entry must be corrected?

- A. Seasonal movement to a favourable habitat - Migration
- B. Number of deaths during a given period - Mortality
- C. Photosynthetic pathway that lets desert plants keep stomata closed by day - Symbiosis
- D. Lichen partnership - Fungus and alga or cyanobacterium

BIO-12-CH13-0075 | Easy | Population attributes

Which biological term or result is specifically identified by the description 'Number of births during a given period'?

- A. Regulators
- B. Opuntia
- C. Mortality
- D. Natality

BIO-12-CH13-0076 | Easy | Age pyramids

Which description is correctly represented by Expanding population in this chapter?

- A. State of reduced activity during an unfavourable period
- B. Broad-base age pyramid
- C. J-shaped population growth under unlimited resources
- D. Individuals before reproductive maturity

BIO-12-CH13-0077 | Medium | Negative interactions

Complete the statement accurately: One is inhibited while the other is unaffected corresponds to _____.

- A. Biochemical cold adaptation
- B. Population density
- C. Amensalism
- D. Water-conservation adaptations

BIO-12-CH13-0078 | Easy | Adaptations

Which biological term or result is specifically identified by the description 'Concentrated urine and reduced sweating in desert mammals'?

- A. CAM
- B. Water-conservation adaptations
- C. Biochemical cold adaptation
- D. Regulators

BIO-12-CH13-0079 | Hard | Positive interactions

A reviewer compares several entries while checking 'Close association often benefiting both partners'. Which entry must be corrected?

- A. Close association often benefiting both partners - Amensalism
- B. Organisms whose body temperature changes with surroundings - Conformers
- C. Broad-base age pyramid - Expanding population
- D. Proportion of individuals in different age groups - Age structure

BIO-12-CH13-0080 | Hard | Negative interactions

A reviewer compares several entries while checking 'One organism kills and consumes another'. Which entry must be corrected?

- A. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation
- B. One organism kills and consumes another - Pre-reproductive group
- C. Close association often benefiting both partners - Symbiosis
- D. State of reduced activity during an unfavourable period - Dormancy

BIO-12-CH13-0081 | Hard | Population attributes

Two conclusions must both be valid to support the reference concept 'Number of individuals per unit area or volume'. Which option contains two accurate associations?

- A. Close association often benefiting both partners - Biochemical cold adaptation; Production of antifreeze compounds in cold-water organisms - Symbiosis
- B. One benefits while harming a host - Symbiosis; Close association often benefiting both partners - Parasitism
- C. Number of individuals per unit area or volume - Population density; Narrow-base age pyramid - Declining population
- D. Concentrated urine and reduced sweating in desert mammals - Parasitism; One benefits while harming a host - Water-conservation adaptations

BIO-12-CH13-0082 | Easy | Abiotic factors

Which biological term or result is specifically identified by the description 'Seasonal movement to a favourable habitat'?

- A. Exponential growth
- B. Migration
- C. Population density
- D. Competition

BIO-12-CH13-0083 | Easy | Positive interactions

Which biological term or result is specifically identified by the description 'Lichen partnership'?

- A. Logistic growth
- B. Fungus and alga or cyanobacterium
- C. Symbiosis
- D. Mortality

BIO-12-CH13-0084 | Easy | Age pyramids

Which biological term or result is specifically identified by the description 'Individuals before reproductive maturity'?

- A. Natality
- B. Mutualism
- C. Pre-reproductive group
- D. Declining population

BIO-12-CH13-0085 | Easy | Abiotic factors

Which description is correctly represented by Conformers in this chapter?

- A. Organisms whose body temperature changes with surroundings
- B. Both species are adversely affected while sharing a limiting resource
- C. Lichen partnership
- D. Desert plant whose leaves are reduced to spines

BIO-12-CH13-0086 | Easy | Positive interactions

Which description is correctly represented by Mutualism in this chapter?

- A. Number of births during a given period
- B. Number of deaths during a given period
- C. Lichen partnership
- D. Both species benefit

BIO-12-CH13-0087 | Challenging | Negative interactions

During error analysis, four proposed associations are reviewed.

- I. One organism kills and consumes another corresponds to Predation.
- II. Both species are adversely affected while sharing a limiting resource corresponds to Competition.
- III. One is inhibited while the other is unaffected corresponds to Mortality.
- IV. Number of deaths during a given period corresponds to Mortality.

Which set contains all and only the valid conclusions?

- A. I, III and IV only
- B. II and IV only
- C. I, II and IV only
- D. II only

BIO-12-CH13-0088 | Medium | Population growth

Complete the statement accurately: Intrinsic rate of natural increase corresponds to ____.

- A. Biotic potential
- B. Biochemical cold adaptation
- C. Logistic growth
- D. Symbiosis

BIO-12-CH13-0089 | Medium | Population growth

Complete the statement accurately: S-shaped growth under resource limitation corresponds to _____.

- A. Regulators
- B. Exponential growth
- C. Logistic growth
- D. Commensalism

BIO-12-CH13-0090 | Medium | Age pyramids

Complete the statement accurately: Broad-base age pyramid corresponds to _____.

- A. Stable population
- B. Exponential growth
- C. Pre-reproductive group
- D. Expanding population

BIO-12-CH13-0091 | Easy | Population attributes

Which biological term or result is specifically identified by the description 'Number of individuals per unit area or volume'?

- A. Age structure
- B. Parasitism
- C. Population density
- D. Symbiosis

BIO-12-CH13-0092 | Easy | Age pyramids

Which biological term or result is specifically identified by the description 'Narrow-base age pyramid'?

- A. Biochemical cold adaptation
- B. Declining population
- C. Age structure
- D. Pre-reproductive group

BIO-12-CH13-0093 | Challenging | Adaptations

A researcher checks whether four linked observations have been interpreted correctly.

- I. Production of antifreeze compounds in cold-water organisms corresponds to *Opuntia*.
- II. Desert plant whose leaves are reduced to spines corresponds to Water-conservation adaptations.
- III. Concentrated urine and reduced sweating in desert mammals corresponds to Conformers.
- IV. Organisms whose body temperature changes with surroundings corresponds to Conformers.

Which combination survives the biological review?

- A. I, III and IV only
- B. IV only
- C. I only
- D. I and II only

BIO-12-CH13-0094 | Medium | Adaptations

A source table contains one accurate entry involving 'Production of antifreeze compounds in cold-water organisms'. Which option preserves a correct biological association?

- A. Broad-base age pyramid - Amensalism
- B. Number of individuals per unit area or volume - Declining population
- C. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation
- D. State of reduced activity during an unfavourable period - Expanding population

BIO-12-CH13-0095 | Hard | Negative interactions

Two conclusions must both be valid to support the reference concept 'One is inhibited while the other is unaffected'. Which option contains two accurate associations?

- A. One is inhibited while the other is unaffected - Amensalism; State of reduced activity during an unfavourable period - Dormancy
- B. Number of individuals per unit area or volume - Water-conservation adaptations; Concentrated urine and reduced sweating in desert mammals - Population density
- C. Production of antifreeze compounds in cold-water organisms - Biotic potential; Intrinsic rate of natural increase - Biochemical cold adaptation
- D. Organisms maintaining a nearly constant internal temperature - Expanding population; Broad-base age pyramid - Regulators

BIO-12-CH13-0096 | Medium | Population attributes

A learner encounters the observation 'Number of deaths during a given period' while analysing population attributes. Which conclusion is supported by the prescribed biology?

- A. Population density
- B. Biotic potential
- C. Mortality
- D. Age structure

BIO-12-CH13-0097 | Hard | Population attributes

A reviewer compares several entries while checking 'Number of individuals per unit area or volume'. Which entry must be corrected?

- A. Close association often benefiting both partners - Symbiosis
- B. Number of individuals per unit area or volume - Declining population
- C. Production of antifreeze compounds in cold-water organisms - Biochemical cold adaptation
- D. Seasonal movement to a favourable habitat - Migration

BIO-12-CH13-0098 | Challenging | Negative interactions

Evaluate the following source-based biological statements.

- I. One is inhibited while the other is unaffected corresponds to Amensalism.
- II. One organism kills and consumes another corresponds to Predation.
- III. Both species are adversely affected while sharing a limiting resource corresponds to Competition.
- IV. Both species benefit corresponds to Amensalism.

Which option identifies exactly the correct statements?

- A. III only
- B. I and IV only
- C. II, III and IV only
- D. I, II and III only

BIO-12-CH13-0099 | Medium | Adaptations

Complete the statement accurately: Concentrated urine and reduced sweating in desert mammals corresponds to _____.

- A. Water-conservation adaptations
- B. Conformers
- C. Dormancy
- D. Regulators

BIO-12-CH13-0100 | Medium | Abiotic factors

A source table contains one accurate entry involving 'State of reduced activity during an unfavourable period'. Which option preserves a correct biological association?

- A. Maximum population an environment can sustain - Dormancy
 - B. Both species benefit - Expanding population
 - C. Production of antifreeze compounds in cold-water organisms - Carrying capacity
 - D. State of reduced activity during an unfavourable period - Dormancy
-

Standard 12 | Chapter 14

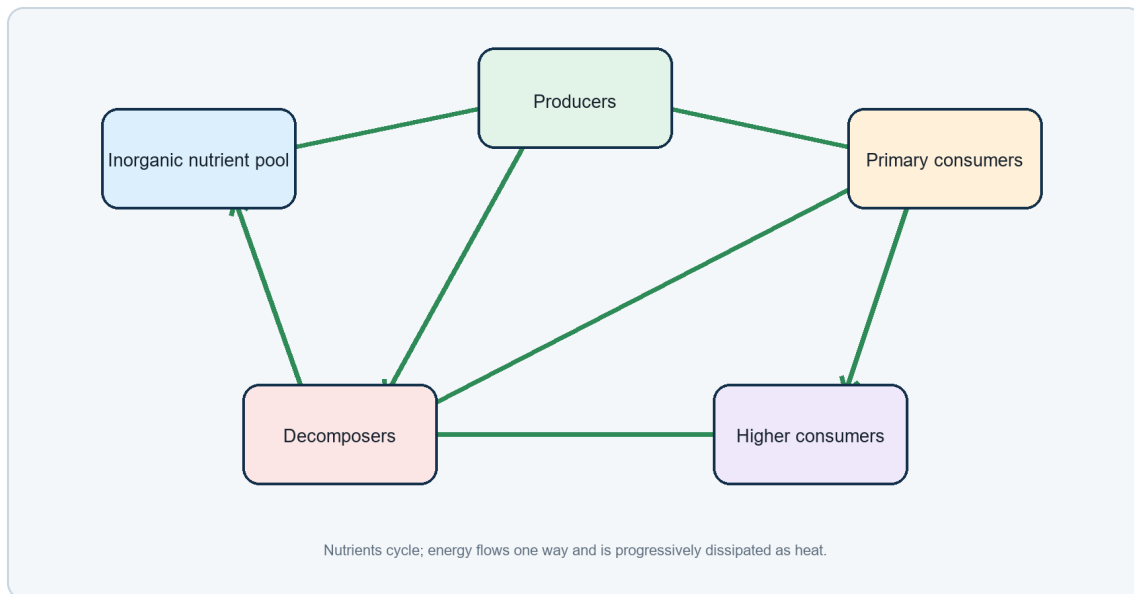
Ecosystems and Energy Flow

BIO-12-CH14-0001 | Medium | Matter flow through an ecosystem

Which group returns inorganic nutrients to the nutrient pool by breaking down dead organic matter?

Matter flow through an ecosystem

Producers, consumers and decomposers are linked by transfers



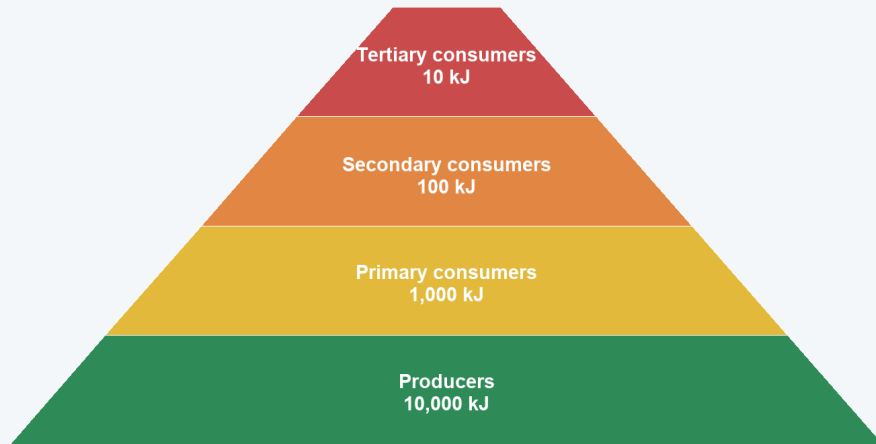
- A. Producers only
- B. Decomposers
- C. Primary consumers only
- D. Higher consumers only

BIO-12-CH14-0002 | Hard | Pyramid of energy

Why is the energy pyramid always upright?

Pyramid of energy

Illustration of the approximate ten-percent transfer rule



Energy is lost as heat and through metabolism at every transfer; the energy pyramid is always upright.

- A. Usable energy is lost at each trophic transfer
- B. Energy is fully recycled by decomposers
- C. Consumers manufacture more solar energy
- D. Tertiary consumers always have the greatest biomass

BIO-12-CH14-0003 | Hard | Succession and services

A reviewer compares several entries while checking 'Succession beginning on a previously lifeless surface'. Which entry must be corrected?

- A. Conversion of organic nutrients to inorganic forms - Mineralization
- B. Pyramid showing standing crop - Pyramid of biomass
- C. Succession beginning on a previously lifeless surface - Pyramid of numbers
- D. Gross primary productivity minus plant respiration - Net primary productivity

BIO-12-CH14-0004 | Medium | Nutrient cycles

Complete the statement accurately: Major reservoir of phosphorus corresponds to _____.

- A. Sedimentary rocks
- B. Decomposers
- C. Primary succession
- D. Energy pyramid

BIO-12-CH14-0005 | Hard | Decomposition steps

Two conclusions must both be valid to support the reference concept 'Enzymatic breakdown by bacteria and fungi'. Which option contains two accurate associations?

- A. Total photosynthetic production before respiratory loss - Pyramid of biomass; Pyramid showing standing crop - Gross primary productivity
- B. Succession beginning on a previously lifeless surface - Detritus food chain; Food chain beginning with dead organic matter - Primary succession
- C. Enzymatic breakdown by bacteria and fungi - Catabolism; Organisms breaking down dead organic matter - Decomposers
- D. Autotrophs converting solar energy to chemical energy - Sedimentary rocks; Major reservoir of phosphorus - Producers

BIO-12-CH14-0006 | Medium | Food chains and energy

Complete the statement accurately: Interconnected food chains corresponds to _____.

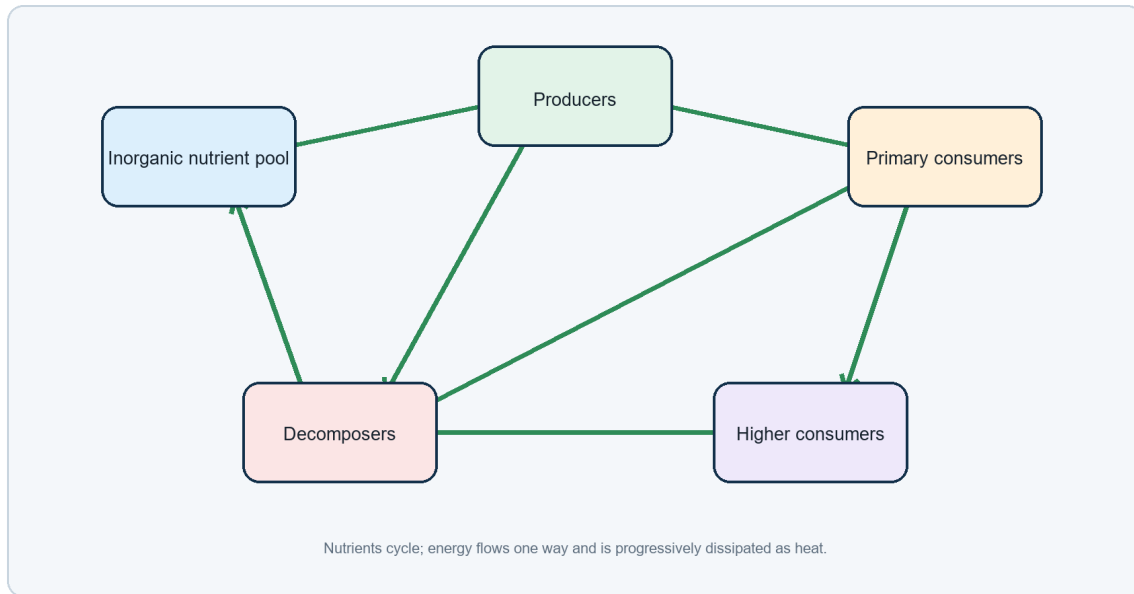
- A. Net primary productivity
- B. Catabolism
- C. Food web
- D. Leaching

BIO-12-CH14-0007 | Challenging | Matter flow through an ecosystem

If primary consumers are abruptly removed, what short-term pattern is most plausible?

Matter flow through an ecosystem

Producers, consumers and decomposers are linked by transfers



- A. Producer biomass immediately becomes zero
- B. Producer biomass rises while higher consumers lose a major food pathway
- C. Higher consumers gain unlimited energy
- D. Nutrient cycling stops permanently in every compartment

BIO-12-CH14-0008 | Medium | Ecosystem organization

A learner encounters the observation 'Organisms breaking down dead organic matter' while analysing ecosystem organization. Which conclusion is supported by the prescribed biology?

- A. Producers
- B. Mineralization
- C. Grazing food chain
- D. Decomposers

BIO-12-CH14-0009 | Medium | Decomposition steps

Complete the statement accurately: Breaking detritus into smaller particles corresponds to _____.

- A. Fragmentation
- B. Climax community
- C. Gross primary productivity
- D. Ten per cent

BIO-12-CH14-0010 | Medium | Ecological pyramids

A source table contains one accurate entry involving 'Pyramid showing standing crop'. Which option preserves a correct biological association?

- A. Total photosynthetic production before respiratory loss - Sedimentary rocks
- B. Benefit such as pollination, climate regulation or purification - Biogeochemical cycling
- C. Major reservoir of phosphorus - Atmosphere
- D. Pyramid showing standing crop - Pyramid of biomass

BIO-12-CH14-0011 | Hard | Ecosystem organization

A reviewer compares several entries while checking 'Organisms breaking down dead organic matter'. Which entry must be corrected?

- A. Major reservoir of phosphorus - Sedimentary rocks
- B. Interconnected food chains - Food web
- C. Organisms breaking down dead organic matter - Mineralization
- D. Conversion of detritus into inorganic nutrients - Mineralization

BIO-12-CH14-0012 | Medium | Nutrient cycles

A source table contains one accurate entry involving 'Major reservoir of atmospheric nitrogen'. Which option preserves a correct biological association?

- A. Major reservoir of atmospheric nitrogen - Atmosphere
- B. Food chain beginning with dead organic matter - Producers
- C. Pyramid that is always upright - Secondary succession
- D. Living components of an ecosystem - Leaching

BIO-12-CH14-0013 | Hard | Decomposition steps

Two conclusions must both be valid to support the reference concept 'Breaking detritus into smaller particles'. Which option contains two accurate associations?

- A. Food chain beginning with dead organic matter - Primary succession; Succession beginning on a previously lifeless surface - Detritus food chain
- B. Non-living physical and chemical components - Biogeochemical cycling; Movement of nutrients between organisms and environment - Abiotic components
- C. Succession beginning on a previously lifeless surface - Gross primary productivity; Total photosynthetic production before respiratory loss - Primary succession
- D. Breaking detritus into smaller particles - Fragmentation; Approximate energy passed to the next trophic level - Ten per cent

BIO-12-CH14-0014 | Medium | Food chains and energy

A learner encounters the observation 'Approximate energy passed to the next trophic level' while analysing food chains and energy. Which conclusion is supported by the prescribed biology?

- A. Mineralization
- B. Ten per cent
- C. Pyramid of biomass
- D. Net primary productivity

BIO-12-CH14-0015 | Hard | Nutrient cycles

A reviewer compares several entries while checking 'Major reservoir of phosphorus'. Which entry must be corrected?

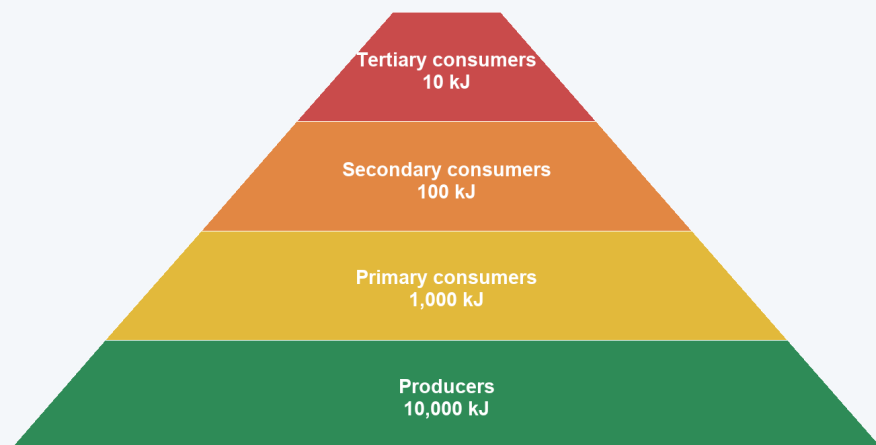
- A. Major reservoir of phosphorus - Energy pyramid
- B. Interconnected food chains - Food web
- C. Succession beginning on a previously lifeless surface - Primary succession
- D. Organisms breaking down dead organic matter - Decomposers

BIO-12-CH14-0016 | Easy | Pyramid of energy

How much energy reaches primary consumers in the illustrated pyramid?

Pyramid of energy

Illustration of the approximate ten-percent transfer rule



Energy is lost as heat and through metabolism at every transfer; the energy pyramid is always upright.

- A. 10,000 kJ
- B. 100 kJ
- C. 10 kJ
- D. 1,000 kJ

BIO-12-CH14-0017 | Medium | Succession and services

A source table contains one accurate entry involving 'Relatively stable final community'. Which option preserves a correct biological association?

- A. Relatively stable final community - Climax community
- B. Food chain beginning with dead organic matter - Net primary productivity
- C. Gross primary productivity minus plant respiration - Catabolism
- D. Conversion of organic nutrients to inorganic forms - Climax community

BIO-12-CH14-0018 | Easy | Decomposition steps

Which description is correctly represented by Fragmentation in this chapter?

- A. Food chain beginning with dead organic matter
- B. Total photosynthetic production before respiratory loss
- C. Succession beginning on a previously lifeless surface
- D. Breaking detritus into smaller particles

BIO-12-CH14-0019 | Medium | Decomposition steps

A learner encounters the observation 'Downward movement of soluble inorganic nutrients' while analysing decomposition steps. Which conclusion is supported by the prescribed biology?

- A. Biogeochemical cycling
- B. Leaching
- C. Energy pyramid
- D. Ecosystem service

BIO-12-CH14-0020 | Easy | Productivity and decomposition

Which description is correctly represented by Net primary productivity in this chapter?

- A. Microbial conversion of detritus to dark amorphous matter
- B. Succession after disturbance where soil remains
- C. Total photosynthetic production before respiratory loss
- D. Gross primary productivity minus plant respiration

BIO-12-CH14-0021 | Challenging | Ecological pyramids

During error analysis, four proposed associations are reviewed.

- I. Pyramid showing organism counts at trophic levels corresponds to Pyramid of numbers.
- II. Pyramid that is always upright corresponds to Pyramid of biomass.
- III. Pyramid showing standing crop corresponds to Pyramid of biomass.
- IV. Conversion of detritus into inorganic nutrients corresponds to Mineralization.

Which set contains all and only the valid conclusions?

- A. I, II, III and IV
- B. II, III and IV only
- C. I, III and IV only
- D. I only

BIO-12-CH14-0022 | Easy | Food chains and energy

Which biological term or result is specifically identified by the description 'Approximate energy passed to the next trophic level'?

- A. Pyramid of biomass
- B. Biotic components
- C. Grazing food chain
- D. Ten per cent

BIO-12-CH14-0023 | Easy | Ecological pyramids

Which description is correctly represented by Pyramid of biomass in this chapter?

- A. Food chain beginning with living green plants
- B. Pyramid showing standing crop
- C. Total photosynthetic production before respiratory loss
- D. Major reservoir of atmospheric nitrogen

BIO-12-CH14-0024 | Medium | Succession and services

A source table contains one accurate entry involving 'Succession after disturbance where soil remains'. Which option preserves a correct biological association?

- A. Non-living physical and chemical components - Ten per cent
- B. Succession after disturbance where soil remains - Secondary succession
- C. Enzymatic breakdown by bacteria and fungi - Gross primary productivity
- D. Total photosynthetic production before respiratory loss - Biotic components

BIO-12-CH14-0025 | Easy | Ecosystem organization

Which description is correctly represented by Decomposers in this chapter?

- A. Food chain beginning with living green plants
- B. Autotrophs converting solar energy to chemical energy
- C. Organisms breaking down dead organic matter
- D. Major reservoir of phosphorus

BIO-12-CH14-0026 | Medium | Nutrient cycles

A source table contains one accurate entry involving 'Major reservoir of phosphorus'. Which option preserves a correct biological association?

- A. Major reservoir of phosphorus - Sedimentary rocks
- B. Organisms breaking down dead organic matter - Primary productivity
- C. Interconnected food chains - Sedimentary rocks
- D. Succession beginning on a previously lifeless surface - Biotic components

BIO-12-CH14-0027 | Hard | Ecosystem organization

A reviewer compares several entries while checking 'Living components of an ecosystem'. Which entry must be corrected?

- A. Food chain beginning with dead organic matter - Detritus food chain
- B. Benefit such as pollination, climate regulation or purification - Ecosystem service
- C. Food chain beginning with living green plants - Grazing food chain
- D. Living components of an ecosystem - Biogeochemical cycling

BIO-12-CH14-0028 | Hard | Productivity and decomposition

Two conclusions must both be valid to support the reference concept 'Rate of biomass production by producers'. Which option contains two accurate associations?

- A. Rate of biomass production by producers - Primary productivity; Downward movement of soluble inorganic nutrients - Leaching
- B. Pyramid showing organism counts at trophic levels - Detritus food chain; Food chain beginning with dead organic matter - Pyramid of numbers
- C. Pyramid showing standing crop - Food web; Interconnected food chains - Pyramid of biomass
- D. Interconnected food chains - Fragmentation; Breaking detritus into smaller particles - Food web

BIO-12-CH14-0029 | Medium | Ecosystem organization

Complete the statement accurately: Autotrophs converting solar energy to chemical energy corresponds to _____.

- A. Leaching
- B. Food web
- C. Producers
- D. Decomposers

BIO-12-CH14-0030 | Medium | Productivity and decomposition

A learner encounters the observation 'Gross primary productivity minus plant respiration' while analysing productivity and decomposition. Which conclusion is supported by the prescribed biology?

- A. Humification
- B. Gross primary productivity
- C. Net primary productivity
- D. Secondary succession

BIO-12-CH14-0031 | Challenging | Food chains and energy

A researcher checks whether four linked observations have been interpreted correctly.

- I. Food chain beginning with living green plants corresponds to Grazing food chain.
 - II. Interconnected food chains corresponds to Food web.
 - III. Approximate energy passed to the next trophic level corresponds to Primary succession.
 - IV. Succession beginning on a previously lifeless surface corresponds to Grazing food chain.
- Which combination survives the biological review?

- A. I and II only
- B. III and IV only
- C. II only
- D. I, III and IV only

BIO-12-CH14-0032 | Challenging | Ecosystem organization

Evaluate the following source-based biological statements.

- I. Organisms breaking down dead organic matter corresponds to Decomposers.
 - II. Non-living physical and chemical components corresponds to Abiotic components.
 - III. Living components of an ecosystem corresponds to Primary succession.
 - IV. Succession beginning on a previously lifeless surface corresponds to Primary succession.
- Which option identifies exactly the correct statements?

- A. II only
- B. I, II and IV only
- C. II, III and IV only
- D. IV only

BIO-12-CH14-0033 | Easy | Decomposition steps

Which description is correctly represented by Catabolism in this chapter?

- A. Organisms breaking down dead organic matter
- B. Enzymatic breakdown by bacteria and fungi
- C. Downward movement of soluble inorganic nutrients
- D. Autotrophs converting solar energy to chemical energy

BIO-12-CH14-0034 | Medium | Succession and services

A learner encounters the observation 'Succession after disturbance where soil remains' while analysing succession and services. Which conclusion is supported by the prescribed biology?

- A. Secondary succession
- B. Producers
- C. Primary succession
- D. Abiotic components

BIO-12-CH14-0035 | Medium | Decomposition steps

A source table contains one accurate entry involving 'Enzymatic breakdown by bacteria and fungi'. Which option preserves a correct biological association?

- A. Organisms breaking down dead organic matter - Detritus food chain
- B. Succession beginning on a previously lifeless surface - Producers
- C. Autotrophs converting solar energy to chemical energy - Gross primary productivity
- D. Enzymatic breakdown by bacteria and fungi - Catabolism

BIO-12-CH14-0036 | Medium | Decomposition steps

Complete the statement accurately: Microbial conversion of detritus to dark amorphous matter corresponds to _____.

- A. Fragmentation
- B. Producers
- C. Sedimentary rocks
- D. Humification

BIO-12-CH14-0037 | Hard | Ecosystem organization

Two conclusions must both be valid to support the reference concept 'Organisms breaking down dead organic matter'. Which option contains two accurate associations?

- A. Interconnected food chains - Secondary succession; Succession after disturbance where soil remains - Food web
- B. Organisms breaking down dead organic matter - Decomposers; Conversion of organic nutrients to inorganic forms - Mineralization
- C. Food chain beginning with living green plants - Sedimentary rocks; Major reservoir of phosphorus - Grazing food chain
- D. Conversion of detritus into inorganic nutrients - Climax community; Relatively stable final community - Mineralization

BIO-12-CH14-0038 | Challenging | Succession and services

A student submits four conclusions after integrating related chapter concepts.

- I. Succession after disturbance where soil remains corresponds to Ecosystem service.
- II. Benefit such as pollination, climate regulation or purification corresponds to Climax community.
- III. Relatively stable final community corresponds to Primary productivity.
- IV. Rate of biomass production by producers corresponds to Primary productivity.

Which combination should be accepted without correction?

- A. II only
- B. II, III and IV only
- C. IV only
- D. II and IV only

BIO-12-CH14-0039 | Hard | Productivity and decomposition

Two conclusions must both be valid to support the reference concept 'Gross primary productivity minus plant respiration'. Which option contains two accurate associations?

- A. Benefit such as pollination, climate regulation or purification - Producers; Autotrophs converting solar energy to chemical energy - Ecosystem service
- B. Succession after disturbance where soil remains - Catabolism; Enzymatic breakdown by bacteria and fungi - Secondary succession
- C. Gross primary productivity minus plant respiration - Net primary productivity; Microbial conversion of detritus to dark amorphous matter - Humification
- D. Enzymatic breakdown by bacteria and fungi - Pyramid of biomass; Pyramid showing standing crop - Catabolism

BIO-12-CH14-0040 | Medium | Ecosystem organization

A source table contains one accurate entry involving 'Autotrophs converting solar energy to chemical energy'. Which option preserves a correct biological association?

- A. Microbial conversion of detritus to dark amorphous matter - Producers
- B. Breaking detritus into smaller particles - Ten per cent
- C. Interconnected food chains - Leaching
- D. Autotrophs converting solar energy to chemical energy - Producers

BIO-12-CH14-0041 | Hard | Food chains and energy

Two conclusions must both be valid to support the reference concept 'Interconnected food chains'. Which option contains two accurate associations?

- A. Interconnected food chains - Food web; Enzymatic breakdown by bacteria and fungi - Catabolism
- B. Major reservoir of phosphorus - Leaching; Downward movement of soluble inorganic nutrients - Sedimentary rocks
- C. Downward movement of soluble inorganic nutrients - Net primary productivity; Gross primary productivity minus plant respiration - Leaching
- D. Gross primary productivity minus plant respiration - Abiotic components; Non-living physical and chemical components - Net primary productivity

BIO-12-CH14-0042 | Medium | Productivity and decomposition

Complete the statement accurately: Total photosynthetic production before respiratory loss corresponds to _____.

- A. First trophic level
- B. Fragmentation
- C. Gross primary productivity
- D. Catabolism

BIO-12-CH14-0043 | Medium | Food chains and energy

A source table contains one accurate entry involving 'Food chain beginning with living green plants'. Which option preserves a correct biological association?

- A. Food chain beginning with living green plants - Grazing food chain
- B. Benefit such as pollination, climate regulation or purification - Climax community
- C. Relatively stable final community - Primary productivity
- D. Pyramid showing organism counts at trophic levels - Ecosystem service

BIO-12-CH14-0044 | Medium | Ecological pyramids

Complete the statement accurately: Pyramid that is always upright corresponds to _____.

- A. First trophic level
- B. Secondary succession
- C. Pyramid of biomass
- D. Energy pyramid

BIO-12-CH14-0045 | Easy | Productivity and decomposition

Which description is correctly represented by Primary productivity in this chapter?

- A. Downward movement of soluble inorganic nutrients
- B. Rate of biomass production by producers
- C. Conversion of detritus into inorganic nutrients
- D. Interconnected food chains

BIO-12-CH14-0046 | Hard | Succession and services

A reviewer compares several entries while checking 'Benefit such as pollination, climate regulation or purification'. Which entry must be corrected?

- A. Pyramid that is always upright - Energy pyramid
- B. Total photosynthetic production before respiratory loss - Gross primary productivity
- C. Approximate energy passed to the next trophic level - Ten per cent
- D. Benefit such as pollination, climate regulation or purification - Fragmentation

BIO-12-CH14-0047 | Easy | Nutrient cycles

Which biological term or result is specifically identified by the description 'Major reservoir of atmospheric nitrogen'?

- A. Secondary succession
- B. Atmosphere
- C. Energy pyramid
- D. Detritus food chain

BIO-12-CH14-0048 | Medium | Succession and services

A source table contains one accurate entry involving 'Succession beginning on a previously lifeless surface'. Which option preserves a correct biological association?

- A. Conversion of organic nutrients to inorganic forms - Producers
- B. Succession beginning on a previously lifeless surface - Primary succession
- C. Major reservoir of atmospheric nitrogen - Pyramid of biomass
- D. Pyramid showing standing crop - Net primary productivity

BIO-12-CH14-0049 | Challenging | Ecological pyramids

During error analysis, four proposed associations are reviewed.

- I. Pyramid that is always upright corresponds to Energy pyramid.
- II. Pyramid showing standing crop corresponds to Pyramid of biomass.
- III. Pyramid showing organism counts at trophic levels corresponds to Pyramid of numbers.
- IV. Total photosynthetic production before respiratory loss corresponds to Energy pyramid.

Which set contains all and only the valid conclusions?

- A. I, II and III only
- B. III only
- C. II, III and IV only
- D. I and IV only

BIO-12-CH14-0050 | Hard | Nutrient cycles

Two conclusions must both be valid to support the reference concept 'Major reservoir of atmospheric nitrogen'. Which option contains two accurate associations?

- A. Food chain beginning with living green plants - Secondary succession; Succession after disturbance where soil remains - Grazing food chain
- B. Autotrophs converting solar energy to chemical energy - Leaching; Downward movement of soluble inorganic nutrients - Producers
- C. Major reservoir of atmospheric nitrogen - Atmosphere; Pyramid that is always upright - Energy pyramid
- D. Living components of an ecosystem - Producers; Autotrophs converting solar energy to chemical energy - Biotic components

BIO-12-CH14-0051 | Medium | Nutrient cycles

A learner encounters the observation 'Movement of nutrients between organisms and environment' while analysing nutrient cycles. Which conclusion is supported by the prescribed biology?

- A. Catabolism
- B. Mineralization
- C. Biogeochemical cycling
- D. Sedimentary rocks

BIO-12-CH14-0052 | Challenging | Succession and services

A researcher checks whether four linked observations have been interpreted correctly.

- I. Relatively stable final community corresponds to Primary succession.
- II. Succession beginning on a previously lifeless surface corresponds to Secondary succession.
- III. Succession after disturbance where soil remains corresponds to Secondary succession.
- IV. Pyramid showing standing crop corresponds to Climax community.

Which combination survives the biological review?

- A. II and III only
- B. III only
- C. I and IV only
- D. III and IV only

BIO-12-CH14-0053 | Medium | Productivity and decomposition

Complete the statement accurately: Conversion of detritus into inorganic nutrients corresponds to _____.

- A. Mineralization
- B. Leaching
- C. Pyramid of biomass
- D. Gross primary productivity

BIO-12-CH14-0054 | Medium | Food chains and energy

A learner encounters the observation 'Food chain beginning with living green plants' while analysing food chains and energy. Which conclusion is supported by the prescribed biology?

- A. Climax community
- B. Humification
- C. Ecosystem service
- D. Grazing food chain

BIO-12-CH14-0055 | Easy | Ecological pyramids

Which description is correctly represented by Energy pyramid in this chapter?

- A. Total photosynthetic production before respiratory loss
- B. Succession after disturbance where soil remains
- C. Rate of biomass production by producers
- D. Pyramid that is always upright

BIO-12-CH14-0056 | Hard | Ecological pyramids

Two conclusions must both be valid to support the reference concept 'Pyramid showing standing crop'. Which option contains two accurate associations?

- A. Pyramid showing standing crop - Pyramid of biomass; Total photosynthetic production before respiratory loss - Gross primary productivity
- B. Major reservoir of phosphorus - Ecosystem service; Benefit such as pollination, climate regulation or purification - Sedimentary rocks
- C. Food chain beginning with living green plants - Sedimentary rocks; Major reservoir of phosphorus - Grazing food chain
- D. Benefit such as pollination, climate regulation or purification - Atmosphere; Major reservoir of atmospheric nitrogen - Ecosystem service

BIO-12-CH14-0057 | Medium | Nutrient cycles

A learner encounters the observation 'Conversion of organic nutrients to inorganic forms' while analysing nutrient cycles. Which conclusion is supported by the prescribed biology?

- A. Primary succession
- B. Grazing food chain
- C. Biogeochemical cycling
- D. Mineralization

BIO-12-CH14-0058 | Medium | Ecosystem organization

Complete the statement accurately: Living components of an ecosystem corresponds to _____.

- A. Grazing food chain
- B. Biotic components
- C. Detritus food chain
- D. Abiotic components

BIO-12-CH14-0059 | Easy | Productivity and decomposition

Which description is correctly represented by Gross primary productivity in this chapter?

- A. Breaking detritus into smaller particles
- B. Movement of nutrients between organisms and environment
- C. Total photosynthetic production before respiratory loss
- D. Rate of biomass production by producers

BIO-12-CH14-0060 | Medium | Food chains and energy

Complete the statement accurately: Food chain beginning with living green plants corresponds to _____.

- A. Climax community
- B. Ten per cent
- C. Pyramid of numbers
- D. Grazing food chain

BIO-12-CH14-0061 | Hard | Succession and services

Two conclusions must both be valid to support the reference concept 'Succession beginning on a previously lifeless surface'. Which option contains two accurate associations?

- A. Conversion of organic nutrients to inorganic forms - Net primary productivity; Gross primary productivity minus plant respiration - Mineralization
- B. Non-living physical and chemical components - Pyramid of biomass; Pyramid showing standing crop - Abiotic components
- C. Succession beginning on a previously lifeless surface - Primary succession; Pyramid showing organism counts at trophic levels - Pyramid of numbers
- D. Pyramid showing standing crop - Mineralization; Conversion of organic nutrients to inorganic forms - Pyramid of biomass

BIO-12-CH14-0062 | Hard | Food chains and energy

A reviewer compares several entries while checking 'Food chain beginning with living green plants'. Which entry must be corrected?

- A. Microbial conversion of detritus to dark amorphous matter - Humification
- B. Food chain beginning with living green plants - Pyramid of numbers
- C. Breaking detritus into smaller particles - Fragmentation
- D. Benefit such as pollination, climate regulation or purification - Ecosystem service

BIO-12-CH14-0063 | Medium | Succession and services

A learner encounters the observation 'Relatively stable final community' while analysing succession and services. Which conclusion is supported by the prescribed biology?

- A. Detritus food chain
- B. Fragmentation
- C. Climax community
- D. Primary succession

BIO-12-CH14-0064 | Hard | Ecological pyramids

Two conclusions must both be valid to support the reference concept 'Pyramid that is always upright'. Which option contains two accurate associations?

- A. Rate of biomass production by producers - Biogeochemical cycling; Movement of nutrients between organisms and environment - Primary productivity
- B. Succession after disturbance where soil remains - Catabolism; Enzymatic breakdown by bacteria and fungi - Secondary succession
- C. Total photosynthetic production before respiratory loss - Secondary succession; Succession after disturbance where soil remains - Gross primary productivity
- D. Pyramid that is always upright - Energy pyramid; Breaking detritus into smaller particles - Fragmentation

BIO-12-CH14-0065 | Hard | Nutrient cycles

A reviewer compares several entries while checking 'Major reservoir of atmospheric nitrogen'. Which entry must be corrected?

- A. Major reservoir of atmospheric nitrogen - Energy pyramid
- B. Food chain beginning with living green plants - Grazing food chain
- C. Food chain beginning with dead organic matter - Detritus food chain
- D. Autotrophs converting solar energy to chemical energy - Producers

BIO-12-CH14-0066 | Hard | Productivity and decomposition

Two conclusions must both be valid to support the reference concept 'Total photosynthetic production before respiratory loss'. Which option contains two accurate associations?

- A. Breaking detritus into smaller particles - Catabolism; Enzymatic breakdown by bacteria and fungi - Fragmentation
- B. Enzymatic breakdown by bacteria and fungi - Humification; Microbial conversion of detritus to dark amorphous matter - Catabolism
- C. Succession after disturbance where soil remains - Ecosystem service; Benefit such as pollination, climate regulation or purification - Secondary succession
- D. Total photosynthetic production before respiratory loss - Gross primary productivity; Movement of nutrients between organisms and environment - Biogeochemical cycling

BIO-12-CH14-0067 | Easy | Ecological pyramids

Which description is correctly represented by First trophic level in this chapter?

- A. Breaking detritus into smaller particles
- B. Pyramid showing standing crop
- C. Pyramid showing organism counts at trophic levels
- D. Trophic level occupied by green plants

BIO-12-CH14-0068 | Challenging | Decomposition steps

Evaluate the following source-based biological statements.

- I. Breaking detritus into smaller particles corresponds to Humification.
- II. Microbial conversion of detritus to dark amorphous matter corresponds to Catabolism.
- III. Enzymatic breakdown by bacteria and fungi corresponds to Catabolism.
- IV. Conversion of organic nutrients to inorganic forms corresponds to Mineralization.

Which option identifies exactly the correct statements?

- A. III and IV only
- B. II and III only
- C. I and II only
- D. I, III and IV only

BIO-12-CH14-0069 | Easy | Ecosystem organization

Which biological term or result is specifically identified by the description 'Living components of an ecosystem'?

- A. Grazing food chain
- B. Biotic components
- C. Pyramid of biomass
- D. Detritus food chain

BIO-12-CH14-0070 | Medium | Productivity and decomposition

A source table contains one accurate entry involving 'Gross primary productivity minus plant respiration'. Which option preserves a correct biological association?

- A. Enzymatic breakdown by bacteria and fungi - Ecosystem service
- B. Gross primary productivity minus plant respiration - Net primary productivity
- C. Microbial conversion of detritus to dark amorphous matter - Secondary succession
- D. Pyramid showing standing crop - Producers

BIO-12-CH14-0071 | Medium | Ecological pyramids

A source table contains one accurate entry involving 'Pyramid that is always upright'. Which option preserves a correct biological association?

- A. Pyramid that is always upright - Energy pyramid
- B. Breaking detritus into smaller particles - Secondary succession
- C. Total photosynthetic production before respiratory loss - Catabolism
- D. Movement of nutrients between organisms and environment - Energy pyramid

BIO-12-CH14-0072 | Hard | Decomposition steps

A reviewer compares several entries while checking 'Breaking detritus into smaller particles'. Which entry must be corrected?

- A. Breaking detritus into smaller particles - Ten per cent
- B. Food chain beginning with dead organic matter - Detritus food chain
- C. Non-living physical and chemical components - Abiotic components
- D. Relatively stable final community - Climax community

BIO-12-CH14-0073 | Challenging | Nutrient cycles

A student submits four conclusions after integrating related chapter concepts.

- I. Conversion of organic nutrients to inorganic forms corresponds to Sedimentary rocks.
- II. Major reservoir of phosphorus corresponds to Sedimentary rocks.
- III. Movement of nutrients between organisms and environment corresponds to Catabolism.
- IV. Enzymatic breakdown by bacteria and fungi corresponds to Mineralization.

Which combination should be accepted without correction?

- A. IV only
- B. II only
- C. I and II only
- D. II and III only

BIO-12-CH14-0074 | Easy | Succession and services

Which description is correctly represented by Primary succession in this chapter?

- A. Major reservoir of atmospheric nitrogen
- B. Relatively stable final community
- C. Succession beginning on a previously lifeless surface
- D. Conversion of organic nutrients to inorganic forms

BIO-12-CH14-0075 | Easy | Ecosystem organization

Which biological term or result is specifically identified by the description 'Organisms breaking down dead organic matter'?

- A. Grazing food chain
- B. Decomposers
- C. Sedimentary rocks
- D. Mineralization

BIO-12-CH14-0076 | Easy | Food chains and energy

Which biological term or result is specifically identified by the description 'Food chain beginning with living green plants'?

- A. Pyramid of numbers
- B. Grazing food chain
- C. Food web
- D. Ten per cent

BIO-12-CH14-0077 | Hard | Succession and services

A reviewer compares several entries while checking 'Relatively stable final community'. Which entry must be corrected?

- A. Relatively stable final community - Mineralization
- B. Conversion of organic nutrients to inorganic forms - Mineralization
- C. Breaking detritus into smaller particles - Fragmentation
- D. Food chain beginning with dead organic matter - Detritus food chain

BIO-12-CH14-0078 | Easy | Succession and services

Which biological term or result is specifically identified by the description 'Benefit such as pollination, climate regulation or purification'?

- A. Ecosystem service
- B. Primary succession
- C. Ten per cent
- D. Fragmentation

BIO-12-CH14-0079 | Hard | Ecological pyramids

Two conclusions must both be valid to support the reference concept 'Trophic level occupied by green plants'. Which option contains two accurate associations?

- A. Enzymatic breakdown by bacteria and fungi - Abiotic components; Non-living physical and chemical components - Catabolism
- B. Approximate energy passed to the next trophic level - Net primary productivity; Gross primary productivity minus plant respiration - Ten per cent
- C. Trophic level occupied by green plants - First trophic level; Breaking detritus into smaller particles - Fragmentation
- D. Non-living physical and chemical components - Ten per cent; Approximate energy passed to the next trophic level - Abiotic components

BIO-12-CH14-0080 | Medium | Ecological pyramids

Complete the statement accurately: Pyramid showing standing crop corresponds to ____.

- A. Ecosystem service
- B. Gross primary productivity
- C. Pyramid of biomass
- D. Sedimentary rocks

BIO-12-CH14-0081 | Easy | Food chains and energy

Which description is correctly represented by Food web in this chapter?

- A. Gross primary productivity minus plant respiration
- B. Interconnected food chains
- C. Food chain beginning with living green plants
- D. Major reservoir of phosphorus

BIO-12-CH14-0082 | Easy | Succession and services

Which biological term or result is specifically identified by the description 'Relatively stable final community'?

- A. Ecosystem service
- B. Net primary productivity
- C. Mineralization
- D. Climax community

BIO-12-CH14-0083 | Medium | Productivity and decomposition

A learner encounters the observation 'Rate of biomass production by producers' while analysing productivity and decomposition. Which conclusion is supported by the prescribed biology?

- A. Food web
- B. Primary productivity
- C. Fragmentation
- D. Mineralization

BIO-12-CH14-0084 | Easy | Nutrient cycles

Which description is correctly represented by Atmosphere in this chapter?

- A. Food chain beginning with dead organic matter
- B. Pyramid that is always upright
- C. Major reservoir of atmospheric nitrogen
- D. Food chain beginning with living green plants

BIO-12-CH14-0085 | Easy | Decomposition steps

Which biological term or result is specifically identified by the description 'Breaking detritus into smaller particles'?

- A. Humification
- B. Detritus food chain
- C. Fragmentation
- D. Gross primary productivity

BIO-12-CH14-0086 | Medium | Ecosystem organization

A source table contains one accurate entry involving 'Living components of an ecosystem'. Which option preserves a correct biological association?

- A. Living components of an ecosystem - Biotic components
- B. Food chain beginning with dead organic matter - Grazing food chain
- C. Food chain beginning with living green plants - Atmosphere
- D. Pyramid showing standing crop - Ecosystem service

BIO-12-CH14-0087 | Medium | Decomposition steps

Complete the statement accurately: Downward movement of soluble inorganic nutrients corresponds to ____.

- A. Fragmentation
- B. Ecosystem service
- C. Catabolism
- D. Leaching

BIO-12-CH14-0088 | Easy | Nutrient cycles

Which biological term or result is specifically identified by the description 'Major reservoir of phosphorus'?

- A. Mineralization
 - B. Decomposers
 - C. Sedimentary rocks
 - D. Energy pyramid
-

Standard 12 | Chapter 15

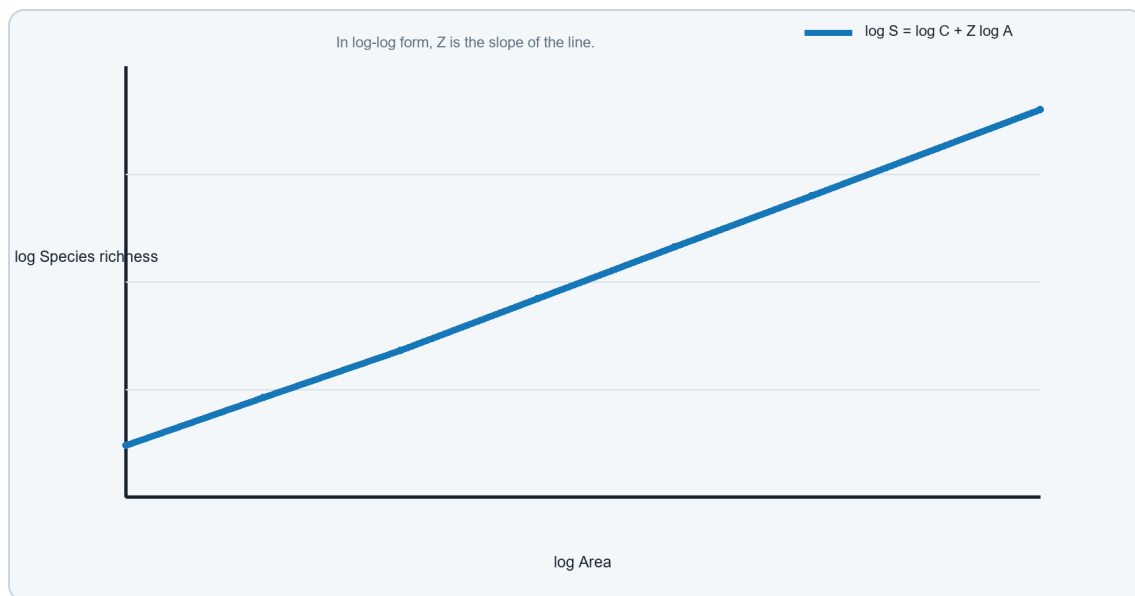
Biodiversity, Conservation and Environmental Issues

BIO-12-CH15-0001 | Hard | Species-area relationship

In $\log S = \log C + Z \log A$, what does Z represent on the plotted line?

Species-area relationship

Species richness increases with sampled habitat area



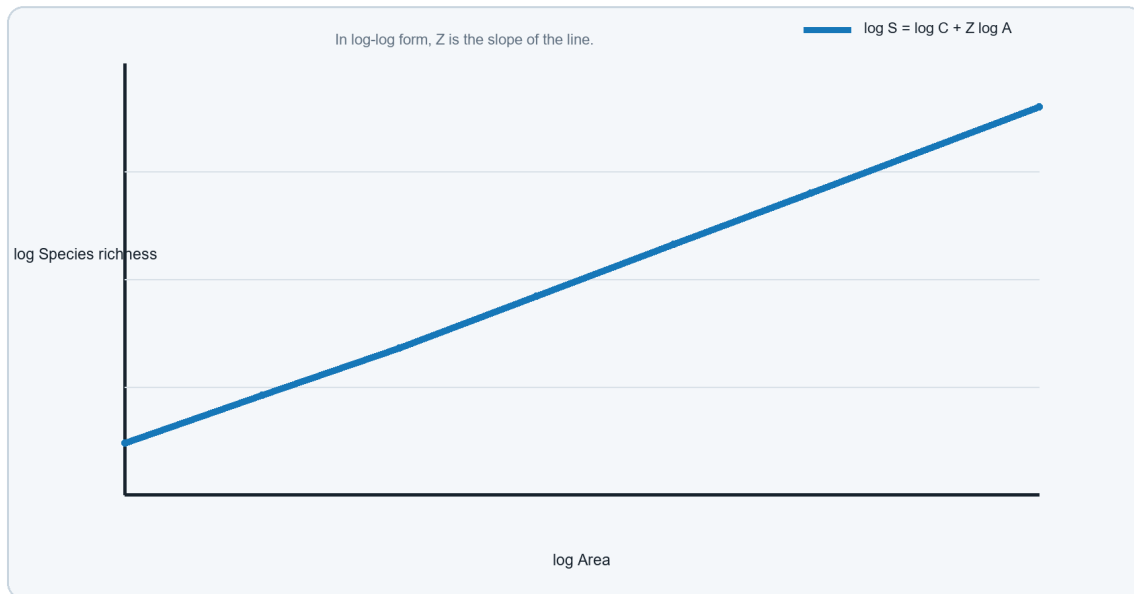
- A. Y-intercept $\log C$
- B. Total habitat area only
- C. Slope
- D. Number of trophic levels

BIO-12-CH15-0002 | Easy | Species-area relationship

What general relationship is shown between habitat area and species richness?

Species-area relationship

Species richness increases with sampled habitat area



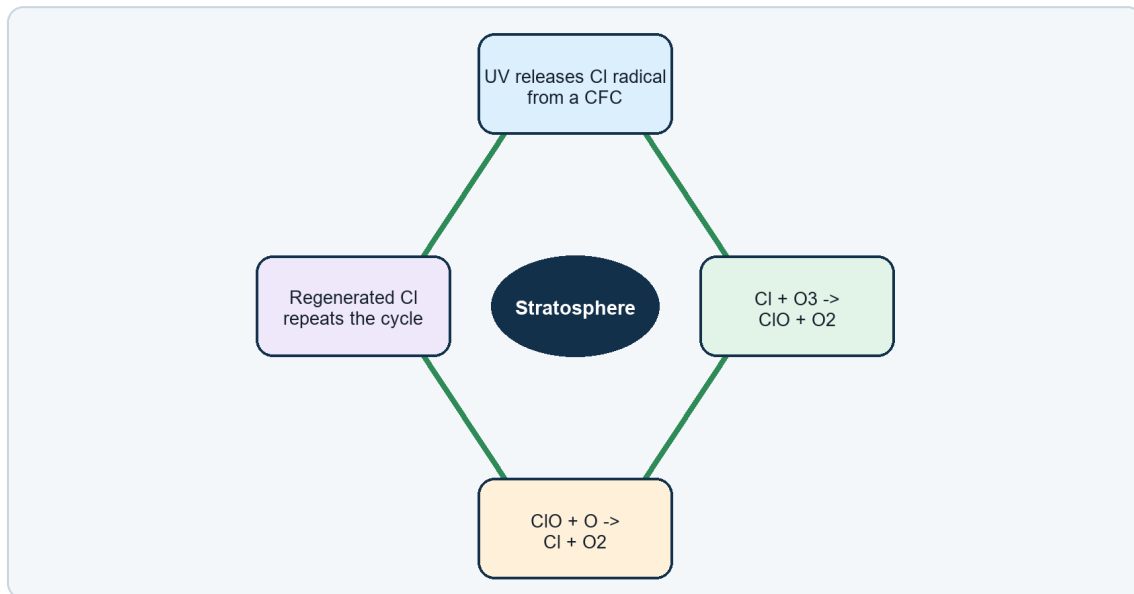
- A. Species richness increases with area
- B. Species richness always falls with area
- C. Area and richness are unrelated
- D. Every area contains exactly one species

BIO-12-CH15-0003 | Medium | Catalytic ozone destruction by chlorine

Which human-made compounds are a major source of ozone-destroying chlorine radicals in the figure?

Catalytic ozone destruction by chlorine

A chlorine radical can be regenerated and attack additional ozone



- A. Glucose
- B. Cellulose
- C. Chlorofluorocarbons
- D. Amino acids

BIO-12-CH15-0004 | Easy | Levels of biodiversity

Which biological term or result is specifically identified by the description 'Number of species present in a community'?

- A. Species diversity
- B. Species richness
- C. Endemic species
- D. It decreases

BIO-12-CH15-0005 | Easy | Forests and law

Which description is correctly represented by Biological Diversity Act 2002 in this chapter?

- A. Pollutant load measured through oxygen consumed by microbes
- B. Destruction and fragmentation of natural habitat
- C. Indian law providing a framework for conservation and benefit sharing
- D. Rise in average global temperature due to enhanced greenhouse effect

BIO-12-CH15-0006 | Easy | Biodiversity patterns

Which description is correctly represented by Species-area relationship in this chapter?

- A. Variation among genes within a species
- B. Ex situ storage of viable seeds
- C. Relationship in which larger areas contain more species
- D. General change in species richness from equator to poles

BIO-12-CH15-0007 | Medium | Levels of biodiversity

A source table contains one accurate entry involving 'Variety of species in a region'. Which option preserves a correct biological association?

- A. Variety of species in a region - Species diversity
- B. Nutrient enrichment causing excessive aquatic plant growth - Overexploitation
- C. Harvesting organisms faster than they recover - Montreal Protocol
- D. Chemicals releasing chlorine that destroys stratospheric ozone - It decreases

BIO-12-CH15-0008 | Easy | Threats and extinction

Which biological term or result is specifically identified by the description 'Secondary extinction following loss of a dependent species'?

- A. Mission Harit Maharashtra
- B. Invasive alien species
- C. Species diversity
- D. Co-extinction

BIO-12-CH15-0009 | Medium | Forests and law

Complete the statement accurately: Maharashtra initiative promoting large-scale tree planting corresponds to ____.

- A. Biomagnification
- B. Montreal Protocol
- C. It decreases
- D. Mission Harit Maharashtra

BIO-12-CH15-0010 | Hard | Forests and law

A reviewer compares several entries while checking 'Large-scale removal of forest cover'. Which entry must be corrected?

- A. Chemicals releasing chlorine that destroys stratospheric ozone - Chlorofluorocarbons
- B. Large-scale removal of forest cover - In situ conservation
- C. Variety of species in a region - Species diversity
- D. Variety of habitats and ecological processes - Ecosystem diversity

BIO-12-CH15-0011 | Medium | Levels of biodiversity

A source table contains one accurate entry involving 'Variation among genes within a species'. Which option preserves a correct biological association?

- A. Non-native organism spreading and causing harm - Carbon dioxide
- B. Gas chiefly responsible for anthropogenic greenhouse warming - Mission Harit Maharashtra
- C. Nutrient enrichment causing excessive aquatic plant growth - Species-area relationship
- D. Variation among genes within a species - Genetic diversity

BIO-12-CH15-0012 | Medium | Forests and law

A learner encounters the observation 'Restoring a forest that once existed but was removed' while analysing forests and law. Which conclusion is supported by the prescribed biology?

- A. Mission Harit Maharashtra
- B. Reforestation
- C. Montreal Protocol
- D. Biodiversity hotspot

BIO-12-CH15-0013 | Medium | Threats and extinction

Complete the statement accurately: Harvesting organisms faster than they recover corresponds to _____.

- A. Overexploitation
- B. Co-extinction
- C. Montreal Protocol
- D. Eutrophication

BIO-12-CH15-0014 | Easy | Threats and extinction

Which biological term or result is specifically identified by the description 'Destruction and fragmentation of natural habitat'?

- A. Invasive alien species
- B. Ecosystem diversity
- C. Reforestation
- D. Major cause of biodiversity loss

BIO-12-CH15-0015 | Medium | Biodiversity patterns

A learner encounters the observation 'Relationship in which larger areas contain more species' while analysing biodiversity patterns. Which conclusion is supported by the prescribed biology?

- A. Genetic diversity
- B. Biosphere reserve
- C. Species-area relationship
- D. Co-extinction

BIO-12-CH15-0016 | Hard | Biodiversity patterns

Two conclusions must both be valid to support the reference concept 'Relationship in which larger areas contain more species'. Which option contains two accurate associations?

- A. Ex situ storage of viable seeds - Genetic diversity; Variation among genes within a species - Seed bank
- B. Relationship in which larger areas contain more species - Species-area relationship; Increase in concentration of persistent toxins at higher trophic levels - Biomagnification
- C. In situ protected area permitting regulated human activity - Co-extinction; Secondary extinction following loss of a dependent species - Biosphere reserve
- D. Variation among genes within a species - Ecosystem diversity; Variety of habitats and ecological processes - Genetic diversity

BIO-12-CH15-0017 | Medium | Threats and extinction

A learner encounters the observation 'Secondary extinction following loss of a dependent species' while analysing threats and extinction. Which conclusion is supported by the prescribed biology?

- A. Invasive alien species
- B. Montreal Protocol
- C. Particulate matter
- D. Co-extinction

BIO-12-CH15-0018 | Hard | Levels of biodiversity

A reviewer compares several entries while checking 'Number of species present in a community'. Which entry must be corrected?

- A. Number of species present in a community - Co-extinction
- B. Species restricted to a particular geographic area - Endemic species
- C. Large-scale removal of forest cover - Deforestation
- D. Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra

BIO-12-CH15-0019 | Easy | Pollution

Which biological term or result is specifically identified by the description 'Increase in concentration of persistent toxins at higher trophic levels'?

- A. Reforestation
- B. Species richness
- C. Biomagnification
- D. Particulate matter

BIO-12-CH15-0020 | Hard | Biodiversity patterns

Two conclusions must both be valid to support the reference concept 'General change in species richness from equator to poles'. Which option contains two accurate associations?

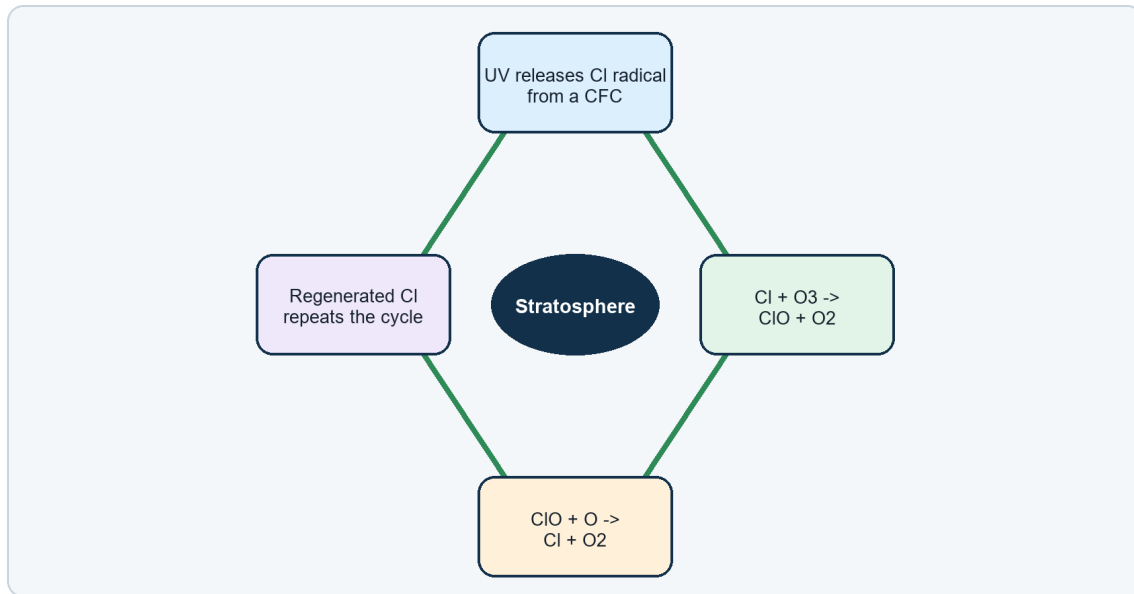
- A. Gas chiefly responsible for anthropogenic greenhouse warming - Biosphere reserve; In situ protected area permitting regulated human activity - Carbon dioxide
- B. General change in species richness from equator to poles - It decreases; Variety of species in a region - Species diversity
- C. Rise in average global temperature due to enhanced greenhouse effect - In situ conservation; Protection of species in their natural habitats - Global warming
- D. Protection of species in their natural habitats - Biological Diversity Act 2002; Indian law providing a framework for conservation and benefit sharing - In situ conservation

BIO-12-CH15-0021 | Challenging | Catalytic ozone destruction by chlorine

Which statement correctly distinguishes ozone depletion from the greenhouse effect?

Catalytic ozone destruction by chlorine

A chlorine radical can be regenerated and attack additional ozone



- A. Ozone depletion and greenhouse warming are exactly the same process
- B. Ozone depletion chiefly increases surface UV-B; it is not itself the direct mechanism of greenhouse warming
- C. Ozone depletion blocks all ultraviolet radiation
- D. Greenhouse gases destroy ozone only by the chlorine-radical cycle shown

BIO-12-CH15-0022 | Hard | Conservation

Two conclusions must both be valid to support the reference concept 'Protection of species in their natural habitats'. Which option contains two accurate associations?

- A. Relationship in which larger areas contain more species - Species diversity; Variety of species in a region - Species-area relationship
- B. Maharashtra initiative promoting large-scale tree planting - Chlorofluorocarbons; Chemicals releasing chlorine that destroys stratospheric ozone - Mission Harit Maharashtra
- C. Variation among genes within a species - Mission Harit Maharashtra; Maharashtra initiative promoting large-scale tree planting - Genetic diversity
- D. Protection of species in their natural habitats - In situ conservation; Nutrient enrichment causing excessive aquatic plant growth - Eutrophication

BIO-12-CH15-0023 | Medium | Biodiversity patterns

A source table contains one accurate entry involving 'General change in species richness from equator to poles'. Which option preserves a correct biological association?

- A. Rise in average global temperature due to enhanced greenhouse effect - Biological Diversity Act 2002
- B. Gas chiefly responsible for anthropogenic greenhouse warming - Seed bank
- C. General change in species richness from equator to poles - It decreases
- D. Ex situ storage of viable seeds - In situ conservation

BIO-12-CH15-0024 | Hard | Levels of biodiversity

A reviewer compares several entries while checking 'Variation among genes within a species'. Which entry must be corrected?

- A. Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra
- B. Variation among genes within a species - Invasive alien species
- C. Relationship in which larger areas contain more species - Species-area relationship
- D. Pollutant load measured through oxygen consumed by microbes - Biochemical oxygen demand

BIO-12-CH15-0025 | Medium | Climate and ozone

A source table contains one accurate entry involving 'Chemicals releasing chlorine that destroys stratospheric ozone'. Which option preserves a correct biological association?

- A. Ex situ storage of viable seeds - Overexploitation
- B. Restoring a forest that once existed but was removed - Species-area relationship
- C. Protection outside natural habitats - Chlorofluorocarbons
- D. Chemicals releasing chlorine that destroys stratospheric ozone - Chlorofluorocarbons

BIO-12-CH15-0026 | Hard | Biodiversity patterns

Two conclusions must both be valid to support the reference concept 'Region with many endemic species under exceptional threat'. Which option contains two accurate associations?

- A. Region with many endemic species under exceptional threat - Biodiversity hotspot; Secondary extinction following loss of a dependent species - Co-extinction
- B. Agreement aimed at controlling ozone-depleting substances - Species diversity; Variety of species in a region - Montreal Protocol
- C. In situ protected area permitting regulated human activity - Deforestation; Large-scale removal of forest cover - Biosphere reserve
- D. Large-scale removal of forest cover - Montreal Protocol; Agreement aimed at controlling ozone-depleting substances - Deforestation

BIO-12-CH15-0027 | Challenging | Forests and law

A researcher checks whether four linked observations have been interpreted correctly.

- I. Large-scale removal of forest cover corresponds to Biological Diversity Act 2002.
- II. Indian law providing a framework for conservation and benefit sharing corresponds to Reforestation.
- III. Restoring a forest that once existed but was removed corresponds to Global warming.
- IV. Rise in average global temperature due to enhanced greenhouse effect corresponds to Global warming.

Which combination survives the biological review?

- A. I, II and III only
- B. IV only
- C. I, III and IV only
- D. I and III only

BIO-12-CH15-0028 | Easy | Conservation

Which biological term or result is specifically identified by the description 'In situ protected area permitting regulated human activity'?

- A. Biosphere reserve
- B. Biodiversity hotspot
- C. Mission Harit Maharashtra
- D. Global warming

BIO-12-CH15-0029 | Medium | Levels of biodiversity

A learner encounters the observation 'Variation among genes within a species' while analysing levels of biodiversity. Which conclusion is supported by the prescribed biology?

- A. Genetic diversity
- B. Carbon dioxide
- C. Species richness
- D. Mission Harit Maharashtra

BIO-12-CH15-0030 | Medium | Pollution

A learner encounters the observation 'Increase in concentration of persistent toxins at higher trophic levels' while analysing pollution. Which conclusion is supported by the prescribed biology?

- A. Particulate matter
- B. Ecosystem diversity
- C. Species richness
- D. Biomagnification

BIO-12-CH15-0031 | Medium | Biodiversity patterns

A source table contains one accurate entry involving 'Relationship in which larger areas contain more species'. Which option preserves a correct biological association?

- A. Increase in concentration of persistent toxins at higher trophic levels - Co-extinction
- B. In situ protected area permitting regulated human activity - Seed bank
- C. Secondary extinction following loss of a dependent species - Genetic diversity
- D. Relationship in which larger areas contain more species - Species-area relationship

BIO-12-CH15-0032 | Medium | Conservation

A learner encounters the observation 'Protection outside natural habitats' while analysing conservation. Which conclusion is supported by the prescribed biology?

- A. Biosphere reserve
- B. Chlorofluorocarbons
- C. Ex situ conservation
- D. Endemic species

BIO-12-CH15-0033 | Medium | Pollution

Complete the statement accurately: Nutrient enrichment causing excessive aquatic plant growth corresponds to _____.

- A. Eutrophication
- B. Co-extinction
- C. Species-area relationship
- D. Carbon dioxide

BIO-12-CH15-0034 | Easy | Forests and law

Which biological term or result is specifically identified by the description 'Maharashtra initiative promoting large-scale tree planting'?

- A. Mission Harit Maharashtra
- B. In situ conservation
- C. Biomagnification
- D. Deforestation

BIO-12-CH15-0035 | Easy | Pollution

Which description is correctly represented by Eutrophication in this chapter?

- A. Pollutant load measured through oxygen consumed by microbes
- B. Nutrient enrichment causing excessive aquatic plant growth
- C. Relationship in which larger areas contain more species
- D. Secondary extinction following loss of a dependent species

BIO-12-CH15-0036 | Hard | Pollution

Two conclusions must both be valid to support the reference concept 'Increase in concentration of persistent toxins at higher trophic levels'. Which option contains two accurate associations?

- A. Number of species present in a community - Reforestation; Restoring a forest that once existed but was removed - Species richness
- B. Region with many endemic species under exceptional threat - Seed bank; Ex situ storage of viable seeds - Biodiversity hotspot
- C. Species restricted to a particular geographic area - Species richness; Number of species present in a community - Endemic species
- D. Increase in concentration of persistent toxins at higher trophic levels - Biomagnification; Variety of habitats and ecological processes - Ecosystem diversity

BIO-12-CH15-0037 | Easy | Conservation

Which biological term or result is specifically identified by the description 'Protection of species in their natural habitats'?

- A. Chlorofluorocarbons
- B. Seed bank
- C. Eutrophication
- D. In situ conservation

BIO-12-CH15-0038 | Medium | Biodiversity patterns

A learner encounters the observation 'General change in species richness from equator to poles' while analysing biodiversity patterns. Which conclusion is supported by the prescribed biology?

- A. It decreases
- B. Global warming
- C. Species diversity
- D. Biosphere reserve

BIO-12-CH15-0039 | Easy | Levels of biodiversity

Which description is correctly represented by Genetic diversity in this chapter?

- A. Relationship in which larger areas contain more species
- B. Non-native organism spreading and causing harm
- C. Maharashtra initiative promoting large-scale tree planting
- D. Variation among genes within a species

BIO-12-CH15-0040 | Medium | Pollution

Complete the statement accurately: Pollutant load measured through oxygen consumed by microbes corresponds to _____.

- A. Biochemical oxygen demand
- B. Species richness
- C. Eutrophication
- D. Particulate matter

BIO-12-CH15-0041 | Hard | Conservation

A reviewer compares several entries while checking 'In situ protected area permitting regulated human activity'. Which entry must be corrected?

- A. Rise in average global temperature due to enhanced greenhouse effect - Global warming
- B. Chemicals releasing chlorine that destroys stratospheric ozone - Chlorofluorocarbons
- C. In situ protected area permitting regulated human activity - Biodiversity hotspot
- D. Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra

BIO-12-CH15-0042 | Hard | Threats and extinction

Two conclusions must both be valid to support the reference concept 'Harvesting organisms faster than they recover'. Which option contains two accurate associations?

- A. Maharashtra initiative promoting large-scale tree planting - Carbon dioxide; Gas chiefly responsible for anthropogenic greenhouse warming - Mission Harit Maharashtra
- B. Protection outside natural habitats - Mission Harit Maharashtra; Maharashtra initiative promoting large-scale tree planting - Ex situ conservation
- C. Nutrient enrichment causing excessive aquatic plant growth - Ex situ conservation; Protection outside natural habitats - Eutrophication
- D. Harvesting organisms faster than they recover - Overexploitation; Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002

BIO-12-CH15-0043 | Hard | Conservation

A reviewer compares several entries while checking 'Ex situ storage of viable seeds'. Which entry must be corrected?

- A. Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide
- B. Secondary extinction following loss of a dependent species - Co-extinction
- C. Ex situ storage of viable seeds - Species diversity
- D. Region with many endemic species under exceptional threat - Biodiversity hotspot

BIO-12-CH15-0044 | Medium | Conservation

Complete the statement accurately: Protection outside natural habitats corresponds to _____.

- A. Seed bank
- B. Ex situ conservation
- C. Biosphere reserve
- D. Montreal Protocol

BIO-12-CH15-0045 | Medium | Conservation

A source table contains one accurate entry involving 'Protection outside natural habitats'. Which option preserves a correct biological association?

- A. Variety of species in a region - Co-extinction
 - B. Chemicals releasing chlorine that destroys stratospheric ozone - Biological Diversity Act 2002
 - C. Protection outside natural habitats - Ex situ conservation
 - D. Species restricted to a particular geographic area - Chlorofluorocarbons
-

BIO-12-CH15-0046 | Easy | Biodiversity patterns

Which description is correctly represented by It decreases in this chapter?

- A. Region with many endemic species under exceptional threat
 - B. Variety of species in a region
 - C. Rise in average global temperature due to enhanced greenhouse effect
 - D. General change in species richness from equator to poles
-

BIO-12-CH15-0047 | Hard | Climate and ozone

A reviewer compares several entries while checking 'Agreement aimed at controlling ozone-depleting substances'. Which entry must be corrected?

- A. Variation among genes within a species - Genetic diversity
 - B. Solid particles and droplets suspended in air - Particulate matter
 - C. Harvesting organisms faster than they recover - Overexploitation
 - D. Agreement aimed at controlling ozone-depleting substances - Invasive alien species
-

BIO-12-CH15-0048 | Medium | Forests and law

Complete the statement accurately: Large-scale removal of forest cover corresponds to _____.

- A. Ecosystem diversity
 - B. Major cause of biodiversity loss
 - C. Deforestation
 - D. In situ conservation
-

BIO-12-CH15-0049 | Medium | Conservation

A learner encounters the observation 'Ex situ storage of viable seeds' while analysing conservation. Which conclusion is supported by the prescribed biology?

- A. Species diversity
 - B. Invasive alien species
 - C. Ex situ conservation
 - D. Seed bank
-

BIO-12-CH15-0050 | Easy | Threats and extinction

Which description is correctly represented by Overexploitation in this chapter?

- A. Maharashtra initiative promoting large-scale tree planting
 - B. Indian law providing a framework for conservation and benefit sharing
 - C. Harvesting organisms faster than they recover
 - D. Nutrient enrichment causing excessive aquatic plant growth
-

BIO-12-CH15-0051 | Challenging | Levels of biodiversity

Evaluate the following source-based biological statements.

- I. Number of species present in a community corresponds to Species richness.
- II. Variety of habitats and ecological processes corresponds to Ecosystem diversity.
- III. Variation among genes within a species corresponds to Genetic diversity.
- IV. Maharashtra initiative promoting large-scale tree planting corresponds to Species richness.

Which option identifies exactly the correct statements?

- A. I, II, III and IV
- B. I, II and III only
- C. II and IV only
- D. I and III only

BIO-12-CH15-0052 | Challenging | Climate and ozone

A student submits four conclusions after integrating related chapter concepts.

- I. Chemicals releasing chlorine that destroys stratospheric ozone corresponds to Montreal Protocol.
- II. Agreement aimed at controlling ozone-depleting substances corresponds to Carbon dioxide.
- III. Gas chiefly responsible for anthropogenic greenhouse warming corresponds to Carbon dioxide.
- IV. Number of species present in a community corresponds to Chlorofluorocarbons.

Which combination should be accepted without correction?

- A. I only
- B. III only
- C. I, II and IV only
- D. I and IV only

BIO-12-CH15-0053 | Medium | Levels of biodiversity

A learner encounters the observation 'Number of species present in a community' while analysing levels of biodiversity. Which conclusion is supported by the prescribed biology?

- A. Species richness
- B. It decreases
- C. Ecosystem diversity
- D. Deforestation

BIO-12-CH15-0054 | Medium | Forests and law

A learner encounters the observation 'Indian law providing a framework for conservation and benefit sharing' while analysing forests and law. Which conclusion is supported by the prescribed biology?

- A. Biochemical oxygen demand
- B. Biological Diversity Act 2002
- C. Mission Harit Maharashtra
- D. Global warming

BIO-12-CH15-0055 | Medium | Pollution

Complete the statement accurately: Increase in concentration of persistent toxins at higher trophic levels corresponds to _____.

- A. Ecosystem diversity
- B. Biomagnification
- C. Eutrophication
- D. Particulate matter

BIO-12-CH15-0056 | Challenging | Forests and law

During error analysis, four proposed associations are reviewed.

- I. Indian law providing a framework for conservation and benefit sharing corresponds to Mission Harit Maharashtra.
- II. Maharashtra initiative promoting large-scale tree planting corresponds to Deforestation.
- III. Large-scale removal of forest cover corresponds to Deforestation.
- IV. Solid particles and droplets suspended in air corresponds to Particulate matter.

Which set contains all and only the valid conclusions?

- A. II and IV only
- B. II and III only
- C. III and IV only
- D. II, III and IV only

BIO-12-CH15-0057 | Easy | Forests and law

Which biological term or result is specifically identified by the description 'Large-scale removal of forest cover'?

- A. Carbon dioxide
- B. In situ conservation
- C. Biological Diversity Act 2002
- D. Deforestation

BIO-12-CH15-0058 | Medium | Levels of biodiversity

Complete the statement accurately: Variation among genes within a species corresponds to _____.

- A. Genetic diversity
- B. Species diversity
- C. Species-area relationship
- D. Carbon dioxide

BIO-12-CH15-0059 | Medium | Pollution

A learner encounters the observation 'Nutrient enrichment causing excessive aquatic plant growth' while analysing pollution. Which conclusion is supported by the prescribed biology?

- A. Co-extinction
- B. Species-area relationship
- C. Particulate matter
- D. Eutrophication

BIO-12-CH15-0060 | Hard | Levels of biodiversity

Two conclusions must both be valid to support the reference concept 'Variation among genes within a species'. Which option contains two accurate associations?

- A. Gas chiefly responsible for anthropogenic greenhouse warming - Species-area relationship; Relationship in which larger areas contain more species - Carbon dioxide
- B. Relationship in which larger areas contain more species - Mission Harit Maharashtra; Maharashtra initiative promoting large-scale tree planting - Species-area relationship
- C. Variation among genes within a species - Genetic diversity; Non-native organism spreading and causing harm - Invasive alien species
- D. Nutrient enrichment causing excessive aquatic plant growth - Carbon dioxide; Gas chiefly responsible for anthropogenic greenhouse warming - Eutrophication

BIO-12-CH15-0061 | Easy | Forests and law

Which description is correctly represented by Mission Harit Maharashtra in this chapter?

- A. Maharashtra initiative promoting large-scale tree planting
- B. Large-scale removal of forest cover
- C. Protection of species in their natural habitats
- D. Increase in concentration of persistent toxins at higher trophic levels

BIO-12-CH15-0062 | Medium | Threats and extinction

A source table contains one accurate entry involving 'Destruction and fragmentation of natural habitat'. Which option preserves a correct biological association?

- A. Nutrient enrichment causing excessive aquatic plant growth - Ecosystem diversity
- B. Agreement aimed at controlling ozone-depleting substances - Seed bank
- C. Protection outside natural habitats - Major cause of biodiversity loss
- D. Destruction and fragmentation of natural habitat - Major cause of biodiversity loss

BIO-12-CH15-0063 | Challenging | Threats and extinction

A researcher checks whether four linked observations have been interpreted correctly.

- I. Harvesting organisms faster than they recover corresponds to Major cause of biodiversity loss.
- II. Destruction and fragmentation of natural habitat corresponds to Major cause of biodiversity loss.
- III. Non-native organism spreading and causing harm corresponds to Biosphere reserve.
- IV. In situ protected area permitting regulated human activity corresponds to Overexploitation.

Which combination survives the biological review?

- A. II only
- B. I and III only
- C. II, III and IV only
- D. I, II and III only

BIO-12-CH15-0064 | Medium | Climate and ozone

A learner encounters the observation 'Rise in average global temperature due to enhanced greenhouse effect' while analysing climate and ozone. Which conclusion is supported by the prescribed biology?

- A. Global warming
- B. Biochemical oxygen demand
- C. Deforestation
- D. Genetic diversity

BIO-12-CH15-0065 | Medium | Biodiversity patterns

Complete the statement accurately: Region with many endemic species under exceptional threat corresponds to _____.

- A. Deforestation
- B. Ex situ conservation
- C. Biodiversity hotspot
- D. Montreal Protocol

BIO-12-CH15-0066 | Easy | Pollution

Which description is correctly represented by Biomagnification in this chapter?

- A. Increase in concentration of persistent toxins at higher trophic levels
- B. Nutrient enrichment causing excessive aquatic plant growth
- C. Species restricted to a particular geographic area
- D. Restoring a forest that once existed but was removed

BIO-12-CH15-0067 | Easy | Threats and extinction

Which description is correctly represented by Invasive alien species in this chapter?

- A. Protection outside natural habitats
- B. Restoring a forest that once existed but was removed
- C. Nutrient enrichment causing excessive aquatic plant growth
- D. Non-native organism spreading and causing harm

BIO-12-CH15-0068 | Hard | Climate and ozone

Two conclusions must both be valid to support the reference concept 'Gas chiefly responsible for anthropogenic greenhouse warming'. Which option contains two accurate associations?

- A. Region with many endemic species under exceptional threat - Major cause of biodiversity loss; Destruction and fragmentation of natural habitat - Biodiversity hotspot
- B. Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide; Harvesting organisms faster than they recover - Overexploitation
- C. Secondary extinction following loss of a dependent species - Biodiversity hotspot; Region with many endemic species under exceptional threat - Co-extinction
- D. Protection of species in their natural habitats - Ecosystem diversity; Variety of habitats and ecological processes - In situ conservation

BIO-12-CH15-0069 | Hard | Conservation

A reviewer compares several entries while checking 'Protection of species in their natural habitats'. Which entry must be corrected?

- A. Protection of species in their natural habitats - Eutrophication
- B. Variation among genes within a species - Genetic diversity
- C. Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra
- D. Relationship in which larger areas contain more species - Species-area relationship

BIO-12-CH15-0070 | Hard | Pollution

A reviewer compares several entries while checking 'Solid particles and droplets suspended in air'. Which entry must be corrected?

- A. Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002
- B. Number of species present in a community - Species richness
- C. Solid particles and droplets suspended in air - Chlorofluorocarbons
- D. Variation among genes within a species - Genetic diversity

BIO-12-CH15-0071 | Challenging | Biodiversity patterns

Evaluate the following source-based biological statements.

- I. Region with many endemic species under exceptional threat corresponds to It decreases.
- II. General change in species richness from equator to poles corresponds to It decreases.
- III. Relationship in which larger areas contain more species corresponds to Deforestation.
- IV. Large-scale removal of forest cover corresponds to Deforestation.

Which option identifies exactly the correct statements?

- A. II and III only
- B. I, III and IV only
- C. I, II and IV only
- D. II and IV only

BIO-12-CH15-0072 | Medium | Threats and extinction

A learner encounters the observation 'Non-native organism spreading and causing harm' while analysing threats and extinction. Which conclusion is supported by the prescribed biology?

- A. Eutrophication
- B. Ex situ conservation
- C. Invasive alien species
- D. Reforestation

BIO-12-CH15-0073 | Easy | Climate and ozone

Which biological term or result is specifically identified by the description 'Agreement aimed at controlling ozone-depleting substances'?

- A. Carbon dioxide
- B. Montreal Protocol
- C. Overexploitation
- D. Ex situ conservation

BIO-12-CH15-0074 | Hard | Threats and extinction

Two conclusions must both be valid to support the reference concept 'Non-native organism spreading and causing harm'. Which option contains two accurate associations?

- A. Chemicals releasing chlorine that destroys stratospheric ozone - Reforestation; Restoring a forest that once existed but was removed - Chlorofluorocarbons
- B. Non-native organism spreading and causing harm - Invasive alien species; Protection outside natural habitats - Ex situ conservation
- C. Nutrient enrichment causing excessive aquatic plant growth - Chlorofluorocarbons; Chemicals releasing chlorine that destroys stratospheric ozone - Eutrophication
- D. Pollutant load measured through oxygen consumed by microbes - Eutrophication; Nutrient enrichment causing excessive aquatic plant growth - Biochemical oxygen demand

BIO-12-CH15-0075 | Medium | Forests and law

Complete the statement accurately: Restoring a forest that once existed but was removed corresponds to _____.

- A. Reforestation
- B. Biodiversity hotspot
- C. Montreal Protocol
- D. Biosphere reserve

BIO-12-CH15-0076 | Medium | Climate and ozone

Complete the statement accurately: Rise in average global temperature due to enhanced greenhouse effect corresponds to _____.

- A. Genetic diversity
- B. Global warming
- C. Chlorofluorocarbons
- D. Particulate matter

BIO-12-CH15-0077 | Hard | Forests and law

Two conclusions must both be valid to support the reference concept 'Maharashtra initiative promoting large-scale tree planting'. Which option contains two accurate associations?

- A. Agreement aimed at controlling ozone-depleting substances - In situ conservation; Protection of species in their natural habitats - Montreal Protocol
- B. Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra; Variation among genes within a species - Genetic diversity
- C. Solid particles and droplets suspended in air - Invasive alien species; Non-native organism spreading and causing harm - Particulate matter
- D. General change in species richness from equator to poles - Montreal Protocol; Agreement aimed at controlling ozone-depleting substances - It decreases

BIO-12-CH15-0078 | Challenging | Conservation

A student submits four conclusions after integrating related chapter concepts.

- I. In situ protected area permitting regulated human activity corresponds to Biosphere reserve.
 - II. Protection of species in their natural habitats corresponds to Seed bank.
 - III. Ex situ storage of viable seeds corresponds to It decreases.
 - IV. General change in species richness from equator to poles corresponds to Biosphere reserve.
- Which combination should be accepted without correction?

- A. IV only
- B. I and IV only
- C. I only
- D. II only

BIO-12-CH15-0079 | Easy | Biodiversity patterns

Which description is correctly represented by Biodiversity hotspot in this chapter?

- A. Agreement aimed at controlling ozone-depleting substances
- B. Protection outside natural habitats
- C. Region with many endemic species under exceptional threat
- D. Secondary extinction following loss of a dependent species

BIO-12-CH15-0080 | Hard | Threats and extinction

A reviewer compares several entries while checking 'Destruction and fragmentation of natural habitat'. Which entry must be corrected?

- A. Variety of habitats and ecological processes - Ecosystem diversity
- B. Destruction and fragmentation of natural habitat - Eutrophication
- C. Restoring a forest that once existed but was removed - Reforestation
- D. Protection outside natural habitats - Ex situ conservation

BIO-12-CH15-0081 | Medium | Climate and ozone

Complete the statement accurately: Agreement aimed at controlling ozone-depleting substances corresponds to _____.

- A. Biosphere reserve
- B. Carbon dioxide
- C. Montreal Protocol
- D. Ex situ conservation

BIO-12-CH15-0082 | Easy | Climate and ozone

Which description is correctly represented by Global warming in this chapter?

- A. Rise in average global temperature due to enhanced greenhouse effect
- B. Large-scale removal of forest cover
- C. Ex situ storage of viable seeds
- D. Chemicals releasing chlorine that destroys stratospheric ozone

BIO-12-CH15-0083 | Medium | Threats and extinction

A source table contains one accurate entry involving 'Secondary extinction following loss of a dependent species'. Which option preserves a correct biological association?

- A. Maharashtra initiative promoting large-scale tree planting - Eutrophication
- B. Variety of species in a region - Montreal Protocol
- C. Secondary extinction following loss of a dependent species - Co-extinction
- D. Pollutant load measured through oxygen consumed by microbes - Particulate matter

BIO-12-CH15-0084 | Challenging | Climate and ozone

During error analysis, four proposed associations are reviewed.

- I. Rise in average global temperature due to enhanced greenhouse effect corresponds to Carbon dioxide.
- II. Gas chiefly responsible for anthropogenic greenhouse warming corresponds to Carbon dioxide.
- III. Chemicals releasing chlorine that destroys stratospheric ozone corresponds to Chlorofluorocarbons.
- IV. Variation among genes within a species corresponds to Global warming.

Which set contains all and only the valid conclusions?

- A. I, II and III only
- B. II and IV only
- C. II and III only
- D. I and IV only

BIO-12-CH15-0085 | Easy | Levels of biodiversity

Which biological term or result is specifically identified by the description 'Variation among genes within a species'?

- A. Invasive alien species
- B. Genetic diversity
- C. Carbon dioxide
- D. Mission Harit Maharashtra

BIO-12-CH15-0086 | Medium | Forests and law

A source table contains one accurate entry involving 'Large-scale removal of forest cover'. Which option preserves a correct biological association?

- A. Chemicals releasing chlorine that destroys stratospheric ozone - Endemic species
- B. Destruction and fragmentation of natural habitat - Species diversity
- C. Large-scale removal of forest cover - Deforestation
- D. Variety of species in a region - Deforestation

BIO-12-CH15-0087 | Hard | Climate and ozone

Two conclusions must both be valid to support the reference concept 'Rise in average global temperature due to enhanced greenhouse effect'. Which option contains two accurate associations?

- A. Large-scale removal of forest cover - Species diversity; Variety of species in a region - Deforestation
- B. Rise in average global temperature due to enhanced greenhouse effect - Global warming; Pollutant load measured through oxygen consumed by microbes - Biochemical oxygen demand
- C. Solid particles and droplets suspended in air - Genetic diversity; Variation among genes within a species - Particulate matter
- D. Variety of species in a region - Biosphere reserve; In situ protected area permitting regulated human activity - Species diversity

BIO-12-CH15-0088 | Easy | Conservation

Which description is correctly represented by In situ conservation in this chapter?

- A. Nutrient enrichment causing excessive aquatic plant growth
- B. Protection of species in their natural habitats
- C. Ex situ storage of viable seeds
- D. In situ protected area permitting regulated human activity

BIO-12-CH15-0089 | Medium | Threats and extinction

A source table contains one accurate entry involving 'Harvesting organisms faster than they recover'. Which option preserves a correct biological association?

- A. Harvesting organisms faster than they recover - Overexploitation
- B. Agreement aimed at controlling ozone-depleting substances - Ex situ conservation
- C. Maharashtra initiative promoting large-scale tree planting - Reforestation
- D. Indian law providing a framework for conservation and benefit sharing - Eutrophication

BIO-12-CH15-0090 | Easy | Pollution

Which biological term or result is specifically identified by the description 'Solid particles and droplets suspended in air'?

- A. Eutrophication
- B. Genetic diversity
- C. Biological Diversity Act 2002
- D. Particulate matter

BIO-12-CH15-0091 | Challenging | Pollution

A researcher checks whether four linked observations have been interpreted correctly.

- I. Nutrient enrichment causing excessive aquatic plant growth corresponds to Eutrophication.
- II. Pollutant load measured through oxygen consumed by microbes corresponds to Biochemical oxygen demand.
- III. Increase in concentration of persistent toxins at higher trophic levels corresponds to Biomagnification.
- IV. Protection outside natural habitats corresponds to Ex situ conservation.

Which combination survives the biological review?

- A. II only
- B. IV only
- C. I, II, III and IV
- D. I and IV only

BIO-12-CH15-0092 | Hard | Pollution

Two conclusions must both be valid to support the reference concept 'Nutrient enrichment causing excessive aquatic plant growth'. Which option contains two accurate associations?

- A. Nutrient enrichment causing excessive aquatic plant growth - Eutrophication; Variety of habitats and ecological processes - Ecosystem diversity
- B. Relationship in which larger areas contain more species - Carbon dioxide; Gas chiefly responsible for anthropogenic greenhouse warming - Species-area relationship
- C. Large-scale removal of forest cover - Overexploitation; Harvesting organisms faster than they recover - Deforestation
- D. Secondary extinction following loss of a dependent species - Deforestation; Large-scale removal of forest cover - Co-extinction

BIO-12-CH15-0093 | Medium | Levels of biodiversity

A source table contains one accurate entry involving 'Number of species present in a community'. Which option preserves a correct biological association?

- A. Large-scale removal of forest cover - Species-area relationship
- B. General change in species richness from equator to poles - Global warming
- C. Secondary extinction following loss of a dependent species - Endemic species
- D. Number of species present in a community - Species richness

BIO-12-CH15-0094 | Hard | Forests and law

A reviewer compares several entries while checking 'Maharashtra initiative promoting large-scale tree planting'. Which entry must be corrected?

- A. Maharashtra initiative promoting large-scale tree planting - Genetic diversity
- B. Protection of species in their natural habitats - In situ conservation
- C. General change in species richness from equator to poles - It decreases
- D. Solid particles and droplets suspended in air - Particulate matter

BIO-12-CH15-0095 | Hard | Forests and law

Two conclusions must both be valid to support the reference concept 'Indian law providing a framework for conservation and benefit sharing'. Which option contains two accurate associations?

- A. Pollutant load measured through oxygen consumed by microbes - Montreal Protocol; Agreement aimed at controlling ozone-depleting substances - Biochemical oxygen demand
- B. In situ protected area permitting regulated human activity - It decreases; General change in species richness from equator to poles - Biosphere reserve
- C. Agreement aimed at controlling ozone-depleting substances - Global warming; Rise in average global temperature due to enhanced greenhouse effect - Montreal Protocol
- D. Indian law providing a framework for conservation and benefit sharing - Biological Diversity Act 2002; Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide

BIO-12-CH15-0096 | Hard | Threats and extinction

A reviewer compares several entries while checking 'Secondary extinction following loss of a dependent species'. Which entry must be corrected?

- A. Solid particles and droplets suspended in air - Particulate matter
- B. Secondary extinction following loss of a dependent species - Species diversity
- C. Maharashtra initiative promoting large-scale tree planting - Mission Harit Maharashtra
- D. Species restricted to a particular geographic area - Endemic species

BIO-12-CH15-0097 | Medium | Conservation

A source table contains one accurate entry involving 'Ex situ storage of viable seeds'. Which option preserves a correct biological association?

- A. Agreement aimed at controlling ozone-depleting substances - Invasive alien species
- B. Ex situ storage of viable seeds - Seed bank
- C. Non-native organism spreading and causing harm - Overexploitation
- D. Secondary extinction following loss of a dependent species - Biodiversity hotspot

BIO-12-CH15-0098 | Medium | Conservation

A source table contains one accurate entry involving 'In situ protected area permitting regulated human activity'. Which option preserves a correct biological association?

- A. In situ protected area permitting regulated human activity - Biosphere reserve
- B. Nutrient enrichment causing excessive aquatic plant growth - Carbon dioxide
- C. Chemicals releasing chlorine that destroys stratospheric ozone - Biosphere reserve
- D. Rise in average global temperature due to enhanced greenhouse effect - Montreal Protocol

BIO-12-CH15-0099 | Easy | Conservation

Which biological term or result is specifically identified by the description 'Ex situ storage of viable seeds'?

- A. Montreal Protocol
- B. Invasive alien species
- C. Ex situ conservation
- D. Seed bank

BIO-12-CH15-0100 | Medium | Climate and ozone

A source table contains one accurate entry involving 'Gas chiefly responsible for anthropogenic greenhouse warming'. Which option preserves a correct biological association?

- A. Gas chiefly responsible for anthropogenic greenhouse warming - Carbon dioxide
- B. Variety of habitats and ecological processes - Biodiversity hotspot
- C. Region with many endemic species under exceptional threat - Eutrophication
- D. Protection of species in their natural habitats - Co-extinction